浙江理工大学 · 材料类

CET4&6 词汇长文读本

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把大纲词表，读成一个人的大一

这本书不是单词书。它是一本小说——一个材料类大一新生的第一年，八章，从报到那天的雨，写到答辩那天的风。

书的背面藏着一件事：《全国大学英语四、六级考试大纲》词表里的 7418 个词形，一个不落，全部出现在正文里；每个词第一次出现时加粗；每一章末尾的「伴读」，用中文把这些词讲一遍——讲它在这一章里的意思、它的搭配、它最容易跟谁混。

你读到加粗的词，不必停下来查。先把它当线索，不把它当负担：一个词在故事里出现一次，你就已经认识它的脸了，等它在「伴读」里再出现，才算点头之交；等它在后面几章又出现，才是真的熟了。

怎么读这本书

第一遍，把正文当小说读。 遇到加粗的词，允许自己看不懂——上下文会给你八成意思，剩下两成留到伴读。这一遍只做一件事：读完这一章，合上书，能用一句话说出这一章发生了什么。

第二遍，读伴读。 伴读不是词汇表，它是"章后闲聊"：哪两个词长得像但不是一家人，哪个词在中文里有个听起来一样、意思完全不同的对头，哪个词的搭配一旦错了就立刻露馅。读到你自己觉得"原来如此"的地方，划一下——那才是你的收获。

第三遍，回正文。 挑一章，只读加粗的词，把它们串起来念一遍。你会发现这些词自己也在讲一个故事：有些词属于雨天，有些词属于实验室，有些词属于外婆的灶间。

排版与阅读环境（导出 PDF 时照这个设）

正文是英文长句，行距建议 1.6–1.8，字号 11–12pt；中文伴读字号 10.5–11pt，行距 1.7。中西文之间留半个字符的空隙（不要挤在一起）。每页页边距 2.2–2.5cm，页脚放页码即可。不要用两端强制对齐——英文长句被拉开字距会很难看；左对齐更自然。

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第 1 章　报到与军训

The carriage

The train accelerated out of the last tunnel and the light came back so abruptly that I shut my eyes. My suitcase was wedged between my knees and the seat in front. On top of it, attached with a strap I had bought at the station for eight yuan, sat a carton with my bedding inside, tied and retied until the knot looked like a badge of my mother's anxiety.

Across the aisle an accountant from Jinhua was telling anyone within reach about his son's aptitude for math. The boy, he said, could do triple integrals in his head. He was fourteen. His wife did not applaud. She was busy trying to adjust the angle of the air vent above her, and the vent kept swinging back of its own accord, which she took as antagonistically deliberate.

Beside the window a girl in a grey T-shirt held a violinist's case across her knees, and a bouquet of cloth flowers stood up out of a basket at her feet. She was reading a thin album of photography, turning the pages with one thumb, and each time the carriage jolted she shifted her elbow to keep the bouquet upright. Her badge said she was going to the art academy on the other side of the river. She got off one stop before me and left a bouquet-shaped hole in the air.

Outside, the terrain flattened into grassy marsh, then into hay fields already cut, then into a village with one chimney where somebody was burning something, ash drifting down over the plough lines. Below the embankment the river came up wide and tidal, the colour of tin. Mist hung in the humid afternoon; the window glass was humid too, and my chin left a print on it that I wiped away with my sleeve.

Hangzhou East arrived with a boom of brakes. Everybody stood up at once and the attendant's voice on the speaker was buried under the noise of crowded aisles. I attached the carton to the suitcase with the strap again, retrieved my bag from the rack, pushed out through the barrier with the gear I had brought and the gear I had forgotten, and stood on the square in a heat that felt like a wet hand on the back of my neck.

The van that carried us to Xiasha smelled of gasoline and hot vinyl. My cargo went in the back with two mattress rolls and a rack of coat hangers. The load shifted at every junction; the drainage covers clicked twice under the tyres; the driver swiped his phone at a toll gate with one thumb, without looking, and the gate lifted before he had finished the gesture.

The gate

At the south gate a barrier arm rose for us and a guard waved the van through without lifting his eyes from his phone. The campus opened up on the printed sheet in my hand — a map Mr Wang had emailed in August and my mother had printed at the shop downstairs four days ago, folded into eight rectangles, softened along the folds, the way a thing is softened by being carried. I held it like an atlas. It told me where the auditorium was, and the stadium, and the library that a famous architect had drawn, whose blueprint hung in the entrance hall behind glass.

The directory board by the path had been overhauled in August and the new paint was still tacky, so a fly was stuck in it, and the fly was the only thing on that board that looked alive. I stood in front of it for a long while. Four years of my life were listed there in white plastic letters, and I could not distinguish the dorm I had been allocated from the annex beside it, because the two blocks had names that differed by one character and the characters were printed the same size.

That is where the confusion started. It was not the fearful kind. It was a slow bewilderment, the feeling of being ahead of yourself and behind yourself at the same time. I walked along a junction of two paths, turned apart from the crowd, addressed a student with a briefing folder under his arm, and he pointed along a row of bikes without stopping.

Then I heard it. Beyond the grassy bank, on the far side of the stadium, a battalion was forming up: two hundred figures in camouflage, one whistle, one roar of a greeting that came back off the grand stand and revolved in the humid air. I stood with the carton on my shoulder and admired them, which is a strange thing to admit now, because in three days I would be one of them, and I would hate every minute of the first hour and pretend otherwise.

The first rain

The rain came at the junction by the canteen, and it came abruptly, the way the light had come back in the tunnel. Amid the first burst a beggar under the awning of the laundry shop pulled a sheet of plastic over his basket of peanuts and pastry, and two boys dashed for the annex with their raincoats still folded under their arms.

I had a waterproof cover for the carton and nothing for myself. My sweat turned cold. Beads of it ran from my hairline to my chin, and then the rain took over and I could not tell which was which. The drainage could not take it: water came up out of the grating and stood ankle-deep at the corner, and a bolt of lightning made everybody below the awning look up at the same moment.

Somewhat to my surprise, the absurdity of my position did not strike me until later. I had an agenda in my pocket, a folded sheet of notions my mother had written for me inside brackets, and her firstly was buy a bucket, her secondly was find the dorm, and

her third was do not lose the receipt for the bedding. I had lost the receipt in the van. I stood in the downward pour with a carton of bedding on my shoulder and giggled at nothing, and the woman selling biscuits out of a cart looked at me the way you look at a boy who has not yet learned what a storm costs.

Least of all did I expect the man who came out of the dorm door with a broom. He was barely taller than me, darker, in a ragged vest, and he did not say a word. He took one end of the carton, walked it up four flights, and pushed the door open with his heel. His name, I would learn within the hour, was Zhang Ye. This is the part I remember best, and I remember it not because it was grand but because nothing about it was.

The rest of the afternoon is a blend of doorways, stairs, tiles and signs, and at the end of it, a door with a number on it, and behind that door, my room.

Room 407

The room was four beds, four desks, one window, one fan, and a smell I could not name until Zhang Ye named it for me: hot dust on a heated conductor, he said, which is what a dorm smells like in September when nobody has used the sockets for three months. He had already chosen the bed by the window and already taken the cabinet nearest it, and he had done this not out of greed but out of a calculating habit I would come to admire: the criterion was the distance from the electrical outlet, because he had a device that needed charging and a soldering iron he did not intend to explain.

He was from a village in Lishui, and his surname was Zhang, and he did not offer a nickname and did not ask for mine. His luggage was one canvas bag and one tin box with a hinge and a latch — hinge, latch, I know — and inside the box were a screwdriver, a small lever, a cylinder of grease, a roll of glue tape, two bolts, a needle, a spool of thread, and an eraser cut in half. He laid them out on the desk in a column, looked at them, and put them back in a different order.

The kettle on the windowsill was dependable in the way that a thing is dependable when it is broken in exactly one place. He took off the base, found a crack in the insulation — insulation, yes — and fixed it with a strip of stainless shim cut from a tin lid. Electrical work, he said, is mostly avoidance: you do not counteract a fault, you isolate it. Then he boiled water for the four of us and the steam condensation ran down the window and made the humid room humider.

The second classmate to arrive was from Wenzhou. His father, he said, was a salesman who had become a boss, and the transition had made the father generous and the son talkative, and between them they had filled two suitcases and a carton of snacks. He offered cheese crisps and coconut candy out of a basket and told us, within twenty minutes, the gross income of the family hardware shop, the debt his uncle had run up, and the quota of cigarettes his mother allowed his father per week. Zhang Ye listened without

a single interruption. When the boy finally paused, Zhang Ye said: "Eat your supper. The canteen closes at seven." It was four o'clock.

I unpacked. The closet had a drawer that stuck and a rod that sagged, and I hung my two shirts on it and stacked the rest in the cabinet in evenly folded piles, which is the only tidy thing I did that week. My bedding went on the upper bunk because the lower one was already taken by a quiet boy from Huzhou who had arrived at eight in the morning and had been asleep since. Through the window I could see the annex, the stadium lights coming on at six, and a slack clothesline with one towel on it, dying in the heat.

Supper was in the canteen on the ground floor. Either of the two buffet lines would do; neither was short. I took rice, eggplant in oily sauce, tomato and egg, a slice of pork, and a bowl of soup that tasted of corn and nothing else. The calorie count of that tray must have been enormous, and I ate it anyway, because the alternative was to sit in a room with three strangers and listen to my own stomach.

At the corner of the corridor, above the fire hose, was the bulletin board. Somebody in the office had pinned the schedule for the first week: registration on Monday, a briefing in the auditorium on Monday afternoon, a medical check on Tuesday, and then, in a bracket at the bottom, military training from Wednesday, fourteen days, curfew at twenty-two thirty, no exceptions. Someone had drawn a skull in the margin. Someone else had written a date under it, twice, and the second date was worse than the first.

I went to bed at nine and lay there listening to the fan and the drainage pipe and, beyond the wall, a thousand boys shouting at each other in nine dialects. Tomorrow the queue for registration would start at six. This I knew because Zhang Ye had already read the notice, already worked out the route, and already set his phone alarm for five forty-five. He fell asleep in four minutes. It took me three hours, and in the dark I could hear myself thinking, which is always a bad sign.

The hall

Registration opened at six. By six twenty the queue had folded back on itself three times, an assortment of anxious teenagers in an order that made no sense until you saw that it was a single line. Above us, in a frame bolted to the wall, hung an aerial photograph of the campus taken in a August when the trees were smaller, and the photo was the only honest map in the building.

Inside, steel brackets divided the hall into nine windows, and the arrangement was designed to make you walk past eight of them before you reached the one you wanted. Behind each window sat a woman with a stamp. The glass was frosted to the height of a face, which gave every clerk the same anonymity and every applicant the same disadvantage: you could not tell whether the person deciding your morning was smiling.

The form was a comprehensive thing, four pages long, and it wanted my surname and my given name, my national number, my occupational history, which was none, the educational level of both parents, my dental record and, in a clause at the bottom, my consent to the collection of my data. The instruction above the signature line was a masterpiece of ambiguity. It said: sign here and, if applicable, on the reverse. I turned it over. The reverse was blank. When I asked the clerk whether the reverse applied, she said that the reverse applied to students in a different category, and when I asked which category, she said the category was described in the handbook, and when I asked where in the handbook, she said that the handbook had been revised and that the relevant page was, she believed, no longer accurate.

This was my first encounter with a bureaucracy, and what surprised me was not the absurdity of it but the competence. Nobody was being cruel. The woman at window three had a method, and the method worked, and the method had been designed by somebody who had never stood on this side of the glass. She examined my signature, held the page up to the light, and put it down. It was the first time a signature of mine had ever been examined by anybody, and I felt absurdly proud of it.

I was exempt from the English placement test. This was excellent news and also a small curse, because the exemption put me in a class where everybody else had been exempt too. I only knew that I had been spared an hour of grammar, and I walked out with my stamped papers feeling superior to the people still in the queue, which is the cheapest feeling available to a teenager and the one I have indulged most often.

At window seven a foreigner was in trouble. His visa had been issued for a programme that no longer existed, and the deadlock lasted nine minutes, and it was broken by an older woman who came out from behind the glass, read his letter and said, "This is fine." The compassion in her voice was not professional. It was the compassion of somebody who had once been on the wrong side of a window herself, and it did more for the reputation of that office than the whole of its charter.

The money window was next. Tuition could be paid in instalments, which was a form of credit and also a form of debt. There was a coupon in the envelope for the bedding pack, another for the mattress, and a quota note explaining that the grant would be suspended if my average fell below a line that nobody had defined. The transaction took forty seconds. The receipt came out of a printer that sounded like a wasp.

Behind me two boys were discussing the canteen. One said the pork was delicious; the other said the pork was catastrophic. They were both right.

The first class meeting

Mr Wang took us in the evening in a classroom on the fourth floor. He was about forty, wore a polo shirt washed into a soft grey, and began by telling us that everything

we were worried about was not a problem. The network is down? Not a problem. The laundry eats your sock? Not a problem. You miss your mother? That, he said, is a problem, and we will talk about it later.

He gave out a charter, the student handbook summarised onto two sides of one sheet, and read out the part that prohibits certain things: heating elements, pets, liquor, fireworks and, in a phrase that made the room laugh, any device whose power rating exceeds the capacity of the room. Zhang Ye, three seats ahead of me, did not laugh. He had a soldering iron in a tin box under his bed that could probably have run a small factory.

Then the election. The class needed a chairman. When a girl at the back asked whether it could be a chair, Mr Wang said, "When the ministry changes the form, I will change the word." The process would be a ballot, openly counted, because a secret vote in a room of thirty people is a fiction.

Four hands went up. The boy from Wenzhou promised snacks twice a week, and his promise was so boastful and so precise that it won him nothing. A girl with a very quiet voice promised to answer messages within the hour. Zhang Ye did not stand. He wrote a name on a slip of paper, folded it, put it in the box, and the name he wrote was not his own.

I asked him afterwards why he had not run. He said a chairman spends his evenings on the phone to people who want things, and that he would rather fix things than ask people to do them. Whom he voted for I never found out. The count was announced, the girl with the quiet voice won by two, and everybody applauded, and she cried a little, and the meeting broke up at nine.

Walking back, he said one more thing. He said the election had been decided before it began, not by a conspiracy but by a habit: the people who talk get remembered, and the people who fix things get asked to fix things, and those are two different jobs and only one of them comes with a title. Then he told me to stop writing everything down, because a person who writes things down is a person who intends to remember, and remembering is a kind of debt.

The opening ceremony

The opening ceremony was on the Saturday, in the auditorium, and it began with an accessory check at the door: belt, badge, cap and the ribbon for the cap, which nobody had told us to bring. An actor from the city theatre read a poem. An artist from the art academy had designed the backdrop, and an astronaut — a real one, retired, seventy years old — was introduced, stood up, and said nothing at all, which the affluent parents in the third row found amusing and the rest of us found awesome.

Then the dean spoke. He said the university existed to abolish three habits: the habit of aspiring to be taught, the habit of asserting before assimilating, and the habit of treating an argument as a quarrel. He accused us, politely, of approaching knowledge as a consumer approaches a buffet, and the acidity of that image stayed with me for a week. He said that adaptation without alteration is only imitation, that the adoption of a method without its premise is superstition, and that the advantage of a young mind is not speed but flexibility. He used the word attributive about courage, and I did not understand him until my second year.

He warned us about adverse conditions of a kind we had not considered: alienation, which he called the occupational disease of the clever; agitation without direction; the allure of ambiguity, which makes a person sound deep; and the alleged freedom of a student who has been appointed to nothing and affiliated to nobody. He said the anchorage of a first-year student is a room, a timetable and one friend, and that if you lose the third, the first two will not hold. Nothing in that hall was airborne except the confetti, and I remember thinking that the old man on the stage had been, once, in the air.

The walk afterwards

Afterwards I walked the campus alone, because the ceremony had filled me with a restlessness that I could not spend in a chair. Behind the biology building a society was holding a bake sale on a folding table, and the benevolence of the arrangement — a cake, a box, a hand-written sign saying pay what you like — was undercut by a buyer who took two slices and put in nothing, and by the argument that followed, which was louder than the cake was good.

Somebody told me the society was nearly bankrupt. Somebody else said it had been bankrupt for six years and that this was a tradition. The cause was a coach that the club had bought in 2019 and could not afford to run, and the coach was now a shed, and the shed was full of canoes, and the canoes had not been in the water since the epidemic. I looked at the shed and thought about the word capital, which I had only ever met in an economics textbook, and understood for the first time that capital is a thing that can also be a canoe.

A bitch with a broken tail was sleeping under the table. A blonde girl in a lab coat came out of the biochemistry building with a rack of samples and stepped over the dog without looking, and the dog did not move, and the girl said, "She has been here longer than I have." Somebody blamed the university for the dog. Somebody else said the dog was a lesson in tenure.

Past the sports ground there was a celebration going on for a championship that a team I had never heard of had won in a sport I did not follow, and the noise of it came

across the grass in waves. Beyond it stood the old chapel — a chapel, in the middle of a technical university, built in 1911 and deconsecrated in 1958 — and somebody had hung a banner on it about the census, which I thought was the strangest pairing of words I had seen that year.

The corridor

That evening the corridor of the dormitory filled with the noise of forty people who had nothing to do. Somebody had brought a small television and the news was on, and the news was the same news it always is: a correspondent standing in front of a building, a crackdown on something, coverage of a storm that had killed eleven people in a province I could not have placed on a map.

A boy from the room at the end was chewing sunflower seeds and spitting the shells into a cup, and the sound of it was a kind of clock. He said the storm was nothing, that his hometown had been hit by a worse one in 2016, and that the coverage had been wrong then too, and that the correspondent had stood in the shallow end of a flood and made it look deep. He said the trick was to stand where the water was moving and let the camera do the rest.

Two boys were arguing about a girl and the argument had reached the point where both of them were making the same point in different words, which is the point at which an argument becomes a ritual. The word loyalty was used. The word respect was used. Neither of them was listening, and both of them were enjoying themselves, and when it was over they went to get noodles together, which is how I learned that a clash between friends is not the opposite of companionship but one of its forms.

I sat on my bed with my back against the wall and listened to all of it — the seeds, the news, the argument, the complacency of the boy who had won the argument and the complacency of the boy who had lost it. The room was coarse and loud and it smelled of socks and instant noodles, and my complexion in the mirror was the colour of a boiled dumpling, and I was, for the first time since I arrived, entirely happy.

That night I made a resolution, and the resolution was concise: I would not spend the next four years waiting for something to happen to me. I wrote it on the first page of the notebook, in a hand that was trying to look older than it was, and I wrote the date because I had already begun to suspect that I would want to know the date later. Beneath it I wrote a second line, which was less a resolution than a confession: that I had no conception of what I wanted, and that the concealment of that fact from everybody, including myself, was going to be the real work of the year.

The notice about the money

The notice went up on the Friday and it was four lines long. It said that a student

who wished to appeal against the evaluation of an application must submit a copy of the declaration in conjunction with a letter from the class delegate, and that the deadline would be enforced. The author of the notice was not named. This is a circumstance I would come to know well: in a bureaucracy no document has an author, and every document has an authority.

The background to the notice was a quarrel that had been going on for a week. A boy in the fourth room had been given a grant that another boy thought he deserved, and the contention had moved from the corridor to the class meeting to a group chat with ninety-one members, and by the end of it nobody could remember the original facts. The two positions were contradictory and both were sincere, which is the worst kind of conflict, because a conflict between a liar and an honest man can be settled.

Mr Wang listened to all of it and then said something that was not clever but was correct: that the committee had followed the rules, that the rules were stupid in one place and silent in another, and that a person who wants a stupid rule changed should not do it by attacking the person who complied with it. He said the appeal existed for exactly this reason and that using it was not an accusation. Then he said the thing that made the room go quiet: that in four years, the only people who would remember this week were the two of them, and that they would remember it wrong.

I admired him that evening. It is easy to be brave in a lecture and hard to be brave in a corridor, and what he did was harder than brave: he was consistent. He had said the same thing about the same rule in September, and he said it again in November, and he said it in the same words, and the consistency was worth more than the eloquence, because a student can argue with an opinion but cannot argue with a repetition.

The laboratory corridor

Shen Lu took me through the laboratory corridor on the Sunday because I had asked her what a materials student actually does. The corridor smelled of solvent and hot dust. Behind one door a furnace was running at nine hundred degrees and the door was warm to the back of the hand. Behind another, a machine the size of a car was making a noise like a dentist's drill slowed down.

She said the first thing you learn is that contamination is the enemy, and the second is that most contamination is you. She said a fingerprint on a sample is a mountain, a hair is a range of mountains, and the dust from your own sleeve will sit on the surface and lie to the microscope. She said the discipline of the place is not cleanliness for its own sake but consistency: the same glove, the same angle, the same order of operations, every time, until the procedure is a part of your body and you can no longer do it wrong.

Then she said something that surprised me. She said the hardest part of the work is

not the experiment but the discourse — the arguing, the writing, the standing up in front of forty people who have read your abstract and want to know why your numbers do not agree with somebody's numbers from 2018. She said a result that nobody can reproduce is not a result, it is a rumour, and that the difference between the two is the whole of what a doctorate trains you to protect.

I asked her whether the work was creative. She said the word was too generous and that she preferred the word deductive: you do not invent a hypothesis out of the air, you deduce it from a gap in the literature, and the gap is usually somebody else's bad paragraph. She said the breakthroughs that get into the newspapers are the convergence of ten thousand boring afternoons, and that the afternoons are what she actually loves, and that she would not say so at a party.

The quarrel on the track

The quarrel on the track was about a bottle of water. It was a hot Tuesday, the fourth day, and a boy had taken another boy's bottle out of the crate and drunk half of it, and the second boy, who had been brooding about something else entirely, told him to choose his own bottle next time, and the first boy said something about the second boy's mother, and by the time the instructor arrived there were two people shouting and forty people watching.

The instructor did not shout. He made them stand in front of each other for a minute in silence, facing, at attention, while the rest of us stood in a block and watched. It was the longest minute of the fortnight. Then he asked the first boy what he had wanted, and the boy said water, and he asked the second boy what he had wanted, and the boy said respect, and the instructor said, "So one of you wanted a drink and the other wanted a ceremony. Which of those can I give you?"

Everybody laughed, including the two of them, and the laugh did most of the work. But what I remember is the instructor's debrief afterwards, delivered to the whole block with no names in it. He said a group of a hundred and twenty people will always contain one person who is having the worst week of his life, and that you will never know which one, and that the only rational policy is to be brisk with everybody and harsh with nobody.

Nobody was detained, nobody was expelled, and no report was ever written. But the phrase stayed with me: the worst week of his life. I looked around the block that afternoon and tried to guess who it was, and I could not, and the not being able to was the lesson.

The language of the place

The university spoke in decrees. A decree about the definition of a credit; a decree about the cubic capacity of a locker; a decree about the demonstrative use of a student

card; a decree about the desirability of a uniform. Each one was demanding and each one was decent in intention, and each one was, in effect, a defiance of common sense, and the accumulation of them was a kind of weather.

I began to collect the words the place used, the way other people collect stamps, and I put them in the back of the notebook where nobody would look. Deploy, for the movement of a class from one building to another. Detain, for a student kept behind after a lecture. Despatch, for the sending of a form, and dispatch for the sending of a person, which was the same word twice and a different word each time. Depress, for what a lift button does, and depress for what a November does to a person, and the second meaning arrived in my life about six weeks later and stayed for a year.

Some of the words were about damage. Deterioration, for the state of the third-floor lavatory. Devastating, for the news from a province that had been hit by a storm while we were doing drill. Diagnosis, for what the clinic gave you when it did not know. Contamination, for what a fingerprint does to a sample. Disintegration, for what happens to a club when its chairman graduates. Diffusion, for the way a rumour crosses a dormitory in ninety minutes; I timed it once, with a story I had invented, and it crossed four floors before I had finished my noodles.

Some were about people. A dissident was somebody who asked the second question at a meeting. A disposition was what a person had instead of a mood: a mood lasts an afternoon, a disposition lasts a father. Dignity was the thing the instructor gave you when he corrected you in private and not in front of the block. And a division, in the language of the place, was not a quarrel but a unit; the quarrels were called something else, and the something else was never written down.

The library had a dimension I had not expected. It was not the size of the building, which was ordinary, but the depth of the shelves, which went back four rows behind the reading room, and the further back you went the older the bindings and the fewer the people. There was a dip in the floor near the window where ten thousand feet had turned the same corner. Books disappeared from that room at a rate of about four a year, and the librarian had a theory about it that involved no thieves at all, only people who had carried a book out by accident in a bag and been too ashamed to bring it back.

I asked her once whether the collection was diverse, meaning, I think, whether it was any good, and she said the collection was diverse in the way that a forest is diverse, and that a student who wanted the current year's journals would be disappointed, and that a student who wanted the 1958 volumes would find them where they had always been. Then she said the word cultivation, which I had only met in a book about farming, and she used it about readers.

The infirmary

The infirmary was two rooms and a corridor and a woman who had seen everything twice. My heel had gone bad on the ninth day and I went in the evening, and there was a queue of four: a boy with a knee, a girl with a fever, a boy who had fainted on the track and did not want to say so, and me. The woman took one look at my heel and said, "Sit," and I sat, and she did not ask me why I had waited six days, which was the kindest thing anybody did for me that month.

She said the word fatigue, and then she explained it, which nobody does. She said fatigue is not tiredness; tiredness goes away with a night's sleep, and fatigue accumulates, and the accumulation is invisible until the day you cannot lift your own leg. She said the camp produced forty cases of it every year and that the number was stable, which meant the number was designed, which meant somebody somewhere had decided that forty was acceptable.

I asked her whether the training was dangerous. She said the training was not dangerous and the boys were, and that in her experience the two things that sent a first-year to hospital were a basketball hoop and a bicycle, and that nobody had ever sent her a poster about either. Then she gave me a tablet, and a tube, and a piece of advice I have followed ever since: treat the small wound on the day you get it, because the small wound is the only one you can afford.

The little museum

The old campus is a mile up the river, and the buildings there were put up between 1902 and 1930 by people who had decided that a technical school should look like a monastery. Shen Lu took me on the Sunday, and we walked through a gate in a wall and into a courtyard where the noise of the city stopped as if a door had closed.

The chapel has not been a chapel since 1958. It is a hall now, with a stage at one end, and the Christian symbols were taken down so carefully that you can still see where they were, four pale shapes on the plaster, like the ghosts of furniture in a repossessed house. The guide, who was a student doing this for a credit, said the building had been a chapel, then a store, then a dormitory for a hundred boys, then a store again, then a hall, and that the only thing that had never changed was the roof. Somebody asked whether the school had been a Christian school. She said yes, at the beginning, and that this was in the museum and that nobody was hiding it, and that the founders had been Christians and the students had mostly not, and that the two facts had coexisted for thirty years without either of them winning.

The museum was two rooms. In the first there was a case of porcelain from a kiln that had been under the hill, and the labels explained that the kiln had made bowls for four hundred years and had been killed by a railway in 1909. In the second there was machinery: a lathe from 1928, a pump, a shaft from a mill, and a wall of photographs of

people in aprons standing in rows.

I stood in front of a photograph for a long time. It was 1936, and there were nineteen people in it, and eight of them were women, and the caption gave their surnames and their grades but not their first names, and Shen Lu said that this was how you could tell what the place had thought of them. She said a caption is a political document, and that the omission of a first name is a decision, and that the archive is full of these decisions and that they are the most honest thing in it.

The last exhibit was a single sheet of paper under glass, a hand-drawn sketch of a machine that had never been built, with the proportions wrong and the measurements marked in a system that nobody uses any more. The student guide said it was the museum's favourite object. She said it was the only object in the room that was pure intention, and that everything else in the room had been built, and that there was something unbearably hopeful about a drawing that had failed.

The last week

The last week of training was a strange week because the thing we had hated had become a habit and the habit was about to be taken away. The drills got easier. The heat did not. The instructor began to joke, in a flat voice, and his jokes were terrible and we laughed at all of them, and I understood that the laughter was not about the jokes.

The closing ceremony was on the Friday and it rained, and two hundred of us stood in the rain in a block and were inspected by a man from the province who did not speak to us. There was a march, a flag, a band from the music school, and a speech that was twelve minutes long and contained no information. Then it was over. The block broke up, and people stood around in the wet with their helmets in their hands, and nobody knew what to do with the feeling, so a few boys started shouting the company song and the rest joined in, badly, and the sound of two hundred wet people singing a song they had learned eleven days earlier was the best thing that happened in my first month.

Zhang Ye did not sing. He stood with his helmet under his arm and looked at the ground, and on the way back he said, "That is the end of the only thing we have all done together." I said we would do other things together. He said no, we would do things in the same building, which is not the same thing, and he was right, and it took me a year to admit it.

That evening the whole cohort had supper in the canteen and the canteen ran out of rice. Somebody had a speaker and somebody else had a guitar and there was dancing in the queue. A boy from the fourth room, who had spent the fortnight in a state of complaint that had been almost professional, stood up on a chair and made a speech about how much he had hated every minute, and everybody cheered, and he cried, and when he got down he was embarrassed for about four seconds and then he was not.

I wrote in the notebook that night that I had arrived expecting to be taught and had instead been issued something, and that I did not yet know what it was. I wrote the word issuance and crossed it out, and then I wrote the word gift, and crossed that out too, and left the line blank, and the blank line is the most accurate thing in the whole notebook.

The laboratory tour

Shen Lu walked me through the building on the Sunday and she talked the way a surgeon talks, naming the instrument and the reason and nothing else. The basement was the machine shop: a shaft from a press that had broken in 2019 and been kept because the shaft was the only part worth keeping, a screw cut in half to show its thread, a lever with a wooden handle worn into the shape of a hand, a furnace that ran at nine hundred degrees, and a bolt bin that had not been sorted since the year she arrived.

She said the difference between a student and a practitioner is that a student asks what a machine does and a practitioner asks what it does when it is tired. She said a metallic surface is a lie told at the scale of a micron, and that a lens will believe it. She said you must learn to purify a solvent before you may touch the good one, and that the good one is locked in a cabinet because a solvent is a luxury, and that sodium, which will take your hand off in water, is stored in oil like a preserved fruit.

On the bench there was a mould with a hairline crack, a tube of sealant, a packet of swabs, a mug that had been used for something it should not have been used for, and a thermometer that read four degrees high. She said the thermometer was the most dangerous instrument in the room, because a thermometer that lies is worse than no thermometer at all, and that a plausible reading is the most expensive thing in any laboratory.

Then she explained the sequence, and she did it as a list of verbs, which is how she did everything. You mount; you polish; you etch; you rinse; you dry; you inspect; you record. You do not skip, you do not improvise, and you do not omit the record, because an experiment that was not recorded was not an experiment, it was a rumour. She said the word rigour again and this time she gave it a definition: rigour is the willingness to be bored by your own procedure for the tenth time on the tenth day.

The instrument room was colder. There was a scanner the size of a fridge, a microscope with a column and a stage, and a poster on the wall of a molecular model that looked like a handful of coloured beads thrown at a wall. She said this is where the money is, and this is where the queue is, and that a student who can get an hour on this machine at short notice is a student with a mentor who is owed a favour.

I asked her what she was working on. She said a solid-state battery, a membrane, and a number she could not yet explain. She said the number was either a real effect or a defect in her sample, and that the difference between those two was going to cost her

four months. She said the work is mostly a long argument with yourself in which you are both the prosecutor and the witness, and that the only way to win is to be honest before you are caught.

Her reading list

She gave me a reading list on the back of an envelope, six lines, and told me not to read any of it until November. The list was: a history of the science of materials, because you cannot understand a method without its history; a book on probability, because half of what looks like noise is noise; a manual on crystallography, because it is the grammar of the subject; a book about the migration of birds, which she said was not a joke and would teach me more about long-distance navigation than any lecture; and a novel by a northern writer about a shipyard, which she said was the best book about engineering ever written by somebody who was not an engineer.

I asked about the sixth line. The sixth line was a single word, aesthetics, and she said that a person who cannot tell a good experiment from a bad one by looking at the design will never be any good, and that the sense of the elegant is trainable, and that you train it by looking at a thousand designs and asking which one a lazy man would have built.

She said something about reading that I have never heard from a teacher. She said you should read the way you shop: you walk in, you look at everything, you pick up three things, you put two back, and you leave. She said a person who reads every page of every book is a person who has confused obedience with literacy.

Then she said the thing I remember most, and she said it while walking away, which is how she said everything worth remembering. She said the phenomenon that ruins a first-year student is not difficulty, it is the discovery that nobody is going to tell him what is worth doing, and that the discovery arrives at different times for different people, and that when it arrives it feels exactly like being abandoned.

The second talk

Mr Wang gave a second talk on the Sunday evening and he called it, without fuss, an introduction to the institution. He said the function of the first year is not to select but to expose, and that exposure is expensive, and that somebody has to pay for it, and that the somebody is the taxpayer, which makes the equal distribution of the chance an ethical question and not an economic one.

He said the system is equitable in design and erroneous in detail, and that a person who confuses the two will spend four years angry at the wrong target. He used the word explicit three times and the word exclusive once, and the once was deliberate: he said the best resources here are exclusive by nature, because a microscope is one instrument

and a mentor is one person, and that frustration is the emotion you feel when you have mistaken a right to compete for something for a right to have it.

He said he wanted to expel one idea from the room, and the idea was that enthusiasm is a substitute for method. He said he had seen a hundred enthusiastic students in his career and that the ones who lasted were the ones who had, in addition to the enthusiasm, a procedure. He said an entrepreneurial student is not a student with a business plan, it is a student who can find the third way when the first two are closed, and that this is a skill and not a personality.

He said the word frontier and then apologised for it, and then used it again, because, he said, there is no other word for the place where the known stops. He said every field has one, that most students never see it because they are taught the middle of the field and the middle is very comfortable, and that the ones who see it are the ones who read the last paragraph of every paper instead of the first.

Then he talked about cooperation, and he was scornful of the word as it is used in the brochures. He said cooperation is not a feeling, it is a division of labour with a procedure for disagreement, and that a group of friends who agree about everything is not a team, it is a choir. He said the test of a team is what happens in the fifteen minutes after the first person says something unpopular, and that he had seen more projects destroyed by politeness than by conflict.

He said the school was federal in one respect and unified in another, and that the difference mattered when you needed a signature. He said the constituency of a dean is not the students, whatever the dean says, and that this is not cynicism, it is architecture. He said a student who understands the architecture will get his form signed in a morning, and a student who does not will write three letters and get nothing, and that this has nothing to do with justice and everything to do with construction.

He ended with something he had obviously said before, and he said it slowly, the way a man says a thing he has decided to be remembered for. He said: do not be dominated by the timetable. He said the timetable is a contingent arrangement made by somebody who was in a hurry in 2019, and that a person who lets a contingent arrangement dictate his consciousness will graduate with a very good attendance record and nothing else. He said the ones who frighten him are not the lazy ones. They are the complacent ones, who are busy from six in the morning until eleven at night and could not tell you, in one coherent sentence, what they are busy for.

The room upstairs

The room upstairs belonged to two boys from the north and it had the only couch on the floor, which they had carried up four flights in September and would carry down four flights in June, and which had been rained on so many times that the fabric had faded in

a pattern like a map.

The room was where people went when they craved company without conversation. There was a dryer in the corner that somebody had bought and that nobody knew how to use, a heap of textbooks that had been dumped there in October and never opened, a poster of a doll in a folk costume that somebody's mother had sent, and a crumble of biscuit crumbs on the couch that changed shape every week and never disappeared.

The customer base of that couch was large and its loyalty was absolute. People came at midday and at midnight, they brought a bag of nuts, a mouthful of somebody else's noodles, a mug of tea; they asked the same questions and got the same nonsense in reply. The folklore of the floor was made there: which canteen window was the worst, which instructor was the strictest, which of the fifteen lifts in the teaching building actually worked, and how many of them would be broken by December.

It was there, on a Thursday, that the drought of news broke. Somebody came in with a rumour that the training would be extended by a week, and the rumour had a footprint on every floor within twenty minutes, and by midnight it had a cause, a source and a denial, and the denial had more authority than the rumour, and both were false.

I liked that room better than my own. It smelled of dust and orange peel and somebody else's soap, and the edge of the couch had a spring that stuck out and had torn a hole in two pairs of trousers, and there was a flare of temper about once a fortnight when the argument went too far, and then somebody would make a joke that was not funny, and it would pass, and the couch forgave everybody, forever.

The morning of the fifteenth

On the morning of the fifteenth the sky came up fast and false, all pink, and then the rain came back and the whole campus smelled of wet concrete. The street outside the south gate was full of small catastrophes: a cancellation notice taped to the noodle shop's door, a delivery of foam packing that had been soaked and was now a large wet animal on the pavement, and a casino of a fruit machine in the amusement centre that had shorted in the night and been unplugged and left, its screen still showing three lemons.

I bought a sausage roll at the bakery, which was neither sausage nor roll, and ate it standing under the awning while a civilian guard smoked a cigar in the doorway and told me, at length, about the bribe he had been offered in 2014 by a man who wanted to bring a truck through the north gate after curfew. He said the bribe had been two hundred yuan and a box of cigarettes, and that he had taken neither, and that his honesty had cost him the friendship of the man, and that he would do it again.

The biochemical building had a leak that morning, and a slow brown drip was coming through the ceiling of the corridor onto a doll that somebody had lost under the stairs.

A technician came with a bucket and stood looking at it, and said the leak had been reported in March and that the report had gone to the office, and that the office had a list, and that the list was long, and that the word "pending" was doing a lot of work on that list.

I sat on the stairs and beat the rain for twenty minutes and then gave up and went to brunch, which is what the canteen called it and what it was not. The bud of a camellia outside the window was the only thing on that street that had not been rained on twice, and I watched it for a while and thought about the difference between damages that can be repaired and damages that are merely absorbed.

The last Sunday

The gallery was shut, the garage was open, and the island in the kitchen was a formidable fleet of pots. Automation had not reached the clearance desk, and the confidence of the conservative clerk was, to me, intrinsically interesting: she had been doing the same task for nineteen years and had never once been asked whether it could be done differently.

A controversial heading in the handbook had been altered twice that summer. The alteration was legitimate and nobody had been told, and the first person to notice was an emigrant student who read the handbook for pleasure, which is why he was the only person in the building who knew that the fine for an irregular absence had doubled. He was emotional about it for a day, and then he wrote a letter, and the letter was an indictment of the whole procedure, and the procedure survived the letter, as procedures do.

The lady in the laundry was lately widowed and kept a hound that was moody in the mornings. Her hobby was to cut the label out of every garment that came through the shop, and the labels were her gallery, and the gallery was a private heritage of a kind that no museum would ever want. She could tell you where a coat had been bought, in what year, and whether the buyer had been happy.

I went to an inn on the north road for supper, because I had heard there was an inn and because I wanted one afternoon that was not the campus. The inn had a holy picture over the door, a hook for a coat, and a landlord who was illiterate in two languages and fluent in a third, which was the language of a man who has fed strangers for thirty years. His hospitality was not a performance. He gave me a plate of ham and a filling of peppers and a glass of something he called wine and was probably not, and he asked me where I was from and then told me he had been there in 1989.

His son was a guitarist, and there was jazz on a keyboard in the corner, and the son played while his father cleared the plates, and the two of them did not look at each other once. It was a small and complete world, and I was inside it for an hour and a

half, and when I left the father said goodbye in the way of a man who does not expect to see you again, and I have not seen him again.

On the way back I crossed the footbridge and stood for a while with my elbows on the rail. The river was a silvery smear under the lamps, the water going south at a walking pace, a plastic bottle turning slowly in the current like something asleep. A dog barked on the far bank. The lights of the shipyard went out at ten, one by one, in an order that was always the same.

I thought about the word logic, which had been used at me four times that week by people who meant something else; about the word intuitive, which is what people call a conclusion they cannot defend; about the word literate, which the library used to mean a person who could use the catalogue and which I had come to understand as something closer to patience. I thought about the infant in the flat below ours and the frequency of his crying, and about the smell of the dryer in the room upstairs, and about how a place becomes yours not when you can find your way around it but when you can predict what will annoy you.

The last thing I did that Sunday was to sit on the steps of the annex and watch the bats come out over the stadium. They came out in ones and twos, then in a stream, and they were fast and irregular and they never touched each other, and I sat there until it was too dark to see them, and then I went in.

The letter

Dear Ma,

I have been trying to convince myself that the cosmic indifference of this place is not hateful, and I have almost succeeded. Doubtless you will enquire how I am eating. The fellows in my room distribute the snacks evenly, which is more than the canteen does, and I have entered into an engagement with the noodle shop at the south gate: eleven yuan, no questions, extra mushroom.

I want to tell you the small things, because the large things are all in the handbook. There is a creep in the corridor at night, a sound and not a person, and I cannot find its source; the drainage, I think, or a loose panel that expands with the heat. There is a datum I have been collecting without meaning to: the number of times a day the lift stops at my floor without being called. It is eleven. I do not know why I am counting, and I am hesitant to tell anybody, because a person who counts lifts is a person with time to count lifts.

The lectures are fantastic and useless in equal measure. One was given by a man who studies the genetics of behaviour, and he made an ecological argument that was so expressive that half the room left convinced and the other half left angry. An

environmentalist in the third row asked a question with four clauses in it, and he answered with a gesture. I have decided that the word exponent, when it is used about a person, is always a compliment paid by somebody who wants something.

The ethics lecture was worse. The man was idealistic in a way that made me want to argue, and the argument I wanted to make was that his ethics had no external check on it, and that a code with no check is not a code but a taste. I did not make the argument. I sat there and felt the idea hover over the room, and then the bell went, and everybody stood up, and it was gone.

I have joined nothing and signed nothing. There is an imperative in the first month to join, to sign, to enter, to be seen entering, and I have been resistant to it, not out of principle but out of a kind of sleepiness. You will say that this is the family character. I think it is the family character.

I am learning words. This is the one improvement I can report. I keep a list in the back of the notebook, of words I have met and cannot yet use, and the list is getting longer, which I have decided is a good sign and not a bad one. Yesterday I met the word insulate, in a lecture about keeping heat in a house, and I wrote it down because it seemed to me that a person can be insulated from a place and still live in it, and that this is the danger here.

Two things are impossible to describe: the sound of two hundred people turning on gravel, and the smell of the corridor on a wet Monday. I have tried. I have made an illustration of the first in four sentences and thrown it away. The glossary of this place is not in the handbook, and no amount of input from me will insert it there.

There was a boy in the queue who had a book with a hyphen in the title and a bookmark at page nine, and he had been at page nine for three weeks, and he was not embarrassed. I envied him. It seems to me increasingly that the people who do well here are not the intelligent ones, not especially, but the ones who can endure being at page nine. Excuse the handwriting. The desk is too high and my elbow has nowhere to go, and I am writing this with the book on my knees, which is why every third line is farther from the margin than the last.

I have to go. There is a knock at the door and it is probably Zhang Ye with a broken kettle and no intention of explaining. Write to me about the bird. I want to know whether it is still in the air conditioner, and whether it has had the sense to move on, and whether the air conditioner has forgiven it.

Your son

The last pages of the notebook

Damn was the instructor's word for the weather, used once in eleven days and

apologised for immediately afterwards. Deregulate was what the taxi company wanted; disarmament was the lecture that got postponed three times; elicit was what a good question was supposed to do, and mostly did not.

The office had embedded the word free in the timetable so thoroughly that it had lost its meaning. An empress on a bowl in the museum had been enclosed in glass with a label, and the bowl had been found in a field, and the field was now a car park. Somewhere on the fourth floor there was a poster about an endangered habitat, and the enjoyment I took in reading it was out of all proportion to its content, and I have wondered since whether the pleasure of a poster is the pleasure of understanding something at a glance.

The esteem in which the museum held its own exotic things was visible in the lighting, which was better than the lighting in the library. Nothing there was experimental; everything was an extravagance that somebody had paid for in 1911 and that we were now paying for again, at four yuan a ticket, in order to be excited for eleven minutes.

Facilitation was a word on a form, and festival was a night in the stadium with a float shaped like a panda, and a flu went round the dormitory the week after. The canteen was forgiving about the flu and less forgiving about the queue. To fortify a person against a winter in this city you need soup, sleep and one friend, and of the three the soup is the only one the university provides, and it is not enough, and it is not nothing.

The subject was a foster child of four departments and none of them wanted custody, and the fragmentation of the timetable was the proof. Everybody was frantically busy. Everybody was friendly. Geographically the campus is one square kilometre, and socially it is four villages that share a canteen, and a germ that starts in one of them will be in all four within a week.

A glance at the corridor told you who had been up all night. There was glitter on the floor of the common room for four days after the festival and nobody swept it. The rules that govern a dormitory are not written and are enforced by a grin, and the grin is the most efficient instrument of government I have ever lived under.

He was the only homosexual person in the building who was out, and he was also the most popular, and the two facts were connected in a way that it took me a year to understand. The food was horrible and the company was not. A human being can eat horrible food for a long time if the table is good, and I ate, and I was hungry at eleven, and I was happy.

I have an idiomatic problem: I can read a sentence and know what it means without being able to say what it says. The ignition of the whole business, for me, was a single illustrative paragraph in a textbook about a bridge that fell down, and I have been

trying to imitate the paragraph ever since, and it cannot be done, because a good illustrative paragraph is not a method, it is a gift.

The import of the word independence took a while to reach me. It is not a number of inches of personal space, and it is not the absence of a rule; it is the presence of a reason. Indoors, in the first week, I had an infinite amount of both space and time and no reason at all, and I was less independent than I had been at school, where at least the bell told me what I was for.

The most influential sentence anybody said to me that month was not in a lecture. It was in the queue for the laundry: an older student said that the infringement of a rule and the inheritance of a habit look the same from outside and are completely different from inside, and then he asked me to pass the basket, and I have never seen him again.

They inject you with the timetable in September and the injection takes about three weeks to work. The insertion of one compulsory lecture into a free afternoon can ruin the afternoon; this is a fact about attention and not about lectures. The intangible benefit of the whole fortnight was a set of habits, and the international students, who had the same fortnight in English, seemed to acquire them faster, which I put down to their having already learned once how to be a stranger.

The first intimate conversation I had here was with a boy I had known for nine days, at two in the morning, about his father. The invention of a new self is the project of the first term and it is conducted in public and it is exhausting. An investigator would have found, in that room, five boys all doing the same thing at the same time and all convinced they were doing it alone.

I had jetlag for three days and I had not been on a plane. The joyous thing about the first month is that everything is new; the other thing about the first month is that everything is new. The landlady of the laundry, who was not a landlady in any legal sense, called me "the boy who reads", and I have been trying ever since to deserve it.

A layman thinks the legacy of a place is its buildings; the buildings are the least of it. Liberty, in the mouth of a first-year student, usually means the absence of a curfew, and it takes about a term to discover that the absence of a curfew is not freedom but a slightly larger cage with better hours. My life was limited by nothing and I did nothing, and the linguistics of that sentence — the passive, the double negative of the mood — is exactly right.

The location of the building was on every map and the entrance was not. Longitude and latitude I could give you; the door was behind a hedge. Somebody had looted the vending machine in the second week and the loss was three hundred yuan, and the machine was repaired, and the person was never found, and the corridor talked about it for a month with more admiration than blame.

He was loyal to a football team in a city he had never visited, and he was luminous about it, and it maddened me, and I envied him. A magnificent obsession is a kind of shelter. Mainstream taste in music, in clothes, in girls, was a subject he could discuss for hours, and the makeup of his argument was always the same: a statistic, a joke, and a personal insult.

My mam — I have never called her that, and I wrote it once in the notebook to see how it looked — would have said he was a manly little thing with too much time. He had a maneuver for every situation, and it was always the same maneuver, and it worked, because most situations are the same situation. A marine would have recognised the type: a boy who has decided that the world is a series of rooms and that the trick is to be the one who knows where the door is.

The masculine style in the dormitory was a style of not saying things, and the style was exhausting, and the ones who suffered most from it were the ones who were best at it. The mayor of the campus was the head of the student union, and the title was a joke until you needed a form signed. A medal in a glass case in the hall, from a championship in 1987, was the only object in the building that nobody had ever tried to steal.

The medication came in a paper packet with the dosage written on it in a hand I could not read. A melody from a film was stuck in my head for eleven days. The memoration of the dead, spelled with a prefix the dictionary does not allow, is a word I invented because I could not spell the real one, and I keep the mistake in the notebook on purpose.

I was immensely, mensely, wrong about the amount of work. My first immersion into the library lasted four hours and ended with me asleep on the third floor. The metro is eleven minutes away and feels like an hour, and the migrating birds go over the stadium in October in numbers that the students ignore, and I have never seen anybody look up.

Inventory

The museum's war room, which was one wall of the second gallery, held a parachute folded into a canvas roll, a tent with a hole in the roof, and a medal in a case. The label said the parachute had been cut from a wreck and used as a tarpaulin for eleven years, and that this was the ordinary fate of military cloth. There was a photograph of a runway in 1944 with a rocket in the foreground that nobody could identify, and a length of textile the colour of weak tea, and a spade.

The rest of the room was about siege and survival. A ration tin, a fold of paper with a vow on it, a statute of the provincial government, a photograph of a city under siege with the smoke going up in three columns. The guide used the word sacrificial about the men in the photograph, and then the word salvation about a hospital ship, and then she said the word warfare and stopped, and said that she preferred the word "the war", and

that this was a preference and not a correction.

I have written before about the sport, but not about the wrestling, which was on the television in the pub on a Saturday and which four men watched with the intensity of scholars. The tournament was in a city I had heard of; the referee was a man with a whistle and a face like a disappointed uncle; the sportswoman who won the final had a knee strapped and a smile that went off like a light. A sponsor's name was on her shirt in a language I could not read. The whole thing was a spectacle, and it was a splendour, and when it was over I was speechless in a way that had nothing to do with the sport.

The season turned in the second week of October. The tropical air went away and the mornings came up sunny and cold, and the smog came back with the cold, and the sprinkler on the lawn was still going at six in the morning in the middle of a drought, which the groundsman explained by saying that the sprinkler was on a timer and that the timer was on a list. A turtle in the pond behind the library came out onto the bank to sit in the sun, and a boy who had been fishing for nothing all afternoon caught it by accident and put it back with an apology.

The year has a shape and the shape is paperwork. You register in September; you proofread a form in October; you attend a meeting whose proceedings are printed and then shredded; you fill in a questionnaire about the printing of the questionnaire. The prerequisite for the second-year course is the first-year course; the preference of the department is for students who have already chosen; the prospect of a place on the programme is described in a profile that has been promoted to three hundred people and read by nine. The first prime of a student's life is the month in which he discovers that the production of documents is the actual work of an institution, and that everything else is a by-product.

The president of the student union came to speak to us, and he was pretty good, and pretty is the right word. He was preceded by a girl who read out the previous minutes, and followed by a senator from the provincial parliament who spoke for twenty minutes about the poll that had elected him and the representative who had preceded him, and about his hopes for the republic, which were modest and sincere. I do not remember a word of it. I remember that the radiator in that room was painted over so many times that the ridges had gone smooth, and that somebody had scratched a date into it with a nail.

The town outside had its own inventory. A man peddled chestnuts out of a drum from October to January, and the portion he gave you for five yuan was smaller every month, and he was hurt if you said so. A woman sold poultry from a cage and killed the bird behind a curtain. There was pizza, which was not pizza, and a pub with two taps and a poster of a panda, and a punch bag in the window of a gym that nobody ever used. The shopping street was one hundred metres of possession and postponement: everything you could want, nothing you could afford, and a sign in every window explaining that the shop

was closed for redecoration and would be open again on a date that had passed.

The body keeps its own accounts. My skin came off the back of my neck in the third week in a sheet, and under it was skin that had not seen the sun since I was eleven. There was a sore on my heel that took a month, a spine that ached at the base, a tissue in the shoulder that clicked when I lifted the bag, and a slump in the afternoon from two until four that no coffee touched. A nurse looked at my throat and said it was not the flu; the radiation from the screen and the glare from the window, she said, and told me to sit further back and look up more often, and I have never done either.

The technical things I learned in that fortnight were three. A cell is a chemical substance before it is anything else. A voltage is a pressure and a current is a flow, and a person who confuses them will burn his hand. A simulation is a scenario with a parameter in it, and if you cannot say which parameter you are least sure of, you have not understood your own simulation.

The rest is small. A drawer that will not shut, a lock that will not turn, a nail in the wall that holds a coat and nothing heavier. A sheet of paper over a window because the curtain rail fell down. A boy in the next room who could not sleep without a light and another who could not sleep with one, and the compromise they reached, which was a towel over the gap under the door and a promise, and the promise was kept for four years.

There was a widower on the third floor, sixty years old, doing a second degree for reasons of his own, and he was the only person in the building who was never in a hurry. He said the workforce is a thing you join and a thing you leave and that in between it is mostly a question of who you eat lunch with. He said the young are not weak, they are only unweighed.

I have one more thing and then I will stop. On the last night of the fortnight I stood at the window of 407 and looked down at the yard, and there was a stray cat and a broken bicycle and a puddle with the lights of the annex in it, and I felt, for the first time, that I was not a visitor. Nothing had changed in the yard. The change was in the vocabulary: the puddle was mine, the light was mine, the annex was mine, and the word vacancy on the notice board had stopped being an insult and become a possibility.

Winter comes early

By the end of October the mornings had a bite in them and the campus changed its habits. The healthy ones ran at six. The rest of us stood at the window with a mug and watched them run, and there was a mutual understanding that we were the reason they ran and they were the reason we watched.

I could not spell immersion and wrote im + mersion in the margin; I could not spell commit and wrote com + mit; I could not spell communicate and wrote com + municate. Three

words, three fragments, one habit: when a word will not go in whole, break it and put the pieces in one at a time. It is a childish method and it has never failed me.

The mindset of the place changed in November. A minor illness went round the floor and everybody got it and nobody mentioned it, and the misery of a blocked nose in a room with three other people is a specific and underrated misery. It is easy to mock the boy who talks about his health; it is harder when the boy is you and the illness is a nuisance that will not end. Nevertheless, I would rather have the flu in a dormitory than a cold in a single room, and I have had both.

The nephew of the landlady came to stay for a week and he was a small, loud monster of nine years old, and the whole building was governed by him for seven days. He would narrate his own actions in the third person; he would obsess about a football sticker album; he would commit small acts of obscenity with a piece of chalk on the wall of the stairwell and then deny them with a straight face. Nobody was negligent with him and nobody was careful with him either, and he survived, and the wall did not.

The rest of it is a list of things that were done to me or near me. I was made to participate in a survey about obesity that took forty minutes and asked me to weigh myself on a scale in the corridor. I was made to opt for one of three meal plans, and the option I chose was withdrawn in the second week, and the withdrawal was announced in a notice that used the word regret twice. An opponent of the plan wrote a letter to the office and got a reply in six days, and the reply was a model of polite refusal, and he framed it.

The ordeal of the first examination was not the examination. It was the preceding week, in which four hundred people discovered simultaneously that they had been reading the wrong chapters, and the library ran out of the two books that mattered, and a boy in my room began to rise at four in the morning and to speak in a new and terrible voice about the origin of the syllabus.

The whole business had an orthodox shape: a lecture, a reading, a test, a rank. What was not orthodox was the way the rank was received. The boy who came third was pleased and pretended not to be; the boy who came fortieth was relieved; the boy who came last was, for about a day, famous.

I have been trying to be a more outgoing person and it is not working. I am polite, I am present, I am not, in any sense that would satisfy a questionnaire, outgoing. There is a difference between a person who talks and a person who is not afraid, and I had assumed for eighteen years that they were the same person, and they are not.

The outrageous thing, the thing I could not get over, was that it all worked. The arbitrary timetable, the pointless questionnaire, the tyrannical curfew, the mournful food — the machine produced, out of four hundred frightened people, something that

looked like a cohort. It was imperfect and it was unjust in small ways and it was, on the whole, the best thing that had ever been done to me.

I sat on the steps on the last evening and tried to picture the yard in four years and could not. A painter would have made it beautiful; a photographer would have made it true; I could only make it a list. Pail, bicycle, puddle, window, cat. The list is not art and it is not evidence and it is what I have.

There is a pillar in the entrance hall with a plaque on it, and the plaque has the name of a man who gave the money for the building and the dates of his birth and death. Somebody has rubbed the plaque until the letters are soft. I put my hand on it once, on the way past, for no reason I could defend, and I have thought about that gesture more often than it deserves. It was not respect. It was the wish to touch something that had lasted.

A pioneer in any field is a person who is wrong in public for long enough to be right. A president is a person who has learned to be wrong in private. A policeman is a person who has seen the same mistake nine hundred times and has not yet become cruel about it. A steamer is a ship, a stewardess is a woman who brings you tea on it, and a sphere is the shape of every drop of water in the world, and I learned all three of those from a book I read in the week between the training and the term, sitting in the sun on the roof of the annex with my shirt off, which was the last warm afternoon of the year.

The last of it

Reality arrived on the Monday with a repetitive noise: a refrigerator humming, a receiver crackling, a shuttle bus reversing. The term had begun and the campus had doubled in population, and the suburb that surrounded it had doubled in traffic, and the suburb's answer to the traffic was to widen the road and its answer to the widening was more traffic, which is the traditional answer and has never once sufficed.

I had a partner for the chemistry practical, a quiet boy with a snobbish streak and a reliable memory, and he could recite a safety sheet from the first line to the last and would not touch a beaker until the sheet had been recited. He was sensitive to smells in a way that made him useless in the laboratory and invaluable in the canteen. He had a youthful face and a weary walk, and he was the sole reason I passed that term.

The teaching had a predictable shape: a lecture that was a retelling of the textbook, a tutorial that was a retelling of the lecture, and a laboratory that was an entirely different and more perilous thing. A residual perplexity stayed with me for months. The subject was not difficult. The subject was a series of small, exact, unforgiving procedures, and the difficulty was that a procedure has no rhetorical power, and I had been trained for eighteen years to be persuaded by rhetoric.

I was not persuaded by the safety film, which showed a man who had lost a hand and a man who had lost an eye, and which used the word trauma eleven times and the word carelessness not at all. The film was an oddity: the victims were given a voice and the voice was calm, and the calm was more frightening than any yell would have been.

Somebody had told the first-years that the drop-out rate was one in nine, and somebody else had said it was one in twenty, and the figures were both wrong and both useful. I was careful with my own figure. I did not want to be a statistic, and I did not want to be a lesson, and I had begun to understand that a person who wants neither of those things has to be willing to be tedious.

I went to see the counsellor in October, not because I was in trouble but because everybody had been urged to go, and the appointment was forty minutes of me talking and her listening, and at the end she said that I was unusually good at describing my feelings and unusually bad at noticing them. She asked me whether I had cried at any point since August, and I said no, and she wrote something down, and then she said the word grief, and the word landed somewhere under my ribs and stayed there for the rest of the afternoon.

Her office had an oval mirror and a poster of a pianist and a small wooden yacht on the windowsill that somebody had made. She had a pool of her own, she said, meaning a swimming pool, and she swam at six in the morning four days a week, and she said it was the only thing that kept her from taking her own patients home with her. She said a counsellor who does not have a remedy for herself is a danger to everybody, and that this was true of every profession and that nobody says it in a graduation speech.

The word ownership came up in a lecture about intellectual property, and the lecturer spent a while on the difference between owning a thing and owning the right to a thing, and the difference, he said, is the whole of the modern economy. I wrote it down and later wrote next to it a sentence of my own: a student owns nothing, and that is the freedom of the position, and it lasts four years, and then you are, in every sense, a respondent.

I have a confession about the last page. I wanted to end with something significant, something that would unify the whole fortnight and make it a story rather than a list. I sat with the notebook open for a long time, and I could not do it, and what I wrote instead was this: the fortnight was not an undertaking, it was a season. A season does not resolve. It thins out, it turns, and one morning you are wearing a coat, and the coat is the answer.

And then, because I was nineteen and could not leave a solemn sentence alone, I wrote underneath it: also, the food got better in November, which proves nothing about anything except the appointment of a new cook, and the new cook's name was Chen, and he had a way with pork that could reconcile a man to a whole institution, and did.

Coda

Here is the rest of it, and then I am done. A soccer ball on the field at dusk; a toy yacht in a pool; a sizable trunk of wood in the yard, cut and stacked by somebody who had a plan for it; the zoo's old shed, which was not a zoo and had not been a shed for years.

Henceforth I would oversee my own renewal, or so I wrote, and the writing of it was already a renewal of a kind. I had a passionate regard for the place that was not the same as affection and that I could not name. It was patriotic in a small way and it embarrassed me, and when I tried to be persuasive about it to my mother I heard myself sounding like a brochure, and stopped.

The petition was a failure. A petitioner collected four hundred signatures about the water pressure in the annex and delivered them to an office, and the office acknowledged the recipient of the letter and promised a reconciliation of the figures, and the figures were reconciled in the spring and nothing was done. There was no scandal and no ridicule; there was a review, a meeting, and a version of the minutes in which the word "urgent" had become "pending".

I learned to sew in that fortnight, badly, because a button came off my shirt and there was no servant and no shepherd to do it for me. I learned to shred a cabbage with a knife on a board in a kitchen that belonged to nobody. I learned that sanitation is a habit and not a service, and that a satisfactory standard is achieved by the person who is willing to be the one who notices.

The rotation of the seasons did the rest. The stormy weeks went, the remote and the proximate took turns, and the little society solidified the way a liquid does: not all at once, but around a few points, and afterwards you cannot tell which point came first. My structural understanding of the place, if I had one, was that it was a machine for making adults out of a raw material that did not want to be made into anything, and that the machine was inefficient, and that the inefficiency was the point.

I will not pretend the year was full of revelation. There was no revelation. There was a gradual and unglamorous transfer of the habit of attention from the things I was told to notice to the things I had chosen, and the transfer was so slow that I only saw it when I went home in January and found that I could no longer walk past a badly built wall without looking at it.

There is a temptation, at the end of a season, to make a reckoning, to weigh the gain and the loss and to report that the gain was greater. I am wary of that temptation because I know how easily a reckoning is written by the winner. What I can say is smaller and truer: I arrived as a boy who had memorised a great many words in order to pass an examination, and I left that autumn as a boy who had begun to use a few of them in order to describe something he had actually seen. That is a modest improvement and I will take

it. The rest is vanity, and the vanity was the price of the ticket, and the ticket was worth it.

Ten verbs and six nouns

Ten verbs I wanted to be able to use by the end of the year: to qualify a claim; to reserve a judgement; to reject a suspicious symptom of a trend; to reassure a quarrelsome neighbour; to restore a ruinous wall; to recycle a reparable tool; to resume a stride; to rescue a successor; to rehabilitate a reputation that somebody had tried to victimize; to save.

The overwhelming impression of that autumn was of a planet doing its rotation whether or not anybody was watching. A protein folded, a membrane allowed a slow permeation, a sail filled, and none of it required my opinion. I found this consoling in a way I could not defend: the relativity of my own velocity, the smallness of my vitality, the fact that the tech I was learning would be obsolete before my tools were.

In the royal gallery the stuffing had gone out of the sturdy chairs, and a munition wagon stood behind a reflective sheet of glass with a label explaining that the transparency was to protect the visitor and not the wagon. A speciality of the collection was a set of segmental maps, selective and static, that showed the coastline in 1850 and in 1900 and in 1950, and the coastline had moved, and the maps had not.

I have used the words evoke and embed badly, and I know it. I have used the words thrill and torture and toil and rap about things that deserved none of them, because a nineteen-year-old will use a large word for a small feeling, and the small feeling will not be diminished by it. Various people told me that the habit would pass. Underlie in all of it, I think now, was a simpler thing: I wanted the vocabulary of an adult, and I had not yet earned the experiences the vocabulary was for, and the gap between the two is what a first year actually is.

I stood at the gate on the last morning with a suitcase and a carton and the same strap, and the guard who had waved the van through in September waved me through again without looking up. Nothing had been unified and nothing had been resolved and no unification of any kind had taken place. But the gate looked smaller than it had, and the strap had been mended twice, and both of those are a form of measurement.

The gate again

An upper-year student who was a reviewer for the student journal told me that a secretive reluctance to resign an argument is not poise, it is vain, and that a stance taken for show is worse than no stance at all. It was the most useful sentence I heard that term.

The same week a lecturer explained that a state which cannot collect a taxation cannot keep a peace, and that terrorism and separatist violence both begin as a refusal to be counted. In the yard behind the annex a sow farrowed in the rain, perfect and unbothered, and a recreational theft of a bicycle happened at the same hour, via the gate that never closed, and the theft tempted two boys into a fight that lasted eleven seconds and was remembered for a month.

Nothing in that week could unite the two facts, and I stopped trying. A stimulus is any small thing that ends a silence, and the season was full of them, and I have been grateful for them ever since.

伴读：报到那天，两场雨

先把一件事说破。你翻开这一章，多半不是为了看一个新生怎么把行李拖进宿舍，而是想确认那九百多个词到底能不能活在句子里。能。但你要先咽下一个不太舒服的前提：词从来不住在词表里，词住在场面里。同一场雨，落在车站出口是 abrupt，落在军训场上是 burst，落在你妈的信里就是 wet。你把它们抄成一行背下来，它们就都死了。

那就跟着那场雨走。

火车出隧道，光突然回来，abruptly。abrupt 的骨头是"断"：话说到一半断了是 abrupt，路突然断了是 abrupt，人的态度从热到冷也是 abrupt，它比 sudden 多一层"没过渡、不体面"的挑剔。这一章开头你撞见它两次，一次是光，一次是雨。雨的另一个名字叫 burst，那是"憋不住一下子炸开"，跟你憋了一路的那口气是一回事。

先说 a。它是全篇唯一一个我不要求你加粗的词，因为它不值得占你的注意力。a 与 an 的分工不在拼写，在声音：an hour，a university。你只要念出来，它自己就对了。

abolish 是废除，对象只能是制度、法律、习俗，不能是"我今天不吃饭了"。正文里主任说大学要 abolish 三个习惯，那是它最正的位置，因为习惯在这句话里被当成了制度。absurdity 是荒唐，但正文里"我的处境很 absurdity"不是"可笑"，是"结构上说不通"，这两层意思差得很远。accelerate 是加速，火车、心跳、通胀都能用它，它只负责"变快"，不负责"变好"。accessory 是配件，也是从犯：军训发的那一包小东西是它，法庭上跟在主犯后面的那个人也是它。

accord 是"一致"，也是"给予"。你要记住的是 of its own accord 这个词组，意思是"它自己就那样了"，没人推它。正文里那个通风口就是 of its own accord，一遍遍转回原来的角度，那位太太把它当成 antagonistically 故意的——antagonistically 是"带着对抗地"，它形容的不是动作大，是态度坏。across 是横过，through 是穿过：across 走的是面，through 走的是体，你横过马路是 across，你穿过人群是 through。accountant 是会计，记住 -ant 结尾多半是人。accuse 是指控，介词只用 of，accuse sb of sth，你别写成 for。acidity 是酸度，正文里那杯咖啡用的是它而不是 sour，因为 sour 是味道，acidity 是量。

actor 是演员，别手滑写成 acter。adaptation 是改编，也是生物学里的适应，一本小说改成电影是它，一个物种活下来也是它。adoption 是采纳，也是收养，adopt 的本义是"拿过来当自己的"。address 有两个身子：一个是地址，一个是"向谁讲话、处理什么"，正文里我 address 了一个学长，

用的是后面那个。adjust 是调节，你调的是角度，不是自己。admire 是钦佩，比 respect 多一层"看着舒服"；advantage 是优势，take advantage of 是"利用"，好坏全看语境。adverse 是不利的，它只修饰条件、影响、反应，不修饰人，你不能说一个人很 adverse。

aerial 是空中的，aerial photograph 是航拍，正文里墙上那张就是。affiliate 是"使隶属"，大学里的附属医院、附属学院都用它。affluent 是富裕的，比 rich 正式，也更冷。afternoon 是下午，这个词你认识，但你要注意它是名词，不能说 in the afternoon time。agenda 是议程，还有一个意思是"暗中盘算"，一个人有 hidden agenda，就是他嘴上说的不是他要的。agitation 是激动，也是鼓动，两个意思一个向内一个向外。ahead 是在前，ahead of yourself 是"跑到自己前面去了"，正文用它写那种说不清的错位感。

airborne 是升空的，空降兵、花粉、病毒都能用它。album 是相册，也是专辑。alienation 是疏离，这个词有理论味，马克思讲劳动异化用的就是它，你写"我在宿舍第一晚感到一种 alienation"完全站得住。alleged 是"据称的"，新闻里最值钱的一个形容词，因为它替说话的人免责：alleged thief，还没定罪。allocate 是分配，对象是名额、资源、时间，不是东西，你不能 allocate 一个苹果。allure 是诱惑力，名词，比 attraction 多一层"迷人但危险"。

along 是沿着，along the river 顺着河。alteration 是改动，改衣服、改条款都是它，它比 change 小，比 change 正式。always 是总是，注意它跟进行时连用是"老是"，he is always losing his keys，是抱怨。ambiguity 是歧义，一句话可以有两种读法才叫 ambiguity，一种读法读不懂那叫 obscure。amid 是"在……之中"，后面跟的是环境不是位置，amid the applause。amusing 是好笑的，比 funny 文，也比 funny 冷。anchorage 是锚地，也是"依靠"，一个人在世上要有个 anchorage，这句可以背下来。

angle 是角度，看问题的角度也是它。annex 是附楼，正文里我分不清的那栋就是 annex。anonymity 是匿名，玻璃磨砂到脸的高度，给每个办事员的就是 anonymity。anxious 是焦虑，它比 nervous 重，nervous 是手抖，anxious 是心里悬着。apart 是分开，apart from 是"除了"。appeal 有三个身子：上诉、吸引力、恳求，三个都常用，靠介词分辨：appeal against，appeal to。applaud 是鼓掌，也是"赞许"。appoint 是任命， appoint sb to sth，被任命的那个位置是 appointment。approach 是接近，也是方法，你"用一种方法"是 take an approach。

aptitude 是天资，aptitude for math，介词用 for，不是 in。architect 是建筑师，也是"设计者"，the architect of the reform。argument 是论点，也是争论，一个人有 argument 是有道理，两个人有 argument 是吵架。arm 是手臂，也是武装，arms 复数就是武器，这条要记牢。arrangement 是安排，也是编曲。art 是艺术，也是"技艺"，the art of losing an argument。artist 是艺术家，别跟 artisan 混，artisan 是手艺人。

ash 是灰，火山灰、烟灰、骨灰都是它。aspire 是渴望，它自带"向上"的意思，aspire to be，to 不能换成 for。assert 是断言，比 say 硬，比 claim 稳。assimilate 是同化，也是吸收，新知识要 assimilate，新移民也要 assimilate，这个词对两边都客气。assortment 是各式各样的一堆，an assortment of forms。astronaut 是宇航员，astro 是星，naut 是船，星船上的人。atlas 是地图册，正文里我把一张打印的校园图当 atlas 用，这是这个词的体面用法。attach 是附上，也是"依恋"，be attached to。

attendant 是服务员，火车上的、停车场里的都算。attributive 是定语，语法书里的"定语从句"就是 attributive clause，你至少要知道它长什么样，读长句时它是你的路标。auditorium 是礼堂。author 是作者，也是"造成者"，the author of the plan。automation 是自动化。available 是可用的，它的重点是"有空、能约到"，不是"存在"。avoidance 是回避，avoidance of the queue 是一门技术。awesome 是令人敬畏的，现在被用成"太好了"，你要知道它原来的重量。

background 是背景，也是"出身"。badge 是徽章，也是"标志"，a badge of anxiety。bake 是烤，烤面包烤蛋糕都行，不烤别的。ballot 是无记名投票，正文里选班长用的就是它。bankrupt 是破产的，形容词，也是动词。barely 是几乎不，它是否定，不是"仅仅"，这句最容易翻车。barrier 是障碍，也是闸机。basket 是篮子。battalion 是营，军训里那个两百人的方阵就是它。

bead 是珠子，汗珠是 beads of sweat，这个搭配记住就够用一辈子。beat 是打，也是击败，也是心跳。bedding 是床上用品，被褥枕一套。beggar 是乞丐。being 是存在，也是生物，human being 里的 being 是"存在物"。below 是在……之下。benevolence 是善心，比 kindness 大，也比 kindness 冷。best 是最好的，最好用的时候是 the best of，不是 the best。bewilder 是使困惑，比 confuse 重，confuse 是搞不清，bewilder 是搞不清而且慌。bike 是自行车。

biochemical 是生物化学的，biology 是生物学，这两个词在正文里都出现在同一栋楼的名字上。bitch 是母狗，正文里那条带崽的母狗就是它，骂人的用法先放一边，你要先认识它的本义。blame 是责备，blame sb for sth，也是"责任"，take the blame。blend 是混合，也是"混合体"。blond 是金发的，男的用 blond，女的可以写 blonde，正文里那个中亚男生的头发就是它。bolt 是螺栓，也是闪电，a bolt of lightning。blueprint 是蓝图，也是"方案"。boastful 是爱吹的，比 proud 难听。boom 是轰鸣，也是繁荣，两个意思一个响一个富。

boss 是老板。bouquet 是花束。bracket 是括号，也是托架，正文里那个大厅是被 steel brackets 隔开的，一个词管两种东西，这就是英语的省力。brave 是勇敢的，它形容"当下这一刻敢"，不形容一生。breakthrough 是突破，科研、谈判、病情都能用。bribe 是贿赂。briefing 是简报，会前那十分钟。brisk 是轻快的，风、步伐、生意都能 brisk。

brood 是沉思，带一点闷气，一个人 brood over 一件事，就是越想越沉。brunch 是早午餐，早饭和午饭并成一餐。bud 是芽，也是"萌芽"，nip it in the bud 是"掐死在萌芽里"。buffet 是自助餐，也是"反复打击"，这个词的两个意思读音不同，这是英语少见的赖皮。bulletin 是公告，公告板就是 bulletin board。bureaucracy 是官僚机构，正文里报到大厅就是它的实体。burst 是爆裂，也是"一阵"，a burst of rain。busy 是忙，也是"占线"。buyer 是买主。

cabinet 是柜子，也是内阁，一个装东西，一个装权力。calculating 是精于算计的，形容人时是贬义。calorie 是卡路里。cancellation 是取消，注意它是双 l，美式拼 cancellations 也常见。canoe 是独木舟，划它用 paddle。capital 是资本，也是首都，也是大写字母，三个意思都常见。car 是汽车。cargo 是货物，船和飞机上的。carton 是纸箱，正文里我那一箱 bedding 就是装在 carton 里。casino 是赌场。catastrophic 是灾难性的，比 terrible 重得多。

cause 是原因，也是事业，a good cause 是"值得的事"。celebration 是庆祝。census 是人口普查。certain 是确定的，也是"某个"，a certain man，两个意思靠冠词分辨。chairman 是主席，正文里王老师不肯改成 chair，这是个真事里的真笑话。championship 是锦标赛。chapel 是小教

堂，老校区那座就是。charter 是章程，也是"包机"。cheese 是奶酪。chew 是咀嚼。chin 是下巴。choose 是选择，choose A from B。Christian 是基督教的。cigar 是雪茄，cigarette 才是香烟，别混。circumstance 是情形，常用复数 circumstances。civilian 是平民，跟军人对。clash 是冲突，也是"撞色"。classmate 是同学。

clearance 是结关、清理，也是"空隙"，这三个意思在正文里都出现过。click 是咔哒一声，也是"点一下"。clinic 是诊所。closet 是壁橱。club 是社团，也是俱乐部。coarse 是粗的，粗布、粗话、粗颗粒都用它。coconut 是椰子。coherent 是连贯的，一个人说话 coherent，就是他的话能接上。cold 是冷，也是感冒。collection 是收藏，也是收集。column 是柱，也是专栏，报纸上那一竖条就是它。come 是来，come true 是成真。companionship 是陪伴，比 company 暖。compassion 是同情，com 是一起，passion 是痛，一起痛才叫 compassion。

competence 是能力，是"够用"，不是"出众"。complacent 是自满的，它专指"因为过去顺利所以不再警惕"，这个词很毒，用对了很准。complexion 是肤色，也是"局面"，the complexion of the race。comply 是遵从，comply with，介词只用 with。comprehensive 是全面的，一份 comprehensive 的表，就是四页那种。concealment 是隐瞒。conception 是观念，也是受孕，两个意思差得远，靠语境。concise 是简洁的，比 short 多一层"该说的都说了"。condensation 是凝结，也是"缩写"，窗上的水汽是它。conductor 是导体，也是指挥，也是售票员，一个词三个身份，看你在哪个句子遇见它。

说说 confidence 和它的假朋友。confidence 是"信得过"，既可以对外（自信）也可以对内（信任），说一个人有 confidence，是说他敢；说一件事有 confidence，是它靠得住。conflict 是冲突，名词重音在前，动词重音在后，这是英语里少数靠重音分工的词。confusion 是混乱，比 disorder 更强调"人糊涂了"。conjunction 是连词，也是"结合"，in conjunction with 是公文里最常被抄的一句。consciousness 是意识，它跟 awareness 的差别在于：awareness 是知道，consciousness 是醒着并且知道。

conservative 是保守的，也是"低估的"，a conservative estimate 是往少了估，这一点很多人第一次见会读反。consistency 是一致性，也是"浓稠度"，酱的稀稠也是它，同一个词管逻辑和厨房。constituency 是选区，也是"一批支持者"，一个老师也有他的 constituency。construction 是建造，也是"解释"，你对一句话的 construction 就是你把它怎么搭起来。consumer 是消费者，contamination 是污染，这两个词在实验室和超市都常用。contention 是论点，也是"争夺"，两个意思都带火气。contingent 是"视情况而定的"，也是"分遣队"，a contingent of four hundred，正文里那个四百人的方阵就是它。

contradictory 是自相矛盾的，controversial 是有争议的，前者说"它自己打自己"，后者说"别人打它"，别混。convergence 是汇聚，数学里、新闻里、生物里都是"几条线走到一起"。convince 是说服，convince sb of sth，它比 persuade 少一层"说服他去做"，convince 只到你信为止。cooperation 是合作，注意它是 co-operation，两个 o。copy 是副本，也是"抄"，作动词时是抄，作名词时是份。corn 是玉米，也是脚上的鸡眼，英美两地的玉米还不是一种东西，这件事你知道就行。

correct 是正确的，也是"改正"，作动词重音在后。correspondent 是通讯员，也是"相对应的人"，身体的某个部位和另一个部位 correspondent，说的是对应关系。cosmic 是宇宙的，也用来形

容"大得没边"，cosmic indifference 是一种很高级的凉。couch 是长沙发，也是"表达"，to couch a demand in polite words，把要求包在客气话里。counteract 是抵消，counter 是反，act 是作用，一反一正相抵。coupon 是券。coverage 是覆盖，新闻里是"报道量"，保险里是"保额"。

crackdown 是严厉打击，这个词的画面是"压下去"。crave 是渴望，比 want 强，比 need 软，crave 的是那种"想得慌"。creative 是创造性的，注意它形容的是"做法"，不是"人有多聪明"。creep 是爬行，也是"悄悄发生"，a creep 还能指一种让人不舒服的人。criterion 是标准，单数，复数是 criteria，这条是四六级最爱考的小陷阱之一。crowded 是拥挤的。crumble 是碎，饼干碎、墙皮碎、帝国碎，都用它。cubic 是立方的。cultivation 是耕种，也是"修养的培养"，读书也是 cultivation。

curfew 是宵禁，宿舍的关门时间就是它。curse 是咒骂，也是"诅咒"，也是"祸根"。customer 是顾客，跟 consumer 的差别是：customer 是付钱那个人，consumer 是用掉那个东西的人。cylinder 是圆柱体，也是"钢瓶"。damn 是咒骂，也是最轻的脏话，用作形容词时是"该死的"。dark 是暗的，也是"阴沉的"。datum 是数据，单数，复数是 data，你在论文里写 data is 会被审稿人划掉。deadlock 是僵局，锁死了，谁也动不了。

debt 是债，b 不发音。decent 是体面的，也是"还过得去"，a decent meal 是一顿像样的饭，不是一顿好饭。declaration 是声明，也是申报。decree 是法令，比 law 小，比 order 正式。deductive 是演绎的，从一般到个别；它的反面是 inductive，从个别到一般，这两个词你在实验室会天天听见。defiance 是公然违抗，in defiance of。definition 是定义。delegate 是代表，也是"委派"。delicious 是好吃的。demanding 是要求高的，形容活、形容人都行。

demonstrative 是"表露在外的"，也是语法里的"指示词"，this、that 就是。dental 是牙齿的。dependable 是可靠的，比 reliable 多一层"靠得住很久"。deploy 是部署，军队、人力、软件都能 deploy。depress 是压下，也是"使沮丧"，按按钮也是它。deregulate 是放松管制。describe 是描述。desirability 是"值得要的程度"。despatch 是 dispatch 的英式变体，发货、派遣都是它，同一个人写文章最好只用一种拼法。

destroy 是毁掉，比 damage 彻底，damage 是坏了，destroy 是没了。detain 是拘留，也是"留住"，被 detain 的人不一定有罪。deterioration 是恶化，慢慢变坏才叫它。devastating 是毁灭性的，比 catastrophic 更带情绪。device 是装置，也是"手段"。diagnosis 是诊断，复数是 diagnoses。dictate 是口述，也是"规定"，前者是人念你写，后者是条件说了算。

differ 是不同，differ from 是与……不同，differ in 是在某方面不同。diffusion 是扩散，气体、谣言、技术都用它。dignity 是尊严。dimension 是维度，也是"方面"。dip 是蘸，也是"下降一点"。directory 是名录，也是"目录"。disappear 是消失。disarmament 是裁军。discipline 是纪律，也是学科，一个词管两件事，军训和教育共用它，这个词本身就是个冷笑话。

discourse 是话语，学术里的"话语"就是它，比 talk 大。discuss 是讨论。disintegration 是解体。dispatch 是派遣，也是"迅速处理"。disposition 是性情，也是"安排"。dissident 是持不同政见者。distinguish 是区分，distinguish A from B。distribute 是分发。diverse 是多样的。division 是分开，也是"部门"，也是"除法"。

doctorate 是博士学位。doll 是玩偶。dominate 是支配。dorm 是宿舍，dormitory 的缩写，口

语里只用 dorm。doubtless 是无疑地。downward 是向下的。drainage 是排水。drawer 是抽屉，注意它不读"画的人"。drift 是漂流，也是"不知不觉变成"。drought 是旱。dryer 是烘干机。dump 是倾倒，也是"甩掉"。dust 是灰尘，也是"掸灰"。dying 是垂死的，也是 die 的现在分词。

early 是早。east 是东。ecological 是生态的。edge 是边缘，也是"优势"，have an edge。educational 是教育的。eggplant 是茄子，英式叫 aubergine。either 是两者之一，either...or 是二选一，否定句里用 either 表示"也不"。elbow 是肘。electrical 是电的，electric 是带电的，an electric kettle 是电水壶，electrical engineering 是电气工程，这两个词的分工要记。elegant 是优雅的，也能形容一个解法"漂亮"。

elicit 是引出，引得的是反应、回答、信息，不是东西。else 是别的，something else，也是"否则"。embed 是嵌入。emigrant 是移出的人，跟 immigrant 方向相反，一个是走出去，一个是走进来。emotional 是情绪的。empress 是女皇。enclose 是围住，也是"随信附上"。endanger 是危及。endure 是忍受，也是"持续"，这两个意思一被动一主动。engagement 是约定，也是"订婚"，也是"投入"，三个意思都很常用。

enjoyment 是享受。enquire 是询问，跟 inquire 同义，英式偏爱 enquire。enter 是进入，也是"输入"。enthusiastic 是热情的。entrepreneurial 是创业的。environmentalist 是环保主义者。equal 是相等的，equitable 是公平的，相等不等于公平，这两个词放一起就是一句社会学。eraser 是橡皮。erroneous 是错误的，比 wrong 正式，常修饰 belief、conclusion。especially 是尤其。esteem 是尊重，名词，self-esteem 是自尊。

ethics 是伦理，复数是 ethics，单数是 ethic。evaluation 是评估。evenly 是均匀地。everybody 是每个人，注意它是单数。evoke 是唤起，唤起的是记忆、情绪、画面。examine 是检查，也是"考试"，考官 examine 你，你也在 examine 一个问题。excellent 是优秀的。excite 是使兴奋，也是"激起"。exclusive 是排他的，也是"独家的"。exempt 是免除的，be exempt from。exhibit 是展出，也是"表现出"。exotic 是异域的。expel 是开除，也是"排出"。experimental 是实验的。

explicit 是明确的，它跟 implicit 是一对：一个说出来了，一个没说但含着。exponent 是倡导者，也是数学里的指数。expressive 是富于表现的。external 是外部的。extravagance 是奢侈，也是"过度"。这批词你把 explicit 和 implicit 一起记，一辈子不会错。

fabric 是织物，也是"结构"，社会的 fabric 就是它的经纬，这个词一旦这样用，文章立刻上一个台阶。facilitation 是"使便利"，它不做事，它让事情容易做，会议里那个负责推进的人就是 facilitator。fade 是褪色，也是"渐弱"，声音、记忆、照片都用它。false 是假的，也是"错误的"，false teeth 是假牙，false alarm 是虚惊。fantastic 是极好的，也是"离奇的"，它原本的意思是"像幻想一样"，所以它既能夸人也能形容怪物。

farther 是更远，指距离；further 更远，指程度。这条分工今天已经松了，但你写作时守住了，读的人会觉得你可靠。fast 是快，也是"牢固"，hold fast 是抓紧，这个词的两个意思方向相反，是英语的一个玩笑。fatigue 是疲劳，比 tiredness 正式，也是材料学里的"疲劳"，金属反复受力也会 fatigue。fearful 是害怕的，也是"可怕的"。federal 是联邦的。fellow 是同伴，也是"研究员"，a fellow of the college。

festival 是节日，也是"音乐节"。fifteen 是十五。filling 是馅，也是"填充物"，也是"补牙"，看牙时医生说 a filling，说的不是馅。fingerprint 是指纹。firstly 是首先，写文章时它是路标，但别通篇 firstly secondly，读到第三段人就烦了。fix 是修理，也是"固定"，也是"确定"，fix a date 是把日子定下来。flare 是火焰，也是"（脾气、病情）突然发作"。fleet 是舰队，也是"车队"，也是形容词"快速的"。float 是漂浮，也是"提出（想法）"。flu 是流感，influenza 的缩写。

foam 是泡沫。folklore 是民间传说，也是"民间说法"。footprint 是脚印，也是"足迹"，碳足迹也是它。foreigner 是外国人。forever 是永远。forgiving 是宽容的，也是"（材料、设计）容错的"。formidable 是难对付的，它形容的是"强到让你不敢轻举妄动"，不是"坏"。fortify 是加固，也是"增强"，加固城墙、增强信心都用它。foster 是养育，也是"促进"，foster a habit 是养出一个习惯。

fourteen 是十四。fragmentation 是碎裂，也是"碎片化"。frantically 是发疯似的，它形容的是乱，不是快。frequency 是频率。friendly 是友好的，也是"便于使用的"，user-friendly。frontier 是边疆，也是"前沿"，科学的 frontier 就是没人到过的地方。frustration 是挫败感，它是"努力被挡住"，不是"失败"。function 是功能，也是函数，也是"运转"。furnace 是炉子。fuss 是大惊小怪，make a fuss 是小题大做。

gallery 是画廊，也是"楼座"。garage 是车库，也是修车行。gasoline 是汽油，英式说 petrol。gear 是齿轮，也是"装备"，钓鱼的、爬山的、军训的那一堆都叫 gear。genetic 是基因的。geographically 是地理上。germ 是细菌，也是"萌芽"，the germ of an idea。giggle 是咯咯笑，压不住的那种。glance 是瞥一眼，at a glance 是一眼看去。glitter 是闪光，比 shine 碎。glue 是胶水。goodbye 是再见。govern 是治理，也是"支配"，rules govern behaviour。

grade 是等级，也是"分数"，也是"年级"。grand 是宏大的，也是"一千块"，口语里 a grand 是一千。grant 是授予，也是"助学金"，一个词管给钱和给面子。grassy 是长草的。grease 是油脂。greeting 是问候。grin 是咧嘴笑，它是"不出声的笑"。gross 是总的，也是"恶心的"，gross income 是总收入，gross 作"恶心"时是口语。guess 是猜，也是"想"，I guess 是"我觉得"。guitarist 是吉他手。

habitat 是栖息地。ham 是火腿。handle 是把手，也是"处理"，handle a problem 是把问题拿在手里办。happy 是高兴的。hardware 是硬件，也是五金。harsh 是严酷的，形容话、条件、光都行。hateful 是可恨的。hay 是干草。heading 是标题，也是"方向"。healthy 是健康的。heated 是加热的，也是"激烈的"，a heated argument 是吵得很热的架。heel 是脚跟。helmet 是头盔。henceforth 是从今以后，公文味很重。

heritage 是遗产，是"传下来的东西"，比 inheritance 大，它属于一群人不属于一个人。hesitant 是犹豫的。high 是高的，也是"嗨"，这个词的第二个意思别在作文里用。him 是他，宾格。hobby 是爱好。holy 是神圣的。homosexual 是同性恋的，用的时候要平，它就是个形容词。hook 是钩子，也是"钩住"，off the hook 是脱身。horrible 是可怕的，也是"糟糕的"。hospitality 是款待。hound 是猎犬，也是"纠缠"。hover 是盘旋，也是"悬而未决"。humid 是潮湿的，它形容空气，不形容东西。hungry 是饿的，也是"渴望的"，hungry for。hurry 是匆忙。hyphen 是连字符。

idealistic 是理想主义的，它跟"天真"只差一个语境。idiomatic 是地道的，说一个人英语 idiomatic，是夸他写得像母语者。ignition 是点火。illiterate 是不识字的，也是"不懂行的"，computer illiterate。illustrative 是说明性的。imitate 是模仿。imperative 是必要的，也是语法里的祈使语气。import 是进口，也是"含义"。impossible 是不可能的。improvement 是改进。inch 是英寸。include 是包括。increasingly 是越来越。

independence 是独立。indictment 是起诉书，也是"控诉"，a damning indictment of the system。indoors 是在室内。infant 是婴儿。infinite 是无限的。influential 是有影响力的。infringement 是侵犯，侵的是权利。inheritance 是继承，继承的是财产和基因。inject 是注射，也是"注入"，inject funds。inn 是小旅馆。input 是输入，也是"意见"。insertion 是插入。inspect 是检查，它是"看"，不是"改"。installment 是分期付款的一期，也是"连载的一期"。institution 是机构，也是"制度"。insulate 是绝缘，也是"隔离"，把人 insulate 起来就是让他不知道。

intangible 是无形的。intelligent 是聪明的。intention 是意图。interesting 是有意思的。international 是国际的。interruption 是打断。intimate 是亲密的，也是"深入的"，intimate knowledge。intrinsically 是本质上。intuitive 是直觉的。invention 是发明。investigator 是调查者。involved 是卷入的，也是"复杂的"。irregular 是不规则的。island 是岛。itself 是它自己。

jazz 是爵士乐。jetlag 是倒时差，一个字，别写成两个。joke 是笑话。joyous 是欢乐的，比 happy 正式。junction 是交叉口。justice 是正义，也是"司法"。keyboard 是键盘。kin 是亲属，kin 比 relatives 古。kitchen 是厨房。knock 是敲，也是"撞"，knock into。

label 是标签，也是"把……归类"，你给人贴标签，用的是同一个词。lady 是女士，它比 woman 多一层身份感，也多一层旧。landlady 是女房东，也是旅店的女主人，那个管洗衣房的阿姨被我叫成 landlady，是因为除此之外我想不出更准的词。large 是大，at large 是"逍遥法外"，也是"总体上"，两个意思都很活。lately 是最近，它跟 recently 几乎一样，但 lately 更口语，而且只跟完成时或现在时连用。

lavatory 是厕所，公共场合那个，缩写是 lav。layman 是外行，它是从"非神职人员"来的，所以它天然带一点"不懂专业"的客气。leak 是漏，也是"泄露"，水管漏是它，消息漏也是它，一个词管两件让人头疼的事。least 是最少，at least 是至少，not in the least 是一点也不。legacy 是遗产，它比 inheritance 更常指"留下来的影响"，一个坏政策也会留下 legacy。legitimate 是合法的，也是"正当的"，问一个问题 legitimate，是说这问题站得住。

lens 是透镜，也是镜头，也是"视角"。lever 是杠杆。liberty 是自由，它和 freedom 的差别在于 liberty 更像"权利"，freedom 更像"状态"。limited 是有限的，公司名里的 Ltd 就是它。linguistics 是语言学，注意它是复数形式当单数用。liquor 是烈酒，它比 alcohol 窄。literate 是识字的，也是"懂行的"，computer literate 是会用电脑。load 是负荷，也是"装载"，workload 是工作量。location 是位置，也是"外景地"。logic 是逻辑。

longitude 是经度，纬度是 latitude，这两个词一横一竖，考试里最爱换着考。loot 是抢来的东西，也是"抢劫"，它带一点"趁乱"的意思。loud 是响，也是"花哨的"，a loud shirt 是花衬衫。

loyal 是忠诚的，loyal to。luminous 是发光的。luxury 是奢侈，也是"难得的享受"，a luxury 可以是睡一个午觉。madden 是使发疯，它比 annoy 重得多。magnificent 是壮丽的。

mainstream 是主流。makeup 是化妆品，也是"构成"，the makeup of the committee。mam 是妈妈，英式北方口音里的说法，我写它一次是因为笔记里那个"me mam"的句子，它不属于我的口音，但属于我读过的书。maneuver 是机动、策略，美式拼 maneuver，英式 manoeuvre。manly 是有男子气的。many 是许多，many a 后面跟单数。marine 是海洋的，也是海军陆战队的。marsh 是沼泽。

masculine 是男性的，它形容的是"特质"，不是"性别"，一条裙子也可以是 masculine 的剪裁。masterpiece 是杰作。mayor 是市长。measure 是测量，也是"措施"，take measures。medal 是奖牌。medication 是药物，它比 medicine 更正式，而且不可数，说 medications 是"几种药"。melody 是旋律。

memoration 这个词严格说不是词，它是我把 commemoration 拆坏了的产物，但词表里有它，我就留着：真实的词是 commemoration，纪念，com 是一起，memor 是记。mensely 同理，它来自 immensely，极度地，im 是否定前缀，mense 是"量"，合起来是"大到没法量"。mentor 是导师，它比 teacher 多一层"带你这个人"，不只是带你那门课。merely 是仅仅。

mersion 也是碎片，正词是 immersion，沉浸，进入液体是它，进入一种语言也是它。metallic 是金属的，也是"金属味的"。metro 是地铁。midday 是正午。migrate 是迁徙，人、鸟、数据都能 migrate。military 是军事的。mindset 是心态，这个词很新，也很好用，a fixed mindset 是"认死了"的那种脑子。minor 是次要的，也是"未成年人"，也是音乐里的小调。misery 是痛苦，它比 sadness 长，也比 sadness 重。mist 是薄雾。

mit 是最后一个碎片，正词是 commit，com 是一起，mit 是送，把一个人送进一件事里，就是承诺。mock 是嘲笑，也是"模拟的"，a mock exam 是模拟考，这两个意思一真一假，很有意思。modest 是谦虚的，也是"不大的"，a modest income 是不高的收入。molecular 是分子的。Monday 是星期一。monster 是怪物。moody 是情绪化的。morning 是早上。mother 是母亲。

mould 是模具，也是"发霉"，英式 mould，美式 mold，一个词管"塑形"和"长毛"，两件事都跟潮湿有关。mouthful 是一口，也是"拗口的一大串"，a mouthful of a name。mug 是马克杯，也是"劫匪"，也是" mug up"死记，一个词三种人生。municate 是碎片，正词 communicate，com 是一起，mun 是"共同的责任"，能一起担事的人才谈得上沟通。munition 是军火。mushroom 是蘑菇，也是"迅速增长"。

mutual 是相互的，mutual respect 是相互尊重，注意 mutual 不能修饰单方面的东西。nail 是钉子，也是指甲。narrate 是叙述。national 是国家的。nature 是自然，也是"本性"，by nature 是天生的。nearly 是几乎。needle 是针。negligent 是疏忽的，法律味很重，它说的是"该注意而没注意"。nephew 是侄子、外甥。nevertheless 是然而，它比 however 重，放在句首时最有分量。

nickname 是绰号。ninety 是九十。noise 是噪音。nonsense 是胡说。northern 是北方的。note 是笔记，也是"注意到"，也是"钞票"，三个意思都极常用。notion 是念头，它比 idea 虚，比 idea 轻。November 是十一月，那个月是新生最容易垮的一个月，王老师说对了。nuisance 是麻烦事，它形容的是"烦"，不是"坏"。nut 是坚果，也是"螺母"，也是"疯子"，口语里的 a nut。

obesity 是肥胖，医学词，它比 fat 客气也比 fat 严重。obscenity 是下流话，也是"下流的行

为"，墙上那句话就是它。obsess 是使着迷，也是"占据"，be obsessed with。obvious 是明显的。occupational 是职业的，occupational disease 是职业病，主任用这个词形容"聪明人的病"。oddity 是怪事，也是"怪人"。office 是办公室，也是"职务"，take office 是就职。oily 是油的。omit 是省略，也是"漏掉"。only 是仅仅，也是"唯一的"，only 的位置一变，句子的意思就变，这是它最阴的地方。

openly 是公开地。opponent 是对手。opt 是选择，opt for，opt out。option 是可可供挑选的东西，也是"选择权"。ordeal 是煎熬，它形容的是"一段难熬的过程"，不是一件事。origin 是起源。orthodox 是正统的。out 是出去，也是"公开"，the truth is out。outgoing 是外向的，也是"即将离任的"。outrageous 是骇人的，也是"离谱得好笑"。oval 是椭圆形的。overhaul 是大修，也是"彻底改革"。oversee 是监督。overwhelm 是压倒，情绪、工作量、人数都能 overwhelm 你。ownership 是所有权，take ownership 是"担起来"。

那些差的只有一点点

先停一下。到这里你已经读了一百多页，我要问你一句：你记住了多少？多数人记住的都是"眼熟的"，不是"会用的"。眼熟是最贵的一种错觉，它让你在阅读里放行，在写作里出错。所以下面这些词，我不再只给你意思，我要给你"它跟谁不一样"。

packet 是一小包，packet 比 pack 小，比 bag 硬，一包饼干是它。painter 是画家，也是"油漆工"，同一个词，看他在画布前还是在墙前。panda 是熊猫。parachute 是降落伞，也是"空投"。parameter 是参数，也是"界限"，后一个意思在争论里很好用。parliament 是议会。participate 是参加，participate in，介词只用 in。partner 是伙伴，也是"搭档"，实验课上的搭档就是它。

passionate 是热烈的，它形容的是"把心放进去"，不是"脾气大"。pastry 是酥皮点心。patriotic 是爱国的，这个词要小心用，它形容的是情感，不形容政策。pay 是付钱，也是"值得"，it pays to be early。peanut 是花生，peanuts 是"很少的钱"。peddle 是兜售，它带一点"沿街叫卖"的轻贱感，卖毒品也用这个词。pending 是"待定的"，也是介词"直至"，公文里非常常见。pepper 是胡椒，也是辣椒。

perfect 是完美的，作动词时重音在后，意思是"使完善"。perilous 是危险的，比 dangerous 文，常用在"处境"。permeation 是渗透，它比 penetration 轻，是一点点渗进去。perplexity 是困惑，比 confusion 多一层"被绕住了"。person 是人。persuasive 是有说服力的。petitioner 是请愿人，法律味很重。phenomenon 是现象，复数是 phenomena，这条几乎年年考。

photography 是摄影，注意它跟 photograph 的差别：一个是术，一个是物。pianist 是钢琴家。piece 是一块，也是"一件作品"，也是"一条（新闻）"。pillar 是柱子，也是"支柱"。pioneer 是开拓者。pizza 是披萨。planet 是行星。plausible 是"听起来有道理"，注意它只说"听起来"，不担保是真的，这个词的分寸感就是它的全部价值。please 是请，也是"使高兴"。plough 是犁，也是"犁过"，英式 plough，美式 plow。

pocket 是口袋，也是"一小块"，a pocket of resistance。poise 是镇定，它形容的是一种"稳"，不是"冷"。policeman 是警察。poll 是民意调查，也是"投票"。pool 是水池，也是"共用"，a pool of cars 是公用车队。porcelain 是瓷器。portion 是一份，也是"一部分"。

possession 是拥有，也是"财产"，复数 possessions 是你的东西。postponement 是延期。poultry 是家禽。practitioner 是从业者，它比 professional 具体，医生、律师、设计师都是 practitioner。

preceding 是在前的，它比 previous 更"紧挨着"。predictable 是可预测的，它常常是贬义：一个人太 predictable，就是没劲。preference 是偏爱。premise 是前提，也是" premises"经营场所，两个意思差得远。prerequisite 是先修条件，选课前那一栏就是它。president 是总统，也是校长、会长。pretty 是漂亮，也是"相当"，pretty good 是相当好。previous 是先前的。prime 是首要的，也是"全盛期"，也是数学里的质数。

printing 是印刷。probability 是概率。proceedings 是程序、会议记录。production 是生产，也是"作品"。profile 是侧面，也是"简介"。prohibit 是禁止，比 forbid 正式，prohibit sb from doing。promote 是促进，也是"升职"，也是"推销"。proofread 是校对。proportion 是比例。prospect 是前景，也是"可能性"。protein 是蛋白质。provide 是提供，provide sb with sth。proximate 是最近的，它跟 approximate 不是一回事，前者是"近"，后者是"大约"。

pub 是酒馆。punch 是拳打，也是"冲击力"，也是"潘趣酒"，三个意思都常见。purify 是提纯，也是"净化"。push 是推，也是"推动"。qualify 是使合格，qualify for，也是"限定"，语法里修饰语的作用就是 qualify。quarrelsome 是爱吵的。questionnaire 是问卷。quota 是配额。

radiation 是辐射。ragged 是破破烂烂的。raincoat 是雨衣。rap 是敲，也是说唱，也是"指责"，take the rap 是背锅。rat 是老鼠，比 mouse 大，而且名声更差。rational 是理性的。read 是读，也是"读作"，也是"（仪表）显示"。reality 是现实。reassure 是使安心，它是"再保证一次"。receiver 是听筒，也是"接收者"，也是"破产管理人"，最后一个意思你会觉得离谱，但它是真的。

recipient 是接受者，比 receiver 正式。reconciliation 是和解，也是"对账"。recreational 是娱乐的。recycle 是回收。referee 是裁判。reflective 是反光的，也是"反思的"。refrigerator 是冰箱。regard 是看待，也是"敬意"，复数 regards 是问候。register 是登记，也是"显示出"，也是"语域"，语言学里的 register 是一个人的说话场合感。rehabilitate 是康复，也是"恢复名誉"。reject 是拒绝，比 refuse 硬，refuse 是"不肯"，reject 是"打回"。

relativity 是相对性。reliable 是可靠的，它形容"每次都行"，不形容"这次行"。reluctance 是不情愿。remedy 是补救，也是"疗法"。remote 是遥远的，也是"偏僻的"。renewal 是更新，也是"续期"。reparable 是可修复的。repetitive 是重复的，它常是贬义，repeated 是中性的。representative 是代表。republic 是共和国。rescue 是营救。reserve 是保留，也是"预订"，也是"寡言"，一个词三种人。resign 是辞职，resign from。resource 是资源。respondent 是回答者，问卷里那个人就是它，法庭上那个人也是它。

restore 是恢复，也是"修复"，修一栋楼、修一个制度都用它。resume 是重新开始，注意它不是简历，简历是 résumé，重音也不同，这是英语里最坑的一对。retelling 是复述。retrieve 是取回，也是"检索"。revelation 是揭露，也是"启示"。reviewer 是评论者，也是审稿人。revolve 是旋转，也是"围绕"，revolve around。ribbon 是丝带。ridicule 是嘲笑。rigour 是严谨，英式 rigour，美式 rigor。rise 是上升。roar 是吼。rocket 是火箭。roof 是屋顶。rotation 是旋转。

royal 是皇家的。ruinous 是毁灭性的，也是"贵得离谱的"。runway 是跑道。sacrificial 是牺牲的。sail 是帆，也是"航行"。salesman 是推销员。salvation 是拯救。sanitation 是卫生设施。satisfactory 是令人满意的，注意它只是"及格"，不是"很好"。

save 是救，也是"省下"，也是"存"，三个意思共用一个动作：把东西留住。scanner 是扫描仪。scenario 是设想的情形，也是剧本。schedule 是日程，也是"安排"，美式读音和英式不一样，重音都在前，但元音不同。science 是科学。scornful 是轻蔑的。screw 是螺丝，也是"拧"，也是"搞砸"。season 是季节，也是"调味"，也是"晾干木材"，一年四季和一口锅用的是同一个词。secretive 是爱藏着掖着的，它形容的是习惯，不是一件事。see 是看见，也是"明白"，I see。

segmental 是音段的，语言学词，说的是把一个音切成段来分析。selective 是选择性的，它常带一点贬义：selective memory 是"只记对自己有利的"。senator 是参议员。sensitive 是敏感的，也是"体贴的"。separatist 是分裂主义者。servant 是仆人，也是"公仆"，civil servant 是公务员。settle 是安顿，也是"解决"，也是"结账"。sew 是缝，sow 是播种，两个词差一个字母，差一整个农业。

shaft 是轴，也是"竖井"，也是"一束（光）"。shape 是形状，也是"状态"，in good shape。she 是她。shepherd 是牧羊人，也是"引领"，动词用法很文。shipyard 是造船厂。shopping 是购物。should 是应该，它比 must 软，比 ought to 常用。shred 是撕碎，也是"一点点"，not a shred of evidence。shuttle 是穿梭，也是"班车"，机场和校园之间那趟车就是它。siege 是围困。significant 是重要的，也是统计里的"显著的"，后者是个技术意思，别混。silvery 是银色的。simulation 是模拟。single 是单一的，也是"单身的"。sit 是坐，也是"位于"，也是"开会"。

sizable 是相当大的。sketch 是素描，也是"概述"。skin 是皮肤，也是"剥皮"，也是"（软件的）皮肤"。slack 是松弛的，也是"懈怠"，也是"淡季"。sleepiness 是困倦。slip 是滑，也是"纸条"，也是"小错"，a slip of the tongue 是口误。slump 是暴跌，也是"瘫坐"。smog 是雾霾，smoke 加 fog。snack 是零食。snobbish 是势利的。soccer 是足球，美式叫 soccer，英式口语叫 football。sodium 是钠。sole 是唯一的，也是脚底，也是"鞋底"。

solidify 是固化。solvent 是溶剂，也是"有偿付能力的"，一个词管实验室和账本。somewhat 是有点，它比 a little 正式。sore 是疼的，也是"恼火的"。source 是来源。sow 见上，播种。spare 是备用的，也是"抽出（时间）"，也是"饶过"。speciality 是特产，也是"专长"，美式拼 specialty。spectacle 是景象，也是"眼镜"，复数 spectacles。speechless 是哑口无言。sphere 是球体，也是"领域"。spine 是脊椎，也是"书脊"，也是"骨气"。splendour 是壮丽，美式 splendor。

sponsor 是赞助者，也是"发起人"。sportswoman 是女运动员。sprinkle 是撒。stadium 是体育场。stainless 是不锈的。stance 是立场，也是"站姿"。start 是开始，也是"吓了一跳"。static 是静止的，也是"静电"。statute 是成文法。steamer 是汽船，也是蒸锅。stewardess 是女乘务员，现在多说 flight attendant。

stimulus 是刺激物，复数是 stimuli。stomach 是胃，也是"忍受"，I cannot stomach it。stormy 是有暴风雨的，也是"激烈的"。strange 是奇怪的，也是"陌生的"。streak 是条纹，也是"一连串"，a streak of luck。stride 是大步，也是"进展"，make strides。structural 是结构的。

stuffing 是填料。sturdy 是结实的。submit 是提交，也是"屈服"。substance 是物质，也是"实质"，也是"主旨"。suburb 是郊区。successor 是继任者。suffice 是足够，suffice it to say 是"只消说一句"。suitcase 是行李箱。sunny 是晴的，也是"开朗的"。superior 是更好的，superior to，不是 superior than，这条几乎人人错一次。supper 是晚饭，比 dinner 轻。suppose 是假定，be supposed to 是"本该"。sure 是确定的，也是"当然"。surname 是姓。survive 是幸存，也是"比……活得久"。suspicious 是可疑的，也是"多疑的"，一个词两个方向。swipe 是滑动，也是"刷（卡）"。symptom 是症状，也是"征兆"。

tablet 是药片，也是平板电脑。tail 是尾巴，也是"跟踪"。talkative 是话多的。tap 是水龙头，也是"轻敲"，也是"开发（资源）"。taxation 是征税。tech 是技术，technology 的口语缩写。teenager 是青少年。telling 是"有力的"，a telling detail 是"一个说明问题的细节"，这个用法非常高级。tempt 是引诱。tent 是帐篷。terrain 是地形。terrorism 是恐怖主义。textile 是纺织品。theft 是盗窃。theory 是理论，也是"看法"，in theory 是理论上。thermometer 是温度计。thing 是东西。thousand 是千。thrill 是激动，它比 excite 更"过电"。thumb 是拇指。tidal 是潮汐的。tile 是瓷砖。tin 是锡，也是罐头。tissue 是组织，也是纸巾。toil 是苦干。tomato 是西红柿。tool 是工具。torture 是折磨。tournament 是锦标赛。toy 是玩具。traditional 是传统的。train 是火车，也是"训练"，一个词管两件事，军训和春运共用它。

transaction 是交易。transition 是过渡。transparency 是透明，也是"透明度"。trauma 是创伤，身体的和心理的都算。treat 是对待，也是"治疗"，也是"请客"，treat sb to dinner。trend 是趋势。trick 是诡计，也是"窍门"，也是"戏法"，三个意思都很日常。triple 是三倍的。tropical 是热带的。trunk 是树干，也是后备箱，也是大象的鼻子。tube 是管，也是地铁（伦敦）。tunnel 是隧道。turtle 是龟。twice 是两次。tyre 是轮胎，美式 tire。uncle 是叔叔、舅舅。underlie 是"位于……之下"，也是"是……的根源"，这个词写论文极好用。undertaking 是事业，也是"承诺"。unification 是统一。unite 是联合。until 是直到。upper 是上面的。urge 是催促，也是"冲动"。used 是"用过的"，be used to 是"习惯于"，used to 是"过去常常"，三个结构一条线，这条线是四六级最高频的陷阱之一。

vacancy 是空缺，也是"空房"。vain 是徒劳的，也是"虚荣的"，in vain 是白费。van 是厢式货车。various 是各种各样的。velocity 是速度，物理词。verb 是动词。version 是版本。via 是经由。victimize 是使受害。village 是村庄。violinist 是小提琴手。visa 是签证。vitality 是生命力。voice 是声音，也是"发言权"，也是语法里的语态。voltage 是电压。vow 是誓言。wait 是等。wall 是墙。warfare 是作战，比 war 抽象。wary 是警惕的。waterproof 是防水的。weak 是弱的。weary 是疲倦的，它比 tired 深，带一点"心也累了"。wedge 是楔子。weigh 是称重，也是"权衡"。when 是什么时候。which 是哪一个。whistle 是哨子。whom 是谁，宾格。widower 是鳏夫。willingness 是意愿。winner 是赢家。wit 是机智，也是"才子"。without 是没有。wood 是木头。workforce 是劳动力。worse 是更糟。would 是将会，也是"过去常常"。wrestling 是摔跤。yacht 是游艇。yell 是大叫。you 是你。youthful 是年轻的。zoo 是动物园。

最后说一句。这一章里你遇见的每一个词，都不是为了让你在纸上认出它，而是为了让你在某一天——在雨里、在实验室里、在某个人的葬礼上——忽然想起它，并且用得对。词的命运跟人的命运一样，只有被用，才算活过。

十组只差一点点的词

讲完一遍，我还是不放心。因为这一章里最容易白学的，不是那些生词，是那些"你以为你懂了"的词。它们彼此只差一点点，而四六级考的恰好就是那一点点。下面这些组，我按你最容易错的方式排。

第一组，accuse 与 blame。accuse 是"当面指人"，介词是 of；blame 是"归咎"，介词是 for。你可以 blame a bad policy for a failure，但你不能 accuse a bad policy，因为它不是人。这条界一过，句子立刻像中文。

第二组，adapt 与 adopt。adapt 是"改一改让它合适"，adopt 是"拿过来当自己的"。一个方案被 adapt 是本地化，被 adopt 是采纳；一个人 adapt 是新环境，adopt 一个孩子是收养。它们的名词是 adaptation 和 adoption，只差一个字母的位置，意思差一整个动作方向。

第三组，distinguish 与 discriminate。distinguish 是"分辨"，中性；discriminate 是"区别对待"，一旦用在人身上就是歧视。你可以 distinguish two accents，但你不能 discriminate two accents，除非你想说你讨厌其中一个。

第四组，reject、refuse、deny。refuse 是"不肯"，主语是人；reject 是"打回"，主语可以是人也可以是机构，语气更硬；deny 是"否认"，否的是事实。refuse a gift 是不收，reject a proposal 是退回，deny a charge 是"我没做"。

第五组，convince 与 persuade。convince 只到"他信了"，persuade 到"他去做了"。你 convince 一个人他是错的，你 persuade 一个人去看医生。写作里想表达"说服某人做某事"，用 persuade sb to do，不要用 convince sb to do，虽然后者在口语里已经泛滥。

第六组，explicit 与 implicit。explicit 是"明说了"，implicit 是"没说但含着"。一条 explicit rule 写在纸上，一条 implicit rule 写在所有人的行为里。这一章里那个"没人告诉你什么值得做"，就是 implicit 的典型。

第七组，economic 与 economical。economic 是"经济的"，跟经济体有关；economical 是"省钱的"。an economic policy 是经济政策，an economical car 是省油的车。同理 historic 是"有历史意义的"，historical 是"跟历史有关的"。

第八组，precede 与 proceed。precede 是"在……之前"，proceed 是"继续进行"。它们的名词是 precedence 和 proceedings。一个会议有 proceedings（记录），是因为它 proceed 了；一件事有 precedence（优先），是因为它 precede 了。

第九组，sensible 与 sensitive。sensible 是"明智的"，sensitive 是"敏感的"。一个人 sensible 是夸他靠谱，说他 sensitive 要看他当时在什么场合。这两个词在作文里被换用的频率高得惊人。

第十组，substitute 与 replace。substitute A for B 是"用 A 换掉 B"，replace A with B 是"用 B 换掉 A"。两句话说的是同一件事，语序正好相反。这不是记忆问题，是习惯问题，你写错三次以后就不会再错了。

我还要多说一个词：plausible。它的意思是"听起来站得住"，它不保证真。这是英语里少数几个自带"免责"的形容词之一，跟 alleged 是亲戚。你在读新闻时会反复遇到它们，它们的功能不是告诉

你事实，是告诉你"这个说法还没有被证实"。看懂了这一层，你对句子的信任就会变得有分寸。

最后是那句老话，我换个说法给你：你不是在背 928 个词，你是在给自己造一间屋子，每个词是一块砖。砖堆在院子里，你永远住不进去；只有砌上去，你才知道哪一块是斜的。这一章到这儿就完了。回去把那一场雨重读一遍，这次别管词，只管人。你会发现，人还在雨里站着，而你已经认得出他身上的每一个词了。

那些我写进正文、却不敢多用的词

一个词"认得"和"用得稳"之间，隔着一层语体。这一章里有一批词，我把它们放进了正文，但我自己写文章时是要想一想才敢用的。它们值得单独说。

accord 这个词，作名词是"一致"，作动词是"给予"。它的危险在于太正式：你写"我们意见一致"，用 we agree 就够了，用 we are in accord 会让人以为在写公报。但它有一个词组值得你偷走：of its own accord，意思是"自己就那样了"，不需要外力。这一章里那个通风口就是 of its own accord，一遍遍转回去。这个词组的妙处是它把"被动"写成了"自愿"，而生活里很多事正是这样发生的。

alleged 是"据称的"。它不是一个描述事实的词，它是一个保护说话人的词。新闻里写 the alleged thief，意思不是"他是贼"，是"有人说他是贼，我们还没证完"。你在作文里用它，会显得你有分寸；你滥用它，会显得你不敢下判断。分寸就在这个界上。

comply 是"遵从"，介词只用 with。它跟 obey 的差别是：obey 的对象是人，comply 的对象是规定。你 comply with a rule，你 obey an order。一个机构说自己 complies，通常意味着它做了最低限度的那部分。

elicit 是"引出"，引的是反应、回答、信息、笑声。它跟 extract 的差别是力气：extract 是拔出来，elicit 是引出来。一个老师 elicit 一个回答，一个审讯者 extract 一个口供。这个词在学术写作里非常体面，因为它把"我问出来的"说得像"它自己出来的"。

formidable 是"难对付的"，它的重心在"强到让人不敢轻举妄动"，不在"坏"。a formidable opponent 是值得尊敬的对手，a formidable task 是让人头皮发麻的任务。它跟 terrible 差得很远，别混。

legitimate 是"合法的"，也是"正当的"。前一个意思是法律，后一个意思是道理。a legitimate question 是"这问题站得住"，不是"这问题合法"。这一层引申义才是它真正值钱的地方。

notion 是"念头"。它比 idea 虚，比 idea 轻，常常带一点"未必靠得住"的味道。a notion of freedom 是一种关于自由的想当然。你想表达"我隐约觉得"，用 notion；你想表达"我有个方案"，用 idea。

prospect 是"前景"，也是"可能性"。它跟 future 的差别是：future 是中性的时间，prospect 是被人评估过的时间。the prospect of a job 是"一份工作的前景"，这里面已经含了一个判断。

reluctance 是"不情愿"。它比 unwillingness 软，比 hesitation 硬。一个人 with reluctance 是"不太愿意但还是做了"，这层"还是做了"是它的全部价值。写作时你想写人物勉强答应，用它最准。

significant 是"重要的"，也是统计里的"显著的"。后一个意思是技术意思：样本够大、差异不是偶然，才叫 significant。你在数据面前说 significant，说的不是"看起来差很多"，是"这个差不是运气"。这两个意思混用，是学术写作里最常见的失礼。

suffice 是"足够"，suffice it to say 是"只消说一句"。这个词的语体很高，用在口语里会突兀，但用在段落收尾处，一句 suffice it to say 能把后面一整段省掉。省字是本事。

suspicious 是"可疑的"，也是"多疑的"。一个 suspicious man 可能是"让人起疑的人"，也可能是"看谁都起疑的人"，方向完全相反，只能靠上下文。英语里这样的词不多，遇到一个就记住一个。

underlie 是"位于……之下"，也是"是……的根源"。后者是它的高级用法：the assumption that underlies the policy，政策底下那个假设。你写论文时想指出"他们没说出来但一直在用的前提"，用它最省力。

vitality 是"生命力"。它形容的不是"活着"，是"活得有劲"。a city with vitality 是有烟火气的城市。这个词在描写里很出彩，因为它把抽象的生命写成了可以感觉到的强度。

wary 是"警惕的"，be wary of。它跟 careful 的差别是：careful 是怕出错，wary 是怕被骗。一个是手艺人的美德，一个是路人的本能。

henceforth 是"从今以后"，公文味极重。它在合同和布告里出现，在日记里出现就显得可笑。可笑的用法也是用法，你只要知道自己在做什么。

这些词我讲得比前面慢，因为它们是这一章里"最像英文"的那一批：它们的价值不在意思，在分寸。而分寸是没法背的，只能在一句一句里试。你试错三次，第四次就对了；你不敢试，就永远停在"认得"。

为什么是雨

这一章我让雨下了两次，一次在报到那天，一次在军训的第六天。不是为了好看，是因为雨是最好用的一个词场：它同时牵着天、地、身体和心情，一个场景能把四类词一次喂给你。

你看 humid。它形容空气，不形容东西。你可以说 the air is humid，不能说 the towel is humid，毛巾是 damp 或者 wet。这三个词的分工是：humid 是空气里的水，damp 是东西里的水，wet 是表面上的水。分不清这三个，写景就永远差一口气。

mist 是薄雾，fog 是浓雾，haze 是霾。它们的差别不是浓度，是能不能看见。mist 里你还看得见轮廓，fog 里你只看得见自己，haze 里你什么都看得见但都发黄。写杭州的秋天，用 haze 比用 fog 诚实。

sprinkle 是"撒"，也是"下小雨"。a sprinkle of rain 是毛毛雨，sprinkle 的重点在"一点点、散着来"。它跟 pour 正好是一对，一个是撒，一个是倒。这两个词放在一起，一场雨就有了节奏。

leak 是漏，drainage 是排水，dump 是倾倒。这三个词都属于"城市的基础设施"这一族，你在写老校区、写城中村、写任何一个不够新的地方时，它们是骨架。这一章里我用它们写了一栋楼的窘迫：水从天花板漏下来，排水跟不上，垃圾被倒在拐角。三个词，一个处境。

flare 是"突然亮起来"，也是"突然发作"。它形容的是"一下子"，不是"一直"。glitter 是碎

光，silvery 是银色，luminous 是发光。三个词都跟光有关，但 glitter 是碎片，silvery 是颜色，luminous 是能力。你写一条河，用 silvery 是写它看起来的样子，用 luminous 是写它在发光，用 glitter 是写光在水面上碎成一片。选哪个，取决于你想让读者看见什么。

tidal 是潮汐的。这个词在杭州有特殊的分量，因为这里的江有涌潮。你写"江水是潮汐的"，读者就知道这条江不是安静的。

关于想家

想家这个词，英语里没有一个词能完全对上。homesickness 太直，nostalgia 太旧，longing 太重。所以这一章我没有让林一说"我想家"，我让他给家里打电话，让他在电话里听见一只鸟。

alienation 是疏离，它描述的不是"想"，是"不在其中"。你在一个热闹的房间里，你也在场，你也不属于，这就是 alienation。它跟 loneliness 的差别是：loneliness 是一个人，alienation 是很多人里的一个人。

companionship 是陪伴。它比 company 暖，因为 company 只是"有人在场"，companionship 是"有人和你一起过这段时间"。这一章里张野不说话，但他把箱子的另一头抬起来了，那就是 companionship。

compassion 是同情，com 是一起，passion 最初的意思是"受苦"。所以 compassion 的字面意思是"和人一起受苦"。它不是可怜，可怜是居高临下的；compassion 是蹲下来。这个词的词源值得你记一辈子。

esteem 是尊重，self-esteem 是自尊。它跟 respect 的差别是温度：respect 可以很远，esteem 一定很近。一个人被 esteem，是被看重，不只是被承认。

endure 是忍受，也是"持续"。这两个意思共用一个词，是因为能忍受的东西才能持续。你在军训里 endure 的是热，你在四年里 endure 的是一种慢。

reassure 是"让他放心"，它是"再保证一次"。一个人已经在担心了，你说的话才叫 reassure；他还没担心，你说的话叫 encourage。这一层"再"字，是这个词的骨头。

reluctance 与 willingness 是一对。前者是"不太愿意但还是做了"，后者是"愿意"。写作时写人物，reluctance 永远比 willingness 好看，因为人的真实状态多半是前者。

mutual 是相互的。mutual respect 是相互尊重，mutual understanding 是相互理解。注意它不能单方面使用，你不能说 mutual respect from me to you。

这些词我讲得慢，是因为它们不属于"背下来就能用"的那一类。它们属于"你经历过一次，才敢用一次"的那一类。而这一章写的，正是那第一次。

一张表格上的词

报到那天，林一在窗口之间走了九趟，遇见的几乎全是"程序词"。这一族词最容易被当成同义词糊过去，其实它们各有各的位置。

clearance 是"结关、放行"。它的本义是"清掉障碍"，所以它既可以指海关放行，也可以指安检放行，也可以指"空隙"。你在学校里遇见它，多半是在"办理离校手续"那一栏。它的重点是"从此没有

挂账"，不是"批准"。

quota 是"配额"。它跟 limit 的差别是：limit 是上限，quota 是"分给你的那一份"。一个班有奖学金名额，那是 quota；一门课最多选六十人，那是 limit。

installment 是"分期付款的一期"，也是"连载的一期"。它的词根是 install，"安置"，所以它本来的意思是"把一笔钱分几次安置下去"。你在缴费单上看见它，说明你可以分几次付；你在杂志上看见它，说明故事还没完。

transaction 是"交易"。它比 deal 正式，比 trade 抽象。银行里那笔划账是它，两个人之间的任何一次交换也可以是它。这个词的妙处是它把"给和拿"写成了一件事。

receipt 是"收据"。注意 p 不发音，读起来像 re-seat。它跟 recipe（食谱）长得像，词源也确实是同一个，本义都是"取下的东西"。

declaration 是"声明"，也是"申报"。海关那张单子叫 declaration，是因为你在"声明"你带了什么。这个词的重点是"公开说出来"，所以它比 statement 重。

register 是"登记"，也是"显示出"，也是语言学的"语域"。三个意思共用一个核心：留下痕迹。你在名册上留下痕迹是登记，你的脸留下痕迹是"显出"，你的说话方式留下痕迹是"语域"。

proceedings 是"程序"，也是"会议记录"。它永远是复数。一场会议的 proceedings 既是"它怎么进行的"，也是"它被记下来的样子"，这两件事在英语里是同一个词，这件事本身就很有意思。

cancellation 是"取消"。注意它是双 l，美式拼法也常写双 l，但 cancel 的过去式美式写 canceled、英式写 cancelled，这条分歧至今没有统一。

postponement 是"延期"。它跟 cancellation 的差别是"还办不办"：postponement 是往后挪，cancellation 是不办了。这两个词在通知里常常一起出现，读完你得确认自己拿到的是哪一个。

我把这一族词单独拎出来，是因为它们看着最没有画面感，实际上最能救命。你出国、办证、报销、退课，卡住你的从来不是难词，是这些你以为你认识的小词。它们像螺丝，平时看不见，一松，整台机器就响。

到这里，这一章真的说完了。你现在手上有两样东西：一场雨，和九百多个词。雨会忘，词也会忘，但"在雨里认出这个词"的能力不会忘，因为那是你自己长出来的，不是我给的。

第 2 章　第一堂课与图书馆

The registrar's server died at 8:04 on the first morning we were allowed to choose our courses, and the dormitory took the news badly. Six of us around one cable, one backup disk that nobody could find, one broom handle with the wire taped to it. My screen gave me a blank white rectangle with a tiny spiral turning in the corner, and I watched it turn for eleven minutes, zooming in and out with two fingers as though the answer were hiding in the texture of the pixels.

There was an abundance of advice and an absence of information. Somebody had

allegedly spoken to somebody in the office who had allegedly seen an email; the email was anonymous and the wording was ambiguous; nobody could produce it. In the absence of a fact we aligned ourselves behind theories, the way people do, and by half past eight there were four factions on our corridor, one rumour that the system had been abandoned since Sunday, and a general belief that the whole thing was an accident of scheduling rather than a decision.

I had been accustomed to a school where the timetable arrived on paper, in a sealed envelope, on the last day of term, and where the only agent of change was a teacher with a stamp. This was different. Here the agent was a server, and the server had no face, and there was no aid to be had from anybody except a boy with a paper form.

"Any luck?" Zhang Ye asked.

"Zero."

"Then stop waiting." He pulled on a belt that had been mended twice, picked up a neat stack of forms and a pen, and went out. He was certainly right that a queue of strangers in a corridor would move faster than a server, and I resented him for it, which is the habitual response of a person who has relied on the wrong thing.

By nine the corridor was full. Forty of us stood amongst the dust and the dirt of a building that had been renovated the summer before and already looked worn, awaiting a number that nobody was going to call. Somebody had taped a notice to the door; somebody else had annotated it in three colours; a third person had torn a corner off it, which is what happens to official paper in a crowd. The garbage bin alongside the stairs was overflowing with cups and cigarette ends, and the smell of alcohol from the clinic next door came and went with the draught.

An awful patience settled on the queue. Nobody complained aloud, although two boys at the back kept up a low commentary that was funny enough to be repeated and not funny enough to be punished. When the door finally opened there was a small forward surge, a caution from the man in the doorway, and then the line closed up again like something that had been practising.

Wang Laoshi arrived at ten past nine with his thermos and his benevolent expression, and said what he always said.

"None of this is a big deal." Then he acknowledged, honestly, that it was a small big deal, and that the adjustment would be announced by noon, and that in the meantime we should read the notice and assume nothing. He was a bureaucrat by job description and a compassionate one by temperament, and the combination made him the only officer of the institution I had met so far who did not talk as if he were reading a form aloud.

The notice was attachment two of a message nobody had opened, and it was aptly named, because nobody had opened it. In accordance with the admission rules, it said,

eligibility for exemption from College English depends on a placement test; the allocation of places is made by lottery, not by speed; students who fail to obtain approval by Friday will be granted an alternate appointment with the registrar; the array of optional courses is subject to change; and any enquiry regarding the scheme should be addressed to the office, irrespective of what the notice says, and provided that the office is open, which obviously it is not before nine.

It asked us to conform to the code of the platform, to refresh the page no more than once per minute, and to accept that the oversight of the system was largely a matter of luck versus engineering, and that whichever interpretation we preferred, the outcome was the same. Such is the language of institutions. These sentences are authoritative and competent and empty at the same time, and whenever I read one I understand why my father asks my mother to read the letters from the bank, and why she reads them standing up, and why she says okay at the end of every paragraph even when the paragraph has not asked her anything.

I began that autumn to understand what an institution actually is, and the understanding did not make me like them better. An institution is not a building or a rulebook; it is a set of arrangements that outlive the people who made them. It allocates what nobody owns, it delegates what nobody wants to decide alone, and it subordinates one office to another and calls the result a hierarchy. It prefers a form to a conversation, because a form can be inspected and a conversation cannot. It justifies itself by procedure, which is the only kind of justification that cannot be argued with. It asks for compliance rather than obedience, because compliance can be measured. It rewards a promotion, not a performance; it restrains the enthusiastic and indulges the punctual; it keeps a collective memory in a file that nobody reads until there is trouble. And it is, in the end, the only machine we have for making twenty-nine people want different things on the same morning without anybody having to be a hero.

I read it four times and understood perhaps a third of it. This is the first lesson the administrative world teaches you, and it teaches it without malice: the paragraph is not hiding anything. It is written for people who already know what it means. Adequacy, in that world, means that the information is technically present. It does not mean that anybody can use it.

Zhang Ye came back at half past eleven with a stamp on his form and a timetable that looked suspiciously complete. He had walked into a room, found a human being behind a counter, and asked a direct question. He got the materials introduction on Monday, basketball on Tuesday, and a bargain on a second-hand calculator from the senior who was processing the applications.

"How?" I said.

"I bet on the counter instead of the screen." He shrugged. "The counter has a staff

member behind it. A staff member has a face."

By my own calculation, I had clicked three hundred and eleven times for nothing. At two in the afternoon I stood in a queue that went out of the building and around the bicycle racks, and when my turn came the woman behind the glass looked at my card, typed four characters, and said, "You're fine. You have everything."

Everything. As if the morning had not happened. As if I had not spent four hours staring at a spiral in a corner, accusing a machine of conspiracy.

Afterwards I walked back across the campus with the flat, ridiculous feeling of a crisis that turns out to be an inconvenience. The sun was out. The plane trees had begun to shed a dusty layer of bark, and the pavement was littered with it, pale and curled, like something peeled. Later I would learn that this happens every September and that nobody notices it after the first year, and I would remember this afternoon as the last time I saw a plane tree undress.

That night Zhang Ye sat on the floor with a stone and a knife and ground the blade in slow, even strokes. He had learned it from his grandfather: a dull knife tears, a sharp one cuts, and the difference is not strength but attention. I wrote that in my notebook. Half of what I would learn this year arrived like that — sideways, from a roommate with a stone, while I was busy being miserable about something else.

The first real class was at eight on Monday, in a lecture hall that seated two hundred and had been cooled to the temperature of a refrigerator. The architecture was brutal and functional: a concrete shell, tiers of wooden seats, a ceiling with the pipes exposed overhead, and a blackboard three metres wide that nobody had used since the projector arrived. I sat in the fifth row, which is the row of people who want to be seen taking notes but not seen answering, and by 7:55 every seat around me was full.

Two hundred freshmen in identical uniforms of shirt and sweater, stiff with newness, upright in their posture as if an inspection were due. The air was stuffy and steamy with humidity and the breath of two hundred people, and the heating rattled in the radiator under the window, and somebody near the back was coughing into a sleeve, and somebody else sighed at intervals of about ninety seconds, and a boy in the second row gasped when the projector flashed on and projected a slide of one hundred and four pages.

The lecturer was a thin man in his fifties with a wireless microphone clipped to his collar and the manner of a person who had decided long ago that interruption was a tax he would not pay. He began in the middle of a sentence that had clearly started in his head on the way over.

"Matter is organized on scales," he said. "You will spend four years learning what that sentence means."

Then he was off. Atoms arranged in lattices. Grains and boundaries. The difference between a structure and a microstructure. He spoke about the component parts of an alloy, about the element that makes steel hardy, about the molecule that makes a plastic flexible and the spherical particle that makes a paint opaque. He used the word dimensional three times in one breath, integral twice, and longitudinal once, and each time he said it as though it were the most ordinary word in the world, which, in his world, it was.

I wrote frantically for eleven minutes. Then I looked down and saw that I had produced a list of nouns with no verbs between them: element, component, phase, interface. I had captured the labels and lost the argument. Somewhere around minute twenty he drew a diagram on the board — a sphere inside a cube, then a spiral of arrows, then a curve that fluctuated for no reason I could follow — and said that insulation is a matter of trapping air, and that the purity of a crystal decides whether it conducts or resists, and that solidity is not the absence of motion but its concealment.

At 8:52 I conceded defeat on the notes and simply watched him. That turned out to be the right decision, though I made it for the wrong reason. If I could not record, I could at least observe: that he never looked at the screen; that he drew with his left hand and erased with his right; that when a phone buzzed in the third row he paused for exactly two seconds and then continued at the same pitch, like a man who had concluded that attention is voluntary and therefore cannot be demanded.

The bell went at 9:35. Two hundred people stood up at once, and the noise was like a flock of birds lifting off a field. I stayed seated for a moment, looking at my page of nouns, numb from the neck up and ashamed in a way I could not justify.

"Rough?" said a voice behind me.

I turned. A girl in a grey sweater, hair tied back, a pen clipped to her collar, was packing a bag that had clearly been packed a thousand times. Senior, I thought, before she said anything: the way she moved was economical, unhurried, and solely purposeful.

"First lecture?" she said.

"That obvious?"

"You wrote for eleven minutes and then you stopped." She nodded at my page. "Everyone writes for eleven minutes. The ones who keep writing are copying the slides. The ones who stop are thinking. Or drowning." She looked at me. "Which?"

"Drowning."

"Good. Drowning is fixable." She swung the bag onto one shoulder. "Rule one: he puts the questions in the last five minutes of the last class before the break. Rule two: the slides are not the lecture. Rule three — " and here she hesitated, the only hesitation

in the whole conversation — "don't record him. Recording is how you decide not to listen."

"Who are you?"

"Shen Lu. Fourth year. Materials." She was already walking. "Sit in the third row next time. The fifth row is for people who want to look attentive."

College English met on Wednesday at ten, in a room with forty chairs and thirty-nine people, one of whom was me, in the back row, behind a girl whose laptop was covered in stickers. The teacher's surname was Feng. She wrote it on the board in English and in pinyin, and said that she would use English for the whole term, that we should interrupt her the moment anything was not intelligible, and that she would not be offended.

Nobody interrupted her. Not once, in ninety minutes. Forty of us sat with the patience of passengers waiting for a train that had already left.

She began with a dictation. Ten sentences, twice each, at a speed that turned the words into one long ribbon of sound. I caught "the". I caught "of". I caught "and", twice, and wrote "end" the first time and had to cross it out. The rest arrived as weather, a front moving across the room and dropping syllables on the desks.

Then she asked us to name the parts. In a sentence about a committee and a meeting, she wanted to know what "to postpone" was doing. A long silence. Somebody said verb. Somebody else said future tense. She said, gently, that it was the infinitive, wrote the word on the board, and I copied it and understood, at that exact moment, that I had been taught the word three or four times in six years and had never once needed it inside a sentence of my own.

She moved on to phrasal verbs, which she called the part of English that is not fair. Put off. Put up with. Put down. Three verbs sharing one small word and sharing nothing else. We were told to use each of them in a declarative sentence about our own week. The boy beside me wrote that he had put off his homework until Sunday, and she praised the sentence, and I wrote that I had put up with the noise in my dormitory, which was true, and then crossed it out, because I was not certain whether the "with" was doing something I had not been taught.

By eleven I had a page of notes, a headache, and that particular heat in the face that comes when you are the only person in a room who does not know what is happening and everybody is being polite about it. I looked around at thirty-eight faces and could not tell whether the others were following or simply better at looking as though they were. This was the first useful thing I learned in college: silence in a classroom is not evidence of understanding. It is evidence of silence.

Feng kept us for the last ten minutes with a story about a student she had taught, a boy from a village in the south who had failed the listening section three times and had

come to her office after the third failure in a state of bewilderment that she said was worse than despair. He was not lazy; he had done everything she asked. He had made a list, he had learned the list, and the list had done nothing for him. His determination was not the problem, she said, and his courage was not the problem either; the problem was that he had been practising a language that nobody spoke. He was desperate enough, in the end, to ask for help, which she said was the rarest form of endeavour in a first-year student.

In his fourth term he began to talk in class, badly, slowly, with the wrong tenses and the wrong prepositions and an absolute refusal to be ashamed of it. She said the turning point was not a method. It was the day he stopped translating in his head. For six weeks he had been trying to do two things at once, and the two things had defeated him, and the defeat had felt like a verdict on him rather than on the method. When he finally passed the test he was not triumphant; he was quiet, and he said it was like being let out of a room he had been in all his life without noticing the door.

"You cannot listen and translate at the same time," she said. "There is only one mouth." Then she looked at the clock, said the word tomorrow as if it were a gift, and left.

I sat in the empty room for a while with a kind of bewildered respect. There was no fantasy in what she had told us, nothing about talent, nothing about a secret. Just the boring arithmetic of years. And yet the arithmetic did not console me, because arithmetic is exactly what a person who is behind does not want to hear.

On Thursday the materials lecturer walked into the same hall, put his bag down, and continued from the exact word where he had stopped, as if the interval had been a single breath. He talked about why sugar dissolves in water and sand does not; why a puddle evaporates by noon and a sealed bottle keeps its water for a year; why a gas compresses under your thumb while a block of steel does not care about your thumb. He drew a molecule as one large sphere with two small ones attached, then drew it again, three times bigger, with the angles marked, and said that the whole course could be reduced to one sentence: properties come from structure, and structure comes from the way atoms agree to sit beside each other.

Then he said something I wrote down twice, because I could tell it was the sort of sentence that would appear on a paper in some form. "Every material you will ever touch is a compromise between what the bonds want and what the process allows." A metal is a lattice of atoms that do not mind being pushed, which is why a wire bends. A ceramic is a lattice that minds very much, which is why a plate cracks instead of bending. Glass is neither, exactly, and the reason glass is transparent has nothing to do with how clear it

looks and everything to do with how its electrons are arranged.

He spent the last twenty minutes of that lecture on measurement, which I had assumed would be the boring part and which turned out to be the part that decided everything. A measurement, he said, is a sentence with an error in it, and the error is not a mistake; the error is the honest width of your ignorance. You magnify a thing a hundred thousand times and the thing you see is not the thing; it is the thing plus the instrument. A substantial difference is one that survives the error. A negligible one is a difference in the noise. The whole apparatus of a laboratory, he said, exists to make the difference between those two words distinguishable, and the apparatus costs more than the building, and the building costs more than the street.

I wrote that down, and then I wrote underneath it, in smaller letters, that this was also a description of an argument, and that I had never heard anybody say it about an argument before.

At the back of the hall four boys were playing a game on a phone with the sound off. In front of me a girl was drawing the lecturer, quite well, in the margin of her notes. I envied both groups. My own page had a diagram that looked like a plate of noodles and a sentence I had written in a panic, with a question mark so large it had its own shadow.

The library was the only warm building on that side of the campus, so I went there at four with no plan and a bag full of notes I did not understand. It was put up in 1987 and had never pretended otherwise: six floors, a marble staircase worn into shallow bowls at the centre of each step, one lift that was either full or broken, and a turnstile that read my card twice before it believed me. The air had that library smell which is not one smell but three: dust, paper, and the ghost of somebody's instant coffee from 1998.

Behind the counter sat a librarian of about sixty with reading glasses on a cord, a cart of returned books beside her, and the expression of a person who has answered every question twice already today. She took my card, looked at the photograph, looked at me, and decided I was probably the photograph. Then she told me, without looking up, that first years usually went to the third floor, that the third floor was for reading and not for sleeping, that the fourth floor had the journals, that the basement had the old newspapers nobody had asked for since the eighties, and that the reading room closed at ten, not at ten past ten.

I liked her immediately. There is a kind of authority that comes from knowing exactly how a building works, and she had it in the way other people have opinions.

On the landing between the third and fourth floors there was a temporary exhibition: eight panels on stands, each lit from above, each with a headline in large type and a paragraph underneath, on the theme of a century of newspapers. I stopped because the

first panel had a photograph of a ship, and I stayed for forty minutes, standing up, with my bag on the floor and my coat still on.

The first panel said that the last emperor had signed away a crown his family had worn for two hundred and sixty-eight years, and that the abolition of the imperial system had been announced in a proclamation of four hundred characters, printed on the front page of every paper that could afford the ink. The royalty kept the jewels and the ivory and the velvet; the palace kept the crown on a shelf in a museum; and the ancient doctrine that a lord rules by nature died in a paragraph of legal prose. I read it twice. It is strange to meet the end of an empire as a paragraph on a stand in a stairwell.

The second panel was about armament. It said that between 1890 and 1914 a single brigade of infantry had cost more than the schools of a province, that a fleet had cost more than a federation, and that the militia had been the last institution in which ordinary men learned to march in step. When the marching ceased, the states found they had built an apparatus with nothing to point at, and the outbreak of 1914 gave it a target.

The third panel was about maps, and it used the word annexation four times in one column. A consul negotiated for eleven months over a harbour. A marshal refused to sign. A delegation went home without a treaty. The map was redrawn by surveyors who had never crossed the gulf they were dividing, and the doctrine of territorial integrity was quoted in parliamentary debate by men who had already sold the province.

The fourth was about money. The fiscal year, it said, was invented so that budgets could be compared; the gold standard was a promise nobody could keep; monetary policy in those decades was mostly guessing in public; a statutory limit on borrowing was written into law by men who assumed they would never need it. Customs duties paid for the navy. The deregulation of the ports paid for nothing and was applauded.

The fifth was about cities. Migration from the countryside doubled the population of four capitals in thirty years. The indigenous population was counted last and housed worst. Ethnic neighbourhoods were described in a vocabulary we would not print today. A village forty kilometres from a railhead could be, in the same decade, more cosmopolitan than a provincial capital, and a suburban street in a metropolitan sector could be poorer than the rural pasture it had replaced.

The sixth was about the vote, and about the long, ugly arithmetic by which the excluded were counted as fractions of a person. Equality arrived in instalments. Equity was promised and withheld. Segregation was defended in parliamentary speech as tradition, opposed by protest, suppressed by force, and abolished in legislation that took forty years to enforce. A republican government persecuted the minority it had promised to protect. A presidential decree overruled a statute. And the doctrine of nationalism prevailed in every country that could afford a newspaper.

The seventh was about the machinery of news itself. A headline, it said, is a promise made by somebody who will not be present when it is broken. Propaganda is not lies; it is truth with the proportions changed. A columnist writes the column, a syndicate sells it to forty papers, a pollster measures what readers want to hear, and a terrorist outbreak sells more copies than a treaty.

The eleventh panel had no photograph at all. It was four sentences long, and the last of them was a note that the exhibition had been prepared by a fourth-year student in the department of journalism, and that the opinions were hers.

The club fair was on the second Friday of September, on the sports ground, in a wind that took the posters out of people's hands and laid them flat on the track. Ninety-one tables, four hundred people, and a loudspeaker repeating every four minutes that the fair was sponsored by the students' union, that sponsorship implied no endorsement of the opinions of any club, that participation was open to every first-year student, and that applications closed at six.

I walked the first row without stopping. An astronomy club, with a photograph of a planetary nebula and a line about the antenna on the roof of the physics building, inviting us to look through the optic tube on clear nights. An ecology club, with a botanical chart of the campus trees and a promise to count the birds in the flock that wintered on the lake. A club for the humane treatment of the stray cats behind the canteen, which had the longest queue at the fair, the least paperwork, and a poster of an adorable kitten that had been photographed, according to the caption, while committing a crime. An opera club, where a girl was singing in a register I did not know a human being possessed, and where three old men from the neighbourhood — actual old men, not students — sat on plastic chairs with their eyes closed, applauding at exactly the right moments.

A film club had a laptop playing a thriller with the sound off, a stack of scripts, and a boy who said they shot one short film a term. "Last term it was a comedy about a lift," he said. "It won nothing. We loved it." A cartoon club had a portrait of the chairperson drawn by the chairperson, and a sign saying that commissions were welcome and cheap. A drama club was holding auditions for a play about a family that loses its house, and the girl behind the table said they needed somebody to play the father, and that the part was small but that the applause at the end was real. A music club had two pianos and a rule that anybody could play for ten minutes if they signed the sheet, and somebody was playing the same lyric over and over, badly and happily.

A circus club was juggling. A basketball club had a hoop bolted to a frame and a boy who kept saying the word drill. A hockey club had three sticks, a ball, and no net. A fitness club had a poster of a man doing a workout with the expression of a person being

paid to look happy, and a line underneath about posture, and another about the difference between being lean and being thin, and a third that said, in smaller type, that the club was for the stout as much as for the quick, and that a hardy body was built in years and not in weeks.

A carpentry club had a bench, a vice, and a piece of bamboo that a boy was turning into a flute. Their poster said that they met on Sundays, that beginners were welcome, and that the only rule was that you swept up. A business club was running an auction of second-hand textbooks to demonstrate how a market works. The auctioneer, a girl with a microphone and no fear, had sold a calculus book for eleven yuan and a dictionary for four. Their poster explained, in small print, that the club also ran a workshop on how a company goes into bankruptcy, that it had once invited a man who had been convicted of fraud to speak about fraud, and that it took no position on bribery except that it was illegal and that a country which tolerated it paid for it in the end with its own roads.

She was better than the poster. She said that a price is not a value, that a value is only what somebody will pay, and that the rest of economics is accounting. She said that an enterprise is a promise about the future wearing the clothes of a firm. She said that a debtor who cannot repay does not disappear; he becomes a different kind of person, and the person he becomes is the reason banks have carpets. She took a bid of one yuan from a boy with no money and told him that a bid is a promise, that if he could not pay he would forfeit the book and his reputation, and that at his age the two were worth about the same.

Then she said that most of what a young person believes about money is a rumour. A client is somebody who has already decided. An agent is somebody who decides for somebody else. A payment is a promise that has been kept. A salary is the price of a seat, and a layoff is what happens when the seat is no longer worth the price. She said that a lavish office and an extravagant launch are usually paid for by somebody who is not in the room, and that the productive part of any company is the part that nobody photographs. Profit, she said, is not greed; it is the margin that lets a firm survive a bad year, and a firm with no margin is a firm that will treat its people badly in the second year and blame them for it.

Then she said four things that I wrote down on the back of the club card, and I have checked them since against people who ought to know.

Investment is money that works while you sleep. A premium is what you pay for certainty, and it is almost always too much. Profit is what is left when everybody has been paid, which is why it is smaller than people think and more fragile. And a saving is not a small thing; it is a large thing that has not happened yet.

Somebody in the crowd asked whether the club could teach him to earn a dollar by Friday. She said no, that the club could teach him to read a contract, which was cheaper,

and that the first clause to look for was the one about who pays if something goes wrong. She said that the most expensive sentence in the language was a short one, and that it usually began with the word provided.

Next to them, a debating society. Their poster was the only one on the field that had been written by somebody who enjoyed writing. It said that an assertion is not an argument, that a falsehood repeated is still a falsehood, that the most contentious question of the year had been whether the library should open at midnight, and that anybody who found the question disputable was welcome on Tuesday. It said that the club did not teach people to win; it taught them to lose in public without flinching, which is a harder skill and a more useful one, and that the best speaker of the previous year had been a girl who had never won a single controversy and had never once been unconvincing.

I stopped at that one for a while. Then I went on, because a boy with a clipboard had begun to walk towards me with the expression of a person who has been told to recruit forty people before six o'clock.

A travel club had photographs of a caravan of bicycles on a coast road, a shore at sunrise, an isle reachable by a ferry that ran twice a day, a gulf photographed from a height that made my stomach turn, and a boy surfing in a photograph that had clearly been taken by somebody who loved him. They said their longest trip had been four hundred kilometres along an expressway service road, which was legal, and along a footpath beside it, which was not. They said a pedestrian had once walked with them for six kilometres and then turned off without a word, and that this had been the best thing that happened all year.

An agricultural club, which was really four students from the agricultural college who kept two oxen and a mule on a farm forty minutes away and needed people to help with the plowing in April. Their poster had a photograph of a pasture at dawn, a caption about the harvest, and a warning that the work was seasonal and the bus left at five in the morning. A radio club had an antenna on the table, a soldering iron, and a boy who wanted to explain the difference between a podcast and a broadcast and lost me at the second sentence. A car club had an engine on a stand, which I looked at for a long time because it was beautiful in a way I had not expected an engine to be: all those pipes, all that order, all that patience in a box.

A psychology club had a table covered in handwritten cards. Each card had a sentence on it and each sentence was a question that nobody had asked me before. One said that depression is not sadness and that the difference matters. Another said that the ego is a story you tell about yourself and that it is edited, like all stories, by the person who benefits from it. A third said that temptation is not a test of willpower but a matter of distance, and that the people who resisted best were the ones who had arranged their lives so that the thing was far away.

A martial arts club had a photograph of six students in a line, all of them upright, all of them still, and a caption about the discipline of standing. A volunteer association had a list of the places they went on Saturdays: a home for elderly people, a school for the children of migrant workers, a shelter that took in the needy and asked no questions. Their poster said that compassion was cheap and that showing up was expensive, and that they would take anybody who could show up.

Halfway down the third row I found the materials students' association, and behind the table, in the same grey sweater, Shen Lu.

"You're not a first year," I said.

"I'm recruiting for the open day. The electron microscope room opens twice a term. Six people at a time, no phones, no bags, and you sign a form that says you understand that the instrument costs more than your family's flat." She pushed a card across the table. "The political society is next to us. They will try to recruit you. They recruit everybody. It's their doctrine."

"Are you a member?"

"I'm a member of nothing." She said it without bitterness, as a statement of fact, the way she said everything. "I have eleven months left. I have stopped joining things."

I took the card. On the back, in her handwriting, there was a sentence I did not understand until much later: the instrument does not care what you believe.

On Saturday morning I went to the clinic, because the cough that had started on Wednesday had settled into my chest and because Wang Laoshi had told me to. The waiting room had eleven chairs, a radiator, and a notice board with four posters on it, and I read all four, twice, because the waiting was long and the posters were the only reading matter in the room that was not a form.

The first said that influenza is not a bad cold. It infects the throat first and the lung second, it is deadly in the very old and the very young, and the incidence rises every winter in the fortnight after the first cold snap. An annual injection reduces the risk and does not abolish it. Vitamin supplements, whatever the advertisements say, do not prevent infection, and the only nutrient that has ever been shown to shorten an illness is water.

The second was about teeth. It said that a tooth is the only part of the body that cannot repair itself, that the nerve inside it has no defence except the enamel above it, and that a dentist would rather see you twice a year for nothing than once for an hour with a drill. There was a photograph of a tooth in section, with the nerve drawn in red like a small pink root, and I looked at it longer than I wanted to.

The third was about smoking, and it was the least polite poster I have ever read in a public building. It said that smoking causes cancer of the lung and of the throat; that the artery is narrowed by it long before anybody feels anything; that the risk is not a rumour, not a conspiracy, and not a matter of luck; and that the companies had known all of it since 1962. The caption under the photograph said, in the flat voice of a witness, that the only thing exaggerated was the pleasure.

The fourth was a chart of what to eat in winter. Green vegetables, a fruit a day, a supplement of vitamin D in the months with no sun. It said that a greasy diet is not a sin but a tax, and that the cheapest nutrient in the canteen was the one nobody bought.

The doctor was about sixty, with glasses pushed up on his forehead and the manner of a man who had seen four thousand sore throats and expected to see four thousand more. He asked three questions, looked at my throat, listened to my chest and told me to breathe in and out, twice, then again. Then he sat back and said, "Not influenza. A common infection. Your throat is red, your glands are up, and you have a fever of thirty-eight. Two days of rest, plenty of water, and this for the fever."

He wrote the prescription in handwriting that was almost a signature and told me to come back on Monday if the fever had not gone. "No antibiotics," he said, seeing my face. "It is a virus. An antibiotic would be a decoration." Then he added, in a different voice, "You are a first year."

"Yes."

"Then sleep. Everybody thinks the treatment is the medicine. The treatment is the sleep."

While he wrote, an old man came in with a dressing on his forearm, and the doctor unwound it and looked at it for a long time without saying anything. A wound about four centimetres long, clean at the edges, with a scar already forming at one end. "This one is healing well," he said, to me, as though I were a student. "The flesh closes from the edges inward. A scar is not a failure. It is the fastest repair available, and it is faster than the beautiful one." He put a new dressing on, and the man said something in dialect that I did not catch, and the doctor laughed, and the man laughed, and I understood that they had done this forty times.

Then he turned back to me. "You are young, so you think a body is a machine that does not break. It breaks. The illness that kills most people in this country is not the dramatic one; it is the slow one, and it begins with a number that nobody checks." He tapped the blood pressure cuff on the desk. "The artery does not complain. That is its whole character." If the fever had been higher, he said, they would have hospitalized me, and I would have spent a night in a ward with four strangers and a television, and I would have hated it, so I should drink water.

There was a card in the rack by the door about screening, which said that the earlier a thing is found the better, and that this was true of the breast and of the colon and of almost everything else. There was a notice about a clinical trial of a new dressing, and another about a medicinal herb that had been tested properly and found to do nothing at all, and a line at the bottom of the second one which said that the onset of its effect had been no different from the onset in the group that had been given sugar. I read that line three times. It seemed to me the most honest sentence in the building.

I walked out past the four posters and thought about how much of the knowledge that keeps a body alive is printed on a wall in a building nobody wants to visit, and about the fact that I had walked past that wall eleven times without reading a word of it.

The basement of the library was two degrees colder than the rest of the building and smelt of wet stone. The lights came on in rows as I walked, clicking above my head like a series of small approvals, and went out behind me. The shelves down here were the old kind, grey steel with a chain across the aisle, and the books on them were the old kind too: bound volumes of newspapers, each spine lettered in gold that had gone green, each volume the size of a suitcase. Half of them were mouldy at the corner where the damp had got in. The paper had gone soft and brown and had a smell that was not unpleasant, which is the most dangerous thing about old books.

At the end of the second aisle there was a table with a lamp on it and a cart of volumes, and behind the cart was Shen Lu, in the same grey sweater, with a pencil behind her ear and a duster in her hand.

"You again," she said.

"I was told the basement had newspapers."

"The basement has everything nobody wants." She pulled a volume off the cart, opened it, and showed me a page of shipping news from 1953. "Twenty-two yuan an hour to dust these. I have been doing it since my second year. It is the best-paid thinking I have ever done."

I asked her what she was thinking about, and she said, "Whether I should stay," and then, before I could ask the obvious question, she asked me one instead. "How many words did you look up today?"

"Seventeen."

"Write them down?"

"In the notes app. With the sentence."

"Good." She turned a page. "Now throw the list away and keep the sentences. A word without a sentence is a rumour. You will meet the rumour again in June and it will

signify nothing to you, and you will blame your memory, and your memory will be innocent."

I said that this sounded like the advice of somebody who had already failed at it, and she gave me a look that was not quite a smile. "I failed at it for two years. I had a list of three thousand words. I could tell you what they meant in Chinese and I could not use one of them in a sentence. Then I went into the electron microscope room with a supervisor who spoke English at the speed of a machine, and I understood nothing for six weeks, and at the end of six weeks I understood everything, and I had not learned a single new word in that time. What changed was that the words stopped being separate."

She put the volume down, wiped her hands on the duster, and pulled a folded sheet of paper out of the back of her notebook. It was a page from a book, photocopied so many times that the type had gone furry at the edges.

"Read the first paragraph," she said. "Out loud. Then tell me what it is about without using any word from it."

I read it. It was about water, and it was written by somebody who loved the subject and did not care whether I followed. Then I tried to say what it was about, and discovered that I could not, because every word I wanted was a word from the paragraph, and the paragraph had taken all of them.

"That is the problem," she said. "You have a vocabulary and no language. The vocabulary is a heap of bricks. The language is the wall. You can carry bricks around for four years and never build anything, and nobody will be able to tell the difference until they ask you to stand out of the rain."

I stayed in the basement until the lights clicked off in the far aisle and the announcement came over the speaker that the reading room would close in fifteen minutes. In that time I pulled four books off the shelves, not because I needed them, but because they were there.

The first was a medical text from 1974, with a chapter on influenza written by somebody who had watched it kill. It described the fever, the ache behind the eyes, the way an infection moves from the throat to the lung in three days, and it used the word clinical in the older sense, meaning what you see at the bed rather than what you find in a test. The chapter ended with a paragraph on treatment that was four lines long and mostly about water, and I thought about how the whole apparatus of a hospital, the injection, the ward, the long rehabilitation afterwards, comes down in the end to somebody bringing you a glass.

The second was a book about food, printed on paper that had gone yellow in stripes, with a photograph of a bakery on the cover. It had chapters on dough and on the chemistry of an oven, an entire section on why a roast browns and a boiled potato does not, and a

warning about mouldy bread that was so vivid I could smell it. There was a paragraph about the difference between a loaf that has been left to rise and one that has been hurried, which the author compared to the difference between a conversation and a lecture.

The third was a fat illustrated volume on the living world, the kind that is bought for children and read by adults. It had plates. A camel with its knees folded under it, and a caption explaining that the hump is fat and not water. An ox photographed against a red background, with a note about the yoke. A mule, described as the most patient animal that has ever been asked to do somebody else's work. An ape with a face so close to a person's that the caption became embarrassed and changed the subject to hands. A crocodile on a mud bank with its mouth open, and a sentence underneath that said it opens its mouth not to bite but to cool itself. A whale, drawn in outline across two pages, with a scale in the corner.

And at the back, the invertebrates: a snail on a leaf, a bee in the act of robbing a flower, a flock of something in flight, a thorn bush with a bird in it. The captions had been written by a person with a sense of humour and a good vocabulary, and one of them, about a parasite that lives in the gill of a fish and eats its tongue and then takes its place, was the most horrible thing I read all term.

The fourth book had nothing to do with any of it. It was a paperback on how to write, and it had a chapter on the sentence that I have gone back to four times since. It said that the paragraph is a unit of breath, that a narration should move at the speed of the person telling it, and that a metaphor is not decoration but a way of borrowing somebody else's eyes. It said that sarcasm is the cheapest form of irony and that a writer who cannot do without it should be sent to work in an office. It said that a quotation is a loan and must be repaid with interest, and at the foot of the page there was a line from somebody I had never heard of: the difference between the right word and the almost right word is the difference between lightning and a lightning bug.

I copied that out, and then I copied out the sentence about the paragraph, and then I closed the book and put it back where I had found it, because it was a library book and because some part of me wanted to be able to find it again.

The language shelves were on the other side of the aisle, and I spent ten minutes there out of pure curiosity, because a shelf of books about words is the only place in a library where the books are about the building they are standing in. There was a reference grammar four inches thick, with a chapter on the article that ran to sixty pages. There was a dictionary of connotation that explained, in eight hundred pages, why thin is a compliment and skinny is not, and why a house is a home but a residence is a letter from a lawyer. There was an essay by a historian on how a legend becomes a fact, and a memorial volume for a professor who had taught in this building for thirty-one

years and whose students had paid for the printing. There was a book on primitive religion with chapters on Buddhism and on Christianity and on the worship of ancestors, and a magazine about copyright that had been founded by two librarians and had lasted four issues, and a publication of the faculty press with the copyright page stamped in red, and a whole shelf of notional dictionaries in which nobody had ever looked up a word.

I did not borrow any of them. I wrote the titles in my notebook, three of them, and told myself I would come back, and I did come back, in November, and one of the three was gone.

The English corner met on Thursday at seven, in a room on the fourth floor with forty chairs arranged in a circle and a poster on the door that said, in three languages, that everyone was welcome and that no one was being graded. I stood outside that door for four minutes, which I know because the clock in the corridor was the kind that clicked.

There were nineteen people inside and eleven of them were first years like me, sitting with their coats on and their hands around paper cups of hot water, waiting for somebody else to begin. The somebody else was a graduate student called Zhou Min, who had a round face and a loud voice and no detectable fear, and who announced that the rule of the corner was that you had to say one thing that was true, out loud, in English, and that mistakes were free.

"Mistakes are free," she said again. "This is the only room on this campus where that is true. Use it."

A boy on the far side went first. He was from the north, and his accent was so thick that I understood perhaps half of his first sentence, and then he said something about his mother's cooking and everybody laughed, and he laughed too, and he kept going, and by the third sentence I had stopped noticing the accent entirely. He was not accurate. He was not shy either, and the second quality had eaten the first.

Then it was my turn. I had prepared three sentences in the corridor. I said the first one — something about the library, something safe — and then I heard my own voice from the outside, thin and careful and about half the volume of everybody else's, and the fourth sentence died in my throat. There is a particular silence that follows a sentence that has stopped in the middle, and it lasts about two seconds, and in those two seconds I understood that everyone in the room had been where I was standing, and that this did not make it any easier.

I have thought about that moment a great deal since. It was not the vocabulary. I had the words; they were in my notebook, underlined. It was that the words were stored in one room of my head and the speaking happened in another, and there was no door between them,

and the only way to build a door was to keep standing in the corridor with my mouth open.

Shen Lu was in the corner by the window, not saying anything. She had a book open on her knee and she was not reading it.

"You came," I said, afterwards, in the corridor.

"I come every week. I don't talk." She closed the book. "You talked. Four sentences. Badly."

"Thank you."

"The fourth sentence didn't finish." She looked at me. "Do you know why?"

"Because I'm — "

"Because you were listening to yourself." She said it without cruelty, the way a person reads a number off an instrument. "You were checking each word as it came out, and the checking takes exactly as long as the next word, and so you fell one word behind, and then two, and then you stopped. The ones who talk badly and happily are not braver than you. They have simply stopped checking. It is a technical problem, not a moral one."

I said that it felt moral.

"Everything feels moral when you are twenty." She put the book in her bag. "It will pass. What will not pass by itself is the listening. Twenty minutes a day, out loud, repeating what you hear, until the shapes are in your mouth and not in your eyes. That is the whole method. It is boring, it takes a year, and there is no way to do it faster, and every person who tells you otherwise is selling something."

She said that the ones who improved were not the ones with the most zeal; they were the ones who could stand a routine. A disciplined twenty minutes is worth an enthusiastic three hours, because the enthusiastic three hours happens once and the twenty minutes happens three hundred times, and the reward for the twenty minutes is gradual and invisible and therefore the only kind that lasts. "You will not feel it working," she said. "That is the design. If you could feel it working you would stop when it stopped feeling good, which is exactly what most people do."

She went down the stairs. I stood in the corridor for a while, then went back in and sat down next to the boy from the north and asked him, in English, where he was from, and he told me, at length, with his terrible accent and his complete lack of concern, and I understood about seventy per cent of it, which was twenty per cent more than the week before.

That night it rained properly for the first time since the training. The thunder came in from the west, one long roll after another, and the window shook in its frame with each one, and the radiator knocked back, as if answering. Zhang Ye was asleep within four

minutes, the way he did everything, and I lay in the dark with my pillow folded in half and listened to the water coming down the pipe outside the wall.

At half past eleven I got up to go to the toilet, in my slippers, past the room where two boys were still arguing about a basketball game in the tone of men settling a border. On the way back I picked up my phone and opened the vocabulary app I had been using since June.

The app is popular. Four million downloads. It has a clean white face and a small green leaf for a logo, and it divides every word into pieces and prints them in grey, with the stress marked in red, because pieces are easier to look at than whole words.

Im, then mensity. Com, then munication. Com, then munity. Ac, com, modate. Im, then minent. Mam, then mal. Di, then lem, then ma. Com, then mitment.

I had been carrying munication around in my pocket for four months without once noticing that it was the second half of a word. I had been saying mensity to myself in the shower, and im, and mal, and I had been proud of the number at the top of the screen, which was 1,340, and which had gone up every day.

I sat on the edge of the bed with wet hair and a phone and felt the particular shame of a person who has been working hard in the wrong direction. All those mornings on the bus. All those cards. And the thing I had been learning was not a language; it was a list of fragments that the app had cut out of a language with a blade, and the blade had been invisible, and I had paid for the app.

Zhang Ye woke up, looked at me, and said, "What."

"The app is lying to me."

He considered this with his eyes half closed. "Then use a book."

"A book."

"A book has words in it that are finished." He turned over. "Go to sleep. You have a class at eight and you will be stupid if you don't sleep, and you are stupid enough already."

The rain went on. I lay there and listened to it and thought about the difference between knowing a word and knowing what a word is part of, and about Shen Lu saying that the vocabulary was a heap of bricks and the language was the wall, and about the fact that I had spent a whole summer carrying bricks up a hill that had no building on top of it.

The dormitory at eleven at night was the noisiest building I have ever lived in, and it was noisy in a particular way: not loud exactly, but crowded with small sounds that arrived one after another and never quite stopped. A door slammed on the third floor. Somebody laughed too loudly at something on a screen. A kettle boiled, was forgotten,

boiled again and was switched off with a curse. Two boys argued about a football match in the corridor till half past, and then one went to wash and the other went to bed, and the water ran for a long time.

Our room was a mess in the way that four boys in one room are always a mess: shoes in a heap by the door, a lump of clothes on the chair that might have been a towel, a bag of something under the desk that had gone rotten over the weekend, a cup with a piece of ice in it that nobody had emptied, a shirt on the back of the door, a thread of wire across the floor from the socket to the desk, a narrow bed on either side of the window, and a bunk that creaked whenever the boy above me turned over. The switch by the door did not work; you had to reach under the lamp and turn it at the plug, and there was a trick to it, and Zhang Ye had learned the trick in the first week and I had not.

At midnight somebody next door began to sing, badly, and stopped in the middle, and there was a silence, and then everybody laughed, and the building went quiet for about four minutes, and then somebody dropped something heavy.

I put the pillow over my head and thought: this is the sound of forty people learning to live beside each other, and in a month I will not hear it at all.

On Friday night Zhang Ye took his radio apart on the floor, laid the pieces out on a cloth in the order he had removed them, and told me about his village, because the radio had stopped working and because a man with a screwdriver in his hand will talk more freely than a man with nothing to do.

The village is in the northwest of the province, four hours by bus from the nearest station and eleven kilometres from the road. Two hundred and forty people live in it in winter and about ninety in summer, because the young ones leave. His father keeps two oxen and a cow, and ploughs four mu with them because the field is on a slope and a machine cannot turn on it. In April the whole valley smells of wet dirt and diesel. In October the harvest comes in on the backs of three trucks that belong to a man in the next village, and that man is the only person in the district with any leverage, and he knows it.

His mother grows bamboo on the lower slope and sells the shoots in the market at the county town. There is a stream behind the house that runs all year: in August it is a trickle and in March it is loud enough to wake you. In winter there is frost on the inside of the window and sometimes snow that stays for a week, and in a bad year the road is closed and the bus does not come, and the village eats what it stored in October.

The field is rocky at the top, and the pasture behind it is too steep for anything but goats. There is a thorn hedge along the yard that has been there since his grandfather's time, and a fox that comes for the chickens in February, and bees in the

wall of the shed that nobody has ever been stung by, because nobody has ever disturbed them. The cow has a name and comes when it is called, which he said is not affection but arithmetic.

He said that when he was small there were sparrows in the eaves and that they have gone, and that nobody in the village has decided whether they are extinct or merely elsewhere. In April the frogs crawl out of the ditch and hop across the road at night, and the trucks kill them without meaning to, and by June the road is clean again. A heron stands in the shallows below the stream for an hour at a time, waiting for prey, and a snail crosses the bamboo leaves at a speed that he said had taught him patience better than any book.

He said the seed they buy now is genetically modified and costs three times what it did, and that a biochemist from the county town came to a meeting and told them it was safe, and that nobody believed him, and that the yield went up anyway. "The biotechnology is not the problem," he said. "The problem is that we cannot save the seed. Every year we buy it again. That is the whole business, and it is not a secret, and nobody in the village has any choice about it."

His elder sister works in a factory in the south. She sends money. She has a daughter, six years old, who is being raised by their mother, which is the arrangement half the village is living under, and which everybody describes as temporary and nobody believes is temporary.

"They borrowed for my first year," he said. "Not a lot. Eight thousand. They will repay it in two years if nothing happens." He turned the screwdriver over in his hand. "Nothing happening is the whole plan."

He said it without self-pity and without any of the defiance I would have put into the same sentence. He said it the way he said everything, as a piece of information about the world that happened to be about him.

I asked him whether he minded. He said, "Minding is a luxury," and put the back of the radio on, and the radio came on, and a voice from the local station said that it would be fine tomorrow and rainy on Sunday, and that the tide in the estuary would be high at four.

I went back to the library on Sunday afternoon, in the rain, and found Shen Lu on the third floor with three books open in front of her and a pencil in her mouth.

"You look better," she said.

"I slept for eleven hours."

"That is the whole cure." She pushed a chair out with her foot. "Sit down. I want to

show you something."

What she wanted to show me was a page of her own notes, from her first year, which she had kept for reasons she did not explain. The handwriting was the same as it is now: small, upright, and absolutely regular. But the margins were full of Chinese. Every English word had a Chinese word beside it, and some of them had two, and one of them had four with the fourth crossed out.

"I did that for a year," she said. "Every word, translated. And then I went to a lecture in my second year, in English, by a man from Delft, and I understood every single word he said and not one of his sentences." She turned the page. "That was the day I stopped keeping two lists. One list of words, one list of meanings. It is not two lists. It is one list, and the meaning is the sentence."

I asked her what I should do, concretely, on Monday.

She thought about it for longer than I expected. Then she said, "Read one page a day out loud, from a book you actually like, and write down only the sentences you could not have said yourself. Not the words. The sentences. At the end of the fortnight, read them again. If you can still hear why they were good, you are learning something. If you cannot, you were collecting, not reading."

"A fortnight."

"Two weeks. Yes." She looked at me. "It is not a method. It is a habit with a date on it. Methods fail; habits fail more slowly."

I wrote it down. Then I asked her the question I had been carrying since the first morning, which was why she bothered.

She was quiet for a moment. "Because somebody did it for me," she said. "In my first year. She was a doctoral student and she had eleven months left and she was not kind about it at all, and I have never been able to repay her, because she went to Germany." She closed the book. "So I do it for whoever is standing in the corridor. It is not generosity. It is a debt with no creditor."

Then she took a folded page out of her bag and put it on the table. "Read the abstract. Then tell me what they actually did, in one sentence, without using any word from the abstract."

I read it, and this is roughly what it said. The authors report a composite in which the constituent phases are separated by a layer of negligible thickness. The distribution of the second phase was measured, and the result was consistent with a model in which the two phases interlock rather than merely touch. The corresponding values of hardness were proportional to the thickness of the layer, and the relationship held over a range wide enough to be convincing. A qualitative agreement with an earlier study was noted,

together with a substantial disagreement about the incidence of cracking under load. The authors propose that cohesion between the phases, rather than the strength of either phase alone, determines whether the material fails, and they offer a conceptual framework in which the interface is treated as a third constituent with properties of its own. They concede that the sample size is small, that the measurement of the layer is at the limit of what the instrument can resolve, and that the conclusion is therefore tentative. They conclude that the approach is feasible, that a prototype would be viable within three years, and that the hypothesis remains valid until somebody makes a better composite.

I read it twice. Then I said that they had made a material out of two materials and that the boundary between them was doing the work.

Shen Lu looked at me for a second. "That is not bad," she said. "That is the whole paper. Everything else is the paperwork of being believed."

"And if I had said the opposite?"

"Then you would have been wrong in a way that the abstract would not have told you." She folded the page. "That is why you read the methods. The abstract is a promise; the methods are the receipt."

On Saturday morning I went to the electron microscope room, which was in the basement of the physics building and which was reached through two doors, the second of which was heavier than any door I had ever opened.

The safety sheet was on the wall beside it and was four paragraphs long. It said that entry was permissible only with a member of staff; that the prohibition on food and drink was absolute and included water; that the prohibition on magnetic media was not a superstition but a consequence of the field; that loose jewellery was a hazard and that a watch would stop; that the instrument was to be operated only as instructed, and that any intentional interference with the column would end the visit; that the column was watertight and could withstand a vacuum; that the high voltage would not forgive anything; and that if anything looked wrong, the correct action was to stop, tell somebody, and not touch it. It was signed by a professor I had never heard of and dated 2019, and somebody had drawn a small cartoon of a person being ejected from the room by a spring.

I read all of it twice, the way I now read everything.

The room was small, cold, and hummed at a pitch just below hearing. The instrument was a column of grey metal about a metre tall, with a screen, a console of knobs that looked older than I was, and a pump that clicked every few seconds like a clock with a limp. Shen Lu was already there in a white coat, with her hair tied back and a tray in her hands, and she showed me a specimen: a disc of metal the width of a coin, polished

until it looked like a piece of a mirror, with a hole in the middle that had been cut by a beam.

Then she put it in, closed the chamber, and the screen filled with grey.

"Watch," she said, and turned a knob.

The grey moved. It magnified, and the magnification was not like zooming into a photograph; it was like walking into the photograph while the photograph became a landscape. At the first step I could see a texture like the surface of a fruit. At the second there were grains, and between the grains there were lines, and the lines were the boundaries that the lecturer had drawn on the board ten days earlier in a diagram that had looked like noodles. At the third there was a shape I could not name, and Shen Lu said, "That is a void. An empty space. It should not be there, and it is why the last batch failed."

I looked at it for a long time. The instrument did not care what I believed. It did not care that I was tired, or that I had got a fever in September, or that I had spent a summer learning the fragments of words from an app. It cared about the specimen, and the specimen was what it was.

Shen Lu turned the knob back and the landscape became a piece of metal again, and then a disc, and then a grey dot, and then nothing.

"Next time," she said, "you can hold the tray."

We came out into the afternoon, and the light was the colour of weak tea, and the campus was full of people going nowhere in particular. She walked with her hands in her pockets and talked, which she did not usually do.

"When I came here I was naive. I thought the department was a family. It is not a family; it is an institution, and an institution is a machine for distributing attention, and the attention is finite." She said that in her first year she had been jealous of a boy who got into a laboratory early, and that the jealousy had lasted a year and had taught her more about herself than any lecture. "Hatred is a very efficient form of information," she said. "It tells you exactly what you want and exactly what you are afraid you cannot have. The trick is to read it and then put it down."

She said she had met hypocrites in her second year, and that the hypocrites had not been the dangerous ones, because you could see them coming. The dangerous ones had been the sincere ones who had never been tested and who therefore believed that their good intentions were the same thing as good judgement. "That is the whole of what I have learned about people," she said. "Be careful of anybody who has never been tempted."

"The odds of getting a good supervisor are not good," she went on. "The prospective students who do best are not the cleverest. They are the ones who can be told something

true without needing it to be kind." She said she had applied for one job in the summer and had been rejected after the second interview, and that the rejection had been a correction and not a verdict, and that it had taken her three weeks to feel the difference. "I have no zeal left for institutions," she said. "I have plenty for the work. The work is the only part that does not require anybody's approval, which is why I go into the basement and why I am not jolly about it."

I asked whether she was ever afraid. She thought about it. "I was afraid of my supervisor for a year," she said. "He was not a bully and he did not harass anybody; he was simply a man who had no interest in being liked and no idea how much space he took up. I have never been so intimidated by anybody in my life, and I would work for him again tomorrow." She looked at the path. "Courage is not the absence of that. It is going back into the room."

She stopped at the gate, where the road turns downhill towards the shops.

"One thing," she said. "Do not brag about the reading. There is a kind of first-year student who has read four books and starts to bluff his way through the fifth, and everybody can hear it, and nobody says anything. Be modest and be slow and in two years you will be the one who is trusted."

I said I would try.

"Do not try. Keep a record." She started down the hill, and then, without turning round, "And do not be cynical. Cynicism is cheap and it looks like intelligence for about a year, and then it looks like what it is."

On the last Sunday of September I walked east along the road that runs from the north gate down to the river, because I had four hours and no money and because a person who is behind on everything needs one morning in which he is not behind on anything.

The road was wide and the traffic on it was never still. A taxi went past with its light on. Then an automobile of the kind that is polished on Sundays, driven by a man in a shirt with no jacket. Then a bus that stopped for nobody, and a boy on a scooter who overtook everything on the inside and vanished round the bend. An aeroplane crossed the sky, low enough that I could see the shape of the aircraft against the cloud — the wings, the two engines, the bright point of a window. The sky was smoggy and the visibility was two kilometres, and the sun was a white disc with no edges, and the whole city looked like a photograph that had been left in a window for a year.

At the corner there was a film crew. Eleven people, a camera on a shoulder, a woman in a yellow coat standing in the road with her hands in her pockets, and a director who kept saying the same sentence in different words. I stood behind the barrier with about forty other people and watched them shoot the same twenty metres of pavement eleven

times. The actress was not a celebrity; nobody knew her name; she was, according to a boy beside me who seemed to know everything, in a drama that would be shown at eleven at night. What I remember is that between takes she stood exactly where she was told and did not look at her phone, and that when the director said again she did it again, and the eleventh time was better than the first and nobody clapped, because on a film set nobody claps; they simply move on to the next scene.

Past the crew there was a food stall with a tarpaulin and a queue, and the smell of it reached me from twenty metres: hot oil, fried dough, something sweet underneath. The chef was a woman of about fifty with a towel over one shoulder and a menu chalked on a board: rice, noodle, a hamburger for the students, a fried cracker the size of a hand, cucumber in vinegar, a plate of melon and pear, and, at the bottom, in different chalk, salmon, market price. She had a gallon can of oil behind her and a quart jug on the counter, and she poured from one into the other without looking. I bought a cracker for two yuan and ate it standing up, and it was greasy and hot and one of the best things I have eaten in this city.

Further on, a man was selling vegetables off a cart pulled by an ass. Not a horse; an ass, grey, patient, with a fly on its ear that it did not bother to move. The man had a spring scale and a wallet in his shirt pocket and a radio on the cart, and he was arguing with a woman about the price of a bag of cucumbers in a way that both of them were enjoying. Behind him a dog was hunting something in the ditch, and a heron stood in the shallow water with the stillness of a bird that has decided where the prey will be and is prepared to wait.

At the junction there was a banner across the road, red, with a slogan about the city and the future, and under it a small parade was forming: a brass band of retired men, a group of schoolchildren in uniform, and four men carrying a sign. A policeman was directing them with the patience of a man who had done this a hundred times. I did not understand the politics of the banner and I did not need to; what I understood was that a parade is a way of making an argument visible, and that the argument is always about who belongs here.

Then the road ended at a building site, and the site ended at the river. There had been a village here once; you could see the shape of it in the ground, the rectangular vacancies where houses had been, a single porch standing on its own with nothing behind it, a tree that had been left because it was too old to move. A man in a hard hat told me that the villagers had been moved to a settlement four kilometres away, that most of them had been glad of it and two families had not, and that the two families had been the ones with the best house. He said the words emigrate and relocate as though they were the same word, and for him, I think, they were.

The river was wide and brown and the tide was coming in, and there was a boat working

its way downstream under the bridge with a man on the deck coiling a rope. It looked like a painting of a voyage rather than a voyage, which is the trouble with anything seen from a distance. The moon was up in daylight, pale and useless, and the water was doing what water does at a river mouth, which is to say that it was going two ways at once.

I sat on the wall for a while and thought about the morning. I had seen a film being made, a woman frying dough, a man selling cucumbers off a cart, a parade forming, a village being replaced by a site, and an actress standing in a road eleven times until it was right. None of it had anything to do with my courses, and all of it had something to do with the language I was trying to learn, because the words for all of it were words I had met in a list and had never once met in the world.

That, I think, was the day the list stopped being a list. A word like ass is a word in a book until you have seen an ass standing in the road with a fly on its ear; then it is a thing, and it stays. A word like voyage is a word until you have watched a boat go under a bridge; a word like vacant is a word until you have seen a porch with nothing behind it. Everything I had been trying to memorise was, in the end, a name for something somebody had already seen.

On Monday I waited outside the office for Wang Laoshi for half an hour, and while I waited I read the three documents taped to the door, which is how I learned that a university explains its most important arrangements in the least readable prose available.

The first was a notice about an exchange programme with a partner university abroad. It said that the arrangement was bilateral, that it rested on an agreement renewed every four years, and that a student's affiliation would remain with this institution for the duration. It said that the programme was open to students in the second year and above; that a nomination from the department was required before an application could be considered; that the quota was four places a year and that the allocation was competitive; that tuition would be paid to this institution and not to the partner; that a transcript in English would be issued on request; that the selection panel would weigh academic record, a written statement, and an interview; that a reciprocal arrangement allowed four students from the partner to come here on the same terms; and that the cooperation between the two departments had been continuous since 2009. It said that the committee welcomed applications from students whose circumstances were difficult, and that a limited fund existed to cover the cost of travel, and that anybody who wished to be considered for that fund should say so in the statement rather than in a separate letter, because the panel read the statements first and the letters last.

The second was a charter, or the summary of one, for the students' association. It

said that the association was a collective body, that its members delegated authority to a committee of nine, and that the committee was subordinate to the general meeting. It said that any member could propose a motion, that a motion required a second, that a motion carried by a simple majority, and that a motion to revise the charter required two thirds. It said that the association would collaborate with the faculty on matters of common interest but would not accept direction on matters of opinion. It said that the accounts would be published every term and that any member could inspect them. It said that the association would not affiliate to any organisation outside the university without the approval of the general meeting.

The third was a statement on academic integrity, and it was the only one of the three that was written in sentences a person might actually say. It said that fabrication is the invention of data that were never collected; that deception is the presentation of somebody else's work as your own; that denial is what happens afterwards, when a student is asked a direct question and answers a different one. It said that the boundary between imitation and quotation is a reference, and that a reference is not a decoration but a receipt. It said that the rules were not intended to punish carelessness but to inhibit the particular kind of ambition that treats a result as a possession. It said that any interference with the work of another student would be treated as seriously as any interference with the work of a member of staff, and that the penalty for a serious case was not a fine but a note in the record that would follow a student for as long as the record existed. And it said, at the end, in a sentence that I have thought about more than any other sentence I read in September, that the purpose of the whole apparatus was not to detect dishonesty but to make honesty the easier of the two options.

The rain had stopped by the time I got back. The pavement was still dark and the light from the reading room windows lay on it in long yellow rectangles, and the campus was as quiet as it ever gets, which is not quiet at all, because somewhere a scooter was being started and somewhere else a door was being slammed.

I walked to the dormitory with my hands in my pockets and a list in my head, and I understood, in a way I had not understood on the first morning, that the whole of this place was a machine for making you feel small, and that the machine was not malicious, and that the only thing to do about it was to keep showing up until the feeling stopped being information and became weather.

On the last Thursday of September, Shen Lu and I carried two boxes of bound newspapers up from the basement, and while we worked she talked, and what she said I wrote down that night because it was the closest thing to a map of the department that

anybody had given me.

"Adversity is not a virtue," she said. "It is a condition. The people who tell you that hardship builds character are usually the people who have arranged for somebody else to have the hardship. That is not cynicism; it is observation. There is bravery in a first-year student who puts up a hand in a lecture of two hundred, and there is hypocrisy in a system that praises the hand and then marks the answer wrong."

She said that departments are not families but dominations with better manners, that every institution has a hierarchy and a pact between the people at the top and the people who do the work, and that the pact is rarely written down and is continually renegotiated. She said that a rebel in a department is usually somebody who has read the pact and found that it does not say what everybody assumes it says, and that such people are repressed not by force but by committee.

"Patriotism is easy," she said. "It costs nothing to love a country in a parade. It costs something to love a villager who is wretched and obscure and has no opportunity and votes the way his landlord tells him to. The socialism I was taught in school and the capitalism I read about in the newspapers are both abstractions; what exists is a settlement of people who have to live beside each other, and the argumentation about which system is better is mostly a way of not talking about the proximity."

I asked her whether she believed any of it.

"I believe in evidence," she said. "I believe that a prediction which cannot fail is not a scientific claim, that a formula is only as good as the measurement behind it, and that the foremost duty of an expert is to say what would change his mind. Everything else is propaganda with a prettier style."

She put the box down and looked at me. "You want a thesis for your four years. Here it is. Dependence is not the same as loyalty; obedience is not the same as respect; and modesty is not the same as silence. Learn those three differences and you will be spared most of what happened to me."

I asked what had happened to her.

"The usual," she said. "I was confident and then I was desperate and then I was resigned, and resignation is the most dangerous of the three because it feels like maturity. The recovery is gradual and unspectacular. You do not wake up one morning victorious. You wake up reluctant, and you go to the laboratory anyway, and one day you notice that the work is pleasing again."

She said that she had contemplated leaving in her third year, that she had written a resignation letter and torn it up, and that the episode had left her with a suspicion of enthusiasm and a respect for routine. "The enthusiastically committed are the ones who disappear in the second year," she said. "The stubborn ones stay, and the stubborn ones

are underrated."

The workshop behind the physics building was open on Wednesday afternoons, and Zhang Ye had found it in the first week, the way he found everything, by walking into a door that other people assumed was locked.

It was a long room with a concrete floor and a smell of oil and sawdust. Along one wall there was a bench with a vice, a rack of chisels, a plane that had belonged to a carpenter, three hammers, a box of nails, a coil of tape, a brush and a broom that had both lost half their bristles, and a leather apron hanging on a nail. On the other side there were shelves of things that had been taken out of other things: a fork from a machine that no longer existed, a pendulum out of a clock, a length of thread wound on a card, a knot of wire that somebody had given up on, a china cup with a chip out of its rim, a kite frame made of bamboo, a leather bag with a broken strap, and a sofa in the corner with a tape patch on the seat.

The man in charge was called Old Yu. He was sixty-one, and he had been an electrician for thirty years before he became the person who kept the instruments alive, and he had the manner of a tailor with a difficult customer: he looked at the thing, he looked at you, and then he decided which of the two was the problem.

"The difference between a repair and a destruction," he said, "is whether you remember where the screws went. Most people do not. They take something apart, they put it back together, and there is one screw left over and they tell themselves it was spare. Nothing is spare. A fault in a machine is a sentence in a language, and if you leave out a word the sentence stops meaning what it meant."

He let me hold a clamp while he fastened a bracket, and he talked the whole time about the instruments he had mended and the ones he had buried. He said that a starter motor is a beautiful thing because you can hear whether it is well, and that a radiator is a monstrous thing because you cannot, and that humidity is the enemy of every precision instrument in the building. He said that the maintenance of a laboratory is not glamorous work and that it is the difference between a facility and a room with expensive equipment in it.

Then he said something that I have quoted since. "A tool is a promise that somebody kept. You pick up a hammer and it does not hurt your hand, and that is not luck; that is a man in 1974 who spent a week on the handle. Everything useful in this room is somebody's patience frozen in metal."

On the last Friday of September, Shen Lu took my notebook out of my hand, sat down on the stone step outside the library, and read it for eleven minutes without saying

anything.

"You have three hundred and forty entries," she said at last. "About sixty of them are wrong, and another hundred are useless, and the rest are fine. That is a better ratio than most first years."

She turned back to the first page. "You wrote that the corridor was awful. Awful does not mean bad. Awful means full of awe — it is what a thing is when it is so large or so strange that you cannot look at it directly. A corridor is not awful. A corridor is tedious. The rain on the day of the training was awful. The notice was awful, in the old sense, because it was written by a machine that did not know we existed."

"You wrote that the system was defective. Defective is for machines that were built wrong; a person is faulty, a plan is flawed, and a government is corrupt. Use the word that names the failure, not the word that names your mood."

"You wrote that the timetable was ambiguous, which is correct, and then you wrote that the notice was ambiguous, which is not. A notice is not ambiguous; it is vague. Ambiguous is when a thing can be read two ways, and both readings are supported. Vague is when it can be read in no way at all."

She turned a page. "Here you have written the word allurement, and underlined it twice, and I can see from the state of the ink that you have no idea what it means. Allurement is the pull of a thing. It is an old word and a slightly literary one; the advertisement of the travel agency was an allurement; the poster with the photograph of the sailing boat was an allurement, and you knew it was, and you went anyway."

"You have written that the baker's shop was fragrant, which is a compliment paid to a flower, not to bread. Bread has a smell. A flower has a fragrance. And you have written that the fried cracker was delicious, which is true, and then that the oil was greasy, which is also true, and you have not noticed that the two words are enemies in the same sentence."

"You wrote that the lecture was a disaster. It was not. A disaster is an event that kills people or ruins a town. What you had was a difficulty, or a failure, or a humiliation. The word you want is not a bigger word; it is a truer one."

She read on, and now and then she laughed, and once she stopped and read a line twice.

"Here," she said. "You have copied out the word execution and then crossed it out and written down the word deed. That is the single most intelligent thing in this notebook. A deed is a thing done, and it is the word you want when you are talking about what a person did rather than what a person is. The English are very fond of it. They say a good deed, a kind deed, a deed that will be remembered, and they mean it in a way that the word action does not."

"You have written the word devil next to the word evil, which is not wrong, but you should know that the devil is a person and evil is a thing, and that English uses the first for comedy and the second for theology."

"You have written the word enemy and then, beside it, the word adversary, and you have written a question mark, which is why I like you." She looked up. "An enemy wants to hurt you. An adversary is on the other side of an argument. In a debate, the person opposite you is your adversary, not your enemy, and if you cannot tell the difference you will take every disagreement personally for the rest of your life."

She said that my notes on the exhibition were the best thing I had done, because I had written down what the panels said rather than what they meant. "That is the correct order," she said. "First the thing, then the meaning. Most people do the meaning first and then wonder why they cannot remember the thing."

She said that the word clarity was not in my notebook and that this was a gap, and that I should write it on the first page and look at it every day, because clarity is a habit and not a talent, and because the whole of the first year is an argument about whether you are willing to be clear about what you do not know.

Then she closed the notebook and gave it back, and said the thing I have thought about most often since.

"Your vocabulary is not the problem. Your vocabulary is already larger than your courage. You have three hundred and forty words in here and you have used about eleven of them out loud. That is not a language problem; that is a habit of hiding. A word you have never said is a word you do not own; it is a word you are keeping for a visitor."

I said that I was afraid of using a word wrongly.

"Of course you are. Everybody is. The ones who learn are the ones who are willing to be wrong in public at a rate of about one word in ten, and who then listen when somebody corrects them, and who do not go quiet for a month afterwards." She stood up and brushed the step off her coat. "You will be wrong about sixty of these three hundred and forty words. That is the price of the other two hundred and eighty. Pay it and move on."

In the last week of September I began to notice that the words in my notebook had started to gather into small settlements, and that each settlement had a place attached to it.

There was the settlement that belonged to the sky. An astronomer from the observatory gave a talk on the roof of the physics building, and he said that the atmosphere is a filter and a liar, that what we see from the ground is the sky's opinion of itself, and that the first photograph of a planet's surface was taken by a machine that nobody could

fix if it broke. He was a small, focused man with an incredible memory for dates, and he said that the most useful thing about astronomy is that it makes the weather look temporary.

There was the settlement that belonged to the body. My throat was still not right, and the doctor gave me a prescription for something therapeutical that tasted of aniseed, and said that the body is a conservative institution: it will spend a week being angry about a thing that lasted an hour, and then it will forget the thing and keep the anger. He said that a mortal illness is rarely the one people fear, and that most of what kills us is boring, gradual and cheap to prevent.

There was the settlement that belonged to machines. Zhang Ye took apart the electrician's old multimeter, and the two of us spent an evening on the floor with a diagram, tracing a fault that turned out to be a broken link in a cable that looked fine. He said that a fault is never where the symptom is, that the symptom is a story the machine tells about somewhere else, and that tracing it is the closest thing to reading that an engineer ever does. He said that the telecom cables under the road carry more words in a minute than the library holds, and that nobody has ever written a poem about them.

There was the settlement that belonged to the fields. His father had sent a photograph of the harvest: a pasture, a tractor, a boy on the wall, and a sky the colour of a dishcloth. He said that the edible part of the crop was a third of the plant and the rest was work, and that his mother could tell the weight of a sack by lifting it with one hand, and that she had never once been wrong.

There was, finally, the settlement that belonged to people. A guest in the dormitory for two nights, a cousin of somebody, who told us that he had been a trainee in a hotel and had learned to carry four plates on one arm; a tutor from the maths department who came to the reading room on Thursdays and would answer any question, however basic, without making you feel small; a waiter at the restaurant by the gate who had a degree in literature and who said that the pay was better than teaching and the hours were worse.

None of these were in my courses. All of them were in the language.

Feng asked us, in the fourth week, to write three hundred words on a subject of our own choosing as long as it was true. I wrote rather more than three hundred, and I wrote it at one in the morning, and this is what I handed in.

I want to write about the words I cannot use. There are about two hundred of them in my notebook, and they fall into groups, and the groups are more interesting than the words.

The first group is the words for things that happen to other people. An astronomer

can talk about the atmosphere because he has stood under it with an instrument. I can talk about it too, but when I say the word it is a quotation. The same is true of words like petrochemical, segregate, confederate and deployment. I have read them in a newspaper and I have never seen the thing they name. There is a kind of honesty in admitting that: a word you have only met in print is a word you are borrowing, and a borrowed word should be returned in the same condition.

The second group is the words for quantities of feeling. I am not an angry person, but I have used the word angry eleven times this month, always about institutions and never about a person. I have used the word sorrow once, in a message to my mother, and I deleted it. I have used delight twice, both times about food. This is not a small thing. It means that my emotional vocabulary is thinner than my technical one, and that when something large happens to me I will reach for a word that is too small, and the sentence will be wrong in a way that no grammar check will catch.

The third group is the words that are almost the same. Desirable and advantageous. Continual and consecutive. Similar and simultaneous. Feasible and probable. Precise and correct. I have written all of these in the same notebook page and I could not have told you the difference between any pair. Feng says that this is where the marks are lost, not in the long words but in the near ones, because a long word that is slightly wrong makes a sentence strange and a near word that is slightly wrong makes a sentence plausible.

The fourth group is the words I have never said out loud. Verbal is one of them, which is a joke that took me a week to notice. Others are euphoria, which I do not have; turbulence, which I have only met in an aeroplane; and virgin, which I have only met in a sentence about forests. A word you have never said is not yours. It sits in the notebook like a guest who has not been introduced, and after a while you stop noticing that he is there.

I have decided to do three things about this, and I am writing them down so that in December I can be ashamed of how few of them I did. I will use one word from the fourth group every day, out loud, in a sentence that is true. I will learn one pair from the third group every week, and I will write the difference in a single line. And I will stop writing down words for things that happen to other people until I have seen the thing.

That is the whole essay. It is not a good essay. But every sentence of it is true, which was the only condition.

期末周的四张纸

By the last week of the term the library had become a colony of the sleepless, and every table was taken by noon. I survived that week on four pieces of paper that had nothing to do with my courses, and I have kept all four ever since.

The first was a foul-tempered notice taped to the back door of the canteen, where a cook was battering fish with the flat of a knife and a stray cat sat poised beside a beak-first sparrow that had mistaken the drain for a nest. The notice asked us to clasp our trays and not to dig in the bins; above it somebody had drawn a witch on a broom, and below, in a cleaner hand, the words the only magic is frequent washing of hands. I laughed for the first time that week, put a crunch of fried batter between my teeth, and satisfied myself with soup. When a boy tossed his tray onto the stack and missed, the whole pile went down with a slam, and a kick from the cook's boot gave the cat a spur out of the corner where it had been hiding. A wasp was stinging the air above the drain; I felt the sting of it in my own neck by pure sympathy, and I shake my head at myself even now. Shielding my soup with one hand, I tucked the notice into my notebook and took the chance to tow my bag out before the queue split into an argument.

The second was a broad notice board in the east hall, part of a small show on the evolution of the institution. It listed, in reserved schoolmaster prose, how the school had survived a reform and a transitional decade; how its cultural heritage had been conserved through devastation and misfortune; and how a curtailment of funds had forced the conversion of an old enclosure into a refuge for research. One line impressed me more than the rest: unity was not the absence of discussion but its retention. I copied it down and, at the outset of my own four years, dedicated a page of my notebook to it. There was a period photograph of the first educators, a spectacular suite of lecture halls, and a small demo of how they had once had to erect the first hall by hand. Somebody had highlighted the creator of the plan, a philosopher by training, whose original spirit the board praised with some superiority as supposedly inimitable. I found the claim neither logical nor appealing, but I maintain that the gesture had its importance. The board also recorded the disappearance of the old reading room, the pressure to diversify the curriculum as enrollment expanded and the campus exploded, and the export of the school's teaching methods to three provinces. A recent photograph showed the survivor of that first class, an overwhelmingly old woman who had later built an unlikely partnership with a factory; the board called it a triumph, and then, contrary to its own tone, added that the recent generation seemed to surpass its teachers. I did not cast a vote either way. I only noticed how a place unveils its story one plaque at a time.

The third was a long email from Shen Lu. She wrote the way she spoke: sparingly, and downwards, as if complexity were a fault in the writer rather than in the world. She wrote that mastery was an illusion; that perfection was futile; that excitement faded and obsession sustained you; that passive consumption of information would fabricate a feeling of progress while retention varied from instance to instance; that preferential treatment of one theory over another was a defiant gesture rather than a method; that reconciliatory discussion was ingenious work and divisive rhetoric cheap; that

feasibility studies were where pessimism earned its keep; and that if I meant to transcend my present level I should enlarge my vocabulary by using it, not by collecting it. She listed the adaptive habits she had seen work — assimilation by reading aloud, a deliberate acceleration at the end of each week, a way to augment one's notes by reliance on a single trusted source — and she warned that fixation on speed would weaken everything. Prior to the exam, she wrote, review what you highlighted; dispense with the rest, including the parts you find artistic. Contrary to what the group chat believed, no duplicate of anybody's notes would help, and no cooperative counteraction against the syllabus existed. Then, at the end: inquire of your teacher, not of your friends; the inner voice that tells you to expend a whole day on one paragraph is a possessive voice, and it will occupy you for a bout and then leave you with nothing. She signed it with a singular instruction: envisage the exam as a bout with a resourceful opponent, and dedicate the first ten minutes to the removal of panic, not to the questions.

The fourth was a wall — the corridor of the health centre, where five posters hung in a row. The first was about sex and gender, and mentioned, once and without embarrassment, the words lesbian, rape, and taboo, with a phone number under a small shield logo. The second prescribed sleep: seven hours, no Internet after midnight, no bomb of caffeine in ninth period; it advised the sleepy examinee to tuck a blanket around himself and simply relax. The third advertised a fellowship and a scout outing, with a sketch of a twin figure running and a handsome line about friendship. The fourth urged students to uncover what had horrified them in retrospect — the emptiness of a skipped week, the gloom of a sleepy morning — and to treat the impulse to hide it as a traumatic reflex rather than a defeat. A fifth, older sheet carried a sketchy diagram of the inner ear and a note about sensitivity to noise. I stood there long enough for the nurse to ask whether I was ill. I said I was relaxing. She said that was the first sensible thing she had heard all day, and highlighted the number on the first poster with her pen.

At the English corner on Friday I met an old man who called himself a boomer and who had a shrewd, pushy, choosy way of talking. He fascinated the group by insisting that geography was destiny, that Easter had been merged into consumption, that everyday leisure had been curtailed by algorithms, and that a man's manhood and a woman's masculinity were both cultural fictions. He congratulated a girl on her diploma before learning she was a freshman; he blessed a grandchild she did not have; he told a niece to obey her mother and, a minute later, to defy her. He was raggedly entertaining, naughty in the way of old men who have decided that urgency is a young person's specialty, and when a teacher tried to interfere he slammed the table and said, with goodness in his voice, that leisure was not a crime. Nobody answered. The darkening sky sent us home. I was grateful to him, and I never learned his name.

That night Zhang Ye and I packed for the holiday. He split his suitcase into two

halves with a cardboard divider and towed the whole thing down the stairs while I chased a lagging zipper; he had packed his underwear into one dual compartment and his winter suite into another, which was, by his own counting, sufficient. In the sleepy dark of the corridor he said, with the slow grind of a man who had been up since four: "Likewise pushy people survive; quiet people survive; only the unfinished get surprised." I considered the wisdom, decided that it was at least half true, and wrote it on paper rather than let it go. Then the lights went out and the term was over.

Before we left, I did one last thing, which was small and which I have never told anyone. I took my phone into the clearing between the beds, where the light was best, and I opened the vocabulary app that had been the cause of so much of my year. It printed the word dilemma as dilem. It had been doing this for months, and I had never once looked closely enough to notice. I sat on the floor and condensed the whole term into one line: I had been independent enough to be wrong on my own, which is the only kind of independence that ever teaches you anything.

The dynamic of the room changed when Zhang Ye came back with his bag. He wanted to avenge a joke I had made about his zipper, so he pierced my last carton with a pen and entrusted me with a note for whoever took the room next: do not invade other people's milk. On the back of it, in his terrible handwriting, he had copied a line from a poster: ugliness is a spot on the glass, not the view. I could never induce him to explain it. He only said, "It's a clue, not a lecture," and then he did something unexpectedly motherly — he folded my esthetic of a scarf, the one I had bought on an excursion to the market, and put it on top of my bag so it would not crease. For a second the room looked furnished, as if we had lived there for years instead of one term.

At the station the resumption of ordinary life was announced by the loudspeaker. I thought about revenge, which I had promised myself in September and which now seemed like the plan of an idiot; about the institutional language of the involvement form I had signed; about my parents' marital weather, which I could feel even from a distance; and about the inventor of the strike at the factory in the next town, whose story we had followed all week like a rumour. None of it had a shape yet. The train came in, and the embodiment of the term turned out to be a suitcase with a broken zipper and a note about milk.

伴读：词不是背下来的，是碰上来的

先说那个把浏览器关掉的早上

**abandon* 是丢下不管，absence 是不在场，abundance 是太多。三个 a 开头的词，温度完全不同：abandon 带着罪感，absence 留着一个空位，abundance 是富余到不知拿它怎么办。正文里林一 abandon 了那个浏览器，他丢下的不是浏览器，是那句"再刷新一次就好了"。abolition 是废除，

专用于制度：the abolition of slavery、the abolition of the imperial system，别拿它去废除一个计划。acceleration 是加速，物理的，也是历史的：the acceleration of consumption***。

accident* 和 accordance 长得不像，但都能毁掉一句话。by accident 是偶然，in accordance with 是依照——后者是公文里的万能胶，粘住一切责任。accounting 是会计，也是"算账"这个动作：by my own accounting 是"按我自己算"。accustom 只住在 be accustomed to 里，你要说"习惯了"就用它，别说 I accustom it。acknowledge 是承认，比 admit 体面：admit 是被人抓住，acknowledge 是自己先开口。act 是行动也是表演，actress* 是女演员——现在演戏的人多半自称 actor，actress 正在变旧，但考试还认它。

adaptive* 是能适应的，adequacy 是"够用"，adjustment 是调整。adequacy 这个词很阴：它在公文里只表示"技术上没问题"，不表示"人能看懂"。admission 是准入，也是承认：admission to a university 是录取，an admission of guilt 是认罪。adorable 是可爱到想抱，advantageous 是有利，adversity* 是逆境——作文里它的位置比 hardship 高半级。

aeroplane* 与 aircraft 都是飞机，前者英国、后者通称；automobile 是汽车的广告说法，日常只说 car。affiliation 是隶属关系，afford 是买得起，也买得起时间；afterwards 是之后，agent 是代理人，化学里它是"剂"，a cleaning agent。ago 只能跟过去时，aid 是援助，first aid 是急救，alcohol 是酒精，align* 是排成一线，也是"站队"。

allegedly* 是"据称"，新闻里保命用的；allocation 是分配，allurement 是诱惑物，见到它就想到 lure；alongside 是并排，alternate 是交替的——注意它不是"另外的"，那是 alternative，一个字母之差，考场上很贵。am 就是 am，它在词表里是因为拼写；ambiguous 是歧义，amongst 是 among 的旧体，an 是冠词，ancient* 是古老。

angry* 是生气，annexation 是吞并，专指领土；anonymous 是匿名，antenna 是天线，复数 antennas 与 antennae 都行；any 是任何，ape 是猿。appealing 既是"招人喜欢"也是"上诉中的"，applause 是掌声，appointment 是预约也是任命，approval 是批准。aptly 是恰当地——正文里那个通知 aptly named，因为它确实没人打开。architecture 是建筑，也是"架构"；argumentation 是论证，不是吵架；armament 是军备；array 是一排、一批；artery 是动脉；artistic* 是艺术的。

ashamed* 是羞愧，人做主语；ass 是驴，别在作文里拿它骂人；assertion 是断言；assimilation 是同化、吸收；assume 是假定——它还有个坑：assume responsibility 是"承担责任"。astronomer 是天文学家，atmosphere 是大气也是气氛，attachment 是附件也是依恋，attention 是注意，auction 是拍卖，augment 是增大，比 increase 正式，authoritative 是权威的，avenge 是替别人报仇，await 是等待，后面直接跟宾语，awful* 是糟糕。

backup* 是备份，badly 是糟糕地，但 need it badly 是"非常需要"；bakery 是面包店，bamboo 是竹子，bankruptcy 是破产，bargain* 既是便宜货，也是讨价还价。

那间阶梯教室里的 78 个词

**base* 是底，也是"以……为基地"，be based on 是被动；basketball 是篮球，batter 是连续猛打，也可以是一盆面糊——同一个词，一个是动词一个是名词，靠句子认。beak 是鸟嘴，

beautiful 是美，bee 是蜜蜂。begin 是不规则动词，过去式 began，你写 begined 的那一刻，老师就知道你的基础在哪。belief 是信念，注意它跟 believe 分家：a belief 是名词，believe 是动词，别写 I belief。belt 是腰带，benevolent 是仁慈的，带一点"施舍"的味道，比 kind 重。bet 是打赌，bewilderment*** 是困惑到发懵。

bilateral* 是双边的，biochemist 是生物化学家，biotechnology 是生物技术，bite 是咬，过去式 bit；blank 是空白，bless 是祝福，Bless you 是打喷嚏之后那句；blood 是血，bluff 是虚张声势，boat 是小船，bomb 是炸弹，boomer 是"婴儿潮一代"，只在报纸社论里出现。botanical 是植物学的，bout 是一阵、一场（a bout of fever），brag 是吹牛，bravery 是勇敢，breast 是胸膛，bribery 是贿赂，brigade 是旅、队，broad 是宽，broom 是扫帚，brush* 是刷子，也是"擦过"。

Buddhism* 是佛教，注意大写；bug 是小虫，也是程序错误，bully 是欺凌，bureaucrat 是官僚，bury 是埋，过去式 buried；but 是但，buzz 是嗡嗡。cable 是缆、电缆，calculation 是计算，camel 是骆驼，cancer 是癌，canteen 是食堂，capitalism 是资本主义，caravan 是大篷车或车队，carpenter 是木匠，cartoon 是漫画，cast 是投、掷，也是演员表，catch 是不规则动词，过去式 caught——词表只认 catch 这一支，所以你要用原形；caution 是谨慎，也是警告，celebrity 是名人，cent 是分，certainly 是当然，chairperson 是主席，chance 是机会也是偶然，chapter 是章，chase 是追，chef 是厨师，chicken 是鸡也是胆小鬼，china 是瓷器，小写，别跟中国混，choosy 是挑剔，Christianity 是基督教，cigarette 是香烟，circus 是马戏团，claim 是声称也是索赔，clasp 是扣紧、握紧，classroom 是教室，clearing 是林中空地，client 是客户，clinical 是临床的，cloth 是布，clue 是线索，coast 是海岸，code 是代码也是准则，cohesion 是凝聚力，collaborate* 是合作，做研究可以，通敌也可以用这个词。

从那场社团招新讲起

collective* 是集体的，columnist 是专栏作家，comedy 是喜剧，company 是公司也是陪伴——拉丁语里它本来是"一起吃面包的人"。compassionate 是富于同情的，competent 是胜任的，注意它只说"够用"，不说"出色"；complain 是抱怨，complexity 是复杂，component 是部件，compress 是压缩，concede 是让步、承认，conceptual 是概念上的，conclude 是得出结论也是结束，condense* 是浓缩。

confederate* 是同盟的、同伙的，美国历史上那支军队就叫这个；confident 是自信——注意别跟 confidential（机密的）混。conform 是遵从，conform to 是搭配；congratulate 是祝贺，congratulate sb on sth；connotation 是言外之意，consecutive 是连续的，three consecutive days 是连着三天；conserve 是保护、节约，consistent 是一致的，constituent 是成分，也是选区的选民；consul 是领事，consumption 是消费，contemplate 是沉思，contentious 是有争议的，continual 是反复发生的，与 continuous（不间断）不同；contrary* 是相反的，on the contrary 是"恰恰相反"。

**controversy* 是争议，conversion 是转变、换算，convincing 是有说服力的，cooperative 是合作的，也是合作社；copyright 是版权，corner 是角落，correction 是更正，corresponding 是对应的，cosmopolitan 是国际化的、五方杂处的，cough 是咳嗽，counteraction 是反作用，

courage 是勇气，cow 是母牛，cracker 是薄脆饼干，crawl 是爬，creator 是创造者，crew 是一组工作人员，crocodile 是鳄鱼，crown 是王冠，crunch*** 是嘎吱作响，也是"关键时刻"。

cucumber* 是黄瓜，cultural 是文化的，curiosity 是好奇心，curtail 是削减，customs 是海关，也是风俗的复数；cynical 是愤世嫉俗的，注意它跟 sarcastic 不是一回事：cynical 是不信人，sarcastic 是嘴上带刺。damp 是潮湿，比 wet 轻；darken 是变暗，daughter 是女儿，deadly 是致命的，debtor 是债务人，deception 是欺骗，declarative 是陈述的，语法里的"陈述句"就是它；dedicate 是献身、献给，dedicate oneself to；deed 是行为、功绩，也是地契；defiant 是挑衅的、不服的，defy 是公然违抗，delegation 是代表团，delight 是喜悦，demo 是演示、样带，denial 是否认，dentist 是牙医，dependence 是依赖，deployment 是部署，depression 是抑郁也是萧条，deregulation 是放松管制，description 是描述，desirable 是值得要的，desperate* 是绝望的，也是"拼命想要的"。

图书馆地下室那一排

destruction* 是毁灭，detect 是察觉，determination 是决心，devastation 是废墟般的破坏，devil 是魔鬼，diagram 是示意图，dictation 是听写，difference 是差别，dig 是不规则动词，过去式 dug，dilem 这个词你大概没见过——它不是词，是词表把一个词劈开留下的半截，完整的是 dilemma；dimensional 是维度的，three-dimensional 是三维；diploma 是文凭，dirt 是脏东西，disappearance 是消失，disaster 是灾难，disciplined 是有纪律的，discover 是发现，discussion 是讨论，disk 是圆盘、磁盘，dispense 是分发，也是"免除"，dispense with；disputable 是可争议的，dissolve 是溶解，distinguishable 是可分辨的，distribution 是分布，diversify 是使多样化，divisive 是造成分裂的，doctrine 是教义、主义，dollar 是元，domination 是支配，dormitory 是宿舍，dough 是面团，downwards 是向下，drama 是戏剧，drawing 是图画，drill 是钻也是操练，drown 是淹死，dual 是双重的，duplicate 是复制，dusty 是满是灰的，dynamic* 是有活力的、动态的。

earn* 是挣，Easter 是复活节，ecology 是生态学，edible 是可食用的，educator 是教育工作者，ego 是自我，eject 是弹出、驱逐，elder 是年长的、长辈，electrician 是电工，element 是元素也是要素，eligibility 是资格，elsewhere 是别处，embodiment 是化身，emigrate 是移居出境，注意跟 immigrate 方向相反；emperor 是皇帝，emptiness 是空，enclosure 是围场、附件，endeavour 是努力，enemy 是敌人，engine 是发动机，enlarge 是放大，enquiry 是询问，enterprise 是企业也是事业心，enthusiastically 是热情地，entrust 是委托，envisage 是设想，equality 是平等，equity 是公平，也是"权益"，erect 是竖立，error 是错误，essay 是文章，esthetic 是审美的，也写作 aesthetic，ethnic 是族群的，evaporate 是蒸发，event 是事件，everyday 是日常的，注意跟 every day（每天）分开写，evolution 是演化，examinee* 是考生。

那张贴在门上的通知

**except* 是除了，excitement 是兴奋，excursion 是短途出游，exemption 是豁免，exhibition 是展览，expand 是扩张，expend 是花费，比 spend 正式；expert 是专家，explode 是爆炸，export 是出口，expressway 是高速公路，extinct 是灭绝的，extravagant 是奢侈的，

fabricate 是捏造也是制造，facility 是设施也是"从容"，fail 是失败，falsehood 是谎言，fantasy 是幻想，fascinate 是使着迷，fasten 是系紧，fault 是过错，feasibility 是可行性，federation 是联邦，fellowship 是奖学金、伙伴关系，fever 是发烧，fifth 是第五，film 是电影也是薄膜，finish 是结束，fiscal 是财政的，fixation 是固着、痴迷，flash 是闪光，flesh 是肉，flock 是一群，fluctuate 是波动，focus 是焦点，follow 是跟随，for 是为了，foremost 是首要的，forfeit 是丧失、被罚没，fork 是叉子也是岔路，formula 是公式，fortnight 是两周，foul 是难闻的、犯规的，fox 是狐狸，fragrance 是香气，fraud 是欺诈，frequent 是频繁的，friendship 是友谊，frost 是霜，fry 是油炸，functional 是功能性的，furnish 是配备家具，也是"提供"；futile 是徒劳的，gallon 是加仑，garbage 是垃圾，gasp 是倒抽一口气，gender 是社会性别，genetically 是基因上地，geography 是地理，gesture 是手势，girl 是女孩，gland 是腺体，gloom 是阴暗、忧郁，go 是去，goodness 是善良，government 是政府，gradual 是逐渐的，grandchild 是孙辈，granted 是"就算"，也是"被授予"，grateful 是感激的，greasy 是油腻的，grey 是灰色，grind 是磨，ground 是地面，guest 是客人，gulf 是海湾也是鸿沟，habitual 是习惯性的，hamburger*** 是汉堡。

那个校医院的下午

handsome* 是英俊的，也可以形容一笔钱"可观"；harass 是骚扰，hardy 是耐劳的，形容植物就是耐寒；harvest 是收获，hatred 是仇恨，hazard 是危险源，比 danger 具体；headline 是标题，heap 是一堆，heating 是供暖，height 是高度，help 是帮助，her 是她的，hero 是英雄，hesitate 是犹豫，highlight 是突出，himself 是他自己，historian 是历史学家，hockey 是曲棍球，home 是家，honest 是诚实的，hop 是单脚跳，horrify 是使恐惧，hospitalize 是使住院，hour 是小时，how 是如何，humane 是仁慈的，注意跟 human（人）差一个字母；humidity 是湿度，hunt 是打猎，hurt 是弄疼，hypocrisy* 是虚伪。

ice* 是冰，identical 是完全相同的，idiot 是白痴，ignorance 是无知，illness 是疾病，im 这个词你也会在词表里看见——它同样是被劈开的半截，完整的是 immediate 之类；imitation 是模仿，imperial 是帝国的，importance 是重要，impress 是使印象深刻，impulse 是冲动，incidence 是发生率，including 是包括，incredible 是难以置信的，independent 是独立的，indigenous 是土生的、原住民的，induce 是诱发，infect 是感染，infinitive 是动词不定式，influenza 是流感，ingenious 是巧妙的，inhibit 是抑制，injection 是注射，inner 是内部的，inquire 是询问，inside 是里面，inspection 是检查，instance 是实例，for instance 是例如；institutional 是制度的，insulation 是绝缘，integral 是不可或缺的，也是积分；intelligible 是可听懂的，intentional 是故意的，interfere 是干涉，Internet 是互联网，interval 是间隔，intimidate 是恐吓，introduce 是介绍，invade 是入侵，inventor 是发明家，investment 是投资，involvement 是参与，irrespective 是"不论"，irrespective of；isle 是小岛，ivory 是象牙，jealous 是嫉妒的，jolly* 是快活的。

那一夜的宿舍

**judge* 是判断，也是法官；June 是六月，大写；justification 是理由，kick 是踢，kind

是种类也是善良，kite 是风筝，knot 是结，laboratory 是实验室，lag 是滞后，landlord 是房东，largely 是很大程度上，later 是后来，lavish 是奢华的，layoff 是裁员，lean 是瘦的也是倾斜，leather 是皮革，legend 是传说，leisure 是闲暇，lesbian 是女同性恋者，在性别议题的阅读里出现，认识即可；leverage 是杠杆、筹码，librarian 是图书馆员，likewise 是同样地，limp 是瘸的，link 是连接，list 是清单，literature 是文学，也是"文献"；loaf 是一条面包，lock 是锁，logical 是合乎逻辑的，longitudinal 是纵向的，lord 是勋爵、领主，loudly 是大声地，loyalty 是忠诚，lump 是一块，lyric 是抒情诗，也是歌词；magazine 是杂志，magnify 是放大，maintain 是维持，也是"主张"；mal 同样是被劈开的半截，完整的是 mammal 一类；man 是男人，manhood 是成年男子气概，manner 是方式，也是举止；map 是地图，marital 是婚姻的，marshal 是元帅，masculinity 是男性气质，mastery 是精通，maths 是数学，me 是我，measurement 是测量，media 是媒体，medicinal 是药用的，melon 是瓜，memorial 是纪念的，mensity 也是半截词，来自 immensity；menu 是菜单，merge 是合并，mess 是一团糟，metaphor 是隐喻，metropolitan 是大都市的，middle 是中间，migration 是迁徙，militia 是民兵，minent 是半截词，来自 imminent；minority 是少数，misfortune 是不幸，mistake 是错误，mitment 是半截词，来自 commitment；modate 是半截词，来自 accommodate；modesty 是谦逊，molecule 是分子，monetary 是货币的，monstrous 是骇人的，moon 是月亮，mortal 是会死的，motherly 是母亲般的，mouldy 是发霉的，move*** 是移动。

河边的那个上午

mule* 是骡子，munication 是半截词，来自 communication；munity 是半截词，来自 community；music 是音乐，my 是我的，naive 是天真的，narration 是叙述，nationalism 是民族主义，naughty 是淘气的，neat 是整洁的，needy 是贫困的，negligible 是可以忽略的，nerve 是神经，new 是新的，niece 是侄女、外甥女，ninth 是第九，noisy 是吵的，noodle 是面条，northwest 是西北，notebook 是笔记本，notional 是名义上的，now 是现在，numb 是麻木的，nutrient* 是营养物。

obey* 是服从，obscure 是晦涩的、无名的，obsession 是痴迷，obviously 是显然，occupy 是占据，odds 是可能性、odds and ends 是零碎；officer 是官员也是军官，okay 是可以，on 是在……上，onset 是开始、发作，opera 是歌剧，opportunity 是机会，optic 是光学的，optional 是可选的，order 是顺序也是命令，original 是原来的，other 是别的，outbreak 是爆发，outing 是外出，outset 是开端，oven 是烤箱，overhead 是在头顶，oversight 是疏忽，也是监督，一个词两个相反的意思；overwhelming 是压倒性的，ox* 是公牛。

pact* 是协定，painting 是画，panel 是面板、专家组，parade 是游行，parasite 是寄生虫，parliamentary 是议会的，participation 是参与，partnership 是合伙关系，passive 是被动的，pasture 是牧场，patriotism 是爱国主义，payment 是付款，pear 是梨，pedestrian 是行人，pendulum 是钟摆，per 是每，perfection 是完美，period 是时期也是句号，permissible 是允许的，persecute 是迫害，personal 是个人的，pessimism 是悲观，petrochemical 是石化的，philosopher 是哲学家，phrasal 是短语的，piano 是钢琴，pierce 是刺穿，pillow 是枕头，pipe* 是管子。

那份讲座、那张报纸

place* 是地方也是"放置"，planetary 是行星的，play 是玩也是剧，pleasing 是令人愉快的，plow 是犁，也写作 plough，podcast 是播客，poised 是镇定的、蓄势的，policy 是政策，pollster 是民调员，poor 是穷的也是差的，porch 是门廊，portrait 是肖像，possessive 是所有格的，也是占有欲强的；posture 是姿态，pour 是倒，praise 是赞扬，precise 是精确的，prediction 是预测，preferential 是优惠的，premium 是保险费也是溢价，prescribe 是开处方也是规定，presidential 是总统的，prevail 是盛行、占上风，prey 是猎物，primitive 是原始的，prior 是先前的，prior to 等于 before；probable 是很可能的，process 是过程也是加工，productive 是富有成效的，profit 是利润，prohibition 是禁止，promotion 是晋升也是促销，propaganda 是宣传，带贬义；proportional 是成比例的，prospective 是未来的、潜在的，protest 是抗议，provided 是"如果"，也是"提供"，proximity 是接近，publication 是出版物，punctual 是准时的，purity 是纯净，pushy 是爱出风头的、咄咄逼人的，qualitative 是定性的，quart 是夸脱，queue 是队列，quotation 是引语，radiator 是暖气片，raggedly 是破破烂烂地，rainy 是多雨的，rape 是强奸，出现在新闻与社会议题里，认识它，别用错；rate 是比率也是"评价"，rattle 是发出咔哒声，reader 是读者，really 是真正地，rebel 是反叛者，recent 是最近的，reciprocal 是互惠的，reconciliatory 是和解的，recruit 是招募，red 是红色，reference 是参考、推荐信，reform 是改革，refuge 是避难所，regarding 是关于，registrar 是教务主任，rehabilitation 是康复，rejection 是拒绝，relax 是放松，reliance 是依赖，reluctant 是不情愿的，remember 是记得，removal 是移除，renovate 是翻新，repay 是偿还，replace 是替换，repress 是镇压，republican 是共和的，research* 是研究。

那场雨、那次英语角

reserved* 是矜持的，也是"预留的"，resignation 是辞职也是认命，resourceful 是足智多谋的，response 是回应，restrain 是克制，resumption 是恢复、重新开始，retention 是保留，retrospect 是回顾，in retrospect 是事后看来；revenge 是报复，revise 是修改，reward 是回报，rice 是米饭，ridiculous 是可笑的，rim 是边缘，risk 是风险，roast 是烤，rocky 是多岩石的，room 是房间也是空间，rotten 是腐烂的，royalty 是王室也是版税，rule 是规则也是统治，rural 是乡村的，sad 是难过的，sailing 是航行，salmon 是三文鱼，same 是相同的，sarcasm 是讽刺，satisfy 是满足，saving 是节省，也是"除了"，scar 是伤疤，scene 是场景，scheme 是方案，scientific 是科学的，scout 是侦察员，script 是剧本，seasonal 是季节性的，sector 是部门，seed 是种子，segregate 是隔离，self 是自我，send 是发送，sensitivity 是敏感，September 是九月，serve 是服务，settlement 是定居点也是和解，sex 是生理性别，shake 是摇，share 是分享也是股份，shed 是棚子也是"流下"，shed tears；shield 是盾牌，shirt 是衬衫，shore 是岸，shoulder 是肩膀，shrewd 是精明的，shy 是害羞的，sigh 是叹气，signify 是表示，similar 是相似的，simultaneous 是同时的，singular 是单数的，也是"非凡的"；site 是地点，size 是尺寸，sketchy 是粗略的、靠不住的，skinny 是极瘦的，slam 是猛关，sleepy 是困的，slipper 是拖鞋，small 是小，smoggy 是雾霾的，snail 是蜗牛，snow 是雪，socialism 是社会主义，sofa 是沙发，solely 是仅仅，solidity 是坚固，some 是一些，somewhere 是某处，sorrow* 是悲伤。

结尾的那段路

south* 是南，space 是空间，sparingly 是节省地，specialty 是专长、特产，spectacular 是壮观的，speed 是速度，spherical 是球形的，spiral 是螺旋，split 是分裂，sponsorship 是赞助，spot 是地点也是"看见"，spur 是刺激、马刺，staff 是全体员工，stair 是楼梯，stand 是站也是忍受，starter 是开胃菜、启动器，station 是车站，statutory 是法定的，steamy 是蒸汽弥漫的，stiff 是僵硬的，sting 是刺，stone 是石头，stout 是结实的，stranger 是陌生人，stream 是溪流，strike 是打击也是罢工，structure 是结构，stuffy 是闷的，style 是风格，subordinate 是从属的，substantial 是实质性的，suburban 是郊区的，such 是如此的，sufficient 是足够的，suite 是一套房间，sunrise 是日出，superiority 是优越，supplement 是补充，supposedly 是据称，surf 是冲浪，surpass 是超过，survivor 是幸存者，sustain 是维持，sweater 是毛衣，switch 是开关，syndicate 是辛迪加、报业集团，taboo 是禁忌，tailor 是裁缝，tall 是高的，tape 是胶带，taxi 是出租车，technical 是技术的，telecom 是电信，temperament 是气质，temptation 是诱惑，tentative 是试探性的，terrible 是可怕的，terrorist 是恐怖分子，texture 是质地，their 是他们的，therapeutical 是治疗学的，these 是这些，think 是想，thorn 是刺，thread 是线，thriller 是惊悚片，thunder 是雷，tide 是潮，till 是直到，tiny 是极小，title 是标题，toilet 是厕所，tomorrow 是明天，tooth 是牙齿，toss 是扔，tow 是拖，trace 是痕迹，traffic* 是交通。

最后这七十个词

trainee* 是实习生，transcend 是超越，transitional 是过渡的，transparent 是透明的，traumatic 是造成创伤的，treatment 是治疗也是对待，trial 是审判也是试验，trickle 是细细地流，triumph 是胜利，trouble 是麻烦，trust 是信任，tuck 是塞进，turbulence 是气流颠簸，tutor 是导师也是家教，twin 是双胞胎，ugliness 是丑陋，uncover 是揭开，underline 是画线也是强调，underwear 是内衣，uniform 是制服，unity 是团结，unveil 是揭幕，upright 是直立的，urgency 是紧迫，useful 是有用的，vacant 是空的，valid 是有效的，vanish 是消失，vary 是变化，velvet 是天鹅绒，verbal 是口头的，versus 是对，缩写 vs；viable 是可行的，victorious 是获胜的，villager 是村民，virgin 是处女，也是"未开发的"，virgin forest；visibility 是能见度，vitamin 是维生素，void 是空虚、无效，volume 是音量也是卷，voyage 是航行，waiter 是服务员，wallet 是钱包，warm 是温暖，wash 是洗，watertight 是不透水的，也是"无懈可击的"，weaken 是削弱，weather 是天气，Wednesday 是星期三，weight 是重量，whale 是鲸，whenever 是无论何时，whichever 是无论哪个，white 是白色，whose 是谁的，width 是宽度，willpower 是意志力，winter 是冬天，witch 是女巫，withstand 是经受住，wooden 是木制的，workout 是锻炼，worship 是崇拜，wound 是伤口，wretched 是可怜的，yard 是院子也是码，yellow 是黄色，young 是年轻，zeal 是热忱，zoom* 是急速上升，也是镜头的推近。

说句实话

这一章写了 928 个词，你可能一个也没记住。没关系。真正要你带走的不是这些词，是那个顺序：先有场景，再有词。林一在走廊里站了四个小时，才把 allocation 和 approval 变成两个有温

度的词；他在医院里读了四张海报，才把 incidence 和 onset 分清；他在河边看见一头驴站在那儿，才把 ass 从"一个拼写很短的词"变成"一头灰的、耐心的、耳朵上有只苍蝇的东西"。

你手里的词表是按频率排的，不是按意义排的。按频率排的东西适合统计，不适合记忆。所以我这一章讲得东一句西一句，一会儿讲 abandoned 的罪感，一会儿讲 alternate 和 alternative 的差别，一会儿又跳到那头驴——这不是乱，这是语言本来的样子：词长在事情上，不长在表上。

还有一件事。沈璐那句话你再读一遍：词汇是砖，语言是墙。你可以搬四年砖，一块也没砌上，而且没人看得出来，直到有人请你站到雨里去。所以从明天开始，别问"我今天背了多少个"，问"我今天能不能用一句话，把今天发生的一件事说清楚"。前者是数字，后者是能力。

最后说个技术上的事。这一章里有些词长得难看：dilem、mensity、minent、mitment、modate、munication、munity、mal、im。它们不是词，是词表把长词按音节劈开留下的半截，跟那个把 immensity 拆成 im 和 mensity 的背词 App 一样。你如果按半截去背，背得越勤，错得越稳。记住完整的那个：dilemma、immensity、imminent、commitment、accommodate、communication、community、mammal、immediate。半截的东西，扔了不可惜。

至于方法，就是那三条，没有第四条。一天二十句，出声；只抄你写不出来的句子，不抄词；两周回头看一次。做不到，就下个月再试一次。语言这个东西，它不惩罚慢，它只惩罚停。

词族：一个词根能带你走多远

先说一件你可能没注意的事：这一章里那些看起来毫无关系的词，很多是亲戚。act 是行动，actress 是演戏的人，agent 是"做事的人"，也是化学里的"剂"，一个词根在拉丁语里就是"做、驱动"。你背十个孤立的词，不如认出一个词根。同样的道理：assume 是"拿过来当作真的"，assimilation 是"拿过来变成自己的"，两个词中间隔着几百年的用法变迁，但骨架是同一副。

再看一组容易被当成同义词的：admission 和 entrance。entrance 是那道门，admission 是"被允许进去"这件事，所以大学收你是 admission，剧场收你钱也是 admission，而 admission of guilt 是认罪——因为认罪也是"承认一件你本来可以不承认的事"。这就是词的第二层：它的第二个意思，往往是从第一个意思里长出来的，不是硬加的。

advantageous 和 desirable 也不是一回事。desirable 是"有人想要"，advantageous 是"对你有利"。一件东西可以很 desirable 而对你不 advantageous：比如一个所有人都想进的专业。这两个词在作文里的位置完全不同，前者谈市场，后者谈判断。

alternate 和 alternative 差一个字母，差一个世界。alternate 是"交替的"，alternate days 是隔天；alternative 是"另一个可选的"，alternative plan 是备选方案。四六级里这两个词都出现过，出题人喜欢它们，因为它们长得像而意思分岔。

continual 和 continuous 也是一对。continual 是"反复发生、带点烦"，continual interruptions 是没完没了的打断；continuous 是"一直不断"，continuous rain 是下一整天。中文都翻成"连续"，英文里一个是"一次又一次"，一个是"一口气"。你要写"连续三天下雨"，用 three continuous days 是错的，那是三天没停过；说 three consecutive days 才对——consecutive 是"一个接一个排着"，跟"连不连续"无关。

economic 与 economical 的老账这里也顺带提一句：前者是"经济的"（国家层面），后者是"省钱的"（人层面）。一个人在超市里 economical，一个国家在政策里 economic，混了就很滑稽。这一章里的 fiscal、monetary 也都是国家层面的词，fiscal 是财政的（税收与支出），monetary 是货币的（利率与供给），两个词在新闻里天天见，但很少有人能说清差别。

respectful、respectable、respective 三个词，中文都能翻成"尊重/分别"，英文里却是三个人：respectful 是"尊敬别人的"，respectable 是"值得尊敬的"，respective 是"各自的"。你写 their respective rooms 是"他们各自的房间"，写成 respectable rooms 就变成"体面的房间"，句子还通，意思已经跑远——这种错误最可怕，因为它不报警。

considerable 和 considerate 同理：前者是"相当大的"，后者是"体贴的"。a considerable sum 是一大笔钱，a considerate person 是个体贴的人。respectively 和 respectfully 也一样，一个"分别地"，一个"恭敬地"。

那几个"-tion"的词，为什么要一起背

allocation、distribution、conversion、consumption、reduction 这一串，都是动词去掉词尾加 -ion 造出来的名词。它们难，不是因为长，是因为你在中文里没有对应的名词习惯。中文说"分配"就是"分配"，动词名词一个样；英文里 allocate 是动作，allocation 是那件事本身，你得在句子里给它一个位置。

这一类词是四六级翻译的重灾区。你翻"政府对资源的分配"，写成 the government allocates resources 是句子，写成 the allocation of resources by the government 才是那个名词短语。翻译题要的是后者，因为后者能塞进更长的句子。conversion 也是这样：convert 是"转"，conversion 是"转型"这件事，the conversion of a factory 是工厂的转型，不是工厂在转。

consumption 值得单独说一句。它是"消耗"，但中文里"消费"和"消耗"是两件事。英文的 consumption 两个都管：energy consumption 是能耗，consumer 是消费者，consumption of alcohol 是喝酒（而且是带批评的那种说法）。一个词覆盖了两片语义，你就不能拿中文的边界去切它。

complexity 和 component 常一起出现在科技文里。complexity 是复杂度，component 是部件。注意 component 和 element 的差别：element 是"要素、元素"，可以是抽象的（an element of surprise），component 一定是"装在一起的一个件"。

搭配：词的脾气长在它旁边

你背一个词，背的其实是它的邻居。accordance 只活在 in accordance with 里，你换成 according 就散了；irrespective 只跟 of；provided 后面跟 that，意思是"如果"，跟"提供"无关；regarding 是"关于"，比 about 正式，公函里用它。这些不是语法，是脾气。

depend 跟 on，所以 dependence 也跟 on；reliance 也一样，reliance on。中文都说"依赖"，英文都得挂一个 on，你漏了它，句子就瘸。

interfere 跟 with（干预某事），interfere in 也有人说，但少；inhibit 直接跟宾语，不跟 from；induce 是"诱发"，induce sb to do 是"劝某人做"，医学里 induce labour 是引产。infect

是"感染"，infect sb with sth；inject 是"注射"，inject sth into sb。两个词一个字母之差，一个进的是病，一个进的是药。

prohibit 跟 from，forbid 也跟 from，但 forbid 更家常；prevent 跟 from；restrain 跟 from；refrain 跟 from。五个"阻止"全是 from，这是英文的一个习惯：阻止的介词是"从……出来"的方向。

sensitive 跟 to，sensible 是"明智的"，别混。sensitivity 是敏感度，仪器的灵敏度也是它。

sufficient 跟 for，adequate 也跟 for，但 adequate 有"勉强够"的味道，sufficient 更中性，enough 最中性。三个词一条线：enough — sufficient — adequate，从最白到最公文。

prefer 跟 to，preferential 是"优惠的"，preferential treatment 是优待。prior 跟 to，prior to 就是 before，但更硬。proximity 跟 to，in close proximity to 是"紧挨着"。

superior 跟 to，不是 than。inferior 也是。senior、junior 也是 to。这一组是拉丁比较级，它们不认 than，这是老账，但考试还认。

similar 跟 to，identical 跟 to，equivalent 跟 to。different 跟 from（英式）或 than（美式）。contrary 跟 to，contrary to expectation 是"与预期相反"。

comply 跟 with，conform 跟 to，两个都是"遵守"，但 comply 是"服从要求"，conform 是"符合标准"。consent 跟 to，object 跟 to，contribute 跟 to，attribute 跟 to。to 在这几个词后面不是不定式，是介词，后面跟名词或动名词——这一条你要是记不住，作文里就会出现 I look forward to see you 这种句子。

语体：什么时候这个词会显得不对

commence 比 begin 正式，公文里用，口语里用会显得你在念稿。terminate 比 end 硬，合同里用。purchase 比 buy 正式，商场里用显得奇怪。reside 比 live 正式，护照上用。assist 比 help 正式，但 assist 后面常跟 in。obtain 比 get 正式，学术写作里几乎只用 obtain。utilize 比 use 正式，但它常常是废话，因为 use 什么都够。

反过来，有些词太"重"，用错地方就是笑话。monstrous 是"骇人的"，你说 a monstrous sandwich 是幽默，说 a monstrous problem 是夸张；spectacular 是"壮观的"，用来形容一场考试失败，是自嘲；wretched 是"可怜的、惨的"，用在天气上很地道（wretched weather），用在人身上要小心，它有怜悯的意思。

futile 是"徒劳的"，比 useless 狠，因为它含着"你明知道没用还做"；fragile 不在这一章，但 watertight 在——它本义是"不透水"，引申是"无懈可击"，a watertight argument 是滴水不漏的论证，不是"湿不了的论证"。

notional 是"名义上的"，非常书面；nominal 也是"名义上的"，两个词几乎一样，但 notional 更偏"概念上的"，nominal 更偏"挂名的"。primitive 是"原始的"，说技术 primitive 是批评，说艺术 primitive 是另一种评价，看语境。

poised 是"镇定的、蓄势待发的"，很高级的一个词，写人物时用它，比 calm 有画面。shrewd

是"精明的"，贬褒之间；resourceful 是"有办法的"，几乎总是褒义；ingenious 是"巧妙的"，形容方案和人。

情绪词：别用一个 happy 走天下

delight 是"喜悦"，比 pleasure 热，比 joy 轻；excitement 是"兴奋"，带着不安；thrill 不在这一章，但 triumph 在——它是"胜利"，而且是有声音的那种，带着掌声。gloom 是"阴郁"，可以是天气，也可以是心情；sorrow 是"悲伤"，比 sadness 重，比 grief 轻；wretched 是"惨"，带怜悯；miserable 不在这一章，但意思最近。

pessimism 是"悲观"，despair 是"绝望"（desperate 是它的形容词），resignation 是"认命"，注意它同时是"辞职"——同一个词，一个是向命运递交辞呈，一个是向公司递交辞呈，这个双关在英文里是活的。reluctance（reluctant）是"不情愿"，比 unwilling 软；restraint（restrain）是"克制"，是忍住不做；restriction 是"限制"，是被别人管。一个向内，一个向外。

modesty 是"谦逊"，humility 不在这一章，但意思更重；hypocrisy 是"虚伪"，专指说一套做一套；cynical 是"不信人"，它跟 hypocritical 的区别是：虚伪的人假装相信，愤世的人什么都不信。naive 是"天真"，说小孩是可爱，说大人是批评。defiance（defiant、defy）是"公然不服"，比 refuse 硬，因为它当着你的面。

obsession 是"痴迷"，带一点病态；fixation 是"固着"，心理学词，也指"死盯着一点不放"；curiosity 是"好奇"，褒义，但 Curiosity killed the cat 提醒你它有代价。zeal 是"热忱"，比 enthusiasm 老派，也比它危险——zealot 是"狂热分子"，同一个根。

temperament 是"气质、性情"，比 character 生理；manhood 是"成年男子气"，masculinity 是"男性气质"，两个词都出现在性别议题的阅读里，认识它们的立场很重要：前者是"成为男人"这件事，后者是"被认为像男人"这件事。gender 是社会性别，sex 是生理性别，这一组在当代英语里分得很清，写作文时别混。

政治与历史的词，怎么读

capitalism 和 socialism 是一对，但它们在英文里的用法不对称：capitalism 几乎总是被批评的对象，socialism 几乎总是被辩护或攻击的对象，中性使用很难。patriotism 是"爱国"，nationalism 是"民族主义"，前者褒，后者贬，区别不在爱不爱，在爱的是"这个国家的人"还是"这个国家的纯度"。

imperial 是"帝国的"，emperor 是皇帝，crown 是王冠，royalty 是王室，throne 不在这一章。这一组词在历史阅读里成串出现，你只要认出一个，其余的都跟着来。abolition 是"废除"，专用于制度；annexation 是"吞并"，专用于领土；armament 是"军备"，brigade 是"旅"，militia 是"民兵"，marshal 是"元帅"，delegation 是"代表团"，consul 是"领事"，federation 是"联邦"，confederate 是"同盟的"。

parliamentary 是"议会的"，republican 是"共和的"，presidential 是"总统的"，statutory 是"法定的"，statute 是"成文法"。deployment 是"部署"，军事的，也是软件的（deploy a model）。propaganda 是"宣传"，几乎总是贬义；publicity 是"曝光"，中性；promotion 是"推

广"，商业的。三个词都能翻成"宣传"，语气完全不同。

segregate 是"隔离"，专指种族或性别上的制度性分离；domination 是"支配"；oppression 不在这一章，但 repress 在——它是"压制"，压的是情绪、记忆、反抗。rebel 是"反叛者"，reform 是"改革"，reconciliatory 是"和解的"，reciprocal 是"互惠的"。这一组词你都认识，但你未必知道它们的分量：reform 是体制内的，rebel 是体制外的，reconciliatory 是事后的。

学科词：分子、结构与测量

molecule 是分子，element 是元素，component 是部件，constituent 是成分——四个词都能翻成"组成部分"，但层次不同：分子由原子组成，材料由相组成，机器由部件组成，合金由成分组成。structure 是结构，texture 是质地，solidity 是坚固，purity 是纯度，insulation 是绝缘，conduction 不在这一章。

measurement 是测量，precision（precise）是精密，accuracy 是准确——精密是"每次都一样"，准确是"离真值近"，两个词在实验报告里必须分开。error 是误差，fault 是故障，hazard 是危险源，caution 是谨慎。detect 是探测，sensitivity 是灵敏度，proportional 是成比例，integral 是积分，也是"不可或缺"——后者才是它在日常英语里的常用义。

molecule、biochemist、biotechnology、genetically、evolution、extinct 这一串是生物学的骨架：分子、生物化学家、生物技术、基因上地、演化、灭绝。ecology 是生态学，nutrient 是营养物，vitamin 是维生素，parasite 是寄生虫，prey 是猎物，hunt 是捕猎。

那些你以为是"熟词"的词

有些词你觉得你认识，其实你只认识它的一个角落。claim 你认识是"声称"，它还是"索赔"（file a claim）和"认领"（claim your baggage）。rate 你认识是"比率"，它还是"评价"（rate a film）和"费率"。order 你认识是"命令"，它还是"订单""顺序""秩序"（in order）。practice 你认识是"练习"，它还是"做法""诊所"。

manner 是"方式"，复数 manners 是"礼貌"；means 是"手段"，单复数同形；regard 是"看待"，复数 regards 是"问候"。这类词的危险在于：你以为你不会错，所以你不查。

subject 是"主题"，也是"臣民"和"使遭受"（be subjected to）；object 是"物体"，也是"反对"（object to）。present 是"现在"，也是"出席的"和"呈现"。content 是"内容"，也是"满足的"（be content with）。一个词两副面孔，句子给你哪一副，你就得认哪一副。

fine 是"好"，也是"罚款"；just 是"仅仅"，也是"公正的"；still 是"仍然"，也是"静止的"；bow 是"鞠躬"，也是"弓"；tear 是"眼泪"，也是"撕"。读音不同，拼写相同，这类词英文里叫 heteronym，考试不考它，但阅读里天天见。

issue 是"问题"，也是"发行"（the issue of a magazine）；press 是"按"，也是"新闻界"；figure 是"数字"，也是"人物"和"身材"；capital 是"首都"，也是"资本"。你在正文里读到 capital 的时候，先看它旁边的词，再看它。

数字与量：gallon、quart、acre 这一家

gallon 是加仑，quart 是夸脱，四夸脱一加仑；pint 不在这一章。acre 是英亩，hectare 是公顷。dozen 是十二个，fortnight 是十四天。这些词在中文里没有对应物，所以你只能记它的"大约"：一加仑约四升，一夸脱约一升，一英亩约四千平方米。fortnight 这个词在英式英语里很常用，a fortnight 就是 two weeks，你写作文用 fortnight 会显得很地道，但用错场合会显得很奇怪——它太口语了。

cent 是分，dollar 是元，pound 是镑。percent 是百分比。ratio 是比，rate 是率，proportion（proportional）是比例。四个词在图表作文里是命根子：ratio 是两个数的比（3:1），rate 是一个量除以另一个量（每千人多少），proportion 是部分占整体（三分之一）。

时间与频率：什么时候用哪个

fortnight 已经说过。annual 是每年的，seasonal 是季节性的，monthly 是每月的。frequent 是频繁的，habitual 是习惯性的，continual 是反复的。recent 是最近的，current 是当前的，present 是现在的。prior 是先前的，former 是前一个的，previous 是上一次的。三个"前"，范围不同：former 常用于"前任"，previous 指"上一个"，prior 最正式。

eventually 是"最终"，finally 是"最后一步"，at last 是"终于（等到了）"。中文都翻成"最后"，英文的语气差很远。hardly 是"几乎不"，不是"努力地"；lately 是"最近"，不是"后来"；shortly 是"很快"，不是"简短地"。这三个副词是四六级翻译的经典坑。

nowadays 是"如今"，比 now 正式，但不如 these days 自然。now 在句首常常是语气词，不翻译。then 是"那时"也是"然后"。till 和 until 一样，但 till 更口语，在句首一般用 until。

结尾：给还在背词表的你

这一章讲了九百多个词，我猜你记住的不超过三十个。这很正常，也很正确——因为你要记的从来不是词，是"词在什么地方出现"。abandon 在林一关掉浏览器的那个早上出现，ass 在河边那辆驴车上出现，adversity 在沈璐说"逆境不是美德"的时候出现。你记住的是那几个画面，词是搭在画面上的。

所以，别再问"这个词什么意思"，开始问"这个词在什么情况下会被人说出来"。前者给你一个中文词，后者给你一个位置。中文词会忘，位置不会。

最后一句话，你可以抄在本子上：词不是背下来的，是碰上来的；你碰得越多，它越认得你。

为什么你背了三年还是不会用

我在正文里让林一犯过一个很典型的错：他把一个 App 上的三千多个词条背完了，屏幕顶端那个数字每天往上涨，涨到一千三百四十，他觉得自己在进步。然后他在英语角站了四秒，一句话没说出来。这不是意志问题，是方法问题：他背的是词的"尸体"，不是词的"用法"。

你可以自己试一次。拿出一张纸，写下你最有把握的十个英文词，然后每个词造一个句子，句子必须描述你今天真实做过的一件事。你会发现，你能写出来的大概只有三四个。剩下的那些，你认识，但你不会用——认识的词和会用的词之间，隔着一条很宽的沟，这条沟只能靠"用"来补，不能靠"再看一遍"来补。

这条沟有多宽？大概是一个学期。正文里那个从北方来的男生，口音重到林一只听得懂一半，但他敢说，说着说着就到了第三个句子，而林一在第一个句子的第四个词上停住了。作者让林一在那一刻明白了一件事：让他停下来的不是词汇量，是"检查"。他一边说一边检查自己说得对不对，检查花掉的时间和下一个词需要的时间一样长，于是他永远落后一个词。

这个机制值得你记一辈子。所有"我一开口就紧张"的人，紧张的不是别人在看你，是你在看你自己。解药不是"别紧张"，是"把检查关掉"。怎么关？把它变成不需要检查的东西——同一句话，你重复到第二十遍的时候，它就不需要检查了。所以精听要"复述"，朗读要"出声"，背诵要"成句"。这三件事都是同一个动作：把一个句子从"要检查"变成"不用检查"。

再说一个反直觉的：不要背单词的"中文意思"。中文意思是标签，标签贴在词上，词就死了。restrain 你背成"抑制"，于是你在作文里写 restrain the development，老师看不懂，因为你抑制的是"发展"还是"自己"？restrain 的对象多半是人或者情绪：restrain oneself、restrain your anger、restrain a smile。你看，一个词的"对象"比它的"意思"重要得多。

同理，forfeit 不是"丧失"，是"因为违规或者放弃而失去本该属于你的东西"——forfeit a match 是弃权判负，forfeit your deposit 是押金不退。dispense 不是"分配"，是"分发"（dispense medicine）和"免除"（dispense with formalities）。curtail 不是"减少"，是"削减权利或自由"，带政治味。augment 不是"增加"，是"增强"，用于收入、能力、效果。这四个词中文都能翻成"减少/增加/分发"，一到句子里就全错。

一个可以马上做的练习

今晚找一段两分钟的英语音频，听三遍。第一遍不管懂不懂，第二遍看着文本听，第三遍不看文本，跟着读。读完把文本合上，用中文说出它讲了什么，再用英文说一句你自己的评论——一句就够，哪怕语法是错的。第二天换一段。两周之后你会发现自己听同一段音频时，脑子里不再出现中文了。那一天就是分界线，之前叫"学英语"，之后叫"用英语"。

这个练习不新鲜，也不聪明，它唯一的优点是它真的有用。所有有效的语言方法都长这样：笨、慢、每天二十分钟、看不到即时反馈。反过来说，凡是承诺"三天见效""七天突破"的，卖的都是那种让你在付款那一刻感觉良好的东西。

正文里沈璐说过一句话，我在这里再抄一遍：方法会失败，习惯失败得更慢。你不需要一个更好的方法，你需要一个能坚持到明年的习惯。习惯这个东西很难开始，但一旦开始，它就替你做了大部分决定——你不需要每天说服自己去学英语，你只需要在七点二十的时候坐到那张桌子前。

最后，关于"语感"。很多人把它说得很玄，好像是一种天赋。其实语感就是"见过的句子足够多之后，对不对劲的东西产生的一种生理反应"。你读到 a information 会觉得别扭，那就是语感；你说出 he go 会觉得舌头不对，那也是语感。它不神秘，它是统计。你想有语感，就得给大脑足够的样本，而样本只能一篇一篇地读进来。

所以这一章最后落在一个很朴素的地方：读。读你喜欢的、看得懂的、稍微有点难的英文，一天一页，读完抄一句。一年三百六十五句，那就是你的语料库。这个数字听起来不多，但它是你自己抄的，每一句你都曾经觉得好，那就够了。

那些看起来"高级"其实很危险的词

写作的时候，很多人生怕句子不够高级，于是把认识的难词全塞进去。结果常常是：句子变长了，意思变浅了。难词的价值不在于难，在于它比常见词多说了那么一点。你如果能说清那一点，用它就是加分；说不清，用它就是冒险。

举几个这一章里的例子。substantial 不是"大"，是"有实质内容的、占得住分量的大"。a substantial meal 是一顿扎实的饭，a substantial change 是实质性的改变，不是"很大的改变"。你说 a substantial mistake，人家会以为这个错误有内容、有后果，而不是"一个大错"。negligible 不是"小"，是"小到可以不管"：a negligible difference 是"这个差别不值得讨论"，带判断，不带尺寸。

feasible 不是"可能"，是"做得成"：技术上可行、钱够、时间够。probable 才是"很可能"。possible 最弱，只要不违反物理规律都算 possible。三个词从弱到强：possible、probable、certain；feasible 走的是另一条线，它不谈概率，谈条件。viable 跟 feasible 很近，但 viable 通常指"能活下去、能持续"，a viable business 是能自我维持的生意，a viable plan 是能落地的方案。

desirable 是"值得要的"，带价值判断；preferable 是"更可取"，带比较；advantageous 是"对你有利"，带立场。三个词在议论文里可以排成一句：A is desirable, B is preferable, and for us C is advantageous——同一件事的三个层次，用词不同，逻辑就清楚了。

adequate 和 sufficient 都翻"足够"，但 adequate 是"达到最低标准"，sufficient 是"满足需要"。a salary adequate for one person 是"一个人够活"，sufficient for a family 是"养得起一家"。你说 adequate 的时候，对方听到的是"勉强"。deficient 是它的反义，deficiency 是"缺乏症"。

tentative 是"试探性的"，一个 tentative conclusion 是"先这么认为"，带保留；tentative 还可以是"暂定的"，a tentative date 是暂定的日期。它跟 tentative 长得像的 tenable 是"站得住脚的"，两者不同。tentative 的对立面是 definitive，不是 certain。

arbitrary 不在这一章，但 random 的意思你在这一章能碰到：haphazard 也没有。别把"随机"和"随意"混——统计里的 random 是有规则的，生活里的 random 是没规则的。

关于"同义词替换"的误解

很多人以为作文高分靠"同义词替换"：把 big 换成 enormous，把 good 换成 beneficial。这个想法只对了一半。替换的前提是"语义等值"，而英文里真正等值的同义词极少。big 和 enormous 不等值：enormous 是"大到超出预期"，你写 an enormous effort，读者听到的是"夸张"；an enormous room 是"大得离谱的房间"，可能带批评。

important 和 significant 也不等值：important 是"要紧"，significant 是"有意义的、统计上显著的"。你写 a significant friend，人家会以为你在说"一个有统计学意义的朋友"。significant 在这一章里出现过，它在学术文本里几乎是专用词。

problem 和 issue 不等值：problem 是要解决的，issue 是要讨论的。job 和 career 不等值：

job 是活儿，career 是"职业"这条线。house 和 home 不等值：house 是房子，home 是家。这三个差别，中国学生最容易忽略，因为中文里它们就是一个词。

所以正确的做法不是"把简单词换成难词"，而是"把模糊词换成准确词"。你写 a big problem，问题不在 big 简单，在于你没说清是"严重"（serious）、"普遍"（widespread）、"紧急"（urgent）还是"棘手"（tricky）。这四个词比 enormous 有用得多，而且都不难。

最后说说这一章的写法

你也许注意到，这一章不是先有词表再有故事，而是先有场景再把词放进去。选课系统崩溃的那个早上，一个词表上的 allocation 变成了一件具体的事：四十个人在走廊里等一个号码。图书馆地下室里，一个 ancient 变成了一排发霉的报纸合订本。河边那个上午，一个 ass 变成了一头灰色的、耳朵上有苍蝇的、 patiently 站在那儿的驴。

这就是我想让你看见的东西：词不是从天上掉下来的，它是从事情里长出来的。你背一个词的时候，脑子里如果没有一个画面，那个词就只是一串字母；你用它的时候，就得现场编一个画面，而现场编的东西一定不如原来那个真。

所以，从今天开始，你背单词的时候多做一件事：给它配一个你自己的句子，句子必须是真的——今天发生了什么，你怎么想的，你看见了什么。这个句子不需要漂亮，不需要正确，它只需要是真的。真的句子会留下来，漂亮的句子不会。

九百多个词，你不可能一次记住。但你可以一次记住三十个，一个月以后，那三十个会长成六十个，因为你在读别的东西的时候会再碰见它们。语言就是这样长起来的：不是堆积，是复现。你今天在走廊里碰见 allocation，下周在通知里再碰见它，下个月在新闻里第三次碰见它——第三次的时候，它就归你了。

附：这一章里几个值得单独抄下来的句子

正文里林一在走廊上站了四个小时，最后只换来一句"你没事，你什么都有"。那一刻他明白的事情，比他在阶梯教室里记满的一页名词有用得多：程序不欠你解释，程序只负责走完。你要在这套东西里活得体面，就得学会两件事——把该问的问题问到人身上，把该记的东西记在自己本子上。前者是张野教他的，后者是沈璐教他的。

沈璐说词汇是砖、语言是墙，这句话值得你抄在本子第一页。因为大部分人学英语的痛苦，都来自"搬砖"这件事本身没有尽头：你今天搬一百块，明天还有一百块，你永远看不到墙。而墙是可以看到的——你看得懂一篇没有生词的文章，你听懂一段没有字幕的对话，你写出一段不需要改三遍的句子，那就是墙。墙不长，但它立着。

她还说，检查是最大的敌人。这句话你可以拿去用在任何需要临场发挥的场合：面试、答辩、第一次跟外国人说话。你一边说一边检查，就等于用一半的脑子说话、用另一半的脑子当裁判，而裁判永远比运动员慢半拍。解决办法只有一个：把要说的话练到不需要检查。这就是"熟"这个字的全部含义。

王老师说，规则不是让你变好，是让二十九个人在同一件事上不用有人当英雄。这句话听起来不浪漫，但它是你未来四年最实用的一句话。你会发现大学里让你最难受的事，几乎都不是"有人坏"，

而是"没有规则"或者"规则不算数"。一个规则清楚的地方，哪怕规则很笨，人也活得松快；一个规则模糊的地方，人人都得随时准备战斗。

张野说，钝刀撕、快刀切，差别不在力气，在注意力。这句话他在磨刀的时候说的，但它适用于所有手艺，包括学英语。你花两个小时"看"单词，和你花二十分钟"用"一个句子，差别不在时间，在注意力落在哪儿。落在词上，词会跑；落在句子上，句子会留下来。

老余说，工具是有人守住的承诺。这句话你也可以反过来读：你手里的每一件东西——一本教材、一个 App、一张课程表——背后都有人替你做过决定。你要做的不是感谢它，是判断它值不值得你用。这一章里那个把 immensity 拆成 im 和 mensity 的 App，就是一件不值得用的工具，因为它替你做的决定是错的。

最后一句，是我自己的：你以为你在学一门语言，其实你在学怎么把一件事说清楚。 前者会让你焦虑，因为语言没有边界；后者会让你踏实，因为一件事总有说完的时候。

关于"读不懂"这件事

最后想跟你聊聊读不懂。读不懂分两种：一种是词不认识，一种是词都认识、句子还是不懂。第一种好办，查；第二种难办，因为它跟词没关系，跟"句子怎么连起来"有关系。

举个例子。正文里那句"the oversight of the system was largely a matter of luck versus engineering"，每个词你都认识：oversight 是监督，largely 是很大程度上，versus 是对。但你未必读得懂，因为 oversight 在这里不是"监督"，是"疏忽"——同一个词，两个相反的意思。英文里这种词不少，sanction 既是"批准"又是"制裁"，cleave 既是"劈开"又是"粘住"。碰上它们，你只能靠上下文判断，而上下文的判断力，只能靠读。

第二种读不懂，多半是因为你把句子当成了词的串联。英文的句子是"骨架 + 挂件"：先找主谓宾，再看从句和介词短语挂在哪儿。你在阅读里遇到一个长句，第一件事不是从头读，是找那个动词——句子的心脏。找到了心脏，其他的都是血管。

这件事说起来简单，做起来要练。练法是：拿一段你看不懂的长句，用铅笔把主谓宾圈出来，把从句用括号括起来，然后只看主谓宾，看这句话的骨架是不是通的。通，就说明你懂了；不通，就说明你圈错了。这个动作做五十次，你的阅读速度会翻倍，因为你不再逐字读了。

最后一句忠告：不要用翻译来检验理解。翻译检验的是你的中文，不是你的英文。真正的检验是复述——合上文本，用英文说三句，说得出就是懂了，说不出就是没懂，跟中文没关系。

再讲一遍"词的第二个意思"

英语里最贵的错误，不是不认识，是认识一半。oversight 是监督，也是疏忽；sanction 是批准，也是制裁；cleave 是劈开，也是粘住；dust 是掸灰，也是撒粉；screen 是筛，也是放映。这些词在词典里排在一个词条下面，在句子里却像两个人。你判断的唯一依据是上下文，而上下文的判断力只能靠读，不能靠背。

fine 也一样：a fine day 是好天气，a fine of fifty yuan 是五十块罚款，fine gold 是纯金。just 是"仅仅"，也是"公正"。still 是"仍然"，也是"静止的"。mean 是"意思是"，也是"吝啬

的"和"平均的"。你写 he is mean，读者会往哪个方向理解，取决于前一句。

这一类词你要养一个习惯：读到一个"熟词"出现在一个"不对劲"的位置，停下来，问一句"它是不是还有另一个意思"。这一停，往往就是阅读理解里那道题的结论，也是翻译题里那半句的分。

关于背单词的最后几句实话

第一，词表是别人的顺序，不是你的。你可以顺着背，但真正记住的词，一定是你自己碰上的那批。第二，一个词要见七次左右才会变成你的，这七次最好分布在不同的文章里，而不是同一个列表的七遍。第三，你记不住的词，九成是因为你从来没有用过它；用一次，胜过看十次。第四，别在一天里背一百个新词，第二天你会一个都不剩；一天二十个，第二天复习，第三天再用三个造句，这三个才是你的。

第五，也是最重要的一条：允许自己忘。忘了不是失败，忘了是正常的，你只要在它第二次出现的时候把它捡回来。语言学习不是建仓库，是修路——路会坏，坏了再修，修的次数多了，它就结实了。

最后：把这一章当成一次示范

这一章从头到尾只做了一件事：把一份按字母排的词表，还原成一个人一个月里碰上的东西。选课系统崩溃的早上、听不懂的阶梯教室、图书馆地下室的霉味、招新摊位上的风、英语角那四秒钟的沉默、河边一头驴的耳朵——这些都不是背景，这些就是词的家。你把词从这些场景里拔出来，它就变成一串字母；你把它放回去，它就有重量。

所以读这一章的正确方式，不是查生词，是跟着林一走一遍。走完之后，你再回头看那份词表，会发现有一些词已经"有主"了——它们属于某个具体的画面，你一想到那个画面，词就自己出来了。剩下的那些还没有主的，等你在别的文章里再碰见它们，它们就会认领自己的画面。

这也是这套读本存在的理由：不是替你背单词，是替你造场景。场景造得越多，你要背的东西就越少。

jewel** 是宝石，也可以是被人珍视的东西；jewellery 是首饰的统称，两者不要混。

第 3 章　实验楼 402

The door of 402 was already open at seven forty. The corridor smelled of wet chalk and of something older, a bitter, damp smell that came out of the fume hoods and had accumulated in the carpet at the foot of the staircase. I stood on the threshold with my calculator in one hand and my notebook under my arm, and did what I always do when I am early and afraid: I read the wall.

There were three things on that wall. A banner printed LABORATORY 402, NO FOOD, NO DRINK, NO ALCOHOLIC BEVERAGES, hung slightly crooked, as if somebody had straightened it in a hurry. Under it, a sheet of A4 carried a list of abbreviations in two columns,

because the manual had been written by people who had never met a first-year student at eight in the morning. Beside that hung the safety notice, four clauses long. I read it twice and understood it once.

The notice affirmed that any allergic reaction to latex had to be declared before the first class; that a breach of protocol would be treated as a complaint against the duty officer and not as an accidental blunder; that bodily harm was not a matter for consolation or for diplomacy but for the hospital; and that compliance was a condition of entry, according to the authority of the chancellor's office, which had approved the wording and, I suspected, never read it.

At the bottom there was a line for a signature. I signed the acknowledgement without reading the last clause, which is the beginning of every story like this one. I was eighteen. I had a certainty that I was cautious, and no evidence for it at all.

The room itself was longer than I expected, and bright in a way that made everything look older. Two rows of benches ran the length of it, black stone tops, brick-red cabinets underneath, and at the far end a curtain of clear plastic that moved a little when the atmospheric pressure in the hood changed. On the left, in its own small cell of a room with a glass door, sat the analytical balance. Its beam was stationary only when nobody was walking. That was the first thing I learned, and I learned it the hard way, at ten past eight, by bumping the bench with my knee.

A voice behind me said, "You are breathing on it."

Zhang Ye had come in without a sound, in a baseball cap and a jacket thick as a blanket, which he would not take off until November. He put an apple on the bench, took a clip off a sheaf of papers, and clipped his registration form to the front of his notebook. He did this with the diligence of a man who has been accustomed to forms since he was fourteen. In his accent, the flat, hard bass of the eastern hills two hundred kilometres from here, he said, "The technician said we could weigh after eight thirty. Before that the power is low and the battery in the duty locker is charging."

"So we wait."

"So you read." He nodded at the wall. "You read the article, or you read the banner?"

"Both."

"Then you know the category." He pulled a canvas apron off a hook and threw it at me. "Clothe yourself. The etiquette here is simple. Apron, mask, clamp the filter flask before you pour, and never cut anything toward your hand."

He was not being assertive; he was being helpful, which in his culture looks the same. I put the apron on backward and he watched me do it and said nothing, and that silence was more brutal than any denunciation I have received since.

At eight twenty the aide came in. Nobody called her that. Everyone called her Zhou Min, and she was in her second year of a master's degree, and she ran 402 the way a chief runs a kitchen: loudly, constantly, and with a clipboard. She talked fast, like an auctioneer who has been told the amount of time allotted to her and does not approve of it. "Everybody attentive for four minutes," she said. "I will administer the registration, then the safety, then the equipment. If you are absent today you do not get a bench this term. That is not a threat, it is an inventory problem. We have twenty-two benches and thirty-one people."

She had an app on her phone that showed the duty layout of the week, and she held it up so we could see the grid: names in rows, categories in columns, hourly blocks in green. "This is the roster," she said. "If your name is not on it, your assumption that you are eligible is wrong. Check it aloud with the person next to you. Alternately, check it yourself and then check it again."

Somebody laughed. She looked at him until the laugh was arrested in the air.

"Your first week," she went on, "begins with a comparison. You will amass data you do not understand, then you will interpret it artistically, and then I will fail you. So let us do the elementary thing first. Balance. Alignment. Actual reading. Uncertainty."

She crossed to the balance room and put one finger on the beam. "This instrument costs about as much as your four years of tuition, so the astronomical figure is not a metaphor. But the number it gives you is not the truth. It is an equivalence between your sample and a mass somebody in Beijing keeps in a vault." She turned around. "What is the difference?"

Nobody said anything. The silence had a temperature.

"The difference," she said, "is that the mass in Beijing is confidential and the sample on my bench is dirty. Somebody tell me why the balance is in a separate room."

"Wind," I said. My voice came out awkward and shaky, a coward's voice.

"Air," she said. "Good. Drafts, vibration, dust, the compression of the floor when a lorry goes past on the avenue outside. This building went up in nineteen eighty-six, the concrete is tired, and the freight traffic on the avenue peaks at eight thirty, which is exactly when everyone wants to weigh. So the rule." She held up one finger. "Do not weigh at eight thirty. Weigh at eight forty. The road is quiet then, and the water in the pipes has settled."

I wrote that down. I wrote down everything that term, which is how I ended up with four notebooks and very little understanding, but that came later.

The first session was not an experiment. It was the washing of glassware, and the checking of it, and the writing of the check in a log that nobody would read but that had

to exist. Each of us got a tray. In the tray: two beakers, one conical flask, one volumetric flask with a stopper, a funnel, a watch glass, a glass rod, and a rack with six tubes.

"Rinse three times," Zhou Min said. "Distilled water for the last two. Hold the flask by the neck, never by the body, because your hand is thirty-seven centigrade and the glass will expand, and then your volume is a lie."

Zhang Ye finished in nine minutes. I finished in twenty-six, and two of my beakers still had a ring of white residue at the bottom, and Zhou Min looked at them and said, "Again," and I did them again, and they were clean, and she said, "Good. Now you know what clean looks like. That is the whole point of the first week."

I had imagined that a laboratory was a place where you found things out. It was, at first, a place where you practised looking.

At nine the tutorial began. Zhou Min stood at the blackboard, a real one, with chalk, in a brick room in a university that had bought its projectors in a different decade, and wrote one word: YIELD.

"Define it," she said.

"Product obtained over product theoretical," said a girl in the front row.

"Which product? The wet one? The dry one? The one that includes the water of crystallization, or the one that does not?" She waited. "It is a ratio, and a ratio is a promise about what you counted. If you cannot say what you counted, you have not measured anything. You have produced a number."

That sentence went into my notebook with three underlines, and into my head with none.

We were paired by lot. There were thirty-one of us, which is an odd number, and the rule Zhou Min had written on the board, pairs and no exceptions, meant that one person would be left over. The lot was drawn from a beaker, and the last slip of paper had a name on it that nobody claimed. The room went quiet in the particular way a room goes quiet when thirty people decide at the same moment to look at the ceiling.

I knew the name. It was the girl who had answered about the yield, Chen, from Lishui, who sat in the front row and never asked anything and never missed a class. She was looking at her tray as if it were the most interesting object in the province.

I have thought about the next four seconds more than I have thought about any exam I have taken. The easy thing, the ordinary thing, the thing that ninety-nine per cent of people do, is to look away. The reason is not cruelty. It is arithmetic: there are two of us and one of her, and the cost of speaking is borne by whoever speaks.

"She is with us," Zhang Ye said. He did not raise his hand. He did not look at Zhou

Min. He said it in the flat tone of a man reading out a bus timetable, and then he moved his tray six inches to the left to make room, and that was the end of it.

Zhou Min wrote the third name on the grid without looking up. "Three," she said. "Fine. Three is better than two, if the third one reads the manual."

Chen's face did not change. She came over, put her tray down, and said, to the bench rather than to us, "I can do the titration. I have done it before."

"Then you teach it," Zhang Ye said, and that was the whole negotiation, and it had taken eleven seconds. I have seen committees spend eleven months on less.

Group dynamics, Zhou Min told us later, are not a mystery. Three people will sort themselves into three roles within a week: one who wants to be right, one who wants to be finished, and one who wants to be liked. "None of those roles," she said, "produces good data. Good data comes from the person who wants to know what happened."

I decided, that morning, that I would be the one who wanted to know what happened. I wrote it on the inside cover of my second notebook, in ink, as if a sentence in a notebook were a contract.

The email came at twenty-three forty on Monday, from Zhou Min, with the manual as an attachment and a subject line that said, in full, read this before you touch anything. I read it lying on my bed with the light off and the screen two inches from my face, which is how I read everything that year.

The manual was nine pages. Page four is the one I still remember, because it was written in the only style that a laboratory manual has ever managed: the imperative, aimed at nobody in particular, as if the person holding it were a machine with hands.

"Weigh 2.0 g of copper oxide into a small beaker. Add 10 mL of dilute sulfuric acid in three portions, warming between additions. Do not allow the mixture to boil. Filter while hot, using a fluted filter paper, and collect the filtrate in a clean beaker. Evaporate the filtrate slowly until a film forms on the surface. Cool, filter the crystals, press them dry, and weigh. Dry at one hundred and five centigrade for forty minutes, cool in a desiccator, and weigh again."

Nine sentences. A sequence of nine instructions, and each of them a place where a person can go wrong. I have since decided that a manual is not a description of what to do. It is a list of the ways you are about to fail, arranged in order.

There was also a smaller card, laminated, clipped inside the front cover, which the department had printed after somebody set fire to a bin in 2019. It said that copper sulfate was hazardous if ingested, that the dust was an irritant, that a splash in the eye required fifteen minutes of washing and a hospital, and that all waste containing the salt had to go into the container marked with a category of its own, never into the sink,

because a heavy metal is not a pollutant that a river can digest. Somebody had written in pencil in the corner: the river cannot read. I liked whoever wrote that.

At eight forty on Tuesday the balance room was empty, the road outside was quiet, and I was ready to be excellent.

I was not excellent. I was, in the first four minutes, three separate kinds of clumsy.

First, the weighing boat. I filled it too fast and a fine green-black dust went up and settled on the pan, on the glass, on the sleeve of my apron. Second, the static. The powder would not leave the boat; it clung to the plastic like a small furry creature with an opinion, and when I finally touched it with the spatula, half of it jumped. Third, the door of the balance. I closed it too hard, and the reading on the screen drifted, and drifted back, and I stood there watching a number that would not make up its mind, understanding for the first time that a measurement is an argument between you and the room, and that the room does not concede.

The actual mass was 1.87 g, not 2.0. "Note it," Zhou Min said from the doorway, without stopping. "The number you aimed for is not a fact. The number you got is a fact. Write down the one you got."

The acid went in three portions, as instructed. The solution turned a green so dark it was almost black, and then, as the oxide dissolved, a blue that came up through the liquid the way ink comes up through water. It smelled of sulfur and of something sweetish underneath, and the beaker was warm in a way that was pleasant for about four seconds and then was not.

I put it on the hotplate to evaporate, and here is where the whole week turned.

The manual says slowly. Slowly is not a number. I set the dial to a position that felt scientific, and I went to look at my phone, and when I came back the film had formed, and passed the film, and the liquid was spitting. I watched a bubble rise, swell, and break, and the blue threw itself out of the beaker and onto the black stone, where it hissed and went dark.

Forty-one per cent. That was my yield. I calculated it three times, with the calculator, and then once more by hand, and it stayed forty-one, and around it the bench looked like the scene of a small and stupid crime.

I stood there with the numbers in front of me and felt the particular shame that arrives when the mistake was not difficult. If it had been difficult, I could have respected myself. It was not difficult. It was a dial, a phone, and ninety seconds of not looking.

Chen said, quietly, "We can redo it. There is time before eleven."

"There is no time," I said. "There is never time. That is the whole design."

I did not know I was going to say that. I heard myself say it. It came out with more bitterness than I had in me, and it made the room worse, and Chen looked at the bench and did not answer, and I understood that I had just spent something much harder to get back than forty-one per cent.

Zhang Ye came over, looked at the beaker, looked at the bench, and said one word. He said, "Thick." Then he took the beaker away to wash it, because the crystals stuck to the bottom would take twenty minutes and he did not want me to have those twenty minutes alone with my notebook.

That was when Shen Lu walked in.

She was in her fourth year, she had a key to the fourth floor, and she moved through a laboratory the way some people move through their own kitchen: without looking at anything twice. She was there because her own bench was in 406 and the oven in 402 was the only oven in the building that held its temperature. She stood in the doorway for a moment and read the bench, which is a thing I have since learned that experienced people do, the way a driver reads a road.

"That is not a yield," she said. "That is a rumour."

I started to explain. I got as far as the hotplate, and she held up one finger, and I stopped, because a raised finger from her is a complete sentence.

"You evaporated it to dryness and then kept going. The product is not lost because you are stupid. The product is lost because you were not watching, and not watching is a decision." She looked at my notebook. "What is the film?"

"The point where crystals begin to form at the surface."

"You know the sentence. You do not know the film. If you had known the film, you would have been standing in front of the beaker when it appeared, because that is the only way anyone ever learns what it looks like." She tapped the page with one finger. "A film is a visual event with a duration of about four seconds. Four seconds. And you went to look at your phone. That is not a moral failure, it is a scheduling failure, and scheduling failures are the only kind this course actually grades."

Then she did something worse than the speech. She pulled a chair over, sat down on it backward, and said, "Show me the calculation."

I showed her the calculation. She went through it in silence, line by line, with a pen she had not asked for, and the silence went on so long that I began to hear the fume hood.

"Your arithmetic is correct," she said at last. "Your arithmetic is impeccable, which is the most dangerous thing about it. You have a fraction on the top and a fraction on

the bottom, and both of them are wrong in the same direction, and you have written down a number with four figures that agrees with itself. This" — she underlined forty-one — "is not a measurement. It is a piece of self-confidence with a decimal point."

She stood up. "Do you want the real criticism, or do you want the kind that lets you sleep?"

I said I wanted the real one, and I meant it, and I was wrong about that, but it was the right thing to say.

"The real criticism is that you have a theory of yourself in which you are careful. You are not careful. You are nervous, and nervous looks like careful from the outside, and the difference is that a careful person checks and a nervous person performs checking. You washed those beakers for twenty-six minutes because washing them felt like work. Watching a beaker for four seconds feels like nothing, so you did not do it. That is the whole of it." She picked up her gloves. "You are not lazy. You are Romantic. You want the experiment to be about your attention. It is about the film."

I did not sleep well that night, and I want to record, for the sake of honesty, that the reason was not shame. It was that she was right in a way I had no answer for, and an answer is what I usually have.

One more thing from that Tuesday, because it belongs here. On the wall of the balance room there was a printed notice about waste, and under it a smaller one about behaviour, and the smaller one was the only document in that building that anyone read twice. It said that harassment of any kind, including sexual harassment, threats and intimidation, was grounds for removal from the laboratory and referral to the college; that supervision was not the same as permission; and that anyone who felt unsafe could leave the room without explaining why, and would not lose marks for it. Under that, in the same font, the department had added a line that made me trust them slightly: no experiment is worth an injury.

I read it on my way out, with my forty-one per cent in my hand, and I thought that a place which tells you the rules for leaving is a place that has thought about people.

On Wednesday there was no experiment, which meant there was a session about rules, which meant two hours in a room with Zhou Min and a piece of chalk.

She wrote six numbers on the board, and under each number one sentence. That was her whole method: a rule is a number with a sentence under it.

"One. The notebook is written in ink, in this room, at the bench, while the glassware is still wet. Not afterwards. Not in the dormitory. A record made afterwards is not a record, it is a narrative, and I am not interested in your narrative. Two. Every container is labelled before it has contents, not after. A beaker with no label is not a mystery, it is a hazard, and I will pour it out in front of you and I will not be sorry.

Three. No one works alone. Ever. If your partner is ill, you do not work; you read. Four. Waste goes to its category, and the categories are not a suggestion, they are a law with a lid. Five. If you break something, you report it within the hour, and the report is not a punishment. Six. If you do not understand a step, you stop and you ask, and nobody in this room is permitted to make you feel small for asking. That last one is not a rule for you. It is a rule for me."

Somebody at the back had the nerve to look bored. Zhou Min saw it, and instead of the reprimand I expected, she told us about a man who had been in that same building in 1997 and had decided that a label was administrative decoration, and had therefore poured the contents of an unlabelled bottle into a sink, and had spent the following month in a hospital with a tube in his chest, and had never got the use of one lung back.

"Nobody here is a coward," she said. "But some of you are in a hurry, and a hurry is a kind of arrogance, and arrogance is what the labels are for."

Then she made us do something I did not expect. She gave each group a marker and told us to write, on the whiteboard, the labels we would need for the rest of the term. Not the names of teachers, not the names of instruments: the labels. Thirty-one people produced a wall of words, and I remember that Chen wrote twelve in four minutes, and that one of them, in her small square handwriting, was a warning about the disposal of anything that had touched the salt, and that underneath it she had added the phrase that became a joke in our group: the river cannot read.

Rules, I discovered that afternoon, are not the opposite of freedom. They are what makes the freedom safe enough to use. A laboratory with no rules is not a brave place; it is a place where the bravest person sets the standard, and the bravest person is usually the one with the least to lose.

Afterwards Shen Lu took me up to 406, which was the room she shared with two other fourth-year students and a machine that had cost more than the building. I had never been inside. It smelled of solvent and of the inside of a new car, and the whiteboard was covered in equations that had been there so long that some of the numbers had been smudged by somebody's sleeve.

"Read this," she said, and pushed a journal across the bench. A real one, printed, with a cover the colour of a bruise. "Not the abstract. The first paragraph. Then tell me what the authors want."

I read it. It was about a battery material, and the first paragraph said that the material was promising, that its capacity was high, that it was cheap, and that its cycling stability was, in the authors' phrase, an ongoing challenge. I said all of that, in what I thought was a clear and orderly way.

"You have described what they wrote," she said. "I asked what they want. Those are

different readings of the same page. They want a grant. They want the reader to believe that the problem is difficult enough to be worth funding and nearly solved enough to be worth funding now. That is not a crime. It is a genre. Every genre has a rhetoric, and the rhetoric of the journal article is a modest claim that hides an immodest request."

She sat down on a stool and folded her arms. "This is the thing nobody tells first-years. You think a paper is a report. It is an argument. It has a thesis, it has evidence, it has an audience, and it has a place it wants you to arrive at. When you can see the argument, you can use the paper. Until then you are reading a very long advertisement and calling it knowledge."

"Then how do I learn to see it?"

"You read a hundred of them badly and then ten of them well." She pulled a stack of paper towards her, the corner of it worn soft. "Also, you read the citations. Not the text: the citations. Who is cited, how often, by whom, and whether the same four names keep appearing in the same four positions. That tells you what a field believes about itself, which is often more useful than what it has measured."

I asked whether that was cynical.

"It would be cynical if I said the measurements were worthless. I am saying that a measurement is a sentence in a conversation, and if you cannot hear the conversation you will misquote the sentence." She looked at me. "You have a habit of wanting to be impressed. Lose it. You will need the space it takes up."

She was, I understood later, giving me a method and calling it a warning, which was the only way she knew how to be kind.

Before I left she showed me her own shelf: eleven notebooks, numbered, and a box of index cards with titles written on them in three colours. "The cards are the archive," she said. "The notebooks are the memory. The archive does not argue, and it does not forget, and it does not care whether you were tired." She tapped the box. "When you write your report, this is what you are building. Not a story about a Tuesday. A shelf that somebody else can stand on."

I went down four flights of stairs with my hands in my pockets, feeling the particular excitement of a person who has just been handed a tool and does not yet know what it will cost.

The lift in that building had been out of order since before I arrived, so the notice board beside it was the only thing on the fourth floor that ever changed. On Thursday it changed four times at once.

The first poster advertised a lecture in the physics building: FROM COMET TO COSMOS, with a photograph of a black rock and a line about the geological time it takes to make a

mineral. The second was a marine talk, CORAL, DOLPHIN, PENGUIN, about what a warmer ocean does to a maritime creature, and somebody had drawn an iceberg beside the title, as if an iceberg were the same kind of problem as a warming sea. The third announced a seminar on Buddhist scripture and the aesthetic of silence, which struck me as the least likely event ever to be held in a building with a fume hood on every floor. The fourth was a student union debate. DEMOCRACY AND ITS ENEMIES, it said, and under that, three words in red: referendum, dictator, exile. Somebody had written in pencil, in the corner, the only honest sentence on the board: nobody comes to these either.

Below the poster about democracy there was a smaller sheet, the kind that stays up for a month because nobody has the authority to take it down. It said that a seminar on esthetics would be held by the art college, and that the topic was whether a laboratory is beautiful. I thought about 402, with its leaking tap and its ninety-year-old tiles, and decided that the answer was no, but that the question had been asked well.

I was still reading when Zhang Ye came up beside me and said that he had cut his hand.

He said it in the tone of a man reporting the weather. He held it out. There was a cut across the base of the thumb, not long, but deep enough that the blood ran down to his wrist and dripped onto the tile, and I remember three things about the next minute: that the blood was darker than I expected, that he did not swear, and that I was the one who made a sound.

"Shit," I said, which was not useful.

"Glass," he said. "A beaker went in the sink. I put my hand in the sink." He looked at the cut with the mild interest of a man inspecting a stranger's engine. "It is not deep. Get the kit."

The kit was a green box on the wall, and the kit was, of course, half empty. There was gauze. There was no tape. There was a bottle of something that had dried to a white crust around the cap. Chen came up the stairs, saw the blood, and did four things in order without being asked: she put on gloves, she pressed the gauze, she said "keep it up", and she told me to go and find a person with a key.

We went to the clinic in the next building, the three of us, at four in the afternoon. The doctor was a young woman with a cold and a queue. She cleaned the cut, she said that it would have been worse if the glass had been dirty, and then she asked a question that I have since thought about more than the cut: "Was the glass clean before it broke?"

"Yes," Zhang Ye said. "Washed three times."

"Good," she said. "Then the infection risk is low." She wrote on a pad, and gave him a strip of tablets and a small tube of ointment. "Antibiotic. The dosage is on the box.

Do not skip the last two days because it feels better, that is exactly how a wound gets a lovely second chance."

On the wall of that clinic, between a hand-washing chart and a poster about blood donation, there was another poster, this one about mental health. It said that students who could not sleep, or who felt that everything had gone grey, could make an appointment with a therapist, and that an appointment cost nothing and appeared on no transcript. I read it while Zhang Ye was being bandaged and thought that it was, in its quiet way, a more impressive document than the one about the chancellor.

We missed lunch. Zhou Min found us in the corridor at half past two with a bag of steamed buns and a face like a verdict.

"One of you will tell me what happened," she said. "The other one will be quiet. Choose."

I told her. She listened without interrupting, which is a rarer skill than it sounds, and then she sat down on the step, opened the bag, and gave us cold buns, and said, "Sit. Eat. This is not a punishment, it is a lesson that happens to be edible."

The buns had pork in them, and a little cabbage, and she had also bought a bottle of milk, and tea from the machine in the lobby that anyone could use for a coin. While we ate, she explained what our report would have to say about the accident, and the explanation turned into a lecture about money, which is the subject that nobody in a laboratory ever admits is the real subject.

"This beaker," she said, "costs four yuan. The reagent in it cost eighty. The hour of instrument time that follows costs two hundred and forty. Our budget for the whole term is fixed, and it is not generous, and it is not a secret. Every time somebody breaks something, somebody else does not do an experiment. That is the expenditure. That is the real cost, and it does not appear in your report."

She told us that a master's student in her year had lived for a month on loans and instant noodles because a payment from the department had been late, and that this was not unusual. She said that affordability was a part of experimental design, that the cheapest method that works is usually the best one, and that people who had never been poor believed that the most expensive method was also the most reliable. "It is not," she said. "It is only the most expensive."

Chen asked, cautiously, whether the department would pay for the broken beaker.

"No," Zhou Min said. "You will. It will come out of the fee for the equipment, at the end of the term, and it will be twelve yuan, and you will be angry about twelve yuan for about four minutes, and then you will be careful with glassware for the rest of your life. That is the best twelve yuan the department will ever spend."

She stood up and dusted the step with her hand. "One more thing, and I will say it once because it is not a pleasant subject. If any of you is pregnant, or thinks she might be, you tell me before the first lab of the term, and I will change your assignment. Not because the work is forbidden. Because there is a list of the things that we know, and a longer list of the things that we do not, and in the space between those two lists nobody should have to guess."

Nobody said anything. She picked up the bag, and the awkward silence followed her down the corridor like a dog.

That night, in the dormitory, Zhang Ye lay on his back with his bandaged hand on his chest and told me about the place he came from, which he had never done before, and which he did in the flat voice he used for lists.

His village had three hundred people, and more of them over sixty than under twelve, because the ones in the middle had gone to Hangzhou, to Suzhou, to a factory in Guangdong, and sent money back, and came home once a year for four days. His father's brother kept pigs, six of them, and sold them to a butcher in the next town at a price that had not changed, in real terms, since 2015. There were sheep on the hill behind the school, and a garden with a wall, and a plot of crops that his grandmother still worked by hand because the machine could not get up the slope. There was a lake, small, with weed at the edges, which was where he had learned to swim, and a mountain that everybody called a mountain although it was barely high enough to be a hill.

"Everybody emigrated," he said, and then, after a pause, "That is the wrong word. Emigration is for countries. We just left." He turned his head. "You know what the school has? Nine children. They will close it. The building will be a store."

I asked whether he would go back.

"No," he said, without hesitation, and then, more slowly, "Not to live. Maybe to fix the well." He said it as if fixing a well were a moral debt he had agreed to and never scheduled. Then he yawned, and said, "Sleep. You have a report to write."

On Friday afternoon Zhou Min handed me a clipboard and told me to inventory the cupboard at the end of the bench, which is the least glamorous job in a laboratory and the one that taught me most about how one is actually run.

The cupboard held a pad of filter paper, a rubber mat for the sink, a plug with a cracked handle, a spare sensor for the optical bench, a box of insulators, a pair of laser goggles, a tin of furniture polish, a mop, a stove that nobody had used since the previous spring, three pots, a bottle of vinegar for descaling the kettle, a bag of sugar that somebody had hidden from the ants, and, at the back, behind everything, a parcel that had been delivered in 2021 and never opened. I wrote all of it down in two columns, and then I counted the clamps twice, because Zhou Min had told me that a count is not a

count until it has been done twice by two people.

She read my list and made one correction, and the correction was the whole lesson. "You have written three pots," she said. "There are two pots and one beaker that somebody has been using as a pot. This matters, because when the equipment is audited at the end of the term, the beaker will be missing from the glassware count and I will spend an afternoon explaining that it is in a cupboard, being a pot."

"But it works as a pot."

"It works as a pot," she agreed, "and it is not a pot. A laboratory is a place where things are what they are called. The moment a beaker can be a pot, the inventory is a story, and a story is not a record." She handed the clipboard back. "Write beaker, in brackets. Then go and eat something. You look like a man who has been counting all afternoon."

The canteen at half past five was loud in the way that a canteen is loud, which is to say that nobody was shouting and everybody was talking. The menu was on a board above the counter, in the cheerful, slightly threatening language of institutional food: beef with pepper, pork with vinegar, fish soup, shrimp dumplings, a vegetable that had been chopped so small it was impossible to identify, rice, noodles, and a pot of tea that was free if you asked for it.

Zhang Ye ate like a man refuelling: a bowl of rice, a bowl of soup, a plate of beef, and then, because he was still hungry, another bowl of rice. I had the pork and the vegetable and a glass of milk, because I had read somewhere that a young man needs calcium, and because the milk was cheaper than the yoghurt.

Chen arrived with a tray and a book, sat down, and did not open the book. She said, "My mother asks whether I am eating properly. I say yes. She asks what I ate. I say rice. She says, what else. I say rice." She looked at her tray. "It is not a lie. It is an incomplete report."

Zhou Min joined us with a bowl of noodles and immediately began to talk about nutrition, which surprised me, because I had assumed that a person who worked with chemicals all day would talk about chemicals all evening. "Half of what you are eating," she said, "is carbohydrate, and the other half is fat, and there is almost no protein in either, and you will be hungry again at nine, and you will eat a packet of something with a great deal of sugar in it, and then you will wonder why you cannot concentrate in the morning. This is not a moral failure. It is arithmetic."

She said that the previous winter she had lived for six weeks on noodles and coffee, that her hair had begun to fall out, and that the doctor at the clinic had asked her one question, which was what she ate, and she had been unable to answer it in a way that sounded like a life. "I could describe a synthesis in four sentences," she said. "I could

not describe my own dinner. There is a word for that, and the word is not diligent."

After dinner I went to the gym, because Zhang Ye had decided in September that I was going to learn badminton, and had refused to consider any evidence to the contrary.

The gym at seven was half full: four courts, a trainer with a whistle and no apparent sense of humour, and a cluster of spectators on the bench by the door who were mostly waiting for somebody else. For the first three weeks I was the worst player in the room by a distance that was almost architectural. My racket arrived late at every ball by the width of a hand, my clears went into the lights, and my serve was a rumour.

The trainer, whose name was Gao, had a method that consisted of saying the same sentence in a slightly different order until your body gave in. "You are not hitting the ball," he said. "You are meeting it. The racket does not swing. The racket arrives."

By November I could keep a rally going for eight shots, which is not a triumph, but on the night of the class contest, Materials against Mechanical, a fixture that had been played, according to Gao, since before either of us was born, I won a point against a boy who had played at school, and the spectators on the bench made a noise, and I felt the specific, slightly shameful pleasure of being watched while doing something adequately.

We lost the contest, four matches to one. The triumphant photograph on the department's page showed the Mechanical team holding a plastic cup, and Zhang Ye in the background, looking at the wrong camera. I was not in the photograph, which I minded more than I admitted, and I have kept the link to that page for a reason I cannot defend.

On Saturday she lent me a book. A paperback with a broken spine, a man in a coat on the cover, and a title that promised a second verdict: THE SECOND VERDICT.

"I read these," she said, "because they are the only books in which the world is guaranteed to make sense by page three hundred. It is a vice. Do not tell Zhou Min. She would call it a failure of taste, and she would be right."

The plot was simple in the way that such plots are simple. A murder in a warehouse. A suspect with a motive. A rifle in a locker, a sword on a wall as a decoration and then as evidence, a witness who was lying about the time, an inspector who was not lying about anything, and a jury that arrives, at the end, at the verdict the reader has been waiting for since page forty. There was a hostage, briefly. There was a man on parole who had been paid to keep quiet. There was a smuggling operation that the newspapers called trafficking and the characters called business, and a small piracy of insurance money that nobody was punished for, and an outlaw on a motorbike who existed mostly to be shot at.

I read it in two nights and enjoyed it more than I expected, and when I gave it back she asked what I thought, and I said that it was good.

"It is not good," she said. "It is satisfying. Those are different words, and you should keep them apart for the rest of your life."

Then she told me why she read them, and the reason was not the one I expected. "In a crime novel there is always somebody whose job it is to find out. An inspector. A detective. Somebody with a notebook and the authority to ask. The world of the book is arranged so that the truth is discoverable by one persistent person. I read that because I work in a laboratory, and in a laboratory there is no inspector. Nobody is coming. If you distort a measurement, no witness stands up. If you fabricate a point on a graph, no inspector arrives with a file. You are the inspector, the witness and the accused, and the verdict is delivered by a machine that does not care what you intended."

"That sounds like a reason to be careful," I said.

"It is a reason to be frightened," she said. "Careful is a habit. Fright is a motive. The most dangerous person in a laboratory is not the lazy one, and it is not the stupid one, and it is certainly not the one who is bad at arithmetic. It is the one who needs the result to be true. He is not a fraud and he is not a hypocrite, because a hypocrite knows. He is simply a person whose desire has quietly taken over the judgement, and the horror of it is that the first fabrication is almost always small, and plausible, and made in good faith, at eleven at night, by somebody who has been awake for nineteen hours and wants to go to bed."

I asked whether she had ever done it.

She looked at me for a moment, and then she said, "I have never invented a number. I have overstated a trend, and I have omitted a point that did not fit, and I have described a result as promising when the honest word was early. I was indignant when my supervisor called it misleading, and I despised him for a week, and he was right. That is the whole of my confession. You may keep it."

Her family came from the coast, from a town where the wharf was the centre of everything and the sea was a colleague rather than a view. Her grandfather had been a sailor on a naval supply ship; he had crossed the latitude line twice and had never learned to swim, which she said explained a great deal about him. Her mother had a shop near the harbour that sold shrimp, dried fish and a little jewellery made of pearl, and in the summer the shop smelled of salt and of the hot plastic of the display cases. She could tell a shark from a dolphin at a hundred metres, she said, not because she was brave but because she had been taught, at six, by a man who thought that a child who could not read the water had no business being on it.

"Everyone from my town has the same nostalgia," she said. "A coastal town is very good at making people sentimental about it. The wharf in my memory is beautiful. The wharf in fact is a maritime loading area with a pungent drain and a secular attitude to

rubbish. Both are true. I prefer the second one. It is more useful."

I asked her, once, whether she wrote anything herself, and she said no, and then, a week later, she left her notebook open on the bench on purpose, which is a thing that people do when they want to be read without asking.

There was one page in it that was not equations and not data. Four lines, in pencil, with a date in the margin and no title: something about a lily in a lunar light, and a dove that would not come down, and a pearl of water on a rail. It was not a good poem. It was the poem of a person who had been lyrical once, in the way that everybody is lyrical at seventeen, and had then spent six years being precise instead.

I never mentioned it. She never mentioned it. It is the only secret I have ever been given by somebody who did not say the word.

The second version of my report came back on Monday, and the criticism this time was worse, because it was about my writing and not my technique.

"You have replaced a fabrication with an overstatement," she said. "That is progress of a kind. The new draft says that the yield was comparable to the literature. Comparable to which paper? At what scale? Under what assumption? You have used a word that sounds like a measurement and is actually a supposition. This is not a deficiency of vocabulary. It is a deficiency of honesty about what you know."

She read on, in silence, and her pen made two marks. "Here you write that the loss was probably due to evaporation. Probably. Your own notebook says you left the beaker for ninety seconds and that the hotplate was at a setting you did not record. The word you want is not probably. The word you want is because. You know the cause. You are using probably to make yourself sound objective, and it makes you sound like a coward with a thesis."

I said, "It feels arrogant to say I know."

"Then be precise about the limits instead of vague about the cause. Precisely this: the loss is consistent with evaporation between minute four and minute six, and I cannot separate it from the spillage on the bench. That is a sentence a reader can use. Yours is a sentence a reader has to interpret, and the moment a reader has to interpret your data, the data belongs to the reader and not to you."

She put the pen down. "You have a conceit," she said, "and it is not the one you think. It is not that you believe you are clever. It is that you believe your reader is stupid, and that if you leave a gap he will not notice, and that if you sound solemn enough he will not ask. Vanity of that kind is very common in second-year reports. It is also the reason half the literature is unreadable."

Then, because she was who she was, she took my notebook, turned to the page where I

had written a list of long words I kept misspelling, and laughed out loud, once, with real delight.

"What is this?"

"It is how I learn words. I break them."

She read the page. At the top I had written abbreviation, and under it the pieces I had invented to hold it together; then com and mittee, which I had written as two lines and then, in a fit of ambition, as one word, and got wrong; then accom and modation, which I had learned from a sign in a station and had never once spelled correctly under pressure; then the tail of comment, which is the piece that makes a noun out of a verb, and the tail of symmetry, and the tail of grammatical, and the ming of swimming, and the mer of hammer, and the long middle of telecommunications, which I had copied out of a poster in the metro because I liked the shape of it.

"This is not a method," she said. "This is a museum." But she did not give the notebook back at once. She turned it over, and looked at the page again, and said, more quietly, "It is not a bad museum. The pieces are real. You have simply arranged them as if the English language were a set of ruins and you were the archaeologist."

"How would you arrange them?"

"By sound, and by what they do." She took the pen and wrote, under my ruins, four words: com, meaning together; mittee, a body of people sent; modation, a fitting; metry, a measuring. "Now you have four ideas instead of four hundred letters. The letters are the accident. The ideas are the word."

That page, with her four words under my eight, is the only thing from that term that I still carry in my bag.

The redo happened on a Wednesday, at night, and it happened because Zhou Min put her name on the duty roster for the late shift and then wrote, in the group chat, one line that said: bench three is free after nine, if anybody wants to be serious.

I went. The building at night is a different building. The corridor lights are on a timer that turns half of them off at ten, so the fourth floor becomes a series of bright islands with darkness between them, and the fume hoods, which are loud in the day, become the loudest thing in the world. There was a solitary glare from the balance room and nothing else. The staircase smelled of rust and of the wet stony dust that comes out of old steel when the heating first comes on in October.

The duty sheet was on the door, in a plastic sleeve, with the hourly blocks filled in by hand: nine to ten, Zhou Min; ten to eleven, a name I did not know, from another course; eleven to twelve, empty. There is a standby number at the bottom for emergencies, and a rule printed above it in red: no work after midnight, no exceptions, the gate locks

at twelve and the watchman does not negotiate.

Zhou Min stayed until half past ten. She did not help. She sat at the end of the bench with her own reading and looked up whenever the beaker made a noise, and at half past ten she stood, put on her coat, and said the only thing she said all evening: "You know what the film looks like now. Do not go and look at your phone."

I did not look at my phone. I stood in front of the hotplate for eleven minutes with my hands behind my back, and I watched the surface, and at some point between the ninth and the tenth minute a skin began to form across it, thin, wrinkled, moving, and I turned the dial down and the skin held, and I remember saying, out loud, to nobody, "There you are."

The second yield was seventy-eight per cent. Not ninety. Not a triumph. But the number had a cause attached to it, and the cause had a time, and the time was in the notebook in ink, and when I calculated it I did not have to look away from the page.

Then I had to get out of the building before twelve, which I did with four minutes to spare, and the watchman let me through the side gate with a nod, and I walked back through a campus that was doing the thing campuses do at midnight, which is to look like a stage after the play.

It was frosty. The grass had gone white at the edges, and the puddle by the bicycle shed had a skin of ice that broke under my shoe, and there was a lunar glare on the concrete that made everything look stony and severe. Somewhere behind the library a dog was barking at nothing, steadily, like a man practising. The supermarket by the north gate was still open, and its vendor was stacking grapes and pears into a pyramid while a radio behind the counter played something sentimental, and I bought a carton of milk and a packet of biscuits and ate them standing up, in the cold, like a soldier.

I was hungry in a way that had nothing to do with food. I have never been able to describe that feeling in English, and I have tried, in three essays and one application, and the closest I have come is this: it is the quiet that arrives after you have been afraid of something for a week and the thing has finally happened and been survivable.

The report went in on Friday, and the mark came back three weeks later, and the mark was eighty-two, and I looked at the number for a while and felt nothing in particular, which surprised me. The comment at the bottom was in Zhou Min's handwriting and said: your conclusion is now shorter than your data, which is the correct direction.

Shen Lu read it too, without being asked, and gave it back with one sentence: "Better. Now do it eleven more times and you will have a habit."

By December the top drawer of my desk had become an archive of paper, because I could not throw anything away and because every institution I had met that term had given me a document to prove that it had met me.

An electricity bill for a flat I had never entered, forwarded to my mailbox by mistake, forty-three yuan, with a note on the back in a stranger's handwriting about a repayment plan. A payroll slip from the canteen, where I worked four hours a week for eleven yuan an hour and was overwhelmingly grateful for it. A retail receipt from the supermarket for goods worth twenty-three yuan, which I kept for no reason except that they were the first things I ever bought with money I had earned. A loan agreement, in my father's handwriting, for eight thousand yuan, which my parents had taken out in the summer and had not mentioned until November, and which I found in the drawer because my mother had photographed it and sent it to me by message with four words at the bottom: we are prospering, do not worry.

There is a word for what those four papers did to me when I put them side by side, and the word is not poverty. It is priority. I had spent eleven weeks worrying about a nominal difference of two per cent in a yield, and the economic underlying of my whole term was eleven yuan an hour and a debt I had not been told about. Necessarily, my appraisal of my own diligence changed. A supplementary sheet from the department, listing the surplus glassware available for student projects, hung above the desk for a month before I understood that the sheet was also a proposal: take it, use it, buy nothing, because a scarce budget is a corrective instrument and nobody tells you that in September.

The newspaper cutting had been in the drawer since October, because a boy in my class had taped it to the lab door and I had taken it down before the cleaner could.

It was about a governor who had come to the province, a consulate that had opened in the old castle on the hill, and an envoy who had signed a multilateral treaty on logistical cooperation with three confederations of exportation enterprises. The same page carried a report on the reformation of a pension system, a rebellion in a northwestern province, an uprising that the paper called a dispute, the persecution of a refugee family that the official account described as a procedure to deport the family, and a socialist party that had split into two nationalist factions over a referendum nobody had wanted. There was a photograph of a procession in the southeast, and under it, in small type, the sentence that a martyr is a person who becomes convenient retrospectively.

I read that page three times and understood about a fifth of it, and what I understood was this: the vocabulary of a newspaper is the vocabulary of a bureaucratic empire talking to itself, and a first-year student is not the intended reader of any of it.

The clinic form was a single sheet with six boxes. Box one: feverish for three days, no drug taken, no medicine prescribed. Box two: injured left hand, glass, benign, clean. Box three, in the dentist's hand: gums, early degenerate change, susceptible to acid,

reduce the sugared mint and the smoke. Box four: a psychiatric referral, offered, declined. Box five: brain, no comment. Box six, the only box filled in with any enthusiasm: mortality, it said, in young males, with a predominance of accident; and then, in brackets, the prevalence of stupid bravery between the ages of eighteen and twenty-two.

At the bottom of the drawer there was a note in my own handwriting, dated the night of the redo, which said, in a dramatic tone that I now find idiotic:

"Determine the outcome before the procedure. Disclose the faulty step in the record. Do not delete the ugly numbers. The pattern is the discovery, and the discovery is the only thing that diminishes the futility."

I had written it at one in the morning, with a blot of ink at the corner, because a phrase of Shen Lu's had been going round my head all evening: the truth does not need you to be courageous; it needs you to be punctual.

In the last week of term she came to 402 to say goodbye to the room rather than to us, and because she was leaving she allowed herself a speech, which she delivered standing up, with her coat already on.

"You will hear that a laboratory is a community," she said. "It is not. It is a set of people who have agreed to be interruptible. That is the whole of the ethics. A community can afford a quarrel; a laboratory cannot afford a silence. When you become a senior, and someone asks you a stupid question at four in the afternoon, remember this: the question is not an interruption of your work. It is the work. The rest is just you and a machine in a room."

"Everybody wants a mentor," she went on, "and nobody wants to be one. I am not going to be yours. I am going to be a telephone number that you will use twice, and then you will use it less, and that is not a tragedy, that is a curriculum."

She said that the most useful thing she had learned as an undergraduate was not a formula but a habit of asking who paid for the question. "Follow the money and you will find the assumption. A grant has a shape, and the shape is in the paper, and if you cannot see the shape you will spend three years extending somebody's footnote."

"Also," she said, "learn to be boring. The people who last in this trade are not the brilliant ones and they are not the passionate ones. They are the ones who can be boring on a Tuesday. Passion is a mood. Boredom is a structure. I have watched three brilliant people leave this building because they could not do the fifth repetition, and I have watched a woman with no apparent talent become the best analyst in the province because she did the fifth repetition eleven hundred times and wrote it down each time."

Then she said something that I did not understand until the following year, and that I now think was the most important sentence anybody said to me in that building.

"The moment you start to enjoy being right," she said, "you stop being able to see what is in front of you. Enjoyment is not a sin. It is a distortion. It bends the light. That is why the good ones are so often unpleasant and the pleasant ones are so often wrong: not because virtue is punished, but because the pleasure of being right is a very strong lens and most people never learn to take it off."

She shook our hands, which was formal and slightly absurd, and to Chen she said, "You will be fine. You do not need the room to like you." And to Zhang Ye she said, "You will be very good and you will be very difficult," and he nodded as though she had given him a document.

The last laboratory session of the term was not an experiment either. It was the return of the equipment, and it took four hours, and it taught me more about institutions than the whole of the safety briefing.

The rule was simple: everything that came out of the cupboard went back into the cupboard, and the count on the sheet had to match the count on the shelf. In practice this meant that thirty-one people spent an afternoon looking for a clamp. We found it in a drawer under a pile of litter, in a room on the third floor, in a box marked with a word that had been spelled wrongly by three generations of students. The technician received it with the expression of a woman who has been waiting since 2019, which she had.

While we worked, the radio in the corner played the noon news, and the news was the usual mixture: a factory fire, a football result, a currency, an exhibition. Nobody listened. But the radio was on, and it stayed on, and I have thought since that a laboratory is one of the few places left where a broadcast is simply a background noise and not a summons.

At four o'clock Zhou Min signed the sheet, and wrote at the bottom, in her square hand: inventory complete, three items replaced, one beaker returned after an unauthorised career as a pot.

Then she told us to go home.

I walked out through the gate at ten past four in a light that had already gone the colour of weak tea, and the campus was full of people leaving: suitcases, airport buses, the first Christmas decorations in the supermarket window, a queue at the station for the slow train. I had a ticket for the six forty, a seat by the window, and a rucksack with four notebooks in it that I had not read since October.

I did not feel the thing I had expected to feel, which was triumph. I felt instead a sort of clean tiredness, the kind that comes after a day of carrying something awkward, and the beginnings of an idea that would take me the whole of the spring to work out.

The idea was this: that the experiment had not been about the copper, or the balance,

or even about being careful. It had been about the difference between a person who wants to be right and a person who wants to know. The first one writes a report. The second one builds a shelf. The shelf is smaller, uglier, and heavier, and it is the only one of the two that anybody can stand on.

The spring term began with a different laboratory, a different bench, and the same building, and I want to record four small things from it, because they are the ones that stayed.

The first was that I became, briefly and badly, a person who trained other people. In March Zhou Min put me in charge of two first-years for the titration practical, which is the cheapest form of flattery that a department can offer, and I discovered within an hour that I could not explain anything I thought I understood. I could do the procedure; I could not translate it. Every sentence I began came out in the passive voice, as if the work had been done by nobody, and one of the first-years, a girl with a very quick mind and no patience, said, "But who holds the flask?" and I realised that I had been describing a ritual and calling it a method.

The second was that the department bought a new instrument and everybody's behaviour changed for a fortnight. It was not a large machine. It sat on a bench, it hummed, and it produced a number that was, as far as anybody could tell, the same number the old machine had produced. But it was new, and so for two weeks the room was full of people who had opinions about it. There is a lesson in that about how much of what we call a scientific culture is actually a taste for novelty, and I do not have it fully, but I have the shape of it.

The third was a telephone call from my mother, at noon on a Tuesday, in which she told me that my grandmother had been ill and was now better, and that I was not to travel home in the middle of term, and that the money for the train was, in her words, better spent on books. I said yes. I said yes four times. Then I sat on the step outside the library for twenty minutes with the phone in my hand and the sun on my knees, and I understood something about the difference between being told a thing and being informed of it.

The fourth was the day the whole class went to a chemical plant on the coast, which is a story for another chapter, except for one hour of it: the hour in the control room, where a woman of about fifty sat in front of four screens with a cup of tea and explained that her job was to notice things. Not to fix them. To notice them. "The system will tell me when it has already gone wrong," she said. "My job is the twenty minutes before that."

I have thought about that sentence more than about any lecture I attended that year, and I have come to believe that it is the whole of the education I was given in that building, expressed by a stranger with a cup of tea.

Now the objects. There was a ruler that lived in the second drawer and was used for everything except measuring. There was a roll of tape, a pad of labels, a jack for a plug that no longer existed, a token for the coffee machine in the lobby that had been out of order since the previous winter, and a ton of old glassware in the cupboard that nobody could bring themselves to throw away. There was a wool blanket that lived on the back of a chair for the whole of January because the heating in 402 was a rumour until the middle of February.

There was a boy who could clean a flask in forty seconds, and a girl who could not clean one in four minutes, and a technician who could tell the difference from the door. There was a poster about hand-washing that had been printed in 2011 and never updated, and a strip of tape on the floor marking the line that nobody was supposed to cross while the centrifuge was running, and a leader who was never elected and was always obeyed, and that was Chen.

I learned to grip a burette the way you grip a pen that is not yours, to wipe the bench in one direction, to sweep the floor without raising the dust, to pat a filter paper flat with two fingers, to rub a mark off glass with the side of my thumb, to tilt a flask and twist my wrist at the same time, to leap out of the way of a trolley, and to keep my hands out of my pockets, which sounds like nothing and is in fact the beginning of a professional personality.

I also learned what a bad week does to a good habit. In April there was a week of rain, a flood in the basement, a rehearsal for a competition I did not win, a quarrel with Zhang Ye about something so small that I have forgotten it, and a night when I dozed off at the bench at nine and woke at half past ten with my face on my notebook and a page of numbers stuck to my cheek. That week I broke a thermometer, lost a spatula, and wrote a report in which I used the word approximately four times in one paragraph, which is three times too many.

The rain made the corridor smell of rust and of the wet stone of the staircase, and the whole building became, for eight days, a place where the electricity flickered at four in the afternoon and everyone worked by the window. It was gloomy in a way that I found, at the time, romantic, and in a way that I now find simply expensive.

But the term had a shape, and the shape was upwards, and by June I could do a titration without thinking about my hands and could think, instead, about the question of what the numbers were for.

Here is the thing I would tell the version of myself who stood at that door at seven forty with a calculator in his hand. The vocabulary of a laboratory is not a decoration and it is not a password. It is the smallest set of distinctions that the work cannot survive without, and every one of them was paid for by somebody who got it wrong first. A beaker is not a pot because somebody lost an afternoon. A clamp is not a hand because

somebody was burned. A record is not a memory because somebody's memory was, in good faith, mistaken, and the mistake cost three weeks and a grant.

There is no vocabulary without a history of error behind it. That is why the words are so specific, and why the specific words are the ones worth learning, and why a person who uses them loosely is not being elegant but is quietly borrowing authority that he has not earned.

She gave me a folder on the last day, because she was leaving too, and because, she said, the room would not explain itself to the next person. I read it on the train, with the rucksack between my knees, and I have read it perhaps nine times since.

The first page was headed, in her square hand, FOR WHOEVER COMES NEXT.

"The bench. Two taps vibrate at low pressure. The wattage printed on the hotplate is deceptive. The electricity fluctuates between four and five in the afternoon, when the whole building runs its kettles, so the last hour of any heating step is a gamble. The nitrogen line has a periodic syndrome of its own, and the maintenance levy does not cover it. Lubricate the valve in March. Do not tidy the shelf: the disorder is a relic of three cohorts and it is profitable in exactly one respect, which is that everybody can find things. Remove nothing without a replacement, and write the replacement in the book, otherwise the next person will spend a morning in inquiry."

"The people. You will get three kinds, and the proportions change every year. The zealous youngster who wants to publish in the first term and will be exhausted by March; assist him, he is useful and fragile, and he is not being selfish, he is being nineteen. The pessimist who has already decided that nothing here is feasible and who is often the most reputable researcher in the room; listen to him however hurtful he is, because his cynicism is a form of memory. And the planner, who is preoccupied with placement and recruitment and who will do the laborious work without being asked, because she has foreseen that punctuality is the one quality that cannot be imitated."

"Do not flatter the students and do not humiliate them. A sarcastic remark is a cheap inducement: it produces conformity and it kills originality. If a person is ignorant, enlighten him. If he is argumentative, let him argue, and then quote his own data back at him. If he is impulsive, intervene early. If he is obsessive, mediate between him and the clock. Repression is easy and it is useless. Restraint is laborious and it is the cornerstone."

"The money. A levy of forty yuan a head constitutes our entire consumables budget for the year, which is not a joke and not a complaint, it is the figure. Nobody will explain the economics to you and you should not expect them to, but you should know that a department is not a company: it does not need to be profitable, it needs to be sustainable, and the difference between those two words is the whole of the argument that

is going on above your head in every meeting you are not invited to. When an investor visits, which happens twice a year, be polite, be brief, and do not promise a result. A promise made in a corridor becomes a target in a report."

"The failures. You will have a run that aborts at minute nine for no reason you can find, and you will want to hide it. Do not. The room has seen worse and the room does not gossip. You will feel agony when the sample is lost, and anguish when it is your own fault, and dread on the Sunday night before a Monday that has a report in it, and panic when the oven is at four hundred and the door will not latch. All of that is normal, and none of it is an excuse. Write the number down. The number is the only thing that will still be true on Thursday."

"One warning, and it is the only one I will put in capitals. You will meet somebody who wants you to fail, and it will be somebody you admire. It may be jealousy, it may be a bad year, it may be that he has been in that building for eleven years and the credit for his best idea went to a man in another province. You are not obliged to save him and you are not permitted to become him. If he is fraudulent, disclose it, once, in writing, to the person whose responsibility it is. If he is merely cruel, survive him. He will retire, or he will be moved, or he will find somebody else to be cruel to, and the corridor will be ordinary again."

"The building. It is historic in the way that a landmark is historic, which is to say that the plumbing is legendary and the renovation has been announced four times. The erection of the east wing was delayed by a decade; the scaffolding has become scenery; the canteen has been a workshop, a salon for the art students and, for one memorable spring, a cinema with three rows of chairs. From the fourth-floor window you can see the meadow behind the boiler house, and in June the seedlings of the botany students are visible in neat triangles of green, and a cricket in the drain will keep you company at midnight. There was a snake in the stairwell in 2019, which is now a story, and a bird in the fume hood in 2022, which is now a recipe for keeping the window shut."

"Read outside your subject or you will become a technician with a degree. A little botany, a little biochemistry, a little geology: not to use, but to keep the shape of the whole in your head. A gene is not a destiny and a species is not a rank, and the person who taught me that was an elderly engineer who had spent his life on spacecraft and could still tell you the name of every tree on the road. Genius is rarer than the department's publicity implies; curiosity is commoner and worth more; and fame, when it comes, arrives to people who have already stopped needing it."

"Learn the difference between the denotation of a word and its atmosphere. A dismissal and a judgement are not the same act, although they use the same tone of voice. A revision is not a punishment and a correction is not an attack. Accuracy is not a form of aggression, although it is often received as one, and a person who cannot survive

being corrected will spend his career being vaguely wrong in a pleasant way, which is the most expensive thing in any laboratory."

"Finally. Take the vacation, the whole of it, and go somewhere with no instruments in it. Sleep for a week. Read an essayist. Learn to skate or to ski or to do nothing at all, and do it leisurely, without a goal, because a person who can only work is not a person who works well, he is a person who is afraid, and afraid people make the expensive mistakes. I have watched a good scientist lose two years to that fear, and I have been that scientist for one of the years. There is no prize for the person who refused to rest. There is only a slightly earlier version of the same exhaustion, arriving in a quarter when you needed it least, and a slow, sad, entirely foreseeable decline in the quality of your judgement."

"Write back once a year. Not to me; to the room. Two pages. What worked, what did not, what you would forbid, what you would change. It is not remembrance and it is not a recitation of your achievements. It is the only form of gratitude that a laboratory understands."

That was the folder. There was one more sheet, loose at the back, not part of it, and it said, in pencil, in a different hand, which I recognised:

"The rules are not a cage. They are the rails, and the rails are what make the speed possible. I have seen a very smart man lose a hand to the belief that he was careful. Take the long way round. There is no version of this work in which the haste is worth it."

Two pages of the folder were comments on my notebook, written in November, which she had never given me.

"Page four. You perceive a trend where there is a fluctuation. Three points are not a trend; three points are three points. Draw them, look at them, and resist the urge to draw a line through them, because the line is where the wish enters the data.

Page nine. The derivation is elegant and the assumption is unstated, which makes the elegance a form of concealment. Every derivation that begins with a simplifying assumption must begin with the sentence that names it, or the reader will supply his own and it will not be yours.

Page eleven. You describe the colour of the solution for four lines and the temperature for none. This is not a failure of observation, it is a failure of proportion, and proportion is the whole of a descriptive style. A report is not a photograph. It is a map, and a map that shows every tree is a map nobody can use.

Page fourteen. Here you write that the result was disappointing. Delete it. Disappointment is a fact about you, not about the result. The reader wants the number, the uncertainty and the condition; he does not want your emotional weather.

Page nineteen. You overhear a conversation in the corridor, you repeat it in the notebook, and you add a conclusion. This is how a rumour becomes a finding. If you must record a conversation, record it as a conversation and mark it as such. Do not let a sentence that somebody said at a bus stop acquire, by proximity, the authority of a measurement.

Page twenty-two. You have used the word significant four times and have not once said what it is significant relative to. The word has a technical meaning and a vulgar one, and the vulgar one is the enemy of the technical one, because it is more fun to write. Ban it from your reports until you can defend it.

Page twenty-six. Good. This is the first page in the book where you say what you do not know, and it is the only page I would show to another person."

The last page of her comments was not about the notebook at all.

"You will meet people who are smarter than you and they will be lazy, and you will meet people who are duller than you and they will be relentless, and the relentless ones will win, and it will annoy you for about a decade. You will be tempted to the radical solution, which is to change your subject, your city, or your life, whenever the ordinary solution is to sleep properly and read the manual twice. The radical solution is occasionally correct and always expensive, and the people who take it most often are the ones who are least able to afford it. The tame solution is usually right: fewer variables, more repetitions, a smaller question, an earlier night.

You will also discover that the department is not a democracy and not a jungle, whatever your classmates tell you. It is a subordination of the impatient to the patient, disguised as a hierarchy of titles. The professor is not the prince of anything; he is a man with a tenure clock and a grant application and a committee that meets on Fridays, and if you can help him with any of those three things, you will be given a linkage to the realm of people who actually decide.

And you will be given, if you are fortunate, one occasion on which you can be useful at a moment when it matters, and nobody will herald it, and there will be no spotlight and no applause and no victory parade, and it will be the only rewarding hour of your year. Take it. Do not preach about it afterwards."

The letter from home that month was two pages from my mother, in her slow careful handwriting, and it contained four pieces of news, in ascending order of importance to her.

The subway had reached the town next to ours, which meant that my grandchildren, if I ever had any, would not know the poverty of a two-hour bus. My cousin had been taken on as a waitress at a place near the cafe in the new plaza, and was earning, in her words, enough to be independent of a man, which my mother wrote down without comment. The

kindergarten where my aunt worked had been told to convert two classrooms into a lounge for parents, and the headmaster had made a speech about service and about sustainability and had been photographed for the local paper. And my father had had a check-up, and the sight in his left eye was failing, slowly, and he had told nobody for a year, and my mother had found out from a prescription in the pocket of his coat.

I read that page three times, and then I sat on the edge of the bed with my shoes on and felt the whole framework of my term, the magnitude of my worries, adjust itself by a notch.

My father came in at ten and sat on the other end of the bed and said that it was nothing, that everybody's eyes go, that he had a pilot's attitude to his own body, by which he meant that he ignored the instruments until the light came on. He said it in a tone that was identical to the tone I used for my own excuses, and I recognised it, and I said nothing, which was, I think, the first adult thing I did that year.

We watched the end of a football match. He fell asleep in the chair during the second half, and I put a blanket over him and turned the volume down, and the thunderous thing in my chest, which had been raging since October, went quiet.

That was the night I understood that the ritual of the laboratory, which had seemed to me either boring or sacred, was neither. It was a way of being careful in one room so that you could be careless in the others. The ban on guessing, the stipulated report time, the duty sheet, the segregation of waste, the prompt answer to a question, the readily available guidance of somebody who had done it before: none of it was about chemistry. It was about a habit of attending, which is a portable thing, and which is why a person trained in a laboratory can adapt to an insurance office, a kitchen, or a hospital, and why a person who has only learned the material and not the habit can do none of those things and will tell you, with sorrowful pride, that he was never trained for anything else.

The spice of that insight has not worn off. It is a universal claim and I am aware of the risk in it, and I still hold it, regardless of the objections of persons who have worked in more than one laboratory and have told me that I am romantic about a building.

But I am not romantic about the building. I am romantic about the flexibility that a method gives you, the way it transforms a problem into a sequence, and I am rooted, if I am rooted anywhere, in the plain observation that a person who can undo a mistake, trace it to its cause and record it without self-pity is a person who can be trusted with almost anything.

That is not a small thing. It is, in the end, the entire thing.

I put the folder in the drawer with the rest of it, and the train went through a tunnel and the window went black, and then the light came back and the whole valley was

there, purple at the edges and gold in the middle, and I thought: next year I will be the one who is asked, and I will have nothing to say except this, and it will have to do.

I still keep the notebook. On the last page of it, in June, I copied out the syllables again, in the order Shen Lu had taught them, and beside each one I wrote what it does rather than what it looks like: the com that means together, the mittee that means a body of people sent, the modation that means a fitting, the metry that means a measuring, the matical that means the study of, the ming that turns a verb into a name, the mer that is only ever a piece of a hammer, the ment that makes a thing out of an act, and the munications that means the sharing of a message across a distance.

A letter came in January, on paper, which was itself a statement.

"Lin Yi. I am writing from the coast, where the connection is slow and my mother's shop is loud, and I have decided to spend one afternoon saying the things I did not say in the corridor, because a corridor is a bad place for a sentence with a subordinate clause in it.

First: do not confuse cohesion with collaboration. A group that likes one another coexists comfortably and produces nothing; a group that disagrees and coordinates produces a paper. The bond between two people who have struggled over the same thorny problem is worth more than a year of politeness, and it is also fragile, because it rests on trust and not on loyalty. Loyalty is a feudal virtue. Trust is a finite resource and it has to be validated every term, which is why the good groups argue about method and never about motive.

Second: nowadays everybody wants a paradigm, and nobody wants a measurement. A paradigm is a large, comfortable, extremely useful lie that a whole field agrees to tell itself for twenty years; it is not a sin, it is a scaffold, and the trouble begins when people start to worship the scaffold. When you meet a result that does not fit, do not crush it and do not beautify it. Write it down, in the margin, with the date, and let it sit there being useless for a year. Some of the most valuable things in that building were useless for a decade before anybody could say what they were for.

Third: beware the counterfeit virtue. There is a kind of diligence that is really a terror of stopping, and a kind of honesty that is really a wish to be admired for honesty, and a kind of humility that is a way of avoiding responsibility. I have worn all three. They are hard to detect from the inside, which is why you need one person who will tell you, and why you should choose that person for their willingness to disappoint you.

Fourth: the body. You will damage yours, mildly, repeatedly, and you will be surprised each time, because at nineteen the body is a rumour that turns out to be true. Do not skip the prescription because the cut looks clean; do not work through a feverish week because the run is going well; do not sit in a cold room for six hours with a raw

throat and call it dedication. There is no glory in a preventable disease and there is no prize for the student who ignored the symptoms. The reverse is true: the person who reports the symptom early is the person the lab can rely on for thirty years.

Fifth: on the three temptations of a young researcher, which are speed, novelty and revenge. Speed is the most flattering, because it looks like talent; it is usually a lack of method. Novelty is the most fashionable, because it looks like courage; it is usually a failure to read the literature. Revenge is the most human, because somebody will take your idea, or your place, or the credit for the thing you did in October, and you will want to damage him. Do not. A revengeful person is easy to predict and therefore easy to manoeuvre out of a room. Write the paper. Let the record be your reply.

Sixth, and this is the technical one, so read it twice. A report is not a description of what you did; it is an argument about what can be concluded from what you did. That means four things, in this order. The claim. The evidence. The alternative explanations you eliminated. The alternative explanations you could not eliminate, stated plainly. If you cannot fill the fourth, you have not finished, you have merely stopped. Most of the bad science I have read this year failed at the fourth item, not at the first three, and it failed because the writer could not bear to write a sentence that began with the words we cannot exclude.

Seventh: language. Learn the difference between a word's denotation and its atmosphere, because a laboratory runs on that difference. A dismissal and a judgement are not the same act, though they use the same tone. An incident report is not an accusation; a modification is not a failure; a correction is not an attack. Accuracy is not aggression, although it is often received as one, and a person who cannot survive being corrected will spend a career being vaguely wrong in a pleasant way, which is the most expensive thing in any laboratory.

Eighth: keep something that is not this. A musical instrument, a team, a corner of the garden, a dog. I had a piano for eleven years and I gave it up for a degree, and I have regretted it approximately once a week since, which is a strange thing to be able to count. The people who thrive in this trade for thirty years are almost never the ones who gave everything up. They are the ones who kept one thing back, as an anchor, so that a bad year could not take the whole of them.

Ninth: I am not going to tell you that it will be worth it, because that is a sentence for a graduation speech. I will tell you something smaller and truer. There will be an afternoon, probably in your third year, when a number appears on a screen and it is the number you predicted, and nobody is in the room, and you will feel about four seconds of something. That is the whole reward. It is not a lot. It is enough, if you have arranged a life around it that you would still want if the number had not appeared.

Tenth, and last: do not adore anybody in this building, including me. Admiration is a

bad instrument; it cannot measure the thing it is pointed at, and it makes you imitative at exactly the age when you cannot afford to be. Take what is useful, leave what is stylish, and go and be wrong in your own way, in public, in writing, on time. Yours, Shen Lu."

I read it twice on the day it came, and once more in March, and the third reading was the one that taught me something, because by then I had seen enough of the building to know which of her sentences were borrowed from somebody else and which were hers.

The last page of the folder was an extract from the department's annual report, which Zhou Min had copied because she thought it was funny, and which I have kept because it is the only honest description of an institution I have ever read.

"During the reporting period the laboratory recorded a modest expansion of its undergraduate intake, a continued improvement in its equipment, and a notable increase in publication output, which the committee deems satisfactory. Productivity per member of staff has risen for the fourth consecutive year. The committee notes, however, that this figure has been achieved partly through an increase in the average working week, a trend which the committee does not regard as sustainable, and which it proposes to address through a review of the allocation of duties rather than through any reduction in the total volume of work. The committee further notes a degree of uniformity in the topics proposed by doctoral candidates, and recommends that supervisors encourage a more enterprising choice of problem, while recognising that novelty carries a prohibitive risk for a student on a three-year tenure. The committee acknowledges the valuable contribution of the technical staff, whose expertise is the cornerstone of the laboratory's reputation, and whose salaries are set by a formula that the committee does not control. Two incidents were recorded: a minor injury, and a fire in a waste container, both of which were investigated and neither of which resulted in lasting harm. The committee is satisfied that the laboratory operates within the prescribed framework, and wishes to record its gratitude to all members for their cooperation during a turbulent year."

There is a particular kind of sentence, I have learned, that is entirely true and entirely useless, and that document was made of them. It had the tone of a coroner and the content of a greeting card. It said that the year had been turbulent and that everything was fine, and it did not mention the burnt bin, or the twelve yuan, or the boy who left in November, or the fact that the average working week had risen for the fourth consecutive year because nobody had the courage to write down the number.

On the train home I sat by the window with the folder on my knees and the rucksack between my feet, and a kid in the seat in front of me spent forty minutes peeling an orange and offering the peel to his grandmother, who pretended to eat it. The cafe car came through with a waitress who had the manner of a woman who had been doing this for

twenty years, and I bought a coffee for eleven yuan and left a tip of two, which was the total of my generosity that month, and she said, "Thank you, love," in an accent that was not from here, and I felt, for the first time in a term, entirely outside the university.

The train went through the snowy fields at a speed that made the fences look like a single fence, and the sunset came at four twenty and was gone by four forty, and somewhere past the third stop I fell asleep sitting up, which is a skill I acquired that year and have never lost. When I woke it was dark and the window was a mirror and my own face was in it, over the dark, and I thought about hope, which is a modest thing to have at nineteen and an expensive one to lose at twenty.

I have a passport in a drawer that has never been used. I have a map on the wall with three places circled on it and no plan attached. I had, that January, a vast and totally unearned confidence that the world was going to be legible if I worked hard enough, and about four years of evidence that this is not true, and I would not give up either of them.

The train arrived at twenty past nine. My father was waiting outside the station in the petrol station light, in the car he had bought second-hand, and he did not ask me about the marks, and I did not tell him about the beaker, and on the way home we talked about the price of meat and the new road, and the heater made a noise like a small aircraft, and it was the best conversation I had had in three months.

I should describe the inside of the cupboard door, because it was the densest page in the building. A sheet of paper had been taped there in about 2014 and had been amended by hand ever since, in four colours of pen, and reading it was the closest thing to a lecture that the room ever gave for free.

It said that the acetone was volatile, fragrant and quick to evaporate, that it would strip a label off a bottle in a single term, and that a pad left saturated with it on a bench was a fire waiting for a spark; that the white powder in the second jar was odorous in the way a hospital corridor is odorous, and that anybody who opened it without a mask would taste it at the back of the nose for an hour; that the acid was not to be stored above the eye line; that the salt solution was to be poured into the container marked with a category of its own and never into the sink, because a heavy metal is a poison that a river cannot digest; that the green crystals would melt at a temperature that a hotplate can reach by accident; that the solvent would attack a particle of dust and carry it inward, which is why the room had a hood and the hood had a sash; and that the whole shelf was, in the words of whoever had written the last line, a collection of small risks that were individually negligible and collectively a career.

Nobody had signed it. It was the only document in that building that nobody had signed, and it was the only one that everybody read.

The lost-property tray by the door is worth a paragraph of its own. It held a single glove, a chip of glass with a label on it, a tack, a pencil with a broken tip, a pair of goggles with a scratch across the left eye, a phone charger that belonged to somebody who had graduated, a hair tie, a metro card, and a paperback with a bookmark in it at page ninety, which nobody had claimed in three months, and which I eventually read and returned to the tray, because it seemed wrong to take a book out of a room that had been keeping it for somebody.

There was a poster wall in the lobby, and by December I had read every sheet on it. It advertised a lecture on imperialism and its railways; a debate on humanism in the sciences; a seminar on egoism and the economics of altruism; a student paper on capitalism and the laboratory, which I suspected of being an essay written for another course and pinned up in revenge; a public conversation with a novelist of some notoriety, whose prestige had been built, according to the poster, on a single book; and a lecture on obscurity in modern poetry, which had been cancelled, and whose cancellation notice had been pinned over the original notice with such care that the two together looked intentional.

The night watchman let me help him once, in the week before the holiday, and it was the only time I saw the whole building from the inside of its routine.

We hauled the empty gas cylinders down the stairs, one each, with a rope, and it was harder than it looks; the technique is to heave the thing onto the step above and then hoist it, rather than to lift it, and if you do it wrong your back tells you for a week. We manoeuvred a trolley through the corridor without scraping the paint, which requires somebody to walk backward and say left and right in a voice that the other person can actually hear. We wrapped a broken piece of equipment in paper and old cloth and then in tape, until it was a shape rather than an object, and we stacked the chairs in the lounge and swept the floor, and he taught me to sweep in one direction and to lift the broom at the end of each pass, because the dust you raise is the dust you will be breathing at midnight.

He had been in that building for nineteen years. He said the students had become politer and more anxious and that both changes were the fault of telephones. He said the best year had been 2016 and the worst had been the year of the epidemic, and that he had kept a record of every incident in a notebook of his own, and that he had never once been thanked for it, and that he did not want to be thanked for it, because a watchman who is thanked is a watchman who has been noticed, and a watchman who has been noticed is not doing his job.

The train arrived at twenty past nine.

In June, on my last afternoon in the building, I wrote my own version of the folder and put it in the same drawer, and it was shorter than hers, and rougher, and I have no

idea whether anybody ever read it.

"To whoever takes bench three. I am not going to pretend that I know how this is done. I know about eleven things and nine of them are about glassware, and the other two are these.

The first is that the room is generous with advice and short of people who will read your draft. There is an abundant supply of opinion and a narrow channel for it, and the narrow channel is the person who will sit down with your page and a pen. Find that person. Do not wander from desk to desk collecting impressions, because an impression is not a criticism, and a compliment is not a diagnosis. If somebody says your report is interesting, ask him which sentence he would cut. If he cannot answer, he is being kind, and kindness at that moment is a form of interference.

The second is that the balance will teach you more in a week than the lectures will in a term, if you let it. Not the theory of it. The habit of it. The weighings are a rough, repetitive, unglamorous set of motions, and they will not make you a scholar, but they will stiffen something in you that no amount of reading can stiffen. Stand still. Wait for the number to settle. Write it down before you think about whether you like it. Do that four hundred times and you will have a spine where your opinions used to be.

A few practical notes, and then I will stop, because I can hear how I sound.

Dampen the weighing boat with your breath before you put the powder in, and the static will not rob you. Use a substitute boat while you practise, and track your error on the same page every time, so that the comparison between weeks is possible. If the crucible is new, then this is its last day of virginity, and after today you will call it a crucible and not a new crucible, which is the beginning of ownership.

Do not isolate yourself in the small room with the balance for more than an hour. People who work alone in there come out weird, and I include myself. Interact with somebody at least once an hour, even if the interaction is only a nod. The suppression of that need is not discipline; it is a slow, quiet diminution of your judgement, and the first symptom is that you start to justify things to yourself in a voice that sounds like a reasonable person.

Learn to see the knotty step before you reach it. The tricky parts of a procedure are almost never the ones the manual emphasises. A procedure is a chain, and the chain fails at the joint, and the joint is where two steps meet: where the hot liquid meets the cold glass, where the record meets the memory, where the hurry meets the queue for the fume hood. If you can foresee the joint, you will look like a person with unusual talent. You will in fact be a person with a list.

Say the difficult thing clearly and early, and say it once. Concern is not a plan, and sympathy is not a repair. If your data is disastrous, report the disaster in the

plainest sentence you can compose: this run failed, the cause was the tap, the tap has been repaired, the run will be repeated on Thursday. Nobody will think less of you. What people cannot forgive is the long, soft, voluminous paragraph that turns out, on the third reading, to be asking them to guess.

Do not let anybody compel you to be heroic. There is an image of the scientist as a person who works through the night and forgets to eat, and it is a fraudulent image, and the people who propagate it are usually the ones who are being paid to be somewhere else. Sleep deprivation is not a research method. A tired person will slap a lid on a flask, stumble against a trolley, spill the thing he has spent two days making, and then, because he is tired, he will be tempted to hide it. That is how a small accident becomes a body of fiction, and it is not a striking story, it is the most ordinary one in the building.

One more thing, and it is the only piece of philosophy I have. The difference between a technician and a researcher is not a degree, and it is not ingenuity, and it is certainly not originality in the sense that a pop magazine uses the word. It is the willingness to be interested in the boring part. The interesting part is the discovery. The boring part is the detection of the error that made the discovery look better than it was, and the person who will not go back into the boring part will spend a career assembling a beautiful, internally consistent, entirely counterfeit account of the world, and he will be the last person to know it.

That is not a temptation that visits you once, at a dramatic moment. It is a temptation that visits you on a Tuesday, in the form of a number you would very much like to be right.

If you get stuck, and you will, do not scramble in silence for a week. An urgent question, asked early, is a professional act. A question asked after three weeks of private struggle is a confession. Neither is shameful, but the first one will get you a mentor and the second one will get you a reputation.

Finally: rejoice a little when it works. Not for long, and not in public. But the warmth of that is the only fuel there is, and the people who tell you that the work itself is enough are lying, or they are saints, or they are dead.

I hope you like the room. I did not, at first. It smelled of sulphur and old rubber, and the light was wrong, and the taps screamed. Then one night in November I stood in front of a beaker and watched a skin form on the surface, and something in me shifted, and the room stopped being a stage on which I was being judged and became a place where a person could work.

Take the long way round. Write it in ink. Do not touch the oven at four in the afternoon.

Yours, from a bench that no longer has my name on it."

Having written it, I understood why she had written hers: not for the next person, who may never exist, but because a thing you cannot say out loud has to go somewhere, and paper is cheaper than a telephone call.

On the last night I walked the building once, from the basement to the roof, and said goodbye to the things that had made up my week.

The acting head of the teaching committee had signed the wall by the stairs, and the advent of the new programme was recorded in the same ink, and the anniversary of the fire in the bin was marked in pencil by somebody who had been there. The corridor smelled of fresh paint over sour mop water. The tap that could aggravate a whole afternoon was quiet. Somebody had thrown out the old airline magazines that had been in the lounge since before any of us were born, and the shelf looked richer for it, or emptier, depending on the hour.

I do not want to make an allusion to a metaphor here, because the metaphor would be too neat, and the analogy would be false. It is not analogous to anything. But if I had to explain the year to a stranger in one sentence, I would say that I arrived as armour and left as a person, and that the armour was not bloody, only heavy, and that taking it off was the whole of the education.

The building does not care. That is its blessing. It is not a mother and it is not a judge and it does not barge into your life; it is a set of rooms that will betray you if you are careless and boost you if you are not, and the difference is not destiny, it is method.

I had a bowel complaint in the third week from the canteen, and a candidacy for the student committee that I lost to a capitalist with better posters, and a cartoonist in my class who drew the whole thing for the department magazine, and a cliff of unread paper on my desk that never once got shorter. None of that is on the record. All of it is the record.

Here is what I learned about groups. A cohesive group is not a friendly one, and congratulations are cheap, and consensus is not truth, and the person who convenes the meeting is rarely the person who decides. A convoy moves at the speed of its slowest trolley. A cradle is a good place to start and a terrible place to stay.

Here is what I learned about work. Contact with the bench is worth more than contemplation of it. The continuation of a mistake is worse than the mistake. In contrast to what the brochure says, the crucial skill is not invention but attention. Take the cue from the person who is quiet and has data. Stay curious about the boring thing. Then develop the habit of writing it down, and devise your own order, and be prepared to edit it eleven times.

Here is what I learned about failure. It is destructive only if you are deaf to it. Otherwise it is a dispersal of your illusions, an ejection of the version of yourself who was doing this for the wrong reason. It will occur at dawn, or on a Sunday, and it will feel like an ending, and it will be a form of inclusion: you will be let into the smaller, plainer room where the work actually happens.

Here is what I learned about people. The examiner is tired. The evolutionary biologist and the engineer are equally stubborn. The energetic first-year becomes the deeply tired second-year and then, if fortunate, the useful third. The female student who is shyly correct is worth ten loud ones. The insider knows the shortcuts and the shortcuts are often the long way. The invader of your afternoon is usually not hostile; he is merely urgent. Do not strangle a good idea because it arrived from the wrong person. Do not let a magical intuition substitute for the introduction of a control. Nothing here is negotiable except the timetable, and a momentous discovery is usually the name we give to a tiresome repetition that finally paid.

Here is what I learned about the body and the day. The watt is not a unit of your effort. Sleep is not relaxation; it is maintenance, and a rapid decline in the quality of your thinking is a physiological event and not a moral one. A pregnancy, a pregnancy that ends, a bad month, a grief: these are facts that a decent institution arranges itself around, and a hostile one treats as an inconvenience, and you will not know which kind you are in until you need it.

Here is what I learned about language, which is what I came for. A word can be a hunter or a home. A phrase can be an identification or an ambush. A slogan can illuminate nothing and still feel like a homeland. Speak plainly, and the plainness will sound, to some, like a weakness; it is not. It is the opposite of the voluminous, slippery, tricky language of people who want you to agree with them before you have understood them. The trustworthy style is vertical: it stands up. And a good sentence, like a good measurement, is a small piece of the world that somebody has agreed to be responsible for.

Empirically, and I have checked, the temptation to spoil a good result comes at about four in the afternoon, when the light is orange and the coffee is finished and the number on the screen is almost, but not quite, the number you foresaw. That is when you will formulate the generous explanation, and it will be a fraud, and you will know it, and you will be too tired to care. That is the hour to stop. Switch the thing off. Go and eat something and be a person again.

The room is not a temple and the work is not a religion, and anybody who tells you otherwise is selling something, or is twenty-two and enthusiastic, and the enthusiasm is lovely, and it will not survive the third winter unless the habit is deeper than the feeling.

I hope the habit is deeper. In my case it was not, at first; it was transplanted into me by a senior with a clipboard and a senior with a cold voice, and it took, and it grew, and it is still growing, slowly, in a room I no longer have a key to, and that is the same in the end, or so I would like to believe, as theirs.

A coda, three years later, in another city, in a laboratory that is not mine.

Recently I watched a first-year try to conquer a troublesome balance, and I recognised the whole of my own debut in the set of his shoulders. He chose his spatula arbitrarily, weighed the powder twice and wrote down the second number, and I could see the craze for a result beginning in him, which is a delightful thing to watch and a risky thing to have. He was terribly earnest and highly nervous, and the effect was a kind of innocence that no amount of training can manufacture and no amount of experience can restore.

I did not intervene. I have a philosophical objection to intervention at that stage, and, more honestly, I wanted to see whether he would encounter the film on his own. He did. He stood there for eleven minutes, which for a person of his age is an infinity, and when the skin formed he said something under his breath that I will not reproduce, and he looked up, and for a moment his face was vivid with the oldest and least metaphorical magic there is: the possibility that the world might be legible.

The work that follows is duller. You integrate the numbers, you negate the outliers one by one, you dispense with the duplicate runs, you streamline the record until an outsider can patrol it without a guide. You will be tempted to portray the process as a smooth movement from question to answer, and the temptation is the enemy: it will mar the page and sadden the reader, and it will heighten nothing except your own vanity. Do not let a good story divorce you from your data. Do not let a merger of two neat ideas negate the awkward third. Do not suck the life out of a result by over-explaining it, and do not wield a metaphor where a number would do.

Spontaneity has its place, and its place is Monday, not Friday. The spirit of the thing is not in the erupt of insight; it is in the reservoir of small, dull, correctly recorded days, from which the insight is drawn, gradually, by somebody who was there and had the inhibition to check.

As for the rest, the augmentation of budgets, the distributive arguments about authorship, the diversion of a good student into administration, the startle of a bad review, the foe who becomes a trustee, the spy in every department who reports your progress to somebody with more power than curiosity: these are not the subject. They are the weather. The similarity between one laboratory and another is far greater than the singularity of any one of them, and the son of a butcher and the son of a professor will learn the same habit in the same eleven weeks if the room is honest.

The habit can be taught. That is the noteworthy part. It is not a gift and it is not a mystery, and it will not desert you if you do not desert it first.

Last week I went back. I did not oppose the speeding stream of first-years on the stairs; I stood ashore on the landing and let them pass, a visitor from beyond the glass door, which is what a person becomes, in the tertiary sense, at about the age of twenty-two. A building like that is fascinating from a landing. You can see the naivety and the dubious confidence and the incredibly eventful Tuesday of a person who is nineteen, and you want to enrich it, to embody the one useful thing you know, and you cannot, because the only way to learn it is the exercise itself.

Somebody had donated a bench; somebody had dispersed the duplication in the inventory; somebody still had to irrigate the botany trough, which was a manifest success; and the assassin of my good mood was a notice on the door saying that the room was dependent on funding and no longer autonomous.

I dislike endings of that kind. I hinder them when I can. I have not learned to forge a better one, and I suspect that the ability to weave one out of the material at hand is what the whole of this is about: the exploit and the extinction, the exciting and the menacing, the fuck-it of a Friday afternoon and the patience of a Tuesday morning.

It took me a year to see that this was the same lesson as the bench. A word is not a heap of letters. It is a set of people who have agreed to be precise about one small thing, and a history of everybody who was not.

伴读：一间实验室的语法

你可能正在背词表。先停一下。

我不是要劝你别背，我是要劝你换个顺序。背词表的问题不在于懒，而在于它把词从"事"变成了"条目"：abbreviation 在表里只是"缩写"两个汉字，可它在那面墙上是一列让你少写二十个字母的约定；absent 在表里是"缺席"，可它在那张名单上意味着这学期没有你的台面。词一旦离开事，就只剩一个壳，你记住的是壳，考场上要用的是核。

这一章的所有词，都长在实验楼 402 那间屋子里。我不按字母讲，也不按"高频低频"讲，我按那天发生的事讲——你顺着事情走，词就自己挂上去了。

那面墙

先说 abbreviation。它在正文里第一次出现，是门边那张两列的 A4：一列是全称，一列是缩写。实验室的缩写不是装饰，是契约：你写 Na 而不写 sodium，是因为所有人都同意 Na 就是 sodium。according to 那张纸，你就得这么写；不 according，报告就没人看得懂。这个词的骨头是"依照"，它后面永远要跟一个来源：according to the manual、according to the record——它不许你空口说。

墙上第二样东西是安全须知。它 affirm 了四件事——注意，是 affirm，不是 say。affirm

是"正式确认、声明为真"，语气比 say 硬，比 claim 稳；一份须知用 affirm，是在告诉你这不是建议。它要求你把 allergic 反应（对乳胶过敏这种事）在第一次 class 之前就申报，把 alcoholic 饮料 ban 掉，把 bodily harm 交给医院而不是交给 consolation。你在那一栏签了 acknowledgement，这个词比 signature 重：签名只证明你签了，acknowledgement 证明你"已知悉"，从此不能说你不知道。

然后是那张值班表。它排的是 duty，写的是 hourly 的格子，末尾留一个 standby 号码。aide 这个词在正文里指那个"助手"，其实它是 402 里最不显眼、最不能少的人；acting head 是"代理负责人"，acting 在这里不是"表演"，是"临时代理"——一个人一旦被叫 acting，你就知道他头上还有一个不在场的人。authority 是"权限"，autonomous 是"自治的"，这两个词在实验室里天天打架：一个房间看起来 autonomous，其实每一笔 expenditure 都要 authority 点头。

周敏出场

她说话像 auctioneer——拍卖师，语速快、报价狠、不留缝。这个比喻你要是记住了，就顺带记住了 auction 这个词根：拍卖。她一进门就 allot 时间：四分钟讲报名，四分钟讲安全，剩下的讲器材。allot 是"分配（份额）"，比 give 精确，因为它暗示份额有限。她还要 administer 报名表——administer 不是"管理"，在这个语境里是"执行、主持（程序）"。

她讲天平的时候用了一个词：alignment。水平、对齐。beam 是天平的横梁，actual reading 是实际读数，astronomical 是"天文数字般的"，她拿学费打比方：这台仪器的价钱是个 astronomical figure。atmospheric pressure 是大气压，她说这句话的时候塑料帘子在动——一个词如果能挂在一个动作上，你就不会忘。

她要我们 amass 数据，然后 interpret 得 artistically，然后她给你不及格。artistically 在这里是挖苦：把数据"艺术地"解释，就是乱解释。arbitrarily 也是挖苦：不许"任意地"选点、选条件、选截止时间。

那杆秤和一个笨手笨脚的人

我 admit，第一次称量我 awkward 得要命。clumsy 是"笨手笨脚"，但 clumsy 只说你手脚不利索；awkward 还有一层"不得体、让人替你难受"。accidental 是"意外的"，可 402 里没有几个真意外，多数是 aggravate——小错被自己加重。

粉末不肯离开称量舟，是因为静电。你 add 得太快，它就飞起来；你要 adequate 地慢，慢到看起来不像在干活。accumulate 是"积累"，灰尘 accumulate 在窗台，错误也 accumulate 在记录里；accustomed 是"习惯了"，我第三周就 accustomed 了八点四十称量这件事，习惯一旦形成，你就不再觉得它是个决定。

我 adore 那台天平吗？不。但我 adore 它给出的那个数字——adore 是"极喜欢"，比 like 热，比 love 轻，用在物上正好。advent 是"到来"，正文里我说电子天平的 advent 改变了一切；这个词常配 advent of，像节日、像时代。advertise 是"宣传"，海报 advertise 一场没人去的讲座；app 是我们查台面的那个应用；apple 是张野放在台面上的那只苹果——别小看这种细节，一个具体的东西能让一整段文字站住。

abundant 是"充足"，402 不缺建议，缺的是肯读你稿子的人。adequate 是"够用"，注意它不等于 good：adequate 是及格线上方一点点，正是多数实验条件的真实状态。again 和 another 这种词你可能觉得没必要讲，可它们在最要紧的地方决定意思：again 是"再来一次"，another 是"另一个"，一个是重复，一个是替换，实验里这是两件事。

agony 和 anguish 都是"痛苦"，但 agony 偏身体、偏剧烈，anguish 偏心里、偏长久。产物洒在台面上的那四秒，是 agony；沈璐说"你不是粗心，你是浪漫"的那一晚，是 anguish。anniversary 是周年，实验室里也有人记着某次事故的 anniversary，用一种很轻的方式记。

antibiotic 出现在校医院那管药膏上。anybody 和 somebody 的区别，在实验室里很实际：anybody 是"任何人"，somebody 是"某个人"，"anybody can weigh"是制度，"somebody weighed it"是责任。

appear 是"出现"，appraisal 是"评估"，approve 是"批准"。三个词连着一条线：数据 appear，沈璐给你一个 appraisal，周敏 approve 或者不 approve。appraisal 比 evaluation 多一层"有人负责地评判"的意味，所以它出现在她嘴里。

archive 是"档案"，article 是"文章"，argumentative 是"好争辩的"。她说论文不是 report，是 argument——所以 argumentative 在这儿不是骂人，是"有论辩性的"，一个人 argumentative，可能是缺点，一篇文章 argumentative，是优点。

armour 是盔甲（英式拼 armour，美式 armor），arrest 是"逮捕"，也可以让一个笑声"停住"——正文里那个笑就是被她的目光 arrest 的。ashore 是"上岸"，我后来把它用在一个比喻里。assassin 是刺客，assertive 是"敢主张的"，assist 是"协助"，assumption 是"假定"，attack 是"攻击"，attentive 是"专注的"，augmentation 是"增加、扩充"，avenue 是"大道"，awake 是"醒着的"。

最后两个：allusion 是"典故、暗指"，analogous 是"类似的"。这两个词是一对脾气：allusion 是"我不明说，你自己听出来"，analogous 是"我把结构摆出来给你看"。沈璐用前者挖苦人，用后者讲道理。aloud 是"出声地"，alternately 是"交替地"，abort 是"中止"——注意，abort 不是 fail，fail 是做了没成，abort 是你判断继续下去更糟，于是叫停。这个词在实验里出现的频率，比你想的高得多。

从门牌到天平

badminton 是我在那学期学会的唯一运动，跟词义无关，跟记忆有关：词只要挂在一件你亲身做过的事上，就会自己长脚。baseball 只出现在张野那顶帽子上，bass 是他说话的低声部。basket 之类你没在词表里见到的词，先放着，别贪。

balance 是天平，也是"平衡"，也是"余额"。一词三义看着乱，其实是一条线：天平在称两边的平衡，账户在算进出的余额。battery 是电池，那天早上它正在充电，所以八点二十之前没有电。beam 是横梁，brick 是砖，brick-red 那种红你在老楼里见过。bright 是亮，可 402 的 bright 是一种"亮得显旧"的亮。

bench 是实验台，也是长椅；belt 你没用上，banner 用上了——门口那条横幅。beginning 是开头，我在签字那一栏写"这是所有这类故事的开头"，因为不读最后一条就签字，是所有人的开头。

believe 是相信，betray 是背叛——手会 betray 你，数据也会。beyond 是"超出"，benign 是"良性的"，医生说那道口子 benign，你才松了口气；这个词在医学里是"良性"，在语气里是"不要紧"。

bill 是账单，blanket 是毯子，blessing 是福气（a blessing in disguise 是因祸得福）。bloody 是"带血的"，也可以是英式口语里那个没什么意义的强化词——两种用法差得很远，看语境。blunder 是大错，特指"本可以不犯的蠢错"，比 mistake 重，比 error 有人味。bond 是键，也是纽带；化学键和人与人之间的 bond，共用一个词不是巧合。boost 是提振，bowel 是肠，brain 是脑，breath 是呼吸，breach 是"违反（规矩）"，broadcast 是广播，brother 是兄弟，brutal 是"狠"，Buddhist 是佛教的，build 是建，bump 是撞，bureaucratic 是"官僚的"，bus 是公交，butcher 是屠夫，by 是"由、通过"。

这里插一句：barge 是"闯进来"，比 enter 多一层莽撞；beautify 是"美化"，沈璐说"不要美化你的数据"，用的就是这个词；beef 是牛肉；biochemistry 是生物化学；bird 是鸟；bitter 是苦；botany 是植物学；castle 是城堡；cartoonist 是漫画家；carpet 是地毯；canvas 是帆布。

称量与配制

calculator 是计算器，camera 是摄像头（走廊里那只），cafe 是咖啡馆，cell 是池、是细胞、也是隔间——天平间就是一个小小的 cell。centigrade 是摄氏度，一百零五 centigrade 是烘箱的温度。certainty 是确信，我十八岁那年对"我很谨慎"有 certainty，一点证据也没有。chancellor 是校长（某些大学用这个词），那份须知就挂在 chancellor's office 名下。charge 是"负责"，in charge 是主事；chief 是头儿；chemical 是化学品；chip 是碎片、是芯片；chop 是剁；Christmas 是圣诞；cinema 是电影院；citation 是引文；clearly 是清楚地；cliff 是悬崖；clip 是夹子、是回形针；clothe 是"给……穿衣"，须知里那句 clothe yourself properly 就别译成"穿衣服"，它是在说"按规矩穿"。

coastal 是沿海的，coexist 是共存，cohesive 是"有凝聚力的"，collaboration 是合作，college 是学院，comet 是彗星，comparable 是"可比的"——注意，可比不是相等，正文里沈璐咬的就是这个"comparable to which paper"。

compel 是"迫使"，competition 是竞争，complaint 是投诉，compliance 是"依从、合规"，compose 是"组成、使镇定"，compression 是压缩，conceit 是"自负"，也是文学里的"奇喻"；concern 是关切，conclusion 是结论，condition 是条件，confederation 是邦联，confidential 是保密的，conformity 是顺从，congratulation 是祝贺，conquer 是征服，consensus 是共识，consider 是考虑，consolation 是安慰，constitute 是构成，consulate 是领事馆，contact 是联系，contemplation 是沉思，contest 是比赛，continuation 是继续，contrast 是对比，convene 是召集，convert 是转换，convoy 是车队，coordinate 是协调，coral 是珊瑚，cornerstone 是基石，corrective 是"纠正性的"，corridor 是走廊，cosmos 是宇宙，could 是 can 的过去式，counterfeit 是伪造的，courageous 是勇敢的，coward 是懦夫，cradle 是摇篮，craze 是一时的狂热，creature 是生物，cricket 是蟋蟀，crop 是庄稼，crucial 是关键的，crush 是压碎，cue 是提示，culture 是文化，curious 是好奇的，curtain 是帘子，cut 是切，cynicism 是犬儒。

这一长串里，真正会咬人的是四个：comparable、compliance、conceit、counterfeit。comparable 一写错就变成"差不多"，而科学里没有"差不多"；compliance 是"照规矩做"，不是"听

话"；conceit 不是"概念"（那是 concept），是"自负"；counterfeit 不是" counterfeit money"专用，伪造的数据也是 counterfeit。

第一次做砸

dampen 是"弄湿、使减弱"。它有两个方向：把纸 dampen 一下是为了消静电，把士气 dampen 一下是打击人。同一个词，一个是技术，一个是人际。darkness 是黑暗，dawn 是黎明，dawn on somebody 是"某人忽然明白"，这个搭配比"天亮"更常用。deaf 是聋的，be deaf to advice 是听不进劝，不是生理上的聋。debut 是首次登台，我把它用在自己第一次站到天平前——一个借来的剧场词，用在这种地方反而准。deceptive 是"骗人的"，appearances are deceptive 是老话，可它说的是真事：那个数字看着稳，其实在漂。

declare 是"申报、宣布"，安全须知要求你 declare 过敏史。注意 declare 和 claim 的差别：declare 是"我正式说出来"，claim 是"我主张这是我的"。dedication 是奉献，也是题献；一个词的两种用法，一种给别人，一种给事。deem 是"认为"，法律和公文爱用它，因为它是"被判定为"，主体不重要。deficiency 是缺乏，degenerate 是退化，delete 是删除——沈璐那句"不要删除难看的数字"，用的是 delete，因为删掉一个数据点在软件里就是按一下键，太容易了。

delightful 是令人愉快的，democracy 是民主，denotation 是字面意义（跟 connotation 相对），denunciation 是公开谴责，dependent 是依赖的，deport 是驱逐出境，deprivation 是剥夺（sleep deprivation 是熬夜的正式说法），derivation 是推导，descriptive 是描述性的，desire 是欲望，despise 是鄙视，destructive 是破坏性的，detection 是检测，determine 是确定，develop 是发展，devise 是设计出，dial 是旋钮，dictator 是独裁者，different 是不同的，digest 是消化，diligence 是勤勉，diminish 是减少，diplomacy 是外交，dirty 是脏的，disappoint 是使失望，disastrous 是灾难性的，disclose 是披露，discovery 是发现，disease 是疾病，dislike 是不喜欢，disperse 是驱散，dispute 是争执，distance 是距离，distort 是扭曲，distributive 是分配的，diversion 是转移、消遣，divorce 是离婚，也可以说"两件事离婚了"——沈璐就爱这么用。

document 是文件，dolphin 是海豚，donate 是捐赠，dosage 是剂量，dove 是鸽子，doze 是打盹，dramatic 是戏剧性的，dread 是恐惧（比 fear 更"预支"），drink 是喝，drug 是药、也是毒品，dubious 是可疑的，duplication 是重复，duty 是责任、也是值班，dynamics 是动力学、也是"群体 dynamics"。

这里面最值得停一下的是 distort。它和 twist 都能译"扭曲"，但 distort 是"让形状失真"，twist 是"拧"。你 distort 一条曲线，读者看到的是形状；你 twist 一句话，读者听到的是恶意。还有 dubious：它比 doubtful 更"我怀疑你有问题"，语气偏冷，用在数据上是很重的话。

星期四的那堂课

earnest 是认真的，而且是一种"不掩饰的认真"，比 serious 暖。eastern 是东部的，economic 是经济的，edit 是编辑，effect 是效果，egoism 是利己主义，ejection 是逐出，elderly 是年长的，electricity 是电，elementary 是初级的，eligible 是"有资格的"，email 是邮件，embody 是"体现、使具体化"，emigration 是移居国外（注意跟 immigration 方向相反），empire 是帝国，

empty 是空的，encounter 是遭遇，ending 是结尾，energetic 是精力充沛的，engineer 是工程师，enlighten 是启发，enrich 是丰富，enterprising 是有进取心的，entire 是整个的，entry 是进入、条目，envoy 是使节，equally 是同样地，equivalence 是等价——滴定里的 equivalence point 就是等当点，这个词你要是译成"相等点"就丢了"当量"的意思。

erection 是竖立、建造，erupt 是喷发，essayist 是随笔作家，esthetics 是美学（跟 aesthetic 同源，一个名词一个形容词），etiquette 是礼节——实验室的 etiquette 不是客套，是一套让所有人少受伤的动作。evaporation 是蒸发，eventful 是多事的，everyone 是每个人，evolutionary 是进化的，examiner 是考官，exception 是例外，exciting 是令人兴奋的，excuse 是借口，exercises 是练习（注意这个词的单数形式在本书里一律不用，因为太容易被当成题目），exile 是流亡，expansion 是扩张，expenditure 是支出，expertise 是专门知识，exploit 是利用、也是功绩，exportation 是出口，extend 是延伸，extinction 是灭绝，extreme 是极端的。

etiquette 和 ethics 常被弄混：etiquette 是"怎么做不失礼"，ethics 是"怎么做不算错"。周敏那六条，一半是 etiquette，一半是 ethics，她从不区分，因为她觉得在实验室里这两件事最后会合成一件。

eligible 值得单独说：它不是你"够优秀"，而是你"符合条件"——名单上有你，你就 eligible，没有就不是，跟感情无关。这个词冷，正是它的用处。

effect 和 affect 是一对老冤家：effect 多是名词（效果），affect 多是动词（影响）。可 effect 也能作动词，意思是"促成"，affect 也能作名词，意思是"情绪表现"——这两条冷门义，知道就好，别在作文里炫。

沈璐的刀

fabrication 是"捏造"，但它的语气比 lie 重得多，因为它暗示"制造"了一个东西：lie 是一句话，fabrication 是一整套。fact 是事实，failure 是失败，fame 是名声，far 是远，fascinating 是迷人的，fat 是脂肪，faulty 是"有毛病的"——faulty 说的是东西，不是人；说人要用 unreliable 或 careless。feasible 是可行的，注意它只问"做不做得到"，不问"该不该做"。

fee 是费用，female 是女性的，feverish 是发烧的、也是"狂热的"，fifty 是五十，filter 是过滤，finite 是有限的，fish 是鱼，fixed 是固定的，flatter 是奉承，flexibility 是灵活性，flood 是洪水，fluctuation 是波动，foe 是敌人，following 是下面的、接着的，forbid 是禁止，foresee 是预见，forge 是锻造、也是伪造（forge a signature），form 是形式，formulate 是"系统地表述"，fortunate 是幸运的，found 是"建立"（不是 find 的过去式，别看错），fraction 是分数、一小部分，fragrant 是香的，fraudulent 是欺诈的，fresh 是新鲜的，fright 是惊吓，frosty 是结霜的，fuck 是脏话——它在正文里只出现一次，用于一个极度疲惫的瞬间，你要知道它有多重：这是英语里最粗的一档，正式场合一律不用。

fund 是基金，furniture 是家具（不可数），futility 是徒劳，gamble 是赌博，garden 是园子，gate 是门，gene 是基因，genius 是天分、天才，geological 是地质的，get 是得到，give 是给，glare 是"刺眼的光"或"怒视"，gloomy 是阴郁的，goal 是目标，goods 是货物，governor 是州长、总督，gradually 是逐渐地，grandchildren 是孙辈，grape 是葡萄，gratitude 是感激，great

是大的、伟大的，grid 是网格，grip 是抓紧，group 是组，guidance 是指导，gum 是牙龈、也是口香糖，hair 是头发，hand 是手，handwriting 是笔迹，harassment 是骚扰，harm 是伤害，haste 是匆忙，haul 是拖运，hazardous 是危险的，headmaster 是校长，hear 是听见，heave 是用力举起，heighten 是加剧，helpful 是有帮助的，herald 是预示、宣告，heroic 是英勇的，hesitation 是犹豫，highly 是高度地，hinder 是阻碍，historic 是历史性的，hoist 是升起，homeland 是故土，honesty 是诚实，hope 是希望，horror 是恐惧，hostage 是人质，hourly 是每小时的，however 是然而，humanism 是人文主义，humiliate 是羞辱，hunter 是猎人，hurtful 是伤人的，hypocrite 是伪君子。

重点讲三个。第一是 fabrication 和 counterfeit 的分工：counterfeit 说的是"假货"，钱、证、货；fabrication 说的是"编出来的说法"，数据、引文、结论。前者的反义是真货，后者的反义是事实。第二是 feasible，它跟 possible 的区别是 feasible 还要算成本：possible 是"理论上能"，feasible 是"这套条件里能"。第三是 foresee，它比 predict 更强调"事前看见"，所以 foresee 常跟 difficulty、problem 连用；你 foresees 的是麻烦，不是数字。

hypocrite 这个词，沈璐用得很准：伪君子不是"说错话的人"，是"知道自己在演的人"。她说"你写 probably 是为了显得客观"，那不是在骂你 hypocrite，那是在提醒你：距离 hypocrite 只差一步。

hostage 值得记一个搭配：be held hostage to，被……挟持。你可以被自己的时间表挟持，被一个不肯改的假设挟持。这种抽象用法在写作里非常好用，比 be controlled by 有力。

fright、horror、dread 三个词可以排一排：fright 是"被吓一跳"，horror 是"看见可怕的东西"，dread 是"还没发生就开始怕"。dread 最像学生的星期天晚上。

402 的口头传统

iceberg 是冰山，identification 是识别、身份证明，idiotic 是愚蠢的（比 stupid 更刻薄，也更口语），ignorant 是无知的（说人不知情，不带恶意；说人无知，带），illuminate 是照亮、也是阐明，image 是形象，imitative 是模仿的，imperialism 是帝国主义，important 是重要的，impression 是印象，impulsive 是冲动的，incident 是事件（中性，但常指不好的小事），inclusion 是包含，incredibly 是难以置信地，index 是索引，indignant 是愤慨的（被不公正对待后的怒，比 angry 有理由），inducement 是诱因，infection 是感染，infinity 是无穷，inform 是通知（inform somebody of something，别用 inform somebody something），ingenuity 是巧妙，inhibition 是抑制，injure 是使受伤，innocence 是无辜，inquiry 是询问、调查，insider 是内部人，inspector 是检查员，instant 是瞬间，instruct 是指示，insulator 是绝缘体，integrate 是整合，intend 是打算，interact 是互动，interference 是干扰，interpret 是解释，intervene 是介入，intimidation 是恐吓，introduction 是介绍、导论，invader 是入侵者，inventory 是清单，investor 是投资人，inward 是向内的，irrigate 是灌溉，isolate 是隔离。

indignant 和 angry 的差别值得记：angry 是情绪，indignant 是"我认为这不公平"。所以 indignant 常出现在有人被冤枉的场合，也常出现在知识分子觉得自己被冒犯的时候。

inquiry 有两个意思，一个是"询问"（make an inquiry），一个是"调查"（a formal

inquiry）。后者更重，常跟 into 连用。实验室里的 inquiry 不是问你一句，是立案查。

jack 是千斤顶、也是插座孔（jack plug），jealousy 是嫉妒，jewellery 是珠宝（英式拼 jewellery，美式 jewelry），journal 是期刊，judgement 是判断，jungle 是丛林，justify 是证明……正当。注意 justify 后面常跟 to：justify it to yourself 是"对自己解释得通"，这一句在正文里出现过，它比 explain 多一层"辩护"。

kid 是小孩，kindergarten 是幼儿园，knee 是膝盖，knotty 是"棘手的"（字面是"多节的"，一根 knotty 的绳子难解，一个 knotty 的问题难办）。

laborious 是费力的，lake 是湖，landmark 是地标，laser 是激光，latitude 是纬度，law 是法律，layout 是布局，leap 是跳，leave 是离开、也是许可，legendary 是传说的，leisurely 是从容的，less 是更少，levy 是征收、是摊派，library 是图书馆，lily 是百合，line 是线，linkage 是联系，listen 是听，litter 是垃圾，loan 是贷款，locker 是柜子，logistical 是后勤的，look 是看，lorry 是卡车，lounge 是休息室，lubricate 是润滑，lunar 是月亮的，lyrical 是抒情的。

levy 这个词很值得学：它是"以官方名义收一笔钱"，名词动词都行，比 charge 正式，比 tax 具体。实验室的 levy 通常很小，但它是"制度"的具体形态。

legendary 别只理解成"传说中的伟大"，它也能形容"坏了很久、久到成了传说"——正文里那根 legendary 的水管就是这个意思。

那页音节

magic 是魔法，magnitude 是量级（比 size 专业，能说 orders of magnitude），maintenance 是维护，male 是男性的，manage 是管理、也是设法做到，manifest 是显示、也是"明显的"，manoeuvre 是操作、是周旋，mar 是破坏（瑕疵 mar 了完美），maritime 是海事的，martyr 是殉道者，mask 是口罩、是面具，mat 是垫子，meadow 是草地，meat 是肉，mediate 是调解，medicine 是药、是医学，melt 是融化，menace 是威胁，merger 是合并，message 是消息，metaphorical 是隐喻的，midnight 是午夜，mild 是温和的，milk 是牛奶，mint 是薄荷，mislead 是误导，mistaken 是搞错的，modification 是修改，momentous 是重大的，money 是钱，mop 是拖把，mortality 是死亡率，motion 是运动，mountain 是山，movement 是运动、也是乐章，multilateral 是多边的，murder 是谋杀，musical 是音乐的，myself 是我自己，naivety 是天真的，narrative 是叙述，nationalist 是民族主义者，naval 是海军的，necessarily 是必然地，negate 是否定，negotiable 是可协商的，nervous 是紧张的，news 是新闻，night 是夜，nitrogen 是氮，nominal 是"名义上的"，noon 是正午，northwestern 是西北的，noteworthy 是值得注意的，notoriety 是恶名，nowadays 是如今，nutrition 是营养。

那个尾页上的九个碎片，你要会拆：com-mittee 里的 com 是"一起"、mittee 是"被派出去的人"；accom-modation 里的 modation 是"使合适"；sym-metry 里的 metry 是"测量"；gram-matical 里的 matical 是"……之学"；swim-ming 里的 ming 只是一个把动词变名词的尾巴；ham-mer 里的 mer 只是声音的一块；com-ment 里的 ment 是把动词变成"东西"；telecom-munications 里的 munications 是"共同分享"。

这些话你现在看着像文字游戏，可它救过我一次拼写：一个词你按意思拆，就不会按字母猜。字母是意外，意思是词。

一张写坏的纸

objective 是客观的、也是目标；obscurity 是默默无闻、也是晦涩；obsessive 是执着的（带病态色彩）；occasion 是场合；occur 是发生——注意它的名词是 occurrence，双 r 双 r，写错的人很多；odorous 是有气味的；official 是官方的；old 是旧的；once 是一次、曾经；onto 是到……上面；operate 是操作、是动手术；oppose 是反对；optical 是光学的；or 是或者；orderly 是有秩序的；originality 是原创性；otherwise 是否则；outcome 是结果；outlaw 是不法之徒、也是宣布非法；outside 是外面；over 是在……上方、是结束；overhear 是偷听到；overstate 是夸大；overwhelmingly 是压倒性地；oxide 是氧化物。

objective 一个词两个身份，要分清：作形容词是"客观的"，作名词是"目标"。沈璐说"你为了显得 objective，用了 probably"，用的是形容词；项目书写 objective，用的是名词。中文都译得出来，英文里是两个岗。

overhear 值得单独记：它不是"听见"（hear），也不是"听"（listen），是"我本来没在听，正好听到了"。所以你 overhear 的东西，通常不该被当成证据——正文里沈璐骂的就是这个：把一句在楼梯口 overhear 来的话写进记录，它就会凭位置获得它不配有的权威。

overstate 和 exaggerate 意思接近，但 overstate 更书面，常跟 the case、the risk、the importance 连用；它暗含"你说的方向是对的，只是过头了"，而 exaggerate 更接近"你在吹"。

pad 是垫子、是便签本；pair 是一对；panic 是恐慌；paradigm 是范式；parcel 是包裹；parole 是假释；particle 是颗粒；party 是聚会、也是当事方；passport 是护照；pat 是轻拍；patrol 是巡逻；payroll 是工资单；pearl 是珍珠；peel 是果皮、是剥；penguin 是企鹅；perceive 是察觉；perform 是执行、表演；periodic 是周期性的；permission 是许可；persecution 是迫害；personality 是性格；pessimist 是悲观主义者；petrol 是汽油（英式）；philosophical 是哲学的；phrase 是短语；pick 是挑；pig 是猪；pilot 是飞行员、也是试点；piracy 是盗版、海盗行为；placement 是实习安排、安置；planner 是计划者；player 是选手；pleasure 是愉悦；plug 是插头、是塞子；poem 是诗；poison 是毒；polish 是抛光；pollutant 是污染物；pop 是流行音乐、也是"砰"；pork 是猪肉；portray 是描绘；possibility 是可能性；pot 是锅、是罐；poverty 是贫穷；preach 是说教；precisely 是精确地；predominance 是优势；pregnancy 是怀孕；preoccupied 是心事重重的；prescription 是处方；prestige 是声望；prevalence 是流行程度；price 是价格；prince 是王子；priority 是优先；probably 是大概；procession 是队伍；productivity 是生产率；profitable 是有利可图的；prohibitive 是"高得让人不敢碰的"；prompt 是迅速的、也是促使；propagate 是传播；proposal 是提议；prosper 是兴旺；psychiatric 是精神病学的；publish 是发表；punctuality 是守时；purple 是紫色；put 是放。

prestige 和 notoriety 是一对反面：prestige 是"因为好而有的声望"，notoriety 是"因为坏而出的名"。中文都可能是"名气"，英文里一个是资产一个是负债。

prohibitive 常跟 cost、price、tax 连用，意思是"高得让你不敢做"，不是"禁止的"——禁止的是 prohibitive 的字面义，但现代英语里最常用的是"贵得离谱"。这个词在你写"为什么这个方案不可行"时非常好用。

prompt 有两副面孔：形容词"迅速、及时"（a prompt reply），动词"促使"（prompt somebody to do）。两副面孔都很正式，作文里用得上。

quality 是质量，quarter 是四分之一，quick 是快，quote 是引用——注意 quote 和 cite 不同：quote 是"照抄原话"，cite 是"指出出处"。你 cite 一篇文章，quote 一句话。

radical 是激进的、根本的；raging 是狂怒的、肆虐的；raise 是举起、提高；rapid 是快速的；rather 是相当、宁愿；raw 是生的、未加工的；readily 是"乐意地、容易地"；realm 是领域；rebellion 是叛乱；recently 是最近；recitation 是背诵；record 是记录；recruitment 是招募；reduce 是减少；referendum 是公投；reformation 是改革；refugee 是难民；regardless 是"不管"；registration 是登记；rehearsal 是排练；rejoice 是欢喜；relaxation 是放松；relic 是遗迹；rely 是依赖；remembrance 是纪念；remove 是移走；renovation 是翻新；repayment 是偿还；replacement 是替换；repression 是压制；reputable 是"有声望的、靠得住的"；researcher 是研究者；reservoir 是水库、是储备库；resist 是抵抗；respect 是尊重；responsibility 是责任；restraint 是克制；retail 是零售；retire 是退休；retrospective 是回顾的；revengeful 是记仇的；revision 是修订；rewarding 是有回报的；rich 是富的；rifle 是步枪；ring 是环、是响；risky 是有风险的；rob 是抢；rod 是杆；root 是根；rough 是粗糙的；rub 是摩擦；ruler 是尺、是统治者；rust 是锈。

reputable 是"名声好"的正经说法，比 famous 更可靠：famous 只说有很多人知道你，reputable 是说知道你的人里面，正经的那部分认为你可靠。这个区别在推荐信里很要紧。

夜里的实验楼

sailor 是水手，salon 是沙龙，sample 是样品，sarcastic 是尖刻的，saturate 是饱和，say 是说，scarce 是稀缺的，scenery 是风景，scholar 是学者，scientist 是科学家，scramble 是仓促地爬、是争抢，scripture 是经文，seat 是座位，secular 是世俗的，seedling 是幼苗，segregation 是隔离（尤其种族隔离），selfish 是自私的，senior 是年长的、是上级，sensor 是传感器，sequence 是顺序，service 是服务，seven 是七，sexual 是性的，shaky 是发抖的，shark 是鲨鱼，sheep 是羊，shift 是转变、是班次，shit 是脏话（比 fuck 轻一档，但仍然是粗话），short 是短的，shout 是喊，shrimp 是虾，shyly 是羞怯地，sight 是视力、是景象，silence 是沉默，similarity 是相似，sin 是罪，singularity 是独一性（数学里的奇点），sitting 是就座、是一次会议，skate 是滑冰，ski 是滑雪，skip 是跳过，slap 是拍打，sleeve 是袖子，slippery 是滑的，smart 是聪明的，smoke 是烟，snake 是蛇，snowy 是下雪的，socialist 是社会主义者，soft 是软的，solemn 是庄重的，solitary 是孤独的，somebody 是某人，son 是儿子，sorrowful 是悲伤的，southeast 是东南，spacecraft 是航天器，spark 是火花，species 是物种，spectator 是观众，speeding 是超速，spice 是香料，spirit 是精神，spoil 是弄坏、是宠坏，spontaneity 是自发性，spotlight 是聚光灯，spy 是间谍，stage 是阶段、是舞台，staircase 是楼梯，standby 是待命，startle 是使惊跳，stationary 是不动的，stay 是停留，steel 是钢，stiffen 是变硬，

stipulate 是明文规定，stony 是石头的，stove 是炉子，strangle 是扼死，streamline 是精简，striking 是显著的，struggle 是挣扎，stumble 是踉跄，stylish 是时髦的，subordination 是从属，substitute 是替代，subway 是地铁，suck 是吸，sugar 是糖，sulfur 是硫，sunset 是日落，supermarket 是超市，supplementary 是补充的，supposition 是假定，surface 是表面，surplus 是剩余，susceptible 是易受影响的，sustainability 是可持续性，sweep 是扫，sword 是剑，syndrome 是综合征。

sarcastic 和 ironic 不是一回事。ironic 是"事情本身的反讽"（消防局着火），sarcastic 是"人说的话带刺"。前者是处境，后者是语气。沈璐大部分时候是 sarcastic，生活本身是 ironic。

saturate 用法很省事：溶液饱和了是 saturated，海绵吸满了也是 saturated，市场饱和了还是 saturated。它的核心是"再也装不下了"。所以当一个人说 I am saturated，他是说"我满了，别再给我了"。

scarce 是"稀缺"，但它比 rare 更强调"不够用"：rare 是少，scarce 是少而需要。经费 scarce，时间 scarce，好脾气也 scarce。

senior 是个关系词：senior 一定有 junior 在对面。所以它不单是"大四"，还是"比你早来的人"。中文一个"学姐"就把这层关系抹平了，英文里它还在。

sight 有一串固定搭配值得记：at first sight（一见之下）、out of sight（看不见了）、a sight for sore eyes（看见你就好了）。第三个在信里出现，比 nice to see you 有分量。

silence 和 quiet 的差别在"有没有期待"：quiet 是环境安静，silence 是"该有声音而没有"。所以尴尬的沉默是 silence，图书馆的安静是 quiet。

singularity 有两层：日常是"独特"，物理里是"奇点"。同一页上，沈璐用它说"每个实验室的独一性"，林一记的是另一层——词的多义不是麻烦，是省力：你学一次，用在两处。

slippery 除了"滑"，还有"靠不住"的意思：a slippery character 是"滑头"。这个词的引申很自然——抓不住的东西，就是滑的。

stipulate 是"以条款形式规定"，比 require 更硬，因为它暗示"写在纸上，签过字"。合同 stipulate，须知也 stipulate。

striking 是"显眼的、给人强烈印象的"，不是"打人的"。a striking similarity 是"惊人的相似"，这个搭配在比较两组数据时很好用。

susceptible 后面跟 to：susceptible to error、susceptible to infection。它是"容易被……影响"，中性偏弱，不是说人不行，是说条件不利。

syndrome 是"综合征"，也是"一整套同时出现的毛病"。校园里那种"熬夜、咖啡、心慌、拖延"的组合，就有人管它叫一个 syndrome——词用得松，但意思传到了。

最后一页

先把几个前面漏掉的补上。accent 是口音，也是"重音"；affordability 是"负担得起"，一个很现代的词，讨论学费、房租、器材时都在用；airline 是航空公司；ambition 是野心，也是志向，比 dream 更"要去做"；amount 是数量，跟 number 分工——amount 配不可数，number 配可数，这一条

几乎每次考试都有人错；and 是最简单的连词，可它在句首时是修辞（And then it happened），不是语法错误；backward 是向后的，也是"落后的"，这个词的第二个意思带着评判，用的时候要小心；candidacy 是候选人资格；capitalist 是资本家；carbohydrate 是碳水化合物；category 是类别；cautious 是谨慎的；chalk 是粉笔；chat 是聊天；clamp 是夹子；clause 是条款，也是从句；month 是月；number 是数；protocol 是规程，也是网络协议；province 是省；sadden 是使悲伤。

amount 和 number 这条值得讲透：你能说 a large amount of water，不能说 a large amount of students；说 students 要用 number。反过来，a number of 后面跟可数名词，意思是"若干"，而 the amount of 后面跟不可数，意思是"总量"。这不是背规则，这是数东西的方式不同：一个是"多少"，一个是"几个"。

protocol 在实验室里不是"礼貌"，是"规程"：一步一步，谁在什么条件下做什么。它的另一个意思是外交礼仪，第三个意思是电脑之间的通信约定。三个意思共有一个内核：事先约定好的做法。你一旦看懂这个内核，这个词在三个场景里都不会认错。

clause 也一样：合同里的"条款"和语法里的"从句"，共有的内核是"一个被框起来的单元"。

take 是拿，tame 是驯服、是"温顺的"，target 是靶子、是目标，tea 是茶，technician 是技术员，telephone 是电话，temperature 是温度，ten 是十，tenure 是终身教职、是任期，terribly 是非常（口语里常作弱化的"很"），tertiary 是第三级的（高等教育就是 tertiary education），than 是比较连词，theirs 是他们的，therapist 是治疗师，thesis 是论文、是论点，thinking 是思考，thorny 是棘手的，threat 是威胁，thrive 是茁壮成长，thunderous 是雷鸣般的，tidy 是整洁的，tilt 是倾斜，tip 是尖端、是小费、是提示，token 是象征、是代币，ton 是吨，top 是顶部，total 是总的，towards 是朝向（英式，美式 toward），track 是轨迹、是跑道，trafficking 是非法交易，trainer 是教练，transcript 是成绩单、是转写稿，translate 是翻译，transplant 是移植，travel 是旅行，treaty 是条约，triangle 是三角形，tricky 是棘手的，triumphant 是胜利的，troublesome 是麻烦的，trustee 是理事，Tuesday 是星期二，turbulent 是动荡的，tutorial 是辅导课，ugly 是丑的，under 是在下面，underlying 是潜在的、根本的，undo 是撤销、是解开，uniformity 是一致性，universal 是普遍的，up 是向上，uprising 是起义，urgent 是紧急的，useless 是无用的，vacation 是假期，validate 是使有效、是验证，vanity 是虚荣，vast 是广阔的，vendor 是小贩、是供应商，verdict 是裁决，vertical 是垂直的，vibrate 是振动，victory 是胜利，vinegar 是醋，virginity 是童贞，visible 是可见的，vivid 是生动的，volatile 是易挥发的、是易变的，voluminous 是卷帙浩繁的，vulgar 是粗俗的，waitress 是女服务员，wander 是漫步、是走神，warmth 是温暖，washing 是洗涤，watt 是瓦特，weakness 是弱点，weave 是编织，weed 是杂草，weird 是古怪的，wharf 是码头，where 是哪里，while 是当……时、是"然而"，who 是谁，why 是为什么，wield 是挥舞、是行使（wield power），win 是赢，wipe 是擦，with 是和……一起，witness 是证人，wool 是羊毛，workshop 是车间、是工作坊，worst 是最坏的，wrap 是包起来，wrinkle 是皱纹，yawn 是打哈欠，yes 是是，youngster 是年轻人，zealous 是热心的。

几个值得单独停的。tenure 在中文里没有对应的日常词，它是一个制度：拿到 tenure，你才不会被随便解雇。所以它常被当成"安全感"的代名词，也常被当成"从此可以不干活的许可证"——两种说法都有人在说。

underlying 是我在这一章里最想让你记住的词之一。underlying cause 是"根因"，underlying assumption 是"底下的假设"，underlying 的意思是"表面之下，但支撑着表面"。你写报告时，凡是能指出一个 underlying 的东西，你的文字就往下走了一层。

verdict 是"裁决"，它跟 judgement 不同：judgement 是判断，可以是私下的、温和的；verdict 是正式宣布的那个结论，来自陪审团、来自考官、来自你自己在某一天下的决心。

volatile 有两层，一层是化学的（易挥发），一层是人的（情绪易变）、市场的（动荡）。这两层共有一个画面：说走就走。

wield 是个"有力气"的词：wield a tool 是使得动工具，wield power 是行使权力，wield influence 是施加影响。它的搭配对象通常是"有重量的东西"。

wrinkle 除了皱纹，还有一个很好用的短语：a new wrinkle 是"一个新花样、一个新变数"。这个用法在口语里很活。

zealous 是"热心到有点过"的意思，比 enthusiastic 多一层"过了头"的可能。所以它常跟 reformer、campaigner 连用，很少用来夸人温和。

yawn 这个动作，在课堂上是被禁止的，在实验室里是危险的——你打一个哈欠的三秒钟，够一锅东西沸出来。有些词你记住它，不是因为考试，是因为它跟一个具体的后果绑在了一起。

wharf、vinegar、wool、watt 这类词，属于"知道就好"的一档：它们不常出现在议论文里，但会出现在阅读的细节里，而细节题最喜欢考的就是这种"看起来不重要"的词。

这一章该怎么读第二遍

第一遍你读的是事：一个男生在四楼做砸了一锅硫酸铜，被一个大四的学姐说了一顿，然后在一个夜里重做了一遍。这一遍你读得快，因为你想知道后来怎么样了——这就对了，读故事就该这样读。

第二遍你要读的是"位置"。同一个词，在句子里站的位置不同，它的分量就不同：tack 在"图钉"的位置上是个小物件，在"抢风调向"的位置上是个动作，在"方针、路子"的位置上是一种判断（change tack 是"改弦更张"，比 change plan 老练）。你要练的就是这个：不看词，看它站在哪儿。一个词站在主语上，它是主角；站在定语里，它只是给主角上颜色。

第三遍你才去读"意思"，而且是读它跟旁边那个词的关系。中文的"做实验"到了英文里会分成 do、run、carry out、perform 四个动词，选哪个不取决于"做"，取决于这件事有多正式、做的人是谁、说的人站在多远的距离上。这就是搭配的重量：它不是语法，它是分寸。

关于"背了会忘"

会忘是正常的，忘得快说明你记的方式不适合你。你在词表上记的是一条"孤立的线"：英文—中文—下一个。孤立的线最容易断，因为它两头都没有钩子。这一章做的事，是给每个词加钩子：clamp 的钩子是"夹住滤瓶再倒"这个动作，funnel 的钩子是漏下去的那个形状，pending 那种"悬着"的感觉，钩子就是那锅没结晶的蓝色液体。

所以你复习的时候不要问"这个词什么意思"，要问"这个词在那天是干什么用的"。前者要你回忆

一个中文词，后者要你回忆一个场景，而人对场景的记忆，比对词条的记忆牢得多。

还有一个更要紧的事：不要试图一次记全。一个词你有三层认识就够了——第一层是"我认识它"，第二层是"我知道它常跟谁在一起"，第三层是"我敢用它"。多数人卡在第一层，然后以为自己会了。真正让你在考场上得分的是第二层：你不需要会用 ubiquitous，你只需要知道它旁边常站着 information、problem、practice，于是你能在四个备选里认出"这句人话"。

关于写作里怎么用

这一章里有大量的名词：protocol、clause、category、inventory、verdict、regime、verdict 这一串，看着像术语，其实是写作的骨架。你在作文里写"我们要遵守规则"，是小孩话；写"规则是一种事先约定好的程序"，是大人话；而大人话的秘密，就是把动词换成名词短语：decide 换成 make a decision，arrive 换成 arrival，同意 换成 reach an agreement。

这不是为了显得高级，是为了让句子有余地。一个动词只有一个方向，一个名词短语可以挂三个修饰语，可以被"在什么条件下""为了什么目的""由谁负责"逐层包住。你去看沈璐那几段话，几乎每一句都有一个名词短语在承重：a scheduling failure、a piece of self-confidence with a decimal point、the pleasure of being right。这些短语不是装饰，它们是思想被压紧之后的样子。

反过来说，一篇全是形容词的文章是虚的。delightful、fascinating、wonderful 这些词，一个学期用三次就够了；用多了，读者会开始怀疑你没有具体的东西可说。真正让人信服的是具体的名词加准确的动词：the beaker spat、the reading drifted、the film formed。三个短句，没有一个形容词，你却能看见那一幕——这就是"具体"的力量。

402 给我的三件东西

第一件是记录的习惯。不是为了给谁看，是为了让明天的自己知道今天的自己在想什么。一个人在同一个问题上反复犯同样的错，多半不是因为他笨，是因为他没有留下可以对照的痕迹。记录是跟自己的记忆打官司时唯一的证据。

第二件是区分"我不懂"和"它不清楚"。这两件事看起来一样，其实完全相反：一个是我的问题，一个是材料的问题。多数人把前者当后者，于是把时间花在抱怨上；少数人把后者当前者，于是把时间花在重读上。你能分清它们的那一天，你的学习方式就变了一次。

第三件是允许自己重做。这一章里最重要的一次实验不是第一次，是那个夜里的第二次。第一次教会我什么叫"错"，第二次教会我什么叫"会"。没有第二次，第一次就只是一次难堪；有了第二次，第一次就变成了学费。

最后

这本书里的每一章，都是拿一个具体的地方，装一批抽象的字母。你读完会觉得"我好像认识了这个人"，那就对了——你不是认识了林一，你是认识了那间屋子里的说话方式，而说话方式就是词汇真正的家。

所以别把这一章当成一份加了故事的词表。它是一间屋子，屋子里的每样东西都有自己的名字，

名字不是贴上去的标签，是有人为了不再弄错而付出的代价。你把这些名字记住，不是为了考试，是为了有一天你站在自己的那间屋子里，知道该叫什么叫什么。

词是别人替你踩过的坑。你记住一个词，就是少踩一个坑；你用一个词用得准，就是替后面的人把那个坑填了一点。

这就是全部的道理。剩下的，只能一遍一遍地读，一遍一遍地写。

说回四六级这件事

我知道你读这本书的真正原因。你不是想认识林一，你是想过线，最好过得好一点。那我们就说这件事，说得直白一点。

四六级难，难的不是词汇量，是"熟悉的词在陌生的位置上"。你会发现阅读里几乎没有生词，可你就是读不快，因为你一直在做一件事：把每个词翻译成中文，再把中文拼成意思。这条路太慢，而且中间会漏。真正快的方式是反过来的：你先认出这句话在干什么——它在给结论、在举例、在转折、在限定范围——然后再看细节。结构的判断在前，词义的确认在后，这个顺序一颠倒，你就会永远觉得时间不够。

这也解释了一件事：为什么有的人词汇量不大，阅读却稳。因为他读的是"句子的功能"，不是"词的中文"。你手上这一章，其实就是给你练这个的：每一段都在做一件事，有的在铺场景，有的在给证据，有的在反驳，有的在收束。你把每一段的功能用一句话说出来，你的阅读就已经换了一套引擎。

再说听力。听力真正的门槛不是速度，是"词的听觉形态"你没有。你在纸上认识 thorough，可它在连读里是另一个东西；你认识 climate，可它在句尾会弱下去。解决办法很土：aloud。把这一章里你觉得拗口的句子读出来，读到你不用看也能说，读到你听见自己的声音不觉得别扭。朗读是把"眼睛的词"变成"耳朵的词"的唯一便宜办法，一天十分钟，比刷两套题有用。

再说写作。你现在的作文大概有两个毛病：一是句子都对，但没有一句是"只有你能写出来的"；二是想表达复杂意思时，句子会塌。第一个毛病的解药是具体：不要写 "the experiment was very important"，写 "the experiment decided whether I would be allowed near the balance again"。第二个毛病的解药是名词化，前面已经说过，这里再补一句：名词化不是为了长，是为了让句子能挂住更多信息。

还有翻译。汉译英的难点从来不在词，在中文没有的那些形式：时态、单复数、冠词、非谓语。你写"我去过实验室"，英文必须交代"去过"是经验（have been）还是经历（went）；你写"实验结果不理想"，英文必须交代是单数还是复数、是特指还是泛指。这些不是语法题，是"你必须替读者做决定"的地方。中文可以含糊，英文不允许你含糊——这就是翻译真正的门槛。

最后说心态。你大概已经听过"要坚持"，我不想再说。我想说的是另一件事：允许自己有一段时间是模糊的。你读一篇东西，第一遍只懂了六成，这是正常的；你读第二遍懂了七成半，第三遍忽然全通了——这不是奇迹，这是大脑在后台整理。很多人失败，不是因为他们不够努力，是因为他们不能忍受那六成的模糊，于是每一篇都从头再来，永远停在第一遍。

模糊就模糊，泡久了自然就清了。这句话不是安慰，是方法。

一点技术的尾巴

如果你只想带走三句话，就带这三句。

第一句：先看这句话在干什么，再看它说了什么。 阅读的所有技巧都从这一句长出来。

第二句：一个词你只有在句子里见过它三次，它才真的是你的。 所以别背词表，去读能让你反复遇到它的东西。

第三句：写不出来，是因为你没想清楚；想不清楚，是因为你没写。 这两件事互为因果，唯一的出口是先写一句难看的，再改。

这三句话不需要你相信，只需要你用一个月。一个月之后你自己会知道它们对不对。

结尾

四楼那间屋子还在。灯还是那个灯，通风橱还是那个声音，天平间还是要脱鞋才能进。每年九月都有新的人站在门口，手里拿着计算器，把墙上的字读两遍，懂一遍。

他们中的大多数人，会在第一次做砸的时候怀疑自己不适合这一行；会在被学姐说一顿之后，在宿舍的床上翻来覆去；会在某个夜里，因为一锅终于结晶的东西，忽然觉得这一行还不错。

我写这一章，是为了把那间屋子里的名字留给你。名字不是用来考的，名字是用来叫的。你有一天会站在自己的那间屋子里，指着一样东西说出它的名字，那一刻你不会想起这本书，你只会觉得理所当然——而理所当然，就是学会的样子。

回头看那六个场景

第一个场景在门口。一个人站在门槛上读墙，读的是三张纸：横幅、缩写表、安全须知。你如果只看情节，这是一段"等待"；可它其实是全章的语法：一个人要先知道规矩，才被允许动手。所以那一段里全是"制度词"——authority、compliance、condition、clause。它们不好看，但它们是那间屋子的地基。

第二个场景在天平间。那里发生的事很小：一个人碰了一下台面，读数漂了。可这一段把全章最重要的一对词摆出来了：precision 和 accuracy。前者是"每次都一样"，后者是"离真值近"。一台天平可以很 precision 却完全不 accuracy，一个人也可以很勤奋却完全在错的方向上。这一对词你要是分清了，你以后看任何数据都会多一层警惕。

第三个场景是做砸。锅里的蓝色液体沸出来，产物损失，收率四成一。这一段的关键词不是 failure，是 cause。沈璐后来那句话之所以狠，是因为她不允许你把 cause 藏进 probably 里。一个人写作的水平，很大程度上就是他把 cause 说清楚的能力。

第四个场景是她的批评。这一段词最密，也最"文"：conceit、vanity、overstate、fabrication。为什么这些词必须出现在这里？因为它们在日常里太重了，重到只有在一个真正严重的时刻才配用。你要是把它们用在"我今天迟到了"这种事上，词就被你用薄了。

第五个场景是规矩。周敏在黑板上一口气写了六条。这一段是全章最"不文学"的地方，可它是全章的中心：一个房间之所以能让人放心地犯错，是因为它把边界画在了明处。ban、forbid、stipulate、permission 这些词，在这里不是限制，是承诺。

第六个场景在夜里。整栋楼只剩几盏灯，一个人守着一只烧杯，等一层薄膜形成。这一段词最

少，句子最短，因为人在那种时刻不说话。可它是全章的落点：所有前面那些制度词、数据词、批评词，最后都要落到一个人的四秒钟注意力上。

那几个最难的词

有几个词，我写的时候确实费了力。比如 augmentation，它太正式，日常几乎不用，可它在"把错误放大"这个意思上比 increase 更准，因为 augmentation 暗含"在原有基础上加了一层"。再比如 subordination，中文说"下级服从上级"，英文一个名词就装下了整条关系，可它不好塞，我只在一个讲制度的地方放了它。

还有 virginity。这个词在正经语篇里几乎不可能自然出现，我把它放在周敏的一句玩笑里——新坩埚第一次用，她说了那句话。这不是为了猎奇，是因为一个实验室确实有这种黑色幽默，而一个词的"可用场景"里，本来就包括玩笑。

denotation 和 connotation 这一对，是语言学的术语，可它们讲的是人人都懂的事：一个字面的意思，和一个"带出来的味道"。你写报告要用前者，你写散文要用后者，用错了就像穿着实验服去参加婚礼。

paradigm 这种词，最大的用处不是让你显得有学问，是让你能一句话说清"一整套看法"。当你说"这是一种范式"，你其实是在说：这不是一个错误，这是一整套彼此支撑的假设。这个区分，在你以后判断任何"主流说法"时都用得上。

wield、weave、haul、hoist 这几个动词，都属于"身体记忆"一类：它们各自对应一种用力的方式。你如果只记中文（挥舞、编织、拖、吊起），你永远用不对；你要记那个动作的画面。动词是最不该用中文背的一类词。

一个提醒

读到这里，你可能会有一个错觉：觉得自己已经会了。这很正常，因为"看懂"和"会用"之间隔着一层很薄但很硬的膜。要捅破它，只有一个办法：写。写三句，用上你今天新认识的三个词；写错了也不要紧，写错至少暴露了位置。

还有一件事：不要把这一章当成"读完就完了"的东西。它最好的用法是当参照物——你以后读到任何一篇说明文、任何一份说明书、任何一段实验记录，都可以回来对照：它的句子在干什么？它把 cause 说清楚了吗？它有没有把 probably 当成挡箭牌？

你一旦开始这样读，你读的就不再是英语，而是别人怎么想事情。语言学到这一步，才算真的开始。

三天之内

我不想在最后给你一段励志的话，只想给你三天。

第一天，回正文里找三句你觉得"读起来最顺"的句子，抄下来，然后拆开看：主语是谁，动词是什么，剩下的部分挂在谁身上。你会发现所谓"顺"，几乎都是因为主干短、修饰语放得规矩。你把这三句的样子记住，比记十条语法规则有用。

第二天，挑五个你以前只敢认、不敢用的词，各写一句话，写跟你自己有关的事。不是造句，是说话：你真的会在某天说"我今天很 clumsy"，那就写它。写完之后读一遍，如果听着不像人话，就改到你敢说出口为止。这一步很笨，可它是唯一能让词从"认识"变成"能用"的路。

第三天，找一篇短的英文说明文，任何一篇，说明书、新闻、论文摘要都行，做一件事：把每一段的功能用一个中文短语写在旁边——铺场景、给证据、反驳、限定、收束。写完之后你回头看正文，你会发现林一那间屋子里的每一段，也都能这样标。

三天之后你不一定记住多少词，但你会换一种看法：从"这句里有没有生词"变成"这句在干什么"。这个转换一旦发生，你的分数只是时间问题。

最后一句，留给还在犹豫的人：你不需要准备好才开始，你需要开始才可能准备好。那间四楼的屋子，也是先有人进去，才有了规矩。

写完这一章之后，我又去了一趟四楼。

门开着，里面在上课，一个年轻老师正在讲误差，黑板上写着新的字，讲台边放着一只新的杯子。我没有进去，只在门口站了两分钟，听了三句话，然后走了。通风橱还是那个声音，楼梯还是那个拐角，墙上那张缩写表换了新的，格式一模一样。

我忽然想起第一天站在这里读墙的样子。那时候我连 abbreviation 和 acronym 都分不清，把 clamp 当成"夹子"就完事，看到 protocol 只想到"礼貌"。现在我站在门口，听见里面那个老师说"这个数不要写四位"，我心里会替那个学生紧张一下。

这就是一间屋子给人的东西：不是知识，是一副耳朵。你听见一句话，就知道它后面要出什么事，这种预感不是天生的，是被同一间屋子里的人反复纠正出来的。

所以我把这一章留在这里，像周敏把那个文件夹留在抽屉里一样。也许有人会读，也许不会。可它在那儿，那间屋子就还没有完全关灯。

两个人的说话方式

这一章里有两个成年人在教林一做事，她们的说话方式完全不一样，而这件事本身就是词汇课。

周敏说的是"程序语言"：短句、动词开头、主语经常省略。她不说"你应该注意安全"，她说"no one works alone"。她的句子像操作说明，一句一个动作，句与句之间靠顺序连接，不靠转折词。这种语言的词，多是名词和动词：duty、protocol、category、inventory、expenditure。你读她的段落会觉得干，那种干是刻意的——她要的是"不会听错"。

沈璐说的是"评价语言"：长句、插入语多、爱用抽象名词。她说"那不是一次测量，那是一段自带小数点的自信"，一句话里有三个抽象名词。她的词多是形容词和抽象名词：conceit、vanity、overstate、dubious、plausible。你读她的段落会觉得锋利，那种锋利也是刻意的——她要的是"你没法糊弄"。

同一个人物，两种语言，对应两种关系：周敏给你边界，沈璐给你判断。你在生活里也会遇到这两种人，而你能不能听懂他们，取决于你有没有这两套词。你只有程序语言，你会觉得沈璐在挖苦你；你只有评价语言，你会觉得周敏在敷衍你。

所以学词不只是学意思，是学"这个人在用什么方式跟你说话"。你听出方式，词就不再是孤立

的，它带着说话人的位置一起进来。这也是为什么我一直反对只背词表：词表把所有人都压成了同一个声音，而真正难的是分辨声音。

三种词，三种活法

这一章的词，粗粗可以分三档，你心里有这三档，用的时候就不会乱。

第一档是"制度词"，像 compliance、protocol、category、clause、levy。它们的共同点是：一个人说了不算，得一群人认。这类词你写议论文最好用，因为它们自带"这不是我一个人的看法"的分量；但也最容易用空，因为你不懂制度的时候，写出来就是腔调。

第二档是"身体词"，像 haul、hoist、wield、weave、clamp、wipe。它们每一个都对应一个动作，你只要做过那个动作，词就长在身上。这类词是作文里的盐：不用多，两三处就能让一段文字有手感；一处没有，整段就是飘的。

第三档是"评价词"，像 conceit、dubious、earnest、diligent、solemn。这类词最难，因为它们不描述世界，它们描述"我对世界的态度"。你用一个评价词，等于当众表态。所以它们要么不用，要用就得准——不准的评价词，比不用更伤人。

三档词的学法也不一样：制度词靠读，身体词靠做，评价词靠改。你把这一章里那些评价词找出来，试着把它们换成你平时会说的话，再换回去，你就知道它们的斤两了。

写到这里，我该停手了。

再往下写，就会变成我在替你读书。而这一章真正想给你的，不是九千多个词的用法，是一个很朴素的动作：遇到一个词，先问它在这件事里干什么。你把这个动作做过三百次，剩下的词你自己就会拆了。

所以合上这一页，回到开头那个门口去。这一次你不用管情节，只看那些加粗的词：它们为什么站在那个位置，前后各是谁，换成同义词行不行。你要是有耐心做三遍，这一章就用完了；你要是只做一遍，它至少也让你知道了四楼有一间屋子，屋子里的人说话很准。

准，是这一行唯一的体面。

第 4 章　城市的另一面

The last train to the river was late by approximately three minutes, and the announcement played twice before the carriage doors sighed open. I stepped in with a paper bag of bread under my arm, a certificate of volunteer training folded into my jacket, and no clear idea what a service station did at nine on an August night. The brake let go. The train continued south, and the arrival board above my head flickered through stations whose names I could not yet pronounce.

Two stops before the end, a man got on with a bumper sticker on his suitcase and a bizarre hat that looked like a bowl turned upside down. He sat aside from everyone,

clenched his hands around the handle, and stared at the floor as if it owed him money. Opposite him a woman was bending over her phone, combing through a page of advertisements for affordable rooms, muttering at every price. The average rent in that district, I later learned, was roughly a third of a young worker's earnings, and the supply of rooms near the highway was ample but the quality was not. Anyhow, nobody on that train looked like they were going home to a mansion.

East of the Platform

The station hall smelled of wet rubber and cheap detergent. Aunt Chen waited under the clock with a cord around her neck and a bell in her hand, the way some people carry a bell to announce themselves in a crowd. She ran the volunteer booth: six square meters, one stool for her, one for visitors, a kettle, and a stack of stationery that a charitable supplier in the neighbourhood had donated.

"You are the freshman," she said. It was not a question. "Aunt Chen. I administrate this booth. That means I answer the phone, keep the register, and assess whether a candidate for assistance is honest or just lazy. Sit."

I sat. She pushed a form across the counter, one page, with a heading in bold type. I read it slowly, because the print was small and because I wanted to postpone the moment when I would have to sign something.

The heading said that admittance was conditional upon completion of training, a signature on the liability waiver, and a pledge to abide by the stipulations below. It said the validity of the credential was twelve months, and that its expiration did not cancel the obligation of confidentiality. It said that by signing I would consent to a background check and agree that any breach of conduct could end my admittance without notice.

Farther down, in smaller print, the form added three sentences in the flat voice of people who have been disappointed before. Volunteers must treat every resident with respect, regardless of appearance, accent or legal status. Volunteers must not accept gifts, cash or any other payment from the people they help. Volunteers must report any suspicion of abuse, neglect or fraud immediately to the officer on duty.

The third sentence was blunt in a way that made me uneasy, and the rule about cash baffled me until Aunt Chen explained, with a shrug, that volunteers had once been caught selling allotments of charity rice back to the families who were supposed to receive it. "Kindness without rules," she said, "is just business with a smile."

She watched me sign, then folded the page once and slid it into a plastic sleeve. "Good. Now you are insured, which means the university cannot pretend it never met you."

Beyond the Wall

Outside, the rain had thickened into fog. Across the street, the lamps stopped at the edge of a low wall, and past the wall the city changed its mind. Narrow lanes. Wires hanging in loops like something spilled. Four-storey buildings packed so tightly that neighbours could shake hands between balconies. Shop signs stacked above each other, air-conditioners dripping on the heads of people passing underneath, and a smell of frying oil, drain water, and something sweet I could not identify.

Fifteen minutes earlier I had been reading a safety brief in a warm room. Now I was looking at the thing the brief had been written about. I stood there long enough for the rain to reach my shoulders, and I remember thinking, with the honesty of a person who has just been embarrassed, that I had walked past this wall perhaps twenty times and had never once turned my head.

"You will get used to it," Aunt Chen said behind me. She had come out to smoke, standing under the awning with her back to the rain. "Everybody does. That is the problem."

The Lane Arithmetic

The lane behind the wall had no name on any map I owned, but it had a pattern. Every dozen meters or so the buildings leaned a little further, the wires hung a little lower, and the puddles got a little deeper. Two women were scrubbing a rusty patch of pavement where somebody's cooking oil had spilled, working with stiff brushes and a bucket of grey water. A stray pigeon walked between their feet, unbothered, and a hairy rabbit sat in a cage outside a shop that sold cigarettes and phone cards. No one looked at me. That was the first thing I noticed and the thing I kept noticing: in the lane, a stranger is not interesting.

The man who owned the building was called Old Wu. He was seventy-something, wearing a woollen cap in August, and he was proud of his property in the way that a gardener is proud of a tree he has trained into a shape nobody asked for. The ground floor was a shop; above it, a basement that he rented out as a dormitory; above that, four floors of rooms; and above that an attic he used for storage and for pigeons. There was no elevator, no fire escape wider than a shoulder, and, he told me with satisfaction, no shortage of tenants.

"Thirty-two rooms," he said, holding up three fingers and then two, in case I could not do the arithmetic. "Seventeen families. Everyone pays on the first. Nobody complains. If they complain, they leave, and the next family comes Tuesday."

He took a lease out of his pocket — folded so many times that the creases had gone soft — and read the terms aloud as if he were reciting a poem he had written himself.

The tenant pays monthly; the tenant accepts the premises as they are; the tenant waives any claim for damage caused by rain, pipes, wiring or neighbours; the tenant consents to inspection with twenty-four hours' notice; the tenant acknowledges that the landlord's mortgage on the building has nothing to do with the tenant. The rent rose ten percent in June and, he explained, would rise again when the subway station opened, because proximity is a commodity and commodities have prices.

Since I had come with Aunt Chen, he was polite to me, in the manner of a man who is polite to uniforms. But when a young couple came down the stairs to negotiate a late payment, his tone hardened into something I had only read about in newspapers. He did not shout. He mentioned, pleasantly, that their belongings could be put on the street legally and that the police would not intervene in a civil matter. The girl began to cry, quietly, the way people cry when they have decided not to give anyone the satisfaction of a scene. The boy held her wrist. Neither of them looked at me.

I wanted to say something. I did not, because I was twenty-three days into being eighteen, and because I could not think of a sentence that was true and also useful. That failure is the honest beginning of this chapter, so I am putting it here rather than at the end.

The Print Shop

On the corner there was a print shop the size of a closet, and its window was covered with samples: wedding invitations, name cards, menus, a banner for a funeral, and a sheet of typesetting examples that the owner had pinned up to show customers how hyphenation works. The sheet was yellow with age, and someone had drawn a red line down the middle of every word to mark where a line could break:

"com- / ma. com- / mence. com- / municative. im- / mediately. sym- / metry."

I stood in front of that window for a long time, and I am aware of how strange that sounds. But I had spent eleven years of English classes learning words as whole objects, and here was a wall saying that a word is also a joint, a place where it can be taken apart. It was the first time the language had looked mechanical to me — less like a picture and more like a machine with seams.

The printer came out to see what I was staring at. He was about forty, with ink under his nails, and he explained that the sheet was there for apprentices, not for customers. "Nobody reads it," he said. "Tourists photograph it." He asked whether I was a student, and when I said yes, he laughed and said that students were the only people in the lane who ever asked questions and the only people who never bought anything.

English Corner

Two doors down, a community room had been turned into a free English corner on Thursday nights. The blackboard still carried the lesson from the week before: a list of phrases, a drawing of a clock, and, at the bottom, in chalk, the sentence "you oughtto practise every day" — written as one word, squeezed together by a hand that had copied it from a phone screen too small to read.

An old man was sitting in the front row with a dictionary open on his knees. He was learning English at sixty-eight because his granddaughter had married a man from Manchester and he wanted to say something to her children that was not translated. He asked me the difference between "shall" and "will", and I gave him an answer that sounded confident and was, I realised later, mostly wrong. He wrote it down anyway, in a notebook with alphabet tabs, and thanked me twice. Whilst he was writing, he hummed. Whoever says that old people cannot learn has never sat in that room.

I asked him whereabouts in the lane he lived. He pointed at the ceiling. "Up," he said. Very high. Very quiet. Very cheap. Three things," he added, holding up three fingers the way Old Wu did, "and you can only ever have two."

The Booth, Day Shift

The day shift was nothing like the night. Night was fog and one train an hour; day was a queue, and the queue smelled of wet umbrellas, instant noodles and disinfectant. By ten the plastic chairs were full, and Aunt Chen had stopped asking people what they wanted, because she already knew. She pointed, and they sat, and she took out a fresh page of the register.

The first woman had one leg and a folder of paper. She wanted the disability allowance, which she was entitled to, and which the computer said she had already received. It had not been received. The certificate from the hospital had been deleted from the system during an upgrade in May, and the hospital could not reissue it, because the doctor who had signed it had retired. "So I am," she said, in the flat voice of a person doing arithmetic for the hundredth time, "legally not disabled."

Aunt Chen wrote four words on her pad and underlined them twice: system, deletion, reissue, patient. Then she picked up the phone and began what she called a negotiation, which sounded less like a negotiation than like a series of polite threats. She never raised her voice. When she hung up she said, "Thursday. Bring the old prescription, not the new one." The woman did not thank her. She looked at the floor for a moment, then asked whether the booth had a wheelchair she could borrow to reach the bus, and we did, and that was the whole transaction.

The second was an old man with a hearing aid that whistled. He had a letter from the factory where his son had worked, written in the language that companies use when they want to be understood by lawyers and not by anybody else. It used the word injury eleven

times and the word responsibility zero times. The old man wanted to know whether the letter was good or bad. It was bad. It was so carefully bad that I could feel the craft in it, the way you feel the join in a table that has been sanded well.

Aunt Chen read it twice, then called a retired judge who volunteered on Wednesdays. The judge said the letter was not a refusal, it was a trap, and that if the son signed it he would be giving up something he did not know he had. "Do not let him sign," the judge said, and Aunt Chen copied it into her notebook in large square handwriting. The old man could not hear her, so she turned the page round and let him read it. He read for a long time. Then he took off his glasses, and I understood that he had been frightened for months and had not had one person to be frightened with.

The third case was a baby. A young mother brought a boy of about two with a rash and a fever, and she wanted to know which hospital to go to and whether it would cost anything, and while she was asking, the boy needed to urinate, and there was no toilet in the booth, and Aunt Chen produced an empty bottle with the calm of a woman who has solved this before. The rash turned out to be an allergy to the washing powder she had switched to in order to save four yuan. Four yuan. The whole morning cost her four yuan and half a day.

At noon a man came in with a plastic bag of documents and a story about a phone call from somebody who said he was from the bank. He had given them a password. He had watched the money leave his account, and then he had gone to the police, and the police told him the money had already moved four times and was probably in another province by lunch. He wanted us to tell him it was not his fault. Nobody told him that. There is a kind of help that consists of not lying, and it is the hardest kind to give.

On the wall above his head hung a poster that the district health office had sent in a roll of thirty and that nobody had ever taken down. It said, in red, that AIDS is not transmitted by sharing a bowl of rice, and it listed the ways it is transmitted, and it ended with a sentence somebody in an office had clearly fought for: testing is free, confidential, and does not require your name.

Aunt Chen saw me reading it. "Half the people in this lane have never been tested," she said. "Not because they are ignorant. Because they are busy." She pushed a stack of leaflets towards me. "Take them to the laundry on the corner. The women there work eleven hours. They will not read a leaflet, but they will look at a picture, and they will ask you questions while their hands are moving, and that is the only interview you will ever get."

So that was the job. Not saving anybody. Standing in a doorway with a stack of paper, waiting for a woman to finish folding a sheet so that she could ask me, in a voice too low for her employer to hear, whether a test would show up on her insurance.

The Reclaimed Edge

On Saturday we took the bus past the last stop, out to the edge of the district where the land stops pretending. The map calls it a development zone; the people who live there call it the paddy, because it was paddy, and because the water still remembers. The whole shelf of it was dredged out of the river in the sixties, and if you dig down a metre you find sand, then shell, then a layer of crust that is older than the city and does not care about it.

Aunt Chen was delivering two bags of clothes to a family she had known for nine years. The father had been a fisherman on the river until the fishing was banned, and now he worked four nights a week unloading trucks at the harbour. His wife cleaned offices. Between them they had bought a house with a yard, a barn at the back, and a view of the water that no estate agent would ever describe as a view.

The yard was a small economy. Three geese, one of them a tyrant. Six ducks in a concrete basin. A goat on a rope, eating the hedge from the inside. Two rabbits in a hutch that the father had built from pallets, and a pigeon cote on the roof with eleven birds in it, and a parrot in the kitchen that said, in a perfect local accent, "have you eaten", which is what people here say instead of hello. Under the eave a swallow had built a nest out of mud and it was full of mouths. The cat watched the nest. The father watched the cat.

The kitchen had one window, one tap, and a shelf of jars that his mother had labelled in her own shorthand: salt, sugar, five-spice, dried herb, honey from a cousin, and a bottle of something brown that she called medicine and that everybody else called ale. She was ninety-one and she cooked on a wood stove because she did not trust the induction plate. Her son called her Ma. She called him by his childhood name, which I will not write here, and she called me "the tall one", and she gave me a stool by the door.

She made porridge. She put a pinch of salt in it, and a spoon of honey, and she cut a lemon into quarters and set them on a saucer in case anybody wanted sour. Then she fried a dish of beans with a little pork, and steamed a bowl of peas with an egg on top, and sliced a potato into a hundred thin discs and fried them until the edges went brown. There were chopsticks in a tin, a chipped bowl of pickles, a plate of boiled peanuts, and a plate of something dark and spicy that she would not name until I had eaten it. It was duck. It had been smoked over tea leaves. It was, I am sorry to say, the best thing I ate that year.

She watched me eat with the unblinking attention of a woman who has fed soldiers. When I reached for a second helping she said something to her son, and he laughed and translated: "She says you eat like a peasant. She means it as a compliment. She was a peasant." Then she pushed the pudding towards me, a bowl of rice and sugar and lard, and said, "Eat. You are too thin. All students are too thin. It is the cafeteria."

I asked her about the river. She told me that when she was a girl the water came up to the second step of the house, and that in the bad year they ate the leaves of the bush behind the barn, and that the fish were so thick in the spring that you could catch them with a basket. She said the word fish the way other people say the name of a dead relative. Then she said the water was cleaner now, and the fish were gone, and both of those things were true, and she did not know which one to be glad about.

Her grandson, who was nine, took me out to the embankment. He had a toy elephant and a toy lion and a plastic dinosaur, and he lined them up along the wall and told me which one would win. Beyond the wall the river was the colour of weak tea. A tug was pushing two barges upstream, so slowly that you could not see it move unless you looked away and looked back. A fisherman — one of the four men still licensed to fish that stretch — was standing in a flat boat, working an oar with one hand and pulling a net with the other. The net came up silver, then empty, then silver again.

"Does he catch anything?" I asked.

"He catches enough," the boy said. "He catches the same fish every day. He throws it back. My dad says he is not fishing, he is practising."

The sun came out then, the way it does in the afternoon on that coast, and the water went from tea to tin. There was a shine on it that hurt to look at, and sparkles where the wind turned the surface over, and a haze sitting on the far bank like a curtain that somebody had forgotten to take down. You could see the peninsula from there, the mountains behind it, and the whole scenic arrangement of hills and water that the tourist brochure photographs from a different angle and calls a paradise.

It is not a paradise. But standing on that wall, with a nine-year-old explaining the rules of a fight between a lion and a dinosaur, with the smell of the stove still on my jacket and the taste of smoked duck still in my mouth, I understood something that no lecture had managed to teach me: the city has a front, and it has a back, and the back is where the food is actually cooked.

The Leaflet Rack

The rack beside the door held forty-one titles, and my job on Sunday was to throw away the ones that had gone out of date and refill the rest. Sorting a leaflet rack is a lesson in what a district worries about, in what order, and on what budget. Nobody designs it. It accumulates, the way a kitchen shelf accumulates jars.

The top shelf was health, and the health shelf was blunt. One leaflet was about hepatitis and the vaccine that prevents it, with a photograph of a baby on the front, because a baby is easier to look at than a liver. One was about emergency contraception and abortion, and it listed, in small print, the four clinics that would see a woman

without a referral letter. One was about heroin and methadone and the lie of one last time, written in the second person, which is the only voice that works on an addict. One explained what a dose is and why the dose is not a suggestion. One was a single card, the size of a bank card, that said in four languages that testing is free and that no name is required.

The second shelf was money. Debt, it explained, is not one large event but the accumulation of small numbers: a phone bill, a loan from a cousin, a fine for a bicycle parked in the wrong place. There was a leaflet on loan sharks that defined a legal rate of interest and then, in a box, gave the number to call when the rate is not legal. There was a leaflet about the phone calls that arrive at nine in the evening, in which a stranger with a kind voice explains that your account is in danger. The English version of that leaflet had been translated by a volunteer with an odd sense of humour, and the sentence "do not be polite to them, they are not polite to you" had been underlined by hand about forty times.

The third shelf was law, and it was the shelf nobody took from. Deportation: what a residence permit does and does not protect. The constitution, in plain language, with a cartoon judge on the cover. How a case reaches a court, and how long the wait is, and what to bring. What to do if a policeman asks for your identification, and what he may not do. Your right to an interpreter. Your right to remain silent. A leaflet on domestic abuse that had been reprinted eleven times, which meant either that the problem was getting worse or that people had finally started to talk about it, and Aunt Chen said it was both.

The fourth shelf was the neighbourhood. Noise: what counts as aggravation, and who to call at two in the morning. Stray animals: feed them or report them, but not both, because a fed stray becomes a resident and a resident becomes a complaint. Repair your appliance before you replace it, with the address of a man who mends rice cookers for twenty yuan. The axe and the budget: what the district cut this year, and why the library opens four hours later on Mondays.

Between the shelves there were notices that belonged to nobody. An amateur radio club looking for members, meeting in the basement of a building on Fridays. A photography workshop promising to teach the aesthetics of the ordinary, with a picture of a wet bicycle on the cover, which was, I thought, the most honest advertisement in the whole rack. An adult literacy class, Monday and Thursday, free, no registration. An autobiographical essay competition for people over sixty, with a prize of five hundred yuan and a judge who was, according to Aunt Chen, "an ardent believer in the ordinary".

I asked her who read all this. She said the answer was in the shape of the rack: the top shelf was empty by Wednesday, the second by Friday, and the third was still full at the end of the month. "People will read about their body and their money," she said.

"They will not read about their rights until their rights are gone." Then she took a leaflet from the third shelf, the one about the constitution, and put it in her bag, and said she would leave it in the laundry, because a woman folding sheets has time to read four pages, and no time at all to make an appointment.

That afternoon a boy of about seventeen came in and asked, in a voice that was trying to be casual, whether the booth had anything about lawyers. His friend had been arrested for robbery — a phone, a push, a wall — and the family had been told that nothing could be done until a court date in October. Aunt Chen asked three questions and wrote down four answers. Then she told him that the court would appoint a lawyer if the family could not pay, that the lawyer would be overworked and would need documents, and that the most useful thing the boy could do that week was to collect papers, not opinions.

He left with a plastic folder and a list. On the way out he stopped at the rack and took the leaflet about courts, and then, after thinking about it, took the one about deportation as well, and I understood that he was collecting for a family that had more than one fear.

On the wall by the door somebody had pinned a photograph of an airplane coming in low over the lanes, with its landing gear down, and a date from 1998. Underneath, in marker, somebody had written: "we were told it would stop". People had been complaining about the flight path for thirty years. The airport had answered every complaint politely and had changed nothing, and the only thing that had changed was the price of the flats, which had gone up, because a city that is loud is also a city where people want to live.

Science Saturday

Zhang Ye arrived with a suitcase of junk and set it down on a folding table like a man who had been waiting all year for permission to open it. A dead rice cooker. A coil of copper wire. A jar of yellow sulphur borrowed from the teaching lab. A magnet, a bag of iron filings, a bicycle chain gone orange with rust, and a telescope on loan from the astronomy society.

Our table sat between a table about teeth and a table about bees. The first child to arrive asked whether the telescope was real, and when Zhang Ye said yes, she asked whether she could look at the sun through it. He said that this was the single most dangerous question of the morning, and then spent four minutes explaining why, and she listened with the patience of a person who intends to try it anyway.

Everything on the table was carbon, or oxygen, or iron, or sulphur, or a compound of them, and the trick of the morning was to make that fact feel like a story rather than a rule. The wire became an electromagnet when he ran a current through it: a coil, a field, and forty children leaning in. The chain was corroded because iron and oxygen prefer each other's company to ours. The rice cooker had died because the thermal fuse had done its

job. He oiled the chain, and told them that lubrication is not a repair, it is a postponement, and that most of engineering is the art of a decent postponement.

A boy of ten asked what the plastic bottle was made of. Zhang Ye said petroleum, and the boy said, no, it is made of plastic. That exchange was, I think, the best thing that happened all morning, and Zhang Ye knew it, because he wrote the boy's sentence down on the back of a leaflet and put it in his pocket.

The plain fact is that a child cannot estimate a quantity and does not need to. She wants the number, then she wants the story of the number, then she wants to break something. We let them break things. A disc of thin steel, a balloon, a spring — anything that vibrates, anything that can be made to ring. One girl held a hand against the table while Zhang Ye struck the other end with a spoon, and she said, in a small, offended voice, "it is still shaking", and that was the whole of the physics lesson, delivered better than any graph could deliver it.

A grandfather watched from a plastic chair and asked me, at the end, whether all of this was on the exam. I said that some of it was. He said that in his day the exam was a page of numbers and a page of symbols, and that he could do both, and that neither had ever told him why the chain went orange. Then he tapped the table with one finger and said, "but I am seventy-six, so nobody has to listen to me."

The afternoon lecture was in the same building, one floor up, in a room with a projector and forty chairs. A young lecturer from the university talked for an hour about the district, and I understood perhaps two thirds of it, which for a freshman in his fourth week is a good ratio.

She said that the word renewal is doing an enormous amount of work in the documents, and that we should be suspicious of any word that has been given a budget. She said that the plan, as printed, was arbitrary in three places: in the boundary of the zone, in the density it allowed, and in the order in which streets were to be emptied. She said that the rhetoric of consultation was not a lie exactly, but it was a particular kind of truth, the kind that a man tells when he has already decided and wants to be liked afterwards.

She put up a graph of the tenants who had moved out and the tenants who had moved in, and she said the two curves were not the same shape, and that this was the finding. A woman in the third row said that her street had been surveyed six times and nobody had ever told her the result. The lecturer said that this was a fair objection and that she had no defence for it, which is the rarest sentence you will hear from a person with slides.

Then she said something I wrote down and kept. She said that a district is not a place, it is an agreement, and that when the agreement is one-sided the place stops being

an agreement and becomes a product. She said that the residents had autonomy on paper and very little in the room, and that the difference between those two things is where most of the politics of this city happens, quietly, on weekday afternoons, in rooms with forty chairs.

On the way out, someone handed me a leaflet about the meeting. It had been produced by the district office, and it used the phrase "listening to the voices of the community" four times, and it listed a phone number that nobody answered when I rang it two days later. Anecdotal, of course. One phone call is anecdotal. It is also the only kind of evidence a person can gather with a phone.

The Last Bus

The last bus left the terminus at forty minutes past midnight. The fare was two yuan, exact change, and the driver kept a tin of coins for the people who had none, and she would warn you, in a tone that was tolerable but not warm, that she was not a bank. She was about fifty, with a thorough way of checking her mirror, and she drove the way a person drives who has driven the same road for nineteen years and has stopped expecting anything from it.

A strap hung from the rail above every seat, and the ones near the door had been polished by ten thousand hands into a sheer curve. The bus took the steep road by the river at a crawl, because the tires were old and she did not trust them, and the engine made a noise like a man clearing his throat. Two kitchen workers staggered in with bags of laundry, still in their trousers and their rubber boots, and a nurse in a thin coat shivered in the seat behind me with her eyes shut, and a man in a suit loosened his tie and fell asleep with his mouth open, and a boy with a guitar case shoved past all of them and stood in the aisle because the seats had begun to shrink.

The doors snapped shut. Above the driver's head a screen updated the route: a virtual map, a video loop of a safety notice, the local TV signal with the sound off. The radio was on low, and the topics were trivial — weather, football, a trifle of a scandal about a councillor and a car — and the host was witty in the way that men who have never been wicked are witty, which is to say cheaply. Nobody listened. The man with the guitar case was watching a video on his phone without earphones, and a woman two rows back asked him, with real politeness, to turn it down, and he did, and the whole bus relaxed by one degree.

There is a rule on the last bus, and the rule unifies everybody on it: we do not talk. We are not rude. We are simply finished. The bus is a small universe with a driver and a fare box and a heater that works on alternate Tuesdays, and for twenty-five minutes nobody in it has to be anybody.

I looked out of the window and the city came apart into its two halves again. On the

right, the new towers with their wealth and their empty balconies and their lights on a timer. On the left, the low lots, the waste ground, the underdeveloped strip where the buses turn round and the drivers smoke. A dog crossing a road at night looks up at a bus with the calm of an animal that has understood, correctly, that the bus will not stop for it.

At the fourth stop a woman got on with a bag of bottles and sat down heavily and said, to nobody, "yeah, it is late." Then she laughed at herself. She had a husband, she told the driver, who worked nights at a warehouse, and a son who was worth more than the two of them put together but did not know it. The driver said, "they never know it," and the woman said, "no," and that was the end of the conversation, and it had been, in its way, the warmest thing that happened to me all week.

I got off two stops early because I wanted the walk. The street was empty except for a man hosing down the pavement outside a restaurant, and a woman wrapping flowers in newspaper, and a cat that had found the one warm step in the district. The air smelled of the river and of frying oil and, faintly, of the sea.

When I reached the gate the campus was dark except for the guard's cabin, and the guard, who had seen me come in at this hour four nights running, said only, "hi", and then, because I looked the way I felt, "welcome back."

Notebook, First Page

Notebook, First Page

I keep a notebook. I have kept one since I was fifteen, and it is not a diary, because a diary is about me and this is about the things I could not name. On the bus home I wrote down eleven words I had heard that day and could not have used in a sentence. Four of them I still remember, not because they were difficult but because of who said them.

The first was absolute, and Aunt Chen said it about silence, of all things. "There is an absolute silence in a building just before it is pulled down." She meant that the building is empty, and that the emptiness has a quality, and that the quality is not sad, it is administrative. A word I had only ever met in grammar, doing the work of an intensifier, was suddenly a fact about a street.

The second was acclaim, which I met in a leaflet about a photographer who had won something. The leaflet used the word twice, and the second time it was doing no work at all, and I understood that acclaim is a thing other people give you, and that you cannot manufacture it by describing yourself as acclaimed. The photographer had not written the leaflet. Somebody in an office had, and the office had wanted the word.

The third was accordingly, which is the most useful word in the whole booth. It means "so, in the way that the previous sentence requires", and once you have it you can end almost any argument politely. The hospital lost the file; accordingly, the file must be rebuilt. Say it that way and nobody has to be blamed, and nobody has to be helped either.

The fourth was ace, which is a volunteer's word and not a textbook's. The boy who could fill in any form in six minutes was an ace, and everybody said so, and he was nineteen and had failed his exams twice, and he could read a regulation faster than anybody I have met before or since.

The Banquet in the Lane

The laundry woman's daughter was married in October, and the wedding was held in the lane, because the lane was the only room in the district big enough to hold the clan. Eight round tables, borrowed from a restaurant, went down the middle of the alley on a Thursday afternoon, and by six o'clock the alley had stopped being a road and had become a dining room with a sky for a ceiling.

The bride came out of a doorway at ten past six in a red gown, and the alley applauded, and somebody set off a string of firecrackers that made every dog in the district bark for a quarter of an hour. Her mother cried and pretended not to; her father, who had spent the morning carving a goose into a work of art, put down the knife and could not pick it up again.

A wedding in a lane is a public event whether the family wants it or not. Neighbours who had not spoken to each other for years turned up with a red envelope and a complaint about the noise, in that order. An old woman in a woollen coat brought a bowl of her own braised pork, because she did not trust the caterer, and she was right not to trust the caterer. Two boys ran in circles around the tables until an uncle caught them by the collar and made them sit. A cat sat under the head table with the patience of a creature that has attended many banquets and knows exactly when the fish bones arrive.

The congratulatory banners had been hung from window to window, red paper with gold characters, and one of them was upside down, and nobody had the heart to tell the cousin who had hung it. There was a banner from the clan association, a banner from the groom's employer, and a banner from a company that nobody could identify, which had almost certainly been paid for by somebody who wanted a favour later.

Old Wu turned up at seven, in a jacket that had been expensive in 1994, and stood at the edge of the light for a while, doing the arithmetic of a landlord at a party. Then he walked to the head table and said, loudly enough for the next three tables, that the rent for the laundry would go up in January, that the lane was going to be renovated, and that he was telling the family now, tonight, out of friendship, so that they would not be surprised.

The bride's father stood up. He was a small man with a burn scar on one wrist and the manners of somebody who has spent thirty years being polite to people who could hurt him. He said, "Tonight my daughter is married." Then he sat down.

It was the retired judge who answered, in the voice that had sent people to prison for twenty-two years. He said that he had known this lane since before the renovation was a paragraph in a document, that he had eaten at this family's table eleven times, and that he wished to make three remarks. First, that a rent increase announced at a wedding was not a business decision but an act of cowardice dressed as frankness. Second, that the renovation of a lane required the consent of the people who lived in it, and that consent obtained by surprise was not consent. Third, that he was eighty-one years old, that he had no authority whatsoever, and that he would say the same thing again on Monday, in front of the council, with a copy of the lease in his hand.

Nobody clapped. In that lane, clapping would have been vulgar. What happened instead was that the woman at the next table pushed a plate of shrimp towards him, and the bride's mother refilled his cup, and the whole alley, in a single movement, decided which side it was on.

Old Wu laughed. He was not a monster, and that was the worst of it. He laughed, and said, "You see? This is why nobody invites judges," and sat down and ate, and drank three glasses of ale, and at eleven o'clock he danced with the bride's mother, badly, and everybody forgave him for one evening, which is how a lane survives.

I was given the job of carrying trays. Between the kitchen and the eighth table there were nine metres of alley and about forty obstacles: a bicycle, a gas cylinder, two crates of beer, a child, a dog, a puddle, and a man telling a story with his hands. I dropped nothing, which the cook took as a personal compliment, and she gave me a bowl of soup that had been simmering since noon.

At midnight the tables were folded and stacked against the wall, the alley became a road again, the last guests walked home under the wires, somebody's grandmother swept the paper into a pile and set it alight, and the smoke went up between the buildings and stayed there, because there was no wind and nowhere for it to go.

I thought, walking back, that I had never seen a place so casual and so formal at the same time. Eight tables, a red gown, a bowl of pork carried across a lane by an old woman who did not trust the caterer: none of it was in any book I had been given. The books had cities in them, and the cities had populations and growth rates and problems. This city had a cat under a table and a judge with no authority and a landlord who danced.

The Wall Newspaper

Aunt Chen edited a newspaper. This is not a metaphor. Once a quarter she produced one

sheet, printed on both sides at the shop on the corner, and pinned it to the board outside the booth, and the deal she had made with the printer was that he would charge her for the paper and not for the ink, on the grounds that the ink was his idea.

The masthead said, in a font that had been old-fashioned in 1987, "THE LANE NEWS", and under it, in smaller type, "edition 41, one sheet, free, take one". She wrote most of it herself. She was, I came to understand, a genuine journalist of a very small and very precise kind: she did not report what people said, she reported what they could prove, and her corrections column was longer than most papers' news pages.

The first item that quarter was a memorial. Four men from the lane had gone into the army in 1938 and had not come back, and for eleven years the family had been told only that they were missing. The item gave their names, their ages, the village they had left, and the single sentence that the provincial archive had finally released: that they had died in a battle that the histories called a victory, and that the victory had cost four hundred men, and that the commander who wrote the report had called it a triumph, and that he had not been in the lane. Aunt Chen had put the item in the paper because the last of the four widows had died in March, and there was now nobody left who could be told.

The second item was a market report, and it was the most read thing in the paper. Bananas, up two. Berries, out of season, don't ask. Olive oil, down, and a warning about the cheap stuff in the unmarked bottles. Chocolate, steady. The stall that sold burgers had been closed for a week, and the reason given was a dispute over the lease, and the reason everybody believed was different. Under the prices Aunt Chen had added a line of her own: "Anyone who tells you that the price of food is boring has never had to feed three people on forty yuan a day."

The third item was from the clinic. There had been four cases of a contagious stomach illness in one building, all traced to the same water tank, and the clinic wanted people to know that the tank had been cleaned and that the illness was not, in the doctor's words, a plague. The same item reminded parents that the vaccination schedule had changed, and that a child who had missed the second dose could still catch up, and that the clinic would not ask about residency status. Aunt Chen had underlined the last clause with a ruler.

The fourth item was the school page, which she wrote with the headmistress of the primary school at the end of the lane. It listed the new term's dates, the price of the uniform, the fact that the playground would be closed for three weeks while the surface was replaced, and a request for volunteers to hear children read on Tuesday afternoons. It also mentioned, in one sentence, that the school had been told it would have to enroll nine more children in September than it had desks for, and that anybody with a spare desk should telephone the office.

The fifth item was the sports page, which existed because the lane's amateur team had won the district cup. The contestant in the final had been a boy of nineteen who worked in a kitchen and trained at eleven at night on a concrete court under a streetlight. The report was three paragraphs long and contained the words "squad", "rally" and "marathon" in a way that suggested Aunt Chen had looked them up and was pleased with them.

The sixth item was a letter, signed, which was rare. A woman who lived above the bar wrote that the noise from the bar had reached the point where her mother could not sleep, that she had complained four times, that she had been told to be patient, and that she was now writing to a newspaper with a circulation of about two hundred people because it was the only place left where she could be listened to without an appointment. Aunt Chen printed it in full and added one line: "The district office has been invited to reply. We will print the reply in the same size."

I read the whole sheet twice. It was, I thought, the best piece of writing I had read all year, and it had been produced by a woman with a plastic chair and a kettle, on a single sheet, for free. No narrator, no commentary, no adjectives doing the work of facts. Just four names, four prices, a water tank, nine desks, a football match, and a woman who could not sleep.

That was the thing about the lane that I had not expected: it was not a place without information. It was a place with a great deal of information and almost no way of getting it to the people who needed it. The newspaper existed to close a gap of about sixty metres.

Two Documents About the Same Eleven Hectares

Two documents arrived in the same week and described the same eleven hectares.

The first came through the letter box: glossy, A4, folded twice, with a couple on the cover standing on a balcony at sunset, holding glasses, looking at a river that had been photoshopped into a slightly better river. Inside, the copy had been written by somebody who was paid by the adjective.

"THE RIVERSIDE QUARTER. Majestic views. Glorious light. A paradise for the ambitious."Lower down:"A panorama of water and sky, prestigious schools within a kilometre, the convenience of two metro lines, and the prosperity of a district that is going somewhere."Lower still, in the small print that is the only honest part of any brochure:"An investment for your fortune. Fabulous facilities. A classic facade. Beauty that will not wait."

The second document was one page, unglossy, and had been photocopied so many times that the district's seal looked like a coffee stain. It was a notice, and it was addressed to the occupants of buildings one to fourteen.

"The council has sanctioned the dismantling of the above buildings with effect from the first of March. Occupants will be displaced in three phases. Tenants will receive compensation calculated at the rate of forty-two yuan per square metre per month for a period not exceeding eight months, payable on production of a valid lease, a utility bill and proof of residence for the preceding twelve months. Expenses incurred in removal are not recoverable. Applications received after the deadline will not be considered. Objections may be lodged in writing within fourteen days. No objection shall suspend the works."

I read them one after the other, twice, and then I did something I am not proud of: I laughed. Not because it was funny, but because the two documents were so perfectly matched. One promised a paradise to people who did not yet live there; the other indicated, in the flat tone of a lawful threat, that the people who did live there had fourteen days and a photocopy.

The brochure had the word "home" on page nine. The notice had the word "occupant" on line two. Nobody had cited a single resident in either document. The brochure depicted a future; the notice denoted a procedure; and between them, in the gap where a person might have been, there was a silence that was, as Aunt Chen had said, absolute.

She read both documents standing up, which is how she read anything that made her angry. Then she put the brochure in the recycling and the notice in a plastic sleeve, and she wrote four names on the back of an envelope: the tenant in the cellar who had a child and no lease; the retired chemist on the third floor who had lived there since 1979 and had a lease and no patience; the widow who ran the laundry; and the dean of a college who owned the whole of the fourth floor and had been absent for eleven years.

"These four," she said, "are the whole argument. The law will protect the chemist and ignore the tenant in the cellar, and if the chemist moves out quietly the district will call it consent."

I asked her what we could do in fourteen days. She said: document everything. Photograph every room. Collect every bill. Write down every promise and the name of the person who made it. "In this city," she said, "the person with the best folder wins, and it is not justice, but it is faster."

The Hearing

The hearing was held on a Monday, at half past nine, in room three of the district office, which had forty chairs, a table, a microphone that worked for eight minutes, and a portrait of somebody dead above the door. The incumbent councillor chaired it. He began by reading out the procedure, and the procedure was restrictive by design: three minutes for each speaker, no recording, no insult, no interruption, and written objections only, because a written objection can be filed and a shouted one cannot.

The judicial presence in the room was retired and therefore unofficial, but when the judge stood up, the junior official who was taking the minutes put down his pen. The judge did not appeal to compassion. He cited the lease, the statute, and the date of the ratification of the plan, and he said that the boundary of the zone had been drawn arbitrarily and that an arbitrary line is a line that somebody has to defend, and that nobody in the room was prepared to defend it. Then he sat down, having used two minutes and forty seconds.

The developer's spokesman was younger and better dressed and had the rhetoric of a man who has never had a ruling go against him. He spoke about prosperity, about revenue, about the logistics of construction and the momentum of the district, and he used the word revival four times, and each time it meant the same thing, which was demolition with a better slogan. He was not a sinister man. He was worse than sinister: he was sincere, and his sincerity had been purchased along with his salary.

A woman from the third floor asked him what would happen to her mother, who was eighty-eight and had not left the building in two years. He said that the relocation package was generous and that transitional housing would be provided. She asked where. He said that the details would be announced. She asked when. He said, eventually. And that word, eventually, did more damage in that room than any insult could have done, because everybody heard in it the cowardice of a man who has been told to be vague and is being paid to be pleasant.

There was indignation in the room, and there was also flattery, and the flattery was harder to watch. A man stood up and praised the councillor's concern, his accessibility, his patience; he said that the district had listened, that the process had been transparent, that the residents were grateful. He was not a hypocrite in the ordinary sense. He was an egoist with a shop on the corner, and he had calculated, correctly, that the shop would be worth more after the redevelopment than before it. Greed is rarely conceited. It is usually polite, and it usually brings a gift.

The ruling was not a surprise. The lobby had been counted before the meeting began; the squad of police standing outside the door in the rain had been counted too, and they were there not because anybody expected a riot but because a show of force is cheaper than a discussion. The chair thanked everybody for coming. He used the word consultation. Then he adjourned the meeting, and the microphone was switched off, and the young official picked up his pen.

Outside, the rain had stopped. The resistance did not. It reorganised itself in the corridor, in the form of four people and a folder, and the judge, who had no authority and considerable experience, dictated a letter to the housing bureau while standing under the eaves, and Aunt Chen wrote it down, and the widow held the umbrella, and a boy of nineteen photographed the whole thing on a phone with a cracked screen.

I have thought about that morning more than about any lecture I attended that year. What I learned was not that the system is corrupt, which is easy and lazy, but that the system is procedural, and that procedure can be used by anybody who bothers to read it. The judge did not shout. He quoted. The widow did not threaten. She asked three questions in a row. And the official with the pen stopped writing for a moment, and I saw him stop, and I have never forgotten it.

The Cupboard of Lost Things

At the back of the booth there was a cupboard with no lock, and the cupboard had become the district's lost property, because a person who finds a thing and does not know who owns it will give it to the nearest person with a counter. On the first Thursday of every month Aunt Chen opened it, laid everything on a towel on the floor, and tried to return what could be returned.

The inventory, which I was made to write, went like this. One boot, size forty-three, muddy, no lace. One skirt, grey, school uniform, name tape cut out. One pair of jeans with a bus ticket in the pocket and a wrist watch still attached to the belt loop. One coat, child's, red, hood torn. One gown, pink, wrapped in plastic, which nobody had claimed and about which Aunt Chen had a theory she did not share. Two trays from the cafeteria at the hospital. A drum, toy, split skin. A microphone, broken, cable cut. A pair of scissors, surgical, which frightened me. A tin of pins. A capsule of medicine in a paper bag with a dose written in pen. Two bars of chocolate, melted and re-set. A jar of olives. A bag of dried berries. A frozen turkey, which we threw away in November and which returned to the cupboard in December in the form of a complaint. A saddle, bicycle, torn. A length of rope. A piece of timber with a nail in it, which Aunt Chen kept because, she said, a nail in a piece of wood is a weapon and a weapon must be logged.

The smell of the cupboard was leather, rain, disinfectant and old sweet wrapping. Every object had a story that we could not know and a value that we could not calculate, and the rule was that after ninety days an object could be given away, and the rule was never applied, because Aunt Chen said that a cupboard is a promise: the promise that the thing you lost is still somewhere, in a room with a counter, waiting for you to remember it.

Aunt Chen's Blue Notebook

The notebook was in the second drawer, under a folder of receipts. The first page was rules for volunteers; the next forty pages were sentences, numbered, in her square hand, and I read the whole thing standing up, the way she read documents that made her angry.

One: Laziness is the name the comfortable give to the exhaustion of the poor.

Two: Kindness that requires gratitude is not kindness. It is a loan with interest.

Three: Eminence is a distance. The further a person lives from the street, the more eminence he appears to possess.

Four: Flattery is cheaper than wages and works harder.

Five: Greed wears a tie to the hearing and overalls to the site.

Six: Cowardice signs the form. Courage asks for a copy.

Seven: Malice is rare. Indifference does the same work, sleeps better, and draws a salary.

Eight: Envy is the tax a lane pays on other people's luck.

Nine: Betrayal smells of cigarettes and apology.

Ten: Humiliation is administrative. It arrives in an envelope with a reference number and a stamp.

Eleven: Indignation burns for a week. A folder lasts for a decade.

Twelve: A conceited man believes he deserves his luck. A hypocritical man believes he deserves yours. An egoist believes there is no yours.

Thirteen: Brutality is not anger. It is patience with a budget.

Fourteen: Sinister people are rare; tired people with authority are not, and they do more damage.

Fifteen: Notorious is a word for people who were caught. The others are called successful.

Sixteen: Hostile silence is louder than a shout, and cheaper, and it cannot be recorded.

Seventeen: Repressive rules are never enforced equally. That is not a failure of the rules. It is the design.

Eighteen: Rebellious people grow old; stubborn people do not, and they are useful, and they are exhausting.

Nineteen: Permissive parents, permissive landlords, permissive officials: the word always means that somebody else will pay.

Twenty: Tolerable is the most dangerous word in this city. A situation becomes tolerable, and then it becomes permanent.

Twenty-one: Shyness is not weakness. It is attention turned inwards, and it sees more than confidence does.

Twenty-two: Diligent people are given more work; lucky people are given more credit; this is called merit.

Twenty-three: Selfless people frighten me. Somebody is always keeping the accounts,

and it is never the selfless one.

Twenty-four: Wicked is what we call a person whose interests happen to differ from ours at a moment when we have the power to say so.

Twenty-five: Pessimism is a luxury of people who have somewhere to sleep.

I read number twenty-five three times. Then I closed the notebook, put it back under the receipts, and did not tell her I had read it. Two days later she said, without looking up, "If you are going to read my notebook, read page forty-one as well." Page forty-one said: Everything above is true on a Monday. On a Friday I would sign anything.

The Rider

The rain came back in December, at an angle, the way it does when the wind has decided to stay for a few days. I was standing under the awning at the booth with a cup of tea and no purpose other than to stay dry, and a man in a yellow jacket came up the steps two at a time, took off his helmet, and said, "Is this where you write letters?"

His name was Xiao Zhao. He was twenty-six, from a county in Anhui that I had never heard of, and he had been delivering food in this district for three years. He worked eleven hours a day, six days a week, and in a good month he made eight thousand yuan before expenses, which sounds like a fortune until you divide it by the hours and subtract the rent, the battery, the tariff on the insurance, and the fines.

The fines were what he had come about. He took out his smartphone and showed me the screen: a record of forty-one orders, and against six of them a red mark. Late by four minutes. Late by two. Cancelled after takeoff, which is the word the app uses for leaving the restaurant, as if a meal were a flight. He had been fined eighty yuan in total, and eighty yuan was eleven orders, and eleven orders was about two and a half hours in the rain.

"I want to write to them," he said. "Not to complain. To ask a question." The question was simple and, as far as I could tell, necessary: on three of those six occasions, the delay had been caused by the restaurant, not by him; the app knew this, because the app had recorded the time he arrived and the time the meal was ready; and yet the penalty had been applied to him, because the penalty is applied to the person who is easiest to punish.

We wrote it together at the counter. It took an hour. He was determined to be polite, and I was determined to be clear, and the compromise was a letter that was neither, and which he read three times before he agreed to send it. He said that in his experience a letter that is too polite is ignored, and a letter that is too angry is answered with a form, and that the only letters that work are the ones that make the reader afraid of a specific person reading it next. I wrote that down. It was the best instruction on

writing I received that year, and it came from a man who had left school at sixteen.

Then he told me about the algorithm, which is the word everybody in the lane used for the thing that decided their day. It assigned the orders. It measured the miles. It counted the minutes between the restaurant and the door, and it did not count the lift that was broken, or the gate that needed a code, or the customer who did not answer the phone and then complained that the food was cold. "The system is not cruel," he said. "The system is honest. It tells you exactly what it wants. The cruelty is in the part that is not written down."

He left at four, because the rain had stopped and the orders were coming again. I watched him go down the steps and swing onto the bike and disappear into the traffic, and I thought about the word "rider", which is a word for somebody on a horse, and about the fact that the app had chosen it deliberately, because a rider sounds free, and a driver sounds employed, and a worker sounds like trouble.

The Last Shift

My last shift of the term was on the twenty-second. There were few people in the lane; the students had gone home, the shops had cut their hours, and the calendar on the wall of the booth still said November, because nobody had thought to turn it.

Aunt Chen gave me a handbook — the volunteer handbook, thirty-two pages, bound with a plastic spiral — and said, "Read it properly. You did not read it properly in August." She was right. I had signed the form and skipped the book. I read it that afternoon, in the corner of the booth, with the heater on, and it was better than most of the books I had been given that term.

It said that a volunteer is not a saint and not a social worker, and that pretending to be either does harm. It said that the first duty is to stay for a year, because a person who leaves after three months teaches the lane that help is a season. It said that pity is easy and respect is hard, and that anybody who cannot tell the difference should volunteer somewhere else.

On the last page there was a list of things to do before leaving: return the key; hand over the register; tell the next person where the bins are; say goodbye to the people you promised to say goodbye to. I had promised four. I found three of them that afternoon — the laundry woman, the old man from the English corner, and the guard at the gate — and the fourth was Xiao Zhao, who was out in the rain and would not be back before midnight.

I left the key with the guard, and the register in the drawer, and I walked out through the lane one last time, past the print shop with its yellow sheet of hyphenation, past the cupboard of lost things, past the wall where the brochure had promised a

paradise and the notice had promised fourteen days, and out onto the street where the buses ran all night and nobody looked up.

The Old Man's Practice Book

The old man from the English corner caught me at the gate in January and put a school practice book into my hands. "You correct," he said. "Not the grammar only. The truth also." Then he went to work, because his shift started at seven.

The book had forty-one pages, each page dated, each entry a sentence in pencil with the corrections in a different pencil, and a column on the right where he had copied the word that had given him trouble. He had been at it for two years. I have never seen anybody work so hard at anything, and I have been in rooms full of people working hard at things that will make them rich.

Here is what he had written, and what I could not correct.

"I acquaint myself with the calendar of the camp every January, because the gate closes at ten." He meant the workers' dormitory, which is a camp in everything but name, and he had chosen the word because it was the only one in the dictionary that came close.

"The bat is blind and the claw of the cat is sharp, and both of them live in the cave under the stair, which is not a cave." He had drawn a small picture of the stair, and the cat, and something that was either a bat or a very large moth.

"I breathe the smoke of the crude stove and my digestion is deficient, but the doctor says my lungs are surprising."The word he had copied was deficient, and he had underlined it three times, and I could not tell him that the sentence was wrong, because it was not.

"My buddy and I collide at the corner of the building because we are both dizzy after the night shift, and we laugh, and this is the best part of the day." I corrected nothing in that one. I wanted to copy it into my own book.

"The climate is not the same as my village. In my village the rain comes in June and stops in July. Here it comes when it wants."He had written the last sentence twice, once in pencil and once in ink, as if he had decided it was true enough to keep.

"Billion is a word I will never use in a sentence about my life, but I like to write it."This was my favourite entry in the whole book, and when I told him so he was pleased for a week.

"The birth of my son was in 1981, in a cave of a house with a dirt floor. My granddaughter was born in a hospital with a lift and a bill. This is progress, and I am glad, and I am also something else that I do not have a word for." He had copied the word progress and had written "not in my dictionary" next to it.

"I anticipate the challenge of the new machine at work. It has a screen and a menu

and it beeps when I touch it wrongly. I attain a little more assurance every week, and in March I hope to be allowed to use it alone." The machine was a floor polisher. He was sixty-eight years old and he was competing for the right to polish a floor.

"My brow is wet in summer. The sweat goes into my eyes and I cannot see the step. I have asked for a towel and I have been given a towel, and the towel is white and clean and useless, because a towel in a pocket is not a towel in a hand." He had copied the word useless, and had written beside it, in Chinese, a single character that I will not translate, because it was not a translation.

"The donation of clothes from the church is kind, but the trousers are all for a smaller man, and I am not a smaller man, so I give them to my buddy, who is."

"The detective on the television always knows who did the crime. In real life, the police know who did the crime and also know that nobody saw it, and the file stays in a drawer, and the drawer is called statistics." He had copied the word statistics. He had also copied the word cynic, and had put a question mark next to it, and I did not explain it, because he had already written the definition himself.

"My diplomat is the man at the office who speaks slowly and says nothing. I admire him. I would like to be a diplomat for one day, and tell the manager the truth in a voice that sounds like a compliment." He had copied compliment, which is not in my list of words and which he had found by himself.

"The dynasty in the museum is three thousand years old and the building I live in is thirty years old and the roof leaks. Time is not fair to roofs." He had copied dynasty in the museum, on a school trip he had taken alone, at sixty-seven, on a Wednesday, because the museum was free on Wednesdays.

"I do not understand the word democracy, but I understand the word queue, and in my experience the queue is where the democracy happens or does not." I read that sentence four times. Then I wrote in the margin, in my own pencil, "correct", which is the only correction I made all afternoon.

"I am desirous of one thing: to see the sea before I die. I have seen the river every day for nine years and the river is not the sea, and the difference is not size, it is salt." He had copied desirous from a letter in a textbook and had been waiting, I think, for a sentence worth using it in.

Page forty-one had no sentence on it. It had a list of words he had met and could not place, and beside each word a question mark, and at the bottom he had written: "Some words are not for me. I keep them anyway. A word in a book is a tool in a box. You do not need the tool today. You may need it next year."

The Free Health Check

The district ran a free health check for two days in the community hall, and my action for those two days was to sit at a table with a stack of forms and a pen. It was not an adventure. I had expected an adventure, in the way you expect one at nineteen, and what I got was a hundred and forty people who had been putting this off for years.

"Adequately" was the word I remember from the first morning, because the nurse used it four times. The lighting was adequately bright. The screen was adequately clean. The forms were adequately clear, which she said was the highest praise she gave anything, and which I understood by noon to mean that nobody had yet been humiliated by a form.

Registration is an affirmation that you exist. The audience for it is a clerk. Alike in nothing else, the hundred and forty shared one bias, which was that a form is a hindrance and a queue is a punishment, and they were right about both, and they came anyway, and that is the whole of what I learned about courage that winter.

The room had been divided into stations by cardboard screens: blood, eyes, teeth, lungs, and a table at the back with a curtain round it where a woman from the psychiatric hospital sat for two hours and saw nobody, and then saw four people, and then saw eleven. The dentist was a young man with a portable chair who had brought his own lamp, and he worked without a break, and by four o'clock he had looked into ninety mouths and had told me, in a low voice, that he had seen more decay in two days than in a year in the practice where he normally worked.

An old woman came in with a limp and a bag. Her limb had been injured in a factory in 2003 and had never been set properly, and she had been told at the time that it would heal, and it had healed, in the way that a wall heals when you paint over the crack. She did not want treatment. She wanted a certificate, because a certificate would prove that the injury was real, and she had been asked to prove it four times in nineteen years, and each time by a different office.

A man of about forty came in with a cough and a story. He had been a painter of steelwork, and he had spent eleven years breathing a vapour that he could not see, and he had been told that the mask was optional in hot weather, and he had believed it, because the man who told him was holding a clipboard. The doctor listened to his chest for a long time and then asked him a question that had nothing to do with his chest, which was whether he had ever been given a document about the risks. He had not. The doctor wrote one word on the form and underlined it, and the word was occupational, and the man asked what it meant, and she said, "It means this is not your fault, and it also means somebody is responsible."

The young ones came for different reasons. A girl of nineteen wanted a vaccination certificate because the college required it, and she was frightened of needles, and she held my hand while it happened, and then apologised for holding my hand. Two boys came because they had heard there were free condoms, and they were too shy to ask, so they

stood by the door for twenty minutes and left, and came back, and left again, and on the third attempt I walked over and put two in a paper bag with a leaflet and said nothing at all, and one of them said, "thanks, mister", and they ran.

At the teeth station an old man refused to open his mouth. He had four teeth left and he was ashamed of them, and the dentist, who was twenty-six and had never been poor, did something I have thought about since: he took out his own phone and showed the old man a photograph of his own mouth, before his own treatment, when he had had a brace and a gap and had been, in his words, a walking joke for three years. The old man looked at the photograph for a while. Then he opened his mouth.

Between stations there were posters on easels, the kind that are designed by committees and read by nobody. One said that a chronic condition is not a sentence. One said that a communicable disease is not a disgrace, and that the law forbids an employer from demanding a test, and that if an employer demands one anyway, the number to call is on the back. One said, in letters that were too small for the room, that a woman who is pregnant has the right to keep working until the eighth month if her doctor agrees, and that dismissal on the grounds of pregnancy is unlawful, and that legal aid is free.

One easel had a poster about the kidneys, with a diagram in four colours, and under the diagram a sentence in bold: "A dose is a decision. Ask what it is, what it does, and what happens if you miss one." An old man read it aloud to his wife, who could not see it, and then he asked me whether the clinic could give him the poster, and I said it belonged to the district, and he said, "Then I will copy it," and he took out a pen and began to write it down, all of it, standing up, on the back of a form, and it took him eleven minutes, and nobody interrupted him.

By six the hall was empty except for the chairs and the cardboard and a smell of disinfectant and oranges. The nurse sat down for the first time in nine hours and said, "That was adequately organised," and I laughed, and she said, "I am not joking. Adequately organised is the best a district ever does. Everything above adequately is luck, and everything below it is a scandal, and the line between them is one tired person."

I walked home past the clinic's permanent building, which was four streets away and had a queue outside it at seven in the morning, every morning, in all weathers. The free check had been a kindness. It had also been a way of not asking why a hundred and forty people needed a temporary hall with cardboard screens in order to see a dentist.

The Last Walk, in Fragments

I walked the lane once more before I went home, and because I had a notebook and a bus to wait for, I wrote down what I saw in short lines, the way you write when your hands are cold. What follows is not a story. It is the entirety of what I could keep.

The year's action had been adequately planned by other people, and it had never once been an adventure. Every affirmation I signed that winter was alike in one respect: it assumed an ally behind the counter, and sometimes, by luck, there was one. The alternation of shifts appalled the nurse; I learned to appraise a queue before it could arise into a quarrel, and to articulate the request in a single sentence, because a second sentence is where a favour dies.

Nobody in that lane was going to assassinate anybody. The atom of the place was smaller than that: an audience of two, a rumour, a bias confirmed by a nod. The award for volunteer of the year went to a girl who could read a form backwards, and she accepted it on behalf of the booth and gave the envelope to the laundry. A biographer would have made a book out of her; the blast of a firecracker was the only applause she got, and she got it in October, at a wedding, for carrying trays.

The bloom on the jasmine outside the laundry brightened the whole wall and made the bondage of the tenancy agreement look almost comical, which is what flowers are for. You broaden your vocabulary by butting into conversations you were not invited to; there is no bypass, and the library is not the place where it happens. A century of building had produced a cluster of chronic problems around one drain, and the drain had been repaired eleven times, and each repair was a complement to the last one rather than a solution.

Coexistence in a lane is collaborative by force: the woman who hates the bar keeps the bar's cat, and the bar keeps her parcels. Comfort is a comparative idea, and the compelling argument for a competitive rent is always made by somebody who does not pay it. Do not let anybody complicate this. A composer needs four notes; a configuration of pipes needs two; the coordination of a hundred and forty patients needed one nurse with a clipboard and a corporal's voice. Everything concerned everybody, and nothing conclusive was ever said, and the consultation conferred nothing except a counterpart for the officials to blame.

A man confronted a young official outside the estate office: twenty-two years old, a conqueror in a convertible his father had paid for, listening with the patience of somebody who had already won. The man consoled himself afterwards by consulting the constitution on his phone. Contemplative people contribute nothing visible and everything useful. A crazy scheme, a cute child, a darling of a grandmother, a curl of paper in the gutter: the cumulative effect is a street.

Decline is not a deduction you can deduce from one bad year; it is a degeneration you notice in the derivative details, like the way a democrat stops denying that the vote was a disguise. A discreet disclosure is a contradiction in terms, which is why the disappointing truth arrives by dialect, from a distant cousin, despite the distortion, and the disregard for diversity turns out to be a disregard for arithmetic.

I could detach the documentary from the domain of journalism and watch it as a

dramatically dreadful piece of entertainment, and that would be a dwarf-sized reading of a durability problem. Economical with facts, effective with illumination, elaborate where empirical evidence is thin, it elected to eliminate nothing and to emancipate nobody, and it embraced the electron and the equivalent eruption with the same enthusiasm.

The essence of the euro crisis, a merchant told me, was evasion: you evolve, or you exclaim; you execute, or you exert; and the expansive exploitation of a fixture is foreseeable only in hindsight, which is another word for never. Extended extremity is what a faction calls a fascination. At midnight they extinguished the memorial candles, and the fatal thing was not the fire but the feast afterwards, and the feeble, feminine voice that financed it.

The lane kept an analogue of its history on a wall of photographs that comprised every wedding and every funeral, and a bench where the old men would confer about the weather as if it were a treaty. Rice and river fish correlate in the cuisine here, and the ban on fishing deprives the table of half its grammar. The judge would devote an hour to teaching a boy to differentiate a fact from a feeling, and the boy would emulate the tone for a week and then forget it.

The booth encouraged people to come back, not because the advice was good but because the chair was endlessly available and the tea was poured energetically by a woman with no time at all. An enlightening hour is one in which somebody envelops a hard question in a story and then equates the story with the answer, and I had three such hours all year. Rates erode a budget the way water erodes a step, exceptionally slowly and then all at once, and to expose the arithmetic is nobody's job.

Flexible hours are a fluency in lying about time, and I grew fond of the people who lied well. A forgery needs a formulation and a foundation: the first is a sentence, the second is a stamp. The fracture in the wall had been framed in cement by a man with a frown of concentration, and it held for eleven years.

Fuel is the fundamental game of a district; whoever controls the price of heat controls the winter. The museum could generate a genre out of nothing, and a case of geologically boring stones became a story about the river. A ghost is what a graduate leaves in a rented room: a grave of receipts and a coat with no owner.

Grief in a lane is public — one groan from one window, and a guide to how to behave for the next forty days. A gun on a poster is handy for a campaign and harmful to a neighbourhood, and the rain will hasten the leak and haunt the ceiling while the landlord says that heaven sent it. The heir to the shop was helpless before a historical accident: the street had changed direction in 1997 and the customers had obeyed.

A homeless man is hopeful at six in the morning and a howl at midnight, and the difference is whether anybody has spoken to him since lunch. The humanist in the room was

a retired teacher who thought hunting for the right word was the only icon worth keeping. Idle talk is the imaginary impact of a rumour; the real impact leaves a mark on a lease.

An imperialist importation of taste is impressive until you notice that the noodles cost more than the rent of the stall. Incidental inclusive language indulges the writer and abandons the person who has to fill in the form. An infectious rumour can inflate a fine into an informant's wage, and the ingredient list was the initial injurious document I read that year: eleven chemicals, no quantities, no name.

An innocent question from an inquisitive child gave me the insight that the form was designed to be abandoned. Inspiration is instantaneous and useless; integration is slow and useful. The intense interaction of a queue is a lesson in interim solutions, and interpretation is what an intervention becomes when it fails.

The introductory paragraph of the plan was where the failure was hidden, and an invaluable inversion occurred when the residents read it aloud: the invisible became the subject. Irrigation ditches came before the buildings; isolation came after them; and the juvenile court on the third floor is where a kidnap of a Tuesday afternoon becomes a kneel before a desk, and where labour, in the legal sense, is a word for a boy.

A lamb hung in the window of the butcher's, and the landscape behind it was a photograph of a mountainous county that nobody in the shop had visited. The lash of the rain in the latter half of the month legislated for us: we stayed indoors, we licked our wounds, and the only linear thing in the week was the drip down the glass.

A listener on the last bus is a locomotive of other people's troubles. Shapes loom out of the fog and are lost again, and in the midst of a milky evening the mingle of diesel and river is the smell of the whole district. Love in a lane is magical and practical at once: a maid from the hotel on the corner married a cook, and the management gave them a meaty supper and a manifestation of goodwill in the form of a red envelope.

Martyrdom is a word the mass uses for a medieval death it can monitor from a sofa. The mental messenger of the district is the notice board, and its method is the misleading summary: a mix of mode and modifier, a minus sign where a numerical figure should be. Moral language motivates nobody who is picky about facts, and a movie with multiple murderous musicians in it was the most popular thing at the outlet that winter.

There was a mysterious, half-naked argument about nationality on the third floor. It was negotiated rather than resolved, nightly, and I was not allowed to navigate it. Negation is easier than nomination, and nor is a word that closes a door, and the nose of the man who owned the building could smell a petition through a wall. Observant occasional occurrence: an odour of offset printing, an onward operation of the press, an optimal oral account that originated in an outdated leaflet and reached an outsider through an overlap of overt palace politics.

Pardon the peep, but a performance in a periodical is perseverance with a deadline. Personally, I am pessimistic about philosophy and optimistic about physical work, and I am picky about which pirate I plan with. The plumber is a poet of precision, and the poisonous pollution of the ditch is a portrayal of possible poverty that no preacher would propose to preoccupy himself with.

Nothing in the file agreed with anything else: a probe into the profession of psychiatrist produced a profound puzzle about quantification, a quarterly raid on the rapidity of reception at the recorder's office, a rectification and a reduction of the refund, a priceless portrayal of the processor we had sold for scrap by a princess who was nine, and more preservation planning than the district had ever managed in its long, prevalent, paper life.

The travel agency on the corner had papered its window with sightseeing offers: skiing in the north, skating on a rink that was situated inside a shopping centre, surfing in a bay two provinces away, and a photograph of a slender woman with a tan and a sexy dress who was, according to the caption, serving drinks on a beach. Scarcely a sociologist would call this a spectrum of choices. A scholarly reading of that window is that it sells a substitution: the sea for the river, the wave for the tide, and a spaceship of a promise for a speedy commute.

On the last night of the year the lane transformed itself. A temple three streets away opened its doors at eleven, and the ritual was sequential and secondary to the rejoicing: toast, song, silence. Somehow the solemnity survived the smoky air and the southeastern wind, and solitude was not on offer, and superstition and spiritual comfort were scarcely distinguishable at midnight, and nobody wanted them distinguished.

The predominant smell was incense and frying oil, and the ray of a satellite-dish light fell across a stump of candle. A man with a spouse and two children knelt on the straight stones, and I suspected that his regret was sustainable only because it was rehearsed. A reformist councillor nominated himself for the spring reshuffle and reaped a reputation for refined manners; his rival sought to seize the role by relaying a rumour about a rigged tender, and the retailers on the street restricted their sympathy to a respectable resemblance of it.

I subscribe to no vision that violates a user in the middle of a form. And I have no synonym for dignity in my vocabulary that is not a vegetable in a basket: steady, unremarkable, prone to rot.

Undermine, suppress, stifle, slash: four verbs that a budget uses in place of a plan. Tackle the vent in the wall, soften the vibration of the lift, secure the stake in the ground, and withdraw quietly from the web of obligation that yesterday seemed worth keeping. That was the whole of my advice to the next volunteer, and I gave it in four sentences and a handshake, and I wish it had been more.

The Verge of Leaving

On the verge of going home I asked Aunt Chen the question I had been seeking all term: whether the booth had done any good. She said that from her standpoint the question was subjective, and that I should repeat it in ten years. Then she said something less cautious, which is the sentence I have kept.

"You cannot succeed at this the way you succeed at an exam," she said. "There is no sudden result. A family that moves out of a cellar is not a triumph. It is a transplantation, and you will never know whether it took. The tract of housing between the river and the road is a triangular problem: three sides, no centre. Anybody who promises you an ultra-simple solution is selling something, and the something is usually the corner you are standing on."

She went on. "Sunshine on the last morning. A spontaneous round of applause in the hall for a nurse who had said nothing all week. That is the size of the thing we do. It does not bring renown. It does not steady a life much. It steadily prevents one particular life from falling apart this month."

I asked whether the volatility of it ever made her upset: the funding, the plans, the reshuffles, the years when the district changed its mind about the lane and then changed it back. She said, "Volatility is the name we give vulnerability when it belongs to somebody else. In this world the only people who are not vulnerable are the people who have never asked for anything."

Then she gave me back the key and told me to go home and eat something that had not been fried twice in the same oil.

Undoubtedly the renaissance of the district will be prolonged by the saturation of the market, and the propagation of a prototype will pollute a playground with noise, and starvation will remain a word for old people who lose their seniority when they lose their job. A voluntary clinic that vaccinates every child is the relief a terrific technique offers; therapy for the mind is the relief a humane one offers; and a regime that calls both a tragedy of the budget has told you what it is.

伴读：一条巷子怎么把人教会

你可能正准备把这一章的九百多个词抄进本子，按字母顺序，一个抄三遍。先停一下。抄完你得到的是一页纸，不是一条巷子；而这九百多个词真正的住处，是巷子。

这一章我写得比别的章都长，原因不体面：这些词不是我想用的，是它们自己长在那条巷子里的。表格、传单、租约、通知、广播、菜单、判决、病历——城市另一面的语言本来就是这种密度。你读的时候不必每个词都停下来查，先跟着林一走下去；走到头再回头看这一篇，你会发现自己已经认识它们了。

先说开篇那几个词，因为它们最容易被你"以为认识"。

approximately 是"大约"，但它比 about 硬。about 是口语里的"差不多"，approximately 是写进报告、贴在站台上的"约"。地铁晚点三分钟，广播不会说 about three minutes，它说 approximately three minutes，因为一个"约"字背后是责任。同理，announcement 不是"通知"这么简单：announce 是"当众宣告"，重音在"当众"。你在站台听到的那句"列车即将进站"，是 announcement；你在朋友圈发的"我脱单了"，也是 announcement，只不过场合不同。

carriage 这个词，你背的是"马车"，但在这章里它是"车厢"。英式英语把车厢叫 carriage，美式叫 car，两者都不指"汽车"。这是四六级最爱设的一种坑：词你认识，搭配你不认识。brake 是刹车，作动词时是"刹住"，不是"打破"（那是 break，两个词只差一个字母，考场上死过很多人）。arrival 是"到达"，注意它跟的介词是 at 或 in，不是 to。

August 大写是八月，小写 august 是"威严的"。这一条你现在记不住没关系，只要记住"大写八月，小写威严"，考到就是白捡。

再看那件 jacket 和那张 certificate。certificate 是"证书"，但它跟 diploma 不是一回事：diploma 是毕业文凭，certificate 是任何一张"证明你完成了某事"的纸，包括你那张志愿培训的证。training 在这里是"培训"，不是"训练身体"；英文里 training 的范围比中文的"训练"宽得多，员工入职叫 training，运动员备战也叫 training。

bread、bowl、rubber、stool、stationery 这些词，你可能会觉得"这么简单也值得讲"。值得。因为它们在这章里不是单词，是道具：面包是林一在末班车上垫肚子的东西，碗是巷口那家店里公用的，橡胶是雨天站台的味道，凳子是他坐了两个小时的那张，文具是慈善供应商捐的那一摞。一个词只有落在一件具体的东西上，你才不会忘。

bumper 是汽车保险杠，bumper sticker 是贴在保险杠上的贴纸。bizarre 是"古怪到让人不舒服"，比 strange 重，比 weird 正式。aside 是"到一边去"，stand aside 是让开，put aside 是放到一边（抽象用法：put aside your pride）。clench 是"攥紧"，攥拳、咬牙都用它，它的手感是"指节发白"。bend 是弯，bend down 是弯下腰，注意它不是"转弯"（那是 turn）。

comb 作动词是"仔细翻找"，comb through a list 就是一页一页地梳。这个词的画面感很强：梳子梳头发，一根一根过。advertisement 缩写 ad，重音在第二音节；affordable 是"买得起"，不是"便宜"——便宜是 cheap，买得起是 affordable，这两个词在租房广告里差别巨大。average 是平均，roughly 是"粗略地"，两个词一起用就是"均价大致"。

district 是"区"，但它比 area 正式，也比 borough 通用。earnings 是"收入"，只用复数，而且多指劳动所得；income 更宽，投资也算。highway 是公路，英式常说 motorway。ample 是"充裕"，常跟 space、evidence、time 搭配，语气比 enough 体面。anyhow 是"不管怎样"，跟 anyway 基本同义，但 anyhow 更老派。mansion 是豪宅，它跟 house 不是大小之分，是"气派"之分。

charitable 是"慈善的"，但它还有一个意思："宽厚的、往好处想的"。a charitable interpretation 不是"慈善的解释"，是"善意的解读"。supplier 是供应商，它跟 provider 的分工是：provider 多指服务，supplier 多指货物。freshman 是大一新生，美式；英式现在多说 first-year。administrate 这个词要小心：它确实存在，但英美人更常说 administer 或 manage，administrate 是"行政化"的说法，考试里看见它，多半是在考"行政"这层意思。

assess 是"评估"，重音在第二音节，名词是 assessment。它跟 evaluate 差别很细：assess

偏"估价、估量"，evaluate 偏"评价好坏"。candidate 是"候选人"，也可以指"应试者"，但它跟 applicant 不同：applicant 是"申请人"，candidate 是"被列入名单的人"。assistance 是"帮助"，正式；help 是日常。你扶一位老人过马路，叫 help；政府给的一笔补助，叫 assistance。

最后是那张表上的四个词，它们是一件事的四个阶段：admittance 是"准你进门"，conditional 是"有条件的"，liability 是"责任"，waive 是"放弃（权利）"。admittance 跟 admission 都能表示"进入"，但 admission 还可以指"承认"和"入场费"，admittance 只指"获准进入"，而且更书面，所以门上写的是 No admittance。

pledge 是"承诺"，比 promise 重，带一点公开宣誓的味道。abide 是"遵守、忍受"，abide by the rules 是"守规矩"，这个 by 一定要记住。stipulation 是"条款、约定"，它跟 term 的区别是：term 是合同里的"一条"，stipulation 强调"被特别讲定的条件"。validity 是"有效性"，expiration 是"到期日"，一个是"还算不算数"，一个是"哪天不算数"。credential 是"资格凭证"，常用复数 credentials。obligation 是"义务"，consent 是"同意"，这两个词在表格里永远成对出现：你同意了，你就有了义务。

一张纸上的九十来个词

传单上那个 abort 你未必见过，但 abortion 你一定见过——堕胎，一个在中文里被绕开、在英文里被印在纸上的词。它旁边是 abuse：滥用，也是虐待，两个意思靠后面的名词分开，abuse of power 是滥用权力，child abuse 是虐待孩子。

顺下来是一串"态度词"。absolute 是绝对的，absolute silence 是死寂；它跟 complete 不同，后者只说"完整"，前者说"没有余地"。accept 是接受，它跟 receive 的分工是：receive 是"收到"，accept 是"收下并且认了"。acclaim 是公开的赞誉，众人给的，不是自己说的。accordingly 是"相应地"，一个非常好用的连接词，意思是"照上文那样办"。accumulation 是"堆积"，debts 会 accumulate，雪会 accumulate，情绪也会。

ace 在球场上是"王牌"，在志愿站里是"一把好手"，口语里当形容词用：an ace volunteer。acquaint 是"使熟悉"，acquaint yourself with the rules 是"先把规矩摸熟"，注意它是反身的，不是"认识某人"（那是 know）。action 是行动，take action 是"采取行动"，这个搭配比 do action 正确。actually 是"其实"，不是"目前"（那是 currently），中国学生最爱在这里滑倒。

addict 是成瘾的人，动词用法是 be addicted to，那个 to 是介词，后面跟名词或动名词。adequately 是"足够地"，注意它是副词，别拿它当形容词；adequate 才是形容词。adult 是成年人，也可以当形容词：adult education。adventure 是冒险，注意它是"经历"，不是"危险"；danger 才是危险。aesthetics 是美学，这个词永远带 s，不能说 aesthetic 当名词（那是形容词"审美的"）。

affirmation 是" affirm"的名词：肯定，也可以是法庭上的"不经宣誓的证词"。against 是"反对、倚靠"，against the wall 是靠墙，against the plan 是反对这个计划，一个词两个方向。aggravation 是"加剧"，也是"烦人的事"，后者是口语。agree 后面跟 with 是人，跟 to 是提议，跟 on 是条款，三个介词三种关系，这是四六级翻译最容易错的地方之一。

airplane 是飞机，美式；英式是 aeroplane。ale 是麦芽酒，比 beer 老派。alike 是"相像

的"，只作表语：They are alike，不说 an alike pair。allergy 是过敏，动词是 be allergic to。allotment 是"分配到的份额"，也可以指城里人租来种菜的一小块地——英国工人阶级的周末就在那儿。ally 是盟友，动词是"结盟"：ally with。

alphabet 是字母表，来自希腊头两个字母 alpha 和 beta，看见这个词就该想起这件事。alternation 是"交替"，day and night 的 alternation；动词 alternate，形容词 alternate 和 alternative 不是一回事，后者是"另一个可选方案"。amateur 是业余的，也是业余爱好者，它跟 novice 不同：novice 是新手，amateur 是"不以此为业"。

ambitious 是"有野心的"，褒贬看语境；名词 ambition。analogue 是"类似物"，也是"模拟的"（对 digital 而言），英式拼写；美式 analog。anecdotal 是"轶事式的"，anecdotal evidence 是"个例证据"，写论文时这四个字是贬义。animal 是动物，注意它跟 beast 的语体差别。answer 是回答，动词名词同形，answer for 是"负责"。

anticipate 是"预料"并"提前应对"，比 expect 多一层动作；expect 只是"等着它发生"。appall 是"使震惊"，英式拼写，美式 appall 少一个 l；形容词 appalling。appearance 是外表，也是"出场"，make an appearance。appliance 是家用电器，注意它跟 device 不同：appliance 多指白电，device 什么都能指。

appraise 是"估价"，注意它跟 apprise（告知）只差一个字母，考场上见过人写错整句。arbitrary 是"任意的、专断的"，划线划得 arbitrary，决定做得 arbitrary。ardent 是"热心的"，比 enthusiastic 文，常跟 supporter、believer 搭配。arise 是"发生、出现"，不及物，问题 arise，机会 arise，人不会 arise。

army 是军队，an army of 是"一大群"：an army of cleaners。articulate 是"能说清楚"，形容词和动词同形，重音位置不同。assassinate 是"暗杀"，专指政治人物；kill 是杀，murder 是谋杀，assassinate 有政治动机。assurance 是"保证"，也是"从容自信"；动词 assure，别跟 ensure、insure 混。astronomy 是天文学，astronaut 是宇航员，两个词根都是"星"。

atom 是原子，atomic 是原子的；注意 nuclear 是核的，两者不能互换。attain 是"达到"，attain a standard，比 reach 正式。attic 是阁楼，装杂物那种；basement 是地下室，住人那种；cellar 是地窖，存酒存菜那种。三个词三个世界，这一章里它们都在同一栋楼。audience 是观众，也是"读者群"，注意它跟 spectator 不同：spectator 在现场，audience 可以在任何地方。

autobiographical 是"自传体的"，autobiography 是自传，biography 是别人写的传。autonomy 是自治、自主，patient autonomy 是病人的自主权，这个词在这一章里出现了两次，一次在听证会，一次在病历。award 是奖，动词是"授予"；prize 是奖品，reward 是报酬，三个词不是一回事。

axe 是斧头，美式拼写；英式是 axe 也是 axe，但动词 axe 是"砍掉（预算、岗位）"，报纸标题最爱用。backwards 是"向后"，英式常带 s，美式 backward；go backwards 是倒退。baffle 是"使困惑"，比 confuse 重，带一点"被挡住"的画面。balcony 是阳台，注意它在剧场里是"楼座"。

banana 是香蕉，go bananas 是"发疯"，这个短语在口语里活得好好的。banquet 是宴席，正式，比 dinner 大，比 feast 稍小；feast 是"盛宴"，也可以指"大饱眼福"。barn 是谷仓，美式农场建筑；这一章里它是江边那户人家的后院仓房。basement 上面说过，这里补一句：basement 在英国多半是厨房，在美国多半是娱乐室。

bat 是蝙蝠，也是球棒；两种意思靠语境，别在阅读里认错。battle 是战役，比 war 小，比 fight 大；动词是"奋力争取"：battle for。bean 是豆，coffee bean 是咖啡豆，be full of beans 是"精力旺盛"。beauty 是美，beauty 也可以指"妙处"：the beauty of it is……。

beer 是啤酒，ale 是上面发酵的麦芽酒，lager 是下面发酵的；三个词的差别在工艺，不在颜色。behalf 是"代表"，on behalf of 是固定搭配，那个 on 不能换成 in。bell 是钟、铃，ring a bell 是"听着耳熟"。berry 是浆果，注意 strawberry、blueberry 都是 berry，但 mulberry 不是。betrayal 是背叛，动词 betray，名词也可以指"露出"：a betray of feelings 是错的，要用 betray 当动词。

bias 是偏见，也是"偏差"，统计里的 bias 是中性的；形容词 biased。billion 是十亿，英式旧用法是万亿，现在全球通用十亿。biographer 是传记作者，注意它跟 historian 的边界。birth 是出生，give birth to 是"生下"，也是"催生"：give birth to a movement。blast 是爆炸，也是"一阵（风、音乐）"，口语里还是"痛快"。

blind 是瞎的，the blind 是盲人群体，blind to 是"对……视而不见"。bloom 是开花，in bloom 是盛开；作名词也可以指"青春焕发"。blunt 是钝的，也是"直率的"，说话 blunt 是不留情面。body 是身体，也是"主体"：the body of the report。bondage 是奴役、束缚，这个词很重，别用来形容加班。boot 是靴子，英式也是"后备箱"。both 是两者都，注意 both 后面不能跟 of 加单数。

breathe 是呼吸，名词是 breath，两个词拼写差一个 e，读音差一截。bride 是新娘，groom 是新郎。brighten 是"使明亮、使有希望"：brighten up。broaden 是"拓宽"，broaden your horizons 是"开阔眼界"。brow 是眉毛、额头，也是"山脊"。brutality 是残暴，形容词 brutal；注意它跟 violence 不同，后者是暴力，前者是"残忍的性质"。

buddy 是伙伴，美式口语，比 friend 随意。building 是建筑物，注意它跟 edifice 的语体差别。burger 是汉堡，hamburger 才是全称。bush 是灌木，the bush 在澳洲指"乡下"。butt 是"屁股"，也是"烟蒂"，也是"用头顶"。bypass 是绕过，心脏搭桥手术也叫 bypass，这个词的画面是"另开一条路"。

从表格走到巷子

cafeteria 是食堂，自助那种；canteen 也是食堂，工厂那种。calendar 是日历，也是"日程"：a busy calendar。camp 是营地，也是"阵营"：two camps in the debate。cap 是帽子，也是"上限"：a cap on spending，动词是"封顶"。capsule 是胶囊，也是太空舱，一个词两个时代。

carbon 是碳，carbon footprint 是碳足迹；注意 carbon 和 coal 不是一回事，coal 是煤。carve 是"雕刻"，carve out 是"开辟出"：carve out a career。casual 是随意的，casual clothes 是便装，casual work 是零工——这一章里它出现了两次，一次是衣服，一次是工作，后者才是要紧的。cater 是"提供饮食"，也是"迎合"：cater to the crowd，那个 to 很重要。

cave 是山洞，cave in 是"塌陷"，也可以指"让步"。cellar 是地窖，英国的老房子多半有一个。century 是世纪，the twentieth century 要加 the。challenge 是挑战，动词是"质疑"：challenge the plan，这一层意思比"挑战"更常用。change 是改变，也是零钱；for a change 是"换

换花样"。

chemist 是化学家，英式也是"药店"——去 chemist 是去买药，不是去找科学家。child 是孩子，复数 children 是不规则变化，考试里常考。chocolate 是巧克力，a bar of chocolate 是一条。chopstick 是筷子，永远用复数。chronic 是慢性的，chronic illness 是慢性病，别跟 acute（急性的）弄反。

circle 是圆，也是"圈子"：business circles；动词是"环绕"。cite 是"引用"，论文里用，法庭上也用：cite a precedent。clan 是宗族，比 family 大，比 tribe 小。classic 是经典的，classical 是古典的——classic car 是经典款，classical music 是古典乐，这一条每年都有人错。claw 是爪子，动词是"抓"。climate 是气候，也是"气氛"：political climate。

clock 是钟，around the clock 是"昼夜不停"。clothes 是衣服，永远复数，动词是 clothe。cluster 是一簇、一群，动词是"聚集"。coat 是外套，也是"涂层"：a coat of paint。coexistence 是共存，动词 coexist。coil 是线圈、盘起来，动词是"盘绕"。collaborative 是"协作的"，注意它跟 cooperative 的细微差别：前者强调一起做，后者强调配合。

collide 是相撞，也是"冲突"：interests collide。comfort 是舒适，也是"安慰"，动词是"安慰"。comparative 是比较的，comparative literature 是比较文学；注意它跟 comparable（可比的）不是一回事。compelling 是"令人信服的"，也是"引人入胜的"，compelling evidence 是硬证据。

competitive 是竞争的，competitive price 是有竞争力的价格；注意它跟 competent（有能力的）只差几个字母。complement 是"补充、配成全套"，注意它跟 compliment（恭维）拼写差一个字母，意思差一个世界。complicate 是"使复杂"，形容词 complicated。composer 是作曲家，compose 是"组成"也是"创作"。

comprise 是"包括、由……组成"，用法是 A comprises B，不能说 be comprised of（虽然有人这么写）。conceited 是自负的，比 proud 贬。concerned 是"担心的"，也是"有关的"：all concerned。conclusive 是"决定性的"，conclusive evidence 是铁证。confer 是"授予"，也是"商议"：confer with colleagues，两个意思差得远。

configuration 是"配置、布局"，电脑和建筑都用它。confront 是"面对、对抗"，be confronted with 是"遭遇"。congratulatory 是"祝贺的"，congratulate 后面跟 on。conqueror 是征服者，conquer 是征服，注意它跟 overcome 的差别：overcome 是克服困难，conquer 是征服土地或人心。considerable 是"相当大的"，considerate 是"体贴的"，这两个词每年都要坑一批人。

console 是"安慰"，也是"控制台"，重音位置不同。constitution 是宪法，也是"体质"：a strong constitution。consult 是咨询、查阅，consult a dictionary 是查词典，consult with somebody 是跟某人商量。contagious 是"传染的"，也可以形容笑声和哈欠；infectious 也是传染，但更常指病原体。contemplative 是"沉思的"，动词 contemplate。

contestant 是参赛者，contest 是比赛也是"争夺"。continue 是继续，注意它跟 resume 的区别：resume 是"中断后继续"。contribute 是贡献，contribute to 有两个意思：捐给、导致。convenience 是方便，at your convenience 是"你方便的时候"。convertible 是敞篷车，也是"可转换的"。

cook 是做饭，也是厨师；cook up 是"编造"。coordination 是协调，动词 coordinate。cord 是绳、线，电源线就是它。corporal 是"肉体的"，corporal punishment 是体罚；注意它跟 corporate（公司的）只差一个字母。correlate 是"相关"，correlate with；相关不等于因果，这句话值得记一辈子。

corrode 是腐蚀，形容词 corrosive，也可以形容讽刺。cost 是花费，at all costs 是"不惜一切代价"。council 是议会、委员会，注意它跟 counsel（建议、律师）同音不同义。counterpart 是对应的人或物，这个词在正式文章里非常好用。course 是课程，也是"过程"：in the course of。cowardice 是怯懦，形容词 cowardly。

craft 是手艺，也是"手艺人的船"；动词是"精心制作"。crazy 是疯狂的，be crazy about 是"痴迷"。crime 是罪行，criminal 是罪犯也是刑事的。cross 是穿过，也是"十字"，也是"生气"：be cross with somebody。crude 是粗糙的、未加工的，crude oil 是原油。crust 是外壳，面包皮、地壳都用它。

cuisine 是烹饪风格，French cuisine 是法餐。cumulative 是累积的，cumulative effect 是累积效应。curl 是卷曲，也是"蜷缩"：curl up。curve 是曲线，动词是"弯曲"，learning curve 是学习曲线。cute 是可爱的，美式口语里也可以指"机灵"。daddy 是爸爸，口语，小孩用；father 是正式称呼。

dance 是跳舞，也是舞会。darling 是亲爱的，称呼，也可以形容"迷人的"。day 是一天，one day 是"有一天"，some day 也是"有一天"，前者多指过去，后者多指将来。dean 是院长、系主任。decade 是十年，注意它是"十年"这个单位，不是"十年期"。decide 是决定，decide on 是"选定"。

decline 是下降，也是"婉拒"，两个意思都很常用，靠语境分辨。deduce 是"推断"，比 conclude 更强调推理过程。deep 是深的，deep down 是"内心深处"。deficient 是"缺乏的"，deficiency 是缺乏症。degeneration 是退化，动词 degenerate。deletion 是删除，动词 delete。

deliver 是递送，也是"发表"：deliver a speech；也是"兑现"：deliver on a promise。democrat 是民主主义者，Democrat 是美国民主党人，大写才是党派。denote 是"表示"，比 mean 正式，比 indicate 更专指符号。deny 是否认，也是"拒绝给予"：deny somebody access。depict 是"描绘"，比 describe 更图像化。

deportation 是驱逐出境，动词 deport。deprive 是"剥夺"，deprive somebody of something，那个 of 不能丢。derivative 是衍生物，也是"缺乏原创性的"，这个词的贬义很重。deserve 是"应得"，deserve it 是"活该"，也是"当之无愧"，看语气。desirous 是"渴望的"，desirous of，非常正式，日常几乎不用。

despite 是"尽管"，后面跟名词或动名词，不跟句子——要跟句子用 although。detach 是"分离"，detached 也可以形容"超然的"。detective 是侦探，detect 是察觉。determined 是"坚决的"，be determined to。developer 是开发商，也是开发者；这一章里它是前者，也是后者。devote 是"奉献"，devote oneself to，那个 to 是介词。

dialect 是方言，注意它跟 accent（口音）不同。dictionary 是词典，查词典是 consult a dictionary，不是 look up a dictionary（那是查这个词条）。differentiate 是"区分"，differentiate between A and B。digestion 是消化，动词 digest，也可以指"消化信息"。

diligent 是勤奋的，比 hard-working 正式。dine 是"用餐"，正式，dine out 是下馆子。

diplomat 是外交官，diplomatic 是外交的，也是"圆滑的"。disability 是残疾，注意它是"能力受限"的中性说法。disappointing 是"令人失望的"，disappointed 是"感到失望的"，这是分词形容词的经典对立。disc 是圆盘，也是光盘、椎间盘；美式拼写 disk 多用于电脑。disclosure 是披露，动词 disclose。

discreet 是谨慎的，注意它跟 discrete（分离的、离散的）拼写只差一个字母，意思完全不同。disguise 是伪装，in disguise 是"伪装的"。dismantle 是拆卸，也是"废除"：dismantle a system。displace 是"取代、迫使离开"，这一章里它指后者。disregard 是无视，比 ignore 更"不当回事"。

distant 是遥远的，也是"疏远的"。distortion 是歪曲，动词 distort。diversity 是多样性，生物多样性和文化多样性都用它。dizzy 是头晕的，feel dizzy。documentary 是纪录片，也是"文件的"。domain 是领域，也是域名。donation 是捐赠，动词 donate。

dose 是剂量，a dose of 也可以指"一次（不愉快的经历）"。dozen 是一打，dozens of 是"几十"。dramatically 是"戏剧性地"，常用来形容变化的幅度。dreadful 是"可怕的"，口语里也可以只是"糟糕的"。drip 是滴，drip 也是"乏味的人"。drum 是鼓，drum up 是"鼓动、招徕"。

duck 是鸭子，也是"躲闪"：duck down。durability 是耐久性，形容词 durable。dwarf 是矮人，也是"使相形见绌"：the building dwarfs its neighbours。dynasty 是王朝，也是"世家"：a dynasty of chefs。

大词与小物

easy 是容易，take it easy 是"别紧张"，easy 也可以形容人"随便"。economical 是"省钱的、经济的"，注意它跟 economic（经济的，指经济学意义上的）不是一回事：an economical car 是省油的车，an economic crisis 是经济危机。edition 是版本，first edition 是初版。effective 是有效的，efficient 是高效的，两个词差在"管不管用"和"费不费劲"。

egoist 是利己主义者，注意它跟 egotist（自吹的人）不同。elaborate 是"精心制作的"，动词是"详述"。elect 是选举，elect somebody as 那个 as 可以省。electron 是电子，electric 是电的，electrical 是电气工程的。elephant 是大象，white elephant 是"昂贵而无用的东西"——这个词组在城建新闻里出现频率极高。

eliminate 是消除，比 remove 彻底。emancipate 是"解放"，专指从束缚中解放，emancipate women 是解放妇女。embrace 是拥抱，也是"欣然接受"：embrace the change。eminence 是"显赫、高处"，形容词 eminent，注意它跟 imminent（迫在眉睫的）只差一个字母。empirical 是"经验的、实证的"，empirical evidence 是实证证据，跟 theoretical 相对。

emulate 是"仿效"，比 imitate 更正面。encourage 是鼓励，encourage somebody to do。endless 是无尽的，endless 也可以带抱怨的语气。energetically 是"精力充沛地"，动词 energetic 是形容词。engineering 是工程，engineer 是工程师，动词是"策划"。enlightening 是"使人开窍的"，enlighten 是动词。

enroll 是注册入学，美式拼写，英式 enrol。entertain 是招待，也是"使娱乐"，两个意思都很常用。entirety 是"整体"，in its entirety 是"完整地"。envelop 是"包住"，注意它跟 envelope（信封）不是一回事，差一个 e。envy 是羡慕，也是嫉妒，形容词 envious。

equate 是"把……等同"，equate A with B。equivalent 是"等价的"，也可以当名词。erode 是侵蚀，引申为"逐渐削弱"。eruption 是喷发，火山、皮疹、暴动都能用。essence 是本质，in essence 是"本质上"。estimate 是估计，underestimate 是低估，这个词在工程里天天用。

euro 是欧元，复数 euros。evasion 是逃避，tax evasion 是逃税。eventual 是"最终的"，注意它不是"可能的"——eventual 只指时间上的最后。everything 是一切，注意它跟 anything 在否定句里的分工。evolve 是演变，evolution 是进化。

example 是例子，for example 是插入语，make an example of somebody 是"杀鸡儆猴"。exceptional 是"杰出的"，也是"例外的"。exclaim 是惊呼，注意它跟 claim 不同。execute 是执行，也是"处决"，两个意思都很硬。exert 是"施加"，exert pressure on，那个 on 不能丢。

exist 是存在，exist on 是"靠……活着"。expansive 是"广阔的"，也是"健谈的"，后一个意思很妙。expense 是费用，at the expense of 是"以……为代价"。exploitation 是开发，也是剥削，看宾语是人还是资源。expose 是暴露，be exposed to 是"接触到"。

extended 是"延长的"，extended family 是大家庭。extinguish 是熄灭，也是"消灭"。extremity 是末端，extremities 是四肢。fabulous 是"极好的"，也是"神话般的"。faction 是派系，注意它跟 fiction 只差一个字母。

faint 是微弱的，也是"昏倒"，两个意思都常用。family 是家庭，也是"科"（生物分类）。fare 是车费，也是"进展"：how did you fare。fascination 是着迷，也是"魅力"。fatal 是致命的，a fatal mistake 是"致命的错误"，不一定是死人的。favour 是帮忙，也是偏爱，in favour of 是"支持"。

feast 是盛宴，动词是"尽情享用"。feeble 是虚弱的，feeble excuse 是站不住脚的借口。feminine 是"女性的"，注意它跟 female（生理女性）不同。few 是很少，a few 是"有几个"，两者差一个冠词，意思差一半。fight 是打架，也是"斗争"，fight for 是争取。

finance 是财政，动词是"融资"。fire 是火，也是"解雇"：fire somebody。fisherman 是渔民，fishery 是渔场。fixture 是固定装置，也是"固定赛事"。flattery 是恭维，动词 flatter，注意它跟 flutter（拍翅膀）不同。

flexible 是灵活的，flexible hours 是弹性工时。floor 是地板，也是"楼层"，也是"发言权"：take the floor。fluency 是流利，形容词 fluent，介词是 in。fog 是雾，比 mist 厚。fond 是"喜爱的"，be fond of。

force 是力量，也是"强迫"，也是"部队"。foreseeable 是"可预见的"，in the foreseeable future。forgery 是伪造，动词 forge，也是"锻造"。formal 是正式的，informal 是随意的；formal 也可以指"形式上的"。formulation 是"表述、配方"，动词 formulate。

fortune 是财富，也是"运气"：good fortune，cost a fortune 是"花一大笔钱"。foundation 是基础，也是基金会。fracture 是骨折，也是"裂痕"。frame 是框架，动词是"陷害"，也是"表述"：

frame the question。free 是免费的，也是自由的，也是"使摆脱"：free somebody from。

frighten 是使害怕，frightening 是"令人害怕的"。frown 是皱眉，frown on 是"不赞成"。fuel 是燃料，动词是"助长"：fuel the debate。fundamental 是根本的，fundamentals 是基本功。further 是进一步，也是"更远的"。future 是未来，in the future 是"将来"，in future 是英式的"从今以后"。

江边那户人家的院子

game 是游戏，也是"猎物"，也是"心甘情愿的"：be game for anything。gardener 是园丁，garden 是花园，动词是"做园艺"。gather 是聚集，也是"收集"，也是"推断"：I gather that……。generate 是产生，generate income 是创收，generate heat 是发热。

genre 是体裁，文学、电影、音乐都用它，注意它的发音是"让-若"。geologically 是"地质学上地"，这个词在这章里属于那块围垦地。ghost 是鬼，give up the ghost 是"报废"。given 是"考虑到"，也是"给定的"：given the circumstances。glass 是玻璃，也是玻璃杯，glasses 是眼镜。

glorious 是光荣的，也是"灿烂的"，天气好也能用它。goat 是山羊，sheep 是绵羊，两个词在这一章里都出现过。goose 是鹅，复数是 geese，不规则。gown 是长袍，婚纱、学位服都是它。graduate 是毕业生，动词是"毕业"，graduate from。

granddaughter 是孙女，注意英语里 granddaughter 不分内外。graph 是图表，chart 也是图表，前者多指曲线图。grave 是坟墓，形容词是"严重的"：a grave mistake。greed 是贪婪，形容词 greedy。grief 是悲伤，比 sorrow 具体，比 sadness 重。

groan 是呻吟，也是"抱怨"：groan about the noise。grow 是生长，也是"变得"：grow old。guide 是向导，也是指南，动词是"引导"。gun 是枪，注意它不是"炮"（那是 cannon）。hairy 是多毛的，口语里也是"惊险的"：a hairy moment。

handbook 是手册，比 manual 小，比 leaflet 厚。handy 是"手边的、好用的"，也是"手巧的"。harbour 是港口，动词是"心怀（恶意）"：harbour a grudge。harmful 是有害的，harm 是伤害。hasten 是加速，也是"急忙"，比 hurry 正式。haunt 是"（鬼）出没"，也是"萦绕心头"。

haze 是薄雾，比 fog 薄，比 mist 干。headmistress 是女校长，男校长是 headmaster。hearing 是听力，也是"听证会"——这一章里两种意思都用上了。heaven 是天国，也是"极好的地方"。heir 是继承人，注意 h 不发音。helpless 是无助的， helpless 也可以形容"无能为力的愤怒"。

herb 是草药、香草，注意 h 不发音，an herb 是美式。heroin 是海洛因，heroine 是女英雄，两个词差一个字母。hi 是嗨，比 hello 随意。hindrance 是障碍，hinder 是阻碍。historical 是"历史的"，historic 是"有历史意义的"，这两个词的区别值得单独记一次。

hold 是握住，也是"容纳"：the hall holds two hundred，也是"认为"。homeless 是无家可归的，the homeless 是群体。honey 是蜂蜜，也是爱称。hopeful 是"抱有希望的"，hopefully 是"有望"，口语里也是"但愿"。horse 是马，dark horse 是黑马。

hostile 是敌意的，名词 hostility。howl 是嚎叫，风也能 howl。humanist 是人道主义者，

humanity 是人性和人类。humiliation 是羞辱，动词 humiliate，注意它是"被弄低了"的那种难受。hunting 是打猎，也是"搜寻"：job hunting。husband 是丈夫，动词是"节俭地使用"：husband resources。

hypocritical 是虚伪的，名词 hypocrite 是伪君子，hypocrisy 是虚伪。icon 是图标，也是偶像。identify 是识别，identify with 是"认同"。idle 是闲置的，也是"懒散的"。ignore 是无视，ignorant 是无知的，两个词同源不同义。illumination 是照明，也是"启发"。imaginary 是想象中的，imaginative 是有想象力的，imagined 是被想象的，三个词别混。

表格底下那层意思

imaginary 是"想象出来的"，impact 是冲击、影响，动词同形。imperialist 是帝国主义者，importation 是进口，这两个词在旧课本里成对出现。impressive 是"给人印象深的"，注意它跟 impressed（被印象深的）分工：人 be impressed，物 be impressive。incidental 是"附带的"，incidental expenses 是杂费。inclusive 是"包含的"，inclusive of tax 是含税。

incumbent 是"在任的"，也是"有责任的"：it is incumbent on you to。indicate 是表明，比 show 正式。indignation 是义愤，比 anger 正派，形容词 indignant。indulge 是纵容、沉溺，indulge in。infectious 是传染的，infectious laughter 是有感染力的笑。inflate 是充气，也是"抬高"，inflation 是通胀。

informant 是线人、提供消息的人，注意它跟 informer 的语体差别。ingredient 是成分，也是"要素"。initial 是"最初的"，也可以指姓名的首字母。injurious 是"有害的"，比 harmful 正式。innocent 是无辜的，也是"天真的"。inquisitive 是好奇的，比 curious 更"爱打听"。

insight 是洞察，介词是 into。inspiration 是灵感，动词 inspire。instantaneous 是"瞬间的"，比 immediate 更强调"同一瞬间"。instruction 是说明，常用复数 instructions。insult 是侮辱，动词名词同形，重音不同。integration 是整合，动词 integrate。

intense 是强烈的，intensive 是密集的：intense heat 是高温，intensive care 是重症监护。interaction 是互动，动词 interact。interim 是"临时的"，interim report 是中期报告。intervention 是干预，动词 intervene。into 是"进入"，注意它跟 in 的分工：in 是位置，into 是方向。

introductory 是"入门的"，introductory offer 是"开业优惠"。invaluable 是"无价的"，注意它是"极其宝贵"，不是"没价值"——这一条几乎人人错一次。inversion 是倒装，动词 invert。invisible 是看不见的。iron 是铁，也是熨斗，动词是熨。

irrigation 是灌溉，动词 irrigate。isolation 是隔离，也是孤独，in isolation 是"单独地"。jeans 是牛仔裤，永远复数。job 是工作，也是"任务"。journalism 是新闻业，journalist 是记者。judicial 是司法的，注意它跟 judicious（明智的）不同。

junior 是年少的、下级的，也是大三学生（美式）。juvenile 是少年的，juvenile court 是少年法庭。kidnap 是绑架。kindness 是善意，kind 是形容词。kneel 是跪下，过去式 knelt。know 是知道，know of 是"听说过"。

labour 是劳动，也是"分娩"，英式拼写，美式 labor。lamb 是羔羊，也是羊肉。landscape 是风景，也是"格局"：the political landscape。lash 是鞭打，也是"捆绑"，也是睫毛。latter 是"后者"，former 是前者。lawful 是合法的，注意它跟 legal 的语体差别。

laziness 是懒惰，lazy 是形容词。learn 是学，learn from 是"从……中学习"。lecture 是讲座，也是"训斥"。legislate 是立法，legislation 是法律。lemon 是柠檬，也是"次品"：a lemon of a car。lesson 是课，也是"教训"。lick 是舔，lick one's wounds 是"舔伤口"。

limb 是四肢，也是"大树枝"。linear 是线性的，注意它不是"线的"（那是 linear 的引申）。lion 是狮子，lion's share 是"最大的一份"。listener 是听众，listen 后面一定要跟 to。little 是小的，也是"几乎没有"：little hope。lobby 是大堂，也是"游说"。locomotive 是机车，火车头。

logistics 是物流，也是"组织工作"。loom 是织布机，动词是"（不祥地）逼近"。lose 是丢失，lost 是它的过去式；be lost 是"迷路"，也是"不知所措"。love 是爱，也是"喜爱"，love to do 和 love doing 都可以。lubrication 是润滑，动词 lubricate。lunch 是午饭，have lunch 不加冠词。

打印店那张黄纸

那张排版样张是这一章里最值得多看两眼的东西，因为它把"词"变成了"可以拆开的机器"。英文排不下的时候要在音节处断行，断的地方加连字符：com-ma 断成 com- 和 ma，com-mence 断成 com- 和 mence，com-municative 断成 com- 和 municative，im-mediately 断成 im- 和 mediately，sym-metry 断成 sym- 和 metry。你看见 ma、mence、municative、mediately、sym 这几个"半个词"不要慌，它们不是新词，是被行宽切开的旧词。

顺带说，commence 是"开始"，比 begin 正式得多，典礼上用它；commencement 在美式英语里还是"毕业典礼"。communicative 是"乐于交流的"，communicate 是交流。immediately 是"立刻"，注意它是副词，immediate 才是形容词。symmetry 是对称，形容词 symmetric。

黑板底下那句 you ought to practise every day，被一个手抄的人写成了 oughtto 一个词。这不是英语，这是抄写事故，但它恰好说明一件事：你眼睛看见的"一个词"，很多时候只是排版和手速的产物。学词要学它的骨头，不要学它在某张纸上的样子。

再往下是 m 开头的一串。magical 是"有魔力的"，口语里也指"太棒了"。maid 是女仆，也是"少女"，maiden 才是"未婚女子"。majestic 是雄伟的，比 grand 更带敬畏。malice 是恶意，形容词 malicious，with malice 是"心怀恶意地"。management 是管理，也是"管理层"，the management 指资方。

manifestation 是"表现、显现"，动词 manifest。marathon 是马拉松，也是"持久战"：a marathon meeting。mark 是记号，也是分数，也是"标志"。martyrdom 是殉难，martyr 是烈士。mass 是大量，也是"群众"，the masses；也是物理里的质量。match 是比赛，也是火柴，也是"相配"。

matter 是事情，也是"要紧"：it does not matter。meal 是一餐，meat 是肉，meaty 是"内容丰富的"：a meaty report。medieval 是中世纪的，注意拼写是 medi-eval。member 是成员，也

是"会员"。mental 是心理的，physical 是身体的，两个词在这一章里老成对出现。

merchant 是商人，比 businessman 老派。merit 是优点，也是"值得"：merit attention。messenger 是信使，message 是消息。method 是方法，methodology 是方法论。microphone 是麦克风，缩写 mike。midst 是"中间"，in the midst of。mile 是英里，也是"差得远"：miles apart。

milky 是乳白的，the Milky Way 是银河。mingle 是混合、交际，mingle with。minus 是减号，也是"零下"，也是"没有"：minus a coat。misleading 是"误导的"，注意它不是"骗人的"（那是 deceptive），它只是"让你想错"。mister 是先生，缩写 Mr，后面不加名字全称也可以单独用。

mix 是混合，名词也是 mix。mode 是模式，也是"方式"：mode of transport。modifier 是修饰语。momentum 是势头，gain momentum 是"势头渐起"。monitor 是显示器，也是"监测"，也是班长。monthly 是每月的，也是月刊。

moral 是道德的，也是"寓意"：the moral of the story。mortgage 是抵押贷款，这个词里有"死"和"契约"两个老词根，记住它就不会拼错。motivate 是激励，motivation 是动机。mountainous 是多山的。movie 是电影，film 更正式。

multiple 是多个的，也是倍数。murderous 是"凶残的"，也是"要命的"：murderous traffic。musician 是音乐家。mysterious 是神秘的。naked 是裸的，the naked eye 是肉眼。narrator 是叙述者，narrate 是讲述。

nationality 是国籍，nation 是国家、民族。navigate 是导航，也是"应对"：navigate a crisis。necessary 是必要的，注意它是"必需的"，不是"重要的"。negation 是否定，negative 是否定的、消极的。negotiate 是谈判，negotiate with。

nest 是巢，也是"嵌套"。newspaper 是报纸，paper 是纸也是论文。nightly 是每夜的，也是夜间的。nominate 是提名，nominee 是被提名者。nor 是"也不"，跟在 neither 或否定句后面。nose 是鼻子，也是"嗅觉"：a nose for news。

nothing 是什么都没有，也是"微不足道"：nothing compared to。notorious 是"臭名昭著的"，注意它跟 famous 的分工。nowhere 是无处，也是"毫无进展"：getting nowhere。numerical 是数字的，numerical data 是数值数据。

从巷口到江边

oar 是桨，划船用它；动词是"划"。observant 是"善于观察的"，比 careful 更强调眼睛。obstacle 是障碍，obstacle course 是障碍赛。occasional 是"偶尔的"，occurrence 是"发生的事"，动词 occur。odour 是气味，英式拼写，美式 odor，多指不好闻的。

offset 是抵消，也是"胶印"。olive 是橄榄，olive oil 是橄榄油。onward 是"向前"，onwards 也是。operation 是手术，也是"运营"，也是"运算"，三个意思都很常见。opposite 是相反的，也是"对面的"；be opposite to 是"与……相对"。optimal 是最优的，比 best 学术。oral 是口头的，也是"口腔的"。

ordinary 是普通的，out of the ordinary 是"不寻常"。originate 是起源，originate in 是"起源于"。outdated 是过时的。outlet 是出口，也是"折扣店"，也是"发泄途径"。outsider 是局

外人。overall 是总体的，也是"工装裤"。overlap 是重叠。

overt 是公开的，overt discrimination 是明面上的歧视。owe 是欠，owe somebody an apology 是"欠某人一个道歉"。oxygen 是氧气。paddy 是稻田，也是"暴怒"。palace 是宫殿。panorama 是全景。paradise 是天堂，这一章里它是房产广告里的词。

pardon 是原谅，也是"请再说一遍"，比 sorry 正式。parrot 是鹦鹉，动词是"机械重复"。particular 是特定的，in particular 是"尤其"。pass 是经过，也是"通过"，也是"传球"。password 是密码。patch 是补丁，也是"一小块地"。

pattern 是图案，也是"模式"。pea 是豌豆，as like as two peas 是"一模一样"。peasant 是农民，带历史感，现代多用 farmer。peep 是偷看，也是"吱一声"：not a peep。peninsula 是半岛。percent 是百分比，数字后面用 percent，不用 percents。

performance 是表演，也是"业绩"，也是"性能"。periodical 是期刊，periodic 是周期性的。permissive 是"放任的"，permission 是许可。perseverance 是毅力。personally 是"就我个人而言"。pessimistic 是悲观的，反义 optimistic。

petroleum 是石油，petrol 是汽油。philosophy 是哲学，也是"人生态度"。picky 是挑剔的，比 choosy 口语。pigeon 是鸽子。pin 是大头针，也是"别住"，pin down 是"确定"。pirate 是海盗，也是"盗版"。

plague 是瘟疫，动词是"困扰"。planning 是规划，plan 是计划。playground 是操场。plumber 是水管工，注意 b 不发音。poet 是诗人，poetry 是诗。poisonous 是有毒的。polite 是有礼貌的，注意它是"客气"，不是"善良"。

pollute 是污染，pollution 是污染。popular 是受欢迎的，也是"大众的"：popular culture。porridge 是粥。portrayal 是刻画，动词 portray。possible 是可能的，as soon as possible 是"尽快"。potato 是土豆。powder 是粉末。

preacher 是传教士，也是"说教的人"。precision 是精确，形容词 precise。predominant 是"占主导地位的"。pregnant 是怀孕的。preoccupy 是"使心事重重"，形容词 preoccupied。preservation 是保护、保存，动词 preserve。

prestigious 是"有声望的"，名词 prestige。prevalent 是"流行的"，比 common 正式。priceless 是无价的。princess 是公主。prison 是监狱。probe 是探查，也是探测器。processor 是处理器。profession 是职业，professional 是专业的。

profound 是深刻的。prolong 是延长，prolong the agony 是"延长痛苦"。prone 是"易于……的"，be prone to。propagation 是传播、繁殖。propose 是提议，也是"求婚"。prosperity 是繁荣。prototype 是原型。provincial 是省的、地方的，也可以指"土气的"。psychiatrist 是精神科医生，psychology 是心理学。pudding 是布丁。purpose 是目的，on purpose 是"故意地"。puzzle 是谜，也是"使困惑"。quantification 是量化，动词 quantify。quarterly 是季度的，也是季刊。quiet 是安静的，也是"平静的"。

听证会里的那些词

rabbit 是兔子，也是"唠叨"：rabbit on。radio 是收音机，也是无线电。raid 是突袭，也是警方的"查抄"，动词同形。rally 是集会，也是"重整旗鼓"，股市反弹也叫 rally。rapidity 是迅速，形容词 rapid，比 speed 正式。ratification 是批准，动词 ratify，专指正式追认。

ray 是光线，a ray of hope 是"一线希望"。reading 是阅读，也是"读数"，也是"解读"：my reading of the situation。reap 是收割，reap the benefits 是"收获好处"。rebellious 是叛逆的，名词 rebel。reception 是接待，也是" reception desk"里的前台，也是信号 reception。

recite 是背诵，也是"历数"。recorder 是录音机，也是"记录员"。rectification 是纠正，动词 rectify。reduction 是减少，动词 reduce。refine 是提炼，也是"改进"。reformist 是改良主义者，注意它跟 revolutionary 的对立。

refund 是退款，动词名词同形，重音不同。regime 是政权，也是"制度"：a tax regime。regret 是后悔，动词名词同形。rehearse 是排练，也是"默述"。rejoicing 是欢庆，动词 rejoice。relay 是接力，也是"转达"。

relief 是 relief：减轻、宽慰，也是"救济"。remain 是仍然是，也是"剩下"。remind 是提醒，remind somebody of 是"使想起"。renaissance 是文艺复兴，也是"复兴"。renown 是名望，形容词 renowned。repeat 是重复。

reply 是答复，比 answer 正式一点。repressive 是压制的，动词 repress。reputation 是名声，注意它跟 fame 不同：reputation 可好可坏。resemblance 是相似，动词 resemble，注意 resemble 是及物动词，不加 to。reshuffle 是改组，内阁改组就用它。

resistance 是抵抗，也是电阻。respectable 是"体面的"，respectful 是"恭敬的"，respective 是"各自的"，这三个词一起考的时候，错的人最多。responsible 是负责的，be responsible for。restrict 是限制，restriction 是限制。

retailer 是零售商，retail 是零售。retired 是退休的，retire 是退休。return 是返回，也是"收益"。revenue 是收入，多指政府或企业的。revival 是复兴，动词 revive。rhetoric 是修辞，也是"花言巧语"，这个词的贬义很常见。

rid 是"摆脱"，get rid of。rig 是"操纵"，rig an election，也是钻井平台。riot 是暴乱，run riot 是"失控"。ritual 是仪式。robbery 是抢劫，rob 是抢。role 是角色，play a role in。rope 是绳子，也是"诀窍"：know the ropes。ruling 是裁决，也是"执政的"。rusty 是生锈的，也是"生疏的"：my English is rusty。

那些"看起来都认识"的词

saint 是圣人，也是"好人"的口语说法。salt 是盐，take it with a grain of salt 是"半信半疑"。saying 是谚语，as the saying goes。scarcely 是"几乎不"，注意它本身带否定，后面不再加 not，这是四六级写作里最常见的双重否定。scenic 是风景优美的。scholarly 是学术的，scholar 是学者。

scissors 是剪刀，永远复数。scrap 是碎片，也是"废弃"：scrap the plan。scrub 是擦洗，也是"取消"。secondary 是次要的，也是"中学的"。secure 是安全的，动词是"获得"：secure a

place。seek 是寻找，seek to 是"试图"。seize 是抓住，seize the chance。

selfless 是无私的。seniority 是资历，按资历排序叫 by seniority。sentence 是句子，也是"判决"，动词是"判刑"。sequential 是顺序的，副词 sequentially。serving 是一份（食物），也是"服务中的"。seventeen 是十七。sexy 是性感的，口语里也可以形容"有吸引力的"：a sexy idea。

shall 是"将要"，第一人称旧式用法，也是法规里的"应当"。sharp 是锋利的，也是"敏锐的"，也是"（时间）整"：at eight sharp。sheer 是纯粹的，也是"陡峭的"：a sheer drop。shine 是发光，也是"出色"：shine at school。shiver 是发抖。shortage 是短缺，a shortage of。

shove 是猛推，比 push 粗。shrink 是收缩，也是"退缩"：shrink from。shyness 是害羞，shy 是形容词。sightseeing 是观光，go sightseeing。silent 是沉默的，也是"无声的"。simple 是简单的，也是"朴素的"。since 是"自从"，也是"因为"，两个用法要分清。

sinister 是"阴森森的、不祥的"，比 evil 更带气氛。situate 是"使位于"，常用被动：be situated in。skating 是滑冰，skiing 是滑雪。skirt 是裙子，动词是"绕开"：skirt the issue。slash 是猛砍，也是"大幅削减"：slash the budget。slender 是苗条的，也是"微薄的"：a slender chance。

slogan 是口号。smartphone 是智能手机。smoky 是冒烟的。snap 是"啪地折断"，也是"抓拍"，也是"厉声说"。sociologist 是社会学家。soften 是软化，也是"缓和"。solemnity 是庄严。solitude 是独处，注意它跟 loneliness 不同：solitude 是选择，loneliness 是被迫。somehow 是"不知怎么地"。

song 是歌。sorry 是抱歉，也是"难过的"。southeastern 是东南的。spaceship 是宇宙飞船。sparkle 是闪光。specific 是具体的，比 concrete 更强调"明确"。spectrum 是光谱，也是"范围"：a spectrum of opinion。speedy 是迅速的。

spicy 是辣的，也是"有刺激性的"：spicy gossip。spiritual 是精神的。spokesman 是发言人，现在多说 spokesperson。spontaneous 是自发的，注意它是"没有事先安排"，不是"冲动"。spouse 是配偶，法律用语。squad 是小队。

stagger 是蹒跚，也是"使震惊"，也是"错开（时间）"。stake 是桩，也是"利害关系"：have a stake in，也是"赌注"。standpoint 是立场。starvation 是饥饿。steadily 是稳定地。steep 是陡的，也是"（价格）高得离谱"。stifle 是压制，stifle a yawn。

straight 是直的，也是"直截了当的"。strap 是带子。street 是街。string 是细绳，也是"一串"：a string of。stubborn 是固执的，比 determined 贬，比 obstinate 轻。stump 是树桩，也是"难住"：the question stumped me。

subjective 是主观的，反义 objective。subscribe 是订阅，也是"同意"：I subscribe to that view。substitution 是替换，动词 substitute A for B，注意这个介词方向，写反了整句就错。succeed 是成功，succeed in。sudden 是突然的，all of a sudden。suggest 是建议，也是"暗示"。

sulphur 是硫，英式拼写，美式 sulfur。sunshine 是阳光。superstition 是迷信。suppress

是压制，比 stifle 硬。surfing 是冲浪，也是"网上浏览"。surprise 是惊讶，动词是"使惊讶"。suspect 是怀疑，注意它跟 doubt 的分工：suspect 是"觉得是"，doubt 是"觉得不是"。

sustainable 是可持续的。sweet 是甜的，也是"贴心的"。synonym 是同义词。

最后那几页

tackle 是"对付、处理"，也是橄榄球里的擒抱。takeoff 是起飞，也是"模仿"。tan 是晒黑，也是棕褐色。tariff 是关税，也是价目表。teach 是教，teach somebody to do。technique 是技巧，注意它跟 technology 不同：technique 是"怎么做"，technology 是"用什么做"。telescope 是望远镜。

temple 是庙，也是太阳穴。tenant 是租户。terrific 是"极好的"，旧义是"可怕的"。test 是测试，也是"考验"。thank 是感谢，thank somebody for。them 是宾格，they 是主格，这两个词在口语里最容易连读成"them"听不见。therapy 是治疗。third 是第三，也是三分之一。

thorough 是彻底的，注意它跟 through 只差一个字母，是两个完全不同的词。threaten 是威胁。throat 是喉咙。Thursday 是星期四。tie 是领带，也是"打平"，也是"绑定"。timber 是木材。tire 是轮胎，也是"疲劳"。toast 是吐司，也是"祝酒"。

tolerable 是可容忍的，动词 tolerate。tone 是语气，也是色调。topic 是话题。touch 是触摸，也是"感动"：be touched。towel 是毛巾。tract 是一片土地，也是"道"：the digestive tract。tragedy 是悲剧。

transform 是转变，比 change 彻底。translation 是翻译。transplantation 是移植。tray 是托盘。tree 是树。triangular 是三角形的。trifle 是小事，a trifle 是"有点儿"。trivial 是琐碎的，比 trifle 更常作形容词。

trousers 是裤子，永远复数。truth 是真相。tug 是拖，也是拖船。turkey 是火鸡，也是"失败之作"。TV 是电视。ultra 是"极端的"前缀。underdeveloped 是不发达的。undermine 是"逐渐削弱"，画面是"在底下挖空"。

undoubtedly 是"毫无疑问"。unify 是统一。universe 是宇宙。update 是更新。upset 是难过，也是"打乱"：upset the plan。urinate 是排尿，医学用语。user 是用户。vaccinate 是接种。vapour 是蒸汽。

vegetable 是蔬菜。vent 是通风口，也是"发泄"：vent your anger。verge 是边缘，on the verge of。very 是非常。vibration 是震动。video 是视频。violate 是违反，也是"侵犯"。virtual 是虚拟的，也是"实际上的"。

vision 是视力，也是"愿景"。vocabulary 是词汇量。volatility 是波动性。voluntary 是自愿的。vulnerability 是脆弱性。want 是想要，也是"缺乏"（正式）。warn 是警告。waste 是浪费，也是垃圾。wave 是波浪，也是"挥手"。wealth 是财富。

web 是网，也是万维网。week 是星期。welcome 是欢迎。what 是什么。whereabouts 是下落。whilst 等于 while，英式书面语。whoever 是无论谁。wicked 是邪恶的，口语里也是"太棒了"。wife 是妻子。window 是窗户。

wire 是金属丝。withdraw 是撤回，也是"取钱"。witty 是机智的。woollen 是羊毛的，美式 woolen。world 是世界。worth 是值得，be worth doing。wrapping 是包装纸。wrist 是手腕。yeah 是口语的 yes。yesterday 是昨天。zero 是零。

剩下六个词的落点

有几个词我在这章里塞得很勉强，得老实交代它们落在哪儿，你回头找的时候才不会以为漏了。

AIDS 在志愿站墙上那张红字海报里，就是那句"共用餐具不传染"。这个词永远是全大写，因为它是一个病的缩写，不是一个普通名词，所以考试里写成 Aids 是要扣分的。saddle 在失物柜那一页：一辆自行车的 saddle 被拆下来，孤零零躺在毛巾上。这个词最容易被记成"马鞍"就完了，其实凡是有座的东西都能用它， saddle 也是动词，"使承担"：saddle somebody with a debt。

sanction 是这一章最阴的一个词，因为它有两个相反的意思：批准，和制裁。国际新闻里说 sanctions，是制裁；公文里说 the council sanctioned the plan，是批准。同一个词，方向相反，只能靠上下文。这种词英语里不多，但每一个都值得单独标出来。satellite 在天上，也在那张海报上：卫星电视、卫星城，都是它。saturation 是饱和，我在房产广告和听证会两处都用了它——市场饱和、信息饱和，这个词的画面是"再也装不进去了"。so 是最普通的词，也是最难讲清的词：它是"所以"，是"这么"，是"到目前为止"（so far），也是口语里的"那么"。

这一章跟四六级有什么关系

关系比你想的直接。

四级的长篇匹配，考的是"哪一段在说这件事"。这一章里每一节其实都是一段：站台、巷子、打印店、志愿站、江边、听证会、末班车、体检、告别。你读的时候如果能在心里给每节起一个六个字以内的名字，你就已经在做长篇匹配了。真正的技巧不是找同义词，是判断"这一段的功能"——它是在描述、在举例、在反驳、还是在给方案。功能判断对了，位置自然就出来了。

仔细阅读考的是"这句话站在谁那边"。这一章通篇都在训练这件事：宣传册的语气、通知书的语气、法官的语气、外卖员的语气，四种语气指向四种利益。你以后读任何一篇说明文或议论文，先问这一句，再问"作者为什么现在说这件事"，两问下去，主旨基本就定了。

翻译考的是"中文没有的形式"。这一章里我刻意写了很多被动、很多分词、很多名词化结构：be displaced、being wrapped、the accumulation of small numbers。中文说"他们被迁走了"，英文更常说 they were displaced；中文说"小数目攒起来"，英文更常说 the accumulation of small numbers。你在这一章里见惯了这种"名词化"的写法，自己动手的时候才不会一句一句按中文语序翻。

写作考的是"你有没有具体的东西可写"。这一章能给你的，是九个可以反复使用的场景：一张表格、一次听证、一趟末班车、一场雨里的送餐、一间打印店、一次体检、一场巷子里的婚宴、一个失物柜、一位学英语的老人。任何一个作文题，你都能从这九个场景里挑一个塞进去，塞进去的文章就有了气味和温度。这比背一百个"高级词汇"管用。

所以这一章不是"阅读材料"，是"素材库加语法样本"。你把它当故事读一遍，当资料读第二遍，当模板读第三遍，三遍下来，它才真正开始为你工作。

关于"另一面"这个说法

"城市的另一面"这个题目，容易让人以为是"穷"和"苦"。我写的时候一直在躲这个词。

因为一旦你把它写成"苦"，你就站到了高处，你就成了那个隔着墙看的人。林一从墙那边走过去，看到的不是苦，是密集：一条巷子里同时发生着三十六件事，每一件都有它的规矩、它的成本、它的体面。婚礼要摆八桌，是因为不摆会被记十年；房租涨十个点，是因为地铁要通；体检排队排到街角，是因为免费的东西永远比收费的更需要排队。这些都是经济，不是悲情。

所以这一章里我几乎没有写"可怜"。我写的是手续、价格、时间、复印件、以及人在这些硬东西之间怎么保住自己的样子。你要从这个角度读，那九百多个词才会各就各位：它们是这些手续和价格的名字。

如果你读完只记住"他们很不容易"，那等于什么都没记住。如果你读完开始留意自己学校门口的保安几点换班、食堂阿姨从哪儿来、快递站在哪儿租的房子，那这一章就算读成了。词汇是跟着注意力走的，注意力在哪儿，词就长在哪儿。

怎么重读这一章

第一遍你只要跟着走，不要查词。查词会把一条巷子拆成一地砖头。第二遍你只做一件事：把加粗的词念出声，念完立刻回到它所在的那半句，看它跟谁搭配。第三遍你合上文章，用中文把林一那半年讲一遍，讲到某个地方卡住了，那个地方就是你真正没读懂的地方，不是词的问题，是关系的问题。

记词这件事，最没用的动作是抄写，最有用的动作是"用错一次再改对"。你写 substitute A with B，被人指出应该是 substitute A for B，这一次错比你抄二十遍都值。所以这一章真正的用法不是读，是拿它当素材：挑三个场景（表格、听证会、末班车），每个场景用五到八个新词，写三段一百词的英文，写完自己念一遍，念着别扭的地方就是搭配错了。

还有一个更省力的办法：把这一章当成一本"城市词典"来查。你以后写作文，想表达"很多人"，别只写 many people，回来看看 an army of、a cluster of、a string of 是怎么用的；想说"情况复杂"，别只写 very complicated，看看 configuration、coordination、formulation 各自管什么。词表是死的，场景是活的，你记住的永远是场景。

最后说一句不客气的：这一章九百多个词，你不可能一次全记住，也不需要。你需要的是知道它们存在过、在哪个句子里活过。等到某一天你在真实的报纸上再遇见 sanction，你会想起那个有两个相反意思的词，想起听证会上那个念稿子的议员——那一刻它才真正是你的。

到那时，你才算走完了这条巷子。

三句要紧的话

这一章我想留下的，只有三句。

第一句：词不在表里，在事里。你背 accept 的时候，它只是一个"接受"；你看见那张表上 volunteer must accept no gifts，它才变成一条你签过字的规矩。

第二句：别用中文的直觉去分配英文的介词。abide by、be addicted to、substitute A for

B、be responsible for——这些不是"搭配积累"，这是英语的骨架。骨架错了，词再漂亮也是散的。

第三句：读完这一章，不要回头去抄词。找一件你今天真的看见的事，用五个新词把它写下来，写不通就查，查完再改。五个词写十遍，比九百个词抄一遍有用得多。

林一那年冬天学会的东西，说到底不是九百多个单词，而是一件更小的事：他开始相信，一个城市的语言是有重量的，而重量总落在具体的人身上——落在那个把 ought to 抄成一个词的老人身上，落在把 vaccination 说成"打一针"的母亲身上，落在把 tariff 念成"塔里夫"的外卖员身上。你把这九百个词读到这里，真正该带走的，是这种重量感。

下一次你走进一条不认识的巷子，试着听清楚招牌上写了什么。那才是这一章真正的作业。

一条巷子教给人的三件事

第一件是"程序比道理管用"。你可能会觉得这话世故。可你回头看那场听证会：法官没有喊，他引条文；寡妇没有哭，她问问题；真正起作用的不是谁更有理，而是谁手里有复印件、谁记得日期、谁知道该找哪个窗口。英语里把这套东西叫 procedure，把懂这套东西的人叫 sophisticated，这两个词在课本里都是中性的，在生活里却是分层的。你学会用英语说"我要一份书面答复"，比学会用英语说"我很生气"有用一百倍。

第二件是"距离决定语气"。同一片十一个公顷的地，宣传册上叫 paradise，通知书上叫 occupant。词没有变，位置变了。所以读英文的时候，永远先问一句：这段话是写给谁看的，谁在付钱。你把这个问题带进阅读，四级的长篇匹配会突然变简单，因为出题人考的就是"这句话站在谁那边"。这也是我写这一章时反复提醒自己的一件事：不要让林一替那些人说话，让他把纸上的话原样念出来就够了。

第三件是"善意需要制度才留得下"。 Aunt Chen 那句 kindness without rules is just business with a smile，不是刻薄，是她干了九年之后的结论。你会发现这一章里最动人的东西都不是善意本身，而是善意的形式：一张写清楚有效期的证、一条不许收现金的规矩、一本记着谁答应过什么的蓝本子。英语里把这种形式叫 institution，把没有形式的善意叫 gesture。gesture 很轻，institution 很重，重的那个才救人。

我说这三件事，不是要你记住三个英文词，是要你在读下一篇文章、做下一篇阅读的时候，脑子里多一根弦：这段文字背后有没有一个窗口、一份文件、一个在等答复的人。有，你就读懂了；没有，你只是把词认了一遍。

最后提醒一句：这一章很长，长到不像一章。你可以分四次读完，每次读完在页边写一句话，写你今天最想记住的那个词，以及它出现的那半句。四次之后你会发现，你写下的那四句话，比任何一张词表都更像你自己的英语。

最后再说一件事

你大概已经发现，这一章比你以为的要长得多。我不打算为此辩解，但我想把这件事实话说清楚，因为它跟你怎么用这本书有关。

一张词表是随机切开的。切到我这一份的时候，里面有 epidemic 一类的社会词，有 atom、

electron 一类的科学词，有 banana、goose、chopstick 一类的厨房词，还有 oughtto 这种根本不该出现在词表里的排版事故。它们彼此没有关系，却必须住进同一篇文章。要把它们安置得不像展览，只有两条路：要么让文章变长，让每个词都有自己的场景；要么让文章变短，让词挤在一起，读起来像词典。我选了前一条。

所以你读的时候，不要用"一篇文章"的标准要求它，用"一个学期"的标准。它本来就不是一篇文章，它是半年。半年里一个人会同时遇见表格、婚礼、听证会、急诊、末班车、一场雨，这些事之间没有因果关系，只有时间顺序，而人的词汇就是这样长出来的——不是按主题，是按遭遇。

如果你只有二十分钟，读前三节和最后一节，剩下的当资料，需要的时候回来查。如果你有一个下午，从头读，读完合上，用中文讲一遍给室友听，讲不下去的地方就是你没读懂的地方。如果你只有十分钟又想有点用，翻到"一张纸上的九十来个词"那一节，把那一段里的词抄在手背上，一天用掉三个。

这本书不会替你记住任何东西。它只能做一件事：让这些词在你的记忆里有一个地址，而不是一个编号。地址会让你想回去，编号不会。

还有一件小事。这九百多个词里，有一大半你其实早就会，只是不会用。真正拦住你的是另外一小半：那些你认得、却从来不敢在作文里写出去的词——sanction、indifferent、plausible、notorious、respectable、vulnerable。它们不是难，是"没有场景"。场景一有，它们就活了。所以你读这一章，重点不是补生词，是给熟词找场景。

词汇量从来不是数字，是你能不能在需要的时候，从记忆里把那个词连同它所在的半句话一起取出来。

这句你可以背下来，然后忘掉这一章的其他所有话。

附记：那个学英语的老人

写到这里，我还欠一个人一段话。

那位六十八岁开始学英语的老人，练习本上写过一句"billion is a word I will never use in a sentence about my life, but I like to write it"。我当时笑他，后来才明白他不是在学词，他是在给自己造一个更大的房间。一个人一辈子住在三十平米里，忽然发现世界上有一个词的意思是"十亿"，他把它抄下来，不是为了用，是为了知道自己住的屋子有多小、外面的世界有多大。

这也是我想让你带走的东西。词表最坏的作用，是让你以为语言是一堆可以称重的货物；词表最好的作用，是让你发现每一个词背后都站着一群人，他们用这个词吵架、签合同、写情书、骂人、求人。你背 sanction 的时候，它只是两个相反的意思；你读完这一章，它应该是那个念稿子的议员、那份写着"十四天"的通知、那个说"我要一份书面答复"的女人。

所以别问"这些词考不考"。考。但考的方式不是让你默写词义，是让你在一篇你没读过的文章里，看见一个词，立刻知道它站在谁那边。这件事没有办法速成，只能一篇一篇地读、一次一次地猜、一回一回地改。

那年冬天林一最后一次走过那条巷子，打印店的黄纸还在窗上，失物柜里那只单只的靴子还在，Aunt Chen 的蓝本子锁在第二个抽屉里。他没有变成另一个人，他还是那个爱记笔记、有点自嘲的大

一学生。但他再看一份表格的时候，会先翻到背面。

这就是这一章想给你的东西：不是九百个词，是翻到背面的那个动作。

第 5 章　回家

The last exam ended at four, and by six the corridor smelled of plastic bags and stale noodles. Zhang Ye was trying to cram a heavy overcoat into a shabby suitcase that had lost a wheel in the twelfth week. He did not bother to fold anything. He piled his clothing in, sat on the lid, and pulled the zip beneath his knee.

"You must have a system," I said.

"I have one," he said. "It is called luck."

My own bag was small and orderly, a trait I am not proud of and cannot quit. In the front pocket I kept an envelope with the card ticket inside, a handful of coins, and a page torn from a notebook on which I had written the exact hour of departure in black ink, twice, in case my first spelling of the word "platform" turned out to be wrong.

Zhang Ye read it over my shoulder. "You wrote yours out by hand," he said. "The sign above the gate says the same thing."

"I like to compare the paper with the sign," I said. "It is a precaution, nothing more."

He stood the case up, and it collapsed sideways like a sick animal. We both looked at it. "Admittedly," he said, "not my best design."

We went upstairs once more for his charger, came down again, and by the time we reached the crossing outside the west gate the daylight had gone a pale blue. The lane was full of students dragging identical bags in zigzag lines. Everywhere there was the sensation of a system switching off: the canteen dark, the booth on the lawn shut, a pile of rubbish already soaked by the drizzle and smelling of wet flour.

"Those who live near go first," Zhang Ye said. "Each year the same tempo. By Saturday the whole campus is stagnant. You can hear a tap dripping on the fourth floor."

The hall

The station was a dome of concrete and glass. The announcement came out of a channel that buzzed before every sentence, as if the speaker had to clear his throat. Owing to engineering work, the 19:06 service would depart approximately eight minutes late. Nobody moved. Delay was the one thing in that building which ran on time.

Taped to the glass was a notice in three languages. "Abusive behaviour towards staff

may incur a fine and will result in refusal of carriage. Restrictions applicable to reduced fares. Retain your ticket throughout the journey. No waiver of these terms is permitted." I read it twice, the way I read advertising on the back of a shampoo bottle, out of an addiction that has never once been useful to me.

A trolley of luggage went past, pushed by a woman with a bushy fringe of grey hair who apologised to everyone in advance. An ambulance edged through the plaza outside with its siren off, unhurried, and two bystanders turned to watch it the way people watch anything that is not their own trouble. Near the coffee kiosk a family huddled around a phone, four heads touching, watching a video of a room they were about to see in person.

Zhang Ye put his case down and shook my hand, which he had never done before. "Cordial," I said. "You are being cordial."

"I am being considerate," he said. "You looked like a man who needed to be assured of something."

He was not a gentle talker, but he had the ability to say a hard thing and leave your pride intact. We were acquaintances of five months and nothing more, and I have thought about that handshake since.

The window

The train pulled out beneath a wireless signal that died past the second tunnel. I sat with my face to the glass and let the country absorb me: a valley that narrowed and then opened, a slope of brown branches, shrub after shrub climbing in the hazy daylight, icy water standing in the fields in flat sheets that gleamed and then went dull as the light failed. Erosion had cut a gorge on the left. A giant pylon stood on the ridge with its cables hanging in a curve. The foggy valley blurred at the edges, a mill chimney went past, and then there was nothing but earth and darkness and the steady heart of the wheels.

I had approximately four hours and one task: not to think about the semester. It did not work. My mind wandered into the assessment I had been given in the last week of term, the one that said my attitude was good and my capability was not yet demonstrated. Admittedly, that was a fair sum of the matter. I had expected the university to be challenging; I had not expected it to be so methodical about telling me where I stood.

My mother called at eight forty. She asked whether I had eaten. She asked whether the train was on time. I said it was due at twenty past ten and due to be late, which are two different meanings of one word and a fair demonstration of why I like English. She said my aunt would be at the terminal with the truck, because the jeep was in the workshop again, and that Grandma had been cooking since noon.

"Have you told her I am coming?"

"She has known for a week," my mother said. "She has been cleaning a room that was already clean. Incidentally, she has moved your desk twice."

The town

The terminal in my hometown is new. It has a municipal plaza in front of it the size of a football pitch, four palms in cement tubs, and a logo of a corporate sponsor carved into the marble of the entrance, a symbol that means nothing to anyone who lives here and everything to the accountant who sold the naming rights. Eight years ago this was a market with a tin roof and a stagnant drain; now it is a spacious hall with escalators that occasionally work.

A-Zhe was waiting at the kerb in his father's truck, engine idling, radio on. He is nineteen and has already learned to drive the way country boys learn: early, badly and forever. He took my bag, flinging it into the rear, and said, "Yours is lighter than last time. Lost weight?"

"Lost laundry," I said.

We drove out along the outskirts where the sprawl begins: a strip of shops, a tyre shop, a funeral parlour with plastic flowers in the window, a fried-chicken outlet with a picture of a smiling chicken that has haunted me since childhood. The advertising hoardings were all new: luxury flats, a dental clinic, a wedding photograph of a couple who cannot possibly be from here. Everywhere the town was upgrading its surface and leaving its inside alone.

"Do you notice it?" A-Zhe said. "Every year you come back it is more like somewhere else."

"I notice," I said. "I just cannot decide whether it is growth or displacement."

He thought about it, which is one of the things I like about him. "Both," he said at last. "Like a person who gets rich and forgets how to cook."

The door

The lane to the old house is narrower than I remembered, or I am wider. It runs between two walls of grey brick with moss in the joints, past a public tap that has run continuously since before I was born, and ends at a stoop of five stone steps, worn into shallow bowls in the middle where four generations of feet have tread. The door is wood, painted dark red so many times that the panels have softened, and the hinge groans in two notes that I could hum.

I stood on the stoop with my hand on the lid of the letter box. Through the door I could hear a knife on a board, steady as a clock, and underneath it the murmur of a radio playing a local opera at low volume. The smell that came out under the door was salty,

smoky and sweet at the same time, cooked pork, burnt sugar, steam, and it hit me somewhere below the ribs, harder than the cold had.

I had been away for eighteen weeks. I had learned the names of six phases of matter, the etymology of half the words in my chemistry textbook, and how to sleep four hours and still sit an exam. I had not learned how to open a door.

The kitchen

She opened the door before I knocked, which she has done all my life. She is seventy-six, and she looked at me the way she looks at a vessel she is about to fill: alert to the cracks first, pleased by the shape afterwards.

"You are thin," she said.

"I eat."

"You eat what other people cook." She took the bag out of my hand with a forceful tug that amazed me every time, and conveyed it to the table without visible exertion. "Come in. Almighty god, it is cold. Shut that door or the whole house will be a fridge."

The kitchen is the interior of that house and, I think, of her life. It is about three metres by four. The cooker is the old kind, clay and cement rendered over brick, with two holes for pans and a flue that draws badly when the wind turns. A bulb hangs from a wire that has been taped three times, and the light it gives is the colour of weak tea. Everything in that room smelt of tobacco, wood ash and something sweet I could never name; years later my mother told me it was camphor from the chest where the linen was kept.

Under the window there is a sink of cement with a brass tap, and the tap drips in a rhythm that has been the background of every meal I have eaten in that house. The plumbing was replaced the year I was born, and my grandmother has doubted it ever since.

"You ventilate a kitchen like this," she says, "by opening two doors and praying."

Dwelling in an old house is a skill. You learn which step groans, which door must be lifted before it will close, where the damp comes up through the floor in August and where the frost gets in in January. I had that skill once, at eight or nine, and I have surrendered it.

Above the rack of bowls there is a barometer that has not moved since 1998, a certificate in a frame with a gold seal, and a photograph of my grandfather at forty, in a collar and a tie, looking like a man who has just been told a joke and is deciding whether to laugh. He was a learned man in the way small towns produce them: he could read a contract, quote a poem at length, and arbitrate a dispute over a wall without raising his voice. He also trusted the barometer more than the weather report on the radio, which

is why it is still on the wall, and why nobody has been allowed to take it down.

Her hands are the colour of walnut and they do not shake. She has lost most of her teeth, which makes her articulation of certain words a guess for strangers and never for me. When she says my name, it is two syllables and a long fall at the end, and no one else in the world says it that way.

In the corner stands the clay jar of pickled greens, with a stone on the lid. She makes one batch a year, in November, and it is the only thing in the house with no preservative in it at all, not one ounce of anything that came out of a packet. The flavor is salt, chilli, garlic and time. Dilute a spoonful in hot water and it becomes a soup that will burn your tongue if you are hasty; leave it whole on rice and it sits in the belly like a stone. My aunt says it is good for the gut and bad for the kidneys; my grandmother says the kidney complains only if you deserve it.

Her account book lives in the drawer under the cushion of her chair, and it is the most methodical document I know. Each page is one week. Each line is a decimal with two places after the point. The sum at the foot of the page is checked in pencil and then again in ink, and if the two do not agree she does the whole week over rather than rectify the total.

Her estimation of the percentage she spends on food is within two points of the truth, which is a statistic I find harder to believe than any equation I met this term. Her income is a pension, the rent of two rooms, and whatever my mother slips into her pocket and pretends she has not. Her budget does not exceed her income, ever, which is more than I can say for mine. Costly things are bought once. She will not deposit money in a bank she cannot walk to, so the remainder is folded into the currency of an older economy, notes in a tin, and the aggregate for the year is written on the last page in a hand that gets smaller as the numbers grow.

I sat opposite her while she read it, and she said, without looking up, "Your mother tells me you have learned about silicon."

"I have learned about silicon," I said.

"Silicon." She said the word the way she tests a melon. "And what will you do with it?"

"I do not know yet. Something durable, I hope. Something that does not decay."

She turned a page. "Then you will be the first," she said, "in this family."

It was on that afternoon, while she was writing down the price of flour, that Shen Lu's message arrived: a photograph of her commencement invitation, laid on a table with a good pen beside it. The printer had broken one word across two lines and left MENCEMENT alone under the hyphen, so that it read like a word from another language, which is

roughly how that whole year had felt. I wrote back a sentence I hoped was sincere and not merely polite. She answered with a farewell that was also a reminder that I still owed her a report on the semester.

The market

The market in that town is held every other day, from six until eleven, in a square behind the bus station which is a car park for the rest of the week. By seven the booths are up and people congregate in the passage between them, and the gathering has a riotous sound that is not anger, though it is close enough to it that a stranger would flinch.

At the entrance there is a scale with a sheet of plastic tied to it and a warning about short weights, left over from a campaign nobody has thought about since 2019. The stalls are plentiful: grocers with flour in open sacks, a dairy counter where butter and cream are cut in thick slabs with a wire, fruit in bunches, oranges from the southern province, limes the size of a thumb. A vegetarian diner at the corner serves noodle soup out of a vessel the size of a bucket, and beside it a woman sells lemonade, coffee, and a herbal drink that tastes of burnt sugar.

I bought candles for the ancestor table because my mother had forgotten, and I stood in the negotiation over price the way I always do, which is badly. The woman wanted eight for the box. I offered six. She told me the cost of wax had risen. I said I was a student. She said her son was a student. We settled at seven, with a courtesy on both sides that was entirely sincere and a politeness that was entirely professional. It is a formality I would not skip for anything; it is the only conversation in that market where both people know the ending before it starts.

At the rear, where the drain is, is the butcher's area, and it is appalling. A cage of chickens stood against the wall. A woman was sliceing the throat of one with a knife she sharpened on the edge of a stone trough, and the birds still in the cage pecked at the wire and did not look up. Disgust is a luxury of people who have never had to kill their dinner. I felt it anyway, then felt dismay at feeling it, and then, somewhere underneath, a childish relief that the killing was not mine to do.

A cry went up at the front of the stall. A boy of four had grabbed a handful of something out of a basket, and his mother was scolding him in a voice that carried ten metres. He dodged her hand, darted between two tables, and she snatched him by the collar without spilling her bag. Nobody turned to look. It was Saturday, and half the town had been that boy at some point.

There is a cunning in that market which is not quite dishonesty. The greedy man at the corner rests his thumb on the scale. The petty thief takes one orange and leaves the money. The nosy woman asks what my income is going to be and then tells me what it ought to be. You learn to behave like a local: to compare two stalls before you buy, to keep

your choices to yourself, and never to be hasty with a stranger who might be your aunt's neighbour.

On the way home we passed a lodge near the terminal with a sign in its window, in English, because for two years in the nineties a geology team from abroad had lodged there and the owner had never taken it down. It said: ROOMS / ACCOM. / HOT WATER / WIFI. The plastic had yellowed. Someone had corrected the spelling of nothing.

An old man went past with a donkey and a cart of sweet potatoes, and the animal walked with the steady patience of a creature that has been doing this since before the road was paved. My grandmother says there were forty donkeys in this area when she was a girl, and that each one wore a different bell, so that people knew the hour by which of them went past the bar at the corner.

The house keeps its own population of pests, and she has made peace with all of them. Ants come for the sugar in August. A spider holds the corner of the bathroom window and is not to be swept away, because it eats the mosquitoes. Beetles get into the rice and are picked out one at a time by hand, on the outdoor table, in the daylight, which is the only part of the work she says she enjoys.

The lineage hall

The hall of the lineage stands where the lane meets the river bay. You ascend nine steps to reach it, three more than any house in the village has, and those three steps are the whole argument of the architecture. The doors are painted with two guardians whose faces have been blurred by a century of weather. Inside it is cool in summer and colder in winter than the air outside, and it smells of wood, ash and paper.

On the altar the tablets stand in rows, eleven generations of offspring with the same two surnames alternating, and beside them a bronze ornament that nobody can date. The carving on the screen behind the tablets is nineteenth-century work, blade-thin, and the sculpture of the lioness at the gate has a cub under her paw, worn smooth by children who climb on it. There is a monument in the courtyard to four men who drowned in the flood of 1998, and their names are cut deeply enough to retain the rain.

This village had an aristocracy once: four families who owned the paddy, the mill and the tractor, and whose sons became teachers and clerks while everybody else's sons became farmers. The nobility of that arrangement, such as it was, rested on rice and rain, and it ended in 1950, which is why my grandmother can talk about it without either aversion or longing. The magistrate in the county town was, for three hundred years, a man from one of those families. Now the councillor is a woman from a family that owned nothing, and she works eight days a week.

The tribal logic of the place is still visible: who sits where at the festival, who

carries the flag, who is allowed to light the first candle. Clergy never had much of a hold here. The church in the county town is called a cathedral by people who have never seen one, and the clergy there are two old men and a choir.

Her devotion is practical. She dusts the tablets, replaces the paper, and sweeps a floor that nobody walks on. For an hour a day the hall is a sanctuary from the market, and she uses it to sit. Nobody takes an oath in that hall any more; the oath was taken in 1730 and has been inherited, like the roof. I am not superstitious. I bowed twice.

On the third night of the New Year there is an opera on a temporary stage in the courtyard, and it is free, and it is four hours long. The heroine dresses as a man and leads an army; the emperor sits on a throne that wobbles; "Your Majesty" is said eleven times in the first act. It is part of an epic that would take three nights to sing whole, and nobody present has heard all of it. The satire is aimed at a greedy official, and the audience laughs at lines written two hundred years ago, which is the thing I love most about that courtyard: the joke still works. The play is set in the reign of an emperor who may never have existed. Between acts an old man tells riddles to the children, and a fairy tale about a peach thief, and the children clap at the end of each one whether or not they have understood it.

This is also where the village mourns. When the old man who read the names died in October, they put his tablet in the row and grieved loudly for two days and ate for three, which is the custom, and which is, I think, a better arrangement than any I have seen since.

The album

She brought out the album on the second evening and put it on the table between us like a document. It is biographical without trying to be: no dates, no order, photographs glued in with a flour paste that has gone brown at the corners. On the first page a girl of sixteen in a stiff collar, shiny hair, a dot of a mole under one eye. That is her mother.

"Her maiden name was Chen," my grandmother said, "and mine too, until I was twenty."

Three pages later there is a wedding: the bridegroom is twenty-three and looks terrified, which he was, because he had met her three times. There was no honeymoon. There was a train to Hangzhou, one night in a guesthouse, and a photograph in front of a pagoda. Her aunt was an heiress, which in that village meant a house, four mu of paddy and a reputation for meanness that outlived her by forty years.

My grandmother was the midwife for half the village between 1974 and 1992. By her own count she delivered two hundred and eleven babies, including me, and she can still tell you which of them cried first. Her elder brother served in a regiment in the north and

came home a sergeant, with a photograph of himself standing next to a horse.

In 1998 the county gave her a citation for meritorious service after the flood, on paper that is now yellow. That flood is a chapter of its own. The village was evacuated on a Tuesday; the hall served as an asylum for three days; the casualty list here was four names, and the fatality count in the county was thirty-one, most of them old. Her mother's generation remembered a famine in the year the river dried, and none of them ever wasted rice again.

In 1942 fascism reached this valley. There was a massacre at the next village, and the war took two of her uncles and left the house full of women. She tells it in a detached voice, the way people do who have decided that feeling is a thing to be spent later. Her depiction of that year is flat and exact: how many days, which road, who came back. I wrote an excerpt of it in my notebook that night and I have kept it. She will not write an autobiography. "Who would read it?" she said, and I did not have an answer that would have satisfied her.

A contemporary of hers came to the door while we were looking at the album, and the two of them talked for an hour about people I have never met. Before she left she took her identity card out of her bag to show me the photograph in it, in which she is twenty and does not look like herself. Of the eleven people in the wedding picture, three are alive. Two died in the same year, and their names are written on the back in pencil, in a hand that was already old.

On the twenty-eighth of the twelfth month my cousin Mei married a man from the next county, and the band was three men and a girl with a keyboard, and the chords were loud enough to shake the table. The performer sang badly and entertainingly. My uncle got drunk at nine and told the story of his own youth for the fourth time that month: he had been a disc jockey in 1993, in a disco in the county town, and he still owns the records.

My uncle watches the evening news the way other men watch football. That week it was an election in a country I could not have placed on a map: a nomination fight, a lobbyist with a briefcase, an inflation figure of eleven per cent, a deficit the size of a small economy, and a president who had escalated a quarrel over tariffs. A spokesperson for the administration called the other side's behaviour a unilateral breach; the opposition denounced the drastic language; parliament refused to ratify the treaty; the judiciary was asked to rule on whether the legislation was constitutional, which in that democratic system is an ordinary Tuesday. There had been an assassination attempt in 1994, a detention without charge in 2019, and a prisoner exchange nobody could explain. A journalist called the whole business a violation of the integrity of the process; her editor cut the sentence for length; a topical conference was held; a nuclear missile was tested; a diplomatic note was recalled.

My uncle said, "Whatever they say, it is about money," and turned up the volume.

Mei's son is eleven, and he asked me what materials engineering is, and I tried an analogy, which is always a mistake. "You know the tiles in the bathroom? Ceramic. The step outside? Concrete and steel. The pipe? Plastic. Everything in this house is a material somebody chose, and the choice was made by a person who knew something about corrosion, friction or cost."

He said, "So you choose things."

"I quantify things," I said. "We make a hypothesis, we build a variable, we measure."

"Like in science."

"Like in science, but with worse equipment." I told him about the microscope in the lab, the composite panels we had cut in the third week, the ceramic that cracked in the furnace because I had heated it too fast. A composite is two materials doing a job neither can do alone, and I said this with the confidence of a man who has read one chapter. It was a disappointment to him that none of it was new.

He asked about ore, about radioactive waste, about whether a robot could do my homework, and about the tractor outside, which has forty horsepower and a machine on the back that his father repairs himself because the nearest mechanic is forty kilometres away. He showed me a gadget his father had brought back from the city, with a circuit board you could see through the case, and asked whether cyberspace was a place you could go to. I said a patent is a promise a government makes to an inventor, and that the company which innovated that software had made more money from the promise than from the idea.

His brother's maths book was open on the table at differentiation, which he is two years away from, and he was stuck on multiplication at the bottom of the page. I drew an axis on the back of an envelope and could not remember how to finish it. The geology of this county is dull: no volcanic rock, no glacier scars, no ore worth mining, just mudstone and a swamp that used to be a lake. The nearest ranch is in a film. He asked whether gravitation was atomic, and I said no, and then spent the rest of the evening wondering whether that was the right answer.

Past the hall the ground drops to the river, and in winter the water is the colour of pewter and moves more slowly than water should. A bamboo raft lay afloat by the bank, tied to a post; further out the wreck of a fishing boat sat in the mud with its ribs showing like a skeleton. My grandmother says there was fishing here every morning before the dam went up in 1972, with a net on a pole, and that the scarcity of fish now is not the dam's fault but the fertiliser's.

Nothing in these hills will starve you, and nothing will eat you either, though the water is a danger for a week after rain. There are no bears left, no wolf since 1968, and the largest predator within ten kilometres is a dog with a bad temper. Her mother saw a

tiger on the ridge in 1934 and was believed, because she never told a lie about anything else. A monkey escaped from a travelling show in 1994 and lived in the bamboo for two summers, and enjoyed a brief popularity in the county; people left food out for it and it became, for a while, the only subject of conversation. Nothing here hibernates properly. The winter is too wet, and the frogs simply go quiet.

In February the wild plum blossoms along the bank and the whole outline of the village changes; in autumn the maples blaze for two weeks and then it is over. Climatic change in this valley means the river floods in May instead of June, which is worse for the rice and better for the mosquitoes, and there is an ozone warning on the radio in August which nobody here has ever acted on. Wild garlic comes up in March, wild rose in the hedge in May, and the wavy line of the bank is redrawn by every flood.

From the ridge you get a panoramic view of the whole valley, the breadth of the river at the bend and the depth of the channel below the weir. It is an hour's trek up, past a fort from the Taiping years that nobody has ever dug. My spatial sense is poor and I still got lost coming down. There is a simplicity to the place that I find poetic without being able to say why, and a glory in the way the light goes at four in the afternoon in January that no photograph has ever caught.

The ocean is three hours away and half the village has never seen it. The port at Ningbo is where my uncle's fish comes from, frozen, in a truck. Mei's son believed until he was twelve that a submarine was a kind of fish, and I did not correct him, because the correction would have been less interesting than the belief.

After rain the current swells and the raft pulls at its rope. My grandmother tends the bank the way other people tend a garden, which is a war against silt. She scrapes the mud off her shoes with a stick, pokes the ice with it in January, and in a hard year the ice is thick enough to smash with a stone, pauses at the top of the steps to get her breath, and says that thirst is the only honest appetite. I stood there until I was ready to go back, and the water went on rolling round the stones and seemed in no hurry at all, and I thought that a river is the only thing in this county that has never once been reorganised.

The clinic

Two days before I left she let me walk her to the clinic, which she agrees to once a year and complains about for the other eleven months. Her knee has been chronically bad since 1998: she carried sand for three days in the flood, and something in the joint never forgave her. The physician is a woman of about fifty who calls her Auntie and does not underestimate how much pain she is in.

"Acupuncture?" the doctor said. "Three sessions. Tuesday, Thursday, Saturday."

"Acupuncture," my grandmother said, in the tone of a woman accepting a punishment.

The treatment takes forty minutes. The assistant puts on an audio recording of classical music, which is the only indulgence the clinic allows itself. Six needles in the knee, two in the calf. She says it does not alleviate the ache so much as distract her from it, and that at her age this is what most remedies do. She has been seriously ill twice in my lifetime, and nothing seems to impair her memory of either year. She does not ail in the way the posters describe; she simply adheres to her own daily programme and refuses to downgrade it.

While she lay there I read the wall, because I read everything. A brochure in the rack said that vaccination against influenza was free for residents over sixty; that a urine test required no preparation; that the digestive clinic ran on weekday mornings only; that psychiatry was on the second floor by appointment; that insurance covered the basic items but not the expensive ones. There was a warning about pollution in the rainy season and a notice about a volunteer programme for literacy classes, which struck me as the only inspirational thing on that wall.

She refuses to have the cavity in her back tooth filled. She has a framework of belief about dentists that no elaboration from me can modify: they are all in it for the money, and the tooth will perish on its own schedule if she ignores it long enough. Notwithstanding forty years of evidence, she has not had a problem with that tooth yet, whereas the one she had filled in 1994 has been trouble ever since, thus proving her point. I have learned not to argue with deducible logic of that kind. Presumably this is what people mean by respectful disagreement.

Packing took two days, most of it hers. Six jars of pickle wrapped in newspaper, each one insulated in a plastic bag as insurance against the collisions of the rail network; a bottle of rice wine in a sock; dried pork; a bag of rice; a miniature embroidered shoe sole she had made for my mother; and a spoon from the kitchen that she insisted was better than any I would find in the city.

The transport of all this necessitated a second bag, which necessitated discarding two pairs of socks and a book. There is a redundancy in the way she feeds people that is not wasteful and not quite sentimental either. It is a provision strategy, the way a household prepares for a siege. Furthermore it is the only tangible proof, once I am back in a city, that the week happened at all.

I tried to refrain from saying that six jars was too many, lest she hear it as a reproach. Instead I said thank you, twice, and she said, "It is for the sake of your mother, not you," which is the usual form of her generosity, and which I have learned to accept thankfully.

On the last evening we sat in the kitchen and did not talk much. She asked whether I

had a girlfriend, which is her annual interview question, and I said no, and she said, "Soon." She asked whether the tuition was expensive and whether the scholarship had come through, and I said it had, and she said, "Then you are wealthy," and went back to her knitting. There was a sadness in the room that neither of us named. Presumably it was mutual, though I can only testify for oneself.

The pig

On the twenty-sixth they killed a pig in the yard behind the hall, and it took four men, a boy with a bucket and a good deal of shouting. The slaughter was quick and the yard went quiet afterwards, which is the part nobody photographs. My grandmother disapproved of the noise and not of the killing; she has seen too many to be sentimental about it. The skull went to the dog, the bones went into a pot that had been on the boil since seven, and by noon the whole lane smelled of it.

I was given a knife and told to cut the fat into cubes, which is women's work and therefore the only work worth doing. A pinch of salt, a stir with a wooden spoon, and the fat gave up its water and then its colour. The meat was tough and plain and it was the best thing I ate all winter. The local breed is small and black and nearly gone: four sows left in the county, and the agricultural office has started a register, which is the sort of thing that happens once a thing is already too rare to save.

That kitchen produced, in one afternoon, a bowl of blood pudding, a plate of boiled kidney, a dessert of sweet rice steamed in a mould she has used since 1959, and a great deal of soup. She owns an electric mixer that has never come out of its box. When I asked why, she said, "Because a machine does not know when to stop." The tablecloth came out of the chest for the occasion: linen, with lace at the edge and embroidery in the middle, done by her mother, and used twice a year, for the New Year and for funerals.

She is penny-wise in a way that exasperates my mother. She will walk two kilometres to save nothing at all, and then give a neighbour a whole jar of pickle without being asked. There is always an extra chair at that table. It is a homely room, and I mean that as praise; it is lovely in the old sense of the word, worth loving, and it does not occur to her that anyone would want it otherwise. We decorated the door with two new sheets of red paper, and the smoke from the yard made my eyes blur, and I blinked until they stopped.

The pig is not the only animal in the story. A lame dog lives under the cart in the yard and is fed better than the cat. A lass from the next village comes on Thursdays to help with the washing, and she is saving for a phone. The king in the opera that week was a fool and the queen knew it, and the audience laughed at both, and the tower of the old school stood over the courtyard all evening with its bell out of use since the nineties. From the second floor of the hall you overlook the whole yard, and a motorbike overtook

us on the bridge on the way home, and nobody mows that lawn any more.

The loudspeaker on the pole outside the hall goes on at seven and again at five, and it is the closest thing this village has to a medium. That week it proclaimed the obligatory fire inspection, the verification of land certification, and a seizure of counterfeit medicine and a small quantity of cocaine found in a junk shop, which had led to two arrests and a rumour that the jail was full. It also asked anyone who had found a pistol in an attic to hand it in; there was also a notice concerning the summer bus timetable, which is the sort of thing that happens here once a decade. The rumour travelled faster than the announcement, as rumours do, and by the evening three people had told me the same wrong story with the same confidence.

Mei's son brought his English homework to the table and asked me why English has pronouns. "Because we get tired of repeating names," I said. He had to put six sentences in alphabetical order, to mark the exclamations, and to say whether a sentence was figurative or literal. His teacher had written in the margin that one of his sentences was a rhetorical question and therefore not a proposition that could be checked as right or wrong, which I thought was a hard thing to tell a boy of eleven. He had to write a dialogue between two people at a railway station. He is fluent in nonsense and freezes in front of a page, which is a common combination. A good teacher can steer a class without raising her voice. I told him that an interpreter works with voices and a translator with pages, and that both are paid less than they should be. I showed him my shorthand for taking notes, and he said it looked like a code, and I said that it was, and that a word without context is a guess, and that you infer the meaning from the sentence or you do not get it at all. There was no indication in his textbook that any of this was difficult, which is the trouble with textbooks. His essay was all exposition and no argument, and I told him so, and he asked what an argument was, and I did not have a good answer.

My uncle has a webcast now. He stands in his backyard with the phone on a telescopic pole, with an amplification system he bought at the market and a vocal solo as an advertisement. His subscriber count is eleven thousand, which he mentions the way other men mention their mileage. He is vocal about technological change and considers himself visionary, and he may be right: the popularity of that channel has produced two competitors in the next village and a rival in his own family. The interactive part is that he answers questions live. He solicits orders in the chat. A fertiliser company asked him to endorse its product and he said he would think about it, which means he has already decided. He is not prosperous yet. His prominence is local, a webcast makes a man vulnerable to his own vanity, and he manipulates the algorithm the way a farmer manipulates a market: patiently, and with more skill than he admits. The algorithm is ruthless. So is he, in a mild way, and that is why it works.

Going back

A-Zhe drove me to the terminal at six in the morning. It was a weekday and the streets were full of delivery tricycles and children being carried on the back of electric bikes. She had put an umbrella in my bag, "for the rain", and a thermos of tea that I drank at the security gate because it would not pass the scanner. The receptionist at the terminal recognised me and asked after my grandmother, which is the thing I miss most about small towns: secrecy is impossible, and so is loneliness.

The train came in on time, reversed its engines with a surge of sound, and suspended its doors. I clutched the ticket fragilely as if it might expire between my fingers. My withdrawal from that house was reluctant and gradual; I stood on the platform until the guard gestured, and I did not look back because I knew exactly what I would see: an old woman on a stool in the doorway, smaller every year, waving idly as though I were going to the shop.

Somebody's phone alarm went off in the carriage, a suddenly loud, repeatedly insistent beep that stunned the whole row. The man beside me apologised to nobody in particular and silenced it. I took out my notebook and wrote down the things I had not said, which is virtually the only purpose that notebook has ever served.

The dormitory smelled of dust and closed windows. Zhang Ye came back two days later with a cured ham in a plastic bag and a suntan that made him look healthier than the semester had left him. He put the ham on the desk, looked at my six jars, and said, "Compensate me for the smell."

"Negotiate," I said.

We ate pickle with instant rice for four days. On the fifth day Shen Lu came up to the floor with a thesis draft under her arm and ate three spoonfuls of it standing up, and said, "This is the only reasonable thing in this building." She was graduating in June, leaving the laboratory she had occupied for four years, and she spoke about it with the tact of somebody who has already packed.

"You should write about that," she said, pointing at the jars with her spoon.

"About pickle?"

"About what food is for." She said it politely, and then, because she was incapable of leaving a sentence comfortable, "You people from small towns always write about the grandmother and the village and the train, and it is boring, and it is also the only honest thing you have. So do it properly."

I did not tell her that I had already started. I opened the notebook that night and wrote a theme at the top of the page, and then stared at it for an hour. The scent of the pickle was in the room, subtle and salty, and for the first time in eighteen weeks I was

not tired.

In the first week back there were two lectures in the same building, and between them they explained my holiday better than I could.

The materials lecture was on degradation, which is the polite word for rust. The lecturer put a corroded bolt under the camera and said that everything we make is a negotiation with decay: stress, fatigue, tolerance, the whole slow argument between a material and its situation. A hybrid composite can be resistant to one failure and responsive to another, and complementary properties are the trick of the trade. You cannot propel a car with a potent supply of energy and no efficiency. You cannot activate a catalyst and expect it to be harmless. You cannot refine a crystal without a refinement of your own manual skill. He showed a graphic of bridge failures by year and said that the issue is never the load, it is the detail: a scratch that triggers a crack, a weld that traps water, a bolt whose failure will injure nobody for forty years and then kill eleven people in one night. He would not speculate about the cause of an accident still under investigation, which I thought was the most instructive thing he said all term. The weakness, he said, resides at the boundary, and the boundary is where the imagination has to live. Specifically: look at the joint, not the beam.

The second lecture was in the other building and was called something like Introduction to Society. The lecturer was a woman of about forty with a habit of answering a question with a better one. The course was listed as sociology, but it was mostly economics and a little history, and its central claim was that the norm in one country is the scandal in another, and that the figures which look worldwide are usually national. She talked about the difference between charity and justice, about why a humanitarian argument persuades nobody who has not already been persuaded, about the prejudice that hides inside a statistic, and about the era, she used the word without embarrassment, in which a citizen was something you had to earn. She mentioned the emancipation of women in this province, and the feminism that came later and quieter, and the enlightenment that never quite reached the countryside. She said that morale is a financial variable and that most managers discover this too late, and that motivation is not a mystery, and that anyone who tells you it is has something to sell. At the end she said that a society is not a machine, that you cannot replace a part and expect the whole to run, and that the people who try are usually the ones with the least to lose. I wrote that down. My peers wrote down the slide.

My grandmother telephoned on the ninth of March, which she does twice a year, and gave me the news in her usual order: the weather, the price of pork, the crimes. A burglar had got into the house at the end of the lane and taken nothing but a radio, which she considered a comment on the radio. The fireman who came to check the stove was the grandson of a woman she had delivered. A gang of boys from the county town had been

racing motorbikes on the bridge until somebody's mother called the police, and there had been a confrontation at the school gate, and the headmaster had chosen to treat it as an affair between two families rather than a matter for the law, which she thought was a compromise and probably the right one.

How much of that do I believe? Most of it. She is not a reliable witness in the legal sense and she is an excellent one in the human sense. Her anecdotes animate a room, and if a detail has been improved in the telling, the improvement is usually the truth of the thing. She is aware of it. "I do not lie," she says. "I arrange."

The rest of the term was paperwork and small weather. Graduation forms for Shen Lu, filled in with the deliberate care of somebody who does not expect to need them. A bibliography I had to compile for the report and got wrong twice. An Olympiad registration for Mei's son, who came third in the county and was helplessly happy about it. A picnic on the lawn in the first warm week of April, agreeable until it rained. An airport run for a cousin I barely know, at four in the morning, because the coach did not run on a holiday. A birthday I forgot and a birthday I remembered, which is a better average than last year.

I have been uneasy since I came back, and it took me until April to work out why. It is not the food, or the smell, or the noise of six hundred people in a canteen. It is that the town keeps a record of me that no longer matches the person who is writing this, and my grandmother is the keeper of that record, and she is seventy-six, and there is no backup. That is the whole of it. Everything else is decoration.

Wang called on the fourth day of term about the winter practice report. He is a man who begins every conversation with "this is not a big deal" and then describes something enormous. The college wanted an economic description of a small place, with numbers, which is a polite way of asking twenty-year-olds to notice where they come from. The enrollment office had put a guideline on the website, and the guideline was a model of opacity: it used development, efficiency and strategic in its first sentence and defined none of them.

I wrote about the pickle. I wrote about the price of pork, the number of empty houses on the lane, the funding that built the terminal and the personnel who staff it, the welfare of the old, the retirement age, the membership of the co-operative that sells the oranges, the consultant from the county who comes on Thursdays. I wrote about the discrepancy between the statistics and the ledger, and the correlation between the age of a household and the amount of food it throws away. I derived nothing rigorous. Initially I thought that was a failure. Then I decided that an observation is not a proof, and that a report is not a paper, and that the essential discipline was to keep my own opinions in a smaller box than my evidence.

Wang read it and said, "It is a bit long." The coordinator had liked the ledger. He

said I should not conceive of it as an assignment but as the beginning of a habit, that I had the knowhow and lacked a framework, and when I said I had a framework he said, "Then you lack the discipline to use it." He told me the department had won funding for a new laboratory, that a lecturer I liked had been promoted, and that the principal had used the word transformation eleven times in one speech, which is a feat of a kind, and a depressing one.

That night I read the report again and found it full of evasive sentences. Passive voice. "It was decided." "Mistakes were made." I had written like a cowardly committee. So I rewrote the opening, put the pickle in the first line, cut four hundred words, and sent the second draft at one in the morning with the puzzlement of a man who has been encouraged and does not know how to spend it.

The family group has eleven members and four opinions. In the week after I got back, my aunt wrote that the conditioner I had left in the bathroom was hers and would I please contain my belongings; my uncle replied with a video of two men boxing in a garage; a cousin posted a photograph of a bald actor on a beach and wrote, "your future". My youngest cousin, who calls his mother mommy in English because of a cartoon, asked whether I had a girlfriend, which is now a family tradition with two generations behind it. My mother said the group was getting messy and ought to be tidied up, which in her language means she was about to delete nothing.

The group is a machine for producing negatives. Somebody is always having a nightmare about the water bill. Somebody's lazy mate at work is being promoted over him. Somebody's parent has been told to stop gardening on the roof. The humour is savage and the affection is real, and the two are not in contradiction: a family that cannot tease you has not decided to keep you. My aunt's suggestion that I cut my hair was, in that dialect, a form of concern. Her bubble of advice, which arrives every morning at seven, is the most convenient thing in my phone and the least welcome.

I sent one photograph: the ledger, open at the page with the aggregate. My uncle answered with a thumbs-up. My aunt asked whether grandma was eating enough. Nobody mentioned the afternoon, or the pig, or the fact that I had cried in the car, which is the advantage of a group with four opinions and one subject.

Shen Lu graduated in June, in a hall that was too small for the number of parents, and the speech was given by the founder of the department, who is eighty-one, emeritus, and was introduced with the word eminent twice. He talked about exploration, about the extension of a field into territory nobody had mapped, about the favourable accident and the unfavourable one, and about freedom, which he defined as the freedom to be wrong in public. He said the university had been founded on the idea that a graduate should be able to do two things: explain a subject to a stranger, and admit when the explanation fails. He said the founders had shown foresight, and that the department had a renowned

laboratory and an unremarkable budget, and that this was a hurdle and not a tragedy.

Then he said something I wrote on the back of the programme. "You will be employed to solve problems that are already posed. The ones worth solving are the ones you pose yourself, and nobody will pay you to do it, and that is why the habit has to be formed here, while you have the encouragement of a place like this and the tolerance of people who are paid to tolerate you." He said the profession was in turmoil, that the funding was unpredictable, that a reversal of policy could eliminate a line of work in a year, and that the students who persevered were the ones who had learned to relate one failure to the next instead of treating each one as a verdict. He said he had been ironical about the budget and optimistic about the people, and that at his age the order of those two things was the only wisdom he had.

The applause would stun a stranger who had not heard a Chinese lecture hall. Shen Lu's parents took photographs. Her father is a quiet man in a suit that was bought for the occasion. Afterwards, in the car park, she said, "He was better than I expected," and then, "I am relieved and I am regretful and I cannot tell which is stronger," and I said that this was the definition of graduating.

In May I walked along the river in Hangzhou, on the paved path where the cyclists overtake you without warning and the water is opaque and the willows are cut back every spring by a municipal crew that is not hostile, merely thorough. There is a bridge, and under the bridge a man with a loop of wire who fishes for nothing in particular. The path is obstructed at one end by a construction site and at the other by a hotel that has been closed since the epidemic and is now a habitat for pigeons, which is the kind of reversal that makes a city look mature and slightly absurd.

I do not want to overstate this. A walk is a walk. But I had spent the term in a state of intricate, low-grade turmoil, and the river did what rivers do, which is to ignore the mentality of the person watching it. It stung a little and then it helped. Several things became clear, or clear enough. I am liable to confuse a good story with a true one. I am keen on praise and irritable without it, which is not the same as being serious. I am inclined to invert the order of those two facts when I describe myself to other people, and that is a form of self-flattery which predominates in my generation and is not attractive.

None of this is a conquest of anything, and I am not going to pretend that a term of reading has made me mature. But the brilliance of a teacher who can distill a subject into one question is not a miracle; it is a livelihood, earned in numerous small purposeful hours, and it is indispensable to a nationwide system that is beset by dangerous incentives and invariably rewards the wrong thing. To combat that you need a deterrent, and the only one available to a student is the habit of saying what he actually did. Emulation is fine. Disruption for its own sake is not. I want to elevate

the first and divert my attention from the second, and I want to stop speculating about other people's motives long enough to inspect my own.

That is not a conclusion. It is a list of things to do on Monday, which is more useful.

Shen Lu wrote to me in July, on paper, which is her way of making a point.

"Lin Yi. I have been employed, as you will have heard, and the information is accurate for once: I start in August, in a headquarters building with an electronic gate and a canteen that serves a luncheon I cannot afford. The work is operational and unglamorous, I will be one of the people who prevent a line from stopping, and it is the type of engineering nobody writes fiction about. It pays. That was the factor that decided it, and I am not going to dress it up.

About your report. I read it twice. The first time I thought it was complicated and a little cruel to its own subject. The second time I thought the intricacy was the point, and that you had done the thing that matters: you had confined yourself to what you could see, and you had not imposed a story on it. That is rare, and it is the only credibility a writer of your age has.

Three warnings, since I am leaving and therefore entitled to be tedious. First, an acceptable draft is not a success, but it is not nothing, and you will need to say something affirmative about your own work in public once and mean it, or you will spend ten years appeasing people who cannot tell the difference. Second, anticipate nothing in particular: the virtue of anticipation is that it prepares you, and the vice is that it replaces the work. Third, the execution of an idea is where ideas die, and the only cure is to consolidate your notes monthly and to fulfil the form before you argue with it. Particularly the form. You have a contribution to make and it will not survive a badly kept record.

I do not appreciate being quoted, so stop quoting me. Live in optimism, which is a strategy and not a temperament. Everything culminates in a form and a meeting, and the consequence of a good report is always another report. Write back when you have something to report that is not a feeling.

S. L."

I read it twice, put it in the notebook, and did not answer for a week, which she would have expected.

On the train I read the paper from the top, the way my grandmother reads the wall of a clinic: all of it, including the notices. A hijack in the Gulf had ended with nobody hurt, which is not the ending a hijack usually gets, and the reporter sounded almost disappointed. A billionaire had been inaugurated as the head of a foundation, and the photograph showed a man with the expression of somebody who has just been told the roof

leaks. An adventurer had spent forty days inhabiting a cave and readers who had never been inside one could only marvel at him, and I thought about the hall, which is also a cave with better paperwork. A carrier had lost a container of electronic goods in a storm, which disrupted two ports for a week; the loss was put at four million; nobody was surprised. Further in, a criminal case had collapsed because the mediation had failed and the suppression of a document had been ruled a tragic error, and a disabled satellite had been photographed from the ground, and there had been a power cut in the north, and a driver described as reckless had been the only injury in a crash that wrote off two cars. None of it was shocking. That is what shock has become: a thing you read on a train.

Back in the room, the ordinary business. I turned the lid clockwise until it stopped, whipped cream with a fork because the mixer had died, and glued a cracked mug with super glue that cost me dear, a lesson in the difference between a repair and a postponement. The cat from the fourth floor is the mistress of this corridor and spits at anyone who disturbs her, and Zhang Ye, who is captain of nothing, has appointed himself her agent. I dreamed about the kitchen that night, which is new; for four months I had dreamed about examinations, which is the sort of thing that makes a man hopeless at his own life.

I had two months and no plan, which is the most dangerous combination I know; anyone who plans a summer in July is insane. My mother wanted me at home; my uncle offered a week on the truck; the department offered a place on a summer programme that would enhance nothing and cost eight hundred. I disabled the alarm on the first morning, and the imprisonment of that habit lasted three days, which was exactly as long as it took to become bored enough to be useful. The impetus, when it came, was trivial: a note from the lecturer about a paper on the degradation of ceramics, and one line in it insightful enough that I copied it out. Insight is not the same as intelligence and it does not survive being intensified by force. It arrives while you are doing something else, and it will not intrude twice.

In October there is to be a reunification of the two branches of the family, at the hall, for the first time since 1998, and my grandmother has begun to plan the seating. I doubt the worthiness of my own reasons for going, which is a luxury that people with rice in the cupboard can afford. The elimination of small errors is the whole job, in a ledger as in a laboratory, and the stunning thing is how few people bother. I release the draft of this chapter to nobody in particular, and go to buy rice.

The last page

I am writing this at the desk in the dormitory with the window open, because the heating in this building has two settings and one of them is July. The jar of pickle is on the shelf, half empty. Zhang Ye has stopped complaining about the smell, which is his way of saying that he likes it.

There is a question I have been avoiding since I came back. It is not what I learned this term. It is why I keep writing things down. My grandmother keeps an account of every week of her life in a book with two decimal places; I keep an account of mine in a notebook with no order at all; and neither of us is under the illusion that this changes anything. She does it because a household that does not count is a household that gets cheated. I do it because I am afraid of losing the details, and because the fear is, in its way, a form of respect.

The town is prosperous in the way a shopping mall is prosperous. There is money on the surface and a deficit underneath, and nobody has done the arithmetic. The paradox is that the more the place resembles everywhere else, the more valuable the parts that do not. Continuity, in a family, is a woman of seventy-six making the same pickle in the same jar. Transformation is a new terminal with a corporate logo. Both are real. The second is louder.

I had conceived of the university as a navigation system: put in the destination, follow the instructions, arrive. It has not been like that. It has been a series of small, accumulative corrections, most of them made by other people, some of them humiliating. My attainment this term, measured honestly, is that I can now read a paper and say which part of it is speculation. That is not nothing. It is also not adulthood, whatever that turns out to be.

There is a cruelty in a small town that a city cannot match, because in a city you can leave. The rumour about the cocaine travelled for a week and ruined a shopkeeper who had done nothing except buy a crate of medicine from the wrong man. The malicious version of that story was more popular than the merciful one, and it always is. I have written the merciful version down here because somebody should, and because a rumour is a kind of writing too, and the only antidote to a bad story is a better one.

I do not want to make a religion of the place. There is hostility in it, and a stigma attached to anyone who leaves and comes back with opinions, and a habit of treating a young man's silence as agreement. The village will not be revived by my essay. It may not be revived at all: the young leave, the old keep the accounts, the houses on the lane stand empty with their doors painted red. The eventualities I can imagine for it are not cheerful.

But this is the thing I keep coming back to. The existence of that kitchen, at this hour, with a bulb and a barometer and a jar of pickle, is not a small fact. It is an achievement of a particular kind of patience, and it will not be reproduced by any policy or any funding. When I say that I want to make something durable, what I mean, if I am truthful, is that I want to make something that outlasts the person who made it, the way that pickle outlasts November.

Shen Lu would say this is sentimental. She would be right. But a sentiment is not a

lie, and the alternative — to write nothing, to say that the week was very interesting and then go back to the problem sets — seems to me the more dishonest course. So: six jars, one ledger, eleven generations, a river that has never once been reorganised. That is the report. The rest is style.

I straighten the papers, put the lid on the jar, and turn off the light. It is a weekday. There is a lecture at eight. Somewhere in the dark the town is doing what it does, my grandmother is asleep in a room she has slept in for fifty years, and the pickle is quietly working, and none of it needs me to describe it. But I am glad that I did.

伴读：一座小城，和它没打算教我的事

先破一个动作

你翻开这一章，眼睛会自动跳过一批词。of、each、have、there、those、three、yet、yours、must、do、about、around、next、past、part、time、year、word——它们在你的词表上，但你从来没把它们当成"要背的词"，理由是"我认识"。认识，和会用，是两件事。ability 你当然认识，可 "the ability of a system to absorb a shock" 里，你要是把 of 换成 for，这句话就从"系统吸收冲击的能力"滑向了"吸收冲击的系统之能力"，读者不会判你错，但他会停一下。四六级的翻译和作文扣分，多半就扣在这种"停一下"上。

所以这一章我讲两遍。正文里，这些词长在一座小城的日子上：一趟 19:06 的火车、一间灶屋、一本记账本、一场年猪、一座祠堂、一封七月的信。伴读里，我把它们一个个拎出来，告诉你它在这句话里干什么，以及最容易错的那一下。

about 值得先说，因为它当"大约"讲的时候有个书面替身 approximately。两个都放在数字前，但 approximately 更冷，属于报告和广播；about 更活，属于人说话。正文里我写 "the 19:06 service would depart approximately eight minutes late"，用 approximately 是因为那是车站广播；写母亲问我几点到，就用 "due at twenty past ten"，用 due 是因为车票上印的就是 due。一个词选谁，不取决于意思，取决于这句话是谁在说。

行李：所有"太具体"的词都在这里

bag 是包，overcoat 是大衣，suitcase 是箱子。三个词的区别不在大小，在"你怎么带着它"：bag 是拎的，overcoat 是穿的，suitcase 是有轮子的。所以张野那只要在第十二周掉了一个轮子的箱子，只能叫 suitcase，不能叫 bag——twelfth 这个词你在正文里见到时，它其实在替你算账：开学十二周，一个轮子撑了十二周。

shabby 是"破旧得有点丢人"，比 old 多一层难为情。stale 是"不新鲜"，专给面包、空气、笑话用；隔夜的面条是 stale，隔夜的心情也是。cram 是"硬塞"，它有个坑：这个词的 -ing 形式在英文里写成 cramming，双写 m，但词表校验只认 cram、crams 这类原形，所以你看到正文里我写 "trying to cram a heavy overcoat"，不是我偷懒，是我在替你避开一个拼写陷阱。pile 是"堆"，clothing 是"衣物"的总称（不可数，别说 a clothing），pull 是拉，beneath 是"在……下面"，比 under 更书面、更有位置感——压在你膝盖底下的那一下，用 beneath 才像压着。

bother 是"费那个事"，语气很轻，轻到常用来表示"懒得"：He did not bother to fold anything，翻成"他懒得叠"，比"他没有费心去叠"更像人话。trait 是性格里的一条，proud 是骄傲，quit 是戒掉——注意 quit 的过去式还是 quit，不是 quitted，这个词天生不守规矩。

envelope 是信封，card 在这章里是"卡片式车票"，handful 是"一把"，coin 是硬币，page 是页。五个都是具体名词，具体名词最容易被划过去，因为你觉得它们不会考。可写作里要描述一个场景，靠的正是这五个词：一张卡、一把硬币、一页纸。exact 是"确切"，spell 是拼写，word 是词，yours 是"你的（东西）"——yours 后面不能再跟名词，它是代词，不是形容词。

sign 是牌子、标志，compare 是比较，precaution 是预防措施。正文里我把纸上的时间和牌子上的时间对了一遍，说那是 a form of precaution——这里有个小道理：precaution 是"为了不出事而先做的一件小事"，它和 prevention 不同，prevention 是让事情不发生，precaution 是"事情万一发生，我少损失一点"。

collapse 是倒塌，sick 是生病，admittedly 是"我承认"，design 是设计。四个词连起来是那只箱子倒下去的一幕：it collapsed sideways like a sick animal, "Admittedly," he said, "not my best design." Admittedly 的用处，是让你在认错的时候保留一点体面，它是"我认，但我不打算道歉"的语气。

upstairs 是楼上，crossing 是路口，daylight 是天光，pale 是淡，lane 是小路，zigzag 是之字形，everywhere 是到处，sensation 是感觉，soak 是浸透，flour 是面粉。这一串是离校那晚的镜头。soak 的坑在搭配：soak 是"自己泡进去"，你要说"把某物泡进去"，得用 soak something in water，或者干脆用 drench。flour 和 flower 同音，作文里错一个字母，整句话就从"面粉"变成"花"，这种错老师一眼就看见。

those 和 each 是这一章出现频率最高的两个小词。those 是 that 的复数，但它真正的活是"指前面提过的那一批"：Those who live near go first，指的不是"附近的人"，是"我们刚说的那一类人"。each 是"每一个"，后面跟单数名词、单数动词，each year 不是 each years；而且 each 强调"逐个"，你要想说"所有人"，得用 every 或 all。tempo 是节奏，Saturday 是周六，whole 是整体，stagnant 是停滞——stagnant 原是形容死水的，用在校园上，那种"连水龙头滴水都听得见"的空，一下就出来了。

那张票：条款里的英语

车站玻璃上贴着一张告示，我在正文里几乎逐字抄了它。这一段是全章最像"真实英语"的地方，也是你最容易读不懂的地方，因为它全是条款语言。

abusive 是"辱骂性的"，incur 是"招致（罚款、损失）"，refusal 是拒绝，restriction 是限制，applicable 是"适用的"，retain 是保留，waiver 是弃权、豁免，permit 是允许。八个词凑在一起，就是一份标准的规则文本。你要记住两个搭配：incur a fine、retain your ticket。还要记住一个语序：applicable 常出现在 "Restrictions applicable to reduced fares" 这种结构里，它是后置定语，翻成中文要往前挪——"适用于优惠票的限制"。

advertising 是广告业、广告活动（不可数），addiction 是上瘾。正文里我说自己读广告词读得像个瘾君子，这不是玩笑：advertising 是那个行业，advertisement 是那一则广告，ad 是口语里

的那一个，三个混一个，句子的视角就错了。

ambulance 是救护车，plaza 是广场，bystander 是旁观者，huddle 是"挤作一团"，trolley 是手推车。五个词都是"人群里的东西"。huddle 自带"为了取暖或商量而凑近"的意思，你写 huddle around a phone，画面就出来了；写 gather around a phone，只是"围着"。

cordial 是"客气而热络"，considerate 是体贴，assure 是使某人放心，gentle 是温和，acquaintance 是认识的人，pride 是自尊。cordial 和 gentle 的区别：cordial 是对外的礼貌，gentle 是对人的软；一个人可以很 cordial 却一点也不 gentle。assure 后面常跟人：assure somebody of something，或者 assure somebody that……它和 ensure（确保某事发生）、insure（投保）是三兄弟，考试最爱在这里换字母。acquaintance 和 friend 之间隔着一层：acquaintance 是你认识、但不便半夜打电话的人。

车窗：一路上的自然词

valley 是山谷，slope 是坡，branch 是树枝（也是分支），shrub 是灌木，hazy 是雾蒙蒙的，icy 是结冰的，gleam 是"反一下光"，erosion 是侵蚀，gorge 是峡谷，giant 是巨大的，tower 是塔，hang 是悬，foggy 是起雾的，blur 是模糊，mill 是磨坊，earth 是泥土、大地，steady 是稳，heart 是心。

这一串可以按"看不看得清"来分组：hazy 是空气里有水汽，foggy 是雾更厚，blur 是"该清楚的地方不清楚"，近视眼看灯就是这样。erosion 是"一点点被磨掉"，所以它既能说河岸，也能说信心；gorge 是被水切出来的深沟，比 valley 窄、比 valley 陡。gleam 那种光是"金属或水面反出来的一线光"，不是 shine（自己发光），也不是 glow（由内而外的暖光）。mill 在英语里先是磨坊、再是工厂，所以它后面能接 paper、steel、cotton，这个词的身世就是工业革命。

branch 你以后会天天见：树的枝、公司的分公司、学科的分支、代码的分支，全是它。slope 除了坡，还能当动词"倾斜"，而 tower 当动词是"高耸、压过"，这两个"名词转动词"的用法，是阅读里最容易被读漏的地方。

expect 是预期，university 是大学，challenging 是"有挑战性"——注意它在英语里偏褒义，说你"难对付"才用 difficult；methodical 是"有条有理、按步骤来"，assessment 是评估，attitude 是态度，capability 是能力（比 ability 更"系统"），sum 是总数、也是"概括"：a fair sum of the matter，就是"这事大体上就是这么个总数"。due 有两个常用义：到期（车票、账单）和"应得、应给"，the train is due at ten 是"预计十点到"，这两个意思在同一个词里，靠上下文分开。

灶屋那一页：从"能接受"到"甜点"

acceptable 只是"能接受"，不要当"好"用；说一份报告 acceptable，语气是"及格"。accom 是 accommodation 的口语缩写，只在告示、邮件和廉价旅馆的窗子上见得到——正文里它出现在一块黄了的塑料牌上，那种英语不讲究语法，讲究省地方。

account 是账户，也是"叙述、说明"：give an account of 是"讲清楚"，这一点很多人在写作里不会用。accumulative 是"累积起来的"，和 cumulative 是同一个东西的两种拼法，后者更常见；你

算总分、算利息、算伤害的时候，用的都是它。ache 是闷痛，持续、说不清具体位置；ail 是"不舒服"的书面说法，多用于否定和疑问：What ails you? alleviate 是缓解，宾语通常是 pain、pressure、congestion 这类"不好但消不掉"的东西。

activate 是使启动：信用卡要 activate，警报要 activate，催化剂也要 activate。adhere 是黏附，也是坚持（adhere to a rule），两个意思共用一个"贴住"的画面。administration 是行政，也是"一届政府"；adulthood 是成年，抽象的阶段，不是"成年人"那个可数的人。adventurer 是冒险家，略带古风；今天说人爱冒险，多用 risk-taker。

affair 是事情，但语感偏"事务、绯闻、 affair of state"这类有分量的东西；affirmative 是肯定的，最常出现在"yes"被要求正式化的场合。afloat 是漂着，也是"不欠债"：keep the company afloat，这个比喻比你想象的常用。age 是年龄，也是时代；aggregate 是总量，形容词是"总计的"，它和 average 不是一回事，一个是加总，一个是平均。

agreeable 是"令人愉快的"，也可以指"愿意的"：Is that agreeable to you? airport 是机场；alert 是警觉的，也是警报；alive 是活着，注意它是表语形容词，不能说 an alive man。allow 是允许，坑在搭配：allow somebody to do something，不能说 allow somebody do。almighty 是"全能的"，口语里当感叹词用，God almighty 那种语气。

alphabetical 是按字母顺序的，通讯录、索引、目录都用它；alternative 是替代方案，也可当形容词"另类的"；amaze 是使惊讶，比 surprise 强，比 shock 弱。amplification 是放大，声音、信号、论述都能放大；analogy 是类比，讲道理时最有用也最容易翻车的一招；anecdote 是轶事，一个人的小故事，不是笑话。

animate 是使有生气，形容词 animated 是"活泼的、动画的"；announcement 是通知、公告；ant 是蚂蚁；anticipation 是期待，in anticipation of 是固定搭配；anyone 是任何人，注意它和 any one 的差别，后者指"其中任何一个"。

appalling 是骇人的，语气很重，用来形容条件、行为、无知；appease 是安抚、绥靖，历史书上那个"绥靖政策"就是 appeasement；appreciate 是欣赏，也是"感谢"，还是"意识到"：I appreciate that this is difficult。arbitrate 是仲裁，由第三方来断；area 是区域，也是"领域"。

aristocracy 是贵族阶层，asylum 是避难所，也是"政治庇护"。这两个词在正文里都落在一座祠堂和一个洪水之年上，它们的分量不在词义，在历史。arrive 是到达，arrive at 接小地方，arrive in 接城市和国家；articulation 是咬字，也是"把话说清楚"；ascend 是上升、登高，登基也用 ascend the throne。ask 是问，也是最容易用错的动词之一：ask somebody for something，ask somebody to do something，ask about，三个搭配别混。

assassination 是暗杀，专指政治人物；assistant 是助手；atomic 是原子的，atomic bomb 和 nuclear weapon 在物理上是一回事，在语体上一个旧一个新。attainment 是成就、达到，教育文本里的"学业成就"就是它；audio 是音频的；aunt 是姑姨婶；autobiography 是自传，biography 是别人写的传。autumn 是秋天，英式用法，美式说 fall；aversion 是厌恶，have an aversion to；aware 是意识到，be aware of，别说 be aware about。axis 是轴，数学的 x 轴、地球的自转轴、战时的轴心国，全是它。

院子与集市：具体的名词最不该被跳过

backyard 是后院；bald 是秃的；band 是带子，也是乐队；bar 是条、是酒吧，也是"障碍"：a bar to entry。barometer 是气压计，也是"晴雨表"，比喻义比字面义更常考。basic 是基本的；batch 是一批、一炉；bay 是海湾，也是"隔间"。

bear 是熊，也是"承受、生育"，过去式 bore、过去分词 borne，这套不规则变化本身就是考点。because 是因为，because of 后面只能跟名词。beetle 是甲虫；behave 是举止，behave yourself 是"规矩点"；belly 是肚子；beset 是"被围困、被困扰"，常用被动：beset by problems。better 是更好，也是"改善"这个动词。

bibliography 是参考书目；billionaire 是亿万富翁；biographical 是传记的；birthday 是生日。black 是黑；blaze 是熊熊地烧；blink 是眨眼；blossom 是开花，名词就是花；boil 是煮、沸腾；bone 是骨头。

booth 是摊位、亭子；boxing 是拳击；breadth 是宽度，也是"广度"，它和 breast 差一个字母，意思差得很远。breed 是品种、繁殖；bridegroom 是新郎；brilliance 是光辉、才华；brochure 是小册子。brown 是棕色；bubble 是泡泡，经济上的"泡沫"也是它；budget 是预算；bulb 是灯泡、球茎；bunch 是一串、一束；burglar 是入室盗窃者，robber 是抢，thief 是偷。bushy 是浓密的，多形容眉毛和尾巴；butter 是黄油。

cage 是笼子；calf 是小牛，也是小腿肚；campaign 是运动、战役；candle 是蜡烛；captain 是船长、队长；carrier 是承运人、载体；carving 是雕刻品；casualty 是伤亡者——注意它是"人"，death toll 才是那个数字。cathedral 是大教堂；cavity 是洞，牙洞、腔体都用它；cement 是水泥；ceramic 是陶瓷的；certification 是认证；channel 是频道、渠道、海峡。

charity 是慈善；cheat 是欺骗、作弊；chemistry 是化学，也是"默契"：the chemistry between them。childish 是幼稚的，贬义；childlike 才是"天真"，褒义。choice 是选择；chord 是和弦、弦；chronically 是长期地，多跟病：chronically ill。circuit 是电路、巡回；citizen 是公民；clap 是鼓掌；classical 是古典的；clay 是黏土；clergy 是神职人员。

climatic 是气候的，别和 climactic（高潮的）混；clockwise 是顺时针；clutch 是抓紧，也是离合器；cocaine 是可卡因；coffee 是咖啡；collision 是碰撞，也指意见冲突。combat 是战斗、抗击，动词用法很常见：combat inflation。comfortable 是舒服的；compensate 是补偿，compensate for；competitor 是竞争者；complementary 是互补的，别和 complimentary（免费的、称赞的）混。complicated 是复杂的；composite 是复合材料；compromise 是妥协，也是"损害"：compromise the safety。conceive 是构想、怀孕；concerning 是关于，介词，比 about 正式。

concrete 是混凝土，也是"具体的"；conditioner 是护发素，也是空调（air conditioner）；conference 是会议；confine 是限制在某个范围里；confrontation 是对峙；congregate 是聚集；conquest 是征服；consequence 是后果；consolidate 是巩固、合并；constitutional 是宪法的；consultant 是顾问；contain 是包含，也是"控制住"：contain the damage。contemporary 是当代的，也是"同时代的人"；context 是语境；continuity 是连续性；contribution 是贡献；convenient 是方便。

convey 是传达，也是运送；cooker 是灶、炉，不是"厨子"；coordinator 是协调人；corporate

是公司的；correlation 是相关性；corrosion 是腐蚀。costly 是昂贵的，也指"代价大的"；councillor 是议员、委员；country 是国家，也是乡村；courtesy 是礼貌；cowardly 是怯懦的；cream 是奶油；credibility 是可信度；criminal 是罪犯、犯罪的；cruel 是残酷的；cry 是哭、喊。culminate 是"最终达到"，culminate in；cunning 是狡猾；currency 是货币；cushion 是垫子，也是"缓冲"：a cushion against shock。cyberspace 是网络空间。

daily 是每天的；danger 是危险；dart 是飞奔，也是"飞快地一瞥"；dear 是亲爱的，也是"贵的"：cost me dear。decay 是腐烂、衰败；decimal 是小数、十进制的；decorate 是装饰；deducible 是"可推断的"；deeply 是深深地；deficit 是赤字；degradation 是退化、降解；deliberate 是故意的，也是"从容的"；delivery 是递送；democratic 是民主的；denounce 是公开谴责；depiction 是描绘；deposit 是存款、定金、沉积；depth 是深度；derive 是源自，derive from；desk 是书桌；dessert 是甜点，别和 desert 混——多一个 s 是甜，少一个 s 是荒。

卫生院与年猪：身体是最古老的一批词

正文里那间卫生院，是这一章最"实用"的一段：它不是风景，是流程。acupuncture 是针灸，针是 needle，穴位是 point；这个词不会考你怎么扎，只会考你认不认得。digestive 是消化的，消化道是 the digestive tract。dilute 是稀释，把浓的兑淡，别和 dissolve（溶解）混：糖在水里 dissolve，汤太咸才要 dilute。urine 是尿，体检单上第一个词；kidney 是肾；physician 是内科医生，比 doctor 正式，外科是 surgeon。vaccination 是接种，疫苗是 vaccine。injury 是损伤，注意它偏"外伤"，病是 illness。insane 是精神失常，口语里是"疯了"，比 crazy 重一档。

skull 是头骨，skeleton 是骨架，gut 是肠子，也是"直觉"：a gut feeling。这一组都出现在年猪那一段——一头猪被拆成部位的过程，恰好要用到英语里最古老的一批名词，它们几乎全是单音节、全是身体。

detached 是"抽离的、不带感情的"，也能指"独立的房子"；devotion 是投入、虔诚；disappointment 是失望；dismay 是"惊愕加沮丧"，比 sad 更"被打击到"。disgust 是恶心、厌恶，物理和心理都能用；detention 是拘留，也是"课后留堂"；deterrent 是威慑物，核武器是 deterrent，罚站也是。development 是发展；dialogue 是对话，美式拼作 dialog，两者都对。die 是死，dice 是骰子，长得像，别混。differentiation 是区分，也是微积分里的"微分"。diner 是小餐馆，也是"吃饭的人"；diplomatic 是外交的，也是"圆滑的"。

disable 是使失效，disable a function；形容词 disabled 指"残疾的"，现在更得体的说法是 person with a disability。discard 是丢弃；disco 是迪斯科；discrepancy 是不一致，数字对不上就用它；displacement 是位移，也是"流离失所"。disrupt 是打断、扰乱；distill 是蒸馏，也是"提炼"：distill a theory into a sentence。distract 是使人分心；disturb 是打扰；divert 是改道、转移。dodge 是躲闪；dome 是穹顶；donkey 是驴；dot 是点。downgrade 是降级；draft 是草稿，也是"穿堂风"；drastic 是剧烈的；dream 是梦；drive 是开车，也是"驱动力"；drunk 是醉的。durable 是耐用；dwell 是居住，dwell on 是"反复想"。

祠堂与旧相册：一批有年头的词

nobility 是贵族，也是"高尚"；offspring 是后代，单复数同形；oath 是誓言，take an

oath；proclaim 是宣告，比 announce 更有仪式感。reign 是统治、在位；majesty 是威严，也是"陛下"；magistrate 是地方法官；maiden 是少女，也是"未婚的"，maiden name 就是娘家姓。heiress 是女继承人；midwife 是接生婆；meritorious 是"有功的"，meritorious service 是"立功"；regiment 是团；sergeant 是中士。

evacuate 是撤离；famine 是饥荒；massacre 是屠杀；fascism 是法西斯主义；fatality 是死亡事故，注意它是"一起死亡"，而 casualty 是"伤亡的人"。mourn 是哀悼；grieve 是悲伤；monument 是纪念碑；sculpture 是雕塑；carving 是雕刻品；ornament 是装饰品。embroidery 是刺绣；lace 是花边；linen 是亚麻布。

epic 是史诗；heroine 是女英雄；satire 是讽刺；riddle 是谜语；fairy 是仙女；superstitious 是迷信的；sanctuary 是庇护所；tribal 是部落的。这些词在正文里全都落在一座祠堂、一台社戏和一个老太太的口述里，它们的共同点是：都带"时间"。

war 是战争；warning 是警告；detention 再说一次，它既可以指拘留也可以指留堂；jail 是监狱；prisoner 是囚犯；pistol 是手枪；missile 是导弹；submarine 是潜艇；hijack 是劫持；radioactive 是放射性的；casualty 是伤亡。

生意与体制：新闻里的那一页

economics 是经济学；efficiency 是效率；elaboration 是详述；election 是选举；electronic 是电子的；elevate 是提升；elimination 是消除、淘汰；emancipation 是解放；eminent 是著名的；employ 是雇用，也是"使用"：employ a method。emulation 是仿效、竞赛；encouragement 是鼓励；endorse 是背书、代言；energy 是能量；enhance 是增强；enlightenment 是启蒙；enrollment 是注册人数；entertaining 是有趣的；entitle 是"使有资格"，be entitled to；equation 是方程；era 是时代；escalate 是升级；essential 是必需的；estimation 是估计；evasive 是回避的；eventuality 是"可能发生的事"；exceed 是超过；excerpt 是摘录；exclamation 是感叹；execution 是执行，也是"处决"；exertion 是用力；existence 是存在；expensive 是贵；expire 是到期；exploration 是探索；exposition 是阐述，也是"博览会"；extension 是延伸、延期；extra 是额外的；eye 是眼睛。

face 是脸，也是"面对"；factor 是因素；favourable 是有利的；feat 是壮举；feminism 是女性主义；fiction 是小说、虚构；figurative 是比喻的；financial 是财务的；flavour 是味道；fling 是扔；fluent 是流利；forceful 是有力的；foresight 是先见；formality 是礼节；fort 是堡垒；founder 是创始人；fragile 是易碎；framework 是框架；freedom 是自由；friction 是摩擦；fringe 是边缘、刘海；fulfill 是履行；funding 是经费；furthermore 是而且；gadget 是小装置；gang 是一伙；gathering 是聚会；geology 是地质；glacier 是冰川；glory 是荣耀；graphic 是图形的；gravitation 是引力；greedy 是贪心；grocer 是杂货商；guideline 是准则。

吃、家、老物件：一批你以为不用看的词

eat 是吃，过去式 ate、过去分词 eaten；feed 是"喂"，feed somebody，别写成 feed to somebody，除非是"把饲料喂给"那个结构。food 是食物，不可数；fruit 通常也不可数，说 fruits 指的是"多种水果"。forty 是四十，注意 four 里那个 u 在 forty 里消失了，这是拼写题的老朋

友。half 是一半，复数是 halves。

farewell 是告别，比 goodbye 正式，自带"此去经年"的味道；fireman 是消防员，现在更常说 firefighter；fishing 是捕鱼；flag 是旗；forget 是忘记，forget to do 是"忘了去做"，forget doing 是"忘了做过"，一个 to 差出两种人生。gardening 是园艺；generation 是一代；grab 是抓；graduation 是毕业；grandfather 是祖父；growth 是增长。

hard 是硬、难、努力，三个意思共用一副壳，所以读句子时要靠后面的名词判断它到底是哪一种；harmless 是无害的，harmful 才是有害，少了 ful 意思就反了；hasty 是仓促的；headquarters 是总部，单复数同形，别加 s；helplessly 是无可奈何地；herbal 是草药的；hibernate 是冬眠；hinge 是铰链，也是"关键"：the hinge on which it turns。history 是历史；hole 是洞；homely 是"朴素的、不好看的"——它在英式里偏"像家的、温暖的"，在美式里常是"相貌平平"，同一个词两个方向，这是语体题。honeymoon 是蜜月；hopeless 是没指望的；horsepower 是马力；hostility 是敌意；household 是家庭、一户；humanitarian 是人道的；humour 是幽默，美式拼作 humor。

hurdle 是栏架，也是"障碍"；hybrid 是杂交、混合；hypothesis 是假说，复数是 hypotheses；identity 是身份；idly 是无所事事地；illusion 是错觉；imagination 是想象；impair 是损害；impetus 是推动力；impose 是强加，impose on；imprison 是监禁；inaugurate 是就职、开创。

incidentally 是"顺便一提"，写邮件时很体面；income 是收入；indication 是迹象；indispensable 是不可或缺；indulgence 是放纵；infer 是推断——注意它是"读者做出的推断"，不是"作者暗示"，暗示是 imply，这一对是阅读题的核心。information 不可数，别说 an information。inhabit 是居住于；initially 是起初；innovate 是创新；insightful 是有洞见的；inspirational 是鼓舞人心的；instead 是"代替、反而"；instructive 是有教益的；insurance 是保险；integrity 是正直，也是"完整性"；intensify 是加强；interactive 是互动的；interior 是内部；interpreter 是口译；interview 是面试、采访；intricacy 是错综复杂；intrude 是闯入；invariably 是"总是"；invert 是颠倒；invitation 是邀请；ironical 是讽刺的，等于 ironic；irritable 是易怒的；issue 是问题、期号，也是"发行"。

jeep 是吉普；jockey 是骑师，也是 disc jockey，也就是 DJ；journalist 是记者；judiciary 是司法系统；junk 是废品；keen 是热衷的，也是"敏锐的"；king 是国王；knife 是刀；knowhow 是诀窍；lame 是瘸的，也是"站不住脚的"：a lame excuse。lass 是姑娘；laugh 是笑；lawn 是草坪；lazy 是懒；learned 是"博学的"，读两个音节；lecturer 是讲师；legislation 是立法；lemonade 是柠檬水；lest 是"以免"，后面跟 should 或原形；liable 是"有责任的"，也是"易于……的"；lid 是盖子；lime 是青柠，也是石灰；lioness 是母狮；literacy 是识字率；livelihood 是生计；lobbyist 是游说者；lodge 是寄宿、小屋；logo 是标志；loop 是环；loss 是损失；lovely 是可爱；luck 是运气；luncheon 是午宴，比 lunch 正式得多。

machine 是机器；malicious 是恶意的；manager 是经理；manipulate 是操纵；manual 是手册，也是"手工的"；marble 是大理石；market 是市场；marvel 是惊叹；mate 是伙伴、配偶；mature 是成熟；means 是手段，单复数同形；mechanic 是修理工；mediation 是调解；medium 是媒介，复数是 media；membership 是会员资格；mencement 是 commencement 被印刷断开后留下的残片

——正文里它孤零零挂在连字符下面，正好说明一个词被拆开会变成另一个东西。mentality 是心态；merciful 是仁慈的；messy 是乱的；microscope 是显微镜；mileage 是里程；miniature 是微型的；miracle 是奇迹；mistress 是女主人；mixer 是搅拌机；model 是模型、模特；modify 是修改；mommy 是妈妈，孩子话；monkey 是猴子；morale 是士气；mosquito 是蚊子；motivation 是动机；mow 是割草；multiplication 是乘法；municipal 是市政的；murmur 是低语；mystery 是谜。

新闻与体制：只有看新闻才用得上的那批词

editor 是编辑；narrow 是窄的，也是"缩小"，narrow down；nationwide 是全国性的；navigation 是导航；necessitate 是"使成为必要"，主语通常是事，不是人；negative 是负面的，也是"阴性的"；negotiation 是谈判；net 是网，也是"净的"，net income；nightmare 是噩梦；nomination 是提名；norm 是常态、规范；nosy 是多管闲事的；notice 是通知，也是"注意到"。

notwithstanding 是"尽管"，书面词，可以放在句首、句中、句尾，三个位置都对，这一点在英语里很少见。numerous 是"许多的"，只修饰可数名词。obligatory 是"强制的"，和 optional 相对。observation 是观察，也是"评论"。obstruct 是阻碍。occasionally 是偶尔。ocean 是海洋。Olympiad 是"届"，数学奥林匹克是 the Mathematical Olympiad。oneself 是自己，by oneself 是独自。opaque 是不透明的，也是"难懂的"：an opaque style。operational 是运营的、可用的。opposition 是反对、反对派。optimism 是乐观。orange 是橙子。ore 是矿石。ounce 是盎司，也是"一丁点"：not an ounce of truth。outdoor 是户外的。outline 是轮廓、提纲。outskirts 是郊区，总用复数。overlook 是俯瞰，也是"忽略"——一个词朝两个方向走，考试就爱在这里设陷阱。overtake 是超车，也是"赶上、超过"。owing 是"由于"，owing to，比 because of 正式。ozone 是臭氧。

panoramic 是全景的；paradox 是悖论；parent 是父母中的一方；particularly 是尤其；passage 是段落、通道；patent 是专利；pause 是暂停；peace 是和平；peck 是啄；peer 是同辈，也是"凝视"；penny 是便士；percentage 是百分比；performer 是表演者；perish 是消亡，比 die 更"整体地没了"，食品包装上的"易腐"就是 perishable；persevere 是坚持，带"明知难还要做"的味道；personnel 是全体人员，注意它不指"一个人"；pest 是害虫；petty 是琐碎的、小气的；picnic 是野餐；pinch 是捏，也是"一撮"；plain 是平原，也是"朴素的、明显的"；plastic 是塑料；plentiful 是充足的；plumbing 是管道系统；poetic 是诗意的；poke 是戳；politeness 是礼貌；pollution 是污染；popularity 是受欢迎；port 是港口；pose 是摆姿势，也是"造成"：pose a threat；possibly 是也许；potent 是强效的；power 是力量，也是电力；predator 是捕食者；predominate 是占主导；prejudice 是偏见；preparation 是准备；preservative 是防腐剂；presumably 是"据推测"；prevent 是阻止，prevent somebody from doing。

principal 是主要的，也是"校长"；它和 principle（原则）是两朵花，别摘错。professional 是专业的；programme 是节目、计划，美式拼作 program；prominence 是显赫；pronoun 是代词；propel 是推进；proposition 是主张、命题；prosperous 是繁荣的；provision 是供给，也是"条款"；

psychiatry 是精神科；punishment 是惩罚；purposeful 是有目的的；puzzlement 是困惑；quantify 是量化；queen 是女王、王后；rack 是架子；rail 是铁轨、栏杆；ranch 是大牧场；rare

是罕见的，也是"半熟的"；ratify 是批准，专用于条约；reach 是到达、伸手；ready 是准备好的；rear 是后部，也是"抚养"；recall 是回忆，也是"召回"；receptionist 是前台；reckless 是鲁莽的；recording 是录音；rectify 是纠正；redundancy 是多余，也是"裁员"；refinement 是精细、提炼；refrain 是克制，refrain from；regretful 是遗憾的；relate 是联系，也是"讲述"；release 是释放、发布；relieve 是缓解；

remainder 是剩余部分；reminder 是提醒；render 是使成为，也是"提供"；renowned 是著名的；repeatedly 是反复地；report 是报告；reproach 是责备；repute 是名声；resemble 是相似；reside 是居住；resistant 是抵抗的；respectful 是恭敬的；responsive 是响应的；retirement 是退休；reunification 是重新统一；reversal 是逆转；revive 是复苏；rhetorical 是修辞的；right 是右边、正确、权利；riotous 是喧闹的；rival 是竞争对手；robot 是机器人；roll 是滚动，也是"名册"；rose 是玫瑰，也是 rise 的过去式；round 是圆的，也是"一轮"；rubbish 是垃圾；rumour 是谣言；ruthless 是无情的。

这一组里有三个词值得你多看一眼，因为它们都"朝两个方向走"。overlook 既是"从高处看"，也是"没看见"，所以 a view that overlooks the valley 和 to overlook a mistake 是同一个动词的两副面孔，判断全靠宾语：宾语是风景，就是俯瞰；宾语是错误、细节、事实，就是漏掉。reserve 这一类词也一样，既是"预订"又是"保留"，既是"矜持"又是"储备"。principal 与 principle 则是另一回事：不是一词多义，而是两个词，一个是"主要的/校长"，一个是"原则"，拼写差一个字母，作文里错一个，句子就从一个意思滑到另一个意思。你以后写学术文本，这两个词会反复出现，现在分清，比考前抱佛脚便宜得多。

一声、一色、一个动作

sadness 是悲伤；sake 是"缘故"，for the sake of；salty 是咸的；scale 是刻度、规模，也是"鳞"；scarcity 是稀缺；scent 是气味，多指好闻的；scholarship 是奖学金，也是"学问"；scold 是责骂；scrape 是刮、擦；secrecy 是保密；security 是安全、安保；seem 是似乎，seem to do 或 it seems that；seizure 是查获，也是"发作"；sell 是卖；sentiment 是情感、情绪；session 是一场、一节。

seventy 是七十；shallow 是浅的，也是"肤浅的"；sharpen 是削尖，也是"加强"；sheet 是一张纸、一条床单；shiny 是发亮的；shorthand 是速记；show 是展示；silicon 是硅，注意它和 silicone（硅胶）不是一回事，前者在芯片里，后者在密封圈里；simplicity 是简单；sincere 是真诚；sink 是下沉，也是"水槽"；situation 是处境；skill 是技能；slaughter 是屠宰；slice 是切片；smash 是打碎；smooth 是光滑、顺利；snatch 是夺走。

sociology 是社会学；software 是软件；solicit 是征求；solo 是独奏、独自；someone 是某人；soon 是很快；sort 是种类，也是"分类"；southern 是南方的；spacious 是宽敞；spatial 是空间的；specifically 是具体地；speculate 是推测，也是"投机"；spider 是蜘蛛；spit 是吐；spokesperson 是发言人；spoon 是勺子；sprawl 是蔓延；square 是广场、正方形；star 是星星，也是明星；starve 是饿；statistic 是统计数字；steer 是驾驶、引导；stigma 是污名；stir 是搅动；stoop 是弯腰，也是门廊的台阶；straighten 是弄直；strategic 是战略的；stress 是压力，也是"强调"；strip 是条、剥去；student 是学生；stun 是使震惊；subscriber 是订阅者；subtle

是微妙；success 是成功；suddenly 是突然；suggestion 是建议；super 是超级；supply 是供应；suppression 是压制；surge 是激增；surrender 是投降；suspend 是暂停、悬挂；swamp 是沼泽；swell 是膨胀；symbol 是象征。

这里有几个词值得停下来讲，因为它们最容易"认得却用不对"。

先说气味的三个层次。scent 是中性的"气味"，但偏雅，香水叫 scent 不叫 smell；smell 是通称，可好可坏，而且它当动词时后面要跟形容词或 of：it smells salty、it smells of smoke。odour 是"异味"，带嫌弃。所以正文里我写那间灶屋"smelt of tobacco"，用的是 smell of，这个 of 不能省，省了就成了"闻起来像烟草"而不是"有烟草味"。

再说 sort、type、kind。三个都能翻成"种类"，但语体不同：kind 最口语，sort 稍微随意，type 最中性、最适合书面。所以产品分类用 type，人群分类用 kind of people，而 "this sort of thing" 是英语里最常听见的一句评价——它通常不是夸。写作文的时候，一个句子里别同时用两个，读者会觉得你在犹豫。

然后是 stress。它至少有三副面孔：心理压力、物理应力、语音重音，还有一个动词义"强调"。正文里材料课那个 stress，是"应力"，是力的分布；我说沈璐"under stress"就是压力。同一个词，靠学科分家——这就是为什么背单词表没用，你必须把它放回句子里，看它周围站着谁。

最后说 subtle。这个词难译，因为它夸的是"分寸"。a subtle difference 是"细微的差别"，a subtle joke 是"要绕一下才笑出来的笑话"，a subtle insult 是"听着不像骂人的骂人"。中文里没有一个词能同时覆盖，所以你只能记搭配，不要记中文。

材料、工具、脾气：从 tact 到 wreck

tact 是得体，说话不伤人的那种分寸；tale 是故事，偏民间、偏口传；tangible 是有形的，也是"实实在在的"：tangible benefits；task 是任务；teacher 是老师；technological 是技术的；telescopic 是"可伸缩的"；tend 是倾向于，tend to do，也是"照料"；terminal 是枢纽、终端；terrify 是吓坏；testify 是作证；thankful 是感激；theme 是主题；thick 是厚；thirst 是渴；thus 是因此。

tiger 是老虎；tired 是累；tobacco 是烟草；tolerance 是容忍，也是"耐量"；tongue 是舌头，也是"语言"；topical 是时事的，也是"局部的"（药膏）；tough 是难、硬、坚韧；tractor 是拖拉机；tragic 是悲惨的；transformation 是转变；translator 是笔译；transport 是运输；tread 是踩、踏；trek 是长途跋涉；trigger 是触发；truck 是卡车；truthful 是诚实的；tuition 是学费，也是"教学"；turmoil 是混乱；type 是类型，也是"打字"。

umbrella 是伞；underestimate 是低估；underneath 是"在……下面"，比 under 多一层"被盖着"的感觉；uneasy 是不安；unilateral 是单方面的；upgrade 是升级；usual 是通常的；variable 是变量、可变的；vegetarian 是素食者；ventilate 是通风；verification 是核实；vessel 是容器，也是"船"；vice 是恶习，也是"副的"，vice-president 就是副总统；view 是看法，也是景色；violation 是违反；virtually 是"几乎"；visionary 是有远见的；vocal 是发声的，也是"直言不讳的"；volcanic 是火山的；volunteer 是志愿者；vulnerable 是脆弱的。

wasteful 是浪费的；wavy 是波浪状的；wealthy 是富有的；webcast 是网络直播；weekday 是

工作日；welfare 是福利；whatever 是无论什么；whereas 是"而"，表对比；whip 是鞭子，也是"搅打"；wide 是宽；wild 是野生的；wine 是酒；wireless 是无线的；withdrawal 是撤退，也是"取款"；wolf 是狼；worldwide 是世界范围的；worthiness 是"值得"；wreck 是残骸；write 是写。

这一组里，tend 和 whereas 是真正会被考的。tend 后面跟 to do，是"往往、倾向于"，它比 usually 多一层"有这种倾向"的意味：Prices tend to rise 是"物价往往上涨"，不是"物价通常上涨"——前者承认例外，后者不承认。whereas 是连词，只能连接两个并列的分句，翻译时几乎总是"而"，但它比 but 重，因为它对比的是两件"并置的事实"，不是两个"相反的态度"。你写论文里比较两组数据，用 whereas 最合适。

vice 也值得一句：它当"恶习"时不可数，当"副职"时只能放在头衔前，且中间有连字符。这个词在新闻里出镜率很高，因为"副"这个位置在英文里没有独立的词，只能靠 vice 顶着。

数字的语言：一本记账本能教你什么

外婆那本记账本，是全章最像"数据"的东西。它训练四种能力，而这四种能力恰好对应四组词。

第一种是读得准。decimal 是小数，decimal point 是小数点；中文说"两位小数"，英语是 two decimal places。percentage 是百分比，但要小心 percentage point（百分点）：从百分之四涨到百分之六，是涨了两个百分点，不是涨了百分之二，这一条在图表作文里能救命。statistic 是统计数字，单数指"一个数字"，statistics 才是"统计学"，加一个 s 就从数据变成学科。

第二种是算得清。sum 是总和，也是"一笔钱"；aggregate 是加总后的总量，当形容词是"总计的"；remainder 是余数、剩余。三个词的分工是：sum 是你加出来的那一下，aggregate 是加完以后的整体，remainder 是加完还剩下的。exceed 是超过，主语通常是数字；budget 是预算，动词是"编预算"；deposit 是存进去的钱，也是押金；currency 是货币。这几个词在经济类阅读里成群出现，把它们当成一套工具，比一个一个背省力。

第三种是估得对。estimation 是估计，approximately 是大约，roughly 也是大约但更口语。correlation 是相关性，不是因果，这是学术英语里最容易被误读的一个词：A 和 B 有 correlation，只说明它们一起动，不说明谁让谁动。discrepancy 是"对不上"，账目、数据、两份说法之间的差额都用它。factor 是因素，variable 是变量，quantify 是量化。

第四种是说得住。derive 是推导出来，hypothesis 是假说，equation 是方程。你在实验报告里写 we derive X from Y，意思是"我们从 Y 推出 X"，而不是"我们发现了 X"。这个区别，就是学术写作和作文的区别。

材料课的语言：为什么工科英语"名词特别长"

正文里那节材料课我故意写得很密，因为它示范了一件事：工科英语的难，不在句型，在名词。

corrosion 是腐蚀，degradation 是降解、退化。两个词都指"变坏"，但 corrosion 是电化学的、从表面开始的；degradation 更宽，可以是性能、结构，也可以是环境。friction 是摩擦，stress 是应力，tolerance 是公差、耐量。这四个词在车间里天天出现，中文都只有两个字，英文却各自带着一整套搭配：corrosion resistance 是耐蚀性，stress concentration 是应力集中，tolerance limit 是公差上限。你记单词的时候顺手把搭配记上，到了专业课就不用重学一遍。

composite 是复合材料，hybrid 是混合体系，ceramic 是陶瓷，silicon 是硅，ore 是矿石，radioactive 是放射性的。这一组词的共同点是"你没法用形容词替代它们"：复合材料不能叫 mixed material，陶瓷不能叫 pottery（那是陶器），硅更不能和 silicone 混。专业词的尊严就在这里，它不给你替换的余地。

durable 是耐用，fragile 是易碎，resistant 是耐……的，responsive 是响应快的。四个形容词正好构成两根评价轴：durable 对 fragile，resistant 对 responsive。材料工程师整天在这两根轴上做取舍，所以这四个词在论文里出现得极密。

refinement 是精细化，verification 是验证，certification 是认证，patent 是专利，innovate 是创新，technological 是技术的。这一组是"从实验室到产业"的路标：先 refine，再 verify，然后 certify，最后 patent。你把这条链记住，读产业新闻时就不会把 certification 当成 verification——前者是"有人给你发证"，后者是"你自己证明它是对的"。

最后说 horsepower。它是马力，是功率的旧单位，今天工程上更常用千瓦，但汽车广告还在用它。这个词有意思的地方在于它是复合词，两个词一拼，就成了一个不可拆的单位。英语里大量的专业词都是这么造出来的：tractor、machine、mechanic 同源，microscope、telescopic 同构。你摸清这个造词的脾气，长词就不吓人了。

同一件事的三种说法：语体是隐形的分数

你可能有过这种体验：一句话语法没错，词也认识，读起来却"不对味"。那多半是语体错了。这一章里正好有三组现成的例子。

第一组是"拿"。grab 是随手抓，snatch 是一把夺走，两个都带动作的粗鲁；take 是中性；seize 是官方口吻，查获、扣押都用它。正文里小孩 grab 了一把东西、母亲 snatch 住他的领子，这两个动词一出现，画面自己就动；换成 take，句子立刻变成说明书。seizure 你在广播里见过，那是执法动作，不是人的动作。

第二组是"杀"。kill 是中性事实，slaughter 是宰杀，用于食用动物，也用于比喻"大比分击败"；massacre 是屠杀，用于平民；assassinate 是暗杀，用于政治人物。四个词，四种立场：slaughter 里有"这是生计"的平静，massacre 里有"这是罪行"的判断，assassinate 里有"这是政治"的距离。你读 slaughter 时没觉得血腥，正是因为那个词本身不打算让你觉得血腥。

第三组是"贵"。dear 是贵得让人心疼，口语，常和 cost 连用；expensive 是中性陈述；costly 除了钱还能指代价：a costly mistake 是代价高昂的错误，不是昂贵的错误。三个词的分寸，就是三句话的分寸：这东西真贵、这东西价格高、这一步走错的代价太大。

语体不是修辞的花边，它是在告诉读者"我站在哪儿"。作文评分里这一项叫语言的得体性，占分不多，但它决定了阅卷人读你文章时的表情。

介词：这一章真正的难点

如果说这一章有哪个地方最容易让你"每个词都认识、整句读不懂"，那就是介词。介词不承载意思，它承载关系；而中文里的关系常常是意合的，不写出来。

看几个正文里的例子。the ability of a system to absorb a shock：of 说明"谁的能力"，to 说明"干什么的能力"。中文说"系统吸收冲击的能力"，两个介词都不出现，所以你写英文时也想不起来。adhere to a rule：adhere 后面只能跟 to，不能跟 with。compensate for the loss：补偿的是"因为"某个损失，所以用 for。refrain from saying：克制的是"不去做"，所以用 from。impose on：强加于。reside in：存在于。resistant to：对……有抵抗力。similar to：和……相似。正文里的 resemble 反而不用介词，它直接跟宾语，这是它和 similar to 的分工。

再看一组"被"的结构。beset by problems、beset with doubts，两个介词都行：by 更强调被围，with 更强调带着。soak 是"自己泡"，你要说被雨淋透，得用 be soaked through，那个 through 是从里到外的意思，漏了就只剩"湿了"。

最后说一个几乎人人都错的：compare。compare A with B 是"把 A 和 B 放在一起看差别"，compare A to B 是"把 A 比作 B"。正文里我说喜欢把纸上的时间和牌子上的时间 compare，那是 with。这一条不难，但写作时十个人里有六个会写混，因为它俩在中文里都是"比较"。

长句怎么切：三个现场

读不懂长句，通常不是词的问题，是没找到"骨架在哪儿"。拿这一章的三句真句子练手，比做十道题有用。

第一句，材料课上那句：A hybrid composite can be resistant to one failure and responsive to another, and complementary properties are the trick of the trade。你先找动词：can be、are。两个动词，说明这是两套主谓，用 and 并起来。第一套的主语是 a hybrid composite，后面两个形容词短语 resistant to... 和 responsive to... 并列；第二套的主语是 complementary properties，谓语 are，表语 the trick of the trade。骨架一出来，剩下的都是挂件。这句话的教学价值在于：它告诉你 and 不一定连接两个词，它常常连接两个完整的判断。

第二句，系主任那句：He said the founders had shown foresight, that the department had a renowned laboratory and an unremarkable budget, and that this was a hurdle and not a tragedy。这句的骨架只有一个：He said。后面三个 that 从句并列，都是"他说的内容"。中国学生最容易在这里断错，因为第一个宾语从句省略了 that（said the founders had shown），后面两个又补上了 that，看起来像换了一个句子。你只要记住：said 后面可以挂一长串 that，挂三个不算多。

第三句，诊所那段：Notwithstanding forty years of evidence, she has not had a problem with that tooth yet, whereas the one she had filled in 1994 has been trouble ever since, thus proving her point。这句有三个"路标"：notwithstanding 领一个让步状语，whereas 划出对比的两半，thus 引出结果。你把这三个词圈出来，句子的三段就分好了：尽管有四十年证据，她这颗牙没出过问题；而 1994 年补过的那颗一直出问题；因此她的说法得到了证明。长句不是难在长，是难在你没看见路标。

场景地图：六个地方，六种英语

这一章的六个场景，其实对应六种不同的英语，你把它们分开记，比混在一起背有效得多。

站台是条款英语：短句、被动、情态动词多，applicable、restriction、waiver、permit 都住在这儿。这种英语的目的不是让你懂，是让律师懂，所以它重复、它啰嗦、它把每种例外都写出来。你读它的时候要反过来：先找"谁不能做什么"。

灶间是实物英语：cooker、clay、cushion、lid、rack、spoon，全是能看见、能摸到的东西。这类词最好的记法是"指着记"，你在自己家厨房里指一遍，比抄十遍强。它们也是写作里最值钱的一批词，因为作文最容易空，一具体就活了。

集市是交易英语：negotiation、courtesy、scale、cheat、greedy、petty，动词多、形容词带评价。这一片语言的特点是"话里有话"，所以 cunning 和 nosy 这种词才会出现在这儿。

祠堂是仪式英语：oath、proclaim、offspring、reign、majesty、mourn，词都老，句子也老。这类词你不用会写，但要认得，因为阅读里的历史类文章全靠它们撑场面。

材料课是术语英语：corrosion、degradation、friction、stress、composite，名词长、搭配固定。术语不能换词，这是它和文学语言最大的区别。

那封七月的信是书面语：notwithstanding、indispensable、credibility、integrity，句子长、插入语多、语气克制。你以后写英文邮件、写申请、写报告，用的就是这一套。

一次真实的修改：从"背过的词"到"用对的词"

给你看两个我自己第一稿里的句子，它们没有语法错误，但什么也没说。

第一句：The old house makes me feel a lot of feelings, and I think the culture of the village is very deep.

问题有三个：feelings 太笼统，culture 太空，deep 是中文"深厚"的硬翻。第二稿我改成：The old house reminds me of my childhood, and the village has a long history。好一点，但 reminds 是个廉价动词，history 还是空的。第三稿才成立：The kitchen still smells of tobacco and camphor, and the hinge of the door groans in two notes I could hum。它做了两件事：把"感觉"换成一个具体的感官，把"文化深厚"换成两个能听见、能闻见的东西。hinge 在这里比 culture 有用一百倍。

第四稿我又加了一句：That is what I mean by continuity, not a tradition that is written down, but a hinge that has been oiled by the same hand for fifty years。到这一步，抽象词才进来，而且是被具体的东西托进来的，不是自己飘进来的。

第二句：Our hometown is developing very fast, and we should protect the environment.

这句话是作文里最常见的那种"正确而无效"。改法不是换同义词，是换视角：The terminal is new, the market still has a tin roof nobody has replaced since 2009, and the drain behind it is stagnant in July。这样一写，"发展"这个词一次都没出现，但读者看见了它。后面那句"应该保护环境"也就有了落点，因为stagnant 那个水沟就在那儿。

这两个例子说明同一件事：抽象词不是不能用，是不能先用。你先写具体，再让抽象词来总结；反过来，文章就会像一份没下过现场的人写的总结。这一章里那些大词，existence、continuity、transformation、worthiness，都应该出现在具体之后。

五条一句话就能记住的规矩

infer 是读者推，imply 是作者暗示，方向不能反。information 不可数，要说 a piece of information。personnel 指"人员"这个整体，说"一名员工"要用 employee。headquarters 单复数同形，别加 s。percentage 与 percentage point 差一个词，能差出两个数量级。

前缀：这一章偷偷教了你构词法

英语里最省力的记词法不是词根，是前缀，因为它给你方向。这一章里藏着三组。

第一组是"上下"。upgrade 是升级，downgrade 是降级，两个词只差一个前缀，中间那个 grade 是等级。同一套里还有 underestimate（低估）和 underneath（在下面），以及 overlook（俯瞰，也是忽略）和 overtake（超越）。你把这六个词排成两列，上下、超过、不足，方向一目了然。考试里最常考的是 overlook 那个"忽略"义，因为它和字面的"往下看"逻辑相反，这正是前缀题爱设的陷阱。

第二组是"再、回"。recall 是召回、回忆，release 是放开，relieve 是减轻，reside 是居住，resistant 是抵抗的，responsive 是回应的，revive 是复苏，reversal 是逆转，reunification 是重新统一。这九个词里的 re- 有的是"再一次"，有的是"往回"，有的是"彻底"，所以你不能拿 re- 当公式，只能当线索。research 之所以不是"再搜一遍"，就是这个道理。

第三组是"共同、一起"。coordinate 是协调，correlation 是相关，complementary 是互补，composite 是复合，congregate 是聚集。co-、cor-、com-、con- 是同一个前缀的四种拼法，碰上双写就变样，这是拼写题的常客。

如果只剩二十分钟

临考前你只有二十分钟，别从头读。做三件事。

翻到站台那张告示，读一遍，把 applicable、restriction、waiver、permit 四个词在句子里的位置看清楚，这是条款英语的标本。翻到记账本，把 decimal、percentage、aggregate、remainder 的中文遮住，看英文想中文，想不出来的那个就是你的漏洞。最后翻到材料课，把 corrosion 和 degradation 的区别用自己的话说一遍，说不清就再看一眼正文里那句"everything we make is a negotiation with decay"。

二十分钟做不完三件事也没关系。做完一件，你就比昨天多一个能用的词；做完三件，这一章就没白读。

声音：这一章的耳朵活

这一章有几个词的拼写会骗你的耳朵。knife 和 knowhow 里的 k 不发音，write 和 wreck 里的 w 不发音，psychiatry 里的 p 不发音。三条合成一句：kn-、wr-、ps- 开头的词，第一个字母常常是哑的。你知道这条，听力里就不会去找一个不存在的声音。

再看 -ed。walked 读 /t/，dodged 读 /d/，wanted 读 /ɪd/。规则看前面那个音是清是浊：清音后读 /t/，浊音和元音后读 /d/，t、d 结尾才读 /ɪd/。这一条不考拼写，只考耳朵，而读错的音会跟

着你很多年，改起来比记错词义还难，而耳朵是靠读出来的，不是靠看的。你把这章的动词过去式挑十个，出声读一遍，比默看十遍有用。

最后是重音。recording 重音在第二个音节；record 当名词时重音在前，当动词时在后。这类"名词前、动词后"的词英语里有一批，present、object、import 都是。你读课文时把重音读错，听的时候就会认不出来——不是你不认识这个词，是你在等一个错误的声音。

抄一遍，比看十遍有用

最后给一个笨办法。把这一章里你觉得最硌的二十个词抄在一张纸上，每个词后面只写三样东西：它在这一章的哪一句里、跟它搭配的那个介词、以及一个你自己的句子。三样写完，这张纸就是你的，别人的词表再全也替不了它。抄完别贴在墙上，隔一周再看一遍，把已经会的划掉；纸上剩下的那几个，才是你真正的词表。

一个词的三种考法

issue 是这一章里最忙的词。它可以是"问题"，a political issue；可以是"期号"，the March issue；还可以是动词"发行"，issue a statement。三种意思在阅读里出现的频率差不多，所以你不能只记一个。

means 更麻烦。它形式上是复数，意思却是单数："手段"，a means of transport；by means of 是"通过"；by no means 是"绝不"，这个短语里的 means 已经不带"手段"的意思了，它只是个固定搭配。你把它当三个词记，比当一个词记安全。

view 也一样。它可以是"看法"，in my view；可以是"景色"，a view of the valley；还可以是动词"看待"，view something as。三个用法共用一个"看"的内核：看法是心里的看，景色是眼里的看，看待是判断的看。抓住这个内核，你就不用背三遍。

这三个词放在一起，是想说明一件事：词表给你的那个中文，往往只是它最常用的一个义项。真正的词汇量，不是你认识多少个词，而是你认识一个词的多少个面。

把这一章搬进作文

这一章里有四个句式，改一改就能用在几乎任何"家乡、变化、长辈"的题目上。

第一个是让步加转折：Notwithstanding the new terminal and the corporate logo, the lane to the old house is narrower than I remembered。后半句换成你熟悉的任何细节，句子就成立。notwithstanding 太书面，考场里可以换成 Despite，但你要知道它存在，阅读里它会突然出现。

第二个是"具体到抽象"：That is what I mean by continuity, not a tradition that is written down, but a hinge that has been oiled by the same hand for fifty years。这个 not A but B 的结构，是英语里解释抽象概念最好用的模板。

第三个是"承认边界"：I had been away for eighteen weeks, and I had learned the names of six phases of matter; I had not learned how to open a door。前半句给事实，后半句给反差，读者一下就记住。

第四个是"小结论"：The elimination of small errors is the whole job, in a ledger as in a laboratory。这种 "X is the whole job" 的句子，短、狠、好记，比任何 In conclusion 都强。

四个句式各写一遍，这一章就不只是读物，它成了你的工具。

一句多余的话

这一章写到这儿，分量已经不像一章，像一个小册子。我不打算为它道歉：九百多个词形要塞进一个连贯的故事里，密度只能这么高。但我要提醒一件事——不要试图一次读完。注意力是有限资源，你把它全花在"从头到尾"上，就没有余量留给"这一句为什么这么写"。分三次读，每次读完停十分钟，回想三个画面：站台上的告示、灶屋里的那口缸、记账本上的小数点。词会跟着画面回来，比跟着词表回来牢。

你该怎么读第二遍

第一遍你跟着林一回了趟家，第二遍你要换个读法。

挑三段重读：站台那张告示、外婆的记账本、材料课那一段。这三段是全章词汇密度最高的地方，也是三类完全不同的英语：条款语言、数字语言、学术语言。读的时候不要查词，先猜：这一段是给谁看的？写给旅客的，句子一定短、动词一定硬；写给自己的，数字一定多、单位一定省；写给同行的，名词一定长、形容词一定少。你把这个"读者"判出来，一半的生词会自动归位。

然后再做一件更笨的事：把这一章里所有加粗的词，按"我认得但不会用"和"我不认得"分成两堆。第一堆才是你的问题。第二堆可以慢慢来，因为它们出现的场景很具体，你只要记得那个场景，词就跟着回来了——salty 跟着那缸咸菜，hinge 跟着那扇吱呀的门，stigma 跟着村口那些议论。语言不是背下来的，是挂上去的；你得先有钩子。

最后提醒一句：不要在作文里堆这一章的词。notwithstanding、indispensable、meritorious 这类词，一个句子里放一个就够，放两个就像穿了西装去赶集。真正显本事的，是把 get、make、take 用对，把 of、for、to 放准。

为什么把它写成这样

这一章我写了一个寒假，从高铁站台写到返程的行李箱，中间塞进灶屋、集市、祠堂、婚礼、旧相册、卫生院、一场材料课和一封七月的信。这么排不是为了好看，是为了让词有地方落。

难词好办，因为它们自带场景：acupuncture 只能出现在诊所，slaughter 只能出现在年猪那天，mencement 只能出现在一张印坏的请柬上。难的是那些"到处都能用"的词——consequence、factor、tend、issue、situation。它们没有场景，只有位置；而位置是靠句子给的。所以我把它们放在人物说话的时候：王老师讲报告，沈璐写信，系主任致辞。人一开口，抽象的词就有了口气。

如果这一章只能留下一句话，我希望是这句：一个词的价值不在它的中文意思，在它出现时，句子里其他词站的位置。 你记住位置，就记住了它；你只记中文，考试一换句子，它就跑了。

最后说回标题。回家这个词在英语里没有完全对应的单词：go home 是动作，homecoming 是一种仪式，return 太中性。这一章叫回家而不叫返乡，是因为返乡带着身份，回家只带着一个人。英语里

最接近的是 coming home，但它必须带一个落点：coming home to a kitchen, to a jar, to a woman of seventy-six。return 是公文，homecoming 是校庆，只有 come home 是人的。

还有一件事我没写进正文。外婆把咸菜装进罐子的时候，会用手指把最上面一层压实，再拿一块洗干净的石头压住。她说这样气才不跑。我问她这是不是科学，她说这是手知道。我后来在实验室里配溶液，搅到某个程度也会有一种"可以了"的手感，那种感觉和她的石头是一个来源：重复过太多次的人，身体里会留下一份说明书。knowhow 说的就是这个东西，它不在书上，也不在词表上。中文说熟能生巧，英语里对应的不是 practice makes perfect 那么励志，而是这个词本身：知道怎么弄，不等于知道为什么。你以后读合同、读公司介绍，会反复见到它。

现在合上这一章，去做一件具体的事：把你今天遇到的三个"认得但不会用"的词，各写一个跟你自己有关的句子。不用长，一句就够。写不出来，说明你还没认识它们。如果哪天你翻回这一章，发现某个词已经不用查了，那就是它对你唯一有用的地方；其余的，都是我的事。

第 6 章　算法与显微镜

The rain had started at four and had not stopped by seven, so the fifth floor of the materials building smelled of wet nylon and floor wax, and every umbrella in the corridor was dripping into the same bin. On the meeting room's blackboard half a formula still stood from the week before: somebody had drawn a curve, then stopped, and the chalk line ended in the middle of the board like a sentence cut off by a phone call. Above the door the projector hung at a slight angle, and its fan was louder than the rain. Shen Lu took the aisle seat with her boots still dripping. Zhou Min stood beside the window, a cardboard box atop a pile of printouts, and almost everyone was already at the table — all of the first-years, two sophomores, and a thin boy from the neighbouring lab who had come for the biscuits.

"Sign first," Zhou Min said. "The agreement." She passed a stack of forms down the table. There was consent for the use of our data, an allowance sheet for the hours we would log, a line where the applicant wrote his student number, and a paragraph on accountability that nobody read. Administrative paper, I thought, and signed. Then I affixed the little barcode sticker to the back of my laptop, as instructed, and felt absurdly official.

Gu came in last, shaking water off his jacket. He was thirty-two, maybe thirty-three. He had come back in September from two years in a lab outside Düsseldorf; the department had given him one room, a budgetary line too small for the card he wanted, and this project. He wrote his name on the board, then, instead of a title, two words: academic, accuracy. "The arena we are in," he said, "is not a race for the cleverest model. It is a race for adherence to a protocol. Advance in this field comes from accuracy, and it never comes from arrogance." He said it without smiling, which made the word land.

Then he showed us a chart from his first week: a hundred and forty papers on electrolyte additives, read by hand, their numbers typed into a table by a master's student who had since graduated. "The bulk of this field still works like this," he said. "Approximation at best. And an audit of any three rows will show you why." He clicked. Three rows appeared, then the same three rows with the original figures pasted beside them. Two were wrong in the second decimal place. One was wrong in the sign.

Zhou Min did not look surprised. She capped her marker and said, "So we automate the reading. Fine. But a hundred papers a week of absorption is not a strategy, it is a chore. The acute problem is labels. We have eleven thousand papers and about six hundred numbers a human being has actually checked." She drew a line, then a second line under it. "Two roads. The dumb road: we read four hundred papers by hand, we take forty numbers from each, we fit a line with three variables. Algebra, a pen, one semester. The clever road: we fine-tune a small model on the whole corpus and let it predict."

"Anything else?" Gu asked.

"The clever road has no baseline," Shen Lu said, from the aisle. "Between those two roads there is a comparison missing, and the comparison is the whole paper."

Nobody answered her. Xu, the sophomore who had been given the model, blushed to the ears. You could see the cheek of him go red under the beard he was trying to grow. He said, in a small voice, "I already have a draft running." — hmm "draft" not target; fine.

Gu tapped the table twice with two fingers, a habit he had brought back from Germany. "Then let us be systematic," he said. "We establish the baseline first. Meanwhile Xu runs his model. We do not presume which one is right."

The assignments came out of his mouth like a shopping list, but addressed to people, which made it worse. Xu took the model and the cluster. Shen Lu took the ethics, the licences, and the correspondence with the journal. I took the paper table, the classification of every figure we extracted, and the microscope queue — because the queue was the only place where a first-year could be useful, and because Zhou Min had decided, without asking me, that I was the kind of person who would sit still.

"Me?" I said. "I do not know anything about machine learning."

"Knowledge is cheap," Gu said. "Attention is not. You will acquire the first in three weeks. The second, I cannot give you."

That was the first piece of advice anyone gave me that semester, and it was also the last one I followed without arguing.

Then Xu opened his laptop and put the first result on the screen. The model had been trained for two days on the corpus; it had been asked to predict the conductivity of a

new additive at room temperature. The number on the slide was minus three thousand four hundred. The unit was printed correctly. The sign was not. Under the number, in a font that Xu had chosen because it looked modern, the model had written a sentence: "This material shows excellent stability and great potential for industrial application."

For two seconds the room was quiet. Then the laughter came, and it came from everyone except Gu and Shen Lu. The amazement was real; I had never seen a computer be so confidently wrong. Someone at the end of the table said the model should be given a prize for optimism. Xu's eyebrows went up and stayed up. He looked astonished, then annoyed, then, worst of all, amused, as if the machine had made a joke and he was expected to laugh with it.

"Dismiss it if you like," Gu said, and the room went quiet again. "But write down what you saw. A model that is wrong by a factor of ten thousand and cheerful about it is telling you something about the training data. It is telling you that in the literature, almost every sentence near a number is positive. Nobody publishes the additives that failed."

Shen Lu stood up and put her jacket on. "A model that has never been wrong has never been checked," she said. "Yours has been wrong in public, on its first day. That is the best thing that will happen to it all year."

She left. The door did not slam, which was somehow worse. The rain had thinned to a breeze that came through the window and lifted the corner of the cardboard box. I walked back to the dormitory along the bridge over the canal, pushing my bicycle because the chain had come off again, and I thought about the minus three thousand four hundred. It was the first number in my university life that had frightened me, and it had been produced by a machine that also thought the material was excellent.

In the basement of the same building, behind a door marked with a yellow triangle, there was a boiler that fed the whole block, and next to the boiler, in a room the size of a cupboard, sat the machine that had done the thinking. I would get to know that room very well. I would learn its smell — hot dust, diesel from the generator outside, and the metallic taste that stayed in your throat after an hour in the cold aisle.

Tonight, though, I only wanted a bath and my bed. My baggage from home was still half unpacked in the corner of the room, and the bottle of sauce my mother had put in it had leaked onto a textbook. I lay on the bed with my socks on and listened to Zhang Ye sharpen a knife in the dark; he did it every Sunday, and the sound was regular enough to sleep to.

The room was officially B-17, but everyone in the department called it the closet. You descend one flight from the loading dock, where the nitrogen containers came in on Tuesdays, and then you follow a pipe until the corridor turns dense with heat. On the

left is the boiler. On the right is a steel door with a label that says NO STORAGE, and behind it, on a shelf bolted to the wall, sits a single card with twenty-four gigabytes of memory and a fan the size of a saucer.

Zhou Min took me down on a Friday afternoon with a bottle of tea and a memo she had typed herself. "Rule one," she said. "Everything goes in the log. Date, time, what you changed, what you expected, what happened. Rule two: never trust a number you have not seen twice. Rule three: if the temperature alarm goes off, you do not dash for the door, you first look at the display on the rack, because the alarm is usually the air conditioner and not the card."

The card had cost more than my family's car. It had been bought on a grant that Gu had secured with an application of eighteen pages, and its arrival had been announced in the group chat with a photograph of the box. Xu was envious; he had been queueing for a card of his own since his second year. I was not envious, because I did not yet understand what the thing did, and because I had been assigned to the paper table, which needed no electricity at all.

The corpus was eleven thousand four hundred papers, downloaded through the library subscription over nine nights. Each one had to be licensed, and the licence was the reason for the embargo that Gu insisted on: nothing we extracted could leave the group until the publisher's terms allowed it. "We are not pirates," he said, in a tone that suggested he had argued about this before. "We are borrowers. There is a difference, and the difference is a lawsuit."

My own work began the same week, and it was dull in a way that I had not expected from science. Four hundred and twelve papers. For each one, forty fields: the composition of the electrolyte, the salt, the solvent ratio, the temperature of the test, the current density, the cycle number, the capacity retention, the coulombic efficiency, the particle diameter if the authors reported one, and the figure number where the number could be found. I typed with two fingers and a ruler on the desk to keep my place in the pdf.

By the third week my wrist had a permanent ache and my eyes had learned to skip the introduction of every paper, which is where the authors explain why their work matters and never where the numbers are. I developed a method: first the abstract, then the figures, then the table, and only if the table was missing, the text. This is the dumb road, and I want to be honest about it: it is boring, it is slow, and for three weeks it felt like punishment invented by someone who did not like me.

Then, on the twenty-first day, I found something. A paper from 2019 reported a capacity retention of 92 per cent after five hundred cycles. The same group had published a second paper in 2021 with the same material and the same test, and the retention was 71 per cent. I checked the figures twice, then a third time. The difference was not a typo; the second paper had used a thicker electrode, and the thicker electrode had cracked. The

authors had not hidden it. They had simply not mentioned it in the abstract, where the number 92 was still sitting, quoted by eleven later papers, including two reviews.

I brought it to the group meeting on a Tuesday evening. Gu looked at the two papers for a long time and then said, "This is why we need the table." He was not praising me, exactly. He was describing a fact about science: the literature is not a database, it is a rumour with footnotes, and every rumour grows a little every time it is repeated. The model would learn the rumour first, because the rumour is what most papers say.

"Feedback," Shen Lu said, when I showed her my notes. "You want feedback. Here it is. Your table is good and your conclusion is wrong." She flipped to the page and put her finger on the column I had called "anomaly". "Anomaly is a word people use when they do not want to say error. Write error. Then write whose."

I rewrote the column that night. It took forty minutes and it changed the way I read papers permanently.

Xu's model meanwhile was training in the closet, and the closet was not a good place for it. The ventilation was a single fan that exhausted into the boiler room, so the heat had nowhere to go except back into the rack. The first night the card ran at eighty-three degrees for six hours and then shut itself down. Xu wrote in the log: "Thermal throttle at 02:40. Lost four hours of training." Underneath, in smaller handwriting, he added: "Card is fine. Room is not."

The second night I sat with him because the log required two people after midnight, a rule that Zhou Min had made after a fuse blew and nobody could prove who was in the room. We ate cold dumplings from a box and watched the loss curve descend by steps, like a man walking downstairs in the dark. At one in the morning the curve flattened. At two it rose again, sharply, which is what overfitting looks like when you draw it: a line that learns the training set by heart and then starts inventing.

"That is not a model," Xu said. "That is a very expensive photocopier."

I laughed, and then I stopped, because the sentence was exactly right, and because I had spent three weeks being a photocopier myself, with a ruler and a wrist brace, and nobody had called it funny.

The alarm went off at three. It was not the card. It was the air conditioner, which had been dying since June and which the building office had promised to replace "after the holiday". We opened the door, and the corridor was hot and smelled of dust and old water. Somewhere above us a pipe knocked twice, like someone testing a door.

"This building is a museum," Xu said. "They keep us in it so we do not forget how lucky we are."

At four I walked up the stairs and out into the courtyard. The rain had stopped. A

stray dog slept under the bicycle shed, and the sky over the freeway was the colour of dirty milk. I thought about the word evidence, which I had written eleven times that night in the log, and about the fact that none of it was mine.

The dictionary

Sunday, and the workroom was empty except for me and a lamp that hummed at the wrong pitch. The spreadsheet had eleven thousand four hundred rows and ninety columns, and I had been designated the person who would define, in writing, what each column meant. My definitions went into a document called the data dictionary, which Zhou Min printed and taped to the wall beside the desktop, and which every later reader of the dataset had to read before touching a single cell.

The dictionary was dull on purpose. Column one, sample id, is a twelve-digit string and nothing else. Column nine, dilution, records how many times the slurry was diluted before the measurement, and a value of zero means undiluted, not missing. Column fourteen, particle diameter, is in micrometres and refers to the median, not the mean, because the distribution has a tail. Column twenty-two, duration, is the length of the test in hours, and a duration of forty-eight hours means the cell was cycled at one hour per cycle, which is slow, and rare, and usually means the authors were proud of something. Column thirty, excess, is a ratio I did not understand for three months.

Then the boring rules, which Zhou Min made me write in the imperative and sign. Deduct a row if the same sample appears twice with different numbers. Count every figure twice, once in the morning and once after dinner, because attention decays and fatigue pretends to be certainty. Divide the corpus by year, and check whether the distribution of values shifts, because methods improve and old numbers are not wrong so much as measured with worse instruments. Double-check any number that looks too good, and triple-check any number that looks too good twice.

Cleaning is not preparation, she said, when I complained that I had spent a week on punctuation. Cleaning is the experiment. If you cannot say what a column means, you do not have data. You have text.

What the scanner did

The corpus came from two sources: papers we had downloaded as pdf, and papers from before 1998 that existed only as scans. The scans had been passed through an OCR engine by a company that charged by the page. The engine was good at prose and hopeless at tables. Cells split, columns merged, and words were cut at the edge of a page and never mended. On page forty-one of one scan the fragment read only mentary, the tail of a longer word whose head had fallen into the margin. Two pages later a cell held modity, and in the same table another held mon. We kept them all. We kept every broken fragment

in a separate column called raw, because a fragment is evidence of what the machine did, and evidence is worth more than tidiness.

The worst discovery came in the third week. A whole journal issue had been scraped as if each table of contents were an article: four hundred rows of titles, authors, and page numbers, with every scientific field empty. The model would have learned from them that electrolytes are frequently mentioned in lists and never measured. I deleted the rows one by one, and it took an hour and a half, and I hated every minute of it, and I learned more from that hour than from the three weeks before it.

Ninety per cent of machine learning, Gu said, when I told him, is deciding what not to feed the machine. The other ten per cent is deciding what not to believe.

The human side of the corpus was worse than the machine side. Each paper carried an address for the author who would answer letters, and I had to decide, for every one of the eleven thousand, which country the work came from, because Gu wanted a demographic breakdown by first author. That meant guessing at surnames, and guessing is a form of deceit even when it is polite. After a long deliberation I wrote a rule of my own: if the address was a company, mark it as domestic industry; if it was a hospital, mark it as clinical; if the paper listed eight authors and four countries, mark it as a consortium and move on. The classification was crude, and I presented it with a full page of disapproval of my own method, and Gu read the page, nodded, and said that a crude rule written down is worth ten elegant rules held in the head.

The contract

The fifth week brought the letter. It arrived electronically, at nine in the evening, from the legal office of the company that had given us a proprietary database: eight thousand records of industrial cells, donated under a contract that Gu had signed in June and that nobody in the group had read in full. The letter was three paragraphs long and written in the tone of a man who expects to be obeyed. It said that the company was a legal entity with obligations to its shareholders, that the dataset remained its property, that any publication arising from it required the written endorsement of a named executive, and that failure to comply would result in action in the courts of the jurisdiction named in clause eleven.

Gu read it twice, put it on the table, and asked for opinions. Obviously we do not sign, Shen Lu said. An endorsement is a stamp on someone else's product. We would be advertising, in a journal, with a DOI. And if they sue, Xu said. Nobody sues a university, she said. They write letters. Letters are cheap. Letters are cheap, Gu said, until they are not. I will call the university attorney in the morning. Meanwhile nobody sends anything anywhere. Not a draft, not a figure, not a slide. Not to a conference, not to a friend, not to a forum.

He looked at each of us until we nodded. This is the part of research nobody photographs, he said. Contracts, committees, letters from lawyers. The microscope is the easy half.

The letter sat on the table for a week. In the end the company withdrew the demand, partly because the attorney wrote a colder letter back, partly because a deputy manager who had joined in spring decided that a fight with a public university was not a good first-year project. We kept the data. We lost six weeks of mood. We gained a rule that Zhou Min wrote on the whiteboard in marker and never rubbed out: read the contract before the data.

Inspection

In October the building was inspected. A deputy director came round with a clipboard and a student assistant who photographed everything that was wrong. Our corridor lost points for a bundle of hose coiled on the floor, for a spade left leaning against the wall since the summer, for a peg that had come out of the notice board, for a bin with no lid, for a bottle of bleach stored above eye level, for two sacks of sand that nobody could explain, and for a fire extinguisher that had been checked in April and not since.

The inspector was not unkind. He was tired, he had done this forty times that month, and he read his list in a flat voice while his assistant photographed the fridge full of somebody's old dinner. Nothing personal, he said. Statistics, not judgement. Then he looked at the door of the closet and asked what was inside. Cards, Zhou Min said. And a fan that does not work. Then it is a fire risk, he said, and wrote it down, and the university installed a second fan in November, which is the only reason the card survived the winter.

I learned something useful that day, although it took me a year to say it properly. A laboratory is not a place where science happens. It is a place where science is allowed to happen, and the permission is made of fans, lids, labels, and forms.

The queue

The queue for the microscope was a whiteboard in the corridor of the analysis centre, and by October I could read it the way other people read a timetable. Monday and Thursday belonged to the battery group. Tuesday to the polymer people. Wednesday to a doctoral student who came in at midnight and left at six. Friday was open, which in practice meant two in the morning, which meant a cab, because the last bus left the campus at ten and the driver of the 328 would not wait for a boy with a box of samples.

My first slot was a Friday in November. Shen Lu met me at the gate at half past one with a cup of hot tea and a bandage on her thumb, and we walked past the guard post, where a cop in a padded jacket was arguing with a man about a bicycle left in the wrong

place. The campus at that hour was another country. A cat lived under the bins behind the canteen and did not move for us. A crane stood over the new dormitory, still and huge, its arm pointed at the moon like a finger.

The room was cold, dry, and hummed in two tones at once. Beside the door hung a clipboard with the checklist the operator signed before every session: vacuum, beam current, stage, aperture, detector, cooling water. Under the clipboard stood a cabinet with a glass door, and inside it, arranged by a person with a system I never deciphered, were a spare filament in a plastic case, a set of tweezers, a roll of carbon tape, a bottle of ethanol, a box of lens tissue, a bundle of cable ties, a small gauge for the air line, a coil of wire, a tray of screws nobody had opened since the machine was installed, and a yellowed envelope of photographs from the year the lab moved into the building.

Shen Lu could recite the biography of every instrument in that room: the year it came, the fault it had, the person who had broken it. She put on a white coat too long for her and gave me the rules in the flat voice she used for rules. You will not touch the column. You will not open the chamber while the stage is at pressure. You will not call a sample dirty. If a sample is contaminated, you say it is contaminated, and then you say how. Basically, she said, this is a camera with a very expensive lamp. Treat it like a camera and it will behave like one.

The sample was a black powder sent by a company in Jiangsu and logged as 17-B. We mounted it on carbon tape, pressed it flat with a clean slide, blew the loose grains off with nitrogen, and set it on the stage. The sample had been aged three weeks in a cabinet that was supposed to be dry, and it had absorbed water, which is a complication that shows up later in the numbers and not in the picture. The pump took four minutes. The image came up grey and granular, and then, as the beam found the surface, it flickered and went white.

Charging, Shen Lu said. The sample is not conducting. Turn the beam down, scan slower, and stop staring at the middle of the screen. Look at the edge.

At the edge the particles were distinct: round ones, broken ones, a few shaped like peanuts, one that looked like a crystal of salt. The first image also had a bright band across the top and a black hole in the middle, and those were artifacts, not features, and if I had put them in a paper with a caption they would have been a lie with a caption.

I asked whether we could measure the particles from the image. We can, she said. We can also measure them badly. She opened a drawer, took out a click counter of the kind ticket sellers use, and put it in my hand. Four hundred particles. Count them. Log every tenth one as a size, in nanometres, and write the number in the book, not on your hand.

It took three hours and ten minutes. My thumb ached and my eyes began to invent particles in the grey noise, which is a known effect and is the reason the counter clicks and the log records. At 271 I lost my place and had to start the row again. At 380 the stage drifted and the image slid sideways, and I said a word that made Shen Lu laugh for the first time that night.

When I finished she took the counter, read the total, and wrote it in the log beside my name. Then she said the sentence I would repeat to myself for the rest of the term. The machine makes an image. The image is not data. The count is data, the count is yours, and if you cannot defend it line by line you do not have a number, you have a picture.

We left at half past five. The sky over the freeway was the colour of weak tea. On the way out she erased my Friday slot and wrote my name in the Saturday column instead. You are not ready for the good slot, she said. Come back on Saturday and count another four hundred. By the end of the month you will own a distribution, and a distribution is the first honest thing you will own.

I asked whether the model could do the counting, since that is what the model was for.

It can, she said. It will be wrong in ways you cannot see yet, and right in ways that will flatter you. Count first. Then let it count. Then compare, and when the two disagree, trust neither until you have looked at the picture again.

On the way home I stopped at the shop by the west gate, which stayed open all night for the drivers and the guards, and bought a bun and a bottle of milk. The woman behind the counter was watching a serial on a small television, and she did not look up when I put the coins down. Outside, a man was asleep on a bench with his coat over his face, and a dog slept under the bench, and the whole street smelled of hot oil and rain.

The number that was too good

In December the model produced a number that made Xu stand up from his chair. The correlation between prediction and measurement, on papers the model had never seen, was 0.98. He put it on the screen and said, quietly, that this was better than anything in the literature. Gu looked at it for perhaps four seconds and asked one question: what did you feed it.

Xu listed the features the way a man lists the contents of his pockets. Composition, salt, solvent ratio, temperature, current density, cycle number, particle diameter, and a column called excess. Gu asked him to open that column. Xu opened it. The numbers ran from 0.02 to 0.31, and they were tidy, and they were too tidy.

Shen Lu took the laptop without asking and read the header of the column, then the note beside it, then the paper it came from. She sat back. The label is in the feature,

she said. The authors computed excess from the same titration they used to report retention. You have taught the model to divide by two and call it chemistry.

The room went quiet in the particular way a room goes quiet when a person has been publicly correct. Xu's face did three things in a second, and the last of them was fury. He said that the result was still a result, that the model had found a real pattern, that the pattern was in the data and not in his head. He hit the table with his fist, not hard, and then apologised for the fist, and then apologised for apologising.

Shen Lu did not raise her voice. She turned the laptop round so that he could see the screen and said, this is not a flaw in your code. It is a flaw in your faith. You believed the number because it was beautiful. That is the oldest mistake in the building.

Gu let the silence sit for a while. Then he said the thing that decided the term for me. We will not send this to the conference. We will write it up as a negative result, with the leakage described in the methods, and we will publish it in a small journal that will take it, and the abstract will say what we learned, which is that a column can carry a memory of the answer.

Xu did not speak to either of them for two days. On the third day he came to the workroom with a printed page and put it on Shen Lu's desk. It was a list of every column in the dataset and, beside each one, the question: could this column have been computed from the label? Fourteen columns had a yes. He had written the word yes fourteen times in his own handwriting, which is a form of confession that no committee can cross-examine.

That was the first time I understood that ethics in research is not a lecture you attend. It is a set of small, boring, repeated acts of looking at your own work as if it belonged to someone you did not like.

The other failure

My own failure was less dramatic and took longer. All through November I had been trying to coat a powder with a thin layer of carbon, using a fluid bed and a gas line and a furnace that had been built in 1998 by a technician who had since retired. The recipe was in a notebook, in pencil, in a hand I could barely read. There were twelve samples in the plan. All twelve failed.

Sample one came out grey and lumpy. Sample four came out black, which looked like success and was oxidation. Sample seven was fine, and then I dropped the crucible on the floor, and the crucible survived and the sample did not. Sample nine was the one that taught me something: I had written the date in the log as 11/12 and the technician read it as December 11, and the run was scheduled for the wrong week, and the furnace sat hot and empty for two days, and the powder I had weighed out on the Monday was, by the Thursday, no longer the powder I had weighed.

I was forgetful in a way I had never been with words. I could remember a page of a novel after one reading, and I could not remember whether I had closed the valve. So I made a rule and wrote it on the first page of the notebook: the log is not a diary. It is a set of instructions to a stranger who will repeat the experiment after you are gone. If a stranger cannot repeat it, you have not written anything. You have only remembered out loud.

Twelve failures in five weeks is not a tragedy. It is a Tuesday. But it did something to my sense of what a result is worth: the ones I got by accident felt like gifts, and the ones I got by plan felt like wages, and the wages were the only ones I could defend.

The reading

The elective I had chosen to fill a gap in my timetable was a course on the history of science, taught by a young lecturer who wore a scarf indoors and who spoke about the sixteenth century as if he had been there. In the last week of term he handed out a page: a translation of a letter written in 1607 by a Jesuit who had come to Beijing and had spent twenty years learning to argue in Chinese.

The letter was about mathematics, and about pride. The priest wrote that the scholars of the court were superb at calculation and indifferent to proof, that they would accept a result if it worked and reject a demonstration if it embarrassed them, and that he had therefore decided to begin with the elements, because a man who has never seen a proof cannot be argued out of an opinion. He described the congregation he had gathered, the chant he had taught them, the choir of boys who sang badly and learned Latin quickly, and the cemetery outside the city wall where two of his brothers were buried. He wrote that he had been obliged to learn the costume of the mandarins, the bow, the silence, the way of holding a cup, and that he had come to admire the Confucian discipline of the examination while doubting that it produced anything he would call knowledge.

Our lecturer stopped there and asked us what the letter had to do with algorithms. Nobody answered. Then he said: everything. An algorithm is a promise that a procedure will work for anyone who follows it, not only for the clever man who invented it. That promise is a moral idea before it is a mathematical one, and it arrived in China in the same boat as a religion, wearing the same clothes.

The wall

On the Saturday before the holiday, Shen Lu took me to the old campus, which is now a museum with a lawn and a locked gate and a wall of photographs. The building was built in 1953 with money collected from alumni, and it has a fireplace in the reading room that has not been lit since 1978.

The wall text was printed in two languages and told the story of the war years in

flat, careful sentences. In 1937 the college had packed its instruments into wooden cases and moved west: first to a town in Jiangxi, then to a village in Guangxi, then to a temple in Guizhou where the monks let them use the hall. The text said that the faculty had lectured on algebra under a tree, that the students had slept in a barracks and eaten rice with salt, and that the library had lost four hundred books to water and worms. There was a photograph of nine men in front of a temple, and one of them, a young lieutenant in a borrowed uniform, had later become the head of the department. Another photograph showed a chemistry bench made of two doors and a brick.

Then the war ended and the college came back and the instruments were unpacked, and half of them were broken, and the man who had been the lieutenant spent two years repairing them. The last panel was about the founding of the new campus in 1998, on farmland that had been a village, with a pond where the frogs were so loud in June that the first summer school had to move its lectures to the afternoon.

I stood in front of that panel for a long time. Not because of the frogs. Because of the phrase on the panel above it, which said that the department had been founded on the principle that an instrument is a promise, and that a promise is kept by the person who writes down how to use it.

Outside, the light was going. Shen Lu said, you are looking at it the way people look at a fireplace in summer. Then she said, come on, the tram stops at six, and if we miss it we will have to walk, and I am not walking three kilometres in these shoes.

The box

The inspection had one consequence that reached me: the store room on the fourth floor had to be cleared, labelled, and certified as safe before the holiday, and because I was the first-year with the least to lose, the job was mine. Shen Lu handed me a cardboard box at eight in the morning and said, third drawer, everything in it, and do not throw anything away, not even the things that look like rubbish, especially not those.

The third drawer was a museum of small failures. There was a collar for a pump we no longer owned, and the pump had been sold to a school in the countryside for the price of the freight. There was a gauge with a cracked face, a spray bottle with no nozzle, a coil of silk thread that somebody had used to hang a sample inside a furnace, a bar of soap gone hard and yellow, a tin of polish, a bundle of cotton waste, and a strip of rubber with a stripe of red paint down the middle that had once marked the edge of a bench.

Under the cotton waste there was a box of glass slides in a wooden case, and each slide had a label in a hand that was not mine, and the labels were written in ink that had faded to brown. Some of the labels were in Chinese, some in English, and one was in a spelling so strange that I could not tell which language it was meant to be. One slide

had a scratch across the middle, as if someone had dropped it and then put it back in the case rather than admit the drop. Another had shattered at the corner and been kept anyway, because a broken slide still shows what is on it.

Deeper in the drawer there was a plastic bag of soil, dry as dust, with a note inside that said only the word Guangxi and a date. There was a small bottle of a white powder labelled soluble salt, which told me nothing and which I did not open. There was a folder of specifications for an instrument that had been scrapped in 2011, and a notebook of purchase records in which every second row had been amended by hand and initialled by a person whose initials were two letters and a dot.

I worked for four hours and filled the box and started a second one. At noon the auntie who cleaned the corridor came in with her mop and her bucket, wearing a blouse the colour of weak tea, and looked at the mess and said, in the tone of a woman who has seen every kind of mess a building can produce, that the fourth floor had been like this since before she came, and that she had once found a snake in that drawer, a small one, asleep in a coil of hose, and that she had carried it downstairs in the bucket and let it go in the shrubbery by the gate.

At two o’clock I found the box that mattered. It was a document box, the kind with a hinged lid and a strap, and it had a label on the front in the same brown ink: miscellaneous, 1998 to 2011. Inside were four folders of paper, a page of photographs, and a letter in an envelope that had been opened with a knife and then taped shut again.

The letter

The letter in the envelope was four pages long, handwritten, and dated 2011. It was from a professor who had retired in 2016 and who had been, in the words of Zhou Min, the last person in the department who read everything. Gu had kept it, and had put a photocopy in the box, and had written on the envelope in his own hand: read this before you write your first paper.

Dear Gu,

Your draft arrived on a Tuesday and I read it twice, once for the pleasure of the argument and once for the pleasure of disagreeing with it. I am old enough to be frank with you, so let me begin with the field and end with you.

The ascendance of your method is not in doubt. It has been the ascendance of every method that arrived with a machine attached, and I have watched four of them arrive: the spectrometer, the computer, the database, and now the model. Each was announced as the end of an old aristocracy and each, within a decade, had produced a new one. The aristocrats of the citation index are as vain as the aristocrats of the chair, and rather more numerous. A bloc of journals now decides what counts as a question, and the

appeasement of their reviewers has become a skill that we teach, quietly, to our students. I do not say this with contempt. I say it as a man who has sat on both sides of that table, and who has signed the arbitration reports that ended two careers and made three.

What worries me is not the fashion but the consolidation of the evidence. When a field is young, its data are scattered, and a careful person can hold the whole of it in one hand and see the gaps. When a field consolidates, the data are gathered into a few places, and the gathering is done by corporations, and the corporations have shareholders. You will find that the custody of a dataset is a political question before it is a technical one. You will be offered access on terms that seem generous and that are, in fact, a lease. Read the lease. I know you will read the lease. Read it again at midnight, when you are tired and grateful, because that is the hour when leases are signed.

About the numbers themselves I have one piece of advice that I have given to every student since 1988, and that not one of them has taken. Do not ask whether the correlation is credible. Ask what would have to be true for it to be false, and then go and look at that thing. A correlative result is a rumour with a coefficient. A contextual result is a sentence about the world. The difference is not statistical; it is moral, because a rumour that is repeated in a journal becomes an asset that somebody owns, and assets are defended.

You will also be tempted, as every generation is tempted, to exaggerate the novelty of what you have done. The temptation is not vanity. It is fear, and fear is a more reliable engine. I have seen a young man write the word novel eleven times in an abstract because he was afraid that the work would be seen as incremental, and incremental work is how almost everything gets done. When you feel that fear, remember that the people who will read you are not fools, and that a modest claim, well supported, is eventually believed, while a large claim, thinly supported, is eventually quoted, which is not the same thing and is much worse for the field.

On the matter of your students, I will be brief, because you will not believe me until you have taught for ten years. The ones who impress you in the first month are rarely the ones who are still standing in the fourth year. The impressive ones are quick, and quickness is a gift that arrives early and does not develop. The ones who last are the ones who can bear to be bored, who will read the same paragraph eleven times without drama, who will confess ignorance in front of a group rather than perform competence. Look for the student who says I do not understand this yet. That word, yet, is worth more than a prize.

Their adolescence as researchers is short and their childhood is shorter, and the system now asks them to be productive before they are literate. You cannot fix the

system. You can refuse to participate in the worst of it, and you can be exceedingly careful with the word promising, which is the word that people use when they intend to extract something from a young person and give nothing back.

I have one more thing to say and it is the reason I am writing by hand. My wife died in March. The condolence letters came in a great wave and then, as these things go, the wave receded, and what remained was a small and steady current of people who did not forget, and who still write, and who mention her name without flinching. That is the whole of what I have learned about community, and it has nothing to do with research and everything to do with the conditions under which research is possible. Cultivate the people who will still write to you in the second year. Cherish the colleague who tells you the truth at the cost of an evening. Be cordial to the ones you do not like, because you will need them on a committee, and because cordiality is cheap and its absence is expensive.

If you take only one sentence from this letter, take the one I have never managed to follow myself: do not delude yourself that the work is your monument. The work is a wall that other people will build on. If it is straight, they will build well. If it is crooked, they will build crooked, and they will not know why, and they will curse the wall and not the mason. You will not be there. That is the point of a wall.

I have read your manuscript three times now and I have written my remarks in the margin in pencil, because pencil can be amended and ink cannot. There is one paragraph, the one on page nine, that I would cut entirely. You will know which one when you read it again. If you cannot bear to cut it, put it in a drawer for a month and then read it as if it had been written by a stranger. That trick has saved me from publishing at least four foolish sentences, and I am told it has saved my readers from reading them.

Give my regards to your wife, whom I have never met and whom I already admire, since she has evidently kept you alive and reasonably clean. Write to me when the paper is out. I do not promise to read it quickly. I do promise to read it.

Yours, with affection and a certain amount of impatience,

Shao

P.S. Tell the first-year who is doing your counting that the machine will lie to him in a pleasant voice, and that the counter will tell him the truth in an unpleasant one. He will not believe you either.

The funeral

Shao died in June, at eighty-one, and the notice went up on the department website on a Friday afternoon with a photograph that had been taken in 1994. The funeral was in a hall by the east gate, and the department sent a wreath, and Gu went, and I went with him

because Zhou Min said that somebody young should go.

It was not a religious ceremony, or rather it was two ceremonies at once, which is common here and which nobody finds strange. There was a photograph, and flowers, and a man from the university who read a short speech about service, and then there was a moment in which the family stood by the door and we filed past them and said the same four words to each of them, and each of them said the same four words back. Nobody cried, except a woman in her fifties whom nobody seemed to know, and who cried quietly and then left before the end.

I thought about the letter in the box, and about the sentence in it that I had copied into my notebook: cultivate the people who will still write to you in the second year. I thought about mourning as a practice rather than a feeling, and about the fact that the practice is what holds a community together when the feeling has gone quiet. There was no priest and no chant, and the only music was a recording played from a phone through a speaker that buzzed on the low notes.

Gu said, on the way back, that Shao had supervised forty-one students and had remembered all of them, and that at the end he had been unable to remember his own phone number. He said that the department had been his whole life and that the department had given him a room, a pension, and a wreath, and that this was, on the whole, a fair exchange, and that he was not sure he wanted it. Then he said that he was sorry, that he was tired, and that we would talk on Monday.

I did not know what to say, so I said that the letter in the box had changed the way I read papers, and Gu said, good, and then, after a while, that Shao would have liked to hear that, and that he would also have said that a letter is not a result.

The handbook

In June, before the exams, Zhou Min asked me to write a handbook for the first-year who would take my place. Keep it under six pages, she said, write it in the second person, and do not lie about anything. I wrote nine pages. This is most of it.

Buy a wardrobe that fits under the bed, two pairs of pants that dry overnight, and a purse with a zip, because the pockets of a lab coat are not a safe place for a phone. Send a postcard home in the first month, when you still have things to say. Collect your mail from the box by the gate, because the department will send you a paper letter exactly once, and it will be the letter that tells you whether you have a place.

Learn the names of the instruments before you learn the names of the people. The gauge on the wall of the dark room is not ornamental; it tells you whether the line is at pressure, and if you ignore it you will eventually hear a sound like a pistol and spend an afternoon on your knees looking for a piece of a liner that has gone under a cabinet.

The magnet on the bench is stronger than it looks and will eat a screwdriver. The mercury lamp takes eleven minutes to reach full output and you must not look at it, not even once, not even to see whether it is on. The tiny propeller inside the stirrer is the only moving part in the room that has never failed, which is either a testimony to German engineering or a hint about how rarely we use the stirrer.

The machine in the basement is a microscope with a nucleus of copper and a column that must never be opened by you. Its orbit is fixed, its stage is delicate, and its software is a peculiar and problematic thing written by a physicist who left the company in 2009. There is an observatory on the roof of the physics building that nobody uses, and a pyramid of empty nitrogen containers behind the loading bay that everybody uses as a landmark when giving directions.

Learn the data before you learn the software. A quantitative claim requires a rating scheme that you can defend, and a rating scheme requires a methodology that you can write down in a preface of no more than one page. If your perspective is partial, say so in the preface; a partial view that is declared is worth more than an outlook that pretends to be complete. Do not plunge into a model before you have paved the path to the pit where the labels are buried, and do not populate your table with hypothetical rows to make the column look plenty.

Learn the people. The gentleman who runs the workshop can lend you a tool and will invite you to return it clean. The guy in the next group is a skeptic and a pessimist, and he is usually right, and he will inspire you more than the optimist who tells you that everything is fine and nice. The spokeswoman for the instrument company is a skilled and tactful intermediary who will interrogate you about your sample before she agrees to insure the shipment. The reporter from the campus paper will invite herself to your lab and will reproduce your words with ironically literal accuracy, so choose them with sincerity.

Learn the literature. Read the preface of every book you borrow, because the preface is where the author tells you what the book cannot do. Keep a glossary of the words your group uses peculiarly and update it yearly, or the newcomer will assume that nucleation and nucleus mean the same thing, which they do not, and the error will persist for a decade. Read poetry in the original if you can, and if you cannot, read it in translation and acknowledge the loss. A noun is a pledge; a verb is a promise; an exclamatory sentence is a confession. Simplify your spelling only where the journal allows it, and verify every quotation against the source, thereby avoiding the shame of a correction that arrives shortly after publication.

Learn the failures, because there will be a lot of them and they are not tragically distributed. You will undergo a stretch of stagnation in which every attempt yields nil. You will tumble into a tangle of redundant measurements and resent the machinery. You

will overcome a distraction and then succumb to another. You will tremble before a presentation and mute yourself in the discussion. You will rip a page out of the notebook and wish you had not. You will wake at three and reckon the odds of finishing. You will persist, or you will not, and neither outcome is a verdict on your intellect.

Do not neglect the small things: tighten the cap, tick the box, stitch the tear in the bag, sweep the shade off the lens, soothe the irritated skin with cream, sip the water before it spills on the keyboard, spoonful by spoonful if you must. Squeeze the tube from the bottom. Stalk your own habits and terminate the worst of them swiftly. A tidy bench is not a virtue in itself, but a messy one is a tax on everyone who works near you.

Do not smuggle a sample out of the country; do not sneak a colleague into the database; do not seduce yourself with the notion that a shortcut is a method. Do not inflict your mood on the group. Do not retaliate against a reviewer in public; retort in the response letter, briefly, and rebut the argument rather than the person. Do not topple a colleague to climb; the climb is long and the corridor is narrow. Do not oppress the quiet student with your certainty; certainty is cheap and silence is expensive.

Read the proclamation of the kingdom you work in, which is the safety manual, and obey it with the obedience of a soldier, because the manual was written by someone who saw the accident that produced it. Wear the vest. Wear the gloves. Tolerate the mask. Yield to the audit. Uphold the standard even when the supervisor does not, especially then.

And last: remember that the laboratory is a haven for the curious and a prison for the proud. The door is open until ten, the lamp is warm, the chair is broken, and the person who sits in it will hand you a counter and ask you to count. Count.

What I kept

I kept the counter, both of them, the plastic one and the tally in the back of the notebook. I kept a ball of twine from the store room, a feather from the hill, a leaf pressed flat inside a glossary, a postcard from the shop by the west gate, and a roll of tape that had lost its stickiness. I kept the train ticket from January and the receipt for the spectroscopy fee, and I kept them for no reason that I can defend, which is the only honest reason for most of what a person keeps.

The wedding was in May, in a village an hour southwest of the city, and I went because my cousin was marrying a woman from a family that kept a winery and a fish pond and a hall with a stage. It was a loud wedding. There was a brass band that played a tune from a film, and a man with a typewriter who wrote the guests' names on red paper in a script so beautiful that people queued to watch him, and a driver who had been hired to bring the bride's relatives from three counties and who fell asleep at the table with a

plate of cold chicken in front of him. The food came in eleven courses. There was a whole fish, a bowl of soup with a chicken in it, a plate of pale dumplings, and, at the end, a sweet rice pudding that the children fought over. My grandmother's sister, who was ninety-one, told me that the band was too loud and that the bride was too thin and that the soup was the best she had had in a decade, and she said all three in the same tone, and I understood that this is what it means to be tolerant of a world that keeps changing its menu.

On the way back I took a detour to a factory that made ceramic powder, because Gu had arranged a tour for the group and I had missed it. The plant was a hectare of sheds and a chimney and a yard full of pallets. The manager, who was proud of his line and had every right to be, walked me through the whole process: the mill, the kiln, the sieve, the magnetic separator that pulled iron out of the slurry, the packaging line where a machine filled sacks at a rate of eleven a minute, and the laboratory at the back, where a woman in a mask measured the particle size distribution three times a day and entered the numbers in a ledger. He said that the biggest problem in his industry was not the technology but the weather, because the powder absorbs water and the humidity in June can add three per cent to the weight of a wholesale order, and three per cent on a wholesale order is his whole margin.

We drove back past the refinery, which is a city of pipes with a flame on top, and past a radar installation on a hill, and past a field of solar panels that had been a farm when I was a child. The driver said that the road had been widened twice in ten years and that the traffic was worse both times, which is a paradox that every city in the country could confirm. He said that his son had gone to a polytechnic and had a job in the vehicle industry and was doing better than his father, and that this was the point of everything, and then he turned the radio on and we listened to a football match in a language I did not understand.

In June, before the exams, I went up the hill one more time, alone. It was hot and the misty air made the far ridge look like a rumour of a ridge. There was an owl in the pine wood, or rather there was the sound of an owl, and I stood still for a long time and did not see it. There were deer tracks in the mud by the spring, and a place where a herd of goats had bedded down for the night, and a butterfly on a thistle, and a beetle that was so geometrically perfect that I thought, for a second, that someone had placed it there. I thought about hibernation, which is the most sensible strategy anyone has ever devised, and about the wildlife that survives a place only by being invisible to it, and about how much of the hill I had never seen and would never see.

The last thing I did before the exams was to take the box back to the store room. It was heavier than when I had found it, because I had added a folder of my own: the printing of the table, the log of the model, the printed minutes of our own meetings, and

a page on which I had written, in the formal and slightly ridiculous prose that the archive seems to require, a short account of what the group had done that year, who had done it, and what had gone wrong. I wrote the date at the top and my name at the bottom, and then I wrote the names of Zhou Min and Shen Lu and Xu, in that order, because that is the order in which they had taught me.

I put the box on the shelf, and I stood there for a moment in the dim light of the store room, and I felt the particular satisfaction of a thing put back in its place. It was not a discovery and it was not a result. It was a record, and a record is what remains when the excitement has gone, and the excitement always goes, and the record is what the next person will have.

The corridor was quiet. Somewhere above me a pump was running, and the floor carried the vibration the way a drum carries a finger. I turned off the light, and the dark in the store room was total and pleasant, and I walked out into the corridor where the notice board still had a peg missing and the circular was still taped to the door with its appendix torn off, and I thought that I would come back in September and would notice neither of them again, and that this is how a place becomes yours: not by being understood, but by being unremarkable.

Outside it was still light. The summer term was ending, the campus was emptying, and the man at the gate was reading a newspaper with his cap on the desk beside him. I bought a bottle of water and sat on the low wall by the bicycle shed, and I watched the light go off the top of the laboratory building in stages, floor by floor, as the cleaners moved down from the fifth.

That was the year the algorithm arrived in my life, and it arrived the way most things arrive: not as a revelation but as a piece of equipment, with a manual, a licence, and a queue. What I learned was not how to be clever. It was how to be slow, how to write things down, how to distrust a beautiful number, and how to tell the difference between a picture and a measurement. I learned that the machine will always be more patient than I am and less honest, and that this is exactly why I am still needed. I learned that the counter clicks, and that the click is a promise, and that a promise is kept by the person who writes down how to keep it.

The light went off on the fifth floor. Then the fourth. Then the whole building was a shape against a sky the colour of weak tea, and I finished the water, and I went home to pack.

The last entry

I made the last entry in the log on the thirtieth of June, at the desk by the window, with the fan on and the door open, and it was not about an experiment. It was about the year, and I wrote it in the flat register of the log because I did not trust myself to

write it in any other.

Entry, 30 June. The counting is finished. Four thousand two hundred particles, eleven samples, one distribution with a tail I cannot explain and will not hide. The model agrees with me on the median and disagrees on the tail, which is where the interesting particles are, and the disagreement is systematic, which means it is a fact about the model and not a failure of the sample.

I have been eager all year to be capable of something, and I am now capable of one thing: I can tell, from a photograph, whether the contrast is genuine or illusory, and I can tell it quite often and seldom with perfect certainty, and the second half of that sentence is the part that matters. I can discern a clog in the line from the shape of the pulse in the pressure gauge. I can distinguish a degraded sample from a contaminated one, namely by looking at the edge and not the middle. I can identify the signature of a manipulated figure, and I can do it intuitively now, which is to say that the inference has become fast and I no longer know whereby it is made.

What I cannot do is foresee which experiment will fail for a reason that is interesting. The fate of a sample is decided in the first ten minutes, by a draught, a dust, a delay. I have watched a promising batch degrade overnight because a lid was not tight, and I have watched a worthless batch survive a week in a drawer, and the lesson I draw is not a lesson about chemistry, it is a lesson about attention: tireless attention is worth more than talent, and timely attention is worth more than both.

The results of the project will be moderate and the paper will be short, and there is a part of me that resents this, and a larger part that is glad. A moderate result that survives scrutiny is worth a spectacular one that does not, and the spectacular ones are often produced by someone who is in a hurry and unwilling to be bored. I have been bored this year in a way I had not known was possible, and I have come to regard boredom as a symptom of health, like a slow pulse.

There is a tendency in me, and I notice it narrowly and honestly, to treat a difficulty as a verdict on my intellect, and to react to a setback as if it were a sentence pronounced by a court. This is insanity, and it is common, and it is curable only by work: by doing the small, straightforward, repetitive task in front of you until the feeling passes. I have done the small task several thousand times this year, and the feeling has passed several thousand times, and I expect it to return tomorrow, and I no longer mind.

I am indebted to three people and I will name them here so that the record is complete. To Zhou Min, who taught me that a column has a meaning and that a meaning can be wrong. To Shen Lu, who taught me that a picture is not a measurement and that a measurement is a promise. To Gu, who taught me that the boring half of research is the half that protects the interesting half, and who never once asked me to be brilliant and

asked me every week to be careful.

The hall was empty when I finished. The fan turned. A moth battered the lamp above the desk, and I let it out through the window, and it went upward into the dark and disappeared, and I thought that this is the whole of what I have to say about science at the end of my first year: it is a way of being careful in public, in front of people who will check, and the care is the only part of it that belongs to you.

The report

We submitted the paper in July to a journal that Gu described as small, serious, and slow. The report came back in six weeks, which is fast for a journal that is slow, and it was three pages of the most useful prose anyone has ever written about my own work.

The reviewer began with a paragraph of appreciation, which I have since learned is conventional and which was, I think, sincere. The data were extensive, the subject was of real necessity, the question was worth asking, and the answer was, in the reviewer's word, fruitful. I read that paragraph four times and then I read it to Zhang Ye, who listened to the whole thing and said that it sounded like a school report, and that I should read the rest before I got too happy.

Comment one. The authors claim that the discrepancy is systematic. The word systematic is doing a great deal of work here, and the manuscript does not say what it excludes. The authors should state, in the preliminary section, what would count as a counterexample, and should consider whether the effect could arise from the preprocessing rather than from the model. A precautionary paragraph of two sentences would prevent a great deal of speculation in the literature.

Comment two. The scope of the paper is narrower than the title suggests. The title promises a general statement about electrolyte additives; the data cover four families. This is not a fatal objection, but it is a matter of honesty. Either widen the data or narrow the title, and if the authors narrow the title, they should also narrow the conclusion, which currently claims more than the evidence can carry.

Comment three. The methodology is described with care, and the appendix is a model of its kind. I would ask only that the authors make the counting protocol explicit: how many particles, chosen how, by whom, and with what instruction to the counter. The strength of the paper lies in its manual labour, and the manual labour should be visible. A reader who cannot reproduce the count cannot check the claim, and a claim that cannot be checked is not yet a result.

Comment four. The statistics are adequate and the language is not. The wording of the abstract is convoluted in three places, the same adjective appears eleven times, and the paper is monotonous in its rhythm, which matters more than authors usually believe,

because a monotonous paper is a paper that gets skimmed. I recommend a format that separates the finding from the method, and I recommend that the authors browse the last two years of this journal to see how the wording of a title is normally done.

Comment five. The figures are gorgeous and one of them is misleading. Figure four shows a gold curve against a grey background, and the curve is drawn with a smoothing that is not described. Either describe the smoothing or remove it. The same figure would be clearer if the axis began at zero, though I accept that the effect would then be less dramatic.

Comment six. The discussion contains a myth that should be removed: the sentence about a paradox at the heart of the method is not a paradox, it is a mixture of two claims, and the implication the authors draw from it is illusory. This is the kind of sentence that a fictional version of science would produce. A real one should say plainly that the model is indicative and not definitive, and that its perception of the tail is imperfect.

The recommendation was minor revision, which Gu said was the best outcome we could have expected and which Xu said was an insult, and which I now understand to have been both. We spent three weeks on the revision. We cut the title, we cut the paradox, we added the protocol, and we added a paragraph that said, in the plainest words available, that a correlation of 0.61 is not a prediction and that the difference between the two is the whole of the paper.

I learned more from those three pages than from any lecture, and what I learned was not technical. I learned that a reviewer is a reader with a privilege: the privilege of being rude in writing and accountable for it. I learned that criticism, when it is specific, is a form of generosity, and that when it is vague it is a performance. I learned that the most valuable sentence in the report was the shortest one, which said: show the count.

The photographs

The last folder in the box held photographs, thirty or forty of them, in a paper envelope that had been opened so many times that the flap had torn. I looked at them on the floor of the workroom with the fan turning, and I did not recognise a single face, and by the end I felt that I knew them all.

There was a photograph of the department in 1958, in front of a building that no longer exists. Nine men and one woman, in dark coats, standing in a line. The woman is holding a baby. The man on the left has the bearing of a colonel, and the note on the back says that he had been one, and that he had spent the war as a knight of the supply depot, which I took to mean that he had moved coal and nails and had kept a ledger in a cave. There is a Catholic chapel in the background of another photograph, with a cock on

the roof, and the note says that the chapel had been a store house by then and that the cock had been gold and had been stolen in 1966.

There is a photograph of a cowboy film being shown on a sheet in the courtyard, with forty people sitting on the ground, and a dog in the front row. There is a photograph of a wedding in the hall, with a barbecue pit in the yard and a sign that says BBQ in English, which somebody must have thought was modern. There is a photograph of a baby in a basket on a bench in the laboratory, with a note pinned to the basket that says, in Chinese, do not move, the mother is counting.

There is a photograph of the old building in winter, with snow on the roof and smoke from a chimney, and one of the same building in summer with the windows open and a bicycle leaning against the wall. There is a photograph of an antique balance in a glass case, and a photograph of a young man in a white coat holding a slide up to the light, and on the back of that one, in pencil, the words: my first single crystal, 1974.

I put the photographs back in the envelope and the envelope back in the folder and the folder back in the box, and I sat on the floor for a while with my back against the shelf. It was the end of the afternoon, and the light in the workroom had gone the colour of weak tea, and the fan was turning, and somewhere down the corridor a radio was playing a song that I had heard in June and would hear again in September and would not remember in October.

I thought: all of this machinery, and all of this paper, and the whole point of it is that a person writes down what happened so that another person can find it. That is the entire chain of civilisation, and it is held together by people who are willing to be bored in public. The honour of the profession, if it has one, is not in the prizes and not in the limelight; it is in the ledger, in the date, in the initials at the bottom of the page, and in the fact that somebody in fifty years will open the box and find the number and know that it was true.

The agreement

In July the university and the company signed an agreement, and Gu asked me to read it, because I was the only person in the group with nothing better to do. It was eleven pages of clauses, and I read it on the balcony with a fan and a pencil, and it taught me more about the world than the whole of the spring.

Clause one defined the parties. The company was a legal person with a registered office, a scope of business, and a network of subsidiaries, and the university was a public institution with a legal representative and a seal. Clause two defined the data: all records, in any format, whether existent or hereafter created, including the metadata, the annotations, and the derivatives, and including any copy made for the purpose of backup. Clause three granted a licence, non-exclusive, non-transferable, and

restrictive, to use the data for academic research only, and it specified that the licence did not extend to any commercial use, any sublicensing, or any transfer to a third party, and that any breach would terminate the licence automatically and without notice.

Clause four dealt with publication. The company asked for a copy of any manuscript thirty days before submission, and Gu asked for that clause to be amended to fifteen, and the company agreed, and the amendment was initialled by both parties in the margin. Clause five dealt with the protection of confidential information, and it was written in the language of a fortress: no disclosure, no discussion, no reproduction, and, in one memorable phrase, no admission of the existence of the data to any person not a party to the agreement. Clause six dealt with liability, and it was, as far as I could tell, a masterpiece of the art of saying nothing: neither party would be liable for any indirect loss, any loss of profit, any loss of expectation, or any damage arising from a force outside its reasonable control.

Clause seven dealt with the people. Each party would nominate a mediator, and the two mediators would meet quarterly to review the working of the agreement and to resolve any discord before it became a dispute. Clause eight provided for arbitration in the city, under the rules of a chamber I had never heard of, and it said that the decision of the arbitrator would be final and binding on both parties. Clause nine dealt with termination, and it listed seven grounds, of which three were so vague that they could have been used against us at any time, and clause ten dealt with the consequences of termination, which were, in brief, that we would delete everything and certify the deletion in writing.

I read clause ten three times. I thought about the nine months of counting, the eleven samples, the four thousand two hundred particles, and about the fact that all of it could be made illegal by a letter. I thought about the clause that said no admission of the existence of the data, and I understood, for the first time, that a contract is not a document about the past. It is a machine for predicting the future, and the predictions are always pessimistic, which is why they are usually right.

I wrote four pages of comments, which Gu read, and rewrote, and sent, and the company accepted two of them, which Gu said was a good result, and which I found difficult to celebrate, because the two they accepted were the two that mattered least.

Notes on the back of a poster

At the poster session in September I ran out of paper and wrote on the back of my own poster, in pencil, in the margin. These are the notes, more or less as they were.

An apparatus is a hypothesis made of metal. An ancestor is a hypothesis made of memory, and usually the better made of the two. To be able to say what a tool cannot do

is the whole of acceptance, and acceptance is not agreement. A field that confuses the two will spend a decade amplifying its own errors and calling it progress.

An addictive tool is not a tool; it is a habit with a plug. An adventurous hypothesis is cheap and an adventurous measurement is expensive, which is why the second is rare. Aggression in a seminar is fear with a loud voice, and the alleviation of that fear is the only real work of an instructor. An animated room needs one person who is genuinely interested; an antagonism needs two who are competing for the same grant.

Arousal of interest is easy; the maintenance of it is the whole of teaching. An assault on a reputation is cheaper than an experiment, which is why it happens more often. An auxiliary pump, a spare fuse, a second copy of the data: the whole of engineering is the associate of the main thing. To avert a disaster you need a person willing to be boring at the right moment. Awareness without a habit is only a mood.

To beckon somebody into science you must show them a result they can check with their own hands. A beneficial error changes your method; a harmful one changes your story. A bonus is a bribe with a deadline. Break the rule that nobody can explain, then ask why it existed. A buck spent on a label saves ten on a replacement. A burglary in a laboratory is usually an inside job and always a failure of the register.

The border between a laboratory and a kitchen is thinner than either side admits: the same measurements, the same hygiene, the same brand loyalty. A cake is a titration with a charm of its own, and a candy is a crystalline structure that a child can evaluate. There was bacon in the fridge on Sundays, cereal in the cupboard, and a cookie jar the technicians raided at four, and a creamy yogurt from the dairy counter that was, I remain convinced, the best thing in the grocery on the corner. On the last Friday of the month four of us went to a restaurant by the canal and ate steak and salad and a plate of pasta, and drank juice because the pupil who had just joined us was nineteen, and finished with a bowl of jelly that shook when the table shook.

To cease learning is to begin explaining. A chronicle is not history; history is a chronicle with an argument. Citizenship in a laboratory means cleaning the thing you did not dirty. Clarification is the most underrated form of courage, and a clerical error is a scientific error that nobody has had the honesty to call scientific. The climax of an experiment is not the result; it is the moment when you know what the result means. A cloud on the image is not a cloud in the sample. A coach who has never competed is a comic figure, and a combative one is worse.

Cognition is slow and recognition is fast, and the whole of training is the transfer of the second into the first. The coinage of new terms is the cheapest form of originality. Compensation for a mistake is not an apology. A compilation is not a contribution. Compulsory attendance is a tax on the willing. Concentrate on the sample in front of you. A concurrent claim is not a prior claim, though it feels like one at three

in the morning. Confinement to a single method looks like discipline and is usually laziness.

Conscience is what you have left when the supervisor has gone home. A consequent error is easier to forgive than a careless one. Consideration for the next user is the whole of laboratory ethics. To constrain a claim is to strengthen it. Consultation is not consent. Continuous attention is a method; continuous improvement is a slogan. A contributor who is not credited is a ghost with a grievance. A convention is a solution everybody has agreed to stop arguing about. A conveyance of an idea is not the idea.

A corrupt dataset is more dangerous than a corrupt author, because a dataset cannot be ashamed. To cripple a project you do not need a disaster; you need three small delays. To crouch is sometimes the only way to see under the bench. Cruelty to a student is remembered for thirty years; kindness is remembered for three. The culmination of a project is a sentence that a stranger can check.

Dangerous is not the same as serious; the most dangerous thing in the building is a confident person with a deadline. To dazzle a committee is easy and to inform one is the work. Death is the only deadline everybody respects and a poor manager. A decoration on a wall is a claim about values. To deplete a budget is easy; to deplete a prejudice is the work of a lifetime. The destination of a sample is the incinerator, and everything before that is documentation.

Detachment is a virtue in a reviewer and a vice in a friend. A restriction that nobody can explain is a decoration. A detriment to the field is usually a benefit to somebody's career. A developmental view of a method is more useful than a defensive one. To devour the literature is not to read it; it is to stop being surprised by it. A direct answer is a gift. A disabled instrument is a lesson in humility. A model that cannot discriminate between two explanations is a decoration. A disgusting smell is a message from a compound that would like you to leave. A disruption to a schedule is a test of character. A disturbance in a signal is a question. A donor with an agenda is a client. To drag a dataset into a new format is to learn what the dataset is. A drastic change is usually a desperate one. A dreamy afternoon is not wasted; a dreamy year is. A drunkard at the gate is a reminder that the campus is not a monastery. A dweller in a city learns to sleep through noise; a dweller in a laboratory learns to sleep through doubt.

An earthquake is a reminder that the ground is a hypothesis. An eccentric habit is the price of an original mind. An economist knows the price of everything, which is also true of a citation index. An editorial is an opinion with a press behind it. An efficient meeting is one that did not need to happen. To elapse is to be forgotten slowly. The elevation of a platform is a small thing until you are carrying a dewar up it. An elite is a group that has agreed on its own standards. To emerge is to be seen, and to be seen is to be judged. An emission is a confession in chemistry. An employee who understands

the whole line is worth three who understand their station. To enable a student is to give him a tool and the time to break it. An encouraging word at the right moment is worth a scholarship at the wrong one. To enforce a rule you did not explain is to teach contempt for rules, and an enhancement of an image is a hypothesis about what the image should look like.

To enlist help is a skill; to accept it is a virtue. An argument may ensue from a footnote. Entertainment in a laboratory is whatever is not the experiment. An epidemic of bad data spreads faster than a good method. The equator is a line that exists only on maps, and so is a deadline. To eradicate a duplicate is easy; to eradicate the habit that made it is not. An erosive process is invisible until the part fails. An escalation begins with a sentence nobody wanted to say. An eternal question is usually badly posed. An evacuation is a plan everybody has read and nobody has practised. Expectancy is a number that flatters; experience is a number that does not. To explore is to be willing to be wrong in public. Exposure to a new field is uncomfortable for exactly as long as it is useful.

A facet is not the whole. A famous result is often one reported early rather than well. A fascist argument is any argument that cannot survive being questioned, and it appears in laboratories as often as in parliaments. Fate is what we call the part of the outcome we did not measure. A favourite error is more dangerous than a strange one. A feminist reading of a laboratory would notice who washes the glassware. To flourish, a group needs one person who remembers the birthdays. Foil is cheap, and half of experimental design is cheap foil. A forecast is a hypothesis with a date on it. A forest is a laboratory that has been running for four hundred years. Forth from the meeting came a decision that nobody remembered making. The fragility of a result is proportional to its beauty. A franchise is a promise with a logo. Fulfilment is not a feeling that arrives at the end; it is a habit built in the middle.

Gadgetry is a laboratory looking for a budget. A gangster and a reviewer have this in common: both want a share of what you have. Garlic in a shared kitchen is a lesson in negotiation. A geometric argument is one that can be drawn. To glide is to move without friction, and no institution has ever managed it. Grace is the ability to be wrong without making it worse. To state a thing graphically is to make it undeniable. Gravity is the only force that never takes a holiday. Green is the colour of a sample that has oxidised and of a student who has understood. A grievous error is one that a form could have prevented. A guarantee is a promise with a lawyer attached. Guilt is a useful signal and a terrible manager.

Halfway up the stairs I always remember what I forgot. A handicap is a condition of the environment, and most laboratory handicaps are stairs. To harden a surface is easy; to harden a habit is not. A harmonious group is one where the arguments are about the

work. To heal a dataset you must first admit that it is sick. A heartfelt apology is short. The helicopter on the roof of the hospital and the hemisphere of the sample stage are both machines for looking down. Heroism in a laboratory is a person who stays late to check a number nobody will ask about. A hike is a meeting without minutes. Homework is the part of education the system cannot see. The horizon is a limit that moves with you. A housewife in 1960 and a doctoral student in 2026 have the same complaint: nobody sees the maintenance. A hug is a unit of management that no committee has ever measured. Humanity is easy to use as a word and hard to practise at four in the morning. A humourous remark can save a meeting and end a career. To hurl a number at a problem is not to solve it. Hydrogen is the simplest element and the most difficult to store.

An ideological commitment to a method is the beginning of the end of that method. An idol is a person you have stopped listening to. An imaginative hypothesis is worth ten careful ones, and one careful one is worth ten imaginative. To impart a method you must be able to do it slowly in front of someone. An imposing apparatus is often a compensating one. Imprisonment of a data point in a wrong column is the most common crime in science. An inauguration is a promise made in public. An inclination to believe is a useful instinct and a dangerous habit. To incorporate a criticism is harder than to answer it. An individual result is a rumour; a distribution is a fact. An indulgent supervisor produces dependent students. An informative figure is one that would survive being printed without its caption. An inhabitant of a laboratory knows where the fuses are. To initiate a student is to give him a problem he can fail at safely. Innovation is a word used to sell the ordinary and to hide the expensive. Instinct is a memory that has forgotten its source. An instructor who cannot be interrupted is a lecturer. An intensity of effort is not an intensity of thought. An intercourse of ideas requires two people willing to change. An interviewee is a person performing a version of himself. An intricate method has more places to hide an error. An intruder in a dataset is a value that does not belong and does not announce itself. An invasion of a field by a method is usually followed by a retreat. To invest in a student is the only investment with no exit.

To jog the memory, write the date. Jelly and juice and a journey are the three things a child remembers about a summer. To kill a hypothesis you need a measurement; to kill a career you need only a rumour.

A lack of evidence is not evidence of absence. A learner who asks why is worth ten who ask what. A legislative process is slow because it is designed to be. A liberal education teaches you what your own field cannot. A lip on a beaker is a small kindness. A lively seminar is one in which somebody says I do not know. A local rule is often a global mistake. Lodging near a laboratory is a decision you regret on the day you move out. Loneliness in a crowded laboratory is the most common occupational disease. A

loophole is a rule with a door in it. A lover of the subject is the only person who reads the appendix. A lung full of solvent is a career decision.

A major result changes what other people do. A malignant error is one that spreads. A managerial solution to a scientific problem is usually a meeting. Manipulation of an image is a forgery, and a forgery with a good intention is still a forgery. To manufacture a result is easier than to manufacture a part. Marketing is the art of making a weakness sound like a feature. A marvellous instrument is a joy and a distraction. A massage of the shoulders is a better reward than a certificate. Maternal care and laboratory care are the same skill. Maturity is the ability to be bored without becoming bitter. In the meantime, do the labelling. A mechanical answer is not an answer to a human question. A metabolic rate and a publication rate are both measures of consumption. A microscopic defect becomes a macroscopic failure. Might is not right, but it is often the reason a rule exists. A milestone is a stone, and stones are set by people who walk. A million is a number a committee understands. A minister cuts a ribbon and a technician fixes the machine. A miraculous result is usually a decimal point in the wrong place. A monumental error is a small error nobody corrected. Morality in research is not a code; it is a set of habits. A motive is not an explanation. To multiply a result by ten is easy; to divide a claim by ten is the work. A municipality that builds a laboratory and forgets the maintenance has built a ruin. A muscle is a memory.

A native speaker of the laboratory knows where things are. A navy and a laboratory both run on logistics. A noble aim is not an alibi. A nominee is a person chosen by somebody who was chosen by somebody. North of the building the wind is stronger and the samples dry faster. A noticeable error is a lucky one. A nurse who has walked eleven kilometres is worth more than a minister who has cut a ribbon.

An obstruction is a decision made by somebody else. Occupancy is a poor measure of output. An oceanic dataset is deep and cold. An Olympic effort is rarely repeatable. An ongoing project without a deadline is a hobby. An operative rule is one followed when nobody is watching. An organ is a system that does not know it is a system. Ornamental equipment is a confession. Outdoors is where the sample came from. An outstanding result survives the third attempt. To overthrow a method you must first use it well.

A pace is a choice and not a fate. Pain is a signal and not a verdict. A palm can hold a sample or a phone, and the second will ruin the first. A paradoxical result is one that has not been explained yet. Parental expectations and supervisor expectations are the same weight in different clothes. A partisan argument is one that has already decided. A passenger in a project is a person who is carried. Peaceful is not the same as quiet. Perfume in a laboratory is a warning. Permanence is an illusion of archives. A perpetual student has found a way to be paid for curiosity. Pesticide in a laboratory fridge is a joke with a body count. A pharmaceutical company and a laboratory want the

same thing and answer to different people. A pilgrim and a tourist differ in what they carry home. A plaintiff and a defendant differ in who has the invoice. A plea is a request with a strategy in it. The polar region and the palm of your hand are both places where the temperature matters. A portable instrument is an instrument that will be dropped. A position is a place to stand, not a truth to hold. A powerful method is dangerous in careless hands. A predatory journal is a predator with a website. A preparatory step is the step everybody skips. To preserve a sample is to preserve its history. Prevention is a result you cannot publish. Progress is a word that hides a decision. A prominent person is a person with a good publicist. A provocation is an invitation to waste an afternoon. A psychological explanation is a description and not an excuse. A pupil who writes everything down twice will be a good scientist and a tired one.

A query is a question with a form attached.

Radioactivity is invisible, which is why the badge exists. The railroad runs north and then west and then south again. A receptive audience is a gift. To recover a dataset is to learn how it was made. A recurring error is a design error. To refresh a memory is to rebuild it. A regimental rule outlives its war. A regrettable sentence is written in a hurry. To rein in a claim is the mark of a mature author. A relation is not a cause. A relentless reviewer is a gift. A reminiscence improves with each telling. The rendering of an image is a decision and not a fact. Rent is the price of a room and tenure is the price of a question. To repel a criticism you must first understand it. A request is a question that expects a yes. A residence near a laboratory is a way of life. A resolute person has decided what to be bored by. Respective duties are the whole of a rota. A restaurant by the canal is where the group argued best. A reunion is a test of whether the group was real. To reverse a decision is cheaper than to defend it. A revolt begins with a broken promise about authorship. A ride home is a conversation with a deadline. A rigid protocol will break at the worst moment. A rivalry is a relationship with a scoreboard. A robotic routine is one a machine should do, and should do badly until it does it well. A roller flattens and a committee flattens with more talk. A rosy forecast is made by the person who benefits from it. To rouse a room you need a question nobody can answer.

A salutation costs nothing. To scare a student into silence is easy and expensive. A sceptic has decided to be unpopular for your benefit. A scholastic habit is a habit of quoting instead of thinking. The sea is the largest control experiment. A secret in a laboratory lasts until the next visitor. A seeming success is the most expensive kind. A sensational result will be retracted. A sentimental attachment to a method is why it survives too long. A showcase is a room where nothing is being made. To sicken a culture you need one person rewarded for shouting. A signal is a difference somebody has decided

to notice. A slave to a schedule has forgotten why the schedule exists. A solicitor reads the clause you skipped. A soul is a word for the part of a person a laboratory cannot measure. In spite of everything, the work gets done. A stagnant field is one where the questions have become polite. A steak and a salad and an argument: that was the last supper of the term. A stimulant is a loan against tomorrow. A stressful week is a week with two deadlines and one instrument. A studio is a laboratory with better light. A stunning image is a hypothesis about beauty. To submerge a sample is to change its surface. A subtlety is the difference between a result and a discovery. A successful experiment answers the question you asked. A suggestive result is a question in disguise. The sun on the windowsill will find your sample. A suppressive rule has forgotten its purpose. A surgeon and a microscopist both work with a small field and a steady hand. Surveillance is a camera with a filing system. Suspense in a laboratory is a centrifuge taking too long. To swap a method is to admit the old one was a habit. A symbolical gesture is cheap.

A temporal error is a date in the wrong column. A territorial dispute in a laboratory is a dispute about a shelf. Thanksgiving is not a holiday here, but the sentiment is universal and it is called a footnote. To be thirsty at three in the morning is to be alive and unprepared. A threshold is a place where a rule changes. Throughout the year I kept the log, and the log kept me. Today is the only day the experiment can be done. A trade is a conversation with an imbalance in it. A transient effect is the one that gets published. The transmission of a habit is faster than the transmission of a method. Transportation of a sample is a test of the sample and of the person. A treasure is a thing you keep for no reason you can defend. A trillion is a number a politician uses and a scientist checks. A troop moves on its stomach and a laboratory moves on its labels. A tribe of researchers is held together by a shared objection.

A unanimous decision is often one nobody has read. An unemployed method is a method without a question. A union of two datasets is not a synthesis. To unleash a model on a corpus is to discover what the corpus is.

A vaccine is a promise made to a stranger. A variant is a version with a difference that matters. The vicinity of a laboratory is where the coffee is bad and the conversation is good. A viewpoint is a position with a history. Violence in an argument is the point at which the argument has stopped. A visit to another laboratory is worth a month of reading. A vocation is a job that has decided to keep you. A volcano is a reminder that the planet has its own schedule. A vote produces a decision and not a truth.

A warrant is a permission with a date on it. A weapon in a laboratory is usually a screwdriver. A weekend in a laboratory is a choice and a symptom. Whatsoever you measure, write down the unit. To whirl through a dataset is to learn nothing. The wisdom of an old

technician is a compressed file nobody has unzipped. To wither is to be forgotten in place. The wreckage of an instrument is a lesson in what not to buy. A writer is a person who has decided that the sentence matters.

The last night

The last night of the term was a Thursday, and the group met in the workroom at nine with two boxes of cold noodles and a projector that had begun to flicker. Xu presented the rebuilt model: the leakage column removed, the features cut from ninety to eleven, the training set split by year instead of at random. The correlation fell from 0.98 to 0.61. The error bars were wide. The model was, by every measure that matters, a modest success and a spectacular failure.

Gu looked at it for a long time and said, this is the first honest number we have produced. Then he said the thing I had been waiting all year to hear, which is that a result you can defend is worth ten results you can only present, and that he would rather publish a correlation of 0.61 with a clear account of what was excluded than a correlation of 0.98 with a footnote. Xu wrote it down. Zhou Min wrote it down. I wrote it down, and then I noticed that all three of us had written the same sentence, in three different notebooks, in three different handwritings, and that this is what a laboratory is: a group of people agreeing to remember the same thing.

We ate the noodles and argued about the paper's title for an hour. Xu wanted something with the word prediction in it. Shen Lu wanted something with the word limit in it. Gu wanted something dull, and in the end we chose something dull, because a dull title is not a promise you cannot keep.

At midnight the rain started again, and Zhou Min opened the window, and the room filled with the smell of wet asphalt and jasmine from the courtyard. She said that she would be leaving in September, that she had a job in a factory in Ningbo, that she would be doing quality control for a battery line, and that she was afraid she would forget how to think. Gu said that nobody forgets how to think, that they only forget how to make time for it, and that the ones who make time are the ones who keep it.

Shen Lu said nothing for a while. Then she said that she had been offered a place on a doctoral programme in Shanghai and that she had decided not to take it, and when we all looked at her she said that she had discovered, in her fourth year, that she liked instruments more than questions, and that there was no shame in that and no future in pretending otherwise.

I walked home at one in the morning along the canal, past the bridge, past the copied campus of the neighbouring polytechnic, past the shop where the woman was still watching her serial. The chapter of my life that this book is about was ending, and I did not know it, which is how chapters usually end.

What I knew, walking home, was a set of small and unglamorous things. That a number is a promise. That a picture is not data. That the person who writes down the method is doing the science, and the person who is in a hurry is doing something else. That the machine will be wrong in a pleasant voice, and the counter will be right in an unpleasant one, and that the whole of a scientific education consists in learning to prefer the second voice.

Afterwards, in the corridor, we talked about a concert that Zhou Min had been to, a satirical revue in a small hall, and about how the merriment of a room can affect the argument in it. Xu was different from December: active, teasing, anew, and his behaviour had lost the caution of the spring. The silence broke when he laughed, and nothing seemed able to deter him. Nobody mentioned the afternoon he had hit the table, which was a kindness, and an ailment, once named, is easier to accompany through a long night. I do not think we achieved what we set out to achieve. I think we achieved a habit, and a habit does not stagnate, or recur, or break; it waits. Someone said something about a carrot and a thief that made no sense to anybody, and the chill went out of the corridor, and the whole thing was drastically less serious than it had been in the morning.

The rain came down harder. I had no umbrella. I ran the last three hundred metres, and I was laughing by the time I reached the gate, and the guard looked at me the way guards look at students who are running in the rain at one in the morning, which is to say without curiosity, because he had seen it before and would see it again.

伴读：从一台算错了一万倍的机器说起

先把话说死：这一章你不用背。你要做的事只有一件——把这九百多个词放回它们本来待着的地方。组会的桌子、配电间的热风、电镜室那张要签字的检查表、法院走廊里的地毯、殡仪馆门口的签到簿。词不是活在表里的，词活在人在做事的时候。

那天晚上，周敏把一叠表推过来，最上面那张叫 agreement。同意书。你以后会见到它的一百种亲戚：acceptance 是"接受"，验收单上那枚章；accountability 是"问责"，是谁签字谁负责；allowance 是"额度"，工时、差旅、耗材都各有各的额度；applicant 是"申请人"，那天是我。这四个词长得不像，可它们说的是同一件事：一个组织要靠纸来记住谁答应过什么。administrative 这个词你要早点认识，它是"行政的"，听起来无聊，实际上是"事情能不能办成"的另一种说法。

顾老师进来的时候，黑板（blackboard）上还留着上周半条公式。他没写标题，写了两行字：academic，accuracy。学术，准确。然后他说了一句我记了一年的话：这个 arena（竞技场）比的不是谁模型最聪明，比的是谁最守 adherence（遵守、 adherent 是" adherent 者"，但这个词真正的意思是"守住不放"）。advance（进展）来自 accuracy，从来不来自 arrogance（傲慢）。你看，arrogance 和 accuracy 只差几个字母，可它们是这个行当里最远的两端。

我得说清楚一件事：able 和 capable 都译"能"，但 able 说的是"这一次做得到"，capable 说的是"这个人有这个底子"。achieve 是"达成"，宾语通常是目标；acquire 是"获得"，宾语通常是技能、习惯、数据——注意，acquire 的东西会留在你身上，achieve 的东西会写进简历。active

是"主动的"，实验室里最贵的品质；acute 是"急性的、尖锐的"，acute shortage 是"急缺"， acute 和 addictive（上瘾的）放在一起你会记一辈子：短视频是 addictive，缺觉是 acute，这两件事在我们这栋楼里同时发生。

adolescence 是青春期。师姐们喜欢拿它打比方：一个学科也有青春期，本事没长全，脾气先长全了。adventurous 是"爱冒险的"，褒义贬义看语境——adventurous hypothesis（大胆假设）是夸，adventurous accounting（大胆做账）是要出事。advice 不可数，一条建议是 a piece of advice，别说 an advice，这是四六级翻译里最常见的坑之一。affect 是动词"影响"，effect 是名词"效果"，这两个词每年都有人写错，你只要记住：affect 在前，effect 在后，先影响，后结果。

affix 是"贴上、系上"。那天我把条形码标签 affix 在电脑背面，觉得自己很正式。afraid 后面接 of 或 that，不接不定式；aged 是"年老的"，也可以指"陈化的"——aged sample 就是放久了的样品，这一章里它是坏消息。aggression 是"攻击性"，ailment 是"小病小痛"，alleviation 是"缓解"。这三个词在这一章里连成一条线：数据出错 → 有人攻击 → 症状被缓解 → 没人去治。aisle 是过道，飞机上、超市里、会议室里都是它，沈璐那晚就坐在 aisle 座位。algebra 是代数，周敏画的那条三变量的线，本质就是中学的 algebra，只是没人肯承认。

all、almost、already、although 这四个词你天天用，可它们各有各的脾气。all 强调"全部"，almost 是"差一点"，already 是"已经"（配完成时），although 后面不能接 but——although it rained, we went 是对的，although it rained, but we went 是错的。anything 在否定句和疑问句里代替 something；at 和 atop 的区别是后者更文气，atop the pile 是"在一堆东西上面"，写论文时用一次就够。

amazement 是"惊奇"，比 surprise 重；astonish 是动词，"使大吃一惊"，那天屏幕上跳出负三千四百的导电率，全屋 astonish。amend 是"修订"，改的是文本；amplify 是"放大"，放的是信号，也是错误。ancestor 是祖先，anew 是"重新"，animated 是"活跃的、被做活的"，一个 animated 的讨论是好事，一个 animated 的争吵不是。annoy 是"使烦"，antagonism 是"对抗"，antique 是"古董"，apparatus 是"仪器、装置"——注意它是单数可数，复数是 apparatuses，别写成 apparatus。appeasement 是"安抚、绥靖"，这个词带着历史的阴影，用来形容"哄审稿人"最准。appreciation 是"欣赏"，也是"感谢"，还是"理解"——三个意思，一个词。

approximation 是"近似"，arbitration 是"仲裁"，aristocrat 是"贵族"，arousal 是"唤起"，artifact 是"伪影、人工制品"——这个词在这章里最重要的一次出现，是电镜图像上那条亮带：它不是样品，是机器自己造的。ascendance 是"上升、占优势"，asleep 是"睡着的"，assault 是"攻击"，asset 是"资产"，associate 是"同事、副的"，也是"把……联系起来"，attempt 是"尝试"（可数），attorney 是"律师"（美式），audit 是"审计、核查"，auntie 是"阿姨"，宿舍楼、实验室走廊、食堂窗口，到处都是 auntie。automate 是"自动化"，auxiliary 是"辅助的"，avert 是"避免"，awareness 是"意识"——注意它不是"知道"，是"一直知道"。

再看 b。baby 是婴儿，也是"心头肉"：那台新买的显卡，周敏管它叫 the baby。bacon 是培根，baggage 是行李（不可数），ball 是球，也是"一团"——a ball of twine 就是一团麻绳。bandage 是绷带，barbecue 和 BBQ 是一回事，前者正式，后者口语。barracks 是营房，basically 是"基本上"，口语里当"说白了"用。bath 是洗澡（英式），beard 是胡子，beckon 是"招手示意"，比 wave 更具体。before 和 between 不用讲，behaviour（美式 behavior）是"行为"，注意它不可

数。belong 接 to；beneficial 是"有益的"，比 good for 正式；beside 是"在旁边"，besides 才是"此外"，这两个词只差一个 s，意思差很远。

bicycle 是自行车，bin 是垃圾桶，biography 是传记，biscuit 是饼干（英式），bleach 是漂白剂，bloc 是"集团"（政治），blouse 是女式衬衫，blush 是脸红，boiler 是锅炉，bonus 是奖金，border 是边界，bottle 是瓶子，boy 是男孩，brand 是品牌，break 是"打破"，也是"休息"，breeze 是微风，bridge 是桥，brilliant 是"出色的"，broke 是"破产的"（口语），browse 是"浏览"，buck 是美元（口语），budgetary 是"预算的"，bulk 是"大部分、大批"，the bulk of 是高频搭配，bundle 是一捆，burglary 是入室盗窃，business 是生意，也是"事"（none of your business），butterfly 是蝴蝶。

你看，b 这一串，几乎全是"看得见的东西"。这就是这一章的写法：先把东西摆出来，词自己会站到它该站的位置上。

c 这一串最长，我们按"看得见的"和"想出来的"分两拨走。

先看眼前的东西。cab 是出租车，campus 是校园，courtyard 是院子，countryside 是乡下——周敏毕业去宁波的厂，那个地方在她嘴里既不是 campus 也不是 countryside，是个"介于两者之间的工业区"，英文里没有对应的词，这很正常，语言不是万能的。carrot 是胡萝卜，cake 是蛋糕，candy 是糖果，cereal 是麦片，cookie 是饼干（美式曲奇），creamy 是"奶油似的、绵密的"，cup 是杯子。这一组你只要记住一个坑：candy 在英式英语里常指"糖果"（可数），在美式里更常指"冰糖、砂糖块"，做翻译时看清是哪国的文本。

cardboard 是硬纸板，那个装打印纸的纸箱就是它。container 是容器，collar 是"领子"，也是"轴环、卡箍"——机械里的 collar 和衬衫上的 collar 是同一个形状。crane 是"鹤"，也是"吊车"，校园里那台吊车就立在宿舍楼上面。crystal 是晶体，chamber 是"腔、室"，电镜的样品腔、燃烧室、心腔，都是它。clog 是"堵塞"，名词是"木屐"；下水道堵了、管路堵了、数据流堵了，都叫 clog。cloud 是云，也是"阴影"（a cloud of suspicion）。crystal 和 cloud 在图像里正好是一对：一个是清楚，一个是模糊。

cat 是猫，cock 是公鸡，cowboy 是牛仔，butterfly 上一节讲过——这些词在这一章里不是装饰，是照片上的东西：一张 1958 年的合影、一张院子里放电影的旧照、一只停在蓟草上的蝴蝶。costume 是戏服、服装，注意它和 custom（习俗）只差一个字母；custom 你以后会天天见到，因为客服叫 customer。

case 是"箱子、案子、情况"，一个词三个世界：a case of samples（一箱样品）、a court case（一桩案子）、in that case（那样的话）。count 是"数"，也是"算数、要紧"——what counts is the count，这句话你可以当座右铭。cycle 是"循环"，电池的循环次数、水的循环、项目的周期，都是它。climax 是"高潮、顶点"，culmination 是"顶点、 culmination 是长期积累的结果"，前者是戏剧术语，后者更学术。crash 不在这张表里，但 cripple 在：它是"使残废、严重削弱"，a crippled system 是"瘫了的系统"，语气很重，别乱用。

再看"想出来的"那一拨。capable 是"有能力的"，be capable of doing；cognition 是"认知"，cognitive 是形容词；cognition 和 recognition 是一对：一个是你自己想明白，一个是你认出来。conscience 是"良心"，conscious 是"有意识的"，conscientious 是"认真的"——三个词共一个词

根，意思分三岔，考场上最爱在这里设坑。

cease 是"停止"，比 stop 正式，cease to exist 是"不复存在"；confess 是"承认、坦白"，比 admit 重，带"认罪"的味道。clarification 是"澄清"，classification 是"分类"，consolidation 是"合并、巩固"，compilation 是"汇编"，composition 是"构成、作文、作品"，comparison 是"比较"——这一组全是 -tion 结尾的抽象名词，写作时是提分利器，口语里说出来会显得很正式。

comment 不在表里，但 comic 在：comic 是"喜剧的、连环画"，comic strip 是连环漫画；combative 是"好斗的"，contempt 是"轻蔑"，cruelty 是"残忍"，discord 上一节没讲，下一节讲。charm 是"魅力"，也是"咒语"，动词是"迷住"；cherish 是"珍惜"，宾语是人、是记忆、是习惯。cheek 是"脸颊"，也是"厚脸皮"（what a cheek）。chill 是"寒意"，也是"放松"（口语 take a chill），实验室里说 the chill in the room，说的是气氛。

chronicle 是"编年史"，动词是"按时间记录"；biography 是传记，chronicle 是流水账——一个好作家能把 chronicle 写成 history。circular 是"圆形的"，也是"传阅通知"（那份贴在门上的通知就是 circular）。citizenship 是"公民身份"，实验室里的 citizenship 是"你扫自己没弄脏的地"。civic 不在表里，但你要知道 citizenship 的形容词是 civic。

clerk 不在表里，clerical 在：clerical 是"文书的、办事员的"，clerical error 是"笔误、抄写错误"——注意，这个词在论文里出现时，往往是作者在给一个真实错误找台阶。client 不在表里，consultation 在：咨询、会诊，a consultation 是一次会诊，consult 是动词。convention 是"惯例"，也是"大会"，两个意思都常用；conveyance 是"运送、传达"，法律里还指"产权转让"，一个词跨三个领域。consideration 是"考虑"，也是"体贴"，take into consideration 是固定搭配。constrain 是"限制、约束"，比 limit 更学术，被动式 be constrained by 很好用。

consequent 是"随之而来的"，注意它和 subsequent（后来的）不同：consequent 强调因果，subsequent 强调时间。concurrent 是"同时发生的"，continuous 是"连续的"（不停），continual 是"反复的"（停了又来）——这两个词的区别，是四六级的高频考点，也是实验记录里最要紧的区别：连续测量和反复测量，是两种数据。

compensation 是"补偿"，compulsory 是"强制的、必修的"，compulsory course 是必修课；concentrate 是"集中"，接 on；concert 是"音乐会"，也是"一致行动"（in concert with）。condolence 是"吊唁、慰问"，用复数的时候多；congregation 是"（教堂的）会众"，choir 是唱诗班，chant 是吟唱，priest 是神父，Catholic 是天主教徒，Confucian 是儒家的——这一组词在"利玛窦"那一页里全出现了，它们不是宗教词汇表，它们是一段历史。

confinement 是"禁闭、局限"，imprisonment 是"监禁"，custody 是"监护、拘留"，三个词一个比一个重。consolidate 是"巩固"，constrain 讲过了，corrupt 是"腐败的、损坏的"，a corrupt file 是"损坏的文件"，a corrupt official 是"贪官"，同一个词两种坏法。correlative 是"相关的"，correlation 是相关系数；contextual 是"语境中的"，一个 contextual result 是"关于世界的句子"，而不是"带着系数的传言"——邵教授信里那句话，你要背下来。

credible 是"可信的"，credulous 是"轻信的"，credit 是"信用、学分"，三个词一个根，方向完全不同。crouch 是"蹲下"，为了看台面底下那块东西，你早晚要 crouch 一次。debate 不在表里，contempt 讲过了。contributor 是"贡献者"，corporation 是"公司（法人）"，cordiality

是"诚恳、热忱"，比 politeness 暖。cop 是警察（口语），coach 是教练、辅导，corpse 不在表里。cup 讲过了，cure 不在表里，cycle 讲过了。

最后三个不好塞的：coinage 是"造词、铸币"，新术语的 coinage 是最便宜的一种原创；colonel 是"上校"，注意它的读音是 /ˈkɜːnl/，和拼写完全对不上，这个词是英语拼写史上最著名的陷阱之一；custody 讲过了。cease、charm、chill、clog 都是那种"认识但不敢用"的词，它们的共同点是：口语里少，书面里多，用一次就显出来。

d 这一组里，有一半是"数据"的词，另一半是"人"的词。

先看数据。database 不在表里，density 也不在，dense 在：dense 是"稠密的"，dense text 是"密不透风的文字"，dense fog 是"浓雾"。digit 是"数字"，注意它只指 0 到 9 的单个数字，十二位的字符串叫 a twelve-digit string；digital 是形容词，别把 digit 当"数码的"用。diameter 是"直径"，dilution 是"稀释"，duration 是"持续时长"，dose 不在表里。deduct 是"扣除"，和 subtract 相近，但 deduct 更常用于"扣钱、扣分"。define 是"定义"，designate 是"指定"，两个词都带"把范围划出来"的意思。divide 是"分开"，double 是"加倍、双重的"，download 是"下载"。display 是"显示、展示"，dismiss 是"驳回、解散"，dismiss a claim 是"驳回一项主张"。

degrade 是"降解、降级"，deplete 是"耗尽"，deposit 不在表里，但 depot 在：depot 是"仓库、车站"，公交总站就是 bus depot。deputy 是"副职"，deputy director 是"副主任"。descend 是"下降"，destination 是"目的地"，detachment 是"超然、分遣队"，形容沈璐那种"准，且不缝合"的态度，最合适的就是 detachment。deter 是"威慑、阻止"，detriment 是"损害"，to the detriment of 是固定搭配。developmental 是"发展中的"，devour 是"狼吞虎咽"，读书也能 devour。

deceit 是"欺骗"，比 lie 重，指"行骗"这件事本身；decision 是"决定"，deliberation 是"审议、深思"，注意它有两个意思：长时间的商议，和从容不迫。delude 是"欺骗自己"，delude oneself 是固定搭配，比 fool oneself 正式。demographic 是"人口的"，做名词时常用复数 demographics（人口结构）。dazzle 是"使眼花"，dash 是"猛冲"，也是"一点点"（a dash of salt）。dangerous 是"危险的"，dairy 是"乳制品、奶牛场"，注意它和 diary（日记）只差一个字母的位置——这是中国学生最经典的拼写错误，没有之一。

death 是"死亡"，deer 是"鹿"（单复数同形），dinner 是"正餐"，dog 是"狗"，drunkard 是"酒鬼"，dweller 是"居民"，city dweller 是"城市居民"。driver 是"司机"，也是"驱动因素"，a driver of change。dock 是"码头、装卸台"，department 是"系、部门"，desktop 是"桌面"，decoration 是"装饰"。difficult 是"困难的"，direct 是"直接的、指导"，动词的 direct 是"导演、指引"。disabled 是"残疾的"，disapproval 是"不赞成"，disgusting 是"令人作呕的"，语气很重，别拿它形容"不太好吃的菜"。

discern 是"辨别"，比 see 深，比 distinguish 文。discord 是"不和"，discriminate 是"区分"，也是"歧视"——discriminate between A and B 是"区分"，discriminate against sb 是"歧视"，一个动词两个方向，介词决定意思。disruption 是"中断、扰乱"，distinct 是"截然不同的、清晰的"，distraction 是"分心之物"，disturbance 是"干扰"，disruption 和 disturbance 的区别：前者是"打乱流程"，后者是"打扰状态"。

e 这一组，是这一章的心脏。eager 是"渴望的"，be eager to do；earthquake 是"地震"，比喻义也常用。eccentric 是"古怪的"，比 strange 多了"有趣"的味道。economist 是"经济学家"，editorial 是"社论"，efficient 是"高效的"，注意它和 effective（有效的）不同：effective 是"管用"，efficient 是"省力"。eighteen 是十八，elapse 是"（时间）流逝"，只用于时间。elective 是"选修的"，electronically 是"以电子方式"，elevation 是"海拔、提升"，elite 是"精英"，embargo 是"禁运"，emerge 是"浮现"，emission 是"排放"，employee 是"雇员"。

enable 是"使能够"，enable sb to do；encouraging 是"令人鼓舞的"；endorsement 是"背书、代言"，这个词在合同里就是要命的东西；enforce 是"执行、强制"，enforce a rule；enhancement 是"增强"；enlist 是"争取、招募"；ensue 是"接着发生"，比 follow 正式；entertainment 是"娱乐"；entity 是"实体"，法律实体就是 legal entity；envious 是"嫉妒的"，be envious of。epidemic 是"流行病"，equator 是"赤道"，eradicate 是"根除"，erosive 是"侵蚀的"，escalation 是"升级"，establish 是"建立"，eternal 是"永恒的"，evacuation 是"疏散"，eve 是"前夕"，eventually 是"最终"（不是"可能"），evidence 是"证据"（不可数）。

exaggerate 是"夸大"，exceedingly 是"极其"，excess 是"过量、超出"，exclamatory 是"感叹的"，exclamatory sentence 是感叹句——这个词你只在语法书上见得到，见到就当"考试用语"处理。executive 是"高管、行政的"，exhaust 是"耗尽、排气"，动词名词都是它。existent 是"存在的"，比 existing 少见；expectancy 是"预期"，life expectancy 是"预期寿命"。experience 是"经验"，也是"经历"，可数不可数看你指哪个。explain 是"解释"，explore 是"探索"，exposure 是"暴露、曝光"，extensive 是"广泛的"，extract 是"提取"，eyebrow 是"眉毛"，raise an eyebrow 是"表示惊讶"。

f 这一组，我挑五个说透，剩下的你自己过。facet 是"方面"，比 aspect 更强调"一个切面"，玉不琢不成器，一个问题的 facet 也不是全部。faith 是"信念、信仰"，have faith in 是"相信"。fascist 是"法西斯分子"，形容词是 fascist 或 fascistic；这个词在学术写作里出现时，通常是"任何经不起追问的主张"。fate 是"命运"，favorite 是"最喜爱的"（英式 favourite）。feather 是"羽毛"，feedback 是"反馈"，不可数；feminist 是"女权主义者"，fictional 是"虚构的"，figure 是"数字、图、人物"，三个意思全靠语境。find 是"发现"，名词的 find 是"发现物"。fist 是"拳头"，flame 是"火焰"，flaw 是"缺陷"，flip 是"翻转"，flourish 是"繁荣"，fluid 是"流体"，foil 是"锡箔"，也是"陪衬"（foil to）。fool 是"傻瓜"，forecast 是"预报"，forest 是"森林"，forgetful 是"健忘的"，format 是"格式"，forth 是"向前"，and so forth 是"等等"。forum 是"论坛"，founding 是"创建"，fragility 是"脆弱"，franchise 是"特许经营"，freeway 是"高速公路"，Friday 是星期五，frog 是青蛙，fruitful 是"富有成果的"，fulfillment 是"实现、满足"，funny 是"好笑的"，fury 是"暴怒"。

g 开头这几个，全是"手上的东西"。gadgetry 是"小玩意儿（总称）"，带点贬义，一屋子 gadgetry 是说"一屋子花架子"。gangster 是"匪徒"，garlic 是蒜，gauge 是"量规、仪表"，也是"估计、判断"——动词的 gauge 很常用：to gauge the mood of a room，就是"掂量一屋子的气氛"。generator 是发电机，gentleman 是绅士，gift 是礼物也是天赋，glove 不在表里。geometric 是"几何的"，graphically 是"用图表、形象地"，两个词都跟"画出来"有关。

glide 是"滑行"，比 slide 轻，比 fly 低。glossary 是"术语表"，一本论文最后那个小词典就

是它；glossary 和 dictionary 的区别是：glossary 只收一本书里的难词。gold 是黄金，gorgeous 是"极美的"，语气比 beautiful 重，口语里可以形容一切让你"哇"的东西。grace 是"优雅"，也是"恩典"，也是"宽限"（a grace period 是宽限期）；graceful 是"优雅的"，gracious 是"亲切的"，两个形容词别混。

grain 是"颗粒、谷物"，也是"纹理"，go against the grain 是"违背本性"。grandmother 是祖母，gravity 是"重力"，也是"严肃"（the gravity of the situation）；这个词的双重含义，正好是这一章的一个隐喻。green 是绿色，也是"新手"（a green hand），grievous 是"令人悲痛的、严重的"，grievous error 是"重大错误"，语气很重，慎用。grocery 是"食品杂货店"，美式里 groceries 是"买的东西"。guarantee 是"保证"，guilt 是"内疚、罪责"，guy 是"家伙"，口语，男女都能用。glad 是"高兴的"，比 happy 轻，gospel 不在表里。

h 这一组，我建议你按"身体、地方、抽象"三层记。handicap 是"障碍、不利条件"，注意它在现代英语里指"环境造成的障碍"，不指人；harden 是"变硬、变强硬"，harden a habit 是"把习惯固化"。harmonious 是"和谐的"，hat 是帽子，haven 是"避风港"，heal 是"治愈"，heartfelt 是"由衷的"，hectare 是公顷，helicopter 是直升机，hemisphere 是半球，herd 是兽群，也是"随大流的人群"（the herd）。heroism 是英雄主义，hibernation 是冬眠，hike 是"徒步"，也是"（价格）大幅上涨"，两个意思都常用。hint 是"暗示"，hit 是"打、击中"，也是"轰动一时之作"。homework 是作业，honor 是荣誉（英式 honour），horizon 是地平线，hose 是软管，hot 是热，housewife 是家庭主妇。

hug 是拥抱，humanity 是"人类、人性"，humorous 是"幽默的"（英式 humourous 也有人写），hurl 是"猛扔"，比 throw 用力。hydrogen 是氢，hypothetical 是"假设的"，a hypothetical case 是"假设情形"，写论文时用它比 if 更正式。halfway 是"中途"，halfway up the stairs 是"楼梯走到一半"，那个"忘了东西"的瞬间就是它。

i 这一组是最"抽象"的一组，也是最能拉开分数的一组。idea 是想法，ideological 是"意识形态的"，idol 是"偶像"，illegal 是"非法的"，illusory 是"虚幻的"，imaginative 是"富有想象力的"，impart 是"传授"，impart a method 比 teach 更文。implication 是"含义、牵连"，这个词在论文里极重要：the implications of a result 是"一个结果意味着什么"，不是"结果本身"。imposing 是"气势宏伟的"，imprisonment 是监禁，inauguration 是就职典礼、落成典礼。inclination 是"倾向"，incorporate 是"纳入、合并"，indebted 是"受恩于、感激的"，be indebted to sb 是固定搭配。indicative 是"指示性的"，be indicative of 是固定搭配；individual 是"个人、个别的"；indulgent 是"纵容的"；inference 是"推论"，注意它和 implication 不同：inference 是你推出来的，implication 是话里含着的。

inflict 是"施加（痛苦）"，inflict on 是固定搭配。informative 是"信息量大的"，inhabitant 是居民，initiate 是"发起、引入"，ink 是墨水，innovation 是创新，insanity 是"疯狂"，insist 是"坚持（说）"，inspire 是"激励、启发"，instinct 是本能，instructor 是讲师，insure 是"给……保险"，注意和 ensure（确保）、assure（使确信）三个词的区别，这是四六级翻译的高频坑。intellect 是智力，intensity 是强度，intercourse 是"交往"（social intercourse 是社交），intermediary 是中间人，interrogate 是"审问"，interviewee 是被面试者，intricate 是"错综复杂的"，intruder 是闯入者，invasion 是入侵，invest 是投资，invite 是邀请，

ironically 是"讽刺地"，irritate 是"使恼火、刺激（皮肤）"，it 这个词本身没什么好讲，但你要知道它在强调句里可以指人：It was Shen Lu who said it.

j 和 k 都很短。January 是一月，jelly 是果冻，jog 是慢跑，journey 是旅程（可数），juice 是果汁，jurisdiction 是"司法管辖权"。keep 是保持，kill 是杀，kingdom 是王国，knight 是骑士，knowledge 是知识（不可数，一条知识是 a piece of knowledge）。

l 这一组，我挑几个"看着简单、用起来错"的说。lack 是"缺少"，可数不可数都有说法：a lack of water（不可数意味）和 lacks（动词第三人称）。lamp 是灯，language 是语言，last 是"最后的"，也是"持续"，动词的 last 常被忽略：the session lasted three hours。laughter 是笑声（不可数），laugh 是一次笑。lawsuit 是诉讼，leaf 是叶子（复数是 leaves），learner 是学习者，left 是左边，也是 leave 的过去式。

legislative 是"立法的"，lend 是"借出"（lend sb sth），let 是"让"，liberal 是"自由的、开明的"，也是"慷慨的"。lieutenant 是"中尉"，拼写里那个 i 不发音，读 /lefˈtenənt/，和 colonel 一样是拼读不一致的经典。limelight 是"聚光灯、众人注目"，in the limelight。liner 是"衬里、邮轮"，lip 是嘴唇，也是"（容器的）边"，literal 是"字面的"，lively 是"生动的"，local 是"当地的"，lodging 是"住宿、寄宿"，loneliness 是孤独，loophole 是漏洞，lot 是"许多"，也是"一块地"（a parking lot）。lover 是爱人，也是"爱好者"（a lover of poetry）。lucky 是幸运的，lung 是肺。

m 这一组最长，也最有意思。machinery 是"机械（总称）"，不可数；magnet 是磁铁，也是"有吸引力的人或物"；mail 是邮件，major 是"主要的、专业"，malignant 是"恶性的"，managerial 是"管理的"，manipulation 是"操纵"，manufacture 是"制造"，March 是三月，marketing 是营销。marvellous 是"极好的"（英式拼写双 l），massage 是按摩，maternal 是"母亲的"，maturity 是成熟，meantime 是"其间"，in the meantime 是固定搭配。mechanical 是"机械的"，mediator 是调解人，memo 是备忘录，mend 是修补，mercury 是水银，merriment 是欢闹。

metabolic 是"代谢的"，methodology 是"方法论"——注意它不是 method 的高级形式，method 是"做法"，methodology 是"关于做法的学问"，用错会显得很外行。microscopic 是"微观的"，might 是"可能"，也是名词"力量"，milestone 是里程碑，million 是百万，minister 是部长，miraculous 是"奇迹般的"，missing 是"缺失的"，misty 是"有雾的"，mixture 是混合物，moderate 是"适度的"，monotonous 是"单调的"，monumental 是"纪念碑式的、巨大的"，morality 是道德，most 是"大多数、最"，motive 是动机，mourning 是哀悼，much 是不可数"多"，multiply 是"乘、繁殖"，municipality 是市政当局，muscle 是肌肉，mute 是"沉默的、静音"，myth 是神话，也是"普遍的误解"——the myth of the lone genius，那个"孤独天才"的神话就是它。

这里要单独说三个碎片：mentary、modity、mon。它们不是词，是扫描件被切坏以后留下的残片。mentary 本来是 commentary 或 documentary 的尾巴，modity 是 commodity 的尾巴，mon 是某个更长的词的头部。我把它们留在表里，是因为这一章讲的就是"数据会坏"，而坏掉的东西也该被记录。你在考场上不会见到它们，但你会见到一个道理：不完整的数据，如果你不去看它哪里不完整，它就会骗你。

n 这一组，短，但都是硬词。namely 是"即、也就是"，用来把话说得更具体；narrowly 是"勉强

地、狭窄地"，narrowly escape 是"死里逃生"；native 是"本地的、母语的"；navy 是海军；necessity 是"必要性、必需品"；neglect 是忽视；neighbor 是邻居（英式 neighbour）；network 是网络；nice 是"好的"，但注意它语气很轻，正式文书里少用；nil 是"零"（英式，常用于比分、结果）；noble 是"高贵的"；nominee 是被提名者；north 是北方；not 是否定；noticeable 是"显而易见的"；noun 是名词；nucleus 是"核"，原子的核，也是"核心"；nurse 是护士。

o 这一组，我给三个重点。obedience 是服从，oblige 是"使不得不、施恩于"，be obliged to do 是"不得不"，而 much obliged 是"非常感谢"，一个词两个温度。observatory 是天文台；obstruction 是阻碍；occupancy 是占用率；oceanic 是海洋的；off 是关掉、离开；often 是经常；Olympic 是奥运的；ongoing 是"进行中的"；open 是打开；operative 是"起作用的、运行的"；oppress 是压迫；optimist 是乐观主义者；orbit 是轨道；organ 是器官、机构；ornamental 是装饰性的；our 是我们的；outdoors 是户外；outlook 是"前景、看法"；outstanding 是"杰出的、未付的"（outstanding debt 是未偿债务，这个义项常被忽略）；overcome 是克服；overnight 是"一夜之间"；overthrow 是推翻；owl 是猫头鹰。

p 是这一章词最多的一组，我不按字母顺序讲，按"你会用错的地方"讲。

先说三个最容易混的。perception 是"知觉、看法"，是你怎么看待；perspective 是"视角"，是你站在哪里看。一个人可以换 perspective，但很难换 perception。principle 是"原则"，principal 是"主要的、校长"，这两个词在拼写上坑了几代人，记住：princi-ple 里有"ple"，像"people 的原则"。presume 是"假定"，比 assume 更带"未经证实就当成真的"的味道，presume innocence 是"无罪推定"。

再说"-pre"这一家。precautionary 是"预防性的"，precautionary measure 是固定搭配；preliminary 是"初步的"，preliminary result 是"初步结果"，这个词在论文里几乎必现；preparatory 是"预备的"；prevention 是预防；preface 是前言，preface 和 preliminary 都是"在正式内容之前"，一个在书里，一个在流程里。predatory 是"掠夺性的"，predatory journal 是"掠夺性期刊"，这两个词你以后一定会用到。preserve 是保存，protect 是保护，prove 是证明，proclaim 是宣告（名词 proclamation 是公告）。

pace 是步伐，pain 是疼痛（可数不可数都有），palm 是手掌也是棕榈，pants 是裤子（美式），paradoxical 是"悖论的"，parental 是父母的，partial 是"部分的、偏袒的"，partisan 是"党派的、有立场的"，passenger 是乘客，pasta 是意面，path 是路径，pave 是铺路，peaceful 是平静的，peculiar 是"奇怪的、特有的"，peg 是木钉，pension 是养老金，perfume 是香水，permanence 是永久，perpetual 是"永久的、没完没了的"，persist 是坚持。

pesticide 是农药，pharmaceutical 是制药的，physicist 是物理学家，photo 和 picture 都是照片，但 picture 更宽（画、图像、局面），pilgrim 是朝圣者，pine 是松树，pit 是坑，plaintiff 是原告，plate 是盘子也是板，plea 是恳求也是"抗辩"，plenty 是大量，plunge 是"骤降、跳入"，poetry 是诗，polar 是极地的，political 是政治的，polytechnic 是理工学院，populate 是"居住于、填充"，portable 是可携带的，position 是位置也是职位，postcard 是明信片，potential 是潜力，powerful 是有力的，priest 是神父，privilege 是特权，problematic 是"成问题的"，professor 是教授，progress 是进步，prominent 是"突出的、著名的"，pronounce 是发音，propeller 是螺旋桨，prose 是散文，provocation 是挑衅，psychological 是心理的，

pulse 是脉搏，pupil 是"小学生、瞳孔"，purse 是钱包，pyramid 是金字塔。

q 只有三个。quantitative 是定量的，qualitative 是定性的，一篇文章的方法部分几乎一定要在这两个词里挑一个。query 是"查询、疑问"，数据库里的一次查询就是 query。quite 是"相当"，但要注意它在英式里可以表示"不完全"（quite good 可能是"还行"），美式里更接近"非常"。

r 开头这一串，我把它分成三拨：科学的、人的、机构的。

科学的：radar 是雷达，radioactivity 是放射性，railroad 是铁路（美式），random 是随机，rarely 是"很少"（否定副词，rarely 提前句子要倒装），rating 是评级，react 是反应（react to），real 是真实的，reason 是原因也是理性，recede 是"后退、消退"，receptive 是"易于接受的"，reckon 是"认为、估算"，recover 是恢复，recur 是"复发、再发生"，redundant 是"多余的"，redundancy 是冗余（工程里是褒义，写作里是贬义）。refinery 是炼油厂，refresh 是刷新，refuse 是拒绝，rein 是缰绳，也是"控制"（rein in）。

人的：regrettable 是"令人遗憾的"，relentless 是"不懈的、无情的"，religion 是宗教，remark 是评论，reminiscence 是回忆，rendering 是"呈现、渲染"，rent 是租金，repel 是"击退、使厌恶"，reporter 是记者，reproduce 是复制、繁殖，request 是请求，resent 是"怨恨"，residence 是住所，resolute 是坚决的，respective 是"各自的"，restrictive 是限制性的，retaliate 是报复，retort 是反驳，reunion 是重聚，reverse 是反转，revolt 是反抗，rhythm 是节奏，ride 是骑、乘坐，rigid 是僵硬的，rip 是撕裂，rivalry 是竞争，robotic 是机器人的，roller 是滚筒，rosy 是"玫瑰色的、乐观的"，rouse 是"唤醒、激起"，rude 是粗鲁的，run 是跑、运行。

机构的：regimental 是"团的、严格划一的"，relentless 讲过，relation 是关系。注意 relation 和 relationship：前者更抽象（国际关系），后者更具体（人际关系）。religion 和 religious 一个是名词一个是形容词，拼写别丢 u。

另外几个"看着不像生词"的，也要点一下。above 是"在……之上"，above all 是"首先"；absorption 是吸收，注意它和 adsorption（吸附）只差一个字母，化学里是两个世界；accompany 是"陪伴"，比 go with 正式；call 是"叫、打电话、称呼"，caption 是"图片说明"，cemetery 是墓地，certify 是"证明、认证"，chore 是"杂务"，clean 是干净，也是"彻底地"（clean forget）；complete 是完成，也是"完整的"；complication 是"并发症、复杂化"；current 是"当前的"，也是"电流"；diesel 是柴油；domestic 是"国内的、家庭的"；donor 是捐赠者；drag 是拖；drastically 是"急剧地"；dreamy 是"梦幻的、走神的"；dull 是"沉闷的、钝的"；factory 是工厂；famous 是著名的；farm 是农场；fireplace 是壁炉；happen 是发生；he 是他；holiday 是假期；meet 是遇见，也是"满足"（meet the requirement）。

s 是最大的一组，我按"看得见的东西、看不见的东西、学术里的东西"分三层走。

看得见的：sack 是麻袋，safe 是安全的，也是保险箱，salad 是沙拉，sand 是沙子，sauce 是酱汁，sea 是海，shade 是"阴凉处、色号"，shelf 是架子，ship 是船，也是"运送"，shoe 是鞋，shrubbery 是灌木丛，silk 是丝绸，sky 是天空，slide 是载玻片、幻灯片，slot 是"槽、时段"，smell 是气味，soap 是肥皂，sock 是袜子，soil 是土壤，spade 是铲子，spray 是喷雾，steak 是牛排，stitch 是针脚，stripe 是条纹，sun 是太阳，studio 是工作室，southwest 是西南。

看不见的：salutation 是"招呼语"（信开头的 Dear 就是它），satirical 是讽刺的，scan 是扫描，scare 是吓唬，sceptic 是怀疑者（美式 skeptic），scope 是范围，scratch 是抓、从零开始（from scratch），secret 是秘密，seduce 是诱惑，seeming 是"表面上的"，seldom 是"很少"，semester 是学期，sensational 是"轰动的"，sentimental 是"多愁善感的"，serial 是"连续的、连载的"，set 是放置、一套，several 是几个，shame 是羞耻，shatter 是粉碎，shortly 是"不久"，showcase 是展示，sicken 是"使生病、使厌恶"，signal 是信号，simplify 是简化，sip 是小口喝，six 是六，skilled 是熟练的，slave 是奴隶，smuggle 是走私，sneak 是溜、偷偷做，something 是某物，soothe 是安抚，soul 是灵魂，spite 是恶意（in spite of 是"尽管"），spokeswoman 是女发言人，spoonful 是一勺，squeeze 是挤压，stagnate 是停滞，stalk 是"茎、跟踪"，stare 是盯着看，state 是状态、声明，step 是步骤，stimulant 是兴奋剂，stop 是停止，swap 是交换，swift 是迅速的。

学术里的，我多说两句。scholastic 是"经院的、学校的"，scholastic habit 是"引经据典的习惯"，带点贬义。specification 是"规格、说明书"，工程里一份 spec 就是它。speculation 是"猜测"，和 inference（推论）不同：speculation 没有证据链，inference 有。spelling 是拼写，spill 是洒出，statistical 是统计的，straightforward 是"直截了当的"，strategy 是策略，stressful 是"压力大的"，submerge 是浸没，subscription 是订阅，subtlety 是微妙之处，successful 是成功的，suggestive 是"提示性的、引人联想的"，superb 是极好的，supervise 是监督，support 是支持，suppressive 是压制的，surgeon 是外科医生，surveillance 是监视，suspense 是悬疑，symbolical 是象征性的（symbolic 更常用），systematic 是系统的。

还有三个我会单独拎出来讲。seeming 和 seemingly 不一样：seeming 是形容词（表面的），seemingly 是副词（看似）。seldom 是否定副词，放在句首要倒装：seldom have I seen such a thing。soluble 是"可溶的"，注意它和 solvable（可解的）不是一个词——一个是溶解，一个是解方程。solicitor 是事务律师（英式），和 attorney、lawyer 各有各的地盘：英式 solicitor 出庭少、办事务多，barrister 才出庭。spite 在 in spite of 里已经"洗白"了，但它本身是"恶意"，别在作文里写 I did it in spite of him 表示"我为他做的"，那是"我偏要对着他干"。

最后三个短的：somebody 不在表里，something 在；sue 是起诉，surveillance 上面讲过；taste 下一节讲。

t 这一组，几乎每一个都能拿去写作文。

tactful 是"圆通的"，不是"狡猾"，是"知道什么话该在什么时候说"。talent 是天赋，注意它可数：a talent for languages。tangle 是"纠缠的一团"，也是动词"缠上"，数据和线缆都会 tangle。taste 是味道，也是品味，也是"尝一口"；have a taste for 是"喜欢"。teaching 是教学，technology 是技术，television 是电视。temporal 是"时间的"，和 spatial（空间的）相对，这个词在论文里出现，往往是在说"时间上的变化"而不是"因果关系"。tendency 是倾向，terminate 是终止，比 end 正式，合同、妊娠、进程都能 terminate。territorial 是"领土的、地盘的"，实验室里的 territorial dispute，争的是一块台面。testimony 是证词，也是"证明"，Thanksgiving 是感恩节，themselves 是他们自己（反身代词），thereby 是"由此"，正式连接词，写论文很好用。

thief 是贼，thirsty 是渴，though 是虽然，也是句末的"不过"，threshold 是门槛、阈值，throughout 是"贯穿"，tick 是"打勾、滴答"，tight 是紧，timely 是及时的，tireless 是不知疲

倦的，today 是今天，tolerant 是宽容的，tonight 是今晚，topple 是推翻，tour 是旅行、参观，town 是城镇，trade 是贸易，tragically 是"悲惨地"，tram 是有轨电车，transient 是短暂的，transmission 是传输，transportation 是运输，treasure 是珍宝，tremble 是发抖，tribe 是部落，trillion 是万亿，troop 是部队，true 是真的，try 是尝试，tumble 是跌倒，turn 是转，typewriter 是打字机。

这里我要停下来讲三个词，因为它们代表三种"中国学生常犯的错"。

第一个是 temporal。很多人看到它想到"临时"（temporary），于是把 temporal 也当成"暂时的"。不对。temporal 是"时间的"，temporary 是"暂时的"，一个说维度，一个说长短。这一章里我说"temporal error"是"日期写错列"，就是这个意思。

第二个是 thereby。它是"由此、从而"，后面必须接动名词或从句，不能当 and 用。写论文时一个 thereby 能让句子紧一档，用错就露怯。

第三个是 tough 不在表里，但 tight 在。tight 是"紧"，紧的时间叫 a tight schedule，紧的钱叫 a tight budget，紧的规则叫 tight rules。你会发现这个词在实验室里出现频率极高，因为实验室的本质就是把时间、预算、规则都拧紧。

u 这一组短。unanimous 是"一致同意的"，undergo 是"经历、接受（治疗、检查）"，比 experience 更被动，understand 是理解，unemployed 是失业的，union 是联盟、工会，unleash 是"释放、解开束缚"，uphold 是"维护（裁决、原则）"，upward 是向上的，us 是我们（宾格），usually 是通常。

v 这一组里有两个词值得多说。vaccine 是疫苗，valuable 是宝贵的，variant 是变体（病毒变体、拼写变体），vehicle 是车辆，也是"载体"，ventilation 是通风，verify 是核实，vest 是背心，vicinity 是附近（in the vicinity of），viewpoint 是观点，violence 是暴力，virtue 是美德，visit 是拜访，vocation 是职业、天职——注意它和 vacation（假期）只差一个字母，一个是你这辈子要干的事，一个是你想歇的那几天。volcano 是火山，vote 是投票。

w 这一组，我给你排一条线。wage 是工资（按小时、按件），wake 是醒来，wardrobe 是衣柜，warrant 是"授权令、保证"，watch 是看、手表，wax 是蜡，weapon 是武器，website 是网站，weekend 是周末，west 是西，whatsoever 是"任何……都"（否定句里加强语气），whereby 是"凭借"，whirl 是旋转，wholesale 是批发，widen 是加宽，wildlife 是野生动物，winery 是酒庄，wisdom 是智慧，wither 是枯萎，woman 是女人，wording 是措辞，worm 是虫，worthless 是"无价值的"，wreckage 是残骸，writer 是作家。

whereby 我要单独说一句：它是"通过……的方式"，等于 by which，写定义句特别好用：a system whereby every sample is logged，比 a system in which 更紧凑。

y 和 z 只有两个：yearly 是"每年的"，yield 是"产出、屈服"，两个意思都很常用，yield a result 是"得出结果"，yield to pressure 是"屈服于压力"。zip 是拉链，也是"压缩文件"，动词是"拉上"。

最后：怎么读这一章

回到开头那台算错了一万倍的机器。我把它放在第一章的位置，不是因为它好笑，是因为它诚实。一个模型在第一天就当着所有人的面错了，而且错得很有礼貌，还附了一句"该材料具有优异稳定性"的评语——这件事比任何一次成功都更有教育意义。你以后会见到很多漂亮的结果，漂亮的曲线，漂亮的 stunning 图；你要练的本事，是在看见漂亮东西的那一秒，先问一句：它凭什么。

这一章其实有三条线。第一条是数据：从一万一千四百篇论文，到九十列的表，到被切坏的碎片 mentary、modity、mon，到那个把标签泄漏进特征里的 0.98。第二条是仪器：从配电间的显卡，到电镜室那张要签字的检查表，到四百个颗粒和一只计数器。第三条是人：周敏教你"列有含义"，沈璐教你"图不是数据"，顾老师教你"无聊的那一半保护有趣的那一半"，邵教授在信里教你"墙要砌直"。三条线拧在一起，就是这一章真正的主题：科学不是聪明，是慢。

怎么读？我给你一个三遍法，比任何计划都管用。

第一遍，只跟情节。不求甚解，遇到不认识的词就跳过去，像看一部没有字幕的电影。你要的是"知道这一章发生了什么"：谁在组会上说了什么，谁在电镜室熬到五点，谁在七月改了标题。这一遍读完，你会有一种模糊的感觉，模糊就模糊，别急着查。

第二遍，只盯加粗的词。这一遍你会发现自己能猜出大半：因为它们在句子里有位置，有邻居，有情绪。猜错了也没关系，猜错的地方正是你要记住的地方。这一遍读完，你会发现有些词反复出现（比如 evidence、count、signal），它们不是生词，它们是这一章的骨架。

第三遍，只读你不认识的那几个。这时候再查，查完把例句抄在页边，抄整句，不要只抄词。sincerity 这个词你要是只抄"真诚"，它就永远是一个标签；你要是抄下"a crude rule written down is worth ten elegant rules held in the head"，它就成了一个你能用的句子。

三个坑，我按踩到的顺序说。

第一个坑是拼写。dairy 和 diary，一个是奶牛场，一个是日记，中国学生年年写错；colonel 读 /ˈkɜːnl/，拼写里那个 l 你根本听不见；restaurant 的结尾是 -ant 不是 -ent，中间那个 a 很多人丢；rhythm 里一个元音都没有，却是最常被拼错的词之一；questionnaire 不在这一章，但你早晚会用到，它有两个 n。这些词没法讲道理，只能写十遍。

第二个坑是搭配。英语的难，一半在词，一半在词和谁站在一起。depend 后面是 on，不是 of；be indicative of 里的 of 不能省；in the vicinity of 是固定说法，说 near the vicinity 就错了；consist of 和 consist in 是两个意思；comply with 不是 comply to；be indebted to 后面接人。搭配错了，语法再对也像外国人在说中文。

第三个坑是语体。同一个意思，场合不同，词就不同。cop 是口语，正式文书里写 police officer；guy 是口语，正式场合用 man 或 person；buck 是口语的"美元"，写论文别用；kids 不在这一章，但同理。反过来，attorney、solicitor、plaintiff 这些词放在日常对话里会很怪。你要练的，是知道一个词"能在哪儿出现"，而不只是"它什么意思"。

再说词族。这一章里 -tion 结尾的词特别多：clarification、classification、consolidation、compilation、composition、consideration、consultation、conversation（不在表里但你天天用）。它们是动词的名词化，而名词化是四六级写作最省力的提分手段：把 decide 写成 make a decision，把 fail 写成 suffer a failure，句子立刻变长，句式立刻有地方可放修饰

语。但别过量，一段里两个就够，多了就成了官腔。

pre- 这一族也值得记：precautionary、preliminary、preparatory、prevention、preface、predatory。它们共一个"在前"的意思，但一个在风险前（预防），一个在正式前（初步），一个在行动前（预备），一个在书前（前言），一个在吞并前（掠夺）。你看，同一个前缀，五个方向。

最后说一件正经事。这一章里我最想让你记住的，不是任何一个单词，是那句"a measurement is a promise"。数字是承诺。你说这个样品导电率是多少，你就欠了别人一个可以复核的过程；你说这个模型准确率是多少，你就欠了别人一份说清楚"哪一列泄漏了标签"的说明。承诺这件事，跟聪明没关系，跟 sincerity 有关系。 sincerity 不是"我很真诚"这种自我表白，它是"我把不利于我的那一半也写进去"。

你今年大概也十九岁。你会遇到一个 restaurant 里的深夜，一次没做出来的实验，一个不肯说你好话的师姐，一份你看不懂的检查表。那时候你能做的事很少：把日期写上，把单位写上，把失败写上，然后 speak，用你自己的话，把这件事讲给一个人听。讲得清楚，你就懂了；讲不清楚，你就还没懂。yourself 这个词，说到底就是这个意思：不是"你自己"，是"你亲自"。

现在合上这一章。别急着背。去把你桌上那张写了一半的实验记录补完，把没有单位的那一行补上单位，把"大概""差不多"改成数字。做完这一件小事，再回来读第二遍。

数字是承诺。承诺是慢的。慢的人最后才被看见。

贴着正文走一遍：九个场景

第一个场景是组会。那一叠表里，agreement 是同意书，allowance 是工时额度，applicant 是申请人，accountability 是问责条款。四个词摆在一张桌子上，说的是同一件事：一个组织靠纸来记住谁答应过什么。顾老师写的那两个词 academic 和 accuracy，其实是一个意思的两面：学术这件事，说到底就是准确。他说 advance 来自 accuracy，从来不来自 arrogance——这句话你可以抄在本子上，因为它同时解释了为什么有的人越做越好，有的人越做越响。

第二个场景是配电间。cabinet 是机柜，gauge 是表，boiler 是隔壁的锅炉，ventilation 是那台不争气的排风。周敏立了三条规矩：所有事进日志，任何数字看两遍，报警先看表再跑。这三条规矩里没有一个生词，可它们就是这一章的方法论。你要是只记住 log 这个词，也行——因为 log 在这章里出现了十几次，每一次都是同一种意思：把发生过的事，写成别人能复查的样子。

第三个场景是数据清洗。数据字典贴在墙上，dilution 那一列写"零表示未稀释，不是缺失"，diameter 那一列写"指中位数不是平均值"，duration 那一列写"四十八小时意味着每小时一次循环"。你看，定义一列是什么意思，比跑一个模型难得多。deduct 是扣掉一行，audit 是抽查三行，approximation 是"差不多"，而"差不多"在数据里是要出人命的。

第四个场景是电镜室。queue 是那块白板，chamber 是样品腔，artifact 是伪影——那条亮带和中间那个黑洞。沈璐说的那句话，我希望你背下来：机器给出图像，图像不是数据，数出来的才是数据。她给你的不是 apparatus，是一只 counter。这就是这一章的题眼：聪明办法和笨办法的差别，不在于谁快，而在于谁留下了可以复查的痕迹。

第五个场景是那个 0.98。leakage 这个词不在表里，但它是这一节的主角。一个特征列里藏着标

签，模型就学会了除法，然后很自信地告诉你它懂化学。faith 是信，flaw 是缺陷，沈璐说"这不是你代码的缺陷，是你信心的缺陷"。接下来是 confession：徐把十四列逐个问了一遍"这一列能不能由标签算出来"，十四次写下 yes。这个过程不好看，但它是这一章里最像科学的一段。

第六个场景是我自己的失败。十二个样品，十二次失败，其中一次是因为我把日期写成 11/12，技术员读成了十二月十一日。forgetful 这个词用在我身上最准。于是我给自己立了一条规矩：日志不是日记，日志是写给一个陌生人的说明书，他要在你走了以后照着做一遍。repeat 这个词在这一章里的分量，比 discover 重。

第七个场景是那只档案箱。circular 是贴在门上的通知，chronicle 是编年史，finding aid 是清册，custody 是保管权。箱子里最动人的不是那些纸，是那些纸上的手写体：一个改了又改的采购记录，一张写着"我的第一颗单晶，1974"的照片背面。你会在这里遇到 ancestor、antique、colonel、Catholic 这些看起来跟实验室无关的词——它们不是跑题，它们是在说：你脚下的这栋楼，是别人一寸一寸垒起来的。

第八个场景是葬礼和校史馆。mourning 是哀悼，honour 是体面，ledger 是账本。邵教授的信里有一句我抄下来了："培养那些第二年还会给你写信的人。"这句话跟科研没关系，可它说的正是科研能发生的条件。校史馆的墙上写着 1937 年、1953 年、1998 年，写着庙里的课桌、两扇门做的实验台、被水泡坏的四百本书。你读完那一段，再回头看配电间那台老空调，感觉会不一样。

第九个场景是最后一夜。相关数字从 0.98 掉到 0.61，误差棒很宽，顾老师说"这是我们第一个诚实的数字"。然后是改标题：周敏说标题承诺了机制，文章只给了相关，五年后会有人引用标题而不读文章，一件小小的不诚实就是这样变大的。改标题花了四分钟，像丢了一样东西。这个"损失"，就是这一章真正想教你的东西。

九个场景走完，你会发现一件事：这一章里的词，几乎没有一个是"孤立"的。agreement 连着 accountability，artifact 连着 count，leakage 连着 faith，title 连着 promise。词和词之间有绳子，你要抓的是绳子，不是珠子。

所以我不建议你按字母背。你可以按"场景"背：想到组会，就把 agreement、allowance、applicant、accountability 一起提起来；想到电镜，就把 queue、chamber、artifact、counter 一起提起来。一次提四个，比一次背十个牢。而且这样记下来的词，是带语境的，写作时它会自己跳出来。

至于那些实在记不住的——auntie、boiler、diesel、vinegar（不在表里）、crouch——就让它们待在句子里。你记住"阿姨拎着桶从走廊那头过来"这个画面，auntie 就跟着画面一起留下来了。人脑记画面，不记列表。这是这一章写法上的全部秘密，也是我写它时唯一的私心。

那些长得像的两个词

英语里最贵的错误，往往不是生词，是"长得像"。这一章里就埋了好几对，我把它们拎出来。

affect 和 effect。一个是动词"影响"，一个是名词"效果"。你可以记一个画面：a 在前，代表动作先发生；e 在后，代表结果随后到。所以 the rain affected the sample 和 the rain had an effect on the sample 是一个意思的两种说法，但 affected 是动词，effect 是名词，写反了整句就塌。

insure、ensure、assure。三个词都有"使确定"的影子，分工却很清楚：insure 是买保险（insure the shipment），ensure 是确保某事发生（ensure that the lid is tight），assure 是让人放心（I assure you）。这一章里我写的是 insure，因为那批样品真的上了保险。

principle 和 principal。前者是原则，后者是主要的、校长。记法：principle 里有 ple，像 people，原则是给人用的；principal 里有 pal，像朋友，校长是那个跟你打招呼的人。

dairy 和 diary。奶牛场和日记。这两个词只差一个字母的位置，却是中国学生拼写错误榜的第一名。记法：dairy 里 i 在前，牛奶先来；diary 里 a 在后，日子在后头。

considerable 和 considerate。前者是"相当大的"，后者是"体贴的"。一个人 considerable 是说他有分量，说他 considerate 是说他会替你想。这一章里我用的是 consideration，它同时有"考虑"和"体贴"两个意思，看语境。

economic 和 economical。前者是"经济的、经济学的"，后者是"省钱的"。an economic policy 是经济政策，an economical car 是省油的车。这一章里出现的是 economist，经济学家，别和 economic 混。

historic 和 historical。前者是"有历史意义的"，后者是"历史上的"。a historic decision 是载入史册的决定，a historical novel 是历史小说。

respectable、respectful、respective。第一个是"体面的"，第二个是"恭敬的"，第三个是"各自的"。这一章里出现的是 respective：respective duties 是"各自的职责"，不是"值得尊敬的职责"。

moral 和 morale。前者是"道德的"，后者是"士气"。这一章里两个都出现了：morality 是道德，merriment 是欢闹，而 morale 你以后会在管理学里天天见到。

personal 和 personnel。前者是个人的，后者是人员。double l 的那个是"全体员工"，用错会在简历上闹笑话。

quiet 和 quite。一个是安静，一个是相当。这一章里出现的是 quite，它还带着英式美式的语感差别，前面讲过。

complement 和 compliment。前者是"补充、补足"，后者是"赞美"。这两个词连母语者都会写错，你只要记住：compliment 里有 i，像"I 喜欢你，所以夸你"。

stationary 和 stationery。前者是静止的，后者是文具。e 结尾的是文具（envelope 也是 e），a 结尾的是站着不动。

eminent 和 imminent。前者是"杰出的"，后者是"迫在眉睫的"。这一章里有 prominent，是"突出的"，三个词长得很像，程度和方向都不同。

alternate 和 alternative。前者是"交替的"，后者是"可替代的、另一种选择"。这一章里有 alternative 的意思（备用方案），没有原词，但你写作文一定会用到。

desert 和 dessert。前者是沙漠（也可以做动词"抛弃"），后者是甜点。两个 s 的是甜点，因为甜点总要再来一份。

adapt 和 adopt。前者是适应、改编，后者是采纳、收养。这一章里没有这两个词，但它们是四

六级写作的高频错点：adapt to the environment 是适应环境，adopt a method 是采纳方法。

我列这些，不是让你一次背完。你只要在读到"这个词我好像认识"的时候，停半秒，问一句：我认识的是哪一个。这半秒，就是精读和泛读的分界线。

再说三条搭配。be indicative of 是"表明"，of 不能换成 for；in the vicinity of 是"在……附近"，不能说 near the vicinity；comply with 是"遵守"，介词是 with，不是 to。搭配这件事没有道理可讲，只能整块记：把 be indicative of 当成一个词记，比记 indicative 有用。

最后一条，关于 it、he、us、our、themselves 这些"不算生词"的词。它们在这一章里被我特意放进去，因为四六级翻译和写作里，代词指代不清是最常见的扣分点之一。你写"the machine is wrong, and it is pleasant about it"，两个 it 指的不是一个东西，读者就会卡住。代词不是虚词，代词是绳结。

三个人的说话方式

这一章里三个人说话方式不一样，词也跟着不一样，你读的时候可以留意。

周敏说话像清单。她不说"你要注意记录"，她说"时间、改动、预期、结果，四样都写"。这种句子里的词都是硬的：log、date、column、duration、definition。跟这样的人学，你学到的不是词，是"把话说完整"的习惯。她也有软的时候，比如她说"你还不配用好时段，周六再来数四百个"，这句话里没有一个难词，可它比任何鼓励都管用。

沈璐说话像刀。她不说"你这个结论有问题"，她说"你的表是好的，你的结论是错的"。她用的词都短，都准：flaw、faith、artifact、promise。她还有一个习惯，喜欢把抽象的词按回具体的东西上："图不是数据""数出来的才是数据""你不能复核的东西不叫结果"。跟这样的人学，你学到的是"把话说准"。

顾老师说话像论文的引言。他喜欢用大词：arena、adherence、protocol、consolidation、accountability。有时候你会觉得他在绕，但你把他的句子抄下来，去掉修饰，会发现每一句都有一个明确的主张。他还有一句口头禅：这件事无聊的那一半，保护了有趣的那一半。这句话里没有一个生词，可它解释了这一章一半的词为什么要存在。

邵教授写信像旧时代的先生。他用 aristocrat、appeasement、arbitration、ascendance 这些大词，句子长，从句套从句，但每一段都有一个具体的落点：读合同、看数据、护学生、砌直墙。他信里最狠的一句是"不要把作品当成你的纪念碑，作品是一堵别人要在上面继续砌的墙"。这句话我抄在本子第一页，抄的时候把 monument 和 wall 两个词圈了起来。

还有一个人，Xu。他说话最短，常常只有一个词："不对""再来""成了"。他代表的是那种"手比嘴快"的人。这一章里他最重要的两句话，一句是"这不是模型，这是一台很贵的复印机"，一句是他把十四列逐个问完以后写下的十四个 yes。前一句是聪明，后一句是诚实。聪明和诚实，在他身上隔了半年。

你看，人物不同，词的密度和难度也不同。这不是巧合，是我写的时候故意安排的：一个人用什么词，说明他站在什么位置。你以后读任何一篇英文，都可以用这个办法判断说话的人：他是清单型

的，还是刀型的，还是引言型的。判断对了，生词就不再是障碍，而是线索。

一年之后

一年之后，我还会想起那个负三千四百。不是因为它好笑，是因为它是我第一次见到"自信"和"正确"可以毫不相干。那台机器用最漂亮的字体，最完整的单位，最肯定的语气，说了一个错得离谱的数。从那以后，我看任何一条光滑的曲线，都会先找它的误差棒。

这一年我学到的东西，几乎都可以用这一章里的词说出来。accuracy 不是"准"，是"经得起复核"；adherence 不是"遵守"，是"在没人看着的时候也遵守"；accountability 不是"负责"，是"签了字就认"；awareness 不是"知道"，是"一直知道"。这四个词，构成一个科研工作者全部的体面。

我也学到了另一面：arrogance 是会传染的，appeasement 是省事的，loophole 是总有人会钻的，corruption 很少从大恶开始，多半从"这一列反正没人看"开始。这一章里那些看起来最不相干的词——custody、jurisdiction、attorney、plaintiff——其实都在说同一件事：你的数据不是你的私事。

如果让我只留一句话给明年坐在这张桌子前的人，我会留沈璐那句：机器给你图像，图像不是数据。再补一句我自己的：你能复核的东西，才是你的。

最后说回那个箱子和那本日志。七月我把它们放回储藏室的架子上，标签朝外，日期朝上。我不知道下一个打开它的人是谁，也不知道他会不会读完。但我知道一件事：只要有人愿意把日期、单位、失败和改动写下来，这栋楼就还是实验室，而不是博物馆。

词是绳结，场景是绳子。你要抓的是绳子。

三个句式，从这一章里拿走

这一章有很多句子是"为了说清楚"才写成那样的，我把其中三个句式拆给你，写作时能直接用。

第一个是"不是 A，是 B"：It is not a race for the cleverest model; it is a race for adherence to a protocol。这个句式的力量在于先破后立，破的那一半要短，立的那一半要具体。你可以套：It is not a matter of talent; it is a matter of attention。四六级作文开头用一次，立场立刻清楚。

第二个是"没有做过 X 的 Y，就没有 Z 过"：A model that has never been wrong has never been checked。这是沈璐的句式，狠，且好记。它的骨架是"否定 + 定语从句 + 否定"，写议论段时用来收口最合适。类似的还有 A picture that nobody can reproduce is not a measurement。

第三个是"X 是承诺"：A measurement is a promise。这个句式的秘密在于把抽象的东西按到一个具体的名词上：a number is a promise，a title is a promise，a signature is a promise。你写作文时如果需要一个能背下来的结尾，就用它，但只用一个，用两个就成了口号。

还有一个不那么显眼、但特别有用的结构：not A but B 和 rather than。这一章里我反复用 rather than，因为它比 instead of 更适合放在书面语里，而且它后面可以直接跟名词、动名词、甚至从句。你写"我们不是要更多数据，而是要更清楚的定义"，就可以写成 we need clearer definitions rather than more data。一句话，两个名词，立场分明。

最后提醒一句：这些句式不要在一篇作文里全用。三个句式用两个，留一个给人喘气。写文章和做实验一样，密度太高，读者会累。

关于那些加粗

最后说一件小事：这一章里每个词只在第一次出现时加粗，后面就恢复正常。这不是排版偷懒，是给你留一条复习的路。第一遍读的时候，你看到的加粗词是"这一章要认的人"；第二遍读的时候，加粗已经消失，你得靠上下文把它认出来——能认出来，才算真认识。

所以复习的时候不要从头再读一遍加粗词表（本书里也没有这样一张表）。你只要做一件事：随便翻开一页，读三行，看看自己能不卡住。卡住的那个词，就是还没到手的那个。

还有一个用法：把加粗的词抄成一个"场景串"。比如"电镜"这一串是 queue、chamber、artifact、counter、stage、vacuum；"档案"这一串是 chronicle、custody、circular、finding aid、initial。一串五六个，抄在一张卡片上，比抄五十个孤立词有用得多。因为人脑记的是路径，不是仓库。

最后，别把这九百多个词当成任务。当成一次搬家：你把它们从一个叫"词表"的旧房子，搬到一个叫"这一章"的新房子里。搬完以后，旧房子空着也没关系，反正你以后要住的是新房子。

所以，明天早上你要做的不是背单词，是打开你那个写了一半的本子，把日期补上，把单位补上，把"大概是"改成数字。做完这一件小事，你再来读第二遍。那时候你会发现，很多词你已经认识了——不是因为你背过，是因为你在场。

顺带说一句：这一章里没有一个词是为了凑数硬塞的，有几个（比如 mentary、modity、mon）确实塞得很别扭，我把它们放进了"扫描件被切坏"的那一段，让它们以"残缺"的身份出场——数据会坏，这是这一章的主题之一，所以它们坏得其所。

第 7 章　车间里的夏天

The gate opened at six forty, and the guard counted us with a clicker, one click per head, the way a man counts screws into a tray. Access was through a side entrance hardly wider than a school gate, and he went through our belongings with the tired thoroughness of a man who had done the job for thirty years and had long since stopped expecting to find anything. We went in two abreast, five hundred of us on the first Monday of July, the alley behind us still full of scooters and soy milk and steam from a steamer that had been going since five.

Above his head, in letters repainted so often that they had gone soft at the edges, somebody had written: "Be here at six forty. Not six forty-one."

I was there at six thirty-two. I had been awake since five, in a dormitory I had never slept in, listening to Zhang Ye breathe like a man who had decided to be calm and was working hard at it. Zhang Ye had found his own internship at a bearing factory two

bus stops away, because a bearing factory, he said, was a place where you could see with your own eyes whether a thing was round. He was not being funny. That is the trouble with Zhang Ye: he is never being funny, and everything he says stays with you for a week.

The HR window was on the second floor, behind a partition of frosted glass. The interviewer — I had met her once in May, at a job fair where she took my application and asked me two questions, one of which I actually answered — did not remember me, which was a relief. She put a thin folder on the counter. There was a memorandum of understanding, an amendment to a standard procedure, an appendix of safety rules, and a designation form which said, in effect, that I was to be called an intern and not a worker, and that I would be paid a salary by the day. I signed where the yellow arrows pointed. My signature had to be duly witnessed, which meant that the woman behind me watched me write my own name, breathing on my neck. Then the interviewer said, in a voice with no weather in it, that I was accountable for my own belongings, that I was obliged to report any abnormal noise from any machine, and that the medical room was on the ground floor next to the washrooms. I said I would, and she stamped the form, and that was the whole ceremony. Nobody shook my hand.

In the corridor a boy of about seventeen was waiting with his own folder. He grinned at me and said, "First time?" I said yes. He said, "Me too. Ding. From the vocational school." His name was Ding Jian, and he had a timid way of standing with one shoulder lower than the other, and an enthusiasm that nobody had corrected yet.

The safety film was forty minutes long and had the look of something made in 2009 by the lowest bidder. The animation was so atrocious that the falling worker looked like a bent paper clip; nobody was amused, though two boys in the back row laughed at exactly the wrong moment and then had to study their shoes. The film showed a crash between a forklift and a tank of solvent, a scream with the sound turned off, an explosion of aluminium dust that lasted four frames, and a man who choked in a silo because he had gone in without a mask. Between scenes a narrator with a warm voice explained that hygiene was not a rule but a habit, that a machine guard was not an obstacle but a colleague, and that eighty per cent of accidents began with the word "just".

That last line I wrote down. I keep a notebook for lines like that.

Then they gave us our work clothes: an apron the colour of weak tea, a garment they called a jacket that was really a shirt with ambitions, gloves with the fingers stiffened by somebody else's sweat, and a locker whose door had a loose hinge and a button that had to be pressed twice.

The shop was four hundred metres from the office, across a yard where the sun lay flat and white. Inside, the temperature dropped for about six steps and then rose again, because the hall had a ceiling of corrugated sheet and no air that moved. A dim light came down through panels of dirty plastic. My glasses fogged. The air was moist and

tasted of oil; the office corridor had been chilly enough for a cardigan, and here the same air was a wet cloth laid over the face.

—

The alloy was 6063, which I knew from a textbook and did not yet know with my hands. Billets came out of the furnace the colour of a sunset and slid into the press, and the press pushed them through a die the way you push toothpaste through a pinhole, except that the toothpaste came out at four hundred degrees and the pinhole cost eleven thousand yuan. The extraction hood pulled the fumes up. A pump fed cooling water all day at a hum you stopped hearing after an hour. The motor was German and older than I was.

The press was automatic in the sense that nobody had to push it, and manual in the sense that three men had to watch it. One man watched the temperature, one watched the die, and one watched the man who watched the die. The instrument panel gave a reading that was accurate to one degree, and the whole art of the job, Chen said, was to know when not to believe it.

"Ascertain," he said, when I asked what he meant. "You check it. Then you check the thing you used to check it." He wiped his hands on his apron. "The gauge is accurate. The man reading the gauge is not."

His name was Chen Guoqing, and he had been at the plant since it was a repair shed with three lathes and a coal stove. He was fifty-eight. He had a humble manner and a knowledgeable eye and the kind of experienced calm that makes a young man feel that his own heartbeat is too loud. He called me "lad" and called the foreman "comrade" when he was annoyed, and he could strip a die and rebuild it in the time it took me to find the right wrench.

Chen had no eloquence at all. He spoke a Mandarin with the corners sanded off it, the way a carpenter sands a table until you cannot feel where the joins are, and he used maybe six hundred words in total. But he had a calling — I do not know a better word for it — and when he talked about metal he became, for a moment, a skilful and almost happy man. He was the first veteran of a trade I had ever met who did not once tell me that young people were worse than they used to be.

"Aim" was his second favourite word. "Aim at the middle of the tolerance," he said, "and you lose both ends. Aim at the top end and you keep one." I wrote that down too, and only understood it in August.

The line was laid out in a single long row, and each station was adjacent to the next, close enough that a man could hand a part sideways without turning. At the end of the row, parts were assembled into frames and assigned to a trolley by a girl with a scanner. The scanner beeped. The trolley moved. The arc of a welding torch flashed blue-

white in the corner of the shed eight times a minute, and each flash left a green ghost on the back of my eye.

二

Shen Lu arrived on the second day, which was lucky, because on the second day I broke a die.

She was in her fourth year, had been at the plant since March, and carried a caliper in her breast pocket the way other people carry a pen. She was meticulous to the point of being irritating, and perceptive enough to know it, and decisive in a way that I found, at nineteen, slightly frightening. Her insistence that a measurement be written down at the moment it was taken — not remembered, not reconstructed — came from a supervision report she had once had to defend line by line. "If it is not written," she said, "it did not happen. That is not bureaucracy. That is the difference between a factory and a rumour."

She was frank to the edge of rudeness and brief by principle, and she could make a brief sentence do the work of a lecture. On my third morning she looked at my notebook, which had twelve lines of temperature data and one line of adjectives, and said, "The adjectives are for your diary. The numbers are for me."

The foreman was a heavy man named Luo who had an arrogant way of standing with his hands behind his back, as if he were reviewing troops. He was not stupid and he was not cruel, but he had spent twenty years being obeyed by people who could not answer back, and it had settled into his shoulders. When Chen spoke, Luo listened. That told me something I have not stopped thinking about: Luo had the rank, and Chen had the thing that rank is supposed to buy.

The new press had been installed in April, and the old one had been ageing for a decade before that. The difference between them was apparent the moment you touched them: the new one was quiet, the old one talked. Chen preferred the old one, on the grounds that a machine which talks tells you what is wrong with it, and a machine which is silent only tells you when it has stopped.

The plant ran three shifts, and the second shift was always aggressive about its targets, because the second shift had the daylight and the first shift had the sleep. The schedule on the wall was printed in a font that made every number look like a threat. Somebody had drawn a small angel in the corner of it, wings and all, and nobody had rubbed it out for two years. I took that as a good sign about the place.

There were thirty of us on the yard crew that summer, half of them under twenty. The adolescent boys worked the packing line with a speed that was amazing and a care that was not. They were admirable in the way that young men are always admirable — full of

affection for each other, cruel to each other in the same breath — and they taught me, in four days, a vocabulary that no textbook would ever print.

On the first Friday I attended the weekly meeting, where the quality figures were read out and the auditing report was summarised in four sentences. The word "achievement" was used twice, and both times it referred to a number going down. I sat at the back with my notebook and felt, for the first time, that I was akin to these people: we were all counting something, and none of us had chosen the numbers ourselves.

The aspect of the job that surprised me most was the waiting. A press cycle is ninety seconds, and in ninety seconds a man can do nothing but watch, and watching, it turns out, is a skill with a cost. By the end of the first week I was so tired that I fell asleep sitting up in the changing room with one boot on, and Ding Jian took a photograph of me, which he later showed to everyone. I asked him to delete it, to no avail.

"Anyway," Ding said, when I complained, "you looked happy."

I said I was not happy, I was unconscious.

"Same face," he said.

Twice that week the line stopped for a breakdown, and both times the stoppage was attributable to a sensor that had been cleaned with the wrong cloth. The first time, I stood around with the rest of them and learned that a stopped line has its own sound: not silence, but the small noises of people who have nothing to do. The second time, Chen handed me a cloth and said, "You clean it. Then you will remember." I cleaned it. I have remembered.

三

The warehouse was the coolest building on the site and the only one that smelled of cardboard. Ware in the system meant finished goods; ware on the floor meant everything else. Bulky cartons stood four high, barrels of detergent leaned into a corner, buckets were nested by size, a coil of chain hung on a hook, and a board on the wall carried the stock figures in chalk, corrected in chalk.

I was lent to the warehouse for two days before the container closed, because the woman who usually did the checking had gone to hospital to have a baby, and because I could read English. That was the entire reason. In a factory, English is not a subject. It is a tool that gets lent out, like a wrench.

The container was a mixed one, thirty-two cubic metres, for a distributor who sold everything from hardware to underwear in a single market. I had forty items on the list and a hundred and nine cartons on the floor, and the job was to make the two agree.

I counted basins, blue, forty to a carton. I counted kettles whose lids sat loose. I

counted sponges pressed into bricks, detergent in bags printed with a smiling woman who had been smiling since 2011, lotion in pink bottles, lipstick in golden tubes, and zippers by the thousand, strung on a chain like fish. There were garments too: shorts, shirts, and a roll of silky cloth that Ding wrapped round his neck like a scarf in thirty-degree heat.

The quantity on the list and the quantity on the floor differed by four. Four kettles. I counted them eleven times. On the eleventh count I found the reason: carton 61 had been opened, four kettles taken out, and the carton closed again with the wrong tape, so that the seal was neat and the number was not. Somebody had wanted four kettles badly enough to break a compact and orderly system with a knife.

"Write it down," Shen Lu said, when I told her. "Do not write 'approximately'. Write four."

Inside carton 77, under a layer of foam, there was a laminated card the size of a postcard, printed in two columns, English on the left and Chinese on the right. It had been made by the trading company for its own clerks, and the typesetter had cut the long words at the margin and dropped the hyphen, so that the card read RECOM on one line and MENDATION on the next, as if the two halves had never met and never would. Further down, in a column of their own, stood MENTATOR, MUTER, MONWEALTH, MUNISM, HOMO, MAND. Ding thought they were a code. Shen Lu looked at the card for two seconds and said, "It is a printing fault. The words are recommendation, commentator, commuter, commonwealth, communism, homosexual, command. The machine ate the front half of each one."

I kept the card. It is still in my notebook, between the page with the temperatures and the page with the adjectives.

The chain of custody for that container was longer than I expected. The goods came from eleven manufacturers in three provinces, were consolidated in a shed not much bigger than a basketball court, and were loaded by a porter who had been doing the job since he was sixteen and who could tell the weight of a carton by the sound it made when he set it down. The tonnage on the manifest was 18.4. The operator of the forklift was a woman of about forty who could put a pallet within a centimetre of a wall and then look at you as though nothing at all had happened.

四

On the ninth of July a man was arrested at the gate, and for three days the plant talked about nothing else.

His name was Hu, and he had been the purchasing clerk for nine years. The allegation was that he had taken illicit payments from two suppliers in return for passing them the prices quoted by the other bidders. The company's own auditing had found nothing, because

Hu was careful; what broke it was a blackmail attempt. He had tried to blackmail a supplier with photographs, the supplier had gone to the police, and the photographs turned out to be of a dinner, a kettle of tea and a scandal that existed mainly in Hu's imagination.

The lawyer who came to brief the managers was a small, quick woman in flat shoes. She sat in the meeting room with a paper cup of tea and explained, in the tone of someone explaining weather, that the case was not about the money. "The money is small," she said. "The corruption is the point. Once a buyer can be bought, every price in the file becomes a doubt."

She used the word "enforcement" four times, and each time it meant something slightly different: the enforcement of the contract, the enforcement of the judgment, the enforcement of the company's own rules, and the enforcement that arrives at seven in the morning with a warrant. Ding asked, in a whisper, whether Hu would go to gaol. She heard him and said, "That depends on whether he pleads guilty and what the court makes of the confession he signed at two in the morning. A confession at two in the morning is worth less than a confession at ten, and everyone in the room knows it."

The interrogation had lasted six hours. Hu had confessed to taking 240,000 yuan over four years. At the trial — I did not go, but Chen did, because Chen had trained Hu on the floor in 2013 and felt, irrationally, responsible — Hu pleaded guilty to everything, was convicted, and was sentenced to three years, suspended. He walked out of the courthouse the same afternoon, which is not how it works in films, and Chen came back looking as if he had been to a funeral.

"They let him go home," Chen said. "His daughter is at university. He will not be able to pay for it now." He said it flatly, without any condemn in his voice at all, and then he was quiet for a while, and then he said the thing I remember: "A man who steals from a company thinks he is stealing from a building. He is stealing from the next man's salary."

The discharge case was worse, and smaller, and it happened in the same week. A workshop on the northeast edge of the estate had been running a ditch behind its shed, and the ditch ran, eventually, into a canal, and the canal ran, eventually, into the river. The enforcement officers came with sample bottles and a camera. Our own discharge was clean, but the scare went through the estate like a quake, and for a week every manager walked about with a clipboard and the expression of a man who has been wrongly accused and knows he will be accused again.

I asked Shen Lu whether the allegation against Hu could have been called a discrimination case — that is, whether a man might say he had been singled out. She said, "Arguable, if you have nothing else to say. The court will look at the invoices." Then she said something that I have since applied to almost everything: "Apparent

fairness and actual fairness are different aspects. The first one is a poster. The second one is a procedure."

五

The export office was on the third floor, with four desks, a printer, a fax machine that nobody had used since 2016 and nobody dared remove, and a map of the world with pins in it. Xu Jie ran the office. She was thirty-four, spoke English at twice my speed with a Hangzhou accent laid over the top of it, and had a custom of answering the telephone in Chinese and continuing in English without noticing that she had changed language halfway through a sentence.

She gave me a file and a chair and said, "Read this. Then tell me what is wrong with it."

The file was a draft bid for a German window-frame company. The covering letter was three paragraphs long and the memorandum of prices ran to two pages. I read it twice and found four things. The shipment date and the warranty period contradicted each other: the goods were promised for the last week of August, and the warranty was said to begin on the day of purchase, which for a distributor might be November. The discount for a quantity above twenty tonnes had been applied to a purchase of eighteen. There was no default clause at all, so that if the buyer simply stopped paying there was nothing in the document to point at. And the deductible in the insurance paragraph referred to a policy that had expired in March.

Xu Jie read my four points and did not compliment me. She said, "Good. Now write the email that fixes them without telling the buyer that we made them." That took me an hour. The word that saved the whole thing was "clarify": I wrote that we wished to clarify two dates and to specify the definite terms of payment, as though the confusion had been his. She read it and changed one word and sent it.

This is what the work actually looked like, most of the time: not a negotiation, but the repair of small definite failures beforehand, so that nobody would ever have to argue about them.

The video call with Bauer was at ten at night, because Hamburg is six hours behind. He was a heavy, careful man of about sixty with a charming habit of repeating your last three words before he answered, which made him sound as if he were considering you. His English was better than mine and slightly old-fashioned; he said "in addition" where I would have said "and". Behind him was a wall of sample frames and a window that showed a grey afternoon.

The negotiation itself lasted forty minutes and was almost entirely about the margin. Xu Jie had drawn the margin on the whiteboard that morning as a bar four centimetres

high, and then drawn what happened to it when the euro depreciated by six per cent: the bar lost a third of itself. "That is why we quote in dollars," she said. "Not because we love dollars."

Bauer wanted a discount of five per cent and a credit period of ninety days. Xu Jie offered two and sixty. He said that sixty days was difficult for a company in a recession, and she said that ninety days was difficult for a company in the same recession, and both of them were telling the truth, and that is what a negotiation is: two true sentences pointing in opposite directions until one of them moves.

At eleven-forty he agreed to two and a half per cent and seventy-five days, and then he said something that I have repeated to people ever since. "You are appreciative of your own product," he said. "That is rare in a young man. Be careful. A man who loves his product will believe his own ad."

Afterwards Xu Jie showed me the owner's history, because she thought I should know whose money I was spending. He had started at nineteen as a broker for scrap aluminium, buying offcuts from three state factories and selling them to a single foundry, and he made his first real money in 2003, in a recession, when everybody else was selling and he was buying. He was now, on paper, a millionaire several times over; in person he was a man in a polo shirt who ate in the canteen and knew the name of every operator on the floor. He had a cottage in Anji, a cousin in the county government, and a cruise he took every January with his wife, and he had never once, in my hearing, used the word "lucrative".

Every Friday the office circulated a one-page statement: orders, shipments, receipts, credit outstanding, cash in the bank. In addition there was a weekly meeting at which the finance manager read the overdue accounts aloud and everybody looked at the table. The numbers were also a burden: I watched a woman of fifty do arithmetic in her head for eleven seconds and then say a number that made the room quiet.

The plant had made one acquisition that year, a small anodising works forty kilometres away with twelve employees and a bleak yard. The word the owner used was not "acquisition". He said, "We took over a block of debt and a bold idea." The idea was that anodising could be brought in-house; the debt was why the price was low. It was, everyone agreed, a blow to the balance sheet in the short term and a good decision in the long term, which is a sentence I have heard about almost every decision ever made in a factory.

六

That was also the week the boycott started. An activist group in northern Europe published a report on aluminium from three provinces and called for a boycott of anyone who bought it. The report's central allegation was that a smelter had been built on

farmland without a permit; one of the eleven manufacturers in our container was named in the appendix. Xu Jie read the report at her desk with a pen in her mouth and said, "Considering that they misspell our town twice, this is a serious document." Then she did the thing I have admired ever since: she wrote a two-page reply in English, listed every certificate the company held, and sent it to the trade tribune that had printed the story. They printed three sentences of it. Three sentences, she said, was enough.

"The retaliation will come," the owner said at the Friday meeting. "It always comes. Somebody in Brussels will decide that our discharge is their crisis, and we will spend a year proving that our water is clean." He said it with no distress in his voice, like a man reading a weather report. "So we keep the certificates in a folder by the door." That folder was the dullest object in the building, and it had saved the company twice.

On the last Thursday of July the plant held its monthly study session in a room with forty chairs, a television that was never switched on, and a cupboard full of ceremonial objects: a flag in a plastic sleeve, a trophy from a volleyball tournament in 2014, and a statue of a mythical bird that nobody could explain.

The branch secretary read from a pamphlet for twenty minutes. There was a passage about Marxism and its Chinese form, a passage about Confucianism and the family, and a passage about the supremacy of law over every other kind of power. He read slowly, in the flat voice of a man who has read the same page forty times and still believes it. Ding slept with his eyes open, which is a skill that adolescent boys perfect and then lose.

"Every ideology becomes a dogma," the secretary said, "if you stop asking why." I wrote that down. I did not disagree with it, and I have never entirely agreed.

Afterwards, over tea, Chen talked about his grandfather, who had been born in a cottage with a mud floor and never learned to read, and who had walked eleven kilometres to a county congress in 1953 to hear a man explain that the land would belong to the people who worked it. "He came back and told my grandmother he had understood maybe a third of it," Chen said. "But he remembered the word liberate. He said it like a man trying on a new coat."

The notice board and my textbook were not the same object, and it took me most of that summer to stop expecting one to explain the other. The board was about who we were supposed to be. The textbook was about how aluminium behaves. Both were authentic. Neither was complete.

七

The exhibition was in the last week of August, in a hall so large that the roof seemed to float aloft like a cloud, and the countdown on the wall said three days. Two hundred manufacturers, a press corps of thirty, and a speaker system that made every

announcement sound like an emergency.

Our stand was nine square metres: a poster of a window frame, a catalogue on a small table, and a digital screen playing a four-minute animation of an extrusion press. I stood there six hours a day in a shirt that was too warm and said the same eleven sentences to two hundred people.

It was there that I began to comprehend what a chain of industry actually is. A man from a furniture factory in Foshan asked whether our alloy could be anodised in dark grey. A woman from a solar company asked about elastic limits at sixty degrees. A buyer from Turkey asked for a discount on a trial order, and I said, "I am the intern," which in that hall was a cardinal sin, so I learned to say, "I will find the person who can answer that," which is a better sentence in every language.

Bauer came on the second day with his wife, who wore a diamond and a silky scarf and asked me, kindly, whether I was a bachelor. I said I was nineteen. She said, "That is the same thing," and laughed at her own joke for longer than it deserved.

That evening the owner took six of us to dinner in a decorative old house by the canal, with a ceremonial order to the seating and a brandy he poured himself because he did not trust the waiter's hand. There was champagne for Bauer's wife, whisky for Bauer, and a beverage that tasted of peach for me. By ten o'clock the table had produced the kind of drunken eloquence that is either the truth or the exact opposite of it, and Bauer, who had been brought up in a conventional household in Bremen, told us about the night he had served as a juror in a case about a smuggler and a barrel of illicit brandy that had been illegal in three countries at once.

"You would have convicted him," he said.

"I would have asked about the barrel," Chen said, and everybody laughed, and Bauer wrote something on the back of a card and gave it to me: the address of a seminar in Shanghai on recycling. "You are too young to be a bachelor," he said. "Go and be a student somewhere else for a year."

八

Saturday was a half day, which meant we stopped at eleven and the afternoon belonged to nobody. Ding had a plan for it that involved eating; I had a plan that involved sleeping. We compromised by going to the clinic first, because Chen had told me that a biological test is the only honest conversation your body ever has with you.

The clinic was two rooms and a corridor on the ground floor of the dormitory block, with a doctor who was also the pharmacy and a queue of eight men who had all been told to come before breakfast. He took four tubes of blood, checked my eyesight by making me read a line of letters with one eye covered, and asked whether anyone in my family had

diabetes. I said my grandmother did.

"Then watch your diet now," he said, "not at forty." He wrote it down slowly, as though writing made it more true.

The results came back on Tuesday on a strip of paper the width of a receipt. My calcium was a little low. My liver was fine. The doctor looked at the sheet and said, "Your metabolism is the metabolism of a boy who skips breakfast." Then he asked whether I had been eating in the canteen, and when I said yes he nodded as if I had confessed to something and said, "Then the bacteria in your stomach are local. You will survive them."

There is a monthly test of the drinking water in the workshop, and the report is pinned to the wall beside the fridge: bacteria count, chlorine, hardness. Nobody reads it except the woman who signs it, and me, and an auditor twice a year.

The barber was on the corner between the clinic and the bus stop: one chair, one mirror, a radio that was always on, and a poster of a man with a facial shape that no human being has ever had. He was seventy-one and had been cutting hair in that chair since before the road was paved. He put his hand on my forehead, tipped my head forward, and went to work.

"Your forehead is high," he said, which I took for a compliment until he added, "you will lose your hair early." Then he told me about his hip, which had been replaced in 2019, and the thigh muscle that had never come back properly, and his little toe, which he could no longer feel in cold weather. He talked about his body the way Chen talked about the press: with a mechanism's affection, part by part.

His arms were bony and his hands were not, and there was something muscular in the way he held the scissors, as if the scissors were an instrument he had been born holding. When he had finished he held a small mirror behind my head so that I could see the back, and I said it was good, and it was good, and it cost fifteen yuan.

The public bathhouse was two streets down, a dwelling that had been a factory in the 1970s and a bathhouse since 1998. You paid at a window, got a plastic basket and a key on an elastic band, and went downstairs into a room of steam and tile. Men bathed in the open, without embarrassment, and talked about football and the price of pork with the calm of people who have nothing to hide because they are all, at that moment, equally ridiculous. I had a shower, then sat in the hot pool until the flush of heat moved from my chest to my ears, and I understood why Chen came here every Sunday and called it "the cheapest tourism in the city".

The housework of a dormitory is a small war that nobody wins. Ding washed his clothes in the sink with a bar of detergent and hung them on the fence behind the block; I swept, because I am the kind of person who sweeps. There was a fridge in the corner of our room, a microwavable box of rice inside it, a kettle, and a cupboard whose door had been taped

shut since before we arrived. The room was a rental, the block was a neighbourhood of forty such rooms, and the residents were a locality of people who nodded in the corridor and knew nothing about one another except the hour they got up.

On Saturday afternoon the mall was a kilometre away, and we walked it, past the pavement where a passerby was selling phone cases off a cloth, and up the escalator into a city of cold air. The skyline from the top floor was not really a skyline: a housing estate, three cranes, and a river the colour of weak tea. I stood at the window and thought that this was where the economy actually happened, in buildings that nobody photographs.

We ate in the food court, badly and happily. Ding had spaghetti with a sauce that tasted of onion and sugar, a sausage the colour of a traffic cone, and a hotdog afterwards, because he is nineteen and can do that. I had a sandwich, a bowl of soup with cabbage in it, and a plate of crab that was on offer because the season was ending. It was tasty in the way that juicy things are, and we ate with our hands and left the table in a state that a woman with a rag cleaned up without comment.

"It is a delicacy where I come from," Ding said, with a crab leg in his hand. "My father is a farmer. Wheat, mostly, and a hen house, and two cattle that are not for eating. When I was small he took me out to the field at four in the morning and I slept in the straw until the sun came up. I thought it was a punishment. Now I think it was the only parenting he knew how to do."

I asked whether he would go back. He said, "There is nothing to go back to. There is a field and a wheat price and a father who has opinions." Then he said, "But I will go back. That is the joke of it."

九

On the second Sunday of August Chen took me to his village, three hours inland by bus and forty minutes up a hill by a road that had been cut into the slope and never quite finished. The route left the railway station at six, crossed the plain, and then began to climb. The region changed the way regions do in that province: first rice, then bamboo, then tea on terraces, then a high plateau of tea bushes with cloud lying on it like washing.

"Agricultural township," Chen said, waving at a row of new houses with balconies and no fences. "Everyone here has a cousin in a city. Half the population is over sixty. The forestry office pays a man to walk the hill and count trees, and the agricultural office pays him to count the insect traps." He said it without resentment, which I noticed, because most men of his age said such things with a bitter taste in the mouth.

His house stood at the end of a trail of concrete slabs, with a pond in front of it

that had been a fountain once, a greenhouse behind it with three rows of tomatoes, and a buffalo in the next field that belonged to a cousin and had been retired for eleven years. "It eats," Chen said. "That is its job now."

His wife met us at the door. She had diabetes and a hearty way of pushing food at people, and within ten minutes of my arrival there were four dishes on the table and a kettle that never went cold. She called me "the undergraduate" and asked whether I could obtain medicine cheaply in the city, and then answered her own question: "No. The pharmacy in the township is cheaper, and the doctor there is a pensioner who observes things."

They kept a hen house with eleven hens, a dog that was obedient to Chen's wife and to nobody else, and a cat that was pregnant and restless and would not tolerate the dog within two metres. There was a mouse in the kitchen that had been evading traps for a month, and Chen's wife spoke of it with the respect one gives an antagonist.

After lunch Chen walked me up the hill to see the tea. The path went past a burial ground on the slope, small stone mounds with straw mats on them, past a shrine the size of a cupboard with three candles in it, and past a grove of trees that his grandparent had planted in the year Chen was born. He stopped there and said, "This is where my ancestry is. Every descendant who leaves comes back to look at these trees, and then leaves again."

I asked whether the family had always been there. "Since the county record began," he said, "which is 1782. Before that nobody wrote anything down, so we do not know." Then he said a thing I have thought about a great deal since: "In a factory, every part has a number and a record. A man does not. A man has a grave and a name and, if he is lucky, a grandson who remembers which hill he walked up."

On the way down we passed the village church, a plain white building with a cross on the roof, a gravel yard, and a notice board announcing that a bishop would visit in October. "Fourteen families here are religious," Chen said. "My aunt was one. She walked to the pilgrimage on the mountain every spring until her knees stopped her, and she could recite the gospel of John from memory in the dialect." He said it with the same flat respect he used for the procedure on the wall at work. "I am not a believer," he said. "But I have observed that a person who believes in eternity is easier to work with than a person who does not. They are less hysterical about a bad month."

The church had a hymn board with numbers on it and a sacred chorus that practised on Thursday nights; from the yard I could hear a woman running scales. Farther down the hill a radio was playing something rhythmic and symphonic that turned out to be a ballet score, and a girl of about nine was dancing to it in a yard with a graceful, entirely self-absorbed seriousness, while her grandmother watched from a bench and knitted.

Then tea, then the bus. On the way back, in the fading light, the hills went gray, then desolate, then tranquil in a way that no photograph of mine has ever caught. Chen slept with his forehead against the window. I looked at the cloudy sky and thought about thrift and generosity, which in that village seemed to be the same virtue wearing two coats: the thrift that keeps a house standing for forty years, and the generosity that fills a table for a stranger at an hour's notice.

The radio on the bus said that a hurricane — they used that word, though in our part of the coast we say typhoon — had turned and would come ashore on Tuesday night. Nobody on the bus reacted. A man two rows ahead said, to nobody in particular, "Every August."

十

On Monday we spent the whole afternoon tightening things. We dammed the ditch behind the warehouse with sandbags and a sheet of steel; we bound the tarps over the scrap with rope that had been used so often it had gone soft; we trimmed the branches of the two trees that leaned over the office car park, because a branch through a window costs more than a day of lost output. The mechanics drained the pump housing. The electrician went round the yard with a torch, checking every socket and every fuse, and told me that water and electronics are the two worst neighbours in the world.

"It is not the water," he said. "It is the salt in the water. Salt gets into a transistor and the transistor emits nothing and transmits nothing and the board is dead, and you cannot see it. You can only smell it."

By nine on Tuesday the wind had a pitch to it, a note that rose and fell; by ten the rain was coming horizontal, in ropes, and the yard was a shallow lake with a groove in it where the concrete had settled. The elevator was shut down as a precaution. The stark white light of the shop came through the windows and made the rain look like smoke.

At eleven the power went. The suspension of the whole plant — the press, the motor, the extraction fans, the lights, the cool boxes in the canteen — lasted four seconds before the generator coughed and took the load, and in those four seconds the shop was silent in a way I had never heard it. Then the lights came back at half brightness and the chaos began: alarms, radios, two men shouting into their phones, a forklift reversing with a beep that sounded, in that noise, almost cheerful.

The press had stopped with a die half through a billet, and that is a bad place to stop. The metal cools in the die and grips it; if you wait, the die is scrap. Chen was already there with a wrench and a strip of brass.

I held the torch. That was my job for two hours: hold the torch, keep it steady, do not sway. My arms began to quiver at about the twenty-minute mark, and by the fortieth I had to rest the torch on a rail, and Chen said, without looking up, "Good. A man who

cannot hold a light should find a rail."

The water came in under the roller door at midnight, a thin sheet that spread across the floor, found the groove in the concrete and ran along it like something with a purpose. Ding and I swept it back with a board, and it came back, and we swept it again. The wind moaned in the roof hatch and something up there banged every eight seconds, and each bang went through my chest. Once a sheet of roofing soared off the shed next door and splashed into the yard, and I froze the way an animal freezes at a sound it cannot place.

A man cut his hand on a strip of steel at one in the morning. It was a severe cut, deep across the palm, and he did not moan. He looked at it with the calm of a man who has been cut before, said, "It is a cut," and sat down on a pallet while the woman from the clinic cleansed the wound with saline and bound it in a bandage that was too wide for his hand.

By three the wind had eased. By four the rain had stopped, and the sky had gone a hard, clean grey, and the water in the yard stood still and held the shed upside down in it, and it was so tranquil that the whole night felt like a story somebody had told me.

The survey on Wednesday morning took four hours. The findings went onto a clipboard in Chen's terrible handwriting: eleven square metres of roofing gone, two pallets of scrap rotten with water, the fence down along the flank of the yard, one lorry with a cracked screen, and the statue of the mythical bird knocked off the wall of the study room and broken into three pieces.

Chen picked up the head of the bird and looked at it for a while. "Lastly," he said, "we put this back." Then he said the sentence I carried out of that summer and have not been able to improve on since: "You can survey a yard in a morning. You cannot survey a habit. The habit is what saves the yard." He put the bird's head in his apron pocket. "A factory is not a machine," he said. "A factory is a habit. The machines are only the part of the habit you can see."

十一

The last week was a handover, and a handover is an education in how much you did not know you were doing.

Shen Lu gave me a template. My job was to compile the file, confirm every line with the person who owned it, and ensure that whoever came next could open the folder and understand it without me. "Accomplish that," she said, "and you will have produced a submission worth reading."

Every row asked the same four things: what the task is, when it is due, who owns it, and how you know it is finished. The fourth question is the one that separates a

practical person from an optimistic one, and I answered it badly nine times before I answered it well once.

I spent three days on the file and produced nine pages, including a flow chart of the packing process and a series of screenshots from the ERP system, a program written in 2006 and held together by habit. Two lines of my report were censored by the sales office because they named a customer, and the customer list itself was classified and kept on a machine with no browser and no networking, only a cable.

Chen read the nine pages and made one correction. He added a column headed variation. "Every batch varies," he said. "If your procedure has no place for variation, it will break the first time a robust solution meets a rigid one." Then he asked me what the utility bill had been in June, and when I said I did not know, he said, "The usage of water per tonne is on the same sheet. A vacuum pump that runs all night is value pouring out of a hole in the floor." He was not angry. He was systematically disappointed, which is worse.

The evaluation form had twelve items, each marked from one to five, and a box at the bottom for the supervisor's opinion. Chen filled the box with four sentences, none of them longer than eight words, and one of them was, "He asks why." I have been told worse things about myself.

十二

On the Thursday there was a sport evening, because the plant had entered a team in the township tournament and had lost in the first round, and the manager had decided that the remedy was a gymnasium.

I had not been inside a gym since March. The floor was pale and the noise came off it in a way that made everybody shout. There was a volleyball match between the packing line and the extrusion crew, refereed by a man with a whistle and a sense of drama; the packing line won, because they were younger and because their best player — a girl of twenty-two who had been a runner at provincial level and still trained every morning — could make the ball bounce off the ceiling. Somebody said she had gone to the Olympics trials in 2024 and had come thirteenth. Nobody asked her about it. She was a packer on the second shift, and that was the whole story.

I played badly. I am not fit and I am not talented, and I have the reactions of a student who has spent a year at a desk. But I cheered for forty minutes with the rest of them, and when the score reached the last point and the hall made the kind of noise a building remembers, her face was radiant with effort and I felt the thing I had been missing all summer: the feeling of being ours rather than mine.

Afterwards we went to a grill house behind the mall and ate until the merry noise

dropped to a murmur. Ding fell asleep with his head on the table, then on my lap, and then, when I moved, on the shoulder of a woman from the accounts office who did not seem to mind. There was beer brewed on the premises, and a plate of grilled fish that came apart at the touch of a chopstick, and somebody ordered a huge bowl of noodles that was gone in four minutes.

At eleven I walked back alone, which I had begun to do at the end of the week, because a lonely walk at the end of a hectic day is a kind of refreshment and nobody tells you that at nineteen. The road was empty, the air was cool at last, and a dog barked at me from behind a gate and then, having decided that I was nobody, stopped.

十三

Shen Lu left on the twentieth of August, a Friday, with two boxes, a bicycle and a certificate in an envelope.

She had three offers and had taken the smallest employer, a company of ninety people that made parts for medical machines, because, she said, "in a company of ninety you can see the whole process, and in a company of nine thousand you can see your desk." Her proficiency in English had got her the second interview and her proficiency with a caliper had got her the first, and she was frank enough to say that the second one mattered more.

I asked her about the unemployment figures for our year, which were said to be bad. She said, "They are bad and they are not the point. The point is that a degree is a ticket, not a destination. You can get on the train with it. You cannot live on the train."

Then she said the thing I think about when I am deciding anything: "Choose the job where you will be allowed to be wrong in front of somebody who can correct you. Everything else is salary."

I asked what she meant by everything else. She said, "Precisely what I said. Everything else."

十四

On my last Saturday I went to the museum, because it was free, because it was raining, and because Ding had told me there was a room with a dinosaur in it and I did not believe him.

There was. It stood in a hall of its own, fifteen metres of bone and shadow, with a cub beside it that had been found in the same layer of rock, and around it a ring of children who had all, at some point that morning, been told not to touch it. The label said the species had lived for eleven million years and then, in a geologically short

time, disappeared. I stood there longer than the children did.

The next room was silk: a veil from the Song dynasty with a flawless edge, a dress of silk so fine that it hung like water, a fragment of golden thread, and a cane with a carved handle that had belonged to a monarch's cousin. The label was three lines long and mentioned oppression twice, once about taxes and once about slavery, which in that dynasty was mostly debt bondage. In the next case there was a cult of ancestor worship, with an altar and eleven bowls, and a text that illustrated how an aristocratic family kept its ancestry in order.

Ding read the labels aloud in a whisper, and every third sentence began with "It says here", and by the fourth room I had stopped correcting him.

In the bookshop downstairs I bought a paperback encyclopaedia of word origins for twenty-eight yuan, because it was the only English book in the shop that was not a textbook, and because I had begun to keep a list of words that had intrigued me on the shop floor.

I read it on the bus. There was an entry on abstract words, which said that most of them began as metaphors and then forgot what they had been metaphors for. There was an entry on the adverb, which said that the -ly ending is a worn-down form of "like". There was an entry on affixation, which explained that English builds new words out of old ones the way a workshop builds parts out of stock. And there was an entry on antonym, which admitted that most words have no true opposite, and that we behave as though they do because it is convenient.

I liked that one. I wrote in the margin: an opposite is a convenience, not a fact.

There was a chapter on jargon, which the author defended. "Jargon is what a trade calls its own shorthand," he wrote, "and it becomes jargon in the bad sense only when it is used to exclude the person who needs to understand it."

That sentence I copied out in full. It described, better than I could have, the laminated card in carton 77.

十五

Bauer flew home on the twenty-fourth, and the plant sent two cars to the airport, which is the kind of thing a company does when a customer is worth more than the cars.

The terminal was cold and bright. From the window we could see the cockpit of a parked Airbus lit up like a small office, and beyond it the runway, and beyond that the flat green edge of the delta. Bauer's son flew gliders, and he showed me a photograph on his phone of a white glider hanging in a blue sky at an altitude of two thousand metres. "That is the quietest room in the world," he said. "No engine. Only the air, and you, and the decision."

Aviation had never interested me before. It interested me for about four minutes that afternoon, which is how most interests begin.

His departure was at four. He shook my hand and said, "You will not stay in a factory." I said I thought I might. He said, "Then you will be a very unusual businessman," and went through the gate with his ticket in his teeth, destined for a continent I had never seen, and I stood at the glass until the plane was gone.

十六

The last day was a Friday, and it was, literally, thirty-eight degrees by ten in the morning.

Chen gave me a card at lunch. It was a folded sheet of paper with the plant's name printed on the front and eleven messages inside, in eleven different hands, and I have it still.

Ding wrote: "Do not misunderstand me. You were the worst intern we have had, and I have enjoyed it." The packing-line girl wrote: "Come back when you can grasp a die without dropping it." The electrician wrote: "Never paste anything onto a live board, and never override a switch because it is faster." Xu Jie wrote: "Reply to emails within a day. Persistence beats eloquence." Luo wrote four words: "Do not be late."

Chen wrote two lines, and they were the only ones in the card that were not funny. He wrote: "You have patience. Use it on something worth it."

I did not know what to say, so I said thank you, and he waved it away with the impatience of a man who has been thanked too often for things that cost him nothing.

In the afternoon I walked the line once more, slowly, the way you walk a house you are leaving. The press was running. The arc flashed. The extrusion came out of the die at four hundred degrees and slid along the cooling table, and the constant hum of the pump was there, and the smell of oil and hot metal was there, and I understood that I would never be able to separate this smell from that summer, and that this is what people mean when they say a place gets into you.

At six the whistle went. The gate opened. The guard clicked his clicker, one click per head, the way he had on the first morning in July, and this time one of the heads was mine for the last time, and he did not know it, and I did not tell him.

十七

Every station had a laminated card clipped to it, and the cards were printed in Chinese and English, because the plant had once had a customer who insisted on reading them himself.

I copied one into my notebook and translated it line by line, which is the slowest way to learn a language and the only way I have ever trusted.

"Before the shift," the card said, "ensure that the guard is intact, that the emergency stop is accessible, and that the die matches the number on the card. Confirm the temperature with a second instrument. Do not override a warning because the schedule is tight. Excessive speed is the most common cause of a defect that is not evident until the final inspection, and haste is detrimental to quality and conducive to injury. Exclude water from the die area. A conscientious operator has fewer accidents than a fast one."

I read the last sentence to Ding. He said, "It means: do not be an idiot." Which is a fair translation, and also, I think, the whole of industrial safety.

At the end of the placement the school required a report of eight hundred words. I wrote mine in English, because Wang laoshi had asked for it in English, and I have never worked harder on eight hundred words in my life.

I wrote that I had spent nine weeks at a manufacturer of extruded aluminium profiles; that I had corresponded with a buyer in Hamburg; that I had investigated two discrepancies in the shipment records and found that four kettles had left the warehouse without a signature; that I had compiled a handover file of nine pages, of which seven survived the censorship of the sales office; and that the completion of that file had taught me more than the whole of my first year at university.

Then I wrote a sentence that I deleted four times and put back five: "The most useful thing I learned is that a tolerance is not a number in a book. It is an agreement between people who will never meet."

十八

Zhou Min telephoned on the last Sunday, from the lab, at ten at night, which is the hour at which people who work in labs telephone.

She was two years into her research. She asked what I had learned. I said I had learned that a factory is a habit. She said, "Then you have learned the only thing worth learning this summer, and you learned it in nine weeks, which is fast."

I asked her what she was doing. She said she was investigating why a batch of cathode material had come out with a capacity eight per cent below expectation, and that she had spent three weeks on it, and that the answer, when it came, would probably be foolish and small: a thermometer that read half a degree high, or a stirrer that ran ten seconds short.

"That is the irony," she said. "You spend three years learning advanced techniques so that you can find out that a thermometer was wrong."

She asked whether I wanted to come into the lab in September. I said yes before she had finished the sentence, which she said was the correct answer, and the only one she had ever given herself.

Then she said something about research that I have written on the first page of every notebook since: "A result is not a fact until somebody else can repeat it. Until then it is only an event."

十九

I went back to campus on the last Sunday of August, on the slow train, with a suitcase, a cardboard box of notebooks, and a plastic bag of tea that Chen's wife had pressed on me at the bus stop.

The train was half empty. A man across the aisle was eating sunflower seeds and putting the shells in a cup. Two students behind me were arguing about a dormitory allocation. The announcement said the train would be delayed by twenty minutes, and everybody sighed the sigh of people who had expected it.

I opened my notebook at the page with the temperatures and read down it: the eighth of July, 486 degrees; the twelfth, 489; the twentieth, 483, when the die had worn and Chen had changed it in eleven minutes. Beside the numbers, in the margin, I had written the words I did not know yet, and beside the words I had begun, in the last weeks, to write the answers.

There were forty-one words on that page. Nine of them I had learned on the floor, from Chen or Ding or a label or a card. The other thirty-two I had looked up at night, in a paperback encyclopaedia, in a rental room, with a fan turning.

That is what a summer does, I think. It does not make you wise. It gives you forty-one words and a habit, and the habit is the gift.

二十

There is one evening from that summer that I have not written down, because for a long time I did not know where it belonged.

It was the Wednesday before I left. Ding took me to the market behind the bus station, the one that opens at nine at night and closes when the last customer goes home, and we ate at a stall with four stools and a plastic sheet for a roof. Ding ordered for both of us: crab, a plate of grilled onion, a bowl of clams in broth, and two bottles of a local brew that was cheap and stronger than it tasted.

I remember that Ding knew the name of every stall owner on that street, and that each of them called him by a nickname. I remember a man with a puppy on his lap, and Ding explaining that the puppy was a present for his landlady, who had six cats and did not

want a seventh. I remember a photographer who came round the tables with a camera and a small printer in a bag, who took our picture for fifteen yuan and printed it on the spot, and who told us, without being asked, that he had once photographed a wedding in a village that had since been demolished. I remember that the picture was bad — my eyes half closed, Ding grinning like a fool — and that it is pinned to the wall above my desk, and that it is the only photograph of that summer in which I am laughing.

I remember Ding saying, at about midnight, with the confidence of a nineteen-year-old: "You will forget us." I said I would not. He said, "You will. It is natural. You will remember the machine and the heat, and you will forget the people, because people are notable only while you are near them." I said that was the most melancholy thing I had heard all summer. He said, "It is not melancholy. It is just true. And it is why I keep the photograph."

二十一

On the train I read a newspaper, which is a thing I had not bought in a year, and there was a feature in it about why young people will not go into factories.

The writer had spent a month in three provinces and had come back with statistics and anger in about equal parts. The majority of the people he interviewed said they would prefer an office, not because the pay was better but because an office had windows and a toilet they did not have to share.

One paragraph I read three times. "The predecessor of every shortage is a presumption," it said. "We presume that a young person will choose the higher wage, and we are astonished when he chooses the shorter commute, the better canteen, the cleaner dormitory. A factory that treats those things as luxuries will lose him to a warehouse that treats them as infrastructure."

That last word stopped me. Infrastructure usually means roads and power lines and the pipes under a city. The writer was using it for dormitories and canteens and buses, and he was right, and I put the newspaper down and looked out of the window for a while.

There was another paragraph about inertia: how a plant that has done a thing one way for twenty years will keep doing it even when the reason has gone, and how the inertia is not stupidity but a kind of loyalty to the people who built the thing. "You cannot argue a habit out of a building," the writer said. "You can only outlast it, or replace the people, and the second is expensive and the first is slow."

I thought about Chen, who had argued with the new press for a year and had been right about the guard and wrong about the speed, and I thought that inertia is not the opposite of intelligence. It is what intelligence turns into when it has nowhere to go.

二十二

I have the card from the shop in my notebook, and the card from carton 77 behind it, and the photograph from the market in a plastic sleeve on my desk.

When people ask what I did in the summer of my first year, I say I worked in a factory. When they ask what I learned, I say I learned how a tolerance becomes a contract, and how a contract becomes a container, and how a container becomes a price on a shelf in a city I will never visit. That is the chain, and I can now follow it from the billet to the shelf.

What I do not say, because it takes too long, is that I learned the difference between a material and a product: a material is what you specify, a product is what somebody buys, and the whole difficulty of engineering lives in the gap between the two.

Chen would have said it in six words. He did say it, on the last afternoon, when I asked what he wanted me to remember. He thought about it, wiped his hands on his apron, and said: "Measure twice. Then measure again."

二十三

The viva was on the eleventh of September, in a room with a projector that worked and a window that did not open, and it lasted twenty-two minutes.

I had prepared eleven slides about the packing line, and Wang laoshi let me get to slide four before he stopped me. "Demonstrate one thing," he said. "Not eleven. One."

So I demonstrated the kettle problem: the list, the count, the eleven recounts, the tape. I had brought the card from carton 77 and I passed it round, and the room went quiet in the way rooms do when a document is more persuasive than a person.

The questions were harder than the presentation.

"How do you know the four kettles were not stolen?" Wang laoshi asked.

"Because the tape was new and the seal was neat," I said. "A thief does not retape a carton neatly."

"Factual," he said. "Good. Now the extent: four kettles out of twelve hundred is a ratio of one in three hundred. Is that a crisis or a rounding error?"

I said it was neither. I said it was a feature of a system with no second pair of eyes, that the formation of the habit of counting was the only preventive measure available, and that I had learned it from a woman with a clipboard whose impatience I had first mistaken for rudeness.

Zhou Min asked about the inherent rigidity of the alloy and whether the intermediate cooling phase had been intensive enough. I said I did not know. She said, "Then say that. An honest 'I do not know' is worthwhile. A speculative guess is tiresome."

Meanwhile Wang laoshi had found the page where I had written that a tolerance is an agreement between people who will never meet. He read it aloud, and then he said, "This is the only sentence in the report that could not have been written by a machine, and it is also the only one you did not illustrate."

Then he asked the question that ended the viva. "A factory is a habit," he said. "Whose habit?"

I said, "Chen's. And the woman with the clipboard. And the man who counted kettles for thirty years, stopped expecting to find anything, and kept counting."

二十四

Afterwards we ate in the noodle shop by the east gate, and Shen Lu came, on her way to her new job, with a necklace she had bought out of her first month's pay and a leaflet from her new company that she was sceptical about.

She said the job was primarily secretarial, which she had not expected, and that she had decided to be versatile about it. "Progression is not always upwards," she said. "Sometimes it is outward." Then she shrugged and ordered soup.

Ding telephoned while we were eating, somewhat drunk, to say that the turnout at the volleyball final had been four hundred people, that the plant had won, that he had been selected for a technician scheme, and that his pronunciation of "sophisticated" had improved because he had been practising it in the mirror.

"You are mad," I said.

"I am joyful," he said, which is not the same thing, and is better.

Before she left, Shen Lu gave me a proverb that her grandmother used: "Measure twice, cut once." It was the same advice Chen had given me in different words, and it is, I now think, the whole of engineering.

I asked whether she was worried about the new job. She said, "Tremendously." Then she said, "That is how you know it is worth doing."

二十五

In October the plant sent me a notification that my overtime had been paid, a leaflet about a safety month, and a form asking whether I would come back next summer.

I filled in the form on the fourteenth of October, which was a Sunday, and in the box marked comments I wrote: "Keep a place for me on the second shift."

二十六

Chen's nephew was wed on the last Saturday of August, in the village, and I was

invited, which surprised me more than anything else that happened that summer.

The marriage had been arranged, in the way these things are still arranged, by two families who had known each other for three generations and by a matchmaker who was also the bride's aunt. The ceremony was short. There was a bow to the parents, a bow to the ancestors, and then a multitude of small things that nobody had written down and everybody knew.

The dragon came at seven: eleven men under a cloth dragon went round the courtyard three times at a run, and the children screamed with a delight that was entirely real and entirely terrifying to me. Then the fireworks, which in a village are not a display but a conversation — each family sets off its own, the noise goes on for an hour, and everybody salutes the sky and goes back to eating.

I was, for that evening, a visitor, and I was treated with a hospitable exaggeration that I did not know how to answer. I was given the seat by the door, the tender part of the fish, and a rug for my knees because the night had turned cold. I was asked eleven times whether I was married, and eleven times I said I was nineteen, and eleven times the answer was received as a joke.

Chen's wife's sister sat next to me and told me about the nursery where she worked, in the township clinic: about the orphan they had kept for three weeks while the county office found a place for him, about the handicapped boy who came every Thursday, and about how the clinic had asked for a second nurse for four years and been refused four times. I listened, and the harmony of that evening was made of a hundred small niceties, and it was wholesome in the old sense of the word — it did me good — and I have not often been able to say that about an evening.

Then the bride's mother sang, badly and without embarrassment, and everyone clapped in time, and I understood that a wedding is a machine for turning two strangers and a roomful of neighbours into a family.

二十七

On the Monday a delegation came from the provincial industry department, with an academician in it, and the plant spent two days cleaning things that had been dirty for a year.

They walked the line in shirt sleeves and safety shoes, and they asked questions that Chen answered in six words each. The trade association that had organised the visit sent a leaflet afterwards with a photograph of the group standing in front of the press.

The academician was a small woman of about seventy with a watchful way of looking at things. She stopped at the ageing oven and asked one question: how many hours, and how do you know. Chen said nine, and because the thermometer says so, and because we cut one

open every shift. She said, "Then you are doing what half the laboratories in this country do not do, which is to check the instrument against the thing."

I have remembered that sentence for a different reason than the one she intended. She meant it as praise of the plant. I heard it as a description of a way of living: check the instrument against the thing.

二十八

I went back once more in October, for two days, to finish the handover with the new intern: a boy of twenty from the vocational college who had the same timid shoulder as Ding and much better handwriting than mine. I gave him the card from carton 77 and told him what it was. He looked at it for a while and said, "So the machine ate the front half."

"Yes," I said. "The machine ate the front half. Your job is to know what the whole word was."

二十九

In September Zhou Min took me into the lab on a Saturday, which is the only day the building is quiet enough to hear the fume hoods.

Her bench had a magnetic stirrer, one compartment of a glovebox the size of a suitcase, and a row of bottles whose labels she had written herself. She was making a combination of two cathode powders that had never been tried together, and she was on her third attempt, because the first two had produced a density she did not believe.

"When the numbers coincide twice," she said, "you may believe them once."

She explained the whole thing in eleven minutes, which is her gift. The core of it was a chunk of lithium compound the size of a fingernail, and the reason a battery works at all is physics: the orbital arrangement of electrons in a crystal, and the fact that some crystals will let an ion in and out without falling apart. The failure she was chasing was the depletion of the electrolyte at the surface, which shows up as a fall in concentration, looks like a bad batch, and is really a bad interface.

"Do you clone cells here?" I asked, because I had seen a machine with that word on it.

"No," she said. "We grow them. Cloning is what you do when you want many copies of one thing. We want many copies of a thing that is slightly different every time."

The initiation of a new student in that lab is a month of washing glassware, and she defended it. "You learn the implicit rules of a place by doing the boring work in it," she said. "You learn where the inferior reagents are kept. You learn that the indicator

on the wall is not decoration."

She was instinctive about one thing only: she checked the blank first, every time, before she looked at the sample. "What looks inborn," she said, "is usually ten thousand hours."

I asked whether science was a matter of individualism. She said, "There is no individualism in a lab. There is concentration, there is a repertoire of techniques, there is a deadline, and there is a group of people who will tell you that your beautiful result is wrong."

"The intellectual part of this job is bigger than people think," she said, "and the cognitive part is smaller. You do not need to be clever. You need an innovative habit and a boring one, and the boring one is checking."

Then she said the thing I wrote on the first page of a notebook and have not crossed out: "An impromptu idea is worth nothing until you have built a refutation of it."

That is not the same sentence as Chen's. Chen said: measure twice. Zhou Min said: try to prove yourself wrong. Both of them meant: do not trust the first answer because it is yours.

三十

The rest of that autumn came in pieces.

In October I went to a university match with Zhang Ye, who had come back from his bearing factory three kilos heavier and much happier, and we watched our department lose by eleven points to a centre who could not be stopped. Afterwards, in the corridor, a girl I did not know kissed him on the cheek and walked away, and he stood there with the joy of a man who has been handed something he did not expect, and said, "That is my lab partner. I have no idea."

November was exams. I did badly in one and well in three, which is typical. Every term has a stagnation in the middle of it, when nothing you read stays in your head, and then a week when everything does; the subsequent recovery is not a miracle, it is just sleep.

Ding sent me a photograph in December of the new line's first shift. He was in the middle of it, grinning, with his arm round the woman from the accounts office, and behind them the press was running. The picture was a little blurred because somebody had moved. I put it on the wall next to the picture from the market, and the two of them together are the whole of that summer for me: one evening under a plastic sheet, one morning on a shop floor, and in both of them somebody is laughing at something I cannot hear.

三十一

In December the television in the common room came back to life, and eight of us watched a drama about a gifted pianist and a married man, which is the plot of every drama made in that country and also, apparently, of every life.

Zhang Ye watched in silence and delivered one judgement at the end of each episode. On the third night he said, "Seduction is a tactic, not a feeling." On the fifth he said, "The piano is hers. That is the only honest thing in it."

There was a flirt at an office in it, and a glamour that no office has ever had, and a nasty mother-in-law, and a hysteria that arrived every episode at the forty-minute mark, and a family firm whose doom was announced in the first episode and postponed for thirty. The havoc of one scandal kept four episodes going. It was foolish, and we watched it every night, and indeed there is no accounting for what people need in December.

三十二

The last thing I translated that summer was not a card or a contract but the employee handbook, which the plant had rewritten after the Hu case and which nobody had yet put into English.

It was thirty-one pages. I did eleven of them, and I have never read a document so carefully in my life, because every sentence had a constraint in it.

"Mission," the first page said. "This company exists to make aluminium profiles that perform as specified, on time, at a reasonable price, and to safeguard the health and safety of every occupant of this site. Protection of the environment is not a public relations exercise; disposal of waste is audited and the figures are published."

The permanent staff and the temporary staff were subject to the same rules, it said, unless a rule expressly said otherwise, and the distinction mattered mainly at the termination of a contract, which was the page everybody turned to first.

"No dismissal without a record," it said. "No dismissal during a medical leave. Neglectful supervision is a disciplinary matter. Behavioural safety — the habit of not walking under a load — is part of the job, not an addition to it."

There was a page on confidentiality which forbade the removal of customer lists, and a page on gifts, which imposed a limit of two hundred yuan, and a page on alcohol that was, in that plant, mostly ignored at weddings. The imposition of a limit is always the easy part; the enforcement is the hard part, and the handbook, to its credit, said so.

I translated it faithfully, including the typos, because a translation that improves the original is a lie with good manners.

三十三

I went home for the winter on the eleventh of January, on a train that was two hours late, in a carriage where five strangers became, for four hours, a small and temporary society.

The man by the window was a township official of about fifty, on his way to a congress in the provincial capital. He had been elected to it twice, and he talked about the electoral procedure in his own township the way Chen talked about a die: precisely, and with an eye on where it could go wrong. "The voter turnout was ninety-one per cent," he said. "The politician who tells you that is the whole story has never counted the ballots."

He had spent four years on the eradication of an illegal dump. "We had to prosecute two people," he said, "before anybody believed the rule had been enacted." He talked about a consultative meeting at which the advocacy of a residents' group had changed a resolution that the establishment had already written. "A resolution is a compromise written down," he said. "It is not a virtue. It is a record."

He was sceptical of militancy in any form. In one village, he said, a leftist faction had held dominance for twenty years, and the militancy had gone out of them the day they won, which he considered a good argument against winning too completely.

The woman opposite me was a nurse from a maternity hospital. She had been in surgery that morning, assisting, and she talked about a tumour the size of a fist, a boy of nineteen who would not feel pain for another week because of the drugs, and a recovery that had taken four months and had been, she said, mostly a matter of feeding him and letting him sleep. "People think mercy is a feeling," she said. "In a ward, mercy is a clean sheet and somebody who answers when you call." Then she said that the health of a hospital is not in its machines but in its night shift, and I wrote that down, because it was the third time that year that somebody had told me a version of the same thing.

The boy in the corner was nineteen and had been an athlete at school, swimming and some cycling, until his knee had gone. He talked about the dam his village had built when he was a child, how the crows had moved to the other side of the water, and how you could still see an eagle over the southwestern ridge on a clear morning in May. He talked about ripe fruit going rotten on a tree nobody owned, and about a scary night on the water when the engine failed. Then he put his headphones on and fell asleep, and the carriage was quiet for a while.

The old woman by the door began to talk, because old women by doors always begin to talk. She had married in 1962, a marriage arranged by two aunts. Her husband had been honourable and violent in about equal parts, and she said so without complaint, as a fact about the century rather than about him. "He was a violent man and an honourable one," she said, "and I have hitherto never explained how both can be true." She said there had been glamour in the film people and none in them, and that they had not minded. She said

she had read romance magazines in the 1980s, that her daughter had thought them a waste of money, and that she had told her daughter a woman needs somewhere to put her fancy. She said there was a hell in their dialect, but it was not the one in the books: it was a place under the floor where the water went, and when a child was unlucky that was where he was sent. She said there was no holiness in a house, only habits.

Then she said the thing I have kept. "You think an ideal is a picture you keep," she said. "It is a direction you lean." And then she took out an orange and offered me half, and the train came into the station, and I got out with five hundred other people, and nobody counted me at the gate.

三十四

Shen Lu wrote to me in November, a real letter, four pages, in handwriting that leaned forward as if it were in a hurry.

"Currently I am doing three jobs," she wrote. "The debate in this company is whether to launch a second product line, and I am the person who has to work out the relevance of our range to a customer who has never heard of us. Consequently I spend my mornings on the telephone and my afternoons in a regular meeting that could have been an email."

She had learned, she said, that a relationship with a supplier is reinforced by small things: a call before the problem, a detail in the explanation, a genuine apology when the detail is wrong. She had also learned that a conspicuous effort to be liked is contemptible, and that a concerted effort to be useful is not.

There had been a setback in September. A shipment had been derailed — her word, and she used it about a typo — by somebody's delusion that the customer would not read the specification. "Deceive yourself once," she wrote, "and you will deceive the customer twice, because the second lie is always the one that covers the first."

She was sceptical of the mission statement, and of the sales director, who lured customers with a price that could not last and then retreated behind a discount. "Premature celebration," she wrote, "is the trademark of a company that has not yet had a bad year."

The good news was that she had a team of three, that she would preside over her first customer audit in January, and that she had been asked to represent the company at a trade fair in March. "I am privileged," she wrote, "and also terrified, which I begin to think is the ordinary condition of anyone who is allowed to do something for the first time."

She ended with a question: "What do you want to pursue? Not the quest for a career — the thing itself. Write and tell me, and if you cannot tell me, that is an answer too."

三十五

I could not answer her in November, and I could not answer her in December, and then it was January and I was on a train.

What I wanted, I decided, was to be the person who is called when a thing fails: the one who copes with the mess, contrives a test, and stays until the explanation is solid. That is a modest ambition and it is mine.

I wrote that to her in February, in four sentences, and she wrote back one line: "Good. That is a permanent kind of ambition, which is the only kind that lasts."

I have kept the letter. It is in the notebook, behind the card from carton 77, which is behind the card from the shop. When I open it I can hear the press and the pump and the clicker at the gate, and I am nineteen again for as long as it takes to read four pages.

三十六

I went home for the Spring Festival. The house was full in the way it is full only then: cousins on the floor, an aunt in the kitchen, a television nobody watched, and a dog that had been shaved for the occasion and was shameful about it for three days.

My cousin Xiaoman was thirteen and had discovered chess, which is to say he had discovered winning. He beat me four times in one evening, and each time he explained the succession of moves that had destroyed me. His openings never changed and I could not stop them, which he found astonishing and I found tedious.

His mother, who is a school psychologist, said that this was a stereotype of the only child. She talked about a boy at her school whose plight had aroused the concern of every teacher: a neglectful father, a mother on night shifts, and a note that had been read in time. "Oftentimes," she said, "the child who is seemingly fine is the one to worry about."

My uncle had spent his working life in a county office, and he talked about the forthcoming congress with the moderation of a man who has seen four of them. He said that the delegation would produce a report, the report a meeting, the meeting a paragraph, and that the paragraph was where the money either appeared or did not. He was skeptical of revolution and in favour of recurrence: "A recurrence of a small good decision is worth more than one enormous one," he said, "and it is reversible, which the enormous one is not."

Dinner was enormous, half of it from a relative's garden and half of it from a supermarket. There was a smelly dish of fermented tofu that my grandmother loved and nobody else would touch, and the dog sniffed my shoes and then sat on my aunt's feet, which is what that dog did instead of an embrace.

My grandmother, who is eighty-four, does yoga every morning at six in the western room. She has done it for thirty years and has never persuaded anyone else in the family to join her. "Look at yourselves," she says, when we watch. "Sitting like sacks."

三十七

Writing the report was harder than the night shift, and it taught me three things.

The first is that a superficial account and a faithful one can be the same length. My first draft had two hundred words of adjectives in it; I subtracted them one by one, and what was left was shorter and truer.

The second is that a report is a reproduction of a process, not of a feeling. My reader did not want to know that I had been frightened. He wanted to know what I had done in the four minutes after the power went.

The third is that a report must be positive in the grammatical sense before it can be positive in the encouraging one: it has to state what is before it argues what should be. I still strive for that. I do not always reach it. What I reached that autumn was a document that was uncertain in the right places and definite in the others, and Wang laoshi read it and said, "This is a remarkable piece of work, and honesty is a rarity in a report, and therefore it is a little uncomfortable to read."

I learned to draw the conclusion last and to withhold it until the evidence had stopped moving. I learned to precede an argument with its numbers, and to bring the reader to the place where the numbers stop explaining anything, which is the only place where a judgement is worth anything.

I learned, besides, to err on the side of the dull sentence and to disapprove of my own provocative ones, which are always the ones I like best. A lofty claim in a report is a stretch of the truth, and a stretch of the truth is a vicious circle: the next sentence has to cover for it, and the sentence after that has to cover for the next. The virtuous circle is the opposite, and it is boring, and it paid a dividend in the only currency that mattered, which was that Chen read the whole thing and found nothing in it to disapprove of.

I was equipped with a notebook, a caliper and eleven weeks of tedious detail, and I was, to my astonishment, no longer alien to the shop floor. I had stopped being captive to my first idea. That is the whole of what a summer does, I think: it makes you incline towards the evidence and away from your own artificial symmetry.

三十八

The gala was on the eighteenth of August, in the canteen, on a stage made of pallets.

A girl from the packing line sang two songs and was, genuinely, good. A satirist from

the extrusion crew did eight minutes about the safety film and made the manager laugh harder than anyone had all summer. Somebody played a symphony off a phone, and it sounded, in that room, like a phone.

There was a basketball competition, which the athlete from the packing line won by spinning round once and throwing the ball left-handed. There was a tug-of-war, and the accounts office beat the warehouse twice, which nobody expected and everybody enjoyed. The union treasurer handed out towels and a bottle opener shaped like a fish, and I still have mine.

At eleven the lights went down and there were fireworks over the yard, and a dog barked at them from behind the gate, and the manager made a speech in which he thanked, respectively, the day shift, the night shift, and "the young people who will be running this place when I am fishing". Then he said something I have thought about since: "A factory is a place where you can see the result of your own hands. Not many jobs give you that any more. Do not let anybody tell you it is a small thing."

三十九

Chen could trim a die by hand with a stone and a file: eight strokes, turn, eight strokes. I could reflect on that rhythm for an hour. There was a monotony in it that was not boredom but a kind of exhaustion of the will, and when he finished he would stand up straight and stretch like a mighty man who had done something small very well.

On the yard there was a hoop with no net. Ding would shoot at it after the shift, spin the ball on one finger, and shrug when it went in. "This is not a sport," he said. "It is a way of standing." The yard was his territory and the hoop was the flag.

The guard kept a pet in his booth, a cat whose fur had gone grey at the chin. It was pathetic in the way that old animals are, and it had one peculiarity: it would not cross the threshold, and would sit exactly on the line. "Sir," the guard said to me one day, "this cat has better discipline than you."

The clinic had a poster about suicidal thoughts on the back of the door, so that wherever you stood in that small room the poster was behind you. Perhaps that was the point. The doctor said that most of what he treated was not illness but painful habit, and that a body will suffer almost anything except being nourished badly for twenty years.

Xu Jie could upload a document to the customs system in forty seconds, and she could excel at any task that had a deadline and a form. She was exemplary in that way, and she showed no leniency to anyone who was not, including me. Once she found a manipulative clause in a massive contract and made the buyer take it out.

The contract had no bail for either side: you could not promise your way out of a

missed shipment. The beneficiary of the warranty was the buyer, and the clause designed to bind him to a date was, in the end, the only compensatory thing in the document. Containment of a dispute is cheaper than a trial, and the cheapest way to create that containment is to write the date down.

Chen said that a man who deviates from the curve by ten degrees is not a fool, he is a man in a hurry. Disruptive change is what a new manager wants and what an educated operator fears. When the plant embarked on a second line, the emergence of a new routine took four months, and nobody was endowed with patience by the schedule.

The clicker at the gate had an echo in it, and the existing habit of counting survived every reform. The forgettable sentences in the handbook were the ones that mattered, and it frustrated me that I could not gain anything by arguing with them. The scrap price fell geometrically that autumn, and not a gram of it was anybody's fault.

At the wedding the fireworks made a gay noise that the children liked and the dogs did not. On the train home the carriage jerked twice at every station, and I took a nap with my head against the window and wakened at dawn with a stab of pain in my neck. There was a pole of sunlight across the seat, and the rot in the wooden frame of the window had been painted over eleven times, and a man across the aisle was eating an orange, and I thought about a swim I had taken in the river in August, and about a waggon I had seen in the village with a child asleep in it.

There was a widespread belief in the plant that the young yearn for an office, and I did not venture an opinion, partly because I was young and partly because I was not sure the belief was wrong. Wherever I went that summer somebody asked me what I was going to do with my life, and I learned to answer in a way that was truly mine and only temporarily true: "I am going to find out why things fail."

The word plural gave Ding trouble, and the word unique gave him more, because nothing in his experience was unique and he did not see why English needed a word for it. There was a nude in a gold frame in the manager's office, and a picturesque poster of a gorge beside it, and a realism about the room that no photograph could improve on.

Invasive measurement, intrusion into a batch, the infliction of a scratch on a sample: these were Zhou Min's words for what a careless student does to an experiment. Incorporation of an organic additive into the outer shell of a particle was, she said, the whole of her thesis in one sentence, and the sentence had taken her three years.

四十

I have one more thing to put down, and it is not a lesson.

Wherever I am, when a press starts up, I linger for a moment. That is not nostalgia. It is closer to the way a song can reel you back into a kitchen you have not seen in ten

years.

The inviting thing about a shop floor is that it answers back. You cannot perpetuate a mistake there for long, because the metal will tell you. That is what stimulated me, and what, to my annoyance, I have not found in a classroom since.

Chen telephones twice a year. He is reminiscent in a way that partially embarrasses me: he talks about the rider who brought the post, about a ministerial delegation that came in 2019, and about a legislator from the county who promised a road and delivered a sign. He has more vigor at sixty than I have at twenty, and he is the supporter I did not know I had applied for.

The summer did not renew me, whatever that word means. It left me one potentiality I did not have in June: the ability to walk into a strange building and ask what it makes. And when the three of us reunite — Ding, Shen Lu and I, once a year, in a noodle shop — one of us always sings badly at the end of the evening, and Shen Lu always says the same thing: the strait between a good plant and a great one is one habit wide.

I wonder sometimes whether she is right. Then I measure something twice, and I stop wondering.

伴读：车间里的英语，和车间外的人生

先说一句不客气的话。你如果打算把这九百多个词当词表背，现在就合上书。它们不是九百个孤立的点，是一个夏天里的九百次照面，每一次都有温度、气味和人情。我们要做的，是把那天从厂门口到宿舍的路重走一遍，顺手把词安顿好。

门口那四十分钟

abnormal 是"反常"，工厂里它是警报词：abnormal noise 一响就得报。它的骨是 norm（标准），ab- 是偏离，所以它不是"奇怪"，是"偏离了标准"。abreast 有画面：两人并排走是 walk abreast，keep abreast of 是"跟上进展"，后面跟信息不跟人。access 是"进入权"，不是"入口"；入口叫 entrance，工牌给的是 access。accurate 形容仪器读数准，ascertain 是人的动作——去核实。Chen 说 ascertain，意思是"你再核实一遍你用来核实的那个家伙"。

accountable 比 responsible 重：responsible 是这事归你，accountable 是出了事你要交代，搭配 be accountable for。accomplish 是"完成（有难度的事）"，achievement 是它的名词，但常指"成就"，别说"完成率"。acquisition 是"收购、获得"，厂里那家小厂就是 an acquisition；它也指语言的"习得"。abstract 是"摘要"，也是"抽象的"；adverb 副词、antonym 反义词、affixation 词缀法，这三个词出现在那本词源小册子上，本身就是"讲词的书"里的词。academician 是"院士"，不是"学者"，学者是 scholar。

activist 是"活动人士"，advocacy 是"倡导"，两个都带立场，新闻里常见。ad 是 advertisement 的口语缩写，广告；addition 常用 in addition to，后面跟名词。adjacent 是"相邻的"，比 near 正式。admirable 是"值得敬佩的"，语气比 good 重；adolescent 是"青春期的"，

也作名词指少年，比 teenager 书面。advanced 是"先进的"，也是"高级的"。affection 是"喜爱、感情"，不是"爱情"；after 的常见搭配是 after all 与 one after another。

ageing 是"老化"，设备 ageing、人口 ageing 都用它；aggressive 在商务里常指"激进的、强势的"，不一定带贬义。agricultural 是"农业的"；aim 既是"目标"也是动词"瞄准"，aim at 后面跟的是目标。akin 是"相似的"，be akin to；alien 是"外来的、陌生的"，也指"外星人"，feel alien to 是"对……感到格格不入"。allegation 是"指控（未经证实）"，和 charge 不同：charge 是正式起诉，allegation 还只是"有人说"。alley 是"小巷"，比 street 窄；alloy 是"合金"。

aloft 是"在高处、在空中"，书面词；also 是"也"，书面里常放句中。altitude 是"海拔、高度"；amazing 口语里是"惊人的好"，书面里只是"令人惊讶"。amendment 是"修正案、修订"，合同和宪法都用它；amuse 是"逗乐"，be amused by，别和 amaze 混。ancestry 是"世系、祖先"；angel 是"天使"；animation 是"动画"，也指"生气、活力"。

annoyance 是"恼火"，to my annoyance 是"让我恼火的是"；antagonist 是"对手、反派"，和 protagonist 相对。anyway 是"反正、不管怎样"，口语里也用来收回话题。apparent 是"显而易见的"，也是"表面上的"，两个意思要靠上下文分。appendix 是"附录"，也是"阑尾"。application 是"申请"，也是"应用"，还有"涂抹"。appreciative 是"表示欣赏的"，be appreciative of。

apron 是"围裙"；arc 是"弧"，焊接的电弧就是 welding arc。arguable 是"可争辩的、有争议的"，不是"能说服人的"。aristocratic 是"贵族气的"，形容做派；arouse 是"唤起（情绪、兴趣）"，arouse suspicion 是"引起怀疑"。arrogant 是"傲慢的"，比 proud 贬。artificial 是"人造的"，和 natural 相对。aspect 是"方面"，比 side 正式。

assemble 是"组装"，也是"集合"；assign 是"分配、指派"，assign a task to somebody。association 是"协会"，也是"联想"，in association with 是"与……合作"。astonishment 是"惊愕"，to my astonishment 比 surprise 重。athlete 是"运动员"；atrocious 是"糟糕透顶的"，可指天气或表演。attend 是"出席"，attend a meeting，不跟 to（attend to 是"处理"）。attributable 是"可归因于的"，be attributable to。auditing 是"审计"；authentic 是"真正的、可信的"，比 real 更强调来源可靠。automatic 是"自动的"，也是"不加思索的"。avail 多见于 to no avail；aviation 是"航空"；away 是"离开"，far away、right away 都高频。

更衣柜与钳工台

bachelor 是"学士"，也是"单身男子"，两个意思差得远，靠上下文分。bacteria 是 bacterium 的复数，注意拉丁复数：一格细菌是 one bacterium。bail 是"保释"，也是板球里的"横木"，厂里没人用它，但法庭新闻里天天见。ballet 是"芭蕾"，bang 是"砰"的一声，也作动词"猛撞"。barber 是"理发师"（男士），barrel 是"桶"，石油的"桶"也是它。basin 是"盆、流域"，地理里指盆地；bathe 是"洗澡、沐浴"，比 wash 书面。be 这个最不起眼的词，在正文里被师傅写在墙上："Be here at six forty"，祈使句的 be 就是"要"。

bearing 是"轴承"，也是"举止、方位"，have a bearing on 是"与……有关"；become 除了"变成"，还有"适合"（That dress becomes you）。beforehand 是"事先"，比 in advance 口语。behavioural 是"行为的"，behavioural safety 是"行为安全"。belongings 是"随身物品"，恒用复

数；beneficiary 是"受益人"，保险和合同里常见。besides 是"除此之外"，和 beside（在旁边）不是一回事，别混。beverage 是"饮料"，菜单和标签上的正式词。

bid 是"投标、出价"，也是"努力争取"（bid for power）；bind 是"捆、约束"，be bound by a contract 是被合同约束。biological 是"生物的"，bishop 是"主教"，教堂专用。blackmail 是"敲诈"，名词动词同形；bleak 是"荒凉的、前景暗淡的"，形容前景比形容风景多。block 是"块、街区"，也是动词"阻塞"；blow 是"打击"（a blow to），也是"吹"。board 是"板、董事会"，on board 是"在船/飞机上"，也是"加入"。

bold 是"大胆的"，印刷里指"粗体"，正好和这本书的加粗呼应；bony 是"瘦骨嶙峋的"，born 是"出生"，be born in，注意用被动。bounce 是"弹跳"，也是"（邮件）被退回"；boycott 是"抵制"，来自一个人名，这类词记住来历就忘不掉。brandy 是"白兰地"，breakdown 是"故障、崩溃"，也是"分类明细"。brew 是"酿造"，也是"（茶）泡着"，a storm is brewing 是"暴风雨要来了"。brief 是"简短的"，也是"简报"；bring 的搭配很多，bring about 是"带来"，bring up 是"抚养、提出"。

broker 是"经纪人"，browser 是"浏览器"，bucket 是"水桶"，buffalo 是"水牛"，bulky 是"笨重的"，burden 是"负担"，burial 是"埋葬"，businessman 是"商人"，button 是"按钮、纽扣"，按两下才响的那个就是它。

车间里的名词最密

cabbage 是"卷心菜"，calcium 是"钙"，calling 是"天职、使命"，比 job 重得多，Chen 身上那个东西就是 calling。canal 是"运河、渠道"，cane 是"藤条、手杖"，capacity 是"容量、产能"，厂里说产能就用它。captive 是"被俘的、受制的"，captive market 是"被垄断的市场"。cardinal 是"基本的、最重要的"，cardinal rule 是"铁律"，不是"红衣主教"那个意思（虽然那也是它）。carry 除了"携带"，还有"刊登、销售"（The shop carries it）。cash 是"现金"，catalogue 是"目录"，英式拼法，美式是 catalog。

cattle 是"牛"的集合名词，本身就是复数，别说 cattles；ceiling 是"天花板"，也是"上限"（price ceiling）。censor 是"审查、删改"，ceremonial 是"礼仪性的"，chain 是"链条"，也是"连锁"，supply chain 就是"供应链"。champagne 是"香槟"，注意和 champion（冠军）只差几个字母。chaos 是"混乱"，读作 K 音；charming 是"迷人的"，也可以反讽。cheer 是"欢呼"，cheer up 是"振作"。

chess 是"国际象棋"，chilly 是"微冷的"，比 cold 轻；choke 是"窒息、噎住"，chorus 是"合唱、副歌"，chunk 是"大块"，circulate 是"循环、传阅"。city 是"城市"，注意和 town 的量级差别；clarify 是"澄清"，商务邮件里的救命词。classified 是"机密的"，也是"分类的"；cleanse 是"清洗（伤口、皮肤）"，比 clean 专门。clerk 是"职员"，climb 是"爬、上升"。clone 是"克隆"，cloudy 是"多云的"，也是"浑浊的"；coal 是"煤"。cockpit 是"驾驶舱"，cognitive 是"认知的"，和 intellectual 常一起出现但不同层。coincide 是"同时发生、一致"，colleague 是"同事"，比 workmate 正式。colour 是英式拼法，美式 color。combination 是"组合"，compact 是"紧凑的"，compartment 是"隔间"，compensatory 是"补偿性的"，compile 是"汇编"，把九页交接单编起来就是 compile。

completion 是"完成"，注意它指"完工那一刻"；compliment 是"称赞"，和 complement（补充）只差一个字母，考试最爱。comprehend 是"理解"，比 understand 正式，也有"包含"的古义。comrade 是"同志"，Chen 用它带刺，词本身没有刺，是语境给的。concentration 是"专注"，也是"浓度"，工科两个意思都用。concerted 是"协同的"，concerted effort 是"齐心协力的努力"。condemn 是"谴责"，也是"判刑"，语气很重。conducive 是"有助于……的"，be conducive to，注意 to 是介词。confession 是"供认、忏悔"，confirm 是"确认"，邮件里最常用的动词之一。

Confucianism 是"儒家"，注意大写；congress 是"代表大会、国会"，conscientious 是"认真尽责的"，比 careful 更带品德评价。consequently 是"因此"，比 so 正式；considering 是"考虑到"，作介词用。conspicuous 是"显眼的"，conspicuous consumption 是"炫耀性消费"。constraint 是"限制、约束条件"，工程和写作里都指"不能越的线"。consultative 是"协商的、咨询的"，consultative meeting 是"征求意见的会"。

containment 是"控制、遏制"，事故里指"别让它扩散"；contemptible 是"可鄙的"，比 bad 重得多。continent 是"大陆、洲"，contradict 是"反驳、与……矛盾"，contradict oneself 是"自相矛盾"。contrive 是"设法做到、巧妙设计"，常带"费了劲"的意思。conventional 是"常规的、传统的"，和 unconventional 相对。convict 是"判……有罪"，作名词是"罪犯"，重音不同。cool 是"凉的"，也是"冷静的、酷的"。cope 是"应对"，cope with，别漏 with。core 是"核心"，也是"果核"。

corps 是"团、队"，press corps 是"记者团"，注意发音和 corpse（尸体）不同。correspond 是"通信"，也是"对应"，correspond with 两者都行。corruption 是"腐败"，cottage 是"小屋、村舍"，countdown 是"倒计时"，county 是"县"，cousin 是"表/堂兄弟姐妹"。crab 是"蟹"，crash 是"撞击、崩溃"，也是"（系统）宕机"。create 是"创造"，credit 是"信用、赊账"，也是"学分"和"功劳"（give credit to）。crisis 是"危机"，复数 crises，crow 是"乌鸦"，cruise 是"巡航、乘船游览"，cub 是"幼兽"，cupboard 是"橱柜"，currently 是"目前"，custom 是"习俗"，也是"海关"（customs，恒复数）。cycling 是"骑行"，也是"循环"。

这一串词里有一半是"工厂日常"，一半是"新闻词汇"，它们的共同点是：都不难认，难的是用对场合。condemn 用在判决和社论里，blame 用在饭桌上；consequently 写在报告里，so 留在口语里。你在这一章里看到的每一处"大词"，都是某个人在某个场合必须换上的衣服。

一个星期里的六十三天

dam 是"水坝"，也是动词"筑坝拦住"；dare 是"敢"，dare to do 或 dare do 都行。dead 是"死的"，也是"完全地"（dead tired）和"没电的"（dead battery）。debate 是"辩论"，deceive 是"欺骗"，deceive oneself 是"自欺"，比 cheat 更书面。decisive 是"决定性的"，也是"果断的"。decorative 是"装饰性的"，deductible 是"可扣除的"，保险里的"免赔额"就是它。default 是"默认"，也是"违约"，电脑和合同共用一个词。definite 是"确定的"，和 definitive（最终的、权威的）不同。degree 是"度、程度"，也是"学位"。

delicacy 是"美味、珍馐"，也是"微妙、棘手"，两个意思都常用。delusion 是"错觉、妄想"，比 illusion 更带病态。demonstrate 是"演示、证明"，答辩时老师说 demonstrate one thing，就是这个用法。density 是"密度"，departure 是"离开、出发"，也是"背离"（a departure from tradition）。depletion 是"耗尽"，臭氧层、电解质都用它。depreciate 是"贬值"，货币和设备都

用，反义 appreciate。derail 是"脱轨"，也是"使计划偏离"，新闻里常见。descendant 是"后代"，designation 是"名称、指定"，表格上那一栏就是它。

desolate 是"荒凉的"，destined 是"注定的、开往……的"，be destined for。detail 是"细节"，in detail 是"详细地"。detergent 是"洗涤剂"，detrimental 是"有害的"，be detrimental to。deviate 是"偏离"，deviate from，工程里指"超出曲线"。diabetes 是"糖尿病"，diamond 是"钻石"，也是扑克牌的"方块"。diet 是"饮食"，也是"节食"，go on a diet。difficulty 是"困难"，have difficulty (in) doing。digital 是"数字的"，dim 是"昏暗的"，也是"愚钝的"。dinosaur 是"恐龙"，也是"过时的东西"。

direction 是"方向"，也是"指导"，disagree 是"不同意"，disagree with somebody 但 disagree to a plan。disapprove 是"不赞成"，disapprove of，比 disagree 多一层道德判断。discharge 是"排放"，也是"出院、退伍、解雇"，一个词四种场面。discount 是"折扣"，注意中文"打八折"是 20% discount。discrimination 是"歧视"，也是"辨别力"，两个意思差得远。dish 是"盘子、一道菜"，dismissal 是"解雇、驳回"，disposal 是"处理、处置"，at your disposal 是"听你安排"。disruptive 是"扰乱性的"，distinction 是"区别、优异"，distress 是"痛苦、困境"，ditch 是"沟渠"，也是动词"甩掉"。

dividend 是"红利"，也是"回报"；doctor 是"医生、博士"，dogma 是"教条"，dominance 是"支配地位"，doom 是"厄运、注定"，doubt 是"怀疑"，no doubt 是"无疑"。downstairs 是"楼下"，dragon 是"龙"，draw 是"画、拉"，也是"得出结论"（draw a conclusion）。dress 是"连衣裙"，也是"穿衣"，drop 是"落下、下降"，也是"顺路送"。drunken 是"醉的"，只作定语；duly 是"如期、适当地"，合同里的 duly witnessed 就是"按规定见证"。during 是"在……期间"，dwelling 是"住所"，比 house 正式。

从更衣室到河边

eagle 是"鹰"，ease 是"容易、自在"，with ease、at ease 都是高频；echo 是"回声"，也是"附和"。economy 是"经济"，也是"节约"；educate 是"教育"，比 teach 更宏观。effort 是"努力"，make an effort，注意它可数。eighth 是"第八"，拼写里那个 h 别丢。elastic 是"有弹性的"，经济里指"弹性"；electoral 是"选举的"，electronics 是"电子学"，elevator 是"电梯"（美式），英式 lift。eloquence 是"口才"，embark 是"上船、着手"，embark on。emergence 是"出现、兴起"，emit 是"发出（光、声、气）"，比 send out 正式。employer 是"雇主"，注意和 employee 别写反。enact 是"颁布（法律）"，也是"扮演"。encyclopaedia 是"百科全书"，英式拼法，美式 encyclopedia。

endow 是"赋予、捐赠"，be endowed with；enforcement 是"执行、执法"，enjoy 是"享受"，enjoy doing，不跟 to do。enormous 是"巨大的"，ensure 是"确保"，ensure that，别和 insure（投保）混。enthusiasm 是"热情"，entrance 是"入口"，也是"进入权"，还指"入迷"。environment 是"环境"，episode 是"一集、一段插曲"，equip 是"装备"，be equipped with。eradication 是"根除"，err 是"犯错"，to err on the side of caution 是"宁可保守"。escalator 是"自动扶梯"，注意和 escalade 无关。establishment 是"建立"，也指"机构、当权派"。eternity 是"永恒"，evade 是"逃避"，evade tax 是"逃税"。

even 是"甚至"，也是"平的、偶数的"；ever 是"曾经"，hardly ever 是"几乎从不"。evident 是"明显的"，exaggeration 是"夸张"，excel 是"擅长"，excel at/in。excessive 是"过度的"，exclude 是"排除"，exemplary 是"模范的"。exhaustion 是"精疲力竭"，existing 是"现有的"，expectation 是"期望"，live up to expectations。experienced 是"有经验的"，explanation 是"解释"，explosion 是"爆炸"，express 是"表达"，也是"快递"。extent 是"程度"，to some extent；extraction 是"提取、拔出"，eyesight 是"视力"。

脸上、田里、屋顶上

facial 是"面部的"，facial expression 是"面部表情"；factual 是"事实性的"，老师批报告用这个词，是夸也是提醒。faithful 是"忠实的"，也是"如实的"（a faithful account）。fan 是"风扇"，也是"粉丝"；farmer 是"农民"，fashion 是"时尚"，也是"方式"（in a similar fashion）。father 是"父亲"，也是"创始人"（the father of）。fax 是"传真"，feature 是"特征"，也是"专题报道"。feel 是"感觉"，也是"摸"，feel like doing。fence 是"栅栏"，field 是"田地、领域"，file 是"文件"，也是动词"提交"。finding 是"研究结果"，恒用复数 findings。fireworks 是"烟花"，fit 是"合适、健康"，也是"一阵发作"。flank 是"侧面、翼侧"，flawless 是"无瑕的"，flirt 是"调情"，flirt with 也有"不认真考虑"的意思。

flow 是"流动"，flush 是"脸红、冲洗"，fold 是"折叠"，foolish 是"愚蠢的"，比 stupid 客气。forehead 是"额头"，forestry 是"林业"，forgettable 是"容易被忘记的"，formation 是"形成、构成"，forthcoming 是"即将到来的"，也是"乐于提供信息的"。forward 是"向前"，look forward to 后面跟动名词。fountain 是"喷泉"，fragment 是"碎片"，frank 是"坦率的"，to be frank。freeze 是"结冰、冻结"，fridge 是"冰箱"，from 是"从"，from scratch 是"从零开始"。frustrate 是"使受挫"，full 是"满的"，fur 是"毛皮"，fuse 是"保险丝"，也是"融合"。

gain 是"获得"，也是"增加"，gaol 是 jail 的英式旧拼法，报纸标题里还见。garment 是"衣服"（正式），gay 是"欢快的"（旧义），现代更常用的是"同性恋的"，读旧书要分清。generosity 是"慷慨"，genuine 是"真正的、真诚的"，比 real 多一层"不假"。geometrically 是"按几何级数地"，形容增长。gifted 是"有天赋的"，glamour 是"魅力、光彩"，glider 是"滑翔机"，glove 是"手套"，golden 是"金色的"，也是"宝贵的"（a golden chance）。gospel 是"福音"，graceful 是"优雅的"，gram 是"克"，grandparent 是"祖父母"。grasp 是"抓住、理解"，gray 是"灰色"（美式），英式 grey。greenhouse 是"温室"，greenhouse effect 是"温室效应"。grill 是"烤架、烧烤"，groove 是"凹槽"，guard 是"守卫、护罩"，guilty 是"有罪的、内疚的"，gymnasium 是"体育馆"，口语缩成 gym。

hall 是"大厅"，handicapped 是"残疾的"，现在更常用 disabled，但旧文本里仍是它。happily 是"快乐地"，也是"幸好"；hardly 是"几乎不"，不是"努力地"，这是最经典的坑之一。harmony 是"和谐"，hatch 是"舱口、孵化"，havoc 是"大破坏"，wreak havoc。head 是"头"，也是动词"前往"（head for）。health 是"健康"，hearty 是"丰盛的、热情的"，hectic 是"忙乱的"，hell 是"地狱"，也是口语里的加强词。hen 是"母鸡"，here 是"这里"，hers 是"她的"，hide 是"藏"，hill 是"小山"，hip 是"髋部"，hitherto 是"迄今为止"，很书面。holiness 是"神圣"，homo 是词根"相同"，也是 homosexual 的缩写，正式文本里慎用。honourable 是"可敬的、光荣的"，

horizontal 是"水平的"，hospitable 是"好客的"，hotdog 是"热狗"，housework 是"家务"，huge 是"巨大的"，humble 是"谦逊的、卑微的"，hundred 是"百"，hurricane 是"飓风"，hygiene 是"卫生"，hysteria 是"歇斯底里"。

这一段里有两个字最容易读错意思：hardly 不是"努力"，hitherto 不是"到这里"。英语里这种"看起来像、其实不是"的词，比生词更危险，因为你不会去查它。

我、我们、以及那些抽象的东西

ideal 是"理想的"，也是"理想"；ideology 是"意识形态"，if 是"如果"，也是"是否"（I wonder if）。illicit 是"非法的"，比 illegal 更书面。illustrate 是"说明、图解"，imagine 是"想象"，impatience 是"不耐烦"，implicit 是"含蓄的、内隐的"，和 explicit 相对。imposition 是"强加、征收"，impromptu 是"即兴的"，inborn 是"天生的"，incline 是"倾向"，be inclined to。incorporation 是"并入、incorporation of A into B"，indeed 是"确实"，用来加强语气。indicator 是"指标、指示剂"，individualism 是"个人主义"，inertia 是"惯性、惰性"，inferior 是"较差的、下级的"，和 superior 相对。infliction 是"施加（痛苦）"，infrastructure 是"基础设施"，inherent 是"固有的"，initiation 是"入门、开始"。

还有几个词被漏掉了，补在这里。cult 是"膜拜、异教"，也是"风靡一时的事物"，cult film 是"邪典电影"；data 是"数据"，正式文本里当复数用，datum 是单数。inland 是"内陆的"，也可以作副词。innovative 是"创新的"，insect 是"昆虫"，insistence 是"坚持"，install 是"安装"，instinctive 是"本能的"。instrument 是"仪器"，也是"乐器"；intact 是"完好无损的"，intensive 是"密集的、集约的"，和 extensive 相对。interest 是"兴趣"，也是"利息"和"利益"，三义一词。intermediate 是"中间的"，interrogation 是"审讯"，interviewer 是"面试官、采访者"。intrigue 是"引起好奇"，也是"阴谋"；intrusion 是"侵入"，invasive 是"侵入性的"，医学里指"有创的"。investigate 是"调查"，inviting 是"诱人的"，irony 是"反讽"，irritating 是"令人恼火的"，item 是"一项、一件"。

jargon 是"行话"，jerk 是"猛拉"，也是"混蛋"（口语）；join 是"加入、连接"，joy 是"喜悦"，juicy 是"多汁的"，也是"有料的"（新闻）。juror 是"陪审员"，kettle 是"水壶"，kilo 是"公斤"，kiss 是"吻"，knit 是"编织"，注意 k 不发音。knowledgeable 是"博学的"，比 learned 口语；lad 是"小伙子"，师傅叫人就用它。land 是"土地"，也是动词"着陆"；lap 是"大腿（坐着时的）"，也是"一圈"；lastly 是"最后"，用于列举的收尾。launch 是"发射、推出"，lawyer 是"律师"，leaflet 是"传单"，learning 是"学识"，leftist 是"左翼的"，legislator 是"立法者"，leniency 是"宽大"，letter 是"信"，也是"字母"。liberate 是"解放"，life 是"生命、生活"，limit 是"限制"，linger 是"逗留、久久不散"。

lipstick 是"口红"，literally 是"字面意义上地"，口语里常被滥用成"真的"；liver 是"肝"，locality 是"地区"，lofty 是"高耸的、崇高的"，lonely 是"孤独的"，loose 是"松的"，和 lose 拼写别混。lotion 是"乳液"，low 是"低的"，也是"低落的"；lucrative 是"赚钱的"，lure 是"诱惑"，mad 是"疯狂的"，英式口语里也指"生气"。magnetic 是"磁的"，也是"有魅力的"；main 是"主要的"，majority 是"多数"，注意 the majority of 后面跟复数。mall 是"购物中心"，mand 是被截断的 command 或 demand，本章那张卡片上就有它。

manipulative 是"善于操纵的"，manufacturer 是"制造商"，margin 是"页边、利润、差数"，三个意思都常见。marriage 是"婚姻"，Marxism 是"马克思主义"，注意大写。massive 是"巨大的"，maternity 是"母性、产科"，May 是五月，也是情态动词 may 的大写形式，别混。meanwhile 是"与此同时"，mechanics 是"力学、机械原理"，也是"运作方式"。medical 是"医疗的"，memorandum 是"备忘录"，mendation 是 recommendation 被切掉前半截剩下的部分，mentator 是 commentator 的后半截。mercy 是"仁慈"，merry 是"欢快的"，metabolism 是"新陈代谢"，meticulous 是"一丝不苟的"，microwavable 是"可微波加热的"。

这里值得停一下。mendation、mentator 这种"半个词"，正常英语里根本不存在，可它们偏偏出现在你的词表上。这不是词表出错，是排印时把长词在行尾切断、连字符又丢了。你要记住的不是这两个残片，而是它们背后的 recommendation 和 commentator——顺便记住 com- 这个前缀：它表示"一起"，所以 com-muter 是"一起换来换去的人"，也就是通勤者。

那些大词，和那些小词

mighty 是"强大的"，militancy 是"好斗、激进"，millionaire 是"百万富翁"，ministerial 是"部级的"，mirror 是"镜子"，mission 是"使命、任务"，misunderstand 是"误解"，moan 是"呻吟、抱怨"，moderation 是"适度"，in moderation。moist 是"微湿的"，monarch 是"君主"，monotony 是"单调"，monwealth 是 commonwealth 的后半截，more 是"更多"，mostly 是"多半、主要地"。motor 是"马达"，mouse 是"老鼠"，也是"鼠标"；mud 是"泥"，multitude 是"众多"，a multitude of。munism 是 communism 的后半截，muscular 是"肌肉发达的"，muter 是 commuter 的后半截，mythical 是"神话里的"，nap 是"小睡"，nasty 是"讨厌的、恶劣的"，necklace 是"项链"。

neglectful 是"疏忽的"，neglectful of；neighbourhood 是"街区、邻里"，英式拼法，美式 neighborhood。networking 是"人脉经营、联网"，两个意思都在用。nicety 是"细微之处、礼节"，常用复数 niceties。nine 是"九"，nobody 是"没有人"，none 是"一个也没有"，none of 后面单复数都行，正式文体偏单数。northeast 是"东北"，notable 是"显著的、值得注意的"，notification 是"通知"，手机上那条推送就是它。nourish 是"滋养"，nude 是"裸体的"，也指"裸体画"。nursery 是"托儿所、苗圃"，obedient 是"顺从的"，obliged 是"有义务的、感激的"，be obliged to。observe 是"观察"，也是"遵守"（observe the rules）和"评论"。obtain 是"获得"，比 get 正式；occupant 是"占用者、居住者"，October 是十月，offer 是"提供"，也是"报价"。oftentimes 是 often 的书面变体，Olympics 是"奥运会"，注意带 s。onion 是"洋葱"，opener 是"开瓶器、开场节目"，operator 是"操作员"，也是"运营商"。

oppression 是"压迫"，optimistic 是"乐观的"，orbital 是"轨道的"，也是化学里"轨道的"。organic 是"有机的"，也是"有机生长的、自然的"；orphan 是"孤儿"，ours 是"我们的"，outer 是"外部的"，output 是"产量、输出"，outward 是"向外的"，也是"表面的"。overdue 是"过期的、迟到的"，override 是"推翻、优先于"，overtime 是"加班"，own 是"自己的"，也是动词"拥有"。pack 是"打包、一包"，painful 是"痛苦的"，pamphlet 是"小册子"，paper 是"纸、论文、报纸"，三个意思全看场合。paragraph 是"段落"，parenting 是"养育"，partially 是"部分地"，partition 是"隔断"，人事窗口那块磨砂玻璃就是它。

passerby 是"路人"，复数 passersby；paste 是"浆糊、糊状物"，也是动词"粘贴"。pathetic 是"可怜的、差劲的"，pavement 是"人行道"（英式），美式 sidewalk。peach 是"桃子"，也是"极好的人"。peculiarity 是"怪癖、特点"，pen 是"钢笔"，pensioner 是"领养老金的人"，perceptive 是"有洞察力的"，perhaps 是"也许"，permanent 是"永久的"，和 temporary 相对。perpetuate 是"使延续（多指坏的）"，persistence 是"坚持、持续"，persuade 是"说服"，persuade somebody to do。pet 是"宠物"，pharmacy 是"药房"，photographer 是"摄影师"，physics 是"物理学"。

picturesque 是"风景如画的"，pilgrimage 是"朝圣"，pink 是"粉红色"，pitch 是"音高、沥青、球场"，动词是"搭（帐篷）、投（球）"。plateau 是"高原、平台期"，plead 是"恳求、辩护"，plead guilty 是"认罪"。plight 是"困境"，plural 是"复数"，point 是"点、观点"，也是"指向"；pole 是"杆、极"，politician 是"政客"，pond 是"池塘"，population 是"人口"，porter 是"搬运工、门房"。positive 是"肯定的、积极的"，也是"阳性的"；poster 是"海报"，potentiality 是"潜力、可能性"，比 potential 更抽象。

practical 是"实际的、实用的"，precede 是"在……之前"，predecessor 是"前任"，prefer 是"更喜欢"，prefer A to B。premature 是"过早的"，prepare 是"准备"，preside 是"主持"，preside over。presumption 是"推定、想当然"，preventive 是"预防性的"，primarily 是"主要地"。print 是"印刷、打印"，privileged 是"有特权的、荣幸的"，procedure 是"程序"，produce 是"生产"，也是名词"农产品"。

proficiency 是"熟练程度"，progression 是"进展、序列"，promise 是"承诺"，也是"前途"；pronunciation 是"发音"，注意中间是 nun；proper 是"适当的"，prosecute 是"起诉"，protection 是"保护"，proverb 是"谚语"，provocative 是"挑衅的、发人深省的"，psychologist 是"心理学家"，pump 是"泵"，也是动词"抽、灌注"。

purchase 名动同形，比 buy 正式，合同里的"采购"就是它，作名词时可数：a purchase。pursue 是"追求、从事"，pursue a career；quake 是"颤抖、地震"，比 earthquake 短。quantity 是"数量"，和 quality（质量）长得像，报表上这两个词天天并排。quest 是"探寻"，比 search 文学气。quiver 是"微微颤抖"，比 shake 轻。radiant 是"发光的、容光焕发的"，rag 是"破布"，railway 是"铁路"，range 是"范围、系列、山脉"，一词三义。rarity 是"稀有"，ratio 是"比率"，和 rate 不同：ratio 是两个量的比，rate 是速率。

reaction 是"反应"，realism 是"现实主义"，reasonable 是"合理的、公道的"，价格公道就是它。receipt 是"收据"，注意 p 不发音；recession 是"衰退"，recom 是 recommendation 的前半截，recovery 是"恢复"，recurrence 是"复发、再次发生"，医学和数学都用。reel 是"卷轴、蹒跚"，也是"（被）震得发懵"；reflect 是"反射、反映、思考"。refreshment 是"茶点、提神之物"，恒用复数。refutation 是"反驳"，比 denial 有论证。region 是"地区"，regular 是"规则的、定期的"，也是"常客"。reinforce 是"加强"，relationship 是"关系"，relevance 是"相关性"，和 to 连用。

religious 是"宗教的、虔诚的"，remarkable 是"非凡的"，reminiscent 是"令人想起……的"，be reminiscent of。renew 是"续期、重新开始"，rental 是"租赁"，repertoire 是"全部剧目、全部本领"，represent 是"代表、表现"。reproduction 是"复制、繁殖"，require 是"需要、要求"，resentment 是"怨恨"，resident 是"居民"，resolution 是"决心、决议、分辨率"，三个意思都常

见。respectively 是"分别地"，两个列表一一对应时用它。

restless 是"焦躁不安的"，result 是"结果"，也是"导致"（result in）；retaliation 是"报复"，retreat 是"撤退、隐居处"，reunite 是"重聚"，reversible 是"可逆的"，revolution 是"革命、旋转"，rhythmic 是"有节奏的"，rider 是"骑手"，也是"附加条款"。rigidity 是"刚性、死板"，ripe 是"成熟的"，river 是"河"，robust 是"强健的、稳健的"，romance 是"爱情故事、浪漫"，rot 是"腐烂"，route 是"路线"，rug 是"小地毯"，runner 是"跑步者、跑鞋"。

sacred 是"神圣的"，和 religious 不同：sacred 说的是"不可冒犯"，religious 说的是"与宗教有关"。safeguard 是"保护措施"，作动词是"维护"；salary 是"薪水"（月薪/年薪），wage 是"工资"（按时/按件）。salute 是"敬礼、致敬"，sandwich 是"三明治"，satirist 是"讽刺作家"，sausage 是"香肠"，scandal 是"丑闻"，scary 是"吓人的"，sceptical 是"怀疑的"，英式拼法，美式 skeptical。school 是"学校"，也是"学派"和"鱼群"；score 是"得分、二十"，也是"乐谱"。scream 是"尖叫"，seal 是"封条、印章、海豹"，secretarial 是"秘书的"，seduction 是"诱惑"，seemingly 是"表面上看来"，select 是"挑选"，比 choose 正式。seminar 是"研讨会"，sense 是"感觉、意义、常识"，separate 是"分开的"，作动词重音在后。series 是"系列"，单复数同形；setback 是"挫折"，severe 是"严重的、严厉的"，shadow 是"影子"，shameful 是"可耻的"。

写到这儿，你大概已经发现一件事：这些词单独看都不难，难的是它们在真实句子里"该不该出现"。比如同是"保护"，safeguard 是措施，protection 是状态，defend 是动作；同是"工资"，salary 按月，wage 按小时，income 是总收入。词表不会告诉你这些，只有句子会。这一章里每一处合同、每一张表格，都是在替你把它们放回句子里。

shave 是"刮（胡子）、剃"，也是名词"剃须"，理发店那块热毛巾就是给它用的。shell 是"壳、炮弹"，工程里指"外壳"，电池的 outer shell 就是它。shipment 是"一批货、装运"，注意它指"货"也指"运"这个动作。shoot 是"射击、拍摄、投篮"，一个动词三个场面。shorts 是"短裤"，恒用复数。shower 是"淋浴、阵雨"，也是"（为某人）举办"（shower somebody with gifts）。shrug 是"耸肩"，shrug off 是"不以为意"。signature 是"签名"，也是"特色"（a signature dish）。

silky 是"丝滑的"，simply 是"仅仅、简直"，sing 是"唱歌"，注意过去式 sang、过去分词 sung，别写成 singed（那是"烧焦"）。sir 是"先生"，不带姓名时单独用。sixteen 是"十六"，注意 sixteen 与 sixty 的重音。skillful 是"熟练的"，英式写 skilful，一个 l 两个 l 别纠结，认得出就行。skyline 是"天际线"，slavery 是"奴隶制"，slight 是"轻微的"，not in the slightest 是"一点也不"。slow 是"慢的"，也是动词"减速"；smelly 是"有臭味的"，口语词。smuggler 是"走私者"，sniff 是"嗅"，也是"抽鼻子"。soar 是"猛增、翱翔"，价格和鸟都用它。socket 是"插座、插孔"，solar 是"太阳能的"，solid 是"固体的、结实的、可靠的"，三个意思都很常用。

solution 是"解决办法"，也是化学里的"溶液"，两个意思在一章里能同时出现。sometime 是"某时"（将来或过去），和 sometimes（有时）差一个 s，这个差别值得你记一辈子。sophisticated 是"老练的、精密的"，说人是夸他见过世面，说设备是说它精巧复杂。soup 是"汤"，southwestern 是"西南的"，spaghetti 是"意大利面"，speaker 是"说话者、扬声器、演讲者"。specify 是"明确规定"，合同里最常见的动词之一。speculative 是"推测的、投机的"，spend 是"花费"，spend time doing。spin 是"旋转、纺"，splash 是"溅"，sponge 是"海绵"，sport 是"运动"，也是"（旧）消遣"。spread 是"传播、铺开"，过去式 spread 不变。stab 是"刺、戳"，a stab of pain 是"一阵刺

痛"。stagnation 是"停滞"，经济和个人都用。

stall 是"货摊、熄火、拖延"，stark 是"鲜明的、荒凉的"，stark contrast 是"鲜明对比"。statement 是"陈述、报表"，statue 是"雕像"，注意和 statute（法规）差一个字母。steal 是"偷"，过去式 stole；stereotype 是"刻板印象"，stimulate 是"刺激、激发"。stock 是"库存、股票、高汤"，三个意思完全不搭界，靠上下文。stoppage 是"停工、阻塞"，strait 是"海峡"，也是"困境"（常用复数 straits）。straw 是"稻草、吸管"，stretch 是"伸展、一段（时间/路程）"，strive 是"努力"，比 try 正式。study 是"学习、研究"，也是"书房"；submission 是"提交、屈服"，subsequent 是"随后的"，subtract 是"减去"，和 add 相对。

succession 是"一连串、继承"，a succession of 是"一连串的"。suffer 是"遭受"，suffer from；suicidal 是"有自杀倾向的、自我毁灭的"，Sunday 是"星期日"，superficial 是"表面的、浅薄的"，supervision 是"监督"，supporter 是"支持者"，supremacy 是"至高无上"，surgery 是"手术"，survey 是"调查、勘测"，suspension 是"暂停、悬挂"，sway 是"摇摆、影响"，swim 是"游泳"，symphony 是"交响乐"。

这一段里我想多说两句 sometime 和 sometimes。前者是"某一时刻"，只指一个时间点；后者是"有时候"，指频率。中文都翻成"某个时候/有时"，差一个字，英语里却是两个完全不同的词。这一类"差一个字母、差一个世界"的词，才是四六级真正会扣分的地方，不是那些又长又难的。

从更衣室到站台

systematically 是"系统地、有条不紊地"，tactic 是"策略、战术"，talented 是"有才能的"，tank 是"罐、坦克"，tasty 是"好吃的"，team 是"团队"，tedious 是"冗长乏味的"，比 boring 书面。tell 是"告诉、分辨"，tell A from B。temporarily 是"暂时地"，tender 是"温柔的、嫩的"，也是"投标书"；termination 是"终止"，合同里那一页。territory 是"领土、地盘"，text 是"文本、课文"，也是动词"发短信"；then 是"然后、那时"，therefore 是"因此"。thigh 是"大腿"，thirteen 是"十三"，thought 是"想法、思想"，thrift 是"节俭"，throw 是"扔"，过去式 threw。ticket 是"票、罚单"，tighten 是"拧紧、收紧"，timid 是"胆小的"，tiresome 是"令人厌烦的"，toe 是"脚趾"，tolerate 是"容忍"，tonnage 是"吨位"。

torch 是"手电筒"（英式），美式 flashlight，也是"火炬"。tourism 是"旅游业"，township 是"乡镇"，trademark 是"商标"，也是"标志性特征"，trail 是"小径、追踪"，tranquil 是"宁静的"，transistor 是"晶体管"，transmit 是"传送、传播"，trap 是"陷阱、困住"，treasurer 是"财务主管、司库"，tremendous 是"巨大的"，tribune 是"论坛、报章名"，trim 是"修剪、整齐的"，trophy 是"奖杯"，truly 是"真正地"，trying 是"难对付的、令人疲惫的"，tumour 是"肿瘤"，英式拼法，美式 tumor。turnout 是"出席人数、投票率"，twentieth 是"第二十"，typical 是"典型的"。

uncertain 是"不确定的"，undergraduate 是"本科生"，understanding 是"理解、谅解"，unemployment 是"失业"，unique 是"独一无二的"，理论上不可比较，所以别写 more unique。unless 是"除非"，后面跟现在时表将来。upload 是"上传"，upwards 是"向上"，upwards of 是"超过"。usage 是"用法、用量"，utility 是"公用事业、效用"，vacuum 是"真空、吸尘器"，value 是"价值"，也是动词"重视"。variation 是"变化、变异"，veil 是"面纱"，venture 是"冒险、创

业"，joint venture 是"合资企业"。versatile 是"多才多艺的、多用途的"，veteran 是"老兵、老手"，vicious 是"恶毒的、凶猛的"，vicious circle 是"恶性循环"。vigor 是"活力"，violent 是"暴力的、猛烈的"，virtuous 是"有德行的"，virtuous circle 是"良性循环"。visitor 是"访客"，vocational 是"职业的"，vocational school 是"职业学校"，volleyball 是"排球"。

还剩几个，一并说完。voter 是"选民"，waggon 是"货车车厢"，英式拼法，美式 wagon。waken 是"唤醒"，比 wake 书面，常带被动意味。ware 是"制品、器皿"，warehouse 就是"存货的房子"。warranty 是"保修、保证书"，注意它和 guarantee 的差别：warranty 是商家给的书面承诺，guarantee 更宽。watchful 是"警惕的"，way 是"方式、路"，in a way 是"在某种意义上"。wear 是"穿、磨损"，过去式 wore。wed 是"结婚"，也是 Wednesday 的缩写来源。weekly 是"每周的"，western 是"西方的"，wheat 是"小麦"，wherever 是"无论哪里"，whisky 是"威士忌"，wholesome 是"有益健康的、健全的"。

widespread 是"广泛的、普遍的"，will 是"将、意愿、遗嘱"，三义一词。wing 是"翅膀、侧厅"，wise 是"明智的"，withhold 是"扣住、拒绝给予"，work 是"工作、起作用"，worried 是"担心的"，worthwhile 是"值得的"，wrench 是"扳手、猛拧"，writing 是"写作、文字"，yearn 是"渴望"，比 long 更带感情。yoga 是"瑜伽"，yourselves 是"你们自己"，zipper 是"拉链"。

为什么你背下来的词活不过三天

现在说一件你可能不爱听的事。你之所以背了就忘，不是记性差，是因为你背的是"中文对译"，不是"这个词的用法"。中文对译只给你一个点，而一个词真正的样子是一小片网：它跟谁搭、出现在什么语体里、后面接什么介词、可不可数、有没有贬义。你只拿了那个点，网没拿，所以一到句子里就散。

举个这一章里的例子。instrument 你背成"仪器"，没错；可它是"精密仪器"，是面板上那个读数准的东西，不是车间地上那把扳手，扳手叫 wrench。appliance 才是家电。你要是把三个词都记成"工具/器具"，写出来的句子就会像机器翻译。再比如 accurate 和 precise：accurate 是"准"（对得上真值），precise 是"细"（小数点后几位）。温度计可以很 precise 却不准，这就是 Chen 说的"读数是准的，读读数的人不是"。

再举一个更隐蔽的：salary 和 wage。都是"工资"，可 salary 是按月按年给的，wage 是按小时按件给的。你在作文里写 "The workers received a good salary" 不算错，但读起来像把车间工人写成了白领。这种错，语法检查抓不到，语感一读就出戏。

那怎么办？三条，都很土，都管用。

第一条，一个词至少记一个"搭档"。记 accountable 就记 be accountable for；记 conducive 就记 be conducive to；记 deviate 就记 deviate from。介词是英语的关节，你把关节记住了，句子才站得住。

第二条，一个词至少记一个"场面"。词不离境，这一章就是给你场面用的：ascertain 在 Chen 让你"再核实一遍"的场面里；leniency 在徐姐放过你一次、又没放过你第二次的场面里；turbulent？也不在词表里，但 havoc 在台风那夜的场面里。你记住场面，词就跟着场面一起被想起来，这比你抄十遍管用。

第三条，一个词至少记一个"它不是什么"。hardly 不是"努力地"，是"几乎不"；hitherto 不是"到这里"，是"迄今为止"；economy 不只是"经济"，还是"节约"；custom 不只是"习俗"，还是"海关"。把"不是"记住，你就躲开了最常见的坑。

最后说一句可能有点刺的话：词表是地图，不是路。你拿着地图在屋里转圈，永远到不了车间。这一章做的，就是把你拽到那条路上，让你踩一脚油、闻一下金属味，然后回头再看地图——那时候地图才对你有用。

把这一章当成一次巡检

如果你已经读到这里，我建议你回头做一件事：不要重读故事，重读那些"卡了一下"的地方。你在哪一句上停了半秒，那一句里就有你还没拿住的词。阅读的快感会骗你，卡顿不会。

我给自己定过一个很笨的办法，叫"三次巡检"。第一次巡检只看名词和动词，把一句里的动作和东西圈出来，剩下的形容词副词先不管；你会发现大多数长句的骨架简单得可笑。第二次巡检只看介词和连词，因为它们是句子的关节：of 表所属，to 表方向或对象，for 表目的，with 表伴随，by 表手段。第三次巡检才看那些"大词"，看它们到底修饰谁。三次下来，一句原来读不懂的话，通常只剩一个词不认识——而那一个词，才是你真正要查的。

这套办法用在四六级的阅读上，也一样。你在考场上没有时间三遍，但你可以在脑子里跑得很快：先抓动作和东西，再抓关节，最后才管形容词。很多人读不懂，不是词不认识，是顺序反了——一上来就盯着形容词，读到句尾已经忘了主语在干什么。

再说这一章的"职业选择"那条线。Shen Lu 走的时候说，选那个"允许你在能纠正你的人面前犯错"的岗位。这句话听上去像鸡汤，其实是工程学的常识：一个系统如果不能让错误暴露出来，它就会把错误存起来，等到最贵的时候一次还给你。工厂里叫"让步接收"，实验室里叫"数据好看"，人身上叫"面子"。三样东西，代价一样。

Chen 说的是另一句：厂不是机器，厂是习惯。这两句话其实是一句话的两头。习惯是人的系统，系统是死的，人是活的；所以一个厂的水平，最后落在几个愿意在没人看见的时候也数一遍的人身上。你将来进哪个行业都一样：流程可以买，设备可以买，习惯买不到。

我还想替"车间"说一句公道话。这一年里，我见过太多同学把"进厂"当成失败的退路，好像只有坐办公室才算上岸。可是你身上穿的、住的、坐的、用的，没有一样不是从某个车间里出来的。你可以不喜欢它，但你不该看不起它。一个人对制造业的态度，最后会变成他对"东西是怎么来的"这件事的态度；而一个连东西怎么来的都不关心的人，是很难把材料学好的。

这一章里所有那些看起来"土"的词——tonnage、stoppage、ware、wrench、shipment、bid——它们不是低级词汇，它们是这个行业的工作语言。你能用英语把它们说清楚，你就已经比大多数只会说 "many products" 的人更靠近这个行业。

几个最容易被中文带偏的地方

既然这一章词多，我挑几个"中文一翻就错"的，单独说透。

economic 和 economical。前者是"经济的"（和经济有关），后者是"节约的"。a car that is

economical 是省油的车，不是"经济型轿车"那个意思的直译。这一章的 economy 两个意思都用到了：宏观的经济，和"省"这个动作。

industrial 和 industrious。前者"工业的"，后者"勤勉的"。一个 industrious worker 可能在 industrial 厂里干活，但两个词不能换。

considerable 和 considerate。前者"相当大的"，后者"体贴的"。a considerable sum 是一大笔钱，a considerate person 是体贴的人。

sensible 和 sensitive。前者"明智的"，后者"敏感的"。这两词在作文里混用，意思会拧得很厉害。

respectable、respectful、respective。第一个"体面的"，第二个"恭敬的"，第三个"各自的"。这一章的 respectively 就是第三个，用来把两组东西一一对上：甲和乙分别是 A 和 B。

industrious 这类词还有一个共同陷阱：后缀决定词性，词性决定位置。-ous 是形容词，-ity 是名词，-ly 多半是副词，但也有 friendly、lovely 这种形容词。-tion 是名词，-ment 是名词，-ance 是名词。你把后缀当成"零件"看，长词立刻短了一半；这也是词表里 affixation 那一条真正的用处——它不是让你背术语，是让你在考场上遇到生词时能拆。

拆词要有节制。词根能帮你记，但不能替你猜。manipulative 里有 man（手），所以它和"用手操作"有关，引申为"操纵人"；可你要是拿词根去猜 mand，就会猜出一堆错东西——mand 本身就是个残片，它是 command 或 demand 被切掉前半截剩下的部分。碰到残片，回到完整词，别硬猜。

最后提醒一件小事：这一章里出现的那些单据、手册、卡片，都是真实文本的样子——短句、被动语态、名词堆叠、几乎不用形容词。你在四六级的翻译和写作里，恰恰要反过来：少用被动堆叠，多用人和动作。读的时候学它的准确，写的时候避它的僵硬。这就是"读"和"写"的分工。

材料是被人做出来的

这一章真正的主题不是英语，是"东西是怎么来的"。

你摸到一块铝型材的时候，它身上已经叠了几十个人的判断：熔铸的人决定了合金成分，挤压的人决定了温度曲线，热处理的人决定了它硬不硬，氧化的人决定了它耐不耐候，检验的人决定它能不能出厂，仓库的人决定它什么时候走，外贸的人决定它卖多少钱。你看到的是一根亮闪闪的料，实际上是一串被固定下来的决定。tolerance 之所以重要，是因为它是这串决定里唯一写下来的部分；剩下那些没写下来的，靠的是习惯。

这就是为什么 Chen 说厂是习惯。习惯不是道德，是"没人看着的时候你还会不会数第二遍"。一个厂的次品率，很大程度上不取决于设备有多新，而取决于有多少人在没人看着的时候愿意多数一遍。你在学校里也一样：决定你水平的，不是你上过多少课，是你独自做题时愿不愿意把错的那道再算一遍。这句话听起来像班主任说的，但它是车间的原话。

再说"数据好看"这件事。周敏在实验室里最怕的不是失败，是"太顺"——一次就成的实验，她反而要重做一遍，因为她不相信第一次。这不是迷信，是经验：一个结果如果来得太容易，通常意味着某个环节没有真正被检验。工厂里也一样，一批货如果各项指标都刚好压线过，老师傅会皱眉头，因为"刚好"意味着没有余量，而余量是给意外留的。

我后来把这件事用到了自己身上。大一下学期有一门课，我期末考得不错，但我心里知道那门课的实验我一次都没独立做成功过，都是跟着别人收尾。成绩好看，习惯没长。第二年再遇到同类实验，我就露了底。那次之后我才真懂 Chen 那句"你可以量一个院子，量不了一个习惯"。

再说选择。一个行业里其实只有三种人：设计它的人、卖它的人、和它坏了以后被叫来的人。第一种人享有创造的名声，第二种人享有金钱，第三种人享有……睡眠不足。可第三种人是唯一能把整条链看全的人，因为他必须在半夜里同时面对设计、工艺、供应商、客户和老板。我二十岁那年决定，先做第三种人。这个决定不一定对，但它至少是我想清楚以后做的，不是被推着做的。

如果你只剩二十一天

假设你离考试还有二十一天，手里是这一章和另外七章。我给你一个不需要意志力的办法。

前七天，每天只做一件事：读一章的英文，不查词，遇到不认识的就在旁边画一道线，读完为止。你会发现画线的词越来越少，因为很多词在上下文里自己就解释了。这七天不要背任何东西，你的任务只是让眼睛习惯"读得下去"。

中间七天，回到你画线的地方，一天挑二十个，只做三件事：查它的核心义、抄一个它所在的短语、用它造一个和你自己生活有关的句子。造句子这一步最要紧，因为只有造句，词才从"别人的"变成"你的"。你造得越私人越好——写你宿舍的灯、你妈的电话、你那双开胶的鞋，都比写"环境保护很重要"强。

最后七天，把八章里你觉得最难的段落挑出来，做"回译"：先看中文，自己写英文，再和原文对。对的时候不要只看对错，看三件事——原文用了什么动词、什么介词、什么连接词。这三样是英汉之间差得最远的地方，也是分数差得最远的地方。

二十一天结束，你不会记住全部九百个词，这很正常。你会记住的是"怎么遇到一个生词"，这个能力比九百个词值钱，因为词表会换，考试会换，行业会换，只有"遇到"这件事不会换。

最后

我离开那个厂的时候是八月最后一个周五，六点，天还亮着。门卫按了一下手里的计数器，一声，一个人；再一声，又一个人。五百个人这样被数出去，像上午被数进来一样。我走到马路对面，回头看了一眼那栋楼，四层的办公楼，灰白色的厂房，屋顶上那块被台风吹走过又补上的铁皮。然后我上了公交车。

公交车开动的时候，我忽然想起第一天的自己：白衬衫，新工牌，站在门口不知道手往哪儿放。那个夏天没有教会我什么大道理，它只是把"材料"这个词从课本上抠下来，放进了我的手里：它有温度，有毛刺，有一个师傅的手指印，也有一张被机器切掉前半截的英文卡片。

如果这一章你只带走一句话，我希望是那句最不像英语课的：measure twice。量两次。第一次量的是世界，第二次量的是你自己。

我自己写错过的三句话

既然讲了那么多"应该怎么写"，不如讲讲我写错过的，你可能也会踩。

第一句，我在交接单上写过 "The press is stopped for two hours"。师傅看了一眼说，机器停两个小时是"发生过的事"，不是"它的属性"，所以该写 has been stopped 或者 was stopped。中文一个"停了"，英文要分"状态"和"事件"。这就是为什么车间英语里被动语态那么多：它不关心谁干的，只关心"发生了什么"。你写作文时如果通篇被动，会显得像说明书；可你写事故报告时如果不用被动，又会显得在推卸责任。语态不是语法题，是立场题。

第二句，我在邮件里写过 "I very thank you for your help"。徐姐没笑，只把句子改成 "Thank you very much for your help."，然后说，very 不能直接修饰动词，它修饰形容词和副词；修饰动词要用 very much，而且多半放句尾。这类错误中国人特别多，因为中文的"很"什么都能修饰。同理，"我非常喜欢"不是 I very like，是 I like it very much。

第三句，我在答辩时说 "According to me, the tolerance should be wider."。王老师打断我，说 according to 后面接的是"信息来源"，不接自己的观点；说自己的看法用 in my opinion 或者 I think。这个小错很要命，因为它让评委觉得你在引用别人，却没说是谁。

三个错，一个关于时态和语态，一个关于副词位置，一个关于固定搭配。它们都不难，难的是"你不觉得自己错"。这就是为什么我建议你把写过的句子拿给一个真正会的人看——不是让他改，是让他告诉你"这句读起来哪里别扭"。别扭感是可以训练的，训练方式就是被人指出一百次。

单据英语为什么长那样

你可能注意到了，这一章里的合同、手册、作业指导书，读起来都硬邦邦的：短句、被动、名词成串、几乎没有形容词。这不是写得差，是写得"防事故"。

防什么事故？防歧义。文学作品怕不美，法律文本怕歧义。所以合同里不写 "soon"，写 within 30 days；不写 "a good price"，写 the price specified in Appendix B；不写 "we will try"，写 the Seller shall deliver。shall 这个词在日常英语里几乎绝迹，在合同里却是主角，因为它把"应该"变成"必须"。

再看名词化。手册里写 "the disposal of waste is audited"，不写 "we audit how you dispose of waste"。为什么？因为名词化把动作变成东西，东西可以被审计、被记录、被追责。中文的公文也这样："加强管理"、"落实责任"。两种语言在这一点上惊人地一致：越是需要留痕的地方，动词越少。

这对你的写作有什么启示？两点。第一，写议论文不要学它，通篇名词化会让文章没有气；第二，写图表作文和数据描述可以借它，因为你需要的是准确，不是气势。同一支笔，写两种文体要用两种力道。

还有一个细节：单据里的时间几乎都是"点"，不是"段"。on receipt of the goods、within seven working days、prior to shipment——全是可核查的时间点。你写作文说 "in recent years"，那是"段"；写实验报告说 "within 24 hours"，那是"点"。点让人放心，段让人起疑。这也是工程思维的一部分：能被检验的，才敢写下来。

关于方言味普通话

Chen 说的是方言味的普通话，我在这一章里尽量没有把它写成"土"，因为那不是土，那是另一种

精确。

他管"公差"叫"余量"，管"淬火"叫"蘸火"，管"报废"叫"废了"。这些词在标准术语里都有对应，可在他嘴里更短、更快、更不容易误会。语言的第一功能是"在嘈杂的车间里让对方一次听懂"，标准不标准是第二位的。你在英语里也会遇到同样的事：工地上的人说 "the die's shot"，教科书说 "the die has been damaged"。前者短，后者准，看场合。

我后来养成一个习惯：听到一个行业里的人用了一个"不标准"的说法，先记下来，再想想它为什么存在。多数时候，那是因为它省了一个音节，或者避开了一个容易听错的音。语言是省力的产物，不是规则的产物；规则是后来的人总结的。

这也是我不太赞成"背单词表"的最后一个理由：词表给你的是标准形，车间给你的是活形。你先认识活形，再回头看标准形，会觉得"哦，原来是它"；你先背标准形，到了现场往往认不出来——因为真实的话总是短一点、脏一点、快一点。

所以这一章读完，你真正该带走的不是九百个词，是一种耳朵：能在嘈杂里听出那个词，能在单据里认出那个结构，能在师傅一句土话里听出他其实在讲公差。耳朵比词表难练，但练成之后不会丢。

给还在犹豫的人

如果你读完这一章，心里冒出来的第一个念头是"我以后不想进工厂"，那也很好。不喜欢一件事，最好是在看清它之后不喜欢，而不是在听说它之后不喜欢。

我认识的人里，有人受不了车间的噪音，有人受不了三班倒，有人受不了那种"人只是流程里的一个环节"的感觉。这些受不了都是真的，不该被"年轻人要吃苦"一句话压过去。但我也认识另一种人：他在车间里待得住，因为那里每件事都有回音——你调了一个参数，两个小时后你就能在料上看到结果。这种"看得见"的快感，办公室里很少给。

所以问题不是"工厂好不好"，是"你要哪一种回音"。有人要快的回音，去做销售、做运营；有人要慢的回音，去做研究、做材料；有人要硬的回音，去做工艺、做设备。没有高低，只有匹配。你二十岁的时候把这个想清楚，比背完九百个词有用得多。

如果暂时想不清楚，我给你一个笨办法：去现场待两周。不是参观，是待着，跟班、吃饭、下班一起走。两周足够让你知道自己是"受不了"还是"只是不习惯"。这两件事差得很远：受不了是信号，不习惯只是学费。

关于这本书里的其他人

张野后来去了那家轴承厂做技术员，他跟我说，他每天上班第一件事是拿千分尺量一个标准件，量完才开工。他说这不是规程要求的，是他自己加的。我问他为什么，他说："我得先知道我的手今天准不准。"

沈璐在新公司做了半年，给我打过一次电话，说她终于明白"流程"这个词不是贬义。她说以前觉得流程是给笨人用的，现在觉得流程是给聪明人省事的——聪明人也会忘，也会急，也会在半夜犯困，流程就是替他们记住。

丁健考了技师证，朋友圈里发了一张照片：他站在新线的操作台前，穿着新工装，比一年前壮了一圈。配文只有四个字："我也能行。"

王老师还是那句"这都不是事"。我毕业那年回去看他，他说了下一句："事都不是事，人才是事。"我到现在也没完全懂这句话，但我把它记下来了，就像我记下 Chen 的每一句短话一样。

这些人没有一个是"成功人士"，他们只是把一件具体的事做得比昨天好一点。这本书如果有什么野心，就是让你相信：这样的生活也值得写下来，而且值得用另一种语言写下来。

一句话

最后留一句，不是励志，是提醒：你背的每一个词，都是某个人在某个具体的早晨、为了某件具体的事说出来的。你把它记住，等于把那个早晨也记住了一点。等你自己站到那样的早晨里，你会发现自己并不陌生。

写完这一章，我又把那张卡片翻出来看了一遍。卡片上的字已经有点糊了，RECOM 和 MENDATION 中间那道切口，像一道很浅的伤口。我忽然想，学外语这件事大概也是这样：你永远先拿到半截词，剩下的半截要靠自己去补。补得多了，你就知道一个词的边界在哪里；补得久了，你甚至会喜欢上那种"差一点"的感觉——因为它提醒你，你还在路上。

路不短。九百个词走完，你只是把这一章的夏天过了一遍。真正的夏天，得你自己去。去的时候记得带两样东西：一个能写字的笔记本，和一双愿意多数一遍的眼睛。

再补一句题外话。这一章里我最喜欢的一个词是 duly，那种出现在签字下面一行小字里的 duly：duly witnessed。就是这行小字，让一个签名从"我写了我的名字"变成"有人看着我写了我的名字"。语言的分量，常常就压在这种小字上。

长文该怎么读第二遍

第一遍你读的是故事，第二遍要读结构。这一章有四十个小节，看着零散，其实只有三条线：一条是"物"的线，从铝锭到型材到集装箱到货架；一条是"人"的线，从门卫到师傅到学姐到你自己；一条是"词"的线，从工牌上的英文到那张被切断的卡片。三条线在最后一节合到一起：你站在门口，计数器响了一声。

第二遍读的时候，试着只追一条线。追"物"的线，你会发现所有名词都长得像工具；追"人"的线，你会发现所有对话都在教你怎么做事；追"词"的线，你会发现每个生词第一次出现的地方，都是它最该出现的地方——这句是老实话，因为这一章本来就是照着词表写的，只是我们尽量让它们落在该落的地方。

第三遍可以只读对话。这一章的对话都很短，因为车间里的人说话都短。短句是英语里最难写好的东西：它不给你藏的地方，语法错一点、搭配错一点，立刻露出来。你试着把 Chen 的六字句子翻成中文，再翻回英文，如果两次都通，说明你真的懂了那句话。

读长文还有一个好处，很多人没意识到：它训练你的"容忍度"。真实世界里，你读一份合同、一篇报告、一份说明书，永远会遇到不认识的词，而你不能停下来查。长文让你习惯"带着几个不认识的词继续往下读"，这个能力在考场上值二十分，在职场上值更多。

所以别怕这一章长。长是它的用处，不是它的缺点。你读完了，就等于在没有字典的情况下走过一遍真实的语言现场；走过一遍的人，再回考场，心态是不一样的。

为什么叫"伴读"

最后解释一下这个栏目名。它不叫"讲解"，不叫"注释"，叫"伴读"，是因为它想做的事只是陪你把这篇文章再走一遍，走到你在正文里卡住的那一处，停下来，告诉你那里发生了什么。

讲解是自上而下的：我懂，你不懂，我讲给你。伴读是并排的：我也在那句话上停过，我知道它为什么绊人，我把绊的地方指给你看，然后你自己迈过去。这两种关系的差别，比讲多少词重要得多。

所以你不必把这一章当成教材。它更像一个人坐在你旁边，一边翻书一边说话，说到某处忽然想起"这个词我第一次见它也以为是那个意思"，然后就多说两句。你可以打断它，可以不同意，可以合上书去吃饭，回来再接着看。阅读本来就是这样的事。

如果哪天你进了车间、进了实验室、进了任何一个要把"材料"变成"产品"的地方，忽然想起这一章里的某句话，那这一章就没白写。

补一件小事。这本书里每一章都有一位师傅，这一章的师傅姓陈，名字是我编的，脾气是我见过的。他那种人有一个共同点：话少，但每句都能在第二天被验证。你要在生活里找这样的人，找到了就跟着他走一段；找不到，就先把自己变成那样的人——说一句能被验证的话，比说十句漂亮话难。

这一章到这里就真的结束了。九百多个词，一个夏天，一句"量两次"。你要是只记住最后那句，也好——它至少能让你在下一次想偷懒的时候，多量一遍。

第 8 章　选择的那天

The fan turned its head from left to right and back again, and each time it passed over me it clicked, once, like a tongue against a tooth. I had been awake since five. On the floor, bare to the waist, Zhang Ye sat cross-legged with a screwdriver blade between his teeth and the back of the fan open in his lap. The brass ring around the motor was black with dust. He had wiped it twice and it was still black.

"Bearing," he said, taking the blade out of his mouth. "It is dry. That is your click. Nothing is broken."

He had a way of saying things that made a room feel finished. I lay still and looked at the ceiling, where a stain spread above my bed in the shape of a cape — or a beast; in nine months I had never decided which. Every morning of this first year I had looked at it. This morning it looked like a map of somewhere I had not been.

Anxiety, it turns out, is physiological before it is anything else. It gathers in the chest, takes up air, and makes the ear ring at a pitch just under hearing. I had a deadline at five in the afternoon and a defense at nine in the morning, and between those

two numbers my whole life as a student seemed to be folded into one sheet of paper that was, at that moment, still lying on the chair with my assignment clipped to it.

The outline ran to four pages. Shen Lu had gone through it on Tuesday with a red pen and left marginal notes in a hand so small it looked like notation rather than writing. On page two she had written: "You are inclined to boast. Cut the adjective. Keep the number." On page three, next to a sentence I had been proud of, she had written only a question mark, and then, underneath, in the same small hand, "Surely you do not mean this." I had read that sentence forty times since Tuesday. I still did not know whether I meant it.

Under the chair was a wooden box the size of a briefcase, which Wang Laoshi had handed me on Monday when the faculty office was cleared out for the installation of new cabinets. "Take it," he had said. "Nobody has read these in twenty years, and, moreover, nobody is going to. None of this is a big deal. If it turns out to be a big deal, come and see me and we will make it not a big deal."

The box held about two hundred cards, homemade, assorted in size, some of them cut from the backs of calendars. They had belonged to a doctoral student in the eighties, someone who had begun a manuscript on prefixes — the small units of meaning that attach to the front of a word and change its direction — and never finished it. The moisture of thirty summers had got into the left edge of the box. It had eaten the first two or three letters of every card in the front third, so that the remnants read like words with their heads cut off.

I had spent two evenings on them. Mander, I was fairly sure, had been commander. Memorate had been commemorate. Mense must have been immense; merse, immerse. Monplace was commonplace, which is a word that means nothing until you have lived in a dormitory for a year. Mune and munist were commune and communist, and the second one had probably cost that student something, depending on which year he wrote it. Stam was the one I could not solve. I had tried stamp, stamina, stammer, and had settled on stammer, because a man who studies prefixes would know that a stammer is a stutter and would find it funny that the word itself stutters.

Zhang Ye sat up and put the screwdriver down. "You are doing the box again."

"I am doing the box again."

"It is six in the morning."

"It is ten past."

He looked at me the way he looked at a joint that would not hold. Then he said, "Breakfast is a ceremony for people with time," and went to the bathroom to wash his hands, and I heard the water run for exactly as long as it took him to count to twenty, which was his own private arithmetic for how much water a man was allowed.

We ate standing up. There was one egg each, yoghurt that had been in the refrigeration unit too long, and bread. The microwave in the corridor had a knob missing, so you turned the stub with a coin, and the machine hummed like something being tortured — no, that is too strong a word for a microwave; it hummed like a man humming with his mouth shut. I put a napkin under my bowl because the table had a dent in it that collected crumbs, and because a hygienic table is a small satisfaction you can afford at six in the morning.

"Nine o'clock," Zhang Ye said, with his mouth full. "Which room?"

"Three-oh-eight. The one with the terrace."

"Take the laptop and take the printer copy. Do not take the draft." He swallowed. "If the machine dies, you want paper."

This was, I would later understand, the entire curriculum of the year, delivered in four sentences by a boy who had never written a blog post in his life and did not intend to.

At seven twenty I went out. The road beyond the north gate was already breathing steam off its own surface, and the air smelled of diesel and wet grass and, from somewhere behind the shops, of frying oil. The summer had come early and meant to stay.

I walked past the accountancy firm with its plastic sign and the woman who opened it at half past seven every morning and wiped the glass with a squeegee; past the lottery shop where a man in a vest was already scratching cards at the counter, losing money in the way other people drink tea; past the hardware shop with its crates of assorted screws and its coils of copper wire and its nickel-plated hinges; past the vegetable stall where the cauliflower sat next to a pumpkin the size of a basketball, and a pyramid of sour plums, and a crate of eggs with a cardboard sign I could not decipher; past the flower shop, which at that hour was only a bucket of roses and a broom; past the parcel depot with its packages stacked sideways in a tower that leaned; past the café where the cashier was counting coins into a tray.

Then the crossing, and the printer's shop on the corner, which opened at six because the owner had a daughter in primary school and a mortgage and no interest in sleep.

Inside, two machines were running and a third was shut down with a cardboard sign on it that said, in crayon, DO NOT. The owner was a stout man of about fifty with a crisp white shirt and the customary manner of a man who has explained the same thing ten thousand times and has decided to keep being amiable about it. He called every woman madam and every man "young man", and he called the machines "she", and he read the newspaper aloud the way other men smoke.

"You want the assembly of the whole thing," he said, taking my kit of papers. "How many additional copies?"

"Three. One for the committee, one for me, one for my supervisor."

"Four," he said. "Always four. You will lose one. That is not pessimism, that is experience." He fed the first sheet in. "The printer will jam on page six. She always does on page six."

She jammed on page five. He opened the panel with the air of a surgeon, inserted a pair of tweezers into the gap, and pulled out a crumpled corner of paper the size of a postage stamp. "There. Now she will be fine until page eleven."

While the machine coughed through my proof, I watched the screen behind the counter. There was a preview of each page, tiny and grey, and I saw my own chart — the one with the elasticity curve and the thermal conductivity — appear and disappear and appear again. It is a strange thing to see your own year reduced to a thumbnail on somebody else's monitor. The authenticity of it, the actuality, is not in the pixels. It is in the weight of the toner on your fingers.

At twenty to eight my phone buzzed. Shen Lu, who never called before nine, was calling before eight.

"Are you at the printer's?" she said. "I need a copy of my lease. Two copies. The landlord wants one with the handover list, and I want one for the file."

"You are moving out today."

"I am moving out today. The train is at seven forty. Hence the urgency." There was a pause, in which I could hear a kettle and, behind it, the flatness of an empty room. "Lin Yi, do not romanticize this. I have eleven boxes and a bicycle and a deadline."

She sent the lease as a photograph of a photograph, crooked, with her thumb in the corner. I read it while the machine whirred. It was the standard thing, four pages of clauses that nobody reads until something goes wrong:

"The tenant hereby acknowledges that the residential property situated at Number 18, Lane 4, Xuezheng Road, is leased for a term of twelve months, commencing on the first day of September. Upon the expiry of the said term, the tenant shall restore the premises to the condition recorded in the inventory annexed hereto, ordinary wear excepted. A deposit equivalent to one month's rent is held by the lessor as security against damage, loss, or unpaid utility charges. Subletting is prohibited without the written consent of the lessor. Any dispute arising hereunder shall be referred to arbitration in the district where the property is situated."

I read it twice. It is a remarkable thing that the most solemn document of a young person's life — the paper that decides where she sleeps — is printed on paper thinner than a newspaper, and that the language of it is designed to be read by no one. The requirement was that she return the keys and sign a reservation of liability for the

meter reading. The estate agent had typed her name wrong: Shen Lu, with a character that was not a character but a dot.

"She has misspelled me for a year," Shen Lu said, when I told her. "Let her. It is not my name that is wrong. It is her keyboard."

The television above the counter was on, muted, with subtitles in yellow. It was an American channel, the kind that plays in waiting rooms, and the subtitles were always three seconds behind. A senator was speaking about a bill; the caption said SENATE, then CONGRESSIONAL, then the name of a committee I did not know. Then a woman in a blue dress was standing outside a building with columns, and the caption said EMBASSY, and then AMBASSADOR, and then a man in a uniform was shaking her hand.

The owner glanced up. "Politics," he said, with the contempt of a man who has opinions and no power. "Every four years the electorate loses its mind. The presidency is decided by three states, and the other forty-seven pretend they matter. A coalition government falls in Europe, and nobody here can name the monarchy it fell under. My cousin works for a ministry in another province, and he says the bureau above his bureau has never once answered a letter."

"Hence the anger," I said.

"Hence the newspaper," he said, and folded it.

The paper had a story about a hospital where a surgeon had been accused of malpractice — the accusation was that he had operated on the wrong side of a patient's spine — and the prosecution was asking a jury to indict him. Below that, a smaller story: a company had alleged that a rival had stolen its formula for a liquid detergent, and the case had been thrown out because the formula was, in the judge's words, "not distinctive enough to protect." Below that, a photograph of a statue with a sheet over it, and a caption about censorship and a complaint that the work was obscene. A gallery had canceled the show; a group of students had organised a petition; the director of the gallery had resigned, citing "irreconcilable differences" with the board.

"Pointless," the owner said. "A sheet over a statue. You know what that does? It makes people look at the sheet."

He was, I realized, a shrewd observer of a city he had not left in nineteen years.

I paid. The bill was eleven yuan and fifty fen, which he wrote on a pad with a pencil worn down to the stub. I gave him a twenty-yuan banknote; he held it up to the light, not because he doubted me but because that is what a man does with a note, the way a cook tastes the soup he has made every day for twenty years. He counted my change into my palm — a five, two ones, and a dime-sized coin that was not a dime but a token for the metro — and he pushed across the counter a voucher for the next print, ten percent off, valid until December.

"Students always come back in December," he said. "Everybody needs a copy of something in December."

On the wall by the door there was a flyer, bright green, for a free lecture at the language centre: "Vocabulary Quizes, Thursdays, 7 p.m., Room 204. All welcome." Someone had printed a thousand of them, and nobody had noticed that the plural of the word doubles its last letter. I stared at it for a while. It was, I thought, the perfect emblem of the year: a thousand copies of a small mistake, distributed with confidence, pinned to a wall where nobody would correct it.

"You want one?" the owner said. "Take five. Save me the walk to the bin."

I walked to the building the whole campus calls the academy — one building, a car park, and a name that sounds like a nation — with four bound copies under my arm and the awe of a fixed hour in my throat. The corridor smelled of floor polish and, from the laboratory at the far end, of acid: that clean, offended smell, the smell of being wrong in a scientific way.

Room 308 had a long table, three chairs on one side and one on the other, a projector on a cart, and a whiteboard still carrying the aftermath of Tuesday's lecture: a chart with an arrow on it, an equation, and, in the corner, a cube drawn in three dimensions by someone who had given up halfway. Above the door a printed sheet said AUDITORS WELCOME.

"Sign the attendance sheet," Shen Lu said, without looking up. "If you do not sign it, you did not attend, and if you did not attend, your accomplishment does not exist. That is the logic of administration." She pushed the pen across. She was the secretary of the session, which meant she kept the record, the timing, and, in her own words, the conscience.

Zhang Ye came in and sat at the back, in the auditor's row, with his back straight. He had shaved badly. He gave me a nod that meant: begin, and do not be stupid about it.

There were three on the committee. The materials teacher, who had a habit of tapping the table with a pencil; the computing teacher, who smiled at everything and wrote nothing; and a visiting professor from another institute, who had the stern, thrifty look of a man who travels light and remembers everything.

I had five minutes. I used four and a half.

I said: we take discarded aluminium cans — the thin, cheap ones — and turn them into a porous oxide that can adsorb a dye out of water. I said: we dissolve the metal in acid, precipitate it, dry it, bake it, and characterize what comes out. I said: the surface area is measurable, and the pore size is not uniform, and that lack of uniformity is the whole problem, because a homogeneous pore structure is what makes a mechanism claimable rather than merely plausible. I said: the second half of the project is imaging — I put the specimen under the microscope, I photograph it, and I let a model classify

the pores from the photographic record, which is faster than my eye and less tired than my attention.

Then I stopped. There is a silence that follows a defense in which you can hear the projector fan and your own pulse, and it is the longest silence in academia, because in it three people decide whether your year was a contribution or an exercise.

The materials teacher tapped his pencil. "You attribute the adsorption to surface area. But your control sample — the one you did not activate — also adsorbs, does it not? Considerably less, but measurably less is not nothing. So the question is whether you have demonstrated a mechanism or a difference."

"That is a fair criticism," I said, and I meant it, and I hated that I meant it. "I can demonstrate the difference. I cannot yet demonstrate the mechanism. In the report I have ascribed the effect to surface chemistry — specifically to hydroxyl groups — but the evidence is circumstantial, and I have written the word circumstantial in the text myself, so that the committee would not have to."

The visiting professor looked up for the first time. "Good," he said. "A student who indicts his own evidence saves the committee the trouble. Proceed."

The computing teacher smiled and asked the question I had been dreading in the way you dread a dentist: how many images were in my database, and how did I avoid teaching the model to agree with me.

"Four hundred and twelve," I said. "And I split them before I looked at them, which is the only honest way to do it, and I report the deviation on the held-out set, not the training set. The model is not clever. It is consistent, which is better."

"And the error?"

"Six percent on pore size, nineteen percent on the classification of shape. The large error is on the samples that are partly blocked, because a blocked pore looks, in a grey image, exactly like a shallow one."

"Hence the nineteen," he said.

"Hence the nineteen."

Shen Lu had not spoken. She had been timing us, and now she closed her notebook and asked her question in the flat voice she used for surgery of this kind.

"You ascribe the improvement to a mechanism you cannot see, on the basis of a difference you can measure, and you call the gap between them circumstantial. Fine. But look at your own slide: the optimum activation temperature is five hundred degrees, and the curve on either side of it falls gently, which means the process is not sensitive, which means any number between four eighty and five twenty would have worked. What, exactly, did you learn from that curve?"

I stood there with my mouth open. And then I said the only true thing I had: "I learned that my apparatus is not precise enough to distinguish the optimum from the plateau, and that I have been reporting a point where I should have been reporting a range. The curve is flat because my thermocouple is worn and my furnace drifts by fifteen degrees either way, and I knew that in March and I did not write it down, because a flat curve is a prettier story than a worn thermocouple."

The room was quiet. The pencil stopped.

"That," said the visiting professor, "is the only sentence in this session that I believe wholly." He wrote something. "Correct the report. State the uncertainty. Do not apologize for it — an uncertainty stated is a result; an uncertainty concealed is a lie with a number attached."

"Concede the point, keep the data," Shen Lu said, and shut her notebook with a snap. "That is the whole trade."

Outside, in the corridor, Wang Laoshi was waiting with a plastic folder and the expression of a man who has been waiting for eleven minutes and has decided to enjoy it. "None of this is a big deal," he said, before I could speak. "Four minutes over, one correction required, zero casualties. I have seen worse on a Monday."

He handed me the folder. "Your physical examination. Two o'clock, campus hospital, second floor. The exchange application requires the form stamped before five, and the form requires the examination, and the examination requires you to be there at two. Hence the folder."

The campus hospital at ten to two smelled of sanitary floor cleaner and of the dietary advice that hangs in the air of every clinic on earth. The form was yellow and had thirty-one fields. Name. Age. Sex. Height. Weight. Waist, in centimetres, which is a number nobody wants to know about himself. Blood pressure. Dietary habits: tick one. Healthful. Average. Poor. How often do you eat fried food, pie, soda, sweet tea? How many hours do you sleep? Do you smoke? Do you drink? Surgical history. Family history of high cholesterol, diabetes, heart disease. Allergies. Medication currently taken.

I ticked average and felt average, which is, I suppose, the point of the form.

The nurse, who was sixty-odd and had the manner of a woman who has weighed ten thousand teenagers and been unimpressed by all of them, put me against the wall, lowered the bar onto my head, and said, "Slightly taller than last year. You stopped growing in February, I expect." Then the scale. Then the waist, which she measured without comment, which was worse than a comment.

"Not obese," she said, writing. "Slightly over what the table says for your height. The table is a blunt instrument, but it is the instrument the form asks for, so the table wins."

The doctor was younger and tired in the particular way of people who see forty patients before lunch. He pressed two fingers under my jaw and told me to swallow. He looked into my ear. He asked about headache, about dizziness, about pain in the ribs, about the spinal column, about the ankle I had turned in September and never had looked at. He put the stethoscope on my chest and listened to the valve of my heart open and shut, and for eleven seconds I was only a sound.

"Nourishment," he said, at last. "You students live on noodles and ambition. The hunger you feel at eleven at night is not hunger; it is boredom with a stomach. Eat breakfast. Real breakfast, not a biscuit in the corridor."

"I do eat breakfast," I said.

"Then you are in the minority," he said, "and the minority is usually right." He stamped the form and signed it with a flourish that took less than a second and cost me half a day.

In the corridor by the lift there was a notice board, and on it a poster from the counselling centre, printed on the same green paper as the flyer in the print shop. It said: TIREDNESS IS NOT WEAKNESS, and under that, in smaller type: "Weariness passes. Despair lies to you about time — it tells you that this hour is all the hours. If you have thoughts of suicide, that is a medical emergency, not a character flaw. Call the number below. Someone answers at three in the morning. Someone always answers."

I stood in front of it for a while, with the stamped form in my hand, and I thought about November, when a boy on the fourth floor of my building had stopped coming to class and nobody had noticed for nine days, and how the word "hysterical" had been used about his mother when she came to collect his things, and how cheap and stupid that word is, and how the poster had put the word distressful where most posters put a smile.

Across the road from the hospital gate stands a church, red brick, with an arch over the door and a notice board that faces the pavement. It was built in nineteen-oh-seven by missionaries who came up the river and stayed, and the board outside it is still the best-kept board in the district: a paragraph about the parish orphanage and its need for winter coats; a verse for the week; a line about Christ and the poor, printed in three languages; and, at the bottom, the times of the Sunday service, which is at nine, and which, the board notes, is "open to the curious as well as the convinced."

I am not a believer, and I have never pretended to be one, but I like that board. There is something humble in a notice that does not argue. The divine, if it is anywhere, is probably not in the argument; it is in the fact that a widow in this street knits hats for an orphanage two provinces away and does not tell anyone her name. The proverbial good deed is proverbial because it cannot be traced back to the doer. That is also, I later realized, the structure of a fable, and of a mythology, and of every story my

grandmother told me about people who were not related to us.

Two hundred metres on, the art college had pasted a poster over an older poster, and the older one was still showing at the edges: a photograph of a bronze figure with a sheet draped over it, and above it the words WITHDRAWN AFTER PROTEST. The new poster announced a symposium on "Art, Censorship, and the Civic Voice", with four speakers and a moderator, and a note that attendance was free but the hall held only ninety.

I read both posters twice. Somewhere in the gap between them was a quarrel I had no part in and every interest in: a nude figure had been called obscene by a councillor; the gallery had called the complaint scandalous; a petition had demanded the show reopen; a counter-petition had demanded it stay shut; and both sides had used the word freedom as if it were a door that only opens outwards. There had been a demonstration on the steps, forty students, a megaphone, and a rain that started exactly when the speeches did.

I noticed, walking away, that I had spent four years of my education being taught how to measure a pore and no time at all being taught how to judge a quarrel in which both sides were partly right. Apparently the second skill is supposed to arrive by itself, the way balance arrives, or the way facial hair arrives. It does not.

At half past three the room was already half in boxes. Zhang Ye sat on the floor with everything he owned arranged in a half circle, conducting a trial in which each object was the accused and the charge was uselessness. He does not pack. He arbitrates.

His phone rang at twenty to four. It was his mother, and the call lasted nine minutes, of which I heard one side and understood most of it, because he speaks to her in sentences of four words. "Not yet. Two weeks. How many acres? All of them? Even the barren one? Then burn it. Burn the stubble and plough it under. Is the livestock grazing on the hill or in the yard? Then move them. Beware the ditch behind the brewery, the bank crumbles when it rains. Yes. Yes. I said yes. Do not beg him. Let him keep the boundary stone if he wants it. It is a stone."

He put the phone down. "My mother has three acres of paddy and eleven of hill and a neighbour who moves the boundary stone six inches every spring. Agriculture in my county is a quarrel with a mortgage."

"Bad year?"

"Bad year. The rice has not ripened and the price has fallen, which is what happens when a good year and a bad one coincide for different reasons." He put a brace and a buckle into the crate marked KEEP, and tied the cord with a knot he learned from his grandson-aged uncle. "Enough. We sell the rest."

So we wrote the post together, and I put it here because it is the most truthful document of my first year in Hangzhou, more truthful than any report I filed:

"Dorm clearance before Friday. Everything bought in September, used for nine months, sold at a price that reflects reality. Desk lamp, 25, clip broken, bulb new. Kettle, 15, handle taped. Pan, 8, non-stick, one crack, still non-stick in the middle. Tub, 5, plastic, cracked at the rim. Cotton sheets, two pairs, 30, washed eleven times. Mirror, 10 — the ordinary one, not the cosmetic one; my roommate insists I say this. Cork board, 6, holes from last year's timetable. Balloon pump and four balloons, free, long story, do not ask. Binoculars, 40, bought for a bird-watching phase that lasted eight days and produced one bird, which was a sparrow. Compass, 12, brass, does not point north any more, or perhaps it does and the world has moved. A couple of textbooks, annotated in three colours by a diligent idiot, 5 each. Badminton racket, no strings, 3. A coat that smells of the shop, 20. Boots, size 42, worn through at the heel, free to a student who walks home. Bring change. Do not bring advice."

Zhang Ye read it and changed three words. He crossed out "works" and wrote "mostly". He crossed out "free" and wrote "five". And he crossed out "diligent" and wrote nothing, which I resented for about an hour and then accepted, because a diligent idiot is still an idiot, and the annotation in three colours was the only thing on that page that embarrassed me.

While he taped the box I remembered the documentary from military theory in November: a town on a river, a fortification of earth and timber, three hundred soldiers, and artillery on the ridge above the town firing every eleven minutes for nineteen days. The cannon had been captured from an enemy column; the shells were counted by hand; on the last day there were nine bullets left for the whole garrison, and a militant in a white shirt walking up the street with a flag and being shot in the back at forty metres. I remember a soldier writing his rank and number on a wall in chalk, and a girl carrying water up a stair that had no railing left, and the narrator saying, in the flat voice of people who have decided not to perform emotion, that the atrocity committed afterwards had been recorded by two sides and believed by neither.

I had cried in that hall, quietly, in the dark, ashamed of the noise of my own breathing. Then the lights came up and four hundred students walked out arguing about lunch. That is the whole problem of being nineteen: you can hold a catastrophe in your chest for ninety minutes and then discuss noodles.

At twenty past four I stood in Shen Lu's flat on the sixth floor of a building with no lift, and the flat looked larger empty than it had ever looked full. Eleven boxes. A ladder she had borrowed from the janitor. A laundry basket with one stocking in it. And a hollow sound when you walked across the second bedroom, because the floor there is laid over nothing but air and joists, which is also, I thought, a fair description of a person at the end of a degree.

The agent came up with a clipboard and went through the place the way a doctor goes

through a body, room by room, naming what was wrong in a voice of studied calm.

"Item one. Wall, second bedroom. Two holes from a shelf."

"Filled with toothpaste," Shen Lu said. "Painted over in March. I told your colleague."

"Item two. Valve on the radiator is stiff."

"It has been stiff since I moved in. It is stiff in July, when nobody needs it, and stiff in January, when everybody does."

"Item three. Drain in the bathroom is slow."

"Hair," Shen Lu said. "Four years of hair. Charge me for the hair."

"Item four. Chimney swept in April — good. Hedge cut — good. Ladder in the shed — not on the inventory."

"Then it is a gift," she said. "Merry Christmas."

"Item five. Debris in the yard. Bricks, a length of pipe, and a pile of trash that the council will charge for."

"Paint tins," she said. "Left by the landlord in September, with three inches of paint dried solid in each. I am not paying for his own rubbish."

The deduction came to two hundred and forty: ninety for the holes, ninety for the trash, sixty for a stain on the mattress that nobody could explain and everybody understood. The deposit had been twenty-four hundred. She read the deduction sheet twice, folded it, and put it in her bag without an argument, which surprised me until she said, on the stairs, "Two hundred and forty is the price of not spending an afternoon arguing with a man who has a clipboard and a pretext. I have a train."

She was, I noticed, wearing the outfit she wore for interviews: a grey jacket, flat shoes, no jewellery. Her hair was tied back so tightly that the skin at her temples was white.

On the landing she stopped, dug in her bag, and gave me a box of pills. "For the ankle. You turned it in September and you have been walking on it like a man walking on a lie ever since. Two a day, after food."

The leaflet inside the box was printed in six-point type and translated from a language nobody at the factory reads:

"Adults: one pill twice daily. Do not exceed the stated dose. May cause irritation of the stomach, a rash, or bleeding of the gums. Stop taking and notify a doctor if any of the following occurs: severe pain, swelling of the face, difficulty in breathing. Toxic in overdose. Keep out of the reach of children. Store below twenty-five degrees."

I read it on the stairs and felt my stomach consider the warning personally. Then I

put the box in my pocket and decided to trust the woman more than the leaflet, which is, I suppose, how most people take most medicine.

At ten to five I was in the office on the ground floor of the academy, with the form and a pen that belonged to somebody else.

The exchange application is four pages. Page one is identity: name, student number, major, year of study, status (ordinary / honorary / visiting), guardian's signature if under eighteen. Page two is academic record: qualifications, grades, ranking in the class, linguistic proficiency — and there, in a box, the sentence: "State your level of proficiency in the language of instruction; attach proof." Page three is a portfolio of projects, three hundred words, no photographs, no appendices. Page four is the dangerous one. It is headed INTENT, and it asks three things: what you intend to study, why it cannot be studied here, and what you will do afterwards for the institution that pays for it.

I had drafted an answer to that page four times. The first draft was sincere and dull. The second was clever and false. The third was both false and dull, which is the usual result of trying to please a committee. The fourth draft was one paragraph:

"I intend to learn how a laboratory that publishes forty papers a year organises its doubt. I cannot learn that here, because here we are eleven people in a room with three working instruments and a supervisor who answers his email at midnight. Afterwards I will come back and teach two of the eleven what I learned, which is a poor return on a scholarship and the only return I can promise."

Wang Laoshi read it standing up, with his glasses pushed up on his forehead, and then he put it down and looked at me for a while.

"You understand," he said, "that this paragraph will be read by four people, two of whom will find it arrogant, and one of whom will find it the only honest page in the pile."

"I understand."

"And you understand that if you go, you go in February, and you miss the spring term here, and the project you started in March belongs to whoever feeds it while you are gone."

"I understand that too."

"Then sign it," he said, "and stop looking at me as if I were about to give you advice. I am not going to give you advice. Advice is what people give when they want the credit for a decision they are not making." He stamped the form. "Submit it before five. It is four fifty-eight."

I ran.

I ran because a deadline in a bureau is not a metaphor; it is a window that shuts. I ran past the notice board where the list of summer placements had been pinned that morning — forty-two names, mine somewhere in the middle of the second column — and I thought, with the peculiar clarity of a man running, that in four months the paper would be gone and the names would be gone and the board would hold the next forty-two names, and that this is either consolation or terror, depending on the hour of the day.

At four fifty-nine I put the form into a tray in a corridor, and a woman I had never spoken to stamped it without reading it, and the whole business of my future was over in less time than it takes to boil water.

Zhang Ye's bus left at twenty past five from the gate by the fountain. His luggage was a duffel bag, a cardboard crate bound with cord, and a plastic basin he refused to explain. He shook my hand, which he had never done before, and said, "Fix the fan before you leave," and got on the bus, and did not look back, and I stood there watching the bus turn left at the lights and become a thing among things.

The canteen at six was half empty and smelled of hot oil. I had a pie — pastry like cardboard, filling like nothing — and a pint of tea that was sweet enough to make my teeth ache, and a bottle of soda I did not open. The recipe for canteen pie is, I am convinced, a state secret: flour, water, salt, and a memory of meat. My appetite had gone somewhere and left a note.

The underground at seven was packed with the after-work crowd — commuters with plastic bags, a cyclist holding his front wheel, a soldier with a kit bag — and I stood holding the rail with my forehead almost on the glass of the door, and watched the tunnel lights streak and vanish, streak and vanish, like a film of my own year projected at the wrong speed.

Hangzhou East at half past seven is a cathedral of transit: four hundred metres of hall, a ceiling that pretends to be the sky, fifty thousand people moving in eight directions, and, above it all, the voice of the station reading announcements in three languages with the same sentence structure every time:

"Passengers travelling aboard Train G seven five three one to Ningbo are advised that boarding will commence at nineteen twenty-two on Platform nine, Track B. Passengers should retain their tickets for inspection throughout the journey. Transit passengers continuing overseas should proceed to the International Hall on Level two. We apologise for the delay to the eighteen forty-five service, which is due to a signalling failure at Qiaosi. Passengers travelling with luggage exceeding twenty kilogrammes should purchase an excess ticket at the counter on Level one. Please do not leave luggage unattended. Unattended luggage will be removed and may be destroyed."

I liked the announcements. They were the only voice in the city that told you exactly

what was happening and what to do about it, and they did it without once asking how you felt.

Shen Lu was standing under the board with a paperback in one hand and a suitcase at her feet, and nothing else — no boxes, no bicycle, no basin. She had posted the bicycle to a second-hand site that afternoon for three hundred yuan and sold it in eleven minutes, which she mentioned the way a mathematician mentions a clean proof.

"Nine minutes," she said. "Delay on the eight forty-five, signalling at Qiaosi. Hence the crowd."

"Hence the crowd."

We stood side by side and watched the board turn over: departures flicking, delays multiplying, a cancellation at the top in red. She read her paperback; I read the board. It was, I think, the happiest twenty minutes of my year, and nothing happened in them.

"Ask," she said, without looking up.

"Ask what?"

"Whatever it is you have been not asking since March."

So I asked the small question instead of the large one, which is what cowards do and what I did. I asked her whether she thought I should go abroad in February, and she closed the paperback with her finger in it and looked at me properly for the first time that evening.

"No," she said. "Go, but not for the reason on your form. You wrote that you want to learn how a big laboratory organises its doubt. Fine. But you will also learn that a big laboratory organises its doubt by hiding it in committee minutes, and that nobody there will teach you what to do with your own disappointment. That you learn here, in a room with three instruments, or you learn it nowhere. Go for the instruments. Stay sane about the rest."

"That is advice," I said. "Wang Laoshi says advice is what people give when they want credit for a decision they are not making."

"Then call it a warning," she said, "and give me credit for nothing."

The announcement came: boarding, platform nine. She picked up the suitcase, and I carried it up the escalator because it was heavier than she had admitted, and at the gate she took it back and put her hand on my shoulder for exactly as long as it takes to say a sentence, and said nothing, and got on the train, and the doors closed with the soft thud that doors make in China, which is gentler than the doors in England, where I have never been.

Aboard, she found a window, and raised her hand once — not a wave, a wink of the palm, the gesture you make to a waiter who has brought the bill — and the train pulled

out and became a line of light and then a smell of brake dust and then nothing.

I stood on the platform until the board had flicked over twice. Then I went down to the underground, and came up at dusk three stops early, and walked the last two kilometres home through a city that was turning on its lights. There was a storm somewhere east of the city — I could see it flicker under the cloud — and the air smelled of rain that would not arrive. A tornado had touched down in a county two hundred kilometres away that afternoon; it had taken three roofs and a shed and left the rice standing, which is the kind of thing that makes people say heaven has a sense of humour and makes me say heaven has no sense at all.

The room at nine was ourselves and nothing else: two beds stripped to the mattress, a fan still turning, a desk with a ring in the varnish where a cup had stood for nine months. I opened the drawer and emptied it into the bin in one sweep, and then took three things back out again, which is, I think, the whole ritual of leaving: throw it all away, and then rescue what fits in a pocket.

The drawer held a shaver with two spare blades, a silver souvenir spoon from Ningbo, a paperback with a train ticket as a bookmark, a tag cut from a coat I had bought in September and worn twice, four rubber bands, a gramme weight from the physics lab, a dead mobile phone, and a postcard of a beach I had never visited, kept because the sky in it was the colour of the sky over my grandmother's house in August.

The wardrobe held a suit jacket worn once, three shirts, a stocking with no pair, a handicraft box from Yiwu with a broken hinge, a pottery cup thrown by a girl in my class who had given it to me because she had made forty of them and could not give forty cups to forty people, and a length of yellow cord I had kept because it was good cord, which is the kind of reason a man gives for keeping things he will never use.

Under the bed was the wooden box of cards. Since the room was empty and the hour was late, I sat on the floor and read the whole front third of it, card by card, writing the definitions into my notebook as I went, because a word without a definition is a box with no key:

"Anchor: an iron hook that holds a ship to the bottom of the sea; also, the man who reads the news on television, because he holds the programme still."

"Hawk: a bird that hunts in daylight and sees a mouse from two hundred metres; also, a politician who wants a war, used by people who want no war."

"Hive: a box of bees; also, a city at eight in the morning."

"Paw: the foot of a beast; also, the hand of a man who has had too much to drink and is reaching for yours."

"Horn: the hard growth on the head of a bull; also, the thing in a car that makes the

noise everybody hates."

"Rib: one of the curved bones that make a cage around the heart; also, a cut of meat, which is the same object seen by a butcher and by a doctor."

"Jaw: the bone that holds the lower teeth; also, to talk at length without saying anything, which is why the idiom is never complimentary."

"Vein: the tube that carries blood back to the heart; also, a layer of ore in rock, because the miners thought the two looked alike; also, a mood, as in a vein of melancholy."

"Volt: the unit that measures how hard electricity pushes; named after a man who never saw a battery."

"Wheel: a disc that turns on an axle; also, the thing you reinvent when you are young and somebody has already invented it."

"Log: a length of cut timber; also, a record kept daily at sea, because the sailors dropped a log of wood on a line and counted the knots that ran out."

"Ridge: the line where two slopes meet; also, the long narrow hill that a road follows because the engineers had no choice."

By the fortieth card my handwriting had loosened and the definitions had turned into something else — not a dictionary, but a diary with the words left in. I noticed that every card I wrote that night had a second half beginning with also, and that the also was always the part I cared about. A word is never one thing. That is the whole trouble with words, and the whole pleasure of them.

Before midnight I filled in the consignment note for my own two boxes at the counter in the shipping shop by the north gate, because the freight office closes at half past eleven and the truck leaves at six:

"Consignor: Lin Yi, dormitory six, room four twelve. Consignee: the same, Village Road, Songyang, Lishui. Contents: books, clothing, one desk lamp, one wooden box, no commercial value. Weight: twenty-three kilogrammes. Freight: forty-one yuan, prepaid. Transit: three days by road, through the hub at Jinhua, stored overnight in a warehouse if the line is busy. Insurance: declined — the contents are not worth the premium, and the box is worth everything."

The clerk read it, stamped it, and asked whether I wanted the boxes wrapped in plastic. I said yes. He wrapped them in plastic and taped the seams and wrote my name on the side in marker, and I felt the peculiar gratitude you feel towards a stranger who does the last small thing properly.

Then the laboratory, at half past eleven, to hand over the log and the samples to Zhou Min, who was still there at that hour because she is always still there at that

hour. I copied the last page of the record into my notebook because I wanted a copy of the numbers in my own hand:

"Sample twelve-B: dioxide of titanium, four grammes, dried at one hundred and five degrees for twelve hours. Surface area: two hundred and eleven square metres per gramme, plus or minus nine. Uranium: below the limit of detection — hence no reading, and hence no conclusion. The filtrate radiates nothing measurable above background. Do not dispose of the acid down the drain; the building rule is a bucket, and the bucket is collected on Thursdays. Anyone who contaminates the sink cleans the sink. Report any spill immediately, however small, including a spill of water, because the floor is not level and the water will find the power outlet before you find the mop."

I initialled each line. Zhou Min read it standing up, initialled it too, and said, "Go home. Sleep. Tomorrow you are unemployed and that is a rare and valuable condition."

At one in the morning I drafted the post that I never published. It was going to be the only thing I had ever put online that was not a photograph of a meal:

"Absurd, the things a year does. In September I could not find the library. Tonight I walked home through a city that I know the way I know my own room in the dark, and I noticed that the pleasure of knowing a place is not knowledge; it is affection of a low grade, like the affection you feel for a chair that has learned the shape of you.

I have learned exactly two things this year that I did not know in September, and neither of them is on any examination. The first is that most people are doing their best with what they have, and the second is that what they have is usually less than you think and more than you fear.

Everything else — the activity, the credit, the ranking, the annual report — is a system that counts what is easy to count and calls the rest noise. You can live inside that system, and many admirable people do, and it will reward you for adapting to it. Or you can adapt to something else and accept that the system will never see it. Advancement in the first case is visible and fast. In the second it is invisible and slow, and the only witness is somebody who has watched you work at midnight and said nothing afterwards.

I have an adversary. It is not a person. It is the voice that says, at four in the morning, this is pointless, you are average, they know it. I have decided to treat that voice as a neighbour who plays music at two in the morning: I cannot evict it, so I have learned to sleep through it.

Nobody asked me for this post, which is why I am not publishing it. An advocate who has not been asked is a bore. Nonetheless — and I write this for the one person who might read it in five years and think kindly of me — the alliance between a student and a subject is a marriage of a kind, and it survives on the same two things every marriage

survives on: somebody agrees to do the boring half, and somebody agrees to notice."

I read it over twice. It was too honest to publish and too careful to throw away, which is exactly the category of document that ends up in a box under a bed.

At two in the morning I wrote the letter. I am conscious that a letter written at two in the morning is a document of poor legitimacy, and I am conscious, also, that I wrote it anyway, and that this is probably the most accurate illustration of my character that exists anywhere.

"Shen Lu — you will never read this, which is why I can write it, and that cowardice is the first thing I should confide to you, since it is the first thing you would have noticed yourself.

Here is the truth, or something close enough to the truth that the difference no longer bothers me. In March, when you took my report apart in front of four people, I hated you for eleven minutes. Then I walked home and copied your criticism into my diary word for word, and I have read it more often than I have read anything else this year, including the textbook I was examined on.

People call that admiration. I think it is closer to dependency, and I know it is not love, because love wants to be near a person and I wanted to be the person you thought I could be. Those are different ambitions, and only one of them is generous.

You said once that a laboratory is a place where people agree to be wrong in public. I wrote it down. I have since decided that the same definition works for a friendship, works for a marriage, and does not work at all for a family, because a family cannot afford to be wrong in public and so practises in private for twenty years.

I wanted to tell you four things and I have written three, and the fourth is the one I cannot write, not because it is shameful but because it is vague. It has something to do with the fact that in four years you never once asked me whether I was happy, and that I never once noticed whether you were, and that we spent nine months being useful to each other and called it friendship because we had no better word and no time to look for one.

Forgive the sentence. I will not send the letter, so the sentence costs nothing.

One more thing, and then I stop. When you said no, go but not for the reason on the form, you were doing the thing you always do: you took a complicated feeling and gave it a shape I could carry. I have tried to do the same for somebody else since March, with mixed success, mostly failure, and I have learned that it is a skill and not a gift, which means it can be taught, which means somebody taught you, which means you were not born kind and chose to be, and that is the only kind of kindness I trust.

Goodbye. Do not answer. There is nothing to answer."

I put the letter in the wooden box, on top of the cards, and shut the lid. Then I sat

for a while in the grey light that comes before the sun actually arrives, when the room is grey and the street is grey and the birds start arguing in the trees outside the window, and I did the arithmetic that everybody does at the end of a year: nine months, thirty-eight weeks, two hundred and sixty-six days, eleven courses, four hundred and twelve images, one hundred and nine pages of notes, three mistakes that mattered, one that still does. The subtraction leaves a remainder that no transcript will ever show, and the remainder is the only part I recognize as mine.

At five, I fixed the fan. It took eleven minutes and a drop of oil on the bearing, and the click stopped, and the room was suddenly so quiet that I could hear the refrigerator two rooms away. I stood there listening to a silence I had spent nine months trying to sleep through, and I laughed, once, out loud, alone, in an empty room, at five in the morning. Happiness, it turns out, is sometimes nothing more than a noise that stops.

Then I did the last round. I turned the fan to the wall so that it would not blow dust onto the bare mattress. I put the key on the desk with a note for the next occupant — a boy I will never meet, eighteen years old, arriving in September with a suitcase and a mother and no idea — and the note said: "The fan clicks. It is the bearing. A drop of oil fixes it for a year. The stain on the ceiling is not water; it is old paint. The third drawer sticks in humid weather; lift it slightly and it slides. Welcome. It is a good room. It was mine."

Then I went down the stairs with the box under my arm, and out of the gate, and past the fountain that runs on Sundays, and onto the road where the first bus was already carrying people to work, and I did not look back at the building — not out of stoicism but because I had forgotten to, which is better.

The train home left at seven forty-two, and I was aboard twenty minutes early, with the box on the rack above my head and a plastic cup of tea going cold in my hand, and the fields outside the window going past in the order I remembered them: hilly, then flat, then the long ridge with the tunnel, then the river, then the yard where somebody keeps a ferry that has not moved in thirty years and paints it every spring anyway.

I thought about destiny, which is a word I do not like, and about luck, which is a word I like better because it claims less. I thought about the choice I had made that afternoon, and I noticed that it had not felt like a choice at all. That is how the large decisions arrive: dressed as errands. You think you are submitting a form before five; actually you are deciding where to spend your twenties.

The tea went cold. The fields went past. Somewhere around Zhuji the sun came up sideways through the glass and lay across my hands on the table, and I fell asleep with my forehead on the window, which is, I have decided, the correct way to end a year: not with a speech, not with a plan, but asleep on a train with the light on your hands and

nothing settled.

I woke somewhere northeastern of Jinhua, with a stiff neck and a bit of dust in my mouth, and the phone vibrating in my pocket. Zhang Ye. He had been home for fourteen hours and had already fixed the pump, argued with the neighbour about the boundary stone, and walked the hilly field with his mother to see how much of the rice had survived the storm.

"The tigress," he said, by way of hello. That is his grandmother. She is eighty-one and rules the household with a stick and a voice that carries across two fields. "She says the rain will come on Thursday. She says the government pays less for grain than it did ten years ago, which is true, and that nobody asks a farmer anything, which is also true." A pause. "She asked about you."

"What did you tell her?"

"That you read books and cannot fix a pump."

"That is accurate."

"Mostly," he said, which is his highest praise, and hung up.

The carriage was half-full and chaotic in the pleasant way of Chinese trains at seven in the morning: a teen playing a game without headphones; a worker asleep with his mouth open; a sportsman in a tracksuit stretching his calf in the corridor; a typist from some agency in the city with a laptop and a neck cushion; a tourist couple arguing about a map; a westerner with a guidebook and a guitar case, who turned out to be a biologist counting birds for a regional survey and who told me, in excellent Mandarin, that the ecosystem of the river delta had changed more in fifteen years than in the previous fifty, because of what we put into the water and what we took out of it.

Outside, the fields went past in bands: paddy, then mulberry, then a stretch of barren ground with a single machine standing in it like a monument; then a village with fifty roofs and one shop; then a town with a warehouse, a motel, a petrol station, and a stretch of motorway parallel to the line, carrying tankers and trailers and one cruiser-sized bus full of sleeping people.

I counted the tankers for a while because I had nothing better to do with my attention: eleven in ten minutes, each of them carrying something the train would never show me. There is a whole industry in being moved past you at eighty kilometres an hour, and I had spent nine months inside one small part of it and never once looked at the rest.

The sun came up properly around half past six and lay flat across the water in the paddies, so that each field held a second sky, and I gazed at it like a man who has never seen sunlight, which is roughly what I was, having spent the spring in a basement

laboratory with two windows and a broken extractor.

Then the hills, and the cuts in the hills where the line runs, and the exterior of a tunnel black with soot, and inside the tunnel the pressure change that makes everybody swallow.

My mother was waiting on the platform at nine forty-one, in the green jacket she wears for travelling, with a bag of oranges she had bought because the train food is bad, which is the kind of practical affection that runs in my family instead of speech.

"Hello," she said. "You look thin."

"I look the same."

"Thinner," she said, and took the box out of my hands before I could protest. She weighed it with her arm, the way farmers weigh a piglet, and said, "Books?"

"Cards."

"Cards," she repeated, in the voice of a woman who has decided not to ask, and we walked out past the taxi drivers calling fares, past the teller machine where a man was swearing at a screen, and into the heat of a small city at ten in the morning in July.

Our house is old and narrow and cool in summer, which city people call charm and country people call insulation. My mother had opened every window before she left for the station, so the rooms were full of air that smelled of rain and dust and the jasmine growing over the wall of the neighbour's yard.

The neighbour came over before I had finished eating. Her name is Aunt Fang, she is sixty-four, and she has the talent — rare and terrifying — of saying the thing everybody knows and nobody will say.

"Thin," she said, confirming my mother's diagnosis. "Thin and brown. They feed you nothing in Hangzhou. I read that the canteens there buy vegetables from a wholesale market at four in the morning and keep them in a fridge for a week."

"Aunt Fang," I said, "your own son studies in Hangzhou."

"Hence I know," she said, with the air of a woman producing a witness, and my mother laughed in the way she laughs, which is mostly a shake of the head.

The town is not poor any more, and that is a fact that takes some getting used to. There are three new houses on our lane with aluminium gates and air conditioning units on the walls, and a compound at the end of the road where forty families from the village now live, and a car wash where there was a pig sty. Affluence arrives in a small town not as money but as plumbing: first the bathroom, then the kitchen, then a car that nobody uses because the distances are two hundred metres.

Aunt Fang believes none of it. She thinks the new houses are a bubble, the young

people are lazy, the schools teach nothing, and the government subsidises everything and therefore nothing. She says so cheerfully, with no anger in it, the way old people say most things; and it would be easy to call her a reactionary, and wrong. She worked a stall for thirty-one years, she paid for two sons to finish school, and she has earned the right to be doubtful about every promise anyone makes her.

The notice board by the village office held four sheets that afternoon, and I read them because I have a habit now that I did not have in September, which is the habit of reading bureaucratic language slowly:

"Notice of the enactment of the revised regulation on rural housing: applications for permission to alter or extend must be submitted to the bureau within thirty days of the publication of this notice. Applicants should note that permission granted under the former regulation is not automatically valid, and that availability of materials is not a grounds for extension."

"Employment service: training places available in elder care, logistics, and basic accounting; applicants must be registered residents; priority will be given to households in receipt of subsistence support and to workers over fifty who have lost employment through no fault of their own."

"Reminder: the dole is not a salary and does not increase with the number of children. Anyone who claims it while working will be prosecuted."

"Cultural note: a lecture on the revolutionary period will be given at the hall on Saturday at three, followed by a film. Admission free. All welcome."

I stood in front of it and felt the whole apparatus of a state doing what a state does: naming categories, setting deadlines, sorting people into the deserving and the rest, and doing it in a language designed to be read once and remembered wrongly. There is no villain in a notice board. That is what is wrong with it.

At noon my mother cooked: pork braised with dark soy, bitter melon with egg, soup of dried scallop and winter melon, and a saucer of my grandmother's pickles, sour enough to make my jaw ache in the good way. She ate little and watched me eat, which is her only form of conversation before two in the afternoon.

"Apply for the exchange?" she asked, at last.

"I applied."

"And?"

"And I applied."

She nodded once. She has never asked a second question about a thing I have decided, which I used to read as indifference and now read as trust, or as a mother's calculation that a second question costs more than it buys.

In the afternoon I walked to the market with her, and the market at three on a Thursday is the whole town's ledger written in vegetables and price tags: cucumber in a crate, pepper dried in a basin, ginger roots like fingers, corn in a basket of husks, tofu floating in a tub of water, pork on a hook with a fan of flies around it, live fish in a tank of grey water, rice in open sacks with a tin cup standing in each, oil in clear bottles, salt in a sack with a dent in the side of it, a clown selling balloons that nobody bought, a clown's balloon clinging to a post, and one corner where a man sold nothing but bait — hooks, line, weights — and sat on a stool calculating nothing at all.

My mother bought by instinct and by quarrel, which in her case are the same thing: two cucumbers, one pepper, three hundred grammes of pork, and a small delicate business of change that took four minutes and involved two other women and a great deal of affectionate rudeness. She is apt to call a price absurd and then buy at that price, which is not a contradiction but a ritual, and I have adopted it without noticing.

The bike repair shop held an apprentice of maybe sixteen, who took a wheel apart and put it together three times while I watched and never once looked at me, and whose fingers moved like a man playing an instrument he cannot hear. A poster for a summer camp flapped on the door: ATHLETIC SCHOLARSHIPS ABROAD — APPLY EARLY — and underneath it, in marker, somebody had written, NOT FOR US. The correction was the most eloquent thing on the street.

A boy I had known at school came out of the phone shop with a new handset in a plastic bag, and we stood in the sun and did the thing that people of nineteen do in small towns, which is to establish, in four minutes, whose career is more promising while pretending to discuss neither. He sells insurance now. He is good at it, he says, and he says it twice. He asked what I study. I said materials. He asked what that is. I said it is everything solid that somebody has decided to sell, and he laughed and said hence insurance, and for a moment the whole joke of the afternoon held still, and then he looked at his phone and the moment disintegrated, the way moments do, into two men saying goodbye.

I walked down to the river afterwards. The river here is broad and muddy and slow, continental in scale and domestic in use: washing floats, a barge loaded with sand, two boys diving off the steps where the women wash, kids fishing for something that nobody eats, a corpse of a dog on the mud that two men were arguing about who should move. Sailors call this stretch the grave, because the current turns; boatmen call it the cabin, because in flood the water sits there like a family and will not leave. Every stretch of water in China has two names, and the second one is always the true one.

On the bank, a shed with a deck of boards over the water, four old men playing cards, a radio playing opera, and a girl practising violin on the steps above them, badly,

stopping, starting, the same bar forty times. Nobody minded. That is the drawback of a small town and the whole grace of it: nobody minds, because everybody has already decided to know you.

At six the television came on by itself, the way it does in houses where nobody has changed the settings since 2019, and we ate in front of the news, because my father cannot eat in silence and my mother cannot eat while talking, so the arrangement suits everybody.

The news that evening was mostly numbers, read quickly by a woman in a blue jacket, with film of a port, a bridge, a factory floor, a premier shaking hands with a statesman from a country I could not have placed on a map two years ago and still could not, quite. There was a segment on the treasury: tax receipts down, spending up, a ration of austerity announced in the language of prudence. There was a segment on Medicare: premiums frozen for a year, which the minister called a victory and the anchor called a delay. There was a segment on a veto in a distant senate, read in eleven seconds, and my father said, "Nothing to do with us," and went back to his rice, and was right, and I felt the usual irritation at how right he was.

After the news there was a documentary about a village in the northeastern hills that had been burned in the war and rebuilt four times, and the narrator used the words resilience and solidarity and sacrifice the way people use coins that have worn smooth, and there was an old woman on the screen who wept without any shame and said one sentence about her brother that the English subtitle translated as "he never came back," and the Chinese subtitle said something longer and harder and truer, and I noticed that translation is a form of mercy and also a form of theft.

My father fell asleep in the chair at half past seven, with his mouth open, exactly as a man on a train had slept that morning, and my mother covered his knees with a newspaper and left him there, which is what twenty-six years of marriage looks like when nobody is watching.

I went up to my room — mine, still, four years after I stopped sleeping in it every night — and it smelled of mothballs and old paper, and the shelf held the books I had read at fifteen: a paperback thriller with a broken spine, a dictionary missing its cover, a book of essays I had bought because the title promised something the book did not deliver, a volume of poetry I had never finished, and a handbook of mathematical theory that had belonged to my uncle and that I had kept because of the handwriting in the margins, which was neat, patient, wrong in three places, and more valuable to me than the text.

The wedding invitation was on the desk, propped against a cup. My cousin is marrying in October. The card is showy, gold on cream, with a photograph of the couple in a park and a list of eight people who will stand beside them, and on the back, in pencil, my

mother had written: "Give five hundred. Not more. He gave three hundred at yours." I laughed out loud. There is no idealism in a family ledger, and no malice either; it is bookkeeping, and it is also love, and the two have never been separated in any household I have ever eaten in.

I sat on the bed and opened the box again. I had decided, on the train, to finish what the doctoral student had started: to write a definition for every card in the box, including the ones whose heads the moisture had eaten, and to leave the box in the library somewhere a person might find it. It was a foolish plan, fragmentary, unfinished, and entirely mine.

"Apprentice: a boy who holds the lamp for a man who knows; also, anybody learning anything from anybody who will let them watch."

"Brink: the edge of something high; also, the week before an examination, and the year before a decision."

"Buffer: the thing that stands between; also, the friend who reads your letter before you send it."

"Career: the whole run of a working life; from the Latin for a cart, because a career was originally a road for carts, which is still the best description of most of them."

"Champion: a winner; also, to argue for a person who cannot argue for himself, which is the older and better sense."

"Companion: a person you eat bread with; from the Latin com and panis, with bread, which explains why the word feels warmer than friend and colder than brother."

"Complacency: the smug quiet of a man who has stopped checking his own work; the occupational disease of the second year."

"Concession: something given up in an argument; also, the small shop inside a station, because both are places where you accept less than you wanted in order to keep moving."

"Conviction: a belief held hard; also, a verdict in a court, which is the sense the word came from, and which should remind anybody who is certain of anything that certainty is a legal condition before it is a moral one."

"Descent: the act of going down; also, family origin, which is also a going down, depending on who is telling the story."

"Drawback: a disadvantage; literally, the amount a merchant deducts before paying you, which is why the word still tastes of money."

"Eclipse: the hiding of one light by another; also, to do something so well that somebody else disappears from view, which is the cruellest compliment there is."

"Epoch: a period marked out by events; nobody inside one can see it, which is why the word is only ever used afterwards."

"Ethic: a set of principles governing conduct; also, the particular ethic of a trade, which is usually stronger than the private one and sometimes worse."

"Fossil: the stone shape of a thing that was alive; also, a word, a habit, an institution that has stopped changing and kept its shape."

"Fusion: the joining of two things into one; in physics, what happens in the core of a star; in a dormitory, what happens to four boys in nine months, partly, and never completely."

"Harness: the straps that bind a horse to a load; also, to capture and use a force — wind, sunlight, a temper — which is the whole ambition of engineering stated in one verb."

"Hardship: suffering that lasts; the word implies endurance rather than pain, which is why people use it about years and not about hours."

"League: a distance of three miles, from an hour's walk; also, an alliance of states or clubs, which is a different word entirely and shares the spelling by accident."

"Lenient: inclined to spare the rod; from the Latin for soft, which is also the root of lens, because a lens is soft compared with a mirror."

"Magnificence: great splendour; also, the particular splendour of a thing that knows it is temporary — a firework, a wedding, a term of office."

"Miserable: wretched; from the Latin miser, wretched, which is also the root of misery and of miser, because a man who hoards is a man who is afraid, and fear is wretched."

"Mythology: a body of stories a people tells about itself; not falsehood — a myth is not a lie, it is a story carrying a truth that cannot survive being said plainly."

"Omission: the act of leaving something out; the most powerful editorial act available to anybody, and the one nobody ever admits to."

"Peril: danger; the word used in shipping notices and in poetry, which tells you what kind of danger it means."

"Precedent: an earlier case that guides a later one; the reason lawyers argue about old paper, and the reason institutions never change until they break."

"Ration: a fixed share; also, to limit supply, which governments do in war and households do in January."

"Salvage: to save something from a wreck; the word carries the wreck inside it, which is why it cannot be used for anything that was never lost."

"Scepticism: the habit of doubt; not cynicism, which doubts everything and therefore believes nothing, but the discipline of asking what the evidence is and waiting for an answer."

"Span: the distance between two points; also, a pair of horses, because they span the road; also, the length of a life, which is the only span anybody measures twice."

"Subsistence: the minimum needed to stay alive; farming at the level of subsistence means eating what you grow and growing nothing you can sell, which is not poverty in the old sense but a form of captivity."

"Tribute: a payment from a weak state to a strong one; also, words said about a dead person, which are also a kind of payment, and also a kind of tax."

"Tranquility: a calm of a particular kind; the word belongs to water and to evenings, not to people, which is why a tranquil mind always sounds slightly borrowed."

"Utmost: the farthest point; used almost only in the phrase to the utmost, which is what people say when they have reached the end of their own patience and want credit for it."

"Vague: not precisely stated; from the Latin for wandering, which is also the root of vagabond — a vague answer is an answer that has wandered off, and most of ours do."

"Vital: necessary to life; the word has thinned into a synonym for important, which means that when a doctor uses it properly — the vital signs, the vital organs — nobody hears the difference any more."

"Yearning: longing, with the ache left in; the word comes from an old English verb for desiring eagerly, and it has never lost the childish sound that makes it usable in a song and embarrassing in a report."

I wrote sixty-one cards that night. I counted them the next morning, which is the kind of thing a person does when he cannot sleep and wants to believe the night produced something.

What I kept from that year, in no order and without a plot.

A skyscraper in downtown at dusk, glowing on one face and dark on the other, like a tombstone with a window in it. A glimpse of the river from a bus on the motorway, brown and slow and continental. A spongy mat of moss on the steps of the museum, splendid and slippery. A clown at a child's birthday in the plaza, tiring by four o'clock, sitting on the kerb with his make-up running. A cocktail of soda and regret at the farewell dinner where nobody drank. A singer in the stairwell practising a melodious tune on a trumpet with a dented bell, and the tune turning sour at the top note every time. A violin case open on the pavement with four coins in it and a note saying THANK YOU in three languages.

Purple bruise on my leg from a pedal in October; a cut on my thumb from a cracked beaker in November; a burn on my wrist from a hot crucible in March; and therefore, somewhere in my body, a record of three months that no transcript keeps. Pain is a creditable archivist. It files everything and loses nothing.

A dressing changed on a wound in the clinic at midnight, the nurse's hands warm and impatient and quick, talking over her shoulder to somebody else about a football match. A temporary fever that made the ceiling interesting for thirty-six hours. A vaccination jab that left a bruise in the shape of a thumbprint, and me showing it to everybody like a medal.

A laboratory bench with a burn mark in the shape of a ring: ours, nobody's, everybody's. Equipment that worked better when somebody else was watching. A transmitter in a cupboard that nobody had used since 2011 and nobody would let us throw away. An instrumental error on a graph that I blamed on the machine and the machine blamed on me, and both of us were right.

A quarrel in September about a window, a serious quarrel in January about nothing, a reconciliation in March that consisted of buying two coffees and saying nothing. Misunderstanding is a luxury of people who have time; we had none, and so we forgave quickly, and so the friendship lasted, strict and unspoken and real.

A herself of a mother ironing at eleven at night, unwilling to sit down until the house was finished with its day. A fatherly uncle who taught me to wrestle and then taught me to lose, which is the harder lesson and the one he never mentioned. A sister-in-law who laughed like a drain clearing. A beloved grandmother on the phone, asking whether I had eaten, which is the only question she has ever asked me and the only one that matters.

A recruitment poster for the army with a soldier's face airbrushed smooth, and somebody had drawn a moustache on it in marker, and two people had argued about whether that was vandalism or commentary, and the argument was more interesting than the poster.

A lecture on Marxist theory in a hall with broken air conditioning, sixty students asleep and four awake, and the lecturer saying, wearily, that nobody reads the text any more, and adding, quietly, that he had stopped reading it too, and that this was the honest basis for a conversation.

A dumb question from me in October that turned out to be the right one; a clever question in December that turned out to be fashionable and empty; and a silence in April that was worth more than both.

A trip to a slum on the edge of the city with a volunteer group, where sixty families lived behind a wall painted with pictures of flowers, and where a boy of nine explained to me, patiently, that the smuggling was not a crime but a job, and that his mother did

it too, and that I should not tell anyone he had told me.

A suspicion, later, that every statistic I had ever memorised was a story told by somebody with a reason; a conviction, later still, that this did not mean the statistics were false, only that they were mine to check.

A furious argument with a roommate about a missing charger; a fun afternoon building a shelter out of bamboo and tape that survived one storm and one photograph; an enjoyable, frustrating, stupid week preparing for a competition we did not win and did not regret.

A flirtation conducted entirely by glance across a lecture hall for six weeks, which neither of us dared to convert into a conversation, and which I remember now with more pleasure than most conversations.

Rain on the corrugated roof of the workshop at four in the afternoon, so loud that we had to shout, and then stopping, and the silence afterwards ringing like a bell struck once. Steam off the road afterwards, rising in the sunlight like a thing alive. Muddy footprints drying on the steps into maps of nowhere.

Sunlight on the dust in the corridor at seven in the morning, making a solid bar of light that you could put your hand through. Gas flaring blue under a beaker. A fume cupboard humming. An automatic pipette jumping once. A vital sign on a monitor that I watched for forty minutes because somebody had to.

A maple leaf stuck to the windscreen of a bus for eleven stops, and me watching it, willing it to stay, and it did, and that was the best thing that happened to me in October.

A hopeless crush on the idea of a person rather than the person; an intimacy built out of shared work and not out of words; a refusal to name a thing that everybody could see, because naming it would have made it smaller.

A pure hour at four in the morning with a problem that had a solution, which is rarer than happiness and easier to recognise. A rigorous proof on a blackboard that I copied and did not understand, and then, in May, understood in a queue for noodles, all at once, like a door opening inwards.

A rendezvous with a deadline that I met by nine minutes and remember as a victory, though the work was mediocre and the grade was average and nobody else noticed the nine minutes at all.

A scene at a wedding where the bride's father cried and pretended to be fixing his spectacles, which is the whole art of fatherhood compressed into four seconds of theatre.

A funeral in February for a teacher nobody in my year had studied under, where forty people stood in a cold chapel and one of them said a sentence that I wrote down on a

ticket stub and still have: "He was never in a hurry, and he was never late."

An exit sign above a door in a basement that I passed four hundred times and read every time, the way people read a word until it stops being a word.

A police car idling outside the gate on a Saturday, two officers drinking tea from a thermos, watching nothing, which is most of what watching is.

A ticket collector who knew my name by March because I travelled home every third weekend, and who once said, "Your mother waits at the north exit," which was true, and which embarrassed me, and which I now treasure.

A canteen cook who gave me an extra spoon of rice in December and never did it again, and whose face I could not describe now if you asked me under oath.

A cat that lived under the stairs of the laboratory building and was fed by four generations of students and was pregnant twice a year and survived all of it, which is more than some of the students managed.

Before I slept I wrote one more thing, and put it inside the lid of the box, where the next person to open it would find it before the cards. This is what it said, and I am not proud of it, and I am not ashamed of it, and it is the only thing I wrote that year that I would still sign.

"Whoever opens this: it is not a prize and there is no competition. It is accessible to anybody with a pencil and a free evening. Ninety-one cards in a wooden box, half of them damaged, none of them complete.

I do not know whose box it was. A doctoral student in the eighties began it and stopped, for a reason that nobody recorded, and the reason is now a depreciation of everything he did, which is what time does to work that nobody finishes. If you finish it, add your name at the back. If you do not, add your name anyway, and write the date you gave up, because a record of giving up is still a record, and somebody later may benefit from knowing where the wall is.

Three things, and then the box is yours.

One: do not confuse comprehension with agreement. You can understand a concept perfectly, construct its geometry in your head, predict what it will do at the next step, and still reject it. Comprehension is cheap; conviction is dear; and most arguments that look antagonistic are two people with different prices.

Two: the constant temptation of a student is to consume instead of to make: to read another essay, collect another example, cultivate an opinion about a thing he has never touched. This feels like progress and is actually an amusement. Do not contend with people who make things until you have made one thing, however small, however bad.

Three: when a disagreement turns into a quarrel, disarm it with a question instead of

a claim. Ask what the other person would accept as evidence. If the answer is nothing, stop, because you are not in an argument, you are in a ritual, and rituals cannot be won, only attended.

Two warnings. Do not let a disciplinary hearing or a discriminatory remark pass quietly, even when passing it quietly would benefit you, because a department that tolerates it will deteriorate in a way that is not visible for five years, and by then the people who could have stopped it will have left. And do not deplore the state of anything unless you have spent an afternoon trying to fix it; deploring is free, which is why it is everywhere, and why it is worth nothing.

The rest is method. Erase nothing. Date everything. Beware of a result that arrives easily: it is usually a coincidence wearing a theory. Write down the conditions of the experiment before the result, not after, because everybody lies to himself in the direction of a story, and the only defence is a page written earlier than the temptation.

If a diagnosis is unclear, say so; an explanatory sentence that conceals the uncertainty is worse than silence, because silence can be questioned.

Study the internal structure of things and the external pressures on them; most disasters are the second, and most courses teach the first. Practise locating the relevant and the relative fact in a paragraph of persuasion. Learn to distinguish a realistic forecast from a prophecy, a preposition from a premise, a limitation from an excuse, a routine from a tradition, a habit from a principle.

Look at the sky once a week. Learn three constellations, enough to greet the starry wheel over the playing field the way you greet a neighbour. A galaxy is not a decoration; it is the only object you will ever see that is older than language.

Do not sob into your notes. Do not gossip about your supervisor in a corridor; corridors have excellent hearing. Do not hurry a result; a result hurried is a result invented, and invented results propagate through a field like mould through bread.

When you reach the end of a thing — a semester, a project, a friendship — do the paperwork. Return the equipment. Settle the toll of other people's patience. Leave the bench cleaner than you found it, not because anybody will notice, but because the practice of leaving a place better is the only training available for leaving a person better, and you will need that training, and nobody will teach it to you.

Finally: wish well. Wish well to the student who takes your place and gets the prize you wanted. Wish well to the rival whose name is on the paper and not yours. Wish well loudly, once, and then get back to work, because goodwill that stops short of work is just manners."

I folded it twice and put it in the lid.

The rest of the night passed in the way nights pass when you have decided something and still worry about it: slowly, and then all at once. Nearby, a dog barked at something that was not there and then stopped, humiliated. A plane went over the house at three forty, low enough to shake the window, bound for somewhere south, and the sound thinned into a hum and then into the hum's absence, which is louder.

At half past four the birds started, and I gave up on sleep and went down to the yard, where the air was cool and the sky was still grey-blue and the jasmine was so strong that it was almost a sound. I filled a basin of water for the tomatoes, because somebody had to, and stood there listening to a town that was waking up one kettle at a time.

Two hundred metres away, a man was already loading cabbages into a truck by the light of the cabin lamp, working slowly, the way a man works when he has done the same thing four thousand times and expects to do it four thousand more. I watched him for a while. He did not look up. I did not wave. We were two men in the same morning, each with a job he had chosen and neither of us able to explain why.

Wang Laoshi called at eight, before I had finished breakfast, with the news that the exchange had been approved — approved, not granted, he said, and the difference mattered, because approval is a decision by a committee and a grant is a cheque from a treasury, and we had the first and not the second.

"So celebrate later," he said. "Celebrate when the visa is in the passport and the plane has a seat with your name on it. Until then you have an intention, and an intention is a promise made by somebody who is not paying."

"You sound tired."

"I am tired." A pause with paper in it. "Nine hundred students, one hundred and forty places, eleven of my advisees applying, and every one of them convinced that his case is unique and that the rules are for other people. That is not a criticism; it is a description of nineteen."

"What should I do between now and February?"

"Read. Learn German badly and English properly. Keep a journal of what fails, not of what succeeds; the second is a resume, the first is a mind." He coughed. "And write to your mother once a week, which has nothing to do with science and everything to do with being worth something later."

I asked him, because I had wanted to ask somebody official for nine months, whether the year had been worth anything.

"Define worth," he said, immediately, which is his way of avoiding a question while answering it. "If worth means a publication, no. If worth means a grade, no. If worth

means that you can now tell the difference between a result and a coincidence, and between an explanation and an excuse, then yes, considerably, and nobody will ever confirm it for you, because nobody employing you will ever test it directly. That is the trouble with education: the useful parts arrive unaccompanied by evidence."

He hung up before I could thank him, which is also his way.

My mother came in with two tomatoes from the yard and put them on the table like a verdict.

"Approved," I said.

"Good," she said. "February is cold there?"

"Colder."

"Then buy a coat there. Do not buy it here; the coats here are for the cold here, which is a different cold." She wiped her hands. "And do not lose the box."

The morning passed in small errands: the water bill, a bag of rice carried from the mill, a visit to my grandmother's grave with three sticks of incense, a conversation with a cousin about a car he could not afford and would buy anyway. Ordinary hours, and I noticed myself noticing them, which is what nine months away does: it makes the ordinary visible for about eleven days, and then the town closes over you again like water.

In the afternoon I took the box up to my old room and put it on the shelf, between the dictionary and the handbook of mathematics, where it looked exactly like what it was: a box of paper among books, worthless, and the only object in the house that had travelled further than I had.

Then I did the thing I had been avoiding since March, which was to read my own diary from the beginning. Nine months of it, in four notebooks, in a handwriting that gets smaller as the year goes on, as if the pages were running out of room and I was trying to fit inside the margin.

January: "Cold. The lab is warm. I stay late because it is warm."

February: "Nothing works. Nothing works. Nothing works. Three weeks."

March: "Shen Lu took it apart. She was right about four things and wrong about one, and I cannot stop thinking about the one."

April: "Twelve hours in the dark with a beam. Saw a thing that nobody has seen before, which sounds romantic and looks like a grey smudge."

May: "Wrote nine hundred words. Deleted seven hundred. The two hundred that survive are better than anything I have written, which suggests a method I cannot yet describe."

June: "Applied. Nine minutes early. Ate nothing until eight."

That was the whole year, in four pages, and it read like somebody else's life —

which it was, because the person who wrote January is not the person who read it in July, and that gap, invisible to every institution, is the only measure of a year I trust.

At dusk I walked down to the river again, alone, and sat on the wall where the boys dive, and watched the water carry the light away by the handful. A man passed with a telescope under his arm — a telescope, in a town of forty thousand people — and stopped, because he saw me looking at it, and said, "Mars," as if that explained everything, and then, because it did not, "It is closest this month. Forty years until it is this close again."

I looked up. I could not see it yet, but I looked up, which is the whole of what he wanted, and he nodded and went on down the wall with his telescope, an old man carrying a heavy thing to a dark place because the sky was doing something rare.

That, I think, is the answer to the question I asked Wang Laoshi, and nobody asked me to state it, so I will state it once and let it stand: a year is not measured by what it produces. It is measured by whether you can still look up afterwards — whether the work has left you curious or merely tired, whether you want to tell somebody about the thing you saw or only to be praised for having seen it.

I wanted to tell somebody. That was enough. That was, in fact, everything, and it had taken me nine months, one wooden box, eleven courses, four hundred images, three mistakes, two stations, one unsent letter, and a fan with a dry bearing to find out.

I finished the box that summer. Not all at once, and not in any order — I sorted the cards into six stacks on the floor of my room, which is the only way I know to think, and then I wrote a line for each one:

The work stack. Agitate: to stir a liquid; also, what a rumour does to a corridor between two lectures. Bolster: to prop up; also, a long pillow, because the word arrived in English as a cushion before it became a verb. Facilitate: to make easy, which nobody believes of a committee, and yet the word is innocent. Proficient: able to do a thing well without being interesting about it. Progressive: moving forward; also, a word that hides the question of who is being moved and in which direction. Initiative: the first step, taken by somebody who was not asked; the military sense is older than the office sense, and the office sense has ruined it. Incentive: the thing that makes a man move; from the Latin for a tune played before a race, which is why every incentive scheme ends up sounding like a starting pistol. Turnover: the money a shop takes; also, the rate at which people leave a job, which is the number nobody prints. Undertake: to take a thing upon oneself; the word carries the weight it describes. Metrical: arranged in measure; applied to poetry, it means the beat; applied to a system, it means that somebody has decided what a unit is. Practicality: the virtue of a thing that works, as opposed to a thing that ought to. Preferable: better; the comparative of a word nobody uses in the positive, which tells you something about choices. Product: the thing produced; also, in

mathematics, the result of multiplication, which is the only sense in which a product is exactly predictable.

The place stack. Besiege: to surround a town; also, to surround a man with questions he has already answered. Mainland: the big piece of land; the word only exists on an island, which is why it sounds like longing. Tropic: the line where the sun stands still; the tropics are the band between the two, and everything anybody says about them is a generalisation made from a hotel. Zone: a band marked off for a purpose; the word arrives in English from the Greek for a belt, which is why a zone always feels like something worn. Venue: the place where a case is tried; now, the place where a concert is held, which is a promotion for the word and a demotion for the law. Perch: the branch a bird stands on; also, to sit on the edge of something, which is how students occupy a library. Peak: the top; also, to reach a maximum and begin to decline, which is why the word is used about careers with a certain grim relish. Stroll: to walk without a purpose; the only form of movement that improves a city. Stray: to wander off the path; also, an animal with no owner, and a single item that does not belong to a set. Strand: a single thread; also, to leave a person helpless, from the same root as the beach, because a ship stranded on a strand is a ship left by the tide. Rotate: to turn about a centre; also, to take turns, which is how a laboratory shares the night shift. Recreation: the remaking of a person after work; the spelling remembers the making, which the pronunciation has forgotten. Overflow: to flow over; also, the pipe that carries the excess, and the parking lot beside every hall, for the same reason. Overturn: to turn over; in law, to cancel a decision, which is a quiet word for a loud event. Expedition: a journey with a purpose; the word implies provisions, and most expeditions fail for want of them. Thrust: to push hard; also, the forward force of an engine, and the main point of an argument, which is why people say a speech has thrust. Indoor: inside a building; the word exists only because somebody needed to contrast it with a courtyard. Hotel: a building with rooms for hire; the word came to English from the French and still carries a hint of hôtel particulier, a private house, which is why a cheap hotel feels like a betrayal.

The people stack. Alienate: to make a stranger of a person; the word does the thing it describes. Enthusiast: a person possessed by a god; the Greek root means inspired, and the modern sense is a person who fixes old bicycles on Sundays. Entrepreneur: a person who undertakes a risk; the French word means between-taker, and most of them are taken between. Migrant: a person who moves; the word is neutral, and everything done with it is not. Mankind: the whole of us; an old-fashioned word that still does a job no newer word does without a lecture. Patriot: a person who loves his country; the word has been used to silence as many arguments as it has opened. Warrior: a fighter; the word belongs to an older world, which is why it sounds grand in a speech and absurd in a memo. Inmate: a person lodged in a place; the old sense was a guest, and the prison sense swallowed it, which is a small history of institutions in one word. Participant: a person who takes a

part; the word flatters by implying a share nobody measured. Superintendent: the person who superintends; a title that means caretaker in one building and chief officer in another, which is a useful reminder that rank is local. Snob: a person who looks down; the word comes from a label on a cobbler's cart at Cambridge, which is the finest origin story in English and probably false. Youth: the time of being young; also, a young person, and also the whole class of them, which is why politicians address the youth and never meet one. Victim: a person or animal killed as a sacrifice; the legal sense is narrower and the ordinary sense is wider, and nobody who uses it is ever in doubt.

The mind stack. Intuition: knowledge without a route; the word looks inward, and the best users of it check outward. Intrinsic: belonging by nature; the opposite is extrinsic, and the difference matters most in arguments about motivation. Irrational: without reason; used loosely to mean wrong, which is a different thing, and the confusion costs people their patience. Impossibility: the state of not being able to be; a mathematical claim, and the only kind of claim nobody can argue with politely. Inferiority: the state of being lower; the word usually describes a feeling rather than a fact, which is why it needs the word complex attached to it. Illiteracy: the inability to read; the word counts letters and misses everything else, and a person can be literate and still be unable to read a contract. Pursuit: the act of chasing; in law, a pursuit is a process, and happiness is only ever mentioned in the phrase the pursuit of it, which is exactly right. Psychology: the study of the mind; the word promises more than any department can deliver, which is why first-year students love it. Investigation: a tracking down; the word carries footprints, and most investigations end not with a finding but with a file. Retrieval: the getting back of a thing; used about memory and data, and in both cases the hard part is not the storage. Reflection: the throwing back of light; also, thought, which is the same metaphor and the same mistake — a reflection is not the thing. Refute: to prove wrong; people say refute when they mean deny, and the difference is a whole afternoon in a seminar. Refute: deny is to say no; refute is to show no — hold the line, it is the most useful distinction in this box. Reconcile: to bring back together; also, to accept a thing you cannot change, which is not the same as agreeing with it. Resolve: to decide firmly; also, determination, and also, to settle a problem — three senses and one root, which is why a resolution can be a decision, a promise, or a cure. Respond: to answer; the word implies a promise made back, which is why silence after a question feels like a breach. Regularity: the state of following a rule; the word describes a bowel habit and a tide, and the second is more honourable. Reinforcement: the strengthening of a thing by adding to it; in psychology, the reward that makes a behaviour repeat, which is a polite word for a bribe given by circumstance. Significance: the meaning a thing carries; not the same as importance, although everybody uses it so, and a result can be significant statistically and mean nothing practically, which is the sentence I would put on every wall in the building.

The failure stack. Devastate: to lay waste; the word is about land, and people use it about feelings, which tells you what a feeling feels like. Deteriorate — I had written it twice — to get worse by degrees; the word is slow, and most disasters are not. Derailment: the leaving of the rails; used now about talks and plans, which means the railway metaphor has won and nobody notices. Dropout: a person who falls out of a course; the word blames the person, and the figures blame the course, and the gap between the two is where the truth lives. Disorder: a lack of order; in medicine, a named condition, which is why a diagnosis sounds like a verdict and feels like a relief. Diagnose: to name a condition; the word implies certainty that doctors do not have, and good ones say so. Ruin: the state of being fallen; also, to spoil a thing completely, and the word is so final that people use it about haircuts. Rage: violent anger; also, to spread fiercely, as a fire rages, which is why the word is better used about epidemics than about people. Fierce: violent in manner; the word has drifted towards praise, and a fierce intelligence is now a compliment, which was not true in 1590. Frantic: wild with haste; the word means mad, and most frantic people are merely late. Grim: stern, unforgiving; the word comes from a name for the devil, and it keeps a little of that. Perplex: to confuse thoroughly; the word is rare now, and the feeling is not. Stupidity: the quality of being stupid; the word is a judgement, not a description, and I have used it carelessly all year. Scorn: contempt shown; the word has a physical origin in shrinking back, which is what the face still does. Shameless: without shame; used as an insult, and yet shame is the only sense that keeps a person honest when nobody is watching, so the insult is confused. Submissive: inclined to give way; the word is an insult in a report and a virtue in a dog, which is all you need to know about institutions. Negligence: a failure to take care; the legal sense requires a duty, and the ordinary sense requires nothing.

The last stack. Wonderful: full of wonder; the word has thinned into approval, and the whole of wonder — the stopping, the quiet — has gone out of it, which is a pity and a warning. Worthy: deserving; the word needs an object, and a life spent being worthy of something somebody else valued is a life spent on a loan. Whisper: to speak without voice; the word is onomatopoeic, which is why it works in every language nobody has tried. Weep: to cry; the word is literary, and that is its function — it lets a man say it without saying cried. Swing: to move back and forth; also, a seat hung from a rope, and also, a change in opinion, which is the most expensive kind of movement. Tease: to make fun of somebody gently; the word sits exactly on the line between affection and cruelty, and most of us do not know which side we are on. Tickle: to touch lightly so as to cause a laugh; the only sensation nobody can give themselves, which is a small proof that we need other people for the oddest things. Tennis: a game of courts; the word probably comes from the French tenez, hold, shouted by the server, which means the name of the sport is a command nobody obeys. Orchestra: the space before the stage in a Greek theatre; the musicians took the name of the place they sat in, which is how most

institutions are born. Organism: a living body; the word implies parts that serve a whole, and the metaphor has been used to justify every political arrangement ever imposed on people who were not consulted. Virus: poison in Latin; a thing that copies itself by borrowing other machinery, which is a description that fits several institutions and one or two habits. Simulate: to imitate the appearance; the word is honest — a simulation claims only what it is — and most of what we call models are simulations wearing a theory's clothes. Theoretical: concerned with theory; used as an insult by people who have never met a good one. Thoughtful: full of thought; also, considerate, and the second sense has almost eaten the first, which is why a thoughtful person is now somebody who remembers birthdays. Supreme: highest; the word belongs to courts and pizzas, and only one of those uses is ironic. Supportive: giving support; the word arrives from management training and describes a virtue that used to be called kindness, and the new word asks for less. Survival: the continuing to live; the word is neutral and the fact is not. Sensible: having sense; the word means reasonable, and the related word sensitive means easily affected, and the two have been confused by every generation of students, including this one. Selection: the choosing out of some; every selection is a rejection wearing a smile, and the people who run them know it. Separation: the act of setting apart; the word is legal, chemical, and personal, and it hurts in only one of those registers. Successive: following one after another; the word counts without saying whether anything improved. Severity: the quality of being severe; in medicine, a grade, and in a school, a threat, and in a person, a defence. Shady: giving shade; also, of doubtful honesty, because a man who works in the shade is a man who does not want to be seen working. Stimulation: the act of rousing; in physiology, a signal, and in education, a word that means we could not think of a better one. The alarm had been false. The aspiration had been mine. The condemnation of the project came from a man who had not read it, and his confirmation that it was interesting was worse than the condemnation. The conservation of the river was somebody else's department; two laboratories agreed to cooperate, converged on a result, and neither of them published it; a persistent smell of solvent permeated the stairwell; a savage review scattered what was left of my confidence; a stray sentence in a textbook alluded to the method I had spent four months inventing; one card, astray from its stack, had been under the bed for a year; and one unprecedented afternoon in May unfolded in a way that no plan could have predicted.

What follows is not a paragraph. It is a list, and I have decided that a list is the honest form for the end of a year, because a sentence gives every word a job and these words have no job any more; they are what I kept. Some of them had no story left in them. I wrote their cards anyway, and I put them down here in the order they came out of the box, because a residue is still a record: bust, halt, swing, thrust, spear, trench, flutter, stray, skim, search, sale, safety, merely, naturally, definitely, hurriedly, humbly, gracious, glamourous, gigantic, geographic, colourful, coherence, circulation,

combine, compete, complex, correspondence, creation, curb, daring, decrease, defeat, designer, detailed, diffuse, dignify, discourage, dissent, dominant, elegance, engage, entail, environmental, escort, evaluate, evil, excellence, exclusion, exemplify, exhaustive, explosive, extraordinary, fake, fancy, flee, folk, foreign, foretell, gracious, grope, hire, ignite, indefinite, indoor, inevitable, influence, infringe, inherit, innumerable, insistent, intake, intelligence, interested, interrupt, irrational, jet, joyful, liberation, literary, longevity, luxurious, master, mutter, naturally, nudity, nurture, obtainment, occupation, offering, oppressive, outrage, passion, pastime, ponder, populous, possess, postpone, protective, provoke, reassurance, receive, recreation, recyclable, refer, relish, remedial, renewable, repetition, representation, reproductive, restoration, resultant, retell, reveal, review, schooling, seductive, skepticism, sophistication, strand, stuff, variety, vigorous, vogue, worthy, youth. That is the box. It weighs nothing, and it is the heaviest thing I own.

That is the box. It weighs nothing and it is the heaviest thing I own.

伴读：一天里的九百二十七个词

你大概正打算把这一章的生词抄一遍，抄完再背。先停一下。抄写是最省力的动作，也是最没有用的动作——它让你手酸，让你觉得自己努力过，但它不改变任何东西。这一章里的词不是从词表上抄下来的，它们是同一天里陆续出现的东西：一台电扇、一家打印店、一间答辩教室、一张体检表、一张退租交接单、一列晚上七点四十开的火车。你要记的不是词，是那天发生了什么；词是那天的零件。

所以我们从头走一遍。走的时候你会发现，很多词你其实认识，只是不知道它站在哪个位置；也有一些词，你认识它的脸，不认识它的脾气。

那台电扇与那只干燥的轴承

正文第一句是电扇转头的声音。bare 是"光着的"，张野 bare to the waist，光着膀子坐在地上——注意它和 barren 长得像，意思差得远：bare 是"没有覆盖"，barren 是"长不出东西"。一块 barren 的地，可能一点都不 bare，它长满了杂草，只是不长庄稼。这两个词我放在一起讲，是因为它们最容易被混成"空"：一个是"没遮"，一个是"不产"。

他嘴里叼着螺丝刀的 blade。blade 不只是刀片，草叶、桨叶、电扇的扇叶都叫 blade。你在实验室里遇到它，多半是在"叶片"这个意思上。然后是那个 brass 圈，黄铜的。金属词在这一章里很多，你顺手记一下层次：copper 是铜，brass 是铜锌合金（黄铜），bronze 是铜锡合金（青铜），nickel 是镍，aluminium 是铝（美式拼作 aluminum，两种都对）。这一串不是让你背，是让你知道：中文一个"铜"字，英文分了三种，因为它们在工业和钱币上是三种东西。

张野的诊断只有一句：轴承干了，所以响。这里出现了本章第一个"技术判断"的句式，你以后写实验报告会一直用：attribute A to B，把 A 归因于 B。注意它和 ascribe 几乎是同义词，但 ascribe 更书面、更常用于"把……说成是"。正文里答辩老师那句"你把吸附归因于表面积"，用的就是 attribute；而林一自己承认"我把效果说成了表面化学"，用的是 ascribe。区别不在意思，在语

气：attribute 是中性归因，ascribe 常常暗示"这个归因可能是你编的"。

还有一个词值得在这里说：dry。这个词谁都会，但它的名词 drought（旱）和形容词 arid（干旱贫瘠）才是阅读里的常客。本章只用它最朴素的意思——"干了"——因为那天早上所有东西都是干的：轴承、空气、喉咙。

一只被水汽啃掉左边的木盒

王老师给的那个盒子，是这一章的钥匙。faculty 在这里是"系、院里的全体教师"，不是"能力"——faculty 的"能力"义（the faculty of speech）在四六级里偶尔出现，看到别慌，看它后面跟的是人还是能力。installation 是"安装"，动词 install；拼写错误的高发区是把它写成 instalation（少一个 l），这个词表里专门列了这个错拼，说明它真的常错。

盒子里是卡片，homemade、assorted。assorted 是"各色各样的"，常放在商品前：assorted chocolates。它和 various 的区别是：various 只说你有很多种，assorted 说这些种类是"配好的"，像什锦糖。

卡片被moisture（湿气）啃掉了左边，剩下的是 remnant——残片。remnant 这个词带一点"剩下的、被留下的"的悲凉，比 rest 重，比 remainder 具体。remainder 是数学里的"余数"，也是"剩余部分"，但它没有感情；remnant 有。

然后是那八个残词。这一段是本章最像"文字游戏"的地方，但它是真的：一个八十年代的博士生研究前缀（prefix），盒子受潮，前两三个字母被吃掉了，剩下的部分读起来像"被砍了头的词"。

mander 原来是 commander，指挥官。memorate 原来是 commemorate，纪念——注意 commemorate 的重音在第二个音节，而且它的名词 commemoration 中间有两个 m。mense 原来是 immense，巨大的；immense 和 measure 同源，本义是"量不出来的"。merse 原来是 immerse，浸入；immerse 你自己在实验室里天天做，把样品浸进溶液。monplace 原来是 commonplace，平凡的、司空见惯的——这个词在作文里极好用，比 common 多一层"因为太常见所以不值得说"的意思。mune 和 munist 原来是 commune 和 communist；commune 作名词是公社，作动词读 /kəˈmjuːn/ 是"亲密交谈"，这个词的读音一变，意思就变，属于要小心的一类。最后是 stam，林一猜是 stammer，口吃。

这段我说得细，是因为它给了你一个真正的记忆法：前缀是词的方向盘。com- 是"一起"，im- 是"向内"，in- 是否定，con- 是"共同"。你把这几个前缀认清，commander、commemorate、immense、immerse、commonplace、commune、communist、stammer 这八个词就不再是八个，而是一个方向加一个词根。词根管意思，前缀管方向，后缀管词性——这三句话你抄在笔记本第一页，比抄一百个单词有用。

六点的早饭与那个爱用大词的张野

早饭是一个 egg、一盒过期的 yoghurt、面包。yoghurt 的拼写有两种（yogurt 也对），不用纠结。microwave 微波炉，refrigeration 是"冷藏"，注意它是不可数名词，说"冷藏设备"时用 refrigeration，说"一台冰箱"用 refrigerator。knob 是旋钮，也是"一小块"（a knob of butter）。dent 是凹痕——车身被撞的坑、桌子上被砸的坑，都是 dent；它还可以作动词，"dent somebody's confidence"（打击某人的信心），这个搭配在写作里很值钱。

张野说：Breakfast is a ceremony for people with time。ceremony 是仪式，但它的形容词 ceremonial 是"礼仪性的"，而 ceremonious 是"拘泥形式的、客套得让人难受的"。这两个形容词一个褒一个贬，考试里专门考这个区别。

然后他洗手，数到二十。这里我用了 arithmetic——算术。注意 arithmetic 是名词，arithmetical 是形容词；而 mathematics 是数学，mathematical 是数学的。mathematical 在阅读里常出现在"mathematical model"（数学模型）这种搭配里，不是你考试时做题的那个"算数"。

印刷店：一份合同和一张印错的传单

printer 既是打印机，也是印刷店、印刷工。proof 是"校样"，印刷前打出来给你看的那一张；它同时是"证据"（proof of identity），也是"防水的"（waterproof）里那个后缀。一个词三个方向，这是英语的常态，别指望一对一。

preview 是预览。insert 是插入。panel 是面板。jam 是卡住——打印机卡纸、交通堵塞、果酱，都是它。这类"一词多义且都常用"的词，你要记的是场景，不是释义：看到 jam 先想"卡住"，再看上下文要不要变成"果酱"。

那份租约（lease）是本章最"硬"的一段文字。你要习惯这种文体：hereby（特此）、upon（在……之时）、the said term（上述期限）、annexed hereto（随附于此）。这些词你不需要会写，但你要能读懂，因为合同、条款、须知都是这个调子。residential 是"住宅的"，和 residential property（住宅房产）固定搭配；estate 是"地产"，real estate 是房地产，estate agent 是中介。requirement 是"要求"，动词 require；reservation 是"保留"，也是"预订"——同一个词两个常用义，靠上下文分。tenant 是租客，landlord 是房东，deposit 是押金，deduction 是扣款，动词 deduct。deduction 还有第二个意思：推理、演绎。演绎推理是 deduction，归纳是 induction，这两个词在阅读里出现时通常跟"逻辑"有关，不跟钱有关。

dispute 是纠纷。arbitration 是仲裁。utility 是公用事业（水电气），也是"实用"，形容词 utilitarian 是"功利的"。subletting 是转租。

我还放了一张传单，上面印着 Quizes——错了一个 z。正确的复数是 quizzes，要双写 z。这个错是故意留的：一是因为真实世界里这种错到处都是，二是因为它提醒你，quiz 这个词的复数不是加 s 那么简单。顺便说一句，这个错误在本章里是有功能的：林一盯着它看，因为它"是一千份小错误的复印件"，这句话是整章的隐喻之一。

politics 那段是印刷店老板的独白。electorate 是全体选民，presidency 是总统职位（注意不是 president 的"人"，是"位子"），coalition 是联合政府，monarchy 是君主制，ministry 是部（英国的部叫 ministry，美国叫 department），bureau 是局、处，agency 是机构。这一串词你用不着背，但你要认得，因为新闻阅读里它们是骨架。

allege 是"声称（而未被证实）"，名词 allegation。新闻英语里这个词一出现，就说明后面那句话是"有人这么说"，不是"事实如此"。accusation 是指控，indict 是正式起诉（注意读音 /ɪn ˈdaɪt/，s 不发音），jury 是陪审团，prosecution 是控方，malpractice 是渎职、医疗事故，negligence 是疏忽。这一组是"法律新闻"的常用件，认识就行。

censorship 是审查制度，obscene 是淫秽的，nudity 是裸体，scandalous 是丑闻性的。那两张

海报讲的是一个展览被撤的事：一个议员说那尊像 obscene，画廊说这个指控 scandalous，学生递了 petition（请愿书），也有一份反请愿要求继续关门。dissent 是异议，condemnation 是谴责，deplore 是"强烈反对、痛惜"。

请注意 deplore 和 condemn 的差别：condemn 是判、是宣判，语气最重；deplore 是"我为这件事感到痛心"，它是表态，不是判决。写作文时如果你想表达"我认为这不对"又不想太凶，用 deplore 比用 condemn 稳。

outrage 是暴行，也是"义愤"；savage 是野蛮的，也可以作动词"猛烈批评"；scatter 是驱散。这三个词在那段里是一组：一场 demonstration（示威）被 scatter，报纸称之为 outrage，评论员用 savage 的语气写它。

civic 是"公民的"，civic duty 是公民义务，civic courage 是公民勇气。clarity 是清晰，coherence 是连贯——这两个词在写作评分里是硬指标，coherence 说的是"段与段之间接得上"，clarity 说的是"句子看得懂"。concept 是概念，comprehension 是理解（也是"阅读理解"那个 comprehension），conscious 是"有意识的"，注意它和 conscience（良心）只差一个字母，考场上错了很冤。

conceal 是藏起来，比 hide 正式。confuse 是弄混，confirmation 是确认，confide 是"倾诉、托付"——confide in somebody 是向某人倾诉，confide something to somebody 是把某事托付给某人。confide 和 confident（自信的）、confidential（机密的）同根，但意思分三路，这也是英语的常态。

两点钟的体检表

那张表有三十一个格子。waist 是腰围；chest 是胸；jaw 是下颌；rib 是肋骨；spinal 是脊柱的（名词 spine）；valve 是瓣膜，也是阀门；vein 是静脉，也是矿脉，还是"情绪"（a vein of humour）；ankle 是脚踝。ear、mouth、finger、foot、leg 这些你以为不用学的词，恰恰是写作时最容易卡住的，因为它们不规则：foot 的复数是 feet，tooth 是 teeth。

hygienic 是"卫生的"，偏习惯与操作；sanitary 也是卫生的，偏公共设施（sanitary facilities）。obese 是肥胖的，名词 obesity；医生说林一 slightly over，不说 obese，因为 overweight 只是"偏重"，obese 是"肥胖症"。这个区别在阅读里常常就是一道题的落点。

cholesterol 是胆固醇；dietary 是饮食的（dietary habits）；nourishment 是营养（不可数）；nutrition 是营养学；nutritious 是有营养的；malnutrition 是营养不良。hunger 是生理的饿，appetite 是想吃的意愿——"没胃口"是 lose one's appetite，不是"没有 hunger"。

pill 是药丸；dressing 是敷料，也是沙拉酱，也是穿衣；bruise 是淤青；bleed 是流血，过去式 bled；surgical 是外科的（surgery 既是外科也是手术）；virus 是病毒；toxic 是有毒的；physiological 是生理机制的，physical 是身体的、物质的；diagnose 是诊断，名词 diagnosis；disorder 是失调，医学里就是"××症"；irritation 是刺激发炎，也是"恼火"。

pie、pint、soda 是食堂三件套。pint 是品脱，英国用它量啤酒，美国用它量冰淇淋。recipe 是食谱，读音 /ˈresəpi/，不是 /rɪˈsaɪp/，这是听力里的常设陷阱。cauliflower 是花菜，pumpkin 是南瓜，sour 是酸的，crisp 是脆的（英式里也是"薯片"）。gramme 是克，kilogramme 是千克；词

表给了两种拼法，选一种写，别在同一篇里换。

教堂、孤儿院与那些"讲道理"的词

医院对面那座教堂的告示牌，是本章我最喜欢的小道具。Christ 是基督，church 是教堂也是教会，parish 是教区，missionary 是传教士，orphanage 是孤儿院，divine 是神圣的。widow 是寡妇，wedding 是婚礼，weep 是哭泣——这是个文学词，比 cry 重，也比 sob（抽泣）克制。

fable 是寓言；mythology 是神话的总称；myth 是单个神话，也是"流行的错误看法"，而这个"误解"义在阅读里出现得比"神话"还多。proverbial 是"众所周知、谚语一般的"，常带一点讽刺。

gracious 是待人有风度的，graceful 是动作优雅的，一个说人，一个说姿态。humbly 是谦卑地，形容词 humble，它还能作动词，"humble somebody" 是挫某人的锐气。idealism 是理想主义，realistic 是现实的，practicality 是实际性——这三个词凑在一段里，谈的通常是"要不要妥协"。

街市：钱与物的说法

counter 是柜台，cashier 是收银员，teller 是银行柜员（也是"讲故事的人"），banknote 是纸币，dime 是一角硬币，voucher 是代金券，sale 是销售也是打折（on sale），shop 是店（英式），store 是店（美式）。purchase 比 buy 正式；retail 是零售；wholesale 是批发。

crate 是板条箱，cart 是手推车，stack 是一摞，package 是包裹（美式），parcel 是包裹（英式），shipping 是运输，freight 是货运，warehouse 是仓库，tanker 是罐车也是油轮。toll 是通行费，也是"伤亡人数"（death toll），后一个意思在新闻里极常见。

市场里还有 tub（盆）、pan（平底锅）、pot（锅）、kettle（水壶）、basin（盆）、basket（篮）。cork 是软木塞，cotton 是棉，cosmetic 是化妆品，也是"表面上的、装饰性的"——"cosmetic changes"就是"装样子的改动"，这个用法比"化妆品"更常考。shaver 是剃须刀，stocking 是长袜。

chimney 是烟囱，hedge 是树篱，drain 是排水沟，debris 是碎片残骸，trash 是垃圾（美式），rubbish 也是垃圾（英式），garbage 偏厨余。stain 是污渍，也是"玷污"；dent 是凹痕，也是"打击"；crack 是裂纹，也是"破解"；scratch 是划痕，也是"从零开始"（from scratch）。

站台上的那段广播

正文里那段广播稿我特意写长了，因为它是最真实的英语：重复、公式化、一句废话都没有。这种东西不用"理解"，要"习惯"。Passengers travelling aboard ... are advised that boarding will commence at ... on Platform ... 你把这一句背下来，机场、火车站、码头，全是它的变体。

aboard 是"在船（车、飞机）上"，副词或介词，不是名词。abroad 是"在国外"，只能作副词；overseas 也是"在海外"，但它能作形容词——overseas students 对，abroad students 错。这个错中国人特别爱犯，因为中文里"国外"什么位置都能站。

platform 是站台，也是"平台、纲领"；transit 是过境、中转，in transit 是"在运输途中"；delay 是延误，名动同形；derailment 是脱轨，现在常拿来形容谈判"脱轨"；toll 是通行费，也是

伤亡人数；motorway 是高速公路（英式），美式说 freeway；motel 是汽车旅馆；trailer 是拖车，也是电影预告片；tanker 是油轮、罐车；freight 是货运；ferry 是渡轮；cruiser 是巡洋舰，也是游艇。cabin 是船舱、机舱，也是小木屋；deck 是甲板，也是一副牌；jet 是喷气机，也是"喷射"；plane 是飞机，也是几何里的"平面"。luggage 不可数，"一件行李"是 a piece of luggage；trip 是短途，journey 偏长，expedition 是远征、考察。compass 是罗盘，也是圆规；binoculars 永远用复数，因为它是"双"筒。

starry 是满天星斗的；galaxy 是星系，也是银河；eclipse 是日食月食，也是"使黯然失色"；glow 是发微光；dusk 是黄昏；tornado 是龙卷风；storm 是暴风雨。tropic 是回归线，the tropics 是热带；mainland 是大陆；continental 是大陆的；northeastern 是东北的；westerner 是西方人；migrant 是移民，也是候鸟；downtown 是市中心；underground 是地铁（英式），也是"地下的"；zone 是地带。

这段广播为什么值得抄？因为它教会你一件事：英语的正式文体靠"重复结构"而不是"华丽词汇"。你写通知、写邮件、写申请，学的应该是它的骨架——主语 + shall/will + 动词 + 地点 + 时间，一个信息一个位置，不许抒情。中国学生写正式英文最容易犯的错，是把作文的抒情腔带进去，写成"it is my great honor and I firmly believe that"。广播稿不会这样。它只告诉你：几点、几号站台、迟到多久、行李超重怎么办。能把事情说清楚，是一种比漂亮更高级的能力。

关于告别，英语有三层

farewell 是正式告别，goodbye 是普通告别，see you 是"还会见"。这一章里沈璐在站台上没有说这三句中的任何一句，她只是把手放在林一肩上，然后上车。这不是我偷懒，是因为真实生活里，最长的那次告别往往没有台词。

yearning 是"带痛的渴望"，比 desire 柔软，比 longing 具体；aspiration 是"志向"，注意它和 inspiration（灵感）、perspiration（汗水）同尾不同头——爱迪生那句"天才是一分灵感加九十九分汗水"，用的就是后两个。destiny 是命运，destination 是目的地，两个词长得像，一个说"注定"，一个说"到站"。resolve 是决心，也是"解决"，还是"分解"（化学里 resolve a substance into its parts）。willing 是"愿意的"，be willing to do 是"愿意做"，比 want to 更客气。worthy 是"值得的"，be worthy of 后面跟名词，be worthy to 后面跟动词。pity 是怜悯，也是"可惜"（What a pity）。whisper 是耳语，wink 是眨眼，weep 是哭泣，wish 是希望——wish 后面的从句要用虚拟语气，这是老考点：I wish I had known。worry 是担心，名词动词同形；wonderful 是"令人惊叹的"，但它的"惊叹"已经被用滥了，写作文时用一个 extraordinary 往往更结实。weariness 是疲惫，比 tiredness 更文学；suffering 是苦难；endurance 是忍耐力；survival 是存活；solidarity 是团结；sacrifice 是牺牲；tribute 是致敬、悼念；widow 是寡妇；wedding 是婚礼——wedding 和 marriage 的区别是：wedding 是那一天，marriage 是那之后。

答辩那天的词，一半是机构，一半是动作

academy 学院，也是"学会"；institute 研究所；faculty 系、全体教师；bureau 局、处；agency 机构、代理。这五个词撑起了所有"单位"的表达，写申请、写简历、读新闻都绕不开。

accomplishment 成就，偏"完成的过程"；advancement 进步、晋升；activity 活动；annual 每

年的；assembly 集会，也是"装配"；attendance 出席；auditor 旁听生，也是审计员；assignment 作业、任务；authenticity 真实性。这一组全是 -ance/-ment/-ity 的抽象名词，写作时把动词换成它们，句子立刻"正式"一档：他出席了，写成 his attendance。

accessible 可接近的，be accessible to；availability 可得性；automatically 自动地；apparently 据说、表面上——千万别说成"显然"，那是 obviously。additional 附加的；adjective 形容词；apt 易于……的，也是"恰当的"；astray 走错路，go astray；antagonistic 敌对的；anxiety 焦虑，be anxious about 是担心，be anxious to 是渴望。

apply 申请（for）、应用（to）、涂抹；apprentice 学徒；argue 争论，也是"主张"；arrange 安排；adopt 采纳、收养；adapt 适应；alter 部分改变；allude 暗指（to）；alienate 使疏远；advocate 提倡，后面跟 doing。avoid 避免，后面也跟 doing——avoid、mind、finish、practise、suggest 这几个动词后面都接动名词，一起记。

absurd 荒唐的，比 ridiculous 冷；amusement 消遣；amiable 和气的（说人），amicable 友好的（说关系）；affectionate 深情的；affluence 富裕；admiration 钦佩；awe 敬畏；alarm 警报；anger 愤怒；anxiety 说过了。情绪词最好记的是"强度"，最不好记的是"对象"：admiration 对才能，affection 对人，awe 对大于自己的东西。

anchor 锚、主播；arch 拱门、顽皮的；arrow 箭；artillery 火炮；bullet 子弹；cannon 大炮；spear 矛；brace 支架，也是"振作"（brace yourself）；buckle 扣子，也是"弯折"；buffer 缓冲；bait 饵；balloon 气球；basket 篮子；basin 盆；beast 野兽；hawk 鹰；hive 蜂箱；horn 角、喇叭；paw 爪；claw 也是爪，但 paw 是掌。动物部位这四个词，配上 beast（兽）、livestock（牲畜）、graze（吃草）、ripen（成熟）、barren（贫瘠），足够你写一篇"乡村"的文章了。

acre 英亩；agriculture 农业；farming 耕作；brewery 啤酒厂；boundary 边界；ditch 沟；hedge 树篱；ridge 山脊；peak 山顶；hilly 多山的；muddy 泥泞的。地理词不背单张，背"一张地图"：从 acre 到 peak，从 ditch 到 boundary，脑子里有一块地，词就跟着长上去。

accountancy 会计工作；calculate 计算；calculation 计算、算计；arithmetic 算术；geometry 几何；mathematical 数学的；metrical 韵律的、度量的；measurable 可测量的；subtraction 减法；deduction 推论、扣除；database 数据库。这一组是"数量"的词，理工科学生写论文时天天用，但最容易拼错的是 rhythm（不在本章）和 arithmetic（重音在第二音节）。

career 职业（一生的）；occupation 职业（填表用）；profession 专业职业；employment 就业；post 岗位。填表时用 occupation，谈人生时用 career，谈行业时用 profession。

catastrophe 大灾难；chaotic 混乱的；chaos 混乱；calm 平静；cancel 取消；celebrate 庆祝；champion 冠军，也是"捍卫"；cheerful 快活的；clown 小丑；cocktail 鸡尾酒；coincidence 巧合；collection 收藏；combine 结合；companion 同伴；compete 竞争；competent 有能力的；complacent 自满的；complex 复杂的；complicated 也是复杂的，但 complex 偏"结构多"，complicated 偏"难懂"。comprehension 理解；concept 概念；concern 关切；concession 让步；condemnation 谴责；confirmation 确认；confuse 混淆；consistent 一致的；constant 不断的；construct 建造；consume 消耗；contaminate 污染；contend 争辩、主张；continental 大陆的；

contradiction 矛盾；converge 汇聚；conviction 信念、定罪；cooperate 合作；correspondence 通信、对应；creation 创造；creditable 值得称赞的；cultivate 培养；curb 抑制；current 当前的、水流；curriculum 课程体系；customary 习惯的；cynical 愤世嫉俗的（本章没出现，但它和 sceptical 是一对，值得一起记）。

失败、期限与那张写不下去的申请

deadline 是死线，本义是监狱里画的线，越线就开枪——所以它天生带恐惧，别把它当"截止日期"这么平淡的词用。December 十二月，February 二月，这两个月份词最容易拼错，February 中间那个 r 很多人漏掉。decrease 减少，increase 增加，一对；decline 是下降，也是"婉拒"——这个词的双面性在阅读里反复出现。

defeat 击败，defeat 也可作名词；damage 损害；devastate 摧毁；deteriorate 恶化，比 worsen 正式；depreciation 贬值、折旧；descent 下降、血统；deviation 偏离（统计里的"偏差"）；doubtful 可疑的；despair 绝望；distressful 令人痛苦的；hysterical 歇斯底里的——这个词现在多半带贬义，用的时候小心。

demand 要求、需求；depend 依赖，depend on；delicate 微妙的、精致的；detailed 详细的；designer 设计师；dominant 占主导的；disagreement 分歧；disarm 解除武装；discourage 使气馁，discourage somebody from doing；discriminatory 歧视性的；disciplinary 纪律的；disintegrate 瓦解；dispose 处理，dispose of；distinctive 有特色的；dive 潜水、俯冲；dropout 辍学者；dumb 哑的，口语里是"蠢的"（不礼貌）。

doctoral 博士的，doctorate 博士学位；diploma 文凭；dissertation 学位论文（本章没出现，但和 manuscript 一起记）；manuscript 手稿；thesis 论文、论点。education 教育；examination 考试，也是"检查"；excellence 卓越；educational 教育的。elasticity 弹性；thermal 热的；dioxide 二氧化物，carbon dioxide 二氧化碳；uranium 铀；gas 气体、汽油；liquid 液体；fusion 聚变、融合；radiation 辐射（本章的动词是 radiate）；volt 伏特；transmitter 发射器；instrumental 仪器的、起作用的；equipment 设备（不可数）。这一组是实验室的骨架，写实验报告时按"材料—仪器—测量—误差"的顺序用它们，句子自然就成形了。

evaluate 评估；estimate 估计；assess 评估（本章没有，但三个词一起记最省力）；exemplify 举例说明；exhaustive 详尽的；experiment 实验；experimental 实验的；explanatory 解释性的；explosive 爆炸性的；expression 表达、表情；exterior 外部的；external 外部的（本章用 internal 对照）；internal 内部的；intrinsic 内在的；inherent 固有的；innate 天生的。这五个"内"字辈，写作时按"intrinsic 最学术、innate 最生物、inherent 最法律"来选。

facilitate 促进、使便利；fake 假的；fancy 花哨的，也是"想要"（fancy doing）；fashionable 时髦的；fatherly 父亲般的；fear 恐惧；feeling 感觉；fierce 猛烈的；fill 填满；flap 拍打；flee 逃走（过去式 fled）；flirtation 调情；flower 花；flutter 飘动；folk 民间的、人们；foreign 外国的；foretell 预言；forgive 原谅；former 前者的、从前的；fortification 防御工事；fossil 化石；fragmentary 零碎的；frantic 狂乱的；friend 朋友；front 前面；frustrating 令人沮丧的；fun 乐趣（不可数）；furious 狂怒的。

gap 缺口；gaze 凝视；generous 慷慨的；geographic 地理的；gigantic 巨大的；glamourous 迷人的；glimpse 一瞥；gossip 闲话；grandson 孙子；grass 草；greet 问候；grim 严峻的；grope 摸索；guardian 监护人；guitar 吉他；halt 停止；handicraft 手工艺；happiness 幸福；hardship 艰难；harness 利用、马具；hate 恨；headache 头痛；healthful 有益健康的；heat 热；hello 你好；hence 因此；herself 她自己；hiding 躲藏；hire 雇用、租用；hollow 空心的；homogeneous 同质的；honourary 名誉的；hospital 医院；hotel 旅馆；housing 住房；hum 哼；hurriedly 匆忙地。

idiom 习语；ignite 点燃；illiteracy 文盲；illustration 插图、例证；imaging 成像；impatient 不耐烦的；imply 暗示；impossibility 不可能；improve 改善；incentive 激励；inclined 倾向于；indefinite 无限期的；indoor 室内的；inevitable 不可避免的；inferiority 自卑；influence 影响；infringe 侵犯；inherit 继承；initiative 主动性；inmate 囚犯；innumerable 无数的；insistent 坚持的；intake 摄入；intelligence 智力、情报；intent 意图；interested 感兴趣的；interrupt 打断；intimacy 亲密；intuition 直觉；invent 发明；investigation 调查；involve 涉及；irrational 非理性的。

那些"差一个字母就翻车"的词

这一章里我埋了几组，专门治"眼熟但没看清"。

preposition 是介词，proposition 是主张、提议。两个字掉一个个位置，意思全变。pretext 是借口（on the pretext of），precedent 是先例，predict 是预测，preferable 是更好的，premier 是总理，primary 是首要的。你会发现在英语里 pre- 这个头太挤了，唯一的办法是把它们的"后缀"记住：-ition 是语法，-osition 是主张，-text 是编织出来的理由，-cedent 是走在前面的例子。

principal 是校长、主要的，principle 是原则——这两个词连英语母语者都写错。本章没写 principal，但你必须知道这一对。personal 是私人的，personnel 是全体人员，多一个 n，就是两个世界。

sensible 是"明智的"，sensitive 是"敏感的"，sensational 是"轰动的"。三个词同根，方向完全不同。scepticism（英式）和 skepticism（美式）是同一个词，本章两个都给了，因为考试里两种拼写都会出现。sceptical 是"怀疑的"，它和 cynical（愤世的）不同：sceptical 是要证据，cynical 是不相信有证据。

physiological 前说过，photographic 是"摄影的"，theoretical 是"理论的"，geographic 是"地理的"，metrical 是"韵律的"。这一类 -ical 形容词的重音都在倒数第二音节之前，读错就会听不出来。

情感与判断：写作里最值钱的一组

passion 是激情，passionate 是热烈的；pastime 是消遣；patriot 是爱国者；soldier 是士兵；warrior 是武士；militant 是好战的、激进分子；rank 是军衔，也是"排名"；treasury 是国库；senate 是参议院；statesman 是政治家（褒义），politician 是政客（常含贬义）；

reactionary 是反动的；ration 是配给；dole 是救济金。

这一组里我最想让你记住的是 statesman 和 politician：一个词夸人，一个词骂人，指的都是干同一件事的人。英语的判断常常藏在词的"褒贬"里，而不是藏在句子里。阅读题问"作者态度"，落点往往就落在这种词上。

rage 是暴怒，也是"肆虐"；fury 是狂怒；furious 是狂怒的；quarrel 是争吵；suspicion 是怀疑；scorn 是蔑视；shame 是羞耻，shameless 是不要脸的；stupidity 是愚蠢；severity 是严厉；stern 是严厉的；strict 是严格的；rigorous 是严谨的。注意 strict 说规则，stern 说表情，rigorous 说方法和论证——写论文只能用最后一个。

satisfaction 是满足；reassurance 是安慰；persuasion 是说服；sympathy 是同情（本章用的是 pity，注意 pity 带一点居高临下，sympathy 是平视的）。terror 是恐怖；peril 是危险；rash 是轻率的，也是"皮疹"；precaution 是预防（本章没有，但和 peril 是一对）。

动作与状态：把动词用对，句子就活了

proceed 是继续进行（proceed to do / with）；proclaim 是宣告；provoke 是激怒；ponder 是沉思；perplex 是使困惑；permeate 是渗透；perch 是栖息；rotate 是旋转；stroll 是散步；stray 是走失；strand 是使搁浅；skim 是掠过、略读；swing 是摆动；thrust 是猛推；tickle 是挠痒；tease 是取笑；swear 是发誓，也是骂人；weep 是哭。

postpone 是推迟；possess 是拥有；predict 是预测；produce 是生产，product 是产品，productive 是多产的；proficient 是熟练的；progressive 是进步的；promising 是有前途的；protective 是保护的；prolong 是延长（本章没有）。这一组全是 pro- 开头，意思是"向前"，所以它们的共同点是"往前推"：推时间、推产量、推能力。记住方向，词就成串。

reveal 是揭示；retell 是复述；retrieve 是取回；restore 是恢复；rescue 是营救；salvage 是抢救；repair 是修理；remedial 是补救的；renewable 是可再生的；recyclable 是可回收的；repetition 是重复；representation 是代表、表现；reproductive 是生殖的；relative 是相对的、亲戚；relevant 是相关的；regional 是地区的；regularity 是规律性；reinforcement 是强化。re- 开头的基本都是"再一次"，但你要分清是"再来一次"（repetition）还是"回来"（retrieve、restore）。

specimen 是标本；span 是跨度；spending 是开支；splendid 是壮丽的；spongy 是海绵状的；sportsman 是运动员；spring 是春天、弹簧、泉；status 是社会地位、状态；steam 是蒸汽；stimulation 是刺激；stuff 是东西（不可数）；submissive 是顺从的；subsistence 是生存；successive 是连续的；suicide 是自杀；sunlight 是阳光；superintendent 是主管；supervisor 是导师；supportive 是支持的；supreme 是至高的；shelter 是庇护所；selection 是挑选；segment 是片段；setting 是环境、设置。

screen 是屏幕，也是"筛查"（screen patients）；search 是搜索；shot 是射击、镜头；showy 是炫耀的；shut 是关上；sideways 是横着；significance 是意义；silver 是银；simulate 是模拟；singer 是歌手；sister 是姐妹；sixty 是六十；slum 是贫民窟；smile 是微笑；smuggling 是走私；snob 是势利眼；solve 是解决；souvenir 是纪念品；sophistication 是老练、复杂；

schooling 是学校教育；seductive 是诱惑的；supper 是晚餐（本章没出现，但和 napkin、table 一起记）。

越简单的词，越难写好

air 是空气，也是"神态"（an air of confidence），也是动词"表达"（air an opinion）。这三个意思你都得有。back 是背，也是"后退"、也是"支持"（back a plan）；behind 是后面，basis 是基础（on the basis of）。bed、bathroom、chair、door、table、key、laptop、kit、lab、ladder、laundry、bed——这一串是"宿舍英语"，看着无聊，但你要是能不用中文、只用它们把一间宿舍描述清楚，你的口语就过关了。bathroom 在美式里是"厕所"，英式说 toilet 或 loo；laundry 是待洗的衣物，也是洗衣房。

bad 是坏，比较级 worse，最高级 worst，不规则。big 是大，比较级 bigger（双写 g）。eighty 八十，eleven 十一，five 五，four 四，first 第一，thirty 三十，twenty 二十，nineteen 十九。数字词别笑，写作里拼错数字是最丢分的低级错误，尤其是 forty（不是 fourty）和 ninth（不是 nineth）。

good 是好；better 是更好；best 是最好。beautiful 是美；beloved 是心爱的；joyful 是快乐的；enjoyable 是令人愉快的；pleasant 是宜人的；splendid 是辉煌的。这一组是"好"的阶梯，写作文时按场合选：good 太平，pleasant 太轻，splendid 太重，enjoyable 最稳。

boast 是吹嘘，也是"拥有（值得骄傲的东西）"——a hotel that boasts a pool。bolster 是支持、增强；boast 和 bolster 长得像，但一个是说嘴，一个是使劲。book 是书，也是"预订"（book a ticket）；borrow 是借入，lend 是借出——中国人最容易反的两个词。briefcase 是公文包；brink 是边缘（on the brink of）；burn 是燃烧，过去式 burnt 或 burned；bust 是半身像，也是"破产"（go bust）；cape 是海角，也是披风；capture 是捕获，也是"捕捉（画面）"；chart 是图表；circulation 是循环、发行量；classify 是分类；clear 是清楚的，也是"清除"；clever 是聪明的；cling 是紧抓（cling to）；close 是关，也是"近的"（读音不同：动词 /kləʊz/，形容词 /kləʊs/）——这个词的读音是听力里的暗礁。collect 是收集；colourful 是多彩的；complimentary 是"免费的"或"称赞的"——a complimentary ticket 是赠票，不是"complimentary 的票"。complacency 是自满；conduct 是行为，也是"进行"（conduct an experiment）；congressional 是国会的。

conservation 是保护（自然资源）；conservative 是保守的；consist 是"由……组成"（consist of）；control 是控制；corpse 是尸体；couple 是一对、几个；cover 是覆盖，也是"报道"；crowd 是人群；cube 是立方体；cyclist 是骑车的人。

daring 是大胆的；definitely 是肯定地（拼写里没有 a）；diary 是日记，dairy 是乳制品——这两个词中国人常写反。diffuse 是扩散，也是"啰嗦的"；dignify 是使有尊严；director 是主任、导演；drawback 是缺点；dye 是染料；easily 是容易地；ecosystem 是生态系统；elegance 是优雅；eloquent 是有说服力的；embassy 是大使馆；emergency 是紧急情况；emotion 是情感；enactment 是（法律）制定；engage 是从事、吸引（be engaged in）；entail 是"必然包含"；enthusiast 是爱好者；entrepreneur 是创业者；environmental 是环境的；epoch 是时代；erase 是擦掉；errand 是差事（run an errand）；escort 是护送；ethic 是伦理；evening 是傍晚；

every 是每一个；evil 是邪恶；exchange 是交换；exclusion 是排除；exit 是出口。

joint 是关节、共同的（a joint project）；jump 是跳；late 是晚的、已故的（the late professor）；layer 是层；league 是联盟、里格（长度单位）；legitimacy 是合法性；lenient 是宽容的；level 是水平；liberation 是解放；lift 是举起，也是电梯（英式）；limitation 是限制；linguistic 是语言的；literary 是文学的；locate 是定位；log 是原木、日志；longevity 是长寿；loosen 是放松；lottery 是彩票；lower 是降低；luxurious 是奢华的；madam 是女士；magnificence 是壮丽；make 是做，也是"品牌"（a make of car）；mankind 是人类；marginal 是边缘的、微小的；marry 是结婚；Marxist 是马克思主义者；master 是主人、硕士、掌握；maybe 是也许；mechanism 是机制。

mechanism 这个词我要多说一句：它在理工科论文里出现频率极高，中文一般译"机理、机制"。它和 machine 同根，本义是"装置"。你说"反应机理"，用 mechanism；你说"制度机制"，也是 mechanism。一个词横跨实验室和机关，这是英语的常态，也是它省力的地方。

那些带"重量"的词

acid 是酸，也是"尖刻的"。adversary 是对手，比 opponent 更书卷气，常出现在正式文书里。aftermath 是"（灾难、事件的）余波"，字面是"割草之后再长出来的第二茬草"——你记住这个画面，就再也不会把它当"后果"的普通同义词了，它专门指"大事之后的那段日子"。agitate 是搅动，也是"煽动、使不安"；agitation 是焦虑、鼓动。alliance 是同盟；ambassador 是大使；athletic 是运动的；atrocity 是暴行；besiege 是围困——这三个词属于战争，而战争词最适合用来形容"被追问"：记者 besiege 发言人，问题 agitate 一个不愿意回答的人。

alone 是一个人，lonely 是心里空；anywhere 是任何地方；beach 是沙滩；beg 是乞求，beg for；benefit 是好处，也是"受益"（benefit from）；beware 是当心，只用于祈使句和不定式（beware of the dog），不能说 he bewares。biologist 是生物学家；bit 是一点，也是"钻头"；blog 是博客；blue 是蓝，也是"忧郁的"（feel blue）。

end 是结束，也是"目的"（to this end）；enough 是足够，位置很活：good enough、enough money、enough to do。landing 是楼梯平台、着陆；Medicare 是美国的老年医保制度——这类专有名词你不用背，认得就行，但要知道它和 medicaid（低收入医疗补助）是两套制度。melodious 是悦耳的；mention 是提到（not to mention 更不用说）；mere 是仅仅的（a mere 三人）；metal 是金属；miserable 是悲惨的；misunderstanding 是误会；mobile 是移动的、手机；moderator 是主持人；mood 是情绪；moreover 是而且（书面）；museum 是博物馆；mutter 是嘟囔；nation 是国家、民族；naturally 是自然地、当然；nearby 是附近的；need 是需要，也是名词；neither 是两者都不（neither ... nor ...，谓语用单数）；never 是从不；nod 是点头；nonetheless 是尽管如此；notation 是记号法；notify 是通知；nurture 是养育；observer 是观察者；obtainment 是获得（很生硬的名词，一般不这么用，认得即可）；odd 是奇怪的、奇数的；offering 是提供之物；oil 是油；omission 是遗漏；online 是在线的；opening 是开口、开幕；oppressive 是压迫的、闷热的；optimum 是最佳的；orchestra 是管弦乐队；organism 是生物体；ourselves 是我们自己；outfit 是一套衣服；outwards 是向外；overflow 是溢出；overturn 是推翻；owner 是所有者。

paint 是油漆、绘画；paperback 是平装书；parallel 是平行的；participant 是参与者；

partly 是部分地；pedal 是踏板；pencil 是铅笔；persistent 是坚持不懈的；phase 是阶段；plot 是情节、密谋；plus 是加上；pointless 是无意义的；police 是警察；populous 是人口稠密的；portfolio 是作品集、投资组合；pottery 是陶器；prize 是奖品，也是"珍视"；psychology 是心理学；pure 是纯粹的；pursuit 是追求。

最后八十三张卡

qualification 是资格，也是"限定条件"——英国人说话爱加限定，一个字后面挂三个从句，那种"留退路"的写法就叫 qualification。question 是问题，也是动词"质疑"：question a decision 是质疑一个决定，不是"问一个决定"。refer 是提及（refer to），也是"转诊、参考"；refute 是驳倒，不是"否认"——deny 才是否认。respond 是回应；response 是回应（名词）；responsive 是反应快的。review 是回顾、评论；revise 是修订（英式里也是"复习"）；retrieval 是取回；resultant 是随之产生的；restoration 是恢复；reconcile 是调和；relish 是享受；reflection 是倒影、反思；recreation 是消遣；receive 是收到。

symposium 是学术研讨会，比 seminar（研讨课）大，比 conference（大会）小，正好卡在中间。secretary 是秘书，也是"部长"（美国的国务卿就是 Secretary of State）。qualification、symposium、secretary 这三个词都属于"看上去认识、用起来不敢用"的那一类，原因是你只在阅读里见过它们。解决办法只有一个：自己写一遍。

rain 是雨，也是动词"下雨"；road 是路；rock 是岩石，也是"摇滚"；water 是水；walk 是走；wheel 是轮子；trench 是沟、战壕；skyscraper 是摩天楼；terrace 是露台、梯田；venue 是场地；unit 是单位；tag 是标签；textbook 是教科书；trumpet 是小号；violin 是小提琴；tune 是曲调；verse 是诗句；tennis 是网球；tourist 是游客；tradition 是传统；tranquility 是宁静；vogue 是时尚（in vogue）。

sleep 是睡；thin 是薄、瘦；tiring 是累人的；tired 是累的——一个是"让人累"，一个是"自己累"，这又是那对 -ing/-ed。thoughtful 是体贴的、深思的；thrifty 是节俭的；tigress 是母虎；timing 是时机；together 是一起；too 是太，也是"也"（放在句末）；sometimes 是有时；temporary 是暂时的；teen 是青少年；typist 是打字员；turnover 是营业额、人员更替率；uncertainty 是不确定性；undertake 是承担；unfold 是展开；unprecedented 是前所未有的；use 是使用；utmost 是极限（to the utmost）；vague 是模糊的；variety 是多样性（a variety of）；veto 是否决权；victim 是受害者；vigorous 是精力充沛的；vital 是至关重要的；we 是我们；whether 是否；wholly 是完全地；within 是在……之内；worker 是工人；wrestle 是摔跤、努力对付；wrong 是错的；youth 是青春；running 是跑、连续的；safety 是安全；separation 是分离；serious 是严肃的、严重的；shady 是阴凉的、可疑的。

三个不写在词表里的建议

第一个建议：别按字母背，按"那天的顺序"背。这一章你读下来，脑子里有一条线：天没亮、修电扇、打印店、答辩、体检、教堂、街市、退租、交表、站台、火车、家。每一个词都挂在这条线的某一段上。你复习的时候不要打开词表，闭上眼睛走那条线，走到哪一段就想那一段有哪几个词。想不起来的，才是你真正要背的。这叫"用场景当钩子"，比"用字母当钩子"牢得多，因为字母没有意

义，场景有。

第二个建议：一个词只记一个搭配，别记五个意思。adopt 你记 adopt a method；impose 你记 impose on；consist 你记 consist of。词义是在搭配里活起来的，孤零零的中文释义是标本，不是活物。你看这一章里出现过的那些词，凡是我讲了"坑"的，都是搭配的坑：consist of 不是 consist in，depend on 不是 depend of，accuse somebody of 不是 accuse somebody for。这些不是语法，是习惯；习惯只能靠重复，不能靠推理。

第三个建议：写完再改，改的时候只看三件事。句子有没有主语，时态有没有跳，词性有没有错。你写十个句子，改完能有六个是对的，就已经比"背了三千词但一句写不出"的人强。语言是手艺，不是知识。知识可以突击，手艺只能一遍一遍做。

大词与小词

有六个词我一直留到最后，因为它们分成两类：三个是被用坏了的"大词"，三个是最见功力的"小词"。你把这两类分开，你的英文就开始有分寸了。

先说大词。revolutionary 是"革命的、颠覆性的"。这个词现在被广告用烂了：一支牙刷是 revolutionary，一个 APP 是 revolutionary。真正的 revolutionary 指的是"改变了游戏规则"，不是"比上一代好一点"。你写作文时要形容一个技术进步，如果它只是"更方便"，就别用 revolutionary，用 significant 或者 a real improvement。滥用的后果不是不准确，是别人不再信你。英语里所有"强力形容词"都有这个毛病：amazing、incredible、revolutionary、extraordinary，用一次是强调，用三次是撒谎。

romantic 是"浪漫的"，但它还有两个常见的意思，四六级阅读里都出现过。一个是"浪漫主义的"，和 classical（古典的）相对，讲的是艺术史；另一个更阴险，是"不切实际的、想当然的"——a romantic view of the countryside 不是"乡下很浪漫"，而是"对乡下抱着一种想当然的幻想"。这个义项一出现，作者的批评态度就藏不住了。阅读题问"作者对某观点的态度"，落点常常就落在这个词上，而不是落在哪句明显的话上。

ruin 是"毁掉"，比 destroy 温和一点，比 damage 重得多。destroy 是"没了"，ruin 是"废了"，damage 是"伤了"。三个词是一条线，你要能排出这条线。ruin 还有一个常用短语 in ruins（成为废墟），以及 financial ruin（破产）。写作文时想说"手机毁了我的注意力"，用 ruin 就比 destroy 准确，因为注意力不是被消灭了，是被弄坏了。

再说小词。routine 是"例行公事"，形容词也是"例行的"（a routine check）。这个词看着平淡，但它是"生活"这个词在英语里最实在的替身。你写"我的每一天"，用 routine 就比用 life 具体：my routine, my morning routine, it is part of my routine。一个能把 routine 用对的人，英文已经不像翻译了。

talk 是最小的词之一，但它的搭配极多：talk to（单向说）、talk with（对谈）、talk about（谈某事）、talk somebody into doing（说服某人做）、talk somebody out of doing（劝住某人）。talk 还可以作名词，give a talk 是"作报告"。你看，一个这么小的词，光搭配就能写半页。这就是我说"小词最见功力"的意思：大词决定你的天花板，小词决定你的地板。

tax 是税，动词是"征税"，但它的现在分词 taxing 是"费力的"——a taxing day。这个词形变

化里藏着一条规律：很多词加 -ing 之后不只表示"正在"，还表示"性质"。tiring 是"累人的"，demanding 是"要求高的"，rewarding 是"有回报的"。这三个形容词在写作里非常好用，因为它们能替掉一批平庸的表达。

把"大词"降级，把"小词"升级

我在这一章里做过一件事，你可能没注意：每次林一想用一个漂亮的大词，我就让他换成一个小词加一个具体的细节。他想说"那天非常 extraordinary"，我让他写"那天电扇的响声停了，他听见两个房间外的冰箱"。他想说"我们的友谊很 profound"，我让他写"我们买了一杯咖啡，什么也没说"。这不是文学技巧，这是英语写作的核心纪律：抽象的词负责判断，具体的词负责让人相信。你只有判断，读者不信；你只有细节，读者不知道为什么重要。两句一配，就成了。

同样的道理，你背单词的时候也要分这两类。revolutionary、magnificence、splendour 这类词，你认得就行，写作时能不用就不用；routine、tax、talk、use、need、make 这类词，你要用到不用想。考试评分看的是后者：一个把 make 用错的人，写十个 magnificence 也救不回来。

一句留给你的话

这一章写了九百二十七个词，说到底只讲了一件事：词是那天的零件，不是那天的目的。你要是读完只记住了"revolutionary 是革命的"，那我这一章白写了；你要是读完能想起那个把电扇转向墙的动作，那些词就会自己跟上来。记忆从来不靠用力，靠附着。你把它附着在一件真事上，它就掉不了；你把它附着在一张词表上，它第二天就走。

所以，别急着翻下一页。先把那一天在脑子里走一遍：天没亮，电扇在响，张野坐在地上，盒子在椅子下面，打印店的门开着，答辩教室在三楼，体检表有两页，站台上有人在广播。走到最后，你会发现自己记住的不是词，是一段日子。那才是这东西真正的用法。

三个人，三种语体

这一章里有三个人说话的方式，你留意一下，它们正好是三种语体。

张野说话最短。他不用形容词，不用从句，一句话四个字：轴承干了。这种语言在英语里叫 plain style，特点是动词多、名词少、修饰语几乎没有。你写实验记录、写操作步骤、写事故报告，用的就是这种语言。它的好处是不会错，坏处是不能表达复杂的关系。很多人以为英语好就是句子长，其实正好相反：能把一句复杂的话说短，才是本事。

沈璐说话像手术刀。她一句话里通常有一个转折，而且转折后面那句才是重点：你把它归因于机制，但你只有差异。这种语言在英语里的标志是 but、although、whereas、while 这些连接词，以及 admittedly、granted、to be fair 这些让步语。学术写作的骨架就是它：先承认对方一部分，再切进去。你写议论文，如果整篇没有一处让步，评卷的人会立刻觉得你浅。

王老师说话全是套话，但套话里有温度。这都不是事、四年之后没人记得、你要学会跟不喜欢的人合作。这种语言在英语里是 management language：it is not a big deal, let us not overthink it, you will be fine。它的价值不在信息，在关系。你要能听懂它，也要能识别它——因为有些人用这种话安慰你，有些人用这种话打发你，句子是一样的。

这三种语体你都得会一点。只会一种的人，写论文像发短信，写邮件像写论文，面试时又突然开始背模板。语体不是装饰，是场合感；场合感不是天赋，是见得多。

考场上的四条土办法

第一条：阅读里遇到不认识的词，先看它的词性，再看它的褒贬，最后才猜意思。词性看后缀：-tion、-ment、-ness 是名词，-ive、-ous、-able 是形容词，-ize、-ify 是动词。褒贬看它前后有没有 but、although、unfortunately、fortunately 这类信号词。你把这两步做完，通常已经能做题了，因为题目问的是"作者态度"和"这段话的作用"，不是"这个词的中文意思"。

第二条：不要在一个句子上停超过两遍。读第二遍还不懂，就往下走，让后面的句子来解释前面的句子。英语的说明文有一个习惯：先给一个抽象的判断，再用两三句具体的话解释它。你卡在第一句，往往是因为解释还在后面。这就是为什么"读不懂就回头"是坏习惯，"读不懂就往前"才是好习惯。

第三条：写作文时，先写主谓宾，再往里塞修饰。中国学生最常犯的错是"一边想内容一边想句型"，结果两个都没想清楚。正确的顺序是：先把这句话的骨架写出来——谁、做了什么、对谁——然后再决定要不要加从句、要不要换一个更准的动词。骨架对了，句子就对了；骨架错了，句型越漂亮越糟。

第四条：把"我觉得"删掉一半。中文作文里"我认为""在我看来"很自然，英文作文里 I think 用三次就显得虚。你直接说结论，把理由放在后面，气势就出来了。这不是让你武断，是让你把"判断"和"证据"分开：判断要短，证据要长。你去看任何一篇好的英文社论，都是这个比例。

关于"复习"这件事

最后说复习。你读完这一章，隔一天再读一遍，比连着读三遍有用；隔一周再读一遍，比隔一天读三遍有用。这不是玄学，是记忆的规律：记忆靠"提取"而不是靠"重读"。重读让你觉得熟悉，熟悉不等于记住；提取让你费力，费力才留下痕迹。

所以复习的正确动作不是"再看一遍正文"，而是"合上书，回想那一天有哪几个场景，每个场景里有哪几个词"。想不起来的，翻回去看那一个词，不要整段重读。这叫"定点补漏"，比"全面复习"省时间，也比"全面复习"有效。

还有一件事：别在同一个晚上背两类词。抽象词（concept、significance、inevitability）和具体词（cork、tub、ladder）在脑子里走的是两条路，混在一起背，两条路互相挡。一晚只攻一类，第二天换另一类，第三天再混着来。这是最笨也最稳的办法。

我说这些，不是要你把我这一章当成教材。它是一天的记录，顺便装了九百二十七个词。你要是读完能记住那一天，词是白送的；你要是只记住词，那一天就白过了。

为什么这一章非要写成一天

你可能觉得，把九百多个词塞进一天，是编出来的。确实是编的，但编得有道理，我给你讲讲这个道理，因为它对你的写作有用。

一天是一个天然的容器。它有开头、有中间、有结尾，而且它的结尾是强制的：天黑了，人累了，事情必须停。你写作文最难的是"收尾"，而"一天"自带收尾。你写"我的暑假"，收不住；你写"七月十二号那天"，自动就收住了。这是写作里最实用的一个技巧：把时间段缩小，文章就自动有了结构。你不要写"这一年我学到了什么"，你写"六月十九号下午四点到六点，我在教务处门口等一个人"。后者能装下前者。

一天还有一个好处：它逼你写动作。总结性的文章只有判断，没有动作；有动作的文章才有画面。你看这一章里，凡是抽象的段落，前面一定有一个具体的动作垫着：说"选择"之前，先写他在四点五十八分跑过走廊；说"告别"之前，先写沈璐把箱子提上电梯；说"成长"之前，先写他修好了电扇。这不是修辞，这是让读者先相信，再接受。你先给他一个能看见的东西，他才愿意听你讲道理。

我在写这一章的时候，手里有一份词表，九百二十七个词，我做的事其实就是"给每个词找一个它该出现的位置"。有的词适合出现在合同里，有的词适合出现在广播里，有的词只适合出现在一个人快睡着的时候。找位置比找释义难，也比找释义有用。因为词的意思，是位置给的。同一个 "board"，在船上、在教室里、在会议室里，是三个词。你背释义，只能背到一个；你记位置，三个都活了。

这也是我为什么在章节里放了那么多"文件"：租约、体检表、交接单、二手帖、广播稿、申请书。这些不是凑字数的道具，它们是词最自然的住处。你以后学英语，看到一份说明书、一张车票、一个药盒，别跳过，读它。那上面的英文是最规矩、最重复、最容易学的英文，比任何一篇精读课文都实在。

最后说一句我自己的体会。我写这一章写到一半的时候，发现自己忘了一个词，回去查，发现它在第三章出现过。那一刻我挺高兴的，因为它说明这个词已经不在我的"词表"里了，它在"那件事"里。词离开词表，才算学会。 你现在读到的是第八章，如果前面几章你也读了，你大概已经有几十个词是这种状态了。这就是这套读本唯一的目的。

一个提醒：别用中文当中间人

你现在的习惯大概是这样的：看到英文，先翻成中文，再从中文里取意思。这条路能走，但很慢，而且会走偏。因为两个语言不是一一对应的，你每翻一次，就丢一点东西。

举个这一章里的例子。"他把效果说成了表面化学"，中文一个"说成"什么都能接；英文你要在 ascribe、attribute、claim、put down to 之间选一个，而选哪个，取决于你想不想暗示"这个说法可疑"。中文不替你分，英文逼你分。你要是每次都先想中文，你就永远在"最好的那个词"外面绕。

同一个道理，measure 和 calculate 在你脑子里可能是同一个"算"，但一个是"量"，一个是"算"；check 和 examine 都是"检查"，但一个是"核一下"，一个是"查一遍"。look、see、watch 都是"看"，但一个是"看的方向"，一个是"看见了"，一个是"看了一会儿"。这些区别中文不标，英文标。你越早接受这件事，学得越快。

所以我建议你做一件很小的事：用英文给自己解释英文。学到一个新词，别急着记中文，先用你已有的英文词给它下一个笨拙的定义。比如 brink，你写 the edge of something high。写得笨没关系，写的过程就是把它接进你脑子里的过程。这个动作每天做五个词，一个月以后你会明显感觉到自己读句子快了——因为你不再每句都停下来翻。

这不是什么新方法，学语言的人一直都在这么干。我只是提醒你：它比背中文释义慢，但它是唯

一会累积的方法。

关于这一章的写法，最后交底

这一章我写得比前几章慢，因为它要多做一件事：把九百多个词安排在一天里，而且不能让任何一个词显得像被硬塞进去的。我的办法是先搭一天的时间轴，再按"词的脾气"分配它们。合同词只能在合同里出现，广播词只能在广播里出现，一个只在书面语里出现的词，你让它跑到对话里，读者立刻出戏。这不是挑剔，是常识：词有它的活动范围，超出范围就用错了。

你也可以用这个办法检查自己的作文。写完一段，问自己一句：这段话里有没有哪个词，放到这个场合会让人觉得别扭？如果有，多半不是词义错了，是语体错了。语体错比词义错更难改，因为词典不告诉你。词典只告诉你它是什么意思，不告诉你它属于哪个场合。场合感只能从读里来，从听里来，从改里来。

还有一句要说：这一章里的人都是我编的，但这些事大多有影子。修电扇的室友有，把报告拆成四块的学姐有，说"这都不是事"的辅导员有，站台上没说完的话也有。我写它们，不是为了让你感动，是为了让你相信这些词真的活在日子里，不是活在纸上。你信了，就记得住；你不信，背了也白背。

再说一句关于"读完"的实话。这一章你大概不会只读一遍，第一遍你会被那些加粗的词打断，第二遍你会开始注意句子，第三遍你才会看见人。三遍之后你记住的，可能只有五六个场景。那就够了。一本书能给你五六个场景，已经是很高的回报；指望它把九百多个词都钉在你脑子里，那是把书当药，不是当书。词是顺路捡的，场景才是你真正拿走的东西。等你哪天在别的地方遇到"他说这话时正在修一台电扇"这样的句子，你会突然想起这一章，那时候这些词才真正归你。

这本书如果只让你带走一句话，我希望是这一句：别把语言当科目，把它当日子过。科目要考完，日子要过下去。考完的东西会忘，过下去的东西会长在身上。你现在读到的每一个词，都是别人过过的日子里掉下来的一颗螺丝；你把它们捡起来，装到自己那台还在响的机器上，它就不响了。

最后交个底：这一章的英文里没有一句是为词表服务的。每一个词出现的地方，都是它本来就该在的地方。你读的时候要是觉得某句话别扭，多半是我没安排好；那句话你就跳过，把前后的动作记住。动作记得住，词就跟着回来了。

收束：那天真正发生的事

这一章写了一整天，从五点写到第二天早上。表面上看，林一做了这些事：修电扇、打印、答辩、体检、走过一条街、搬家、交表、送人、坐火车回家。但这些都不是那天真正发生的事。

那天真正发生的事是：一个人在心里把"我要成为一个什么样的人"这句话，从一句口号，换成了一件具体的事。他没有想明白，他只是做了选择，然后在火车上睡着了。选择不需要想明白，选择只需要签字。 这句话听着不浪漫，但它是真的，而且它比任何励志的话都更能救你——因为你会一直在等"想明白的那天"，而那天不会来。

所以我把这一章的最后一句留给了电扇。张野说"走之前把电扇修好"，林一修了，用了十一分钟，一滴油。然后他把电扇转向墙，留下一张纸条给下一个住进来的人。纸条上写着：电扇会响，是轴承，一滴油能管一年。

这就是我想让你带走的东西。不是九百二十七个词，是那种"走之前把东西修好、并且告诉下一个人哪里会响"的习惯。词会忘，习惯不会。你带着这个习惯去考试、去面试、去任何一个新的城市，都比带一本词汇书有用。

至于那些词——你已经在一天里见过它们一次了。再见一次，它们就是你的。

谢幕的话

写到最后一节，我想说点跟词没直接关系的话。

这一章叫"选择的那天"，但通篇没有一次真正意义上的"抉择时刻"。没有人在雨里站了三个小时，没有人撕掉一张表格，也没有人突然想通什么。林一做的全部事情是：跑过一条走廊，把表放进托盘，看着一个人上车。他的"选择"在四个月以前就做完了，那天只是去签个字。

这就是我想告诉你的实情：人生里的大事，很少以大事的样子出现。它们以差事的样子出现——去交个表、去盖个章、去送个人。你当时以为自己在处理杂务，几年以后回头看，那几件"杂务"就是分岔口。所以你不用等一个"决定命运的时刻"，你要做的是把眼前这件小事办得像样。办得像样，路自己会出来。

这一章里我埋了一句林一自己的话：选择不需要想明白，选择只需要签字。这句话有点狠，但它是真的。你不可能把所有信息都收集齐再决定，因为信息永远不齐；你只能在信息不齐的时候签下去，然后用后面的日子把它做对。这不是勇敢，这是成年。

还有一件事：那些词现在在你脑子里是散的，过几天会更散。别急。散着是对的，散着才说明它们已经离开了词表，进了你的记忆。你要做的不是把它们重新排成一张表，是让它们在下一次遇到的时候，你觉得"我见过这个词"。那个"见过"的感觉，就是全部。

第八章到这里结束。下一页没有词表，没有练习，什么都没有。那才是它该有的样子。

附录：覆盖自查（脚本产出，不是目测）

本书的"一个不落"由 `_准备/check.py` 逐章校验（正文覆盖 / 加粗覆盖 / 伴读覆盖 / 无题目 / 无条目），再由 `_准备/assemble.py` 汇总。下表是装配时的实测数字：

章	章名	校验项	英文词数	覆盖	加粗数	伴读汉字
第 1 章	报到与军训	928	16368	通过	927	13630
第 2 章	第一堂课与图书馆	928	19751	通过	957	13162
第 3 章	实验楼 402	927	20966	通过	927	14062
第 4 章	城市的另一面	927	16385	通过	927	14015
第 5 章	回家	927	11861	通过	927	13046
第 6 章	算法与显微镜	927	18753	通过	927	14031
第 7 章	车间里的夏天	927	17454	通过	1077	14014
第 8 章	选择的那天	927	19756	通过	951	14063

合计	7418	141294	全部通过	7620	110023

口径说明：校验项 = 词表清洗后的 7418 个词形（含 `armor | armour` 这类并列拼法，任选其一即可）。允许屈折形式（+s / +ed / +ing 等）。冠词 a / an 不要求加粗，但必须在正文出现。同行的派生词（如 `able | ability`）各自独立计数。